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Scholarships and Financial AidEach semester, Brac University is proud to honor academically talented and exceptionally skilled students with a variety of scholarships and awards. The university annually awards more than 100 million takas as scholarships to both undergraduate and postgraduate students.Know More

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Know More

Away from home...Brac University’s Residential Semester (RS) at Savar campus is unique amongst higher education experiences in Bangladesh. The RS offers a holistic curriculum based on the principle of ‘experiential learning’, which cultivates a broad range of soft skills and qualities, to compliment the theoretical development that students undergo.The Residential Semester takes place at a specially designed campus in Savar, which provides a support system that aids students in becoming confident and self-reliant.

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Message From The Founder

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Away from home...

University News and Notices

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

BRAC University plants trees, holds photo exhibition in celebration of Sir Fazle Hasan Abed's 88th birthday

BracU conveys condolences on passing of singer Sadi Mohammad

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The third day of the International Conference on Business and Management (ICBM 2019), organized by Brac Business School, Brac University was held on the 27th of April at The Westin, Dhaka Where 12 parallel sessions were held Like the first day, the session brought together prominent academicians and industry experts as well as promising participants from both home and abroad. The parallel sessions were divided into three time slots with presentations happening simultaneously in four rooms. The sessions were divided into distinct tracks so that each session focused on a particular arena of business and management. A total of 85 papers were presented on that day in front of the distinguished session chairs.

Parallel Session 4A: Operations and Supply Chain Management was chaired by Prof. Dr. Nazrul Islam and Dr. Emi Normalina Omar; 4B: (Theme: Accounting, Banking and Finance), by Prof. Dr. Amit Mittal and Mr. Mahmudul Haq; 4C: (Theme: Marketing Management), by Prof. Dr. Md Aminul Karim and Dr. Mohammad Siddiquee; 4D: (Theme: Human Resource Management), by Prof. Dr. S.M. Murali Krishna and Assoc. Prof. Mashiour Rahman.

For Parallel Session 5A: (Theme: Human Resource Management), the session chairs were Dr. Md. Mahbub Alam and Mr. Ferdoush Saleheen; for 5B: (Theme: Accounting, Banking and Finance), Prof. Dr. Md Rabiul Islam and Mr. Monojit Kumar Ojha FCA; 5C: (Theme: Marketing Management), Assoc. Prof. Dr. Mahbubur Rahman and Assoc. Prof. Lt. Col AHM Yeaseen Chowdhury, 5D: (Theme: Human Resource Management), Prof. Dr. Abhijit Das and Dr. Siti Noorsuriani Maon.

The session chairs of the last Parallel Sessions were as follows: 6A: Operations and Supply Chain Management - Prof. Dr. Anwar Hossain and Prof. Dr. Rahim B. Talukder, 6B: Maritime Management - Rear Adm ASM Abdul Baten and Cdre M Ziauddin Alamgir, 6C: Technology Management - Prof. Dr. Bishnu Prasad Mishra and Assoc. Prof. Dr.

Mohammad Rafiqul Islam and 6D: Marketing Management - Assoc. Prof. Dr. Mir Abdul Sofique and Dr. H.M. Emrul Kays.

Total fourteen (14) papers on Operations and Supply Chain Management, thirteen (13) on Accounting, Banking and Finance, nineteen (19) on Marketing Management, twenty four (24) on Human Resource Management, eight (8) on Maritime Management and seven (7) on Technology Management were presented on the second day. The sessions ended with the honorable session chairs distributing certificates to the participants and receiving certificates as a token for giving their valuable time.

Local and international participants particularly from Canada, Sweden, Singapore, Malaysia, Thailand, Taiwan, Indonesia, India, Nigeria, China, UK, presented their papers in various sessions.

BRAC University allows students from other institutions to transfer their credits. Transfer of credits from an educational institution with a system similar to BRAC University may be considered only after admission. In order to estimate which courses can be transfer to BRAC University, Students need to follow these guidelines:

The advantage of Postgraduate study is that it allows students to enter employment with additional skills and knowledge. One of the benefits of a Postgraduate research degree and the purpose of research is the output of original contributions to knowledge. The degree can demonstrate capability and tenacity to undertake an extended piece of investigative work.

A Postgraduate degree can also help to develop important transferable lifestyle skills and employability skills: such as

The degree works towards producing graduates well prepared to take a leadership role that upholds professionalism, responsibility and motivation. A Postgraduate degree offers the graduate students a continuous higher education with the preparations of scientific and professional skills. It prepares students for the future challenges and provide them with knowledge and skills that expose them to creativity and enable them to contribute to development.

BRAC University strives to enhance the national development process through the creation of a center of excellence in higher education that is responsive to society's needs. BRAC University provides opportunities in enhancing excellence in learning, teaching and research technology to achieve higher position among local and international universities in the field of higher education programs. BRAC University offers specialized programs ranging from absolute sciences to a variety of multidimensional disciplines in Liberal Arts. It provides affordable high quality education in a safe and secure environment, which enables them to excel in life.

The essential role of science and technology in modern times, be it in the realms of physical sciences, biological sciences, medical sciences, social sciences, engineering, agriculture, finance, commerce, business and management cannot be over emphasized. So it was quite natural that the Department of Mathematics and Natural Sciences (MNS) was established as one of the key departments of Brac University when it started its journey in 2001. MNS Department has been providing quality education in basic and applied sciences to the students of different disciplines of BracU including its own.

The last century belonged to physical sciences while the 21st century belongs to life sciences. However, with the discovery of Higgs boson in a Large Hadron Collider experiment at CERN, the interest in physics has been revitalised. Both fields belong to the vast arena of natural sciences. Understanding this importance, the Department of Mathematics and Natural Sciences was established as one of the key departments. It is actively involved in teaching and research in these disciplines. It offers courses in physics, applied physics, chemistry, biology, biotechnology, microbiology, mathematics, statistics, economic geography and environmental sciences, etc.

The department also offers double majors and minors in physics, APE, biotechnology, microbiology and mathematics. Most of the students pursuing the aforementioned programs are under merit scholarship.

A plasma astrophysics laboratory with major facilities has been set up for the first time in Bangladesh under the keen supervision of expert faculty. Research facilities for polymer physics under physics and applied physics disciplines are also being developed. The research programs in life sciences in the department incorporate vital areas of medical, environmental and plant biotechnology with well-equipped laboratories.

The department also has collaborative projects at various reputed research institutes like IdeSHi, BIRDEM, the University of Dhaka, BCSIR, the Bangladesh University of Health Sciences and the Centre National de la Recerche Scientifique in Grenoble, France.

The department has also signed memorandums of understanding on taking up R/D initiatives jointly with BAEC, BRAC ARDC, ICDDR,B and ACI Ltd. Many of the postgraduate students have achieved awards in international and national scientific conferences for their research.

Faculty-wise, the department is the second largest in BracU having a total of 42 full-time teachers along with 12 teaching assistants. The faculty is a blend of young and brilliant teachers with PhDs or Master’s degrees attained from abroad. Choosing teaching and research as their vocation, they bring along years of teaching and research experience at prestigious academic, research and development institutions both at home and abroad.

Programs, Majors, Minors

Faculty and Research Staff

Mahbubul Alam Majumdar, PhD

Md. Firoze H. Haque, PhD

A F M Yusuf Haider, PhD

Tibra Ali, PhD

Aparna Islam, PhD

Mahboob Hossain, PhD

Syed Hasibul Hassan Chowdhury, PhD

Mohammad Rafiqul Islam, PhD

Hasibun Naher, PhD

Sharmina Hussain, PhD

Munima Haque, PhD

Nadia Sultana Deen, PhD

Mahbubul Hasan Siddiqee, PhD

Iftekhar Bin Naser, PhD

Fahim Kabir Monjurul Haque, PhD

Mir Mozibor Rahman, PhD

Maruf Ahmed

Moushumi Zahur

Jebunnesa Chowdhury, PhD

Mahabobe Shobahani

Fardousi Ara Begum

Mohammad Mastak Al Amin

Muhammad Lutfor Rahman

Zayed Bin Zakir Shawon, PhD

Mohammad Hassan Murad

Asim Kumer Dey

Mahdi Moosa, PhD

Sanjeeda Nazneen

Romana Siddique

Mehnaz Karim

Sabrina Shahrin Sharna

Mohammad Mosaddidur Rahman

Md. Mahfuzur Rahman

Md. Saddam Hossain

Kashmery Khan

Md. Azmir Ibne Islam

Akash Ahmed

Md. Mehedi Hasan

Fahmina Akhtar

Md. Salman Shakil

Abdul Quader

Md Hasanuzzaman

Farzana Siddika

Tasmin Kamal Tulka

Sharmin Akter

Anika Ferdous

Moumita Datta Gupta

Fahmida Sultana

Sukriti Kundu

M H M Mubassir

Zubaida Marufee Islam

Mahmud Hasan

Shamira Tabrejee

S M Rakib-Uz-Zaman

Tokee Mohammad Tareq

Tabassum Rahman Sunfi

Shamima Nasrin

Md. Abdullah-Al-Kamran Khan

Saba Fatema

Samanta Saha

Muhammad Shabeeb Ameen

Md. Anwar Hossain

Md. Mohtasir Billah Sheraj

Muhammad Shahnoor Rahman

Masuda Akther

Tasnia Islam

Nourin Ferdausi

Mumtrarin Jannat Oishee

Fariha Nusrat

Abu Nasim Haider

Ahmed Rakin Kamal

Mahfuza Haque Mahi

Tushar Ahmed Shishir

Tushar Mitra

Sarah Umaymah Mahdiyah

Mahmuda Kabir

Afroza Khan

K.M. Mazharul Alam

Most. Salma Yeasmin Jannaty

Md. Rafsanjany Jim

Baby Naznin

Tonima Fairooz Mouly

Subarno Hossain Turja

Afia Maheda Prema

Nowrin Hossain

Nabila Khan

Nabiha Tasneem Khan

Aksa Hossain Nizhum

Durdana Mahin Priom

Md Tawsif Ur Rashid

News and Events

Biotechnology seminar hosts Bangladeshi scientist who developed high-yielding “Panchabrihi” rice

Orientation held for Summer 2024 biotechnology students

Nanobiotechnology seminar organized by BUSB and Biotechnology Program

The biotechnology program and BUSB organized the inaugural Biotechnology Workshop Series on “Understanding and Designing Biosensors”

Celebrating International Women’s Day 2023: Women in Science

A Virtual Webinar Celebrating International Women's Day

Science Exhibition - 2019

Monthly Seminar Talk of MNS Department for January 2019

MNS department of Brac University are going to organize an International virtual Webinar to Celebrate International Women’s Day on 24 March, 2021 (07:00 pm – 8:30 pm) bd Time.

Honorable Lady Syeda Sarwat Abed will deliver her speech as a Chief Guest and International and National Scientists will deliver speeches to motivate and inspire our students and will share important information about grants, scholarship, etc.

Programme(PDF)

A Virtual Webinar Celebrating International Women's Day

Nabiha Tasneem Khan joined as a Lecturer in the Biotechnology Program of the Department of Mathematics and Natural Sciences, BRAC University in January 2024. Before that, she was a Course Contract Faculty at BRAC University for one semester. She also worked as a Research Assistant at the Department of Genetic Engineering and Biotechnology, University of Dhaka. She did her graduation (BS) and post-graduation (MS) from the Department of Genetic Engineering and Biotechnology, University of Dhaka. She received the Dean's Award 2020 from the Faculty of Biological Sciences in recognition of her excellent academic performance. She also received the Government Scholarship 2020 for her undergraduate results. She completed her school and college from Viqarunnisa Noon School and College, Dhaka.Her MS thesis work was a combination of molecular biology, immunology, and bioinformatics. The title of her thesis is ‘Screening and Identification of NAT2 Gene Variants in Bangladeshi Population by Allele Specific Polymerase Chain Reaction Assay’. During her thesis work and graduate studies, she acquired expertise in DNA Extraction and Quantification, Polymerase Chain Reaction, Gel Electrophoresis, Western Blotting, and Bioinformatics analysis: Protein structure analysis, Molecular docking, and Molecular dynamic simulation.

Level: 4Kha-224 Merul BaddaDhaka 1212. Bangladesh

BTE 101: Introduction to Biotechnology and Genetic Engineering BTE 203: Introduction to Molecular BiologyBTE 102: Microbial WorldBTE 103: Plants and People

1. National Science and Technology (NST) Fellowship for MS research, Session 2022-23; Ministry of Science and Technology, Bangladesh2. Dean’s Award 2020, Faculty of Biological Sciences, University of Dhaka3. Government Scholarship for Bachelor of Science degree (2020), University of Dhaka

• Molecular Biology• Precision Medicine• Genetic Diseases• Molecular Diagnostics• Computational Biology

Nabiha Tasneem Khan

News & Events

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EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Accommodation & Transport

Brac University provides international students accommodation facilities and transport to and from the BracU campus.

Medical

An international student will have to undergo a medical checkup in his/her own country and submit a certificate by a physician for General Fitness Test. He/she will be required to submit a report of this medical examination to the University.

An international student will also have to undergo medical screening including a thorough check-up to a few selected clinical investigations and immunizations in his/her own country and submit the report to BracU.

Visa

BracU will assist in expediting the visa process for international students.

Download Logo: .PNG | .AI

The arc of circle suggests the wish to reach education to an international standard and milieu.

The symbolic presentation of three books towards the bottom of the design suggests different stages, levels and subjects of education. An all round broadbased education has been emphasized here. The suggestive drawing of open books stresses a disciplined and dynamic development of education. The three colors used in the design - blue (#253494), silver (#999999) and black (#000000); the size and shape of the letters that spell BRAC University, and the balanced presentation of the three books and the circle suggest that BRAC University is a strong and dynamic institution of international standard, dedicated to achieving the aims of education, peace, equality and discipline.

BRACU Logo Designed by renowned artist Mr. Hashem Khan. 2001

Day 3 of ICBM 2019 came to a close with the prize giving and closing ceremony, which also marked the end of this year’s conference.

The ceremony was graced by the presence of Mr. K M Khalid, Honorable State Minister, Ministry of Cultural Affairs, Government of the People's Republic of Bangladesh as the chief guest, Prof. Dr. Vincent Chang and Prof. Dr. Mohammad Tamim as the Special Guests, and Mr. M. Anis Ud Dowla, as Guest of Honour. Besides, the ceremony was overseen by the Session Chair, Prof. Dr. Mohammad Mahboob Rahman and the Program Chair, Assoc. Prof. Dr. Md. Mamun Habib. The panel also consisted of Prof. Dr. Hui-Ming Wee and Prof. Dr.Tzong-Ru Lee from Taiwan as the Keynote Speakers.

The ceremony started off with a vote of thanks to the invited guests and sponsors of ICBM 2019 by the Dean of BBS, Dr. Mahboob. He then expressed his gratitude to the Program Chair, Dr. Mamun Habib and his team consisting of the conference secretariat and volunteers.

Mr. Anis expressed his heartfelt gratitude towards Brac University for organising the ICBM that allowed the great minds of the country to come together under one umbrella to discuss the future challenges. He also stated that it is very important that the stake holders assemble from time to time to exchange views for the betterment of education.

Honourable Minister, Mr. K M Khalid also expressed his gratitude to Brac Business School for inviting him to the 2nd edition of the ICBM. "The industry-academia talk was a great initiative", said Mr. Khalid. "ICBM will greatly contribute towards development of business and our future. It gives students the platforms to interact with the industry and I am confident that Brac University will take more initiatives like this for the benefit of society". The Minister also thanked Prof. Dr. Mamun for his energy and dedication that kept everyone motivated throughout ICBM 2019.

It was a special attraction of the conference to distribution of certificates among the volunteers, conference secretariat, and the authors of the best papers of ICBM 2019, namely: Dr. Erne Suzila Kassim (Malaysia), Dr. Maria Fregidou Malama (Sweden), Prof. Dr. B. Chandrachoodan Nair (India) and Dr. Ireen Akhter (Bangladesh)

The conference was then brought to a close by the concluding speech of Dr. Md. Mamun Habib, Program Chair of ICBM 2019. On behalf of the organizing committee, he expressed his thanks to Almighty Allah for reaching the terminal point of this conference. It was a marvellous conference that was made possible due to the relentless efforts of its Conference Secretariat, organising committee and volunteers, he added. Dr. Habib provided the entire credit to Fazle Hasan Abed KCMG, Chairperson, BoT of Brac University for his dynamic visionary leadership. He notified the fourteen university partnership that added significant values to that conference. Besides, the event received tremendous support from its high profiled sponsors which included:

ACI Limited – Platinum Sponsor; BRAC Bank, RUNNER Group, IPDC Finance and Abdul Monem LTD. – Gold Sponsors; Mutual Trust Bank Ltd. and bKash – Sliver Sponsors; A.H. Khan & Co. and Better Business Governance (BBG), Singapore – Supporting Sponsors; The Westin, Dhaka – Hospitality Partner

The guests and participants then enjoyed themselves to a scrumptious dinner as the 2nd International Conference on Business & Management (ICBM) came to an end.

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Children and Privacy

BRACU is concerned about protecting children's privacy. We welcome children to learn more about BRAC University and about the issues presented on this website, and we are particularly respectful of the privacy of our young user. We do not collect personally identifiable information from children as well. If a parent or guardian becomes aware that a child under 13 has provided such information to us, s/he should contact us.

Policy Changes

We may, at any time, amend the provisions in this Privacy Policy. Changes to this policy will be reflected by the “Effective Date” as mentioned in the website. If we make any substantial changes in the way we use your personal information, we will notify you by posting an announcement on our website. Your continued use of this web site following the posting of changes to these terms constitutes your acceptance of those changes.

Comments

If you have any questions regarding this policy or BRACU websites in general, please contact us at wmt[at]bracu.ac[dot]bd.

Mahmud Hasan appointed as a full time lecturer of Mathematics in Mathematics and Natural Sciences Department at BRAC University since January 2015 . He has completed his graduation and Post graduation from Jahangirnagar University,Savar, Dhaka, Bangladesh. During his post graduation, he developed his thesis on Facility Layout Planning Problems. He has interest in Applied Mathematics, Operation research and management.

Level: Kha-224 Merul BaddaDhaka 1212. Bangladesh

University Scholarship on the base of Honors result, Department of Mathematics, Jahangirnagar University

Member of JU Science Club

Operation Research, Operation Management

Mahmud Hasan

News & Events

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Engaged: Student Action and Youth Leadership conference

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Workshop on Architecture Model Making Held at BRAC University

I am a mathematician with a wide array of interests. My interests in computer science are development oriented -- how can we use computer science techniques and compuational power to make the lives of Bangladeshis better? My academic interests are in string theory, early universe cosmology, quantum computing and machine learning.

For those wanting to work with me or do a thesis, an outline of my interests is below.

Computer Science

Theoretical Physics

Mathematics

Other

One of my primary goals is the guiding of the best mathematical and scientific talent in Bangladesh and the creation of a community of extremely capable Bangladeshi mathematicians, theoretical physicsists and computer scientists. I am MIT's Regional Coordinator and the National Coach of the Bangladesh Mathematical Olympiad Team. I try to contribute to nation-building by injecting technical sophistication into the business and financial spaces and supporting entrepreneurs by finding them financial support and providing technical expertise to turn their ideas into successful companies to create a modern Bangladesh.

I have taught a wide array of courses ranging from Machine Learning to Number Theory to Quantum Field Theory.

Thesis Students Supervised (B.Sc. and M.Sc.)

2017

o Md. Adib Muhtasim, Syeda Ramisa Fariha, Nabila Islam, Rayhan Rashid, “Secure Data Transaction and Data Analysis of IOT Devices Using Blockchain,” December 2017

o Ashiq Rahman, “Nonperturbative Phenomena in Field Theory and Gravity, and use of Borel Summation to Capture Nonperturbative Information” December 2017

o Reefat, “The AdS-CFT correspondence and the black hole information paradox,” March 2018

o Tareq Ahmed, “Information transmission through a quantum information channel,” December 2018

o Pranjal Chakraborty, Rashad Al Hasan Rony, Ummay Sani Pria, “Predicting Stock Market Movement using Sentiment Analysis of Twitter Feed,” April 2018.

o Paresha Farastu, Rafiduzzaman Sonnet, Saddat Hasan, Sandipon Paul “Analysis of quantum algorithms,” August 2017

o Nishat Shama, “A Machine Learning Approach to Predict Crime Using Time and Location Data,” April 2018

o Tasnim Dewan Orin, “Golpo: implementation of a Bangla chatbot” April 2018

2016

o Md. Nahid Hasan Himel (mnhhimel111[at]gmail[dot]com, Araf Iftekhar (aiadnan1993[at]gmail[dot]com), “IT Infrastructure for Economic Zone,” December 2016.

o Ipshita Bonhi Upoma, “Supersymmetric Gauge Theories as an Application of Group Theory in High Energy Physics and Starobinsky Cosmological Inflation,” April 2016.

o Noshin Nawwar Sadat, “Implementation of Time Series Approaches to Financial Data,” August 2016

Previous Years

o Mishkat Al Alvi, Avik Roy, Moinul Hossain Rahat, “A quantum information theoretic analysis of the black hole information paradox in formulating a nonlocal model for information transfer,” February 2013

o Raza Sabbir Sufian, “One-Loop Effective Action in M(atrix) Theory,” July 2010

My personal webpage is: http://whereismahbub.com . I am very fortunate to be married to Sharmeen Hossain, an investment banker, and someone with a personality orthogonal to my own.

Level: 4, Room: 4G33Kha-224 Merul BaddaDhaka 1212. Bangladesh

Journals

Alvi, M. A., Majumdar, M., Matin, M. A., Rahat, M. H., & Roy, A. (2019). Modifications of the page curve from correlations within hawking radiation. Physics Letters, Section B: Nuclear, Elementary Particle and High-Energy Physics, 797. https://doi.org/10.1016/j.physletb.2019.134881

Al Alvi, M., Matin, M. A., Rahat, M. H., Roy, A., & Majumdar, M. (2014). Comments on information erasure in black hole. Modern Physics Letters A, 29(38). https://doi.org/10.1142/S0217732314501995

Majumdar, M., & Davis, A.-C. (2004). Inflation from tachyon condensation, large N effects. Physical Review D, 69(10), 103504–103508. https://doi.org/10.1103/PhysRevD.69.103504

Majumdar, M., & Davis, A.-C. (2004). Inflation from tachyon condensation, large N effects. Physical Review D, 69(10), 103504–103508. https://doi.org/10.1103/PhysRevD.69.103504

Majumdar, M., & Davis, A.-C. (2003). D-brane anti-brane annihilation in an expanding universe. Journal of High Energy Physics, 2003(12). https://doi.org/10.1088/1126-6708/2003/12/012

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Burgess, C. P., Majumdar, M., Nolte, D., Quevedo, F., Rajesh, G., & Zhang, R.-J. (2001). The inflationary brane-antibrane universe. Journal of High Energy Physics, 2001(07), 47. https://doi.org/10.1088/1126-6708/2001/07/047

Davis, A.-C., & Majumdar, M. (1999). Inflation in supersymmetric cosmic string theories. Physics Letters B, 460(3), 257–262. https://doi.org/10.1016/S0370-2693(99)00801-1

Conferences

Chakraborty, P., Pria, U. S., Rony, M. R. A. H., & Majumdar, M. A. (2018). Predicting stock movement using sentiment analysis of Twitter feed. In 6th International Conference on Informatics, Electronics and Vision and 2017 7th International Symposium in Computational Medical and Health Technology, ICIEV-ISCMHT 2017 (Vol. 2018-Janua, pp. 1–6). Dhaka, Bangladesh: IEEE. https://doi.org/10.1109/ICIEV.2017.8338584

Muhtasim, M. A., Fariha, S. R., Rashid, R., Islam, N., & Majumdar, M. A. (2018). Secure data transaction and data analysis of IoT Devices using blockchain. In 2018 International Conference on Circuits and Systems in Digital Enterprise Technology, ICCSDET 2018. Dhaka, Bangladesh: IEEE. https://doi.org/10.1109/ICCSDET.2018.8821190

Roy, A., Rahat, M. H., Alvi, M. A., Majumdar, M., & Matin, M. A. (2012). A novel parametric bound for information retrieval from black hole radiation. In 2012 7th International Conference on Electrical and Computer Engineering, ICECE 2012 (pp. 767–770). Dhaka, Bangladesh: IEEE. https://doi.org/10.1109/ICECE.2012.6471663

After completing Ph.D., he was a post-doc at Imperial College, London. He came to Bangladesh soon afterwards. He eventually joined North South University. He joined BRAC University in 2014.

Some things people remember him for…

Mahbubul Alam Majumdar, PhD

News & Events

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Workshop on Architecture Model Making Held at BRAC University

Professor Syed Mahfuzul Aziz holds BSc and MSc degrees in Electrical and Electronic Engineering, a Ph.D. degree in Electronic Engineering from the University of Kent (UK) and a Graduate Certificate in Higher Education from Queensland University of Technology. He joined BRAC University in February 2023 as Pro Vice Chancellor and has been the acting Vice Chancellor. Winning the Prime Minister’s Award for Australian University Teacher of the year in 2009 was an important milestone in his distinguished academic career. Professor Aziz has been acknowledged for his national/international leadership and engaging teaching methods, which have been widely adopted. He has pioneered innovative Project-Based Learning and Assessment approaches to address student diversity. He has collaborated with leading European institutions to support the development of educational tools and learning resources that are used in thousands of institutions.

After graduation he worked as a research engineer at Bangladesh Atomic Energy Commission (1985-86), where he led the development of the first batch of single-board computers and embedded systems in Bangladesh for various applications. He mentored colleagues at BAEC on microprocessor-based system development. His full-time academic career commenced in 1986 as a lecturer at Bangladesh University of Engineering and Technology (BUET). He was awarded a Commonwealth Academic Staff scholarship (UK) to pursue his PhD focussing on VLSI Design (1989–1993). On completion of his PhD, he returned to his academic role in BUET and pioneered the introduction of ‘microchip design teaching’ in Bangladesh, initially through courses in Masters and then in Bachelor of Engineering programs. He was promoted to the position of Professor in 1998. During 1998–99, Prof Aziz led the establishment of a networked multi-operating system Computer Aided Design lab in BUET, the first of its kind in the country, to provide students practical training in VLSI Design. He trained and mentored academic staff in BUET and other institutions to help establish VLSI teaching programs across the country and shared his teaching materials widely.

In 1999, Professor Aziz took up the role of Program Director of undergraduate and postgraduate programs in Computer Systems Engineering at the University of South Australia (on leave from BUET) and served in that role for eight years. In January 2002, he relinquished his professorial position at BUET and continued his academic career at the University of South Australia (UniSA), where he has been a professor of Electrical and Electronic Engineering. Professor Aziz also served as academic director of common first year engineering program (2007−2012). His contributions in this role include the introduction of reflective portfolios in engineering education and strategies to support student academic success. He also served as discipline leader of Electrical and Electronic Engineering during 2013−2015 and as acting Dean in 2015. In the above roles he provided leadership for the review and benchmarking of engineering programs, quality assurance and accreditation.

Professor Aziz has been a research-active academic. At UniSA he led research projects in digital system design, low-power embedded processing architectures, and grid integration and energy economy of renewable sources and electric vehicles. His research has encompassed various sectors including energy, defence, health and the environment. He has been involved in many industry projects and attracted substantial research funds from government agencies, Australian Research Council and various industry sectors. He has 180 research publications including Quartile 1 journals which have been highly cited in recent years.

Professor Aziz takes keen interest in promoting academic excellence through strategies to enhance student experience and learning. He served as member of the Australian national teaching awards committee (2018−2021) and as assessor of educational programs and research grants internationally. He was a visiting scholar at the University of Texas at Austin (1996), invited professor at the National Institute of Applied Science Toulouse, France (2006) and visiting professor at University College London (2017). He is a Senior Member of IEEE and a member of Engineers Australia.

Professor Syed Mahfuzul Aziz, PhD

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

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Students doing BS in biotechnology can do a second major in microbiology. For this one has to complete extra 15 credits with 136 credits for a total of 151 credits. The required courses are given below:

Core Courses (39 Credits):

MIC 101 General Microbiology (Eq.-BTE 102)BCH 101 Basic Biochemistry*MIC 102 Basic techniques in Microbiology MIC 201 Microbial ChemistryMIC 202 Microbial Metabolism (Eq.-BTE 202)MIC 203 Environmental Microbiology (Eq.-BTE 304)MIC 204 Medical MicrobiologyBTE 207 Introduction to Molecular Biology (Eq.-BTE 203)BCH 301 Basic Immunology (Eq.-BTE 303)MIC 301 Virology (Eq.- BTE 411)MIC 302 Food Microbiology (Eq.-BTE 309)BTE 401 Bioinformatics*MIC 401 Microbial Genetic Engineering (Eq.-BTE 204)

Elective Courses (12 Credits):

BCH 202 Enzyme and Enzyme Kinetics*MIC 304 Microbial Biotechnology*MIC 307 Microbiology of Frozen Food and FishBTE 312 Computer Application in Biotechnology*BTE 315 Bioremediation and Biodeterioration*BTE 403 DNA Fingerprinting and Molecular Diagnostics*MIC 408 Extremophiles* refers to courses common for undergraduate microbiology and biotechnology programmes.‘Eq.’ refers to equivalent courses for undergraduate microbiology and biotechnology programmes.

Internship (3 credits):The internship should be in line with one of the majors or on some interactive topic involving the two majors.

Thesis/ Project (4 Credits):The thesis/project work should be in line with one of the majors or on some interactive topics involving the two majors.

Lab (6 credits):

As students who are doing major in biotechnology will be doing 4 different lab courses that include most of the major practical oriented problems of microbiology so they are exempted from doing any extra lab course in microbiology.

Dr. Sharmina Hussain received her Ph.D in Mathematics from the University of Dhaka in 2018. Before that she completed her M.Eng. in 2004 (January) in Mechanical Engineering from University of Victoria, British Columbia, Canada. She finished her M.Sc in Applied Mathematics from the University of Dhaka in 2000 and B.Sc (Hons.) in Mathematics from the same institution in 1998.

Level: 4, Room: 4G50Kha-224 Merul BaddaDhaka 1212. Bangladesh

Journals

Roy, N. C., Hossain, M. A., & Hussain, S. (2016). Unsteady laminar mixed convection boundary layer flow near a vertical wedge due to oscillations in the free-stream and surface temperature. International Journal of Applied Mechanics and Engineering, 21(1), 169–186. https://doi.org/10.1515/ijame-2016-0011

Hussain, S., Roy, N. C., Hossain, M. A., & Saha, S. C. (2015). Effect of fluctuating surface heat and mass flux on natural convection flow along a vertical flat plate. Mathematical Problems in Engineering, 2015. https://doi.org/10.1155/2015/258016

Hussain, S. (2012). CFD study of mass transfer in spacer filled membrane module. GANIT: Journal of Bangladesh Mathematical Society, 31, 33–41. https://doi.org/10.3329/ganit.v31i0.10306

Hussain, S. (2009). Computational fluid dynamics study of flow behaviour inmembrane module with multiple filaments. BRAC University Journal, 6(2), 1–10. Retrieved from http://hdl.handle.net/10361/467

Hussain, S. (2008). Turbulent flow over a surface mounted obstacle. GANIT: Journal of Bangladesh Mathematical Society, 28, 81–92.

Hussain, S. (2008). Computational fluid dynamics study of flow behaviour in a single spacer filled membrane module. BRAC University Journal, 5(1), 1–8. Retrieved from http://hdl.handle.net/10361/405

Hossain, M. A., Hussain, S., & Rees, D. A. S. (2001). Influence of fluctuating surface temperature and concentration on natural convection flow from a vertical flat plate. ZAMM Zeitschrift Fur Angewandte Mathematik Und Mechanik, 81(10), 699–709. https://doi.org/10.1002/1521-4001(200110)81:10<699::AID-ZAMM699>3.0.CO;2-3

Hussain, S., Hossain, M. A., & Wilson, M. (2000). Natural convection flow from a vertical

permeable flat plate with variable surface temperature and species concentration. Engineering Computations (Swansea, Wales), 17(6–7), 789–812. https://doi.org/10.1108/02644400010352261

Currently working as an Assistant Professor of Mathematics in the Department of MNS

MAT-101, MAT-102, MAT-103, MAT-104, MAT-105, MAT-110, MAT-120, MAT-215 , MAT-216,PHY-304

Emma Staff exchange Scholarship-2012, University of Victoria Fellowship-2001

Life member of NOAMI, Bangladesh Mathematical Society(BMS)

Fluid Dynamics, Computational Fluid Dynamics(CFD)

Sharmina Hussain, PhD

News & Events

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Dr. Aparna Islam is teaching at the BRAC University in Dhaka as a Professor of the Biotechnology Program. She joined the university back in 2007 as an Assistant Professor. In June 2012, Aparna became an Associate Professor; five years later she assumed full professorship. Prof. Aparna Islam received PhD in 2005 from the Jawaharlal Nehru University for her work at the International Centre for Genetic Engineering and Biotechnology (ICGEB), New Delhi. She was the first Pre-doctoral ICGEB

Fellow from Bangladesh. For her PhD, she worked on proteomics, biosafety, and transgenic crop assessment against biotic stress. Upon returning home, Dr Islam became a post-doctoral fellow for a couple of years under a USDA-funded agricultural project at the University of Dhaka. Aparna got introduced to and learnt plant tissue culture and transformation while doing MSc in Botany at the University of Dhaka. She carried out her research on peanut as a fellow of the National Science and Technology (NST) under the Ministry of Science & Technology of Bangladesh. After completing post-graduation, Aparna worked as a research fellow at the International Jute Organization (IJO) in Dhaka before leaving for the ICGEB in early 2001. Over the past 15 years, Prof. Aparna Islam has explored her passion for teaching, research, and mentoring, at the BRAC University, in three major ways. Along with teaching, she actively conducted research on developing transgenic crops with useful genes ensuring safe use of biotechnology. At the moment she is leading the research group “Plant Biotechnology and Biosafety Group”. So far, she has supervised theses of 31 students who have received their BSc/MS degrees. Dr Islam was also the Coordinator of MS Biotechnology Program for several years. Dr. Aparna Islam was the Principal Investigator of an agricultural research project funded by ‘BAS-USDA Endowment Fund’ (worth of approx. 60 lacs) under which several MS theses were completed and two labs, namely Plant Biotechnology Lab and Molecular Biology Lab, were established at the BRAC University. Finally, a former professional dancer herself, Dr Aparna Islam was also involved in co-curricular activities of her students. She was the Director of the Office of Co-curricular Activities (OCA) and the Advisor of the BRAC University Cultural Club (BUCuC). As a recognition of her contributions to the BRAC University, Dr. Aparna Islam was awarded the ‘Best Teacher 2014’ and the ‘2007-08 Vice Chancellor Award’ as Advisor of the BUCuC. Prof. Aparna Islam has been trained at home and abroad on biosafety. She has also been involved in various national projects aiming at strengthening the legal framework for biosafety. Currently, she is member of several committees of the Government of Bangladesh for biosafety's legal framework development. Aparna has been working on assessing perception of young minds about GM food as Bangladesh is in the post-release era Aparna and her co-workers have so far published 2 books, 2 book chapters, 1 Book Editing, 16 original research papers, 2 review articles, and 16 popular articles along with supervise students to write popular article 13. Prof Islam joined the South Asia Biosafety Program (SABP) as the Country Manager of Bangladesh in November 2018 and lead the USAID funded program till Feb 2021. On Feb 2021 she re-joined Brac University which is leading the online educational system in Bangladesh in this pandemic and still teaching.

Level: 5, Room: 5B14 Kha-224 Merul BaddaDhaka 1212. Bangladesh

Books

Islam, A., Ahuja, V., & Roberts, A. F. (2021). Frequently asked questions: Genetically engineered plants and biosafety. South Asia Biosafety Program, Agriculture & Food Systems Institute (AFSI), Washington, DC. https://bangladeshbiosafety.org/biosafety-book/book2/

Islam, A., & Roberts, A. F. (2020). Biosafety regulation and processes in Bangladesh: A guide for researchers in agricultural biotechnology. South Asia Biosafety Program, Agriculture & Food Systems Institute (AFSI), Washington, DC. https://bangladeshbiosafety.org/biosafety-book/book1/

Book Chapters

Islam, A., & Roberts, A. F. (2019). Global status, needs and possible areas of harmonization in agriculture and biosafety regulations in South Asia. In P. R. Pandey, B. Bajaj, & A. Islam (Eds.), Agricultural Biotechnology and Biosafety in South Asia: Progress and Prospects. SAARC Agriculture Centre (SAC). http://www.sac.org.bd/archives/publications/Agricultural Biotechnology and Biosafety.pdf

Islam, A. (2011). Udbhit gene prokoushol: agrobactariumpoddhoti. In Bongshogoti

Bidyar Mulkotha O Gene Prokoushol (pp. 305–314). Dhaka: Scholars Publications Ltd.

Journals

Datta, A., Ferdous, M.-E.-M., & Islam, A. (2022). Establishment of in planta transformation protocol of tomato (Solanum lycopersicum L.) through antiporter gene for improved salinity tolerance. Plant Science Today, 9, 16–24. https://doi.org/10.14719/pst.1764

Ferdous, M.-E.-M., Datta, A., & Islam, A. (2022). Evaluation of factors influencing in vitro regeneration and transformation protocols to produce salinity tolerant tomato (Solanum lycopersicum L.). Journal of Applied Biotechnology Reports, 9(3), 726–739. https://doi.org/10.30491/JABR.2022.336979.1519

Mimmi, S. F. H., & Islam, A. (2021). A comparative study of environment risk assessment guidelines for genetically engineered plants of developing and developed countries including Bangladesh. Science Heritage Journal, 5(2), 21–28. https://doi.org/10.26480/gws.02.2021.21.28

Datta, A., Chowdhury, M. N., & Islam, A. (2017). Transgenic crops targeting ion homeostasis machinery: Bangladesh perspective for adaptation to climate change to ensure food security. Plant Tissue Culture and Biotechnology, 27(2), 241–256. https://doi.org/10.3329/ptcb.v27i2.35029

Kashmery Khan, & Islam, A. (2017). Screening of agrobacterium mediated transient transformation of peanut (Arachis hypogaea). Journal of Plant Science & Research, 4(2), 1–5.

Riva, S. S., Islam, A., & Hoque, M. E. (2017). In vitro regeneration and rapid multiplication of Dendrobium bensoniae, an indigenous ornamental orchid. The Agriculturists, 14(2), 24–31. https://doi.org/10.3329/agric.v14i2.31341

Parveen, F., Khatun, M., & Islam, A. (2014). Micropropagation of environmental stress tolerant local potato (Solanum tuberosum L.) varieties of Bangladesh. Plant Tissue Culture and Biotechnology, 24(1), 101–109. https://doi.org/10.3329/ptcb.v24i1.19218

Razzaque, S., Mahdi, R., & Islam, A. (2014). Identification of synchronized role of transcription factors, genes, and enzymes in Arabidopsis thaliana under four abiotic stress responsive pathways. Computational Biology Journal, 2014, 896513. https://doi.org/10.1155/2014/896513

Islam, A. (2014). Establishment of a simple efficient in-vitro regeneration protocol for locally grown tomato (Lycopersicon esculentum Miller) cultivers of Bangladesh. GSTF Journal of Biosciences, 1(2), 25–30.

Islam, A., Chowdhury, J., & Seraj, Z. (2010). Establishment of optimal conditions for an Agrobacterium-mediated transformation in four tomato (Lycopersicon esculentum Mill.) varieties grown in Bangladesh. Journal of Bangladesh Academy of Sciences, 34(2), 171–179. https://doi.org/10.3329/jbas.v34i2.6863

Islam, A. (2008). Preliminary risk assessment of a novel antifungal defensin peptide from chickpea (Cicer arietinum L.). Applied Biosafety, 13(4), 222–230. https://doi.org/10.1177/153567600801300405

Islam, A., Hassairi, A., & Reddy, V. S. (2007). Analysis of molecular and morphological characteristics of plants transformed with antifungal gene. Bangladesh Journal of Botany, 36(1), 47–52.

Islam, A. (2006). Fungus resistant transgenic plants: Strategies, progress and lessons learnt. Plant Tissue Culture and Biotechnology, 16(2), 117–138. Retrieved from https://www.scopus.com/inward/record.uri?eid=2-s2.0-33846964858&partnerI...

Sarker, R. H., & Islam, A. (2000). Direct organogenesis from leaflet explants of peanut (Arachis hypogaea L.). Bangladesh Journal of Botany, 29(2), 109–114. Retrieved from https://www.scopus.com/inward/record.uri?eid=2-s2.0-33745965721&partnerI...

Sarker, R. H., Islam, M. N., & Islam, A. (2000). Agrobacterium-mediated Genetic Transformation of Peanut. Plant Tissue Culture and Biotechnology, 10(2), 137–142.

Sarker, R. H., & Islam, A. (1999). In vitro regeneration and genetic transformation of peanut (Arachis hypogaea L.). Dhaka University Journal of Biological Sciences, 8(2), 1–9.

Conferences

Chowdhury, J., & Islam, A. (2012). A comparative study on in-vitro regeneration frequency of four locally grown popular tomato (Lycopersicon esculentum Miller) varieties of Bangladesh. In 2nd Annual International Conference on Advances in Biotechnology (BioTech 2012) Proceedings (pp. 26–30). https://doi.org/10.5176/2251-2489_BioTech42

Islam, A. (2011). Climate change, emerging infectious diseases and food security. In Fifth International Botanical Conference 2011 Souvenir (pp. 44–45).

Newsletters

Islam, A. (2017). SABC 2017: A Platform for sharing experiences and paving future paths. South Asia Biosafety Program Newsletter, 14(10). https://ilsirf.org/wp-content/uploads/sites/5/2017/10/ILSIRF_SABP_Octobe...

Islam, A. (2014). Perspective on the South Asia biosafety conference. South Asia Biosafety Program Newsletter, 11(10).

*received 2007-08 'Vice Chancellor's award for outstanding contribution to co-curricular activities' and advisor of the ‘Best Club’.

*first person to receive the fellowship from Bangladesh.

FUNDED RESEARCH PROJECT

EXPERT/ EVALUATOR

ASSOCIATION WITH PEER-REVIEWED JOURNALS

Editor

Guest Editor

MS level

KEYNOTE SPEAKER

PANELIST OR DISCUSSANT

PAPER PRESENTATIONS

International

* awarded the best speaker

awarded the best speaker

National

UNB

Discussion on role of innovative technologies in addressing agricultural challenges (https://unb.com.bd/category/bangladesh/discussion-on-role-of-innovative-technologies-in-addressing-agricultural-challenges-held/87327)

The Business PostUSDA, USAID hold virtual media tour for Bangladeshi agricultural journalistshttps://businesspostbd.com/national/usda-usaid-hold-virtual-media-tour-for-bangladeshi-agricultural-journalists

Press release from US EmbassyU.S. EMBASSY IN BANGLADESH SUPPORTS VIRTUAL MEDIA TOUR FOR BANGLADESHI AGRICULTURAL JOURNALISTShttps://bd.usembassy.gov/u-s-embassy-in-bangladesh-supports-virtual-media-tour-for-bangladeshi-agricultural-journalists/

Daily SunVirtual media tour for Bangladeshi agricultural journalists heldhttps://www.daily-sun.com/post/604423/Virtual-media-tour-for-Bangladeshi-agricultural-journalists-held?fbclid=IwAR2bHh-j7Wg_btkCoQwqqB_TGIZMakCqs4BjJGsEChLMBt5p1wmaSLk8FYo

Agrilife[illegible]

ACADEMIC AWARDS

PROFESSIONAL AWARDS

NON-ACADEMIC AWARDS

INVOLVEMENT WITH LEARNED SOCIETIES

OTHER INVOLVEMENTS

JURY MEMBER AT SEMINARS/ CONFERENCE

Leading the research group “Plant Biotechnology and Biosafety Group”.

FACILITATION OF WORKSHOPS

Biotechnology

Biosafety

Biodiversity

1. Regional Workshop on Preparation of National Conservation Strategy (NCS) organized jointly by Government of Bangladesh and IUCN, Bangladesh Country Office, Barisal, 8 May 2016.2. The World Wetlands Day organized jointly by the Ministry of Environment and Forests(MoEF) and IUCN, Bangladesh Country Office, Dhaka, 2 February 2011.3. National Inception Workshop on Bangladesh Environment and Climate Change Outlook (ECCO) Report, organized by the Department of Environment (DoE), Dhaka, 28-29 Sept, 2010.4. Biodiversity National Assessment and Programme 2020 organized by the Department of Environment (DoE), Dhaka, 14 July 2010.5. Stakeholder Consultation on the 2010 Biodiversity Target National Assessment organized by the Department of Environment (DoE), Dhaka, 14 October 2009.6. Stakeholder Consultation on the Fourth National Report to the Convention on Biological Diversity organized by the Department of Environment (DoE), Dhaka, 14 October 2009.

Academician development

Others

Aparna Islam, PhD

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design

Competition

Workshop on Architecture Model Making Held at BRAC University

Mohammad Mosaddidur Rahman has completed his BSc (Honours) in Physics from Shah Jalal University of Science and Technology, Sylhet in 2006 and MSc in Sustainable Electrical Power from Brunel University, UK in 2009. He joined BracU in 2012 as a lecturer of Physics and Applied Physics programs of MNS department in 2012. Apart from regular academic duties Mr Mosaddid also working as an Advisor for First Year Advising Team (FYAT). He is also one of the advisors of BRACU Media and Journalism Forum.

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PHY110, PHY111, PHY112, PHY202, PHY210, APE101, APE205, APE407

International Conference on Nanotechnology jointly organized by University of Dhaka and BUET, December 21-23, 2012, University of Dhaka

Life Member, National Oceanographic and Maritime Institute (NOAMI)

Reactor Physics, Renewable Energy, Power Electronics

Mohammad Mosaddidur Rahman

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BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

The Department of Mathematics and Natural Sciences (MNS) in collaboration with Bangladesh Mathematical Society (BMS) organized the Regional Round (Dhaka North Region) of the 8th National Undergraduate Mathematics Olympiad- 2016 on November 12, 2016 at the BRAC University campus in the city. Students of 29 public and private universities and colleges from Dhaka North Region took part in this Olympiad where BRAC University students by their extraordinary performance have secured most of the top-level positions. A total of 5 students, out of the 10 best ones, were selected from BRAC University in this regional round. This was a regional round for selecting the most talented and brightest undergraduate students who would be taking part in the final round of the 8th National Undergraduate Mathematics Olympiad- 2016.

From this Regional Round, the 10 best performers were selected as the regional champions. A total of 60 students would be selected for competing in the final round. Students who got selected from BRAC University in this regional round are Sadat Husain (1st) from Mathematics and Natural Sciences Department, Sumaya Habib (2nd) from Computer Science and Engineering (CSE) Department, Waqqas Iqbal (3rd) from CSE Department of BRACU. The remaining two students who jointly secured the 9th position are Ishtiaq Ahmad Zim from Electrical and Electronic Engineering (EEE) Department and Maesha Armeen from Economics and Social Sciences (ESS) Department.

Bangladesh Mathematical Society (BMS) has been holding Undergraduate Mathematics Olympiad every year since 2009. AF Mujibur Rahman Foundation (a charity devoted to knowledge and education) is supporting BMS in holding the 8th National Undergraduate Mathematic Olympiad 2016.

May 2017 was the busiest and the most exciting time for ENH! The department successfully organized its much-anticipated and widely-circulated international conference on gender and literature, titled Redrawing Gender Boundaries in Literary Terrains on May 18-19, 2017. The conference had seventy-three paper presenters from national and international institutions. The inauguration ceremony was held at 9.30am on Thursday May 18 at BRACU auditorium with a welcome speech by Dr. Rifat Mahbub, Assistant Professor of ENH. The Vice-Chancellor of BRACU, Professor Syed Saad Andaleeb, honoured the guests with Uttoriyos and flowers. The VC graced the ceremony with his valuable speech where he highlighted the various exciting initiatives taken up by BRACU as such this conference. Professor Firdous Azim, Chairperson, ENH, in her welcome address highlighted the thoughts behind this conference, and thanked everyone from the VC, guests, participants, to student volunteers for making it possible. Special mention was made to City Bank and Prime Bank, the sponsors of the conference and Professor Syed Manzoorul Islam and Professor Samina Sultana for their contributions. The keynote session took place after the inauguration and the first tea break. The keynote speakers were Professor Ruth Evans from St. Louis University, Missouri, USA, and Professor Niaz Zaman, Independent University, Bangladesh. Professor Evas's paper was titled "Unrelatability: Reading Literary Works from the Past in the Feminist Present" and Professor Zaman's "The Accidental Feminist." This session was moderated by Dr. Rifat Mahbub.

In addition to academic papers, the conference included a reading session by Gantha, a creative writing platform for women, on the first day, and an open discussion by Centre for Bangladesh Studies, an independent think tank, on the topic of struggles against patriarchy in Bangladesh, on day 2. The key focus of the conference was to look at gender and its multiplicities through topics such as identity politics, gender norms and movements, bending boundaries and transnational women's writings. From literature (both Bangla and English), film and media to politics and society, ideas and thoughts were shared inside the auditorium and classrooms as well as outside over seemingly endless cups of hot tea.

An exciting session titled "S/heis S/he!!!: Queering Gender Performativity and

Transformativity in Indian Myths and Literature" was held by a team of faculty members from Jahangirnagar University. Two lectures were given on Language, Literature and Society- the first by Sowyma Dechamma, Assistant Professor, Centre for Comparative Literature, University of Hyderabad, on "Orality, Sexuality and Caste: A Case of Minority Languages" and another by Azfar Hussain, Associate Professor of Liberal Studies/Interdisciplinary Studies at Grand Valley State University and Professor of English, World Literature, and Interdisciplinary Studies at The Global Center for Advanced Studies, USA, on "Giocanda Belli and Revolutionary Feminism." The conference ended with the talk by our special guest Professor Susie Tharu, from University of Hyderabad, India. In her talk, titled, "Women Writing in India: A Jubilee Reconsideration", Professor Tharu revisited the twenty-five years of publication of her seminal work, Women Writing in India: 600 B.C. to Early Twentieth Century (Vol. I & II). This session was moderated by Rukhsana Rahim Chowdhury, Senoir Lecturer, ENH. In this well-attended conference, it was promised that selected papers would be published either in a book or a special edition of BRACU journal.

Zubaida Marufee Islam completed her school and college from BCIC School and College. She finished her B. Pharm from University of Asia Pacific in 2011. He did a Masters in Biotechnology from BRAC University. Later she did another Masters from Lund University under the Swedish Institution Study Scholarship Program.

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B. Pharm-Graduated in 2011Masters in Biotechnology- Graduated in 2014Masters in Molecular Biology, molecular Genetics and Biotechnology- Graduated in 2019

Former co-advisor of BRACU debate clubFormer FYAT advisorInternational student mentor 2018, Lund University, Sweden

BIO101, BCH102, BCH202, BTE101, BTE201, BTE202, BTE302, BTE304, BTE417, BTE255, BTE155

BLAM conference 2019, Lund University, SwedenGCSTMR (Global Circle for Scientific, Technological and Management Research) Conference 2017BSBMB (Bangladesh Society of Biochemistry and Molecular Biology) Conference 2013

• Scholarship from The Student Welfare Association• Board Scholarship for Secondary and Higher Secondary results• Vice Chancellor’s Gold Medal for outstanding result in B. Pharm• Chancellor’s and Vice Chancellor’s Gold Medal for outstanding result in MS in Biotechnology• Letter of recognition from First Year Advising Team(FYAT) of BRAC University for being a Student Advisor• Letter of recognition from MNS Department BRAC University for being an international conference organizer (Design sub-committee

• Member of Bangladesh Pharmacy Council• Member of ASM (American Society of Microbiology, Member ID:56946221)• Member of Canadian Diabetes Association (Membership ID: 8-10121482)• Member of NOAMI (National Oceanographic and Maritime Institute)

Molecular Biology, Microbiology, Protein Science

 Certificate of Honor from model United Nation 2002 Certificate of Registered A grade pharmacist (REG No. A-9070) Certificate of appreciation from BRAC University Debate club for being a Co-advisor (Effective till date) Certificate of Higher Education in Teaching (CHET) training program 2016 arranged by BRAC University

Zubaida Marufee Islam

News & Events

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Workshop on Architecture Model Making Held at BRAC University

Md. Anwar Hossain completed his BSc. From Bangladesh University of Engineering and Technology in Civil Engineering in 2015. He completed his Master’s in Theoretical Physics from University of Dhaka in 2019. Before joining Brac University MNS department, in Spring 2020 as a Lecturer, he had worked at Stamford University Bangladesh, for four years as a Lecturer in the department of Civil Engineering. His research interest is in high energy theoretical physics.

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• January 2020 – PresentLecturer,Department of Mathematics and Natural Sciences (MNS)Brac UniversityDhaka, Bangladesh.• January 2016 – January 2020Lecturer,Department of Civil EngineeringStamford University BangladeshDhaka, Bangladesh.

• PHY111: Principles of Physics I.• PHY302: Fluid Mechanics.

Civil Engineering:“Trihalomethane Formation Potential at Surface Water Treatment Plants and Effect of Ammonia on THM Formation”, International Conference on Recent Innovation in Civil Engineering for Sustainable, DUET, Gazipur, Bangladesh.

• B. Sc. Degree awarded with honors.• 1st in M. Sc.

High Energy Theoretical Physics.

Md. Anwar Hossain

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The Biotechnology Program under the Department of Mathematics and Natural Sciences and the BRAC University Society for Biotechnology (BUSB) successfully arranged the first session of the informative workshop series on Thursday, 29th February 2024. This interactive session shed light on a revolutionary area of Biotechnology named “Biosensors”. Dr. Munima Haque, Program Director and Associate Professor in the Biotechnology program, inaugurated this event with her insightful words. Dr. Munima introduced the speaker Subarno Hossain Turja. Subarno is a lecturer under the Biotechnology Program at BRAC University who completed his BSc. in Biomedical Engineering from Bucknell University, Pennsylvania, USA and MSc. also, in Biomedical Engineering from Drexel University, Pennsylvania, USA. Subarno's single-minded devotion to the biomedical field is a valuable addition to the biotechnology program as well as BRAC University.

During the session, Subarno enlightened the audience with fascinating contexts of Biosensors and associated applications. Biosensors are considered the key to the future of biotechnological advancement. The research area related to Biosensors assembles intellectuals of different academic backgrounds. BUSB arranged this session for the curious minds of all departments. Apart from the biotechnology students, CSE, and EEE major students also joined this session and quenched their thirst for knowledge on Biosensors.

The workshop provided a comprehensive understanding of biosensors, highlighting their multifaceted applications. Highlighted examples include glucose biosensors for diabetes management, environmental biosensors for pollutant detection, and DNA biosensors for genetic analysis. The workshop also shed light on the engineering principles related to biosensors, which emphasize safety, functionality, design, innovation, and sustainability. Participants explored the critical components of biosensors, primarily transducers and electronic circuits, and gained insights into their functionality in detecting changes. The interactive session not only promoted active participation but also allowed students to apply their newly acquired knowledge creatively.

The workshop on biosensors proved to be an enlightening experience, equipping students with both theoretical knowledge and practical skills, laying a foundation for their future endeavors in biotechnology.

News Archive >>

The biotechnology program and BUSB organized the inaugural Biotechnology Workshop Series on “Understanding and Designing Biosensors”

Related News

Biotechnology seminar hosts Bangladeshi scientist who developed high-yielding “Panchabrihi” rice

Orientation held for Summer 2024 biotechnology students

Nanobiotechnology seminar organized by BUSB and Biotechnology Program

Orientation held for Spring 2024 biotechnology students

Mr. Shobahani has completed his B.Sc.(hon’s) and M.Sc. degree in Physics from Shahjalal University of Science and Technology. He completed his M.Phil. degree in Solid State Physics from Bangladesh University of Engineering & Technology. He worked as a Teaching Assistant when he was a M.Phil student in Physics department of BUET and after that he joined as a Teaching Assistant in the department of Mathematics and Natural Sciences of BRAC University. Later he joined as a full-time faculty member. He carried out research work in experimental solid state physics and his interest of research areas are solid state physics, plasma and astrophysics

Level: 5, Room: 5B01Kha-224 Merul BaddaDhaka 1212. Bangladesh

• M.Phil. (Solid State Physics), BUET• M.Sc. (Physics), SUST• B.Sc. (hon’s) SUST• HSC, Dhaka Board• SSC, Dhaka Board

Journals

Shobahani, M., & Ahmed, M. U. (2014). Plasma astrophysics and experimental study of gas discharge under electric and magnetic fields. In Proceeding of the International Conference on Physics for Energy and Environment (p. 8). Dhaka, Bangladesh.

Conferences

Shobahani, M., & Ahmed, M. U. (2014). Plasma astrophysics and experimental study of gas discharge under electric and magnetic fields. In Proceeding of the International Conference on Physics for Energy and Environment (p. 8). Dhaka, Bangladesh.

Mr. Shobahani worked as a co-investigator of plasma research related to fusion and astrophysics research and is working to set up a well-equipped research laboratory. The Ministry of Education of the Government of Bangladesh is given the financial assistance to the project under a grant for advanced research in science.

• PHY310: Advanced Solid State Physics• PHY210: Quantum Physics of Atoms Solids & Nuclei• PHY204: Classical Mechanics and Special Theory of Relativity• PHY202: Optics• PHY201: Solid State Physics• APE201: Solid State Physics & Material

Sciences• PHY113: Waves, Oscillation and Acoustics• PHY112: Principles of Physics II• PHY111: Principles of Physics I• PHY110: Mechanics & Properties of Matter• PHY101: Introduction to Physics• APE101: Mechanics, Properties of Matter, Waves & Oscillations• PHY203: Physics lab II• PHY307: Physics lab III

Mr. Shobahani attended the International Conference on “Physics for Energy and Environment” and presented a paper titled as “Plasma Astrophysics and Experimental Study of Gas Discharge under Electric and Magnetic Fields” on 7 March, 2014 held at Dhaka. The conference was organized by Bangladesh Physical Society.

Mahabobe Shobahani

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

A seminar titled “Abed Chaudhury: A multi-decade journey in science covering Food, Climate and Genofax” was organized at BRAC University on 6 June 2024. The event was held as a part of a “Biotechnology Seminar Series” under the Biotechnology Program of the Department of Mathematics and Natural Sciences and in cooperation with the BRAC University Society for Biotechnology.

The guest speaker was Dr. Abed Chaudhury, a geneticist based both in Australia and

Bangladesh who is currently working at Soil Company and whose expertise spans across crop breeding, molecular biology, human nutrition and biotechnology. He has previously taught and conducted research at the Massachusetts Institute of Technology, Commonwealth Scientific and Industrial Research Organisation, agrichemical company Syngenta and Vitagrain.

During the seminar, Dr. Abed Chaudhury spoke on his three primary projects, one of which was on ensuring food security through the development of a multi-harvest strain of rice named “Panchabrihi” which he has been cultivating at his birthplace in Kanihati of Sylhet.

This rice plant, with distinguishable roots, needs to be planted just once to provide harvests five times, which is in stark contrast with existing varieties which need to be planted for a harvest each time. Dr. Abed Chaudhury also gave a comparative analysis of the high yield of this variety with other hybrids. He also showed how he used drone technology to observe the growth of rice in his paddy fields.

He also said to have founded an organization to run a crop breeding operation in Bangladesh with innovations like rice varieties with an extended shelf life and which can help maintain low blood glucose.

Afterwards, he talked about his work on fertilization-independent seed development in Arabidopsis Thaliana, a small plant from the mustard family. His other research focuses on methods for the reduction of methane production in ruminants.

Faculty members and curious-minded students asked insightful questions about his works and relevant applications. Prior to the seminar, Dr. Abed Chaudhury conducted an interview on informing that he was looking for enthusiastic minds to engage in internships and research at Genofax, of which he was the co-founder and chief scientist. He informed that Genofax was a global biotechnology company that specializes in developing AI technologies to craft personalized medicine based on individual gut microbiome gene sequences.

This is in tune with the company's vision to see a world where healthcare is tailor-made for every person with maximum precision to promote overall wellness. Several BRAC University students were able to crack the interview.

News Archive >>

Biotechnology seminar hosts Bangladeshi scientist who developed high-yielding "Panchabrihi" rice

Related News

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Nanobiotechnology seminar organized by BUSB and Biotechnology Program

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Orientation held for Spring 2024 biotechnology students

Fahmina Akhtar received both her B.Sc. and M.S. from Bangladesh Agricultural University. She secured First Class First Position in B.Sc. and achieved many awards in recognition of her academic excellence. Before joining Brac University, she spent five years as a faculty in the Department of Biotechnology at Bangladesh Agricultural University.

Level: 5, Room: 5B16Kha-224 Merul BaddaDhaka 1212. Bangladesh

Journals

Alam, M., Nur-A-Hasan, Ahsan, S., Pazhani, G. P., Tamura, K., Ramamurthy, T., … Nair,

G. B. (2006). Phenotypic and molecular characteristics of escherichia coli isolated from aquatic environment of Bangladesh. Microbiology and Immunology, 50(5), 359–370. https://doi.org/10.1111/j.1348-0421.2006.tb03802.x

M.S., H., Chowdhury, E. H., Yasmin, S., Akhtar, F., & Nasirudin, K. M. (2006). Study of genetic diversity of chilli genotypes using RAPD markers. Molecular Biology and Biotechnology Journal, 4(1&2), 27–30.

1. Lecturer, Biotechnology Program, Department of Mathematics and Natural Sciences, Brac University, 15 September, 2019 to Present.2. Teaching Assistant, Biotechnology Program, Department of Mathematics and Natural Sciences, Brac University, 14 January, 2018 to 12 September, 2019.3. Teaching Assistant, Department of Molecular Biology and Microbiology, University of Central Florida, August, 2010 to December, 2010.3. Assistant Professor, Department of Biotechnology, Bangladesh Agricultural University, June, 2006 to December, 2009.4. Lecturer, Department of Biotechnology, Bangladesh Agricultural University, March, 2004 to May, 2006.5. Research Assistant, Department of Microbiology, University of Dhaka, January, 2003 to December, 2003.

1. University Award (2003): Securing First Class First position in B.Sc. in Animal Husbandry.2. Prof. Dr. Abdus Salam Mia Trust Gold Medal (2002): Securing First Class First position in B.Sc. in Animal Husbandry & securing highest mark among the female students of all faculties for the session 1992-93.3. University Grant Commission Award (1998): Securing highest mark from Secondary School to Third year of B.Sc. in Animal Husbandry for the session 1992-93.

1. Bangladesh Association for Biotechnology (Life member)2. Plant Breeding & Genetics Society of Bangladesh (General member)3. Global Network of Bangladeshi Biotechnologists (GNOBB)

Molecular genetics, molecular biology, selective breeding.

Fahmina Akhtar

News & Events

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EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

The monthly seminar talk of the MNS department for the month of January 2019 will be presented by Dr. Fahim Haque, an assistant professor of MNS department on Thursday the 24th January, 2019 at 11:30 a.m. in room number UB 21511. The seminar details are as follows:Seminar Title: Organoids from nephrotic disease-derived iPSCs identify impaired NEPHRIN localization and slit diaphragm formation in kidney podocytes

Published in Stem Cell Reports (11:1-14. 2018 DOI: https://doi.org/10.1016/j.stemcr.2018.08.003

Speaker: Dr. FAHIM KABIR MONJURUL HAQUE

Date and time: 24th January (Thursday), 2019, 11:30am.

Place: UB21511 (MNS multipurpose room)

Abstract:

Mutations in the NPHS1 gene, which encodes NEPHRIN, cause congenital nephrotic syndrome, resulting from impaired slit diaphragm (SD) formation in glomerular podocytes. However, methods for SD reconstitution have been unavailable, thereby limiting studies in the field. In the present study, we established human induced pluripotent stem cells (iPSCs) from a patient with an NPHS1 missense mutation, and reproduced the SD formation process using iPSC-derived kidney organoids. The mutant

NEPHRIN failed to become localized on the cell surface for pre-SD domain formation in the induced podocytes. Upon transplantation, the mutant podocytes developed foot processes, but exhibited impaired SD formation. Genetic correction of the single amino acid mutation restored NEPHRIN localization and phosphorylation, colocalization of other SD-associated proteins, and SD formation. Thus, these kidney organoids from patient-derived iPSCs identified SD abnormalities in the podocytes at the initial phase of congenital nephrotic disease.

Biography of the speaker:

Fahim Kabir Monjurul Haque joined the Department of Mathematics and Natural Sciences (Microbiology Program) at BRAC University as an Assistant Professor in January 2019. Before joining BRAC University, he was working as an Assistant Professor at Wakayama Medical University, Japan. He received his PhD in Medical Sciences from Kumamoto University, Japan in September, 2017 under the HIGO Program scholarship which was funded by MEXT. He completed his Master’s (M.Sc.) and Bachelor’s (B.Sc.) in Microbiology from University of Dhaka. Immediately after obtaining his Master’s degree he joined icddr,b in Virology Laboratory. He worked there as a Senior Research Officer until he left for perusing his PhD in September 2013. Throughout his research career he has worked on developmental biology and public health related projects, such as, urogenital track development, effect of diabetes & low-testosterone in adult penile tissue homeostasis, molecular mechanism of kidney development using animal model (transgenic and conditional knockout mice), congenital kidney disease pathogenesis using patient specific iPS cell derived kidney organoids, child pneumonia etiology, clinical drug trial and HIV surveillance.

Monthly Seminar Talk of MNS Department for January 2019

Tibra Ali is a theoretical physicist with a strong research interests in string theory, quantum field theory, holographic duality and related areas. He has over a decade of experience teaching at highly reputed international educational programs including the

Perimeter Scholars International at the Perimeter Institute for Theoretical Physics, Canada. He is passionate about raising the standard of physics and mathematics education and research in Bangladesh.

Level: 5, Room: 5B02Kha-224 Merul BaddaDhaka 1212. Bangladesh

Journals

Ali, T., Bhattacharyya, A., Shajidul Haque, S., Kim, E. H., & Moynihan, N. (2020). Post-quench evolution of complexity and entanglement in a topological system. Physics Letters, Section B: Nuclear, Elementary Particle and High-Energy Physics, 811. https://doi.org/10.1016/j.physletb.2020.135919

Ali, T., Bhattacharyya, A., Haque, S. S., Kim, E. H., Moynihan, N., & Murugan, J. (2020). Chaos and complexity in quantum mechanics. Physical Review D, 101(2). https://doi.org/10.1103/PhysRevD.101.026021

Ali, T., Bhattacharyya, A., Haque, S. S., Kim, E. H., & Moynihan, N. (2019). Time evolution of complexity: a critique of three methods. Journal of High Energy Physics, 2019(4). https://doi.org/10.1007/JHEP04(2019)087

Ali, T., Haque, S. S., & Murugan, J. (2017). Holographic entanglement entropy for gravitational anomaly in four dimensions. Journal of High Energy Physics, 2017(3). https://doi.org/10.1007/JHEP03(2017)084

Ali, T., Haque, S. S., & Jejjala, V. (2015). Natural inflation from near alignment in heterotic string theory. Physical Review D - Particles, Fields, Gravitation and Cosmology, 91(8). https://doi.org/10.1103/PhysRevD.91.083516

Greenwald, J., Moore, D., Pechan, K., Renner, T., Ali, T., & Cleaver, G. (2011). On a NAHE variation. Nuclear Physics B, 850(3), 445–462. https://doi.org/10.1016/j.nuclphysb.2011.05.001

Devin, M., Ali, T., Cleaver, G., Wang, A., & Wu, Q. (2009). Branes in theMD×Md+×Md−compactification of type II string onS1/Z2and their cosmological applications. Journal of High Energy Physics, 2009(10), 95.

https://doi.org/10.1088/1126-6708/2009/10/095

Ali, T., & Cleaver, G. B. (2008). A note on a standard embedding on half-flat manifolds. Journal of High Energy Physics, 2008(7). https://doi.org/10.1088/1126-6708/2008/07/121

Ali, T., & Cleaver, G. B. (2007). The Ricci curvature of half-flat manifolds. Journal of High Energy Physics, 2007(5). https://doi.org/10.1088/1126-6708/2007/05/009

• Electricity and Magnetism• Modern Physics• Quantum Mechanics• Group Theory• Complex Analysis• Quantum Field Theory• Physics I• Physics II• Quantum Mechanics III• Statistical Mechanics • Atomic and Molecular Physics

Tibra Ali, PhD

News & Events

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Dr. Dey completed his BSc and MS degree in Applied Statistics from University of Dhaka, and he received his doctoral degree in Statistics from the University of Texas at Dallas. Dr. Dey’s research focuses broadly on statistical methods for complex networks, topological and geometric data analysis, extreme value modeling, and their applications to power systems, critical infrastructures, blockchain data analytics, climatology, and finance. Dr. Dey is a member of the American Statistical Association, Institute of Mathematical Statistics, and the Institute of Electrical and Electronics Engineers.

Level: Kha-224 Merul BaddaDhaka 1212. Bangladesh

2019 Ph.D., Statistics, The University of Texas at Dallas, TX, USA 2015 M.S., Mathematics (Statistics), Lamar University, TX 2010/2009 M.S./B.Sc., Applied Statistics, University of Dhaka, Bangladesh

Selected Publications

Book Chapters

Ofori-Boateng, D., Dey, A. K., Yulia, R. G., & Poor, H. V. (2021). Graph-theoretic analysis of power grid robustness. In A. Tajer, H. V. Poor, & S. M. Perlaza (Eds.), Advanced Data Analytics for Power Systems (pp. 175–194). Cambridge University Press. https://doi.org/DOI: 10.1017/9781108859806.011

Journals

Dey, A. K., Tian, Y., & Gel, Y. R. (2022). Community detection in complex networks: From statistical foundations to data science applications. Wiley Interdisciplinary Reviews: Computational Statistics, 14(2). https://doi.org/10.1002/wics.1566

Yuvaraj, M., Dey, A. K., Lyubchich, V., Gel, Y. R., & Poor, H. V. (2021). Topological clustering of multilayer networks. Proceedings of the National Academy of Sciences of the United States of America, 118(21). https://doi.org/10.1073/pnas.2019994118

Dey, A. K., Akcora, C. G., Gel, Y. R., & Kantarcioglu, M. (2020). On the role of local blockchain network features in cryptocurrency price formation. Canadian Journal of Statistics, 48(3), 561–581. https://doi.org/10.1002/cjs.11547

Dey, A. K., Gel, Y. R., & Poor, H. V. (2019). What network motifs tell us about resilience and reliability of complex networks. Proceedings of the National Academy of Sciences of the United States of America, 116(39), 19368–19373. https://doi.org/10.1073/pnas.1819529116

05/2022 - Assistant Professor, Department of Mathematics and Natural Sciences (MNS), Brac University

Courses at BRAC University1. STA 201: Elements of Statistics & Probability2. STA 301: Modern Probability Theory & Stochastic Processes

The 4th Annual Meeting of the SIAM Texas-Louisiana Section, November 2021. “Topological anomaly detection in a temporal transportation network”, (Invited)

The 4th International Conference on Econometrics and Statistics (EcoSta 2021), June 25, 2021. “Topological Clustering of Multilayer Networks”, (Invited).

NSF Conference on COVID19 Modelling, Jan 29, 2021. “Impacts of COVID-19 local spread and Google search trend on the US stock market”.

2020 Joint Statistical Meetings (JSM), Aug 3, 2020. “Topological and Geometric Methods for Resilience Analysis of Power Grid Networks”, (Invited).

“Multivariate Risk Modeling for Precipitation-related Home Insurance Claims”, (Invited Research Seminar) University of Maryland Center for Environmental Science, Oct 30, 2019.

2021 Summer Institute (Statistics and Modeling in Infectious Diseases) Scholarship, Department of Biostatistics, University of Washington, WA.2020, 2019 Best Paper Award, American Statistical Association (ASA) Section for Statistics in Defense and National Security (SDNS).

American Statistical Association, Institute of Mathematical Statistics, and the Institute of Electrical and Electronics Engineers (IEEE).

Statistical foundations for complex network analysis, topological and geometric data analysis, spatio-temporal processes, reliability and risk quantification, in applications to blockchain, power systems, security, environment, epidemiology, and finance.

Asim Kumer Dey

News & Events

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Workshop on Architecture Model Making Held at BRAC University

The inauguration of the 2nd International Conference on Business and Management (ICBM 2019), hosted by Brac Business School (BBS), Brac University, Bangladesh began with a solemn note as the Master of Ceremony, Adiba Naoshin requested all participants and attendees to observe a minute of silence for Fahmida Haque Labannya, a student of the Computer Science and Engineering department of Brac University whose life was tragically cut short due to a road accident on the 25th of April. Prof. Dr. Vincent Chang, Special Guest and Vice Chancellor of Brac University, respected Guest of Honor Mr. M. Anis Ud Dowla, Chairman of ACI Group; Prof. Dr. Mohammad Mahboob Rahman, Session Chair and Dean of Brac Business School, Assoc. Prof. Dr. Md. Mamun Habib, Program Chair and the 3 keynote speakers Prof. Dr. Hui-Ming Wee (Taiwan), Prof. Dr. Tzong-Ru Lee (Taiwan), Prof. Dr. Razamin Ramli (Malaysia) graced the inauguration on April 26, 2019 at The Westin, Dhaka, Bangladesh.

Program Chair of ICBM, Assoc. Prof. Dr. Md. Mamun Habib spoke first and welcomed everyone to ICBM 2019. Dr. Habib reminisced about the inception of ICBM and how his idea had grown into this unique platform. He also hoped that ICBM 2019 would provide a much needed bridge between the corporate and academic worlds. The growth and progress of ICBM was also noted and applauded. ICBM 2017 had 150 papers and 180 participants. Within a short span of 2 years the conference had widened its reach. ICBM 2019 had initially received more than 260 papers, out of which 170 were selected. The number of participants had also grown to around 300 and there were more than 40 foreign delegates present at the conference.

Special Guest and Vice Chancellor of BRAC University Prof. Dr. Vincent Chang talked about the importance of an initiative like ICBM. About the significance of ICBM 2019, Prof. Dr. Chang said, “Particularly from Bangladesh’s perspective I think it is a) timely b) relevant and c) a useful platform”. He also implored the international guests to explore the city of Dhaka and remarked that the people of Bangladesh were some of the most hospitable he had ever met.

In his short speech the Chairman of ACI Group and Guest of Honor Mr. M. Anis Ud Dowla introduced the concept of “Industry 4.0” and lauded the efforts of ICBM 2019 to bridge the gap between academia and industry. The future will be full of challenges and opportunities, he said, and everyone would need to work together to tackle them. In his closing remarks he reinforced the importance of collaboration and said, “In this era of globalization 4.0 it is inevitable for us to collaborate and share ideas to harness the potential of the emerging technology of the industry 4.0”.

In the closing statement of the inauguration, Session Chair Prof. Dr. Mohammad Mahboob Rahman spoke about the importance of being relevant and to design and conduct all business activities in a way that ultimately leads to global prosperity. He hoped all participants would take away something of value from the conference. Finally, Prof. Dr. Rahman expressed his special gratitude to the Platinum sponsor ACI Limited, Gold sponsors Brac Bank, Runner Automobiles, IPDC Finance and Abdul Monem Ltd, Silver sponsors Mutual Trust bank and bKash and the hospitality partner The Westin, Dhaka.

The conference proceedings was launched by the guests in the stage during the inaugural session. This proceedings has two versions, book format (hard copy) and soft copy format in USB drive, consists of ISBN 978-984-344-3540.

Md. Saddam Hossain is a lecturer in Mathematics at the Department of Mathematics and Natural Sciences at BRAC University. He has completed his B.Sc. (Honors) degree in Mathematics and M.Sc. in Applied Mathematics from University of Dhaka, Bangladesh. His areas of interest at present are Mathematical Modeling in Biology, Numerical Analysis, Dynamical system.

Level: 5, Room: 5B18Kha-224 Merul BaddaDhaka 1212. Bangladesh

1. “Global Stability Analysis of a Vector-Borne Disease Model” International Journal of Scientific and Engineering Research - (ISSN 2229-5518)(accepted).2. (ID: 2860048 A Dengue Transmission Model in Bangladesh Considering Migratory Population) (need to revise), Open Journal of Modelling and Simulation (Scientific Research)“Effective Control Strategies on the Transmission Dynamics of a Vector-Borne Disease” IOSR Journal of Mathematics(accepted)

Mathematical Modeling in Biology, Population Estimates and Projections, Numerical Analysis, Dynamical system.

Md. Saddam Hossain

News & Events

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Primary tabs

Faculty & Staff Dashboard

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The Department of English and Humanities aims to promote studies in English and Humanities. It fosters critical thinking and encourages students to think independently. The emphasis at the department is to inculcate reading and writing habits and to prepare students with adequate critical and analytical skills so that they can contribute to society and become an integral part of the contemporary world.

The Department of English and Humanities is the liberal arts core of the University. The Department fosters the imaginative, observational, analytical and communicative skills of its students through its three major streams: Literature, Applied Linguistics & ELT and Media and Cultural Studies. The Department also offers a Minor programme in History, which helps the students to contextualize their learning.

The Literature concentration introduces students to a wide variety of English writing from different historical periods and regions and aims to inculcate a strong analytical orientation, encouraging students to contextualize literary works in relation to their historical and cultural realities. A critical and analytical focus is encouraged through interactions with theoretical materials and writing exercises.The Applied Linguistics and ELT concentration offers an understanding and analysis of language both as an acquired skill and a socio-cultural phenomenon, introducing students to different theories of language and their application in real life for English language teaching and beyond.The interdisciplinary nature of the Media and Cultural Studies concentration equips students with an orientation based on the origin and evolution of “media”, study of cultural theories and an awareness of power-plays and world politics, enabling them to interpret the ever-changing cultural traditions and gain insights into global events and universal or personal experiences.

The undergraduate students excel in their final year, when they have to write a dissertation or a report of the internship done in a relevant organization. This exercise helps them to gain confidence and focus on the application of the knowledge and training they receive in their previous years in the University.The postgraduate programme for the degree of M.A. in English has two streams: Literature and Applied Linguistics & ELT. In their literature concentration, students get exposed to a wide variety of subjects like postcolonialism, feminism, postmodernism and the many dimensions of contemporary literature, making it a unique literature programme in Bangladesh. The Applied linguistics & ELT concentration also equips the students with a deep understanding of English as a language and the different approaches adopted towards English teaching. The introduction of the Minor in History allows students from all departments to take various History courses, making the Department of English and Humanities an interactive one, with a multi-disciplinary student body.

Faculty members represent a wide range of academic interests, imparting valuable knowledge in teaching and mentoring students in the very diverse fields across which this department spans. Specialisations range from translation studies and developing online teaching materials to research on historical and literary material. This is reflected in a wide variety of publications and seminars that have been hosted over the years.

The academic year 2016-2017 has seen talks by visiting scholars and those within the department on topics ranging from medieval Bangla writings to the latest teaching techniques or the phonological aspects of language learning. In keeping with its commitment to foster links between social and literary studies, the department hosted an international conference on “Redrawing Gender Boundaries in Literary Terrains” in May 2017. The conference was indeed well attended by scholars from home and abroad, including India and the US. The two-day conference comprised of over 50 papers, special plenary discussion sessions and readings by writers from their novels and short stories. This was a landmark conference not only for our department, but in the field of literary studies of the nation.

The ENH department is part of the Scholars’ Program, which is an initiative to draw together high-performing students from across disciplines. The program has generated a lot of interest and we are hoping to graduate the first cohort of students by the end of the Fall Semester 2017.

Situated in and fostered by the ENH department, BracU Media Lab is perhaps one of the major pioneering attempts at co-curricular activities taken up by BracU. Its aim is to give students complete access to a world of digital audio and video editing and mastering, helping to acquaint them with the digital world.

Faculty and Research Staff

Firdous Azim, PhD

Nahid Afrose Kabir, PhD

Asifa Sultana, PhD

Sabiha Huq, PhD

Md Al Amin, PhD

Sabreena Ahmed, PhD

Abu Sayeed Mohammad Noman, PhD

Roohi Huda

Gazi Md Mizanur Rahman, PhD

Mahruba Mowtushi, PhD

Rukhsana Rahim Chowdhury

Salma Khan

Mohammad Zaki Rezwan

Nawshaba Ahmed

Anika Saba

Lubaba Sanjana

Seema Nusrat Amin

Jahin Kaiissar

Samirah Tabassum

Atiya Tafannum

Departmental News and Events

ENH lecturer attains Fulbright teaching assistantship

Prof. Asifa Sultana publishes a paper titled 'What About the Indigenous Languages of the Plains?

ENH News Update, June 2024

ENH Update: April- May 2024

ENH Webinar on “A Comparative Study Between the Selected Plays of Shakespeare and Tagore"

ENH webinar on Approaches to Critical Thinking in Education

ENH Webinar on Critical Thinking in Secondary English Classrooms

ENH Event

BDT 80,000/=

Convocation Gown Fee:(refundable after return of gown)

BDT3,500/=

BDT 57,500/=

Revised RS Fee For Students Enrolling from Fall 2022 onwards: BDT 75,000/=

Convocation Gown Fee:(refundable after return of gown)

BDT3,000/=

Enrollment fee, Tuition fee, Semester fee etc.

Payment options:

Afroza Khan is a newly appointed Lecturer in the Microbiology Program of the Department of Mathematics and Natural Sciences of BRAC University from September,

2022. Currently, she is one of the Student advisors of Microbiology Program at Brac University. Before joining BRAC University, she worked as a Research Assistant in Genomics and Bioinformatics Laboratory, Microbiology, Dhaka University and she also worked as a Lab analyst in Centre for Advanced Research in Sciences (CARS). Recently, for Covid-19 Emergency Response Pandemic Preparedness, Afroza Khan has been recommended as a lab consultant officer (Microbiologist) by Directorate General of Health Services (DGHS).

Afroza Khan completed her school and college from Viqarunnisa Noon School and College. She graduated from Microbiology, University of Dhaka securing Second position, in 2019 and completed MSc, securing First class First position in Microbiology from the same institution in 2021. Her thesis was about Metagenomic sequencing Analysis of Microbiome Structure and Functional Profiling of Agricultural and Urban soils of Bangladesh. Along with this, she was involved in multiple projects on assessing Prevalence of Extended Spectrum Beta-lactamase genes in Multidrug Resistant Gram Negative Clinical pathogens along with whole genome sequencing Analysis. She is passionate about Environmental Microbiology, Immunology, Molecular Biology, Virology and Bioinformatics.She has been involved with co-curricular and extra-curricular activities since school till University level. She has been an organizer and active member of the Bangladesh Society of Microbiologists and organized several Scientific conferences, Competitions and Cultural programmes. She received prize for poetry recitation in several competitions and has been an active anchor in school, college, university programmes.

With her teaching and dynamic research ideas, she wants to motivate students, contribute to and add more values in research on life science.

Level: 5Kha-224 Merul BaddaDhaka 1212. Bangladesh

Journals

Khan, A., Ema, N. T., Rakhi, N. N., Saha, O., Ahamed, T., & Rahaman, M. M. (2021).

Parallel outbreaks of deadly pathogens (SARS-CoV-2, H5N8, EVD, Black Fungi) around East Africa and Asia in 2021: Priorities for outbreak management with socio-economic and public health impact. COVID, 1(1), 203–217. https://doi.org/10.3390/covid1010017

• Currently the Student Advisor of the Mathematics and Natural Sciences (MNS) department of BRAC University• Lab Analyst, Centre for Advanced Research in Sciences (CARS) (1 July, 2020- 31 December, 2021)• Research Assistant, Genomics and Bioinformatics Laboratory, Microbiology, Dhaka University

General Microbiology (MIC101), Microbial Metabolism (MIC202)

Oral presentation at 34th BSM International Conference (2019) on “Detection and Characterization of ESBL-Producing Pathogenic Gram-Negative Bacteria from Human Samples in Bangladesh” organized by Bangladesh Society of Microbiologists.

I. Talentpool Scholarship from University of Dhaka for academic excellence in Bachelor of Science in 2022II. Dean's award by Faculty of Biological Science, University of DhakaIII. National Science and Technology (NST) Fellowship, 2020-21.IV. Talent pool Scholarship from Directorate of Higher Secondary Education in 2016V. Best Student of the year, 2013, Viqarunnisa Noon School & CollegeVI. Vice Captain & House Captain, Viqarunnisa Noon School & College (2010-2013)

o Media & Publication Team, General member, Bangladesh Society of Microbiologistso Microbiology Blood Donation Club, Dhaka University,

• Antimicrobial resistance profiling• Virology• Immunology• Molecular Biology• Bioinformatics (Metagenome, whole genome)• Environmental Microbiology• Data Science

Afroza Khan

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

International and Scholarship Office of BracU is at the center of aspects of management and student relations, student-conduct, consultations, disciplinary action and conflict/dispute resolution. The goal of International and Scholarship Office is to establish and heighten mutual trust and acceptance within the campus community.

The team always strives to maintain a campus climate that is conducive to the educational mission of the University. The office closely interacts with University management, administration, OSA and BRAC Head Office and is responsible for the tailored communication and presentation of BracU for certain audiences. Its responsibilities include the brand positioning of BracU in the national and international ranking of universities.

The core responsibilities of International and Scholarship Office are:

To contact the International and Scholarship Office or email info[at]bracu.ac[dot]bdKha 224 Bir Uttam Rafiqul Islam AvenueMerul BaddaDhaka 1212. Bangladesh

Shamima Nasrin has completed both her B.Sc (Hons.) in Mathematics in 2015 and M.Sc in Mathematics in 2016 from Jahangirnagar University . She completed her school and college from Rajbari Govt. Girl’s High School and Rajbari Govt. College. Her teaching career started with working in Brac University under the Department of Mathematics and Natural Sciences as a Teaching Assistant(from the period 6th November, 2016 to 30th April,2017). Then she joined in the same department as a Lecturer on May 7, 2017.

Level: 4Kha-224 Merul BaddaDhaka 1212. Bangladesh

BSc and MS in Mathematics -Undergraduate completed in 2015 and Graduated in 2016 from Jahangirnagar University

• Currentiy working in Brac University as a Lecturer under the Department of Mathematics and Natural Sciences since 7th May, 2017.• Worked as a Teaching Assistant in Brac University under the department of Mathematics and Natural Sciences from the period 6th November, 2016 to 30th April,2017• Certificate in Teaching strategies for higher education, Professional Development Centre(PDC), Brac University.

MAT092 (Remedial Course in Mathematics)MAT101 (Fundamentals of Mathematics)MAT103 (Basic Concepts of Mathematics)MAT 110 (Differential Calculus & Coordinate Geometry)MAT 104 (Mathematics)MAT 105 (Calculus)MAT 120 (Integral Calculus & Differential Equations)MAT 215 (Complex Variables & Laplace Transformations)MAT 216 (Linear Algebra & Fourier Analysis)

• National Science and Technology(NST) Fellowship for M.Sc Thesis from Ministry of Science and Technology of Government Republic of Bangladesh, Dhaka.• Merit based scholarship according to B.Sc result from Jahangirnagar University.

• General Secretary, RJSWA(Rajbari district students welfare association), Jahangirnagar University.

In her thesis work was entitled “ introduce of Lumped parameter model in PDE and its flow simulation for human lung”. She would like to continue her research in this topic. Currently working on High frequency oscillatory ventilation for acute respiratory distress syndrome in adult patients.Computational fluid dynamics and numerical analysis.

Shamima Nasrin

News & Events

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EEE student selected for Huawei program in China

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Workshop on Architecture Model Making Held at BRAC University

Md. Abdullah-Al-Kamran Khan completed his school and college from Rajuk Uttara Model School and College. He finished his BS and MS from University of Dhaka in 2018 and 2019, respectively.

Level:Kha-224 Merul BaddaDhaka 1212. Bangladesh

BS and MS in Genetic Engineering and Biotechnology -Undergraduate completed in 2018 and Graduated in 2019

Journals

Islam, A. B. M. M. K., Khan, M. A.-A.-K., Ahmed, R., Hossain, M. S., Kabir, S. M. T., Islam, M. S., & Siddiki, A. M. A. M. Z. (2021). Transcriptome of nasopharyngeal samples from COVID-19 patients and a comparative analysis with other SARS-CoV-2 infection models reveal disparate host responses against SARS-CoV-2. Journal of Translational Medicine, 19(1). https://doi.org/10.1186/s12967-020-02695-0

Islam, M. S., & Khan, M. A.-A.-K. (2021). Computational analysis revealed miRNAs produced by Chikungunya virus target genes associated with antiviral immune responses and cell cycle regulation. Computational Biology and Chemistry, 92. https://doi.org/10.1016/j.compbiolchem.2021.107462

Khan, M. A.-A.-K., & Islam, A. B. M. M. K. (2021). SARS-CoV-2 proteins exploit host's genetic and epigenetic mediators for the annexation of key host signaling pathways.

Frontiers in Molecular Biosciences, 7. https://doi.org/10.3389/fmolb.2020.598583

Khan, M. A.-A.-K., Turjya, R. R., & Islam, A. B. M. M. K. (2021). Computational engineering the binding affinity of Adalimumab monoclonal antibody for designing potential biosimilar candidate. Journal of Molecular Graphics and Modelling, 102. https://doi.org/10.1016/j.jmgm.2020.107774

Islam, A. B. M. M. K., & Khan, M. A.-A.-K. (2020). Lung transcriptome of a COVID-19 patient and systems biology predictions suggest impaired surfactant production which may be druggable by surfactant therapy. Scientific Reports, 10(1). https://doi.org/10.1038/s41598-020-76404-8

Islam, A. B. M. M. K., Mohammad, E., & Khan, M. A.-A.-K. (2020). Aberration of the modulatory functions of intronic microRNA hsa-miR-933 on its host gene ATF2 results in type II diabetes mellitus and neurodegenerative disease development. Human Genomics, 14(1). https://doi.org/10.1186/s40246-020-00285-1

Turjya, R. R., Khan, M. A.-A.-K., & Mir Md Khademul Islam, A. B. (2020). Perversely expressed long noncoding RNAs can alter host response and viral proliferation in SARS-CoV-2 infection. Future Virology, 15(9), 577–593. https://doi.org/10.2217/fvl-2020-0188

Khan, M. A.-A.-K., Sany, M. R. U., Islam, M. S., & Islam, A. B. M. M. K. (2020). Epigenetic regulator miRNA pattern differences among SARS-CoV, SARS-CoV-2, and SARS-CoV-2 world-wide isolates delineated the mystery behind the epic pathogenicity and distinct clinical characteristics of pandemic COVID-19. Frontiers in Genetics, 11. https://doi.org/10.3389/fgene.2020.00765

Murad, M. W., Khan, M. A.-A.-K., Islam, M. S., & Islam, A. B. M. M. K. (2020). A switch in bidirectional histone mark leads to differential modulation of lincRNAs involved in neuronal and hematopoietic cell differentiation from their progenitors. Journal of Cellular Biochemistry, 121(5–6), 3451–3462. https://doi.org/10.1002/jcb.29619

Ahmmed, S., Khan, M. A.-A.-K., Eshik, M. M. E., Punom, N. J., Islam, A. B. M. M. K., &

Rahman, M. S. (2019). Genomic and evolutionary features of two AHPND positive Vibrio parahaemolyticus strains isolated from shrimp (Penaeus monodon) of south-west Bangladesh. BMC Microbiology, 19(1). https://doi.org/10.1186/s12866-019-1655-8

Islam, M. S., Khan, M. A.-A.-K., Murad, M. W., Karim, M., & Islam, A. B. M. M. K. (2019). In silico analysis revealed Zika virus miRNAs associated with viral pathogenesis through alteration of host genes involved in immune response and neurological functions. Journal of Medical Virology, 91(9), 1584–1594. https://doi.org/10.1002/jmv.25505

Lecturer, Brac University (January, 2020 – present)

BTE-101 (Introduction to Biotechnology and Genetic Engineering)BTE-308 (Plant Biotechnology and Genetic Engineering)BIO-101 (Introduction to Biology)

1) National Science and Technology (NST) Fellowship for MS research, Session 2018-2019; Ministry of Science and Technology, Government of the People’s Republic of Bangladesh2) Dean’s Award for outstanding performance in BS (Honour’s) Examination 2017, Faculty of Biological Sciences, University of Dhaka

Bioinformatics, Oncology, Immunology

Md. Abdullah-Al-Kamran Khan

News & Events

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Workshop on Architecture Model Making Held at BRAC University

On April 26, 2019, the first keynote session of ICBM 2019 was chaired by Prof. Dr. Salehuddin Ahmed, Brac Business School, Brac University and Prof. Dr. Hui-Ming Wee, Department of Industrial and Systems Engineering, Chung Yuan Christian University, Taiwan. The keynote speaker at this session was Prof. Dr. Tzong-Ru Lee, Department of Marketing, National Chung Hsing University, Taiwan and the Industry talk was delivered by Mr. M. Anis Ud Dowla, Chairman of ACI Group.

Prof. Dr. Tzong-Ru Lee discussed his extensive research on Purchasing Manager's Index in agriculture and block-chain technology and its many uses. He also emphasized that these technologies were useful not just for large industries but smaller entrepreneurs and even blue collar workers like farmers could benefit from them.

Mr. Anis Ud Dowla started his speech with some eye-opening facts and statistics - 40% of all jobs that currently exist will disappear in the next 10 years and 65% of the children who are now in primary school will grow up to do jobs that have not even been created. Artificial intelligence, automation and machine learning will also result in a loss of a significant number of jobs by 2030, he said. By outlining these scenarios, Mr. Dowla said that he intended to highlight the fact that the future will be full of challenges and opportunities and only through adapting to the changing times will we be able to survive. He remarked that new and disruptive technologies can take away the competitive advantage of companies and make a sector unviable within an extremely short frame of time. All of these changes are part of the 4th industrial revolution and it is being driven by multiple factors.

Mr. Dowla stressed the importance of embracing Industry 4.0, not fighting it and resetting the playing field by exploring and sharing new ideas. Towards the end of his speech Mr. Dowla detailed some steps that have to be taken by the government and policy makers, academia and industry to survive and flourish in the changing world. His final message to all the participants and attendees was, "Our preparedness to harness the potential of Industry 4.0 will define how our next generation remember us, let's act before it’s too late”.

During that session, the Westin Ball room was full of participants as well as there were a lively Q & A session.

Since its inception in 2001, Brac University has become one of the most reputed educational institutions in Bangladesh. We have focused on generating new knowledge and promoting critical thinking amongst our students, graduating more than 7,000 young men and women during this time. At present, the total number of students is around 8,000.

Our goal at Brac University is not only to provide the highest quality teaching in the region, but also to inculcate the values essential for tomorrow's leaders. Through a liberal education, we endeavour to produce graduates with the attributes needed to achieve personal success and make a valuable contribution to society. Our talented and dedicated faculty play a critical role in both these processes.

I am pleased to report that the construction of the new University campus, comprising a 14-storey building on 5 acres of land, is well under way. We hope to relocate by mid-2020

Sir Fazle Hasan Abed, KCMGChairperson Emeritus, Board of TrusteesBrac University

Ms. Saba Fatema received her B. Sc. in Mathematics and MS in Mathematics (Applied) from the University of Chittagong, Bangladesh in 2006 and 2007, respectively. To her credit, she has published several research articles, focusing on the anisotropic stellar models of electrically neutral/charged compact stars, in internationally prestigious and recognized peer reviewed journals like Astrophysics and Space Science, International Journal of Theoretical Physics, European Physical Journal C, and Annals of Physics. Currently she is continuing her M. Phil study in BUET. Before joining Brac University, in Spring 2020 as a Lecturer, she worked at the University of Information Technology and Sciences (UITS, Chittagong) for one year as a Lecturer of Mathematics, and at Daffodil International University (Dhaka) for nearly two and a half years as a Lecturer, and nearly six years as a Senior Lecturer of Mathematics, respectively.

Level:Kha-224 Merul BaddaDhaka 1212. Bangladesh

M. Phil in Mathematics (ongoing)Bangladesh University of Engineering and Technology (BUET)

Masters of Science in Mathematics (Applied)Year: 2007 (Held in 2009) University of Chittagong, Chittagong, Bangladesh.

Bachelor of Science (Honors) in MathematicsYear: 2006 (Held in 2008)University of Chittagong, Chittagong, Bangladesh.

Higher Secondary School Certificate (HSC)Year: 2001Board: DhakaRifle's Public School and College

Secondary School Certificate (SSC)Year: 1999Board: DhakaArgani School and College(Previously known as Argani Girls School)

January 2020–presentLecturer of MathematicsDepartment of Mathematics and Natural SciencesBrac UniversityDhaka, Bangladesh

February 2014–January 2020Senior Lecturer of MathematicsDepartment of General Educational Development(Previously known as Department of Natural Sciences) Faculty of Sciences and Information TechnologyDaffodil International UniversityDhaka, Bangladesh

September 2011–January 2014Lecturer of MathematicsDepartment of Natural SciencesFaculty of Sciences and Information TechnologyDaffodil International UniversityDhaka, Bangladesh

September 2010–September 2011Lecturer of MathematicsSchool of BusinessUniversity of Information Technology and Sciences (UITS)Chittagong, Bangladesh

Throughout her teaching career, she taught several courses at the undergraduate level at different universities. She is currently teaching the following courses:

Period 2020–

MAT101 Fundamentals of MathematicsAn introductory course for the students of

Business, Economics, Microbiology, Biotechnology, and Pharmacy programs.

MAT215 MATH-III: Complex Variables & Laplace TransformsA second-year applied mathematics course designed for engineering students.

Period 2010–2011

MAT101 Mathematics-IAn introductory mathematics course for business students.

MAT112 Business MathematicsA course on linear algebra and optimization problems using calculus for business students.

Period 2011-2020

MAT101/ MAT111 Mathematics-ISingle variable calculus course for computer science and engineering, electrical and electronic engineering, software engineering, textile engineering, and nutrition & food engineering students.

MAT114 Basic Mathematics and StatisticsA basic mathematics and statistics course designed for pharmacy students.

MAT121/ MAT122 Mathematics-IIAn analytic geometry, complex variable and linear algebra course designed for computer science and engineering, electrical and telecommunication engineering, software engineering, and textile engineering students.

MAT131/ MAT134 Mathematics-IIIAn ordinary and partial differential equations course designed for computer science and engineering, electrical and telecommunication engineering students.

MAT211 Mathematics-IVA vector analysis, Laplace transforms, and Fourier analysis course designed for computer science and engineering, electrical and telecommunication engineering students.

MAT203 Mathematics-IVA vector analysis and statistics course designed for civil engineering students.

Enlisted artist of Rabindrasangit at Bangladesh Betar.

Analysis, Differential Equations, Linear Algebra, Mathematical Physics, Fluid Dynamics.

Saba Fatema

News & Events

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EEE student selected for Huawei program in China

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Workshop on Architecture Model Making Held at BRAC University

An orientation program for first-year undergraduate students of Summer 2024 semester was organized under the Biotechnology Program of the Department of Mathematics and Natural Sciences (MNS) in cooperation with BRAC University Society for Biotechnology (BUSB) at BRAC University on 8 June 2024.

The event started off with Dr Munima Haque, Director of the Biotechnology Program, welcoming the freshers and informing of different teaching, research and academic facilities offered under the program.

Joining virtually, Professor Mahbubul Alam Majumdar, Dean of the School of Data and Sciences, shared opportunities available under the program, ways to collaborate with other programs under the MNS department and how students should always reach out to faculty members for assistance.

A video was screened giving an introduction to the Biotechnology Program. Professor

Dr. Aparna Islam, Assistant Professor Dr. Mahdi Moosa and Associate Professor Dr. Iftekhar Bin Naser shared insights from their own research careers.

An alumnus, Rezwanul Kabir, spoke of his time as a student. BUSB President Md. Nafin Rahman enlightened the newcomers about the diversified aspects of both the Biotechnology Program and the student club.

The event ended with a cultural program comprising delightful songs, poetry recitations, and dance performances.

Related News

Biotechnology seminar hosts Bangladeshi scientist who developed high-yielding “Panchabrihi” rice

Nanobiotechnology seminar organized by BUSB and Biotechnology Program

The biotechnology program and BUSB organized the inaugural Biotechnology Workshop Series on “Understanding and Designing Biosensors”

Orientation held for Spring 2024 biotechnology students

Professor Syed Mahfuzul Aziz, PhDPro Vice-Chancellor and Acting Vice-Chancellor Email: pvc[at]bracu.ac[dot]bd

Professor Mohammad Mahboob Rahman, PhDTreasurerEmail: mahboob.rahman[at]bracu.ac[dot]bd

Dr David DowlandRegistrar & Controller of Examinations (acting)Email: registrar[at]bracu.ac[dot]bd, controller[at]bracu.ac[dot]bd

Professor Fuad H. Mallick, PhDDean, School of Architecture and DesignEmail: fuad[at]bracu.ac[dot]bd

Professor Arshad M. Chowdhury, PhDDean, School of Engg. Email: arshad.chowdhury[at]bracu.ac[dot]bd

Professor Mahbubul Alam Majumdar, PhDDean, School of Data and Sciences Email: majumdar[at]bracu.ac[dot]bd

Samia Huq, PhDDean, General EducationEmail: shuq[at]bracu.ac[dot]bd

Professor Eva Rahman Kabir, PhDDean, School of PharmacyEmail: eva.kabir[at]bracu.ac[dot]bd

Barrister Manzoor HasanExecutive Director, Centre for Peace and JusticeEmail: manzoor.h[at]bracu.ac[dot]bd

Professor K. Shamsuddin MahmoodDean, School of LawEmail: kshamsuddin[at]bracu.ac[dot]bd

Erum Mariam, PhDExecutive Director, Brac Institute of Educational DevelopmentEmail: erum[at]bracu.ac[dot]bd

Imran Matin, PhDExecutive Director, Brac Institute of Governance and Development Email: imran.matin[at]bracu.ac[dot]bd

Ariful IslamChief Financial OfficerEmail: arif.islam[at]bracu.ac[dot]bd

Rubana Ahmed, PhDUniversity ProctorEmail: proctor[at]bracu.ac[dot]bd

Ms. Hasina AfrozUniversity LibrarianEmail: hasina[at]bracu.ac[dot]bd

Brac Institute of Languages (BIL) of Brac University is working with the goal to assist students in developing the language skills they need to become successful in graduate schools and in their professional lives. BIL focuses on implementing student-centered and creative language teaching techniques by developing modules based on students' proficiency level. This not only helps avoid having mixed ability classes, but also ensures small classes with excellent interaction between teachers and learners.

Pre-University is a unique course designed for those students who proved themselves competent in their respective disciplines during the admission test except in English. Besides, MA in TESOL, the flagship programme of BIL, is creating efficient English language teachers and leaders from around the country with its well-organized curriculum, and innovative and practice-oriented teaching approach.

BIL had conducted a three-week long English language development course for 'Medhabikash' and Training for RAI Foundation Scholarship Recipients. It had worked with BRAC's STAR project in collaboration with UNICEF and ILO for the capacity development of the outreach students who are currently involved in Technical and Vocational Education and Training (TVET). Together with BRAC, ILO and Bureau of Manpower (Bangladesh), BIL had worked on establishing the language labs for the project titled 'Promoting Decent Work through Improved Migration Policy and its Application in Bangladesh'. In collaboration with BRAC Institute of Global Health (BIGH), BIL had developed six modules on English for Specific Purposes (ESP) for the 'Diploma in Midwifery' course. BIL had also collaborated with the English in Action project aided by the Open University, UK in their MYTEC programme.

The strengths that the faculty members of BIL reflect are their interests and work experiences in diverse spheres of academia, such as assessment, reflective teaching, peer review, critical and creative thinking, adult teaching, teacher training, experiential learning, English for specific purposes, English for academic purposes, curriculum & syllabus design, material development, education and society, use of literature in language teaching, computer assisted language learning, etc.

ContactBRAC Institute of Languages (BIL)Level: 5Kha-224 Merul BaddaDhaka 1212. Bangladesh

Contact

BRAC Institute of Languages (BIL)Level: 5Kha-224 Merul BaddaDhaka 1212. Bangladesh

Welcome to Brac Institute of Languages

News and Events

BIL News Bulletin, May 2024

BIL Bulletin: March-April, 2024

BIL News Bulletin, February, 2024

BIL organizes workshop "Communicate Yourself into Research: How to Avoid Predatory Journals and Conferences in the Process”

5th Intra University Moot Court Competition- 2024

Exploring New Contextualism 2.0: An Approach to Architecture and Urban Design

LSDB’s Regional Education Award 2023 (LREA-2023)

Certificate Course on ‘Intellectual Property for Innovation, Creativity and Branding’

Faculty and Research Staff

Lady Syeda Sarwat Abed

Professor Shaila Sultana, Ph.D

Ms. Liza Reshmin

Mr. Mohammad Rafiqul Islam, PhD

Mr. Mohammad Aminul Islam

Ms. Nazneen Zafar

Mr. Md. Raisul Islam Chowdhury

Ms. Evita Umama Amin

Mr. Pankaj Paul

Ms. Hasna Khanom

Mr. Md. Abu Sufian

Mr. Imam Zafar Numanee

Mr. Abdul Karim

Ms. Effat Hyder

Mr. Md. Mahbubul Islam

Ms. Farina Haque

Ms. Tanjila Ferdous

Ms. Srabosti Saha

Ms. Fariah Amin

Ms. Marwa Mohammad Masood

Ms. Nipa Nasrin

Ms. Tanzima Baten

Ms. Nafisa Moquit

Ms. Tasnova Humaira

Ms. Shehneela Naz

Mr. Farid Ahmed

Mr. Kazi Abu Bakar Siddique

Ms. Mehetaz Chowdhury

Ms. Sarahnaz Kamolika Rahman

Mr. Kushal Das

Mr. Moshiur Rahman

Mr. Md. Julhas Uddin

Ms. Tarannum Fatema Chowdhury

Mr. Easir Arafat

Ms. Farjana Afrin

Mr. Yusha Alam

Ms. Ghazzala Shabi

Mr. Md. Sohan Shahriar

Ms. Tanjeela Osmani

Ms. Tanjena Sultana Khan

Mr. John Paul Shimanto Sarkar

Ms. Masura Kamal Peuly

Ms. Samira Osmani

Ms. Bineeta Rahman

Ms. Sanzida Pias

Qingshan Huang

Ms. Nazia Masood

Jannatul Ferdous

Ms. Nashrah Sharfuddin

Ms. Sabah Tasnia Rowshon

Mr. Shahnawaz Hossain

The admission information is available in the following places:i) BRACU website (www.bracu.ac.bd).ii) Inquiry, Registration & Admissions Office, BRACU (Phone: +88 09638464646, Call Center ext: 1001) One can also get admission information by sending queries at the e-mail

address:

BRACU offers 3 semesters a year: Spring, Summer & Fall, and these start in January, May & September respectively.

For Pharmacy & LLB: BRACU offers admission for Pharmacy & LLB department twice a year. You can apply in for Pharmacy: Spring or Summer semester and for LLB: Spring or Fall. The classes start from January and July for Pharmacy. January and September for LLB accordingly.

Tentative Schedule - Admission Tests:

Semester

1st Intake

2nd Intake

Spring

November

December

Summer

March

April

Fall

July

August

The available Undergraduate programs with credits are as follows:B. Arch – 207 CreditsBBA – 130 CreditsBSc in CSE – 136 CreditsBSc in CS – 124 CreditsBSc in ECE – 136 CreditsBSc in EEE – 136 CreditsBSS in Anthropology - 120 CreditsBSS in Economics - 120 CreditsBA in English - 120 CreditsLL.B (Hons) - 135 CreditsBSc in Physics - 132 CreditsBSc in Applied Physics and Electronics - 130BSc in Mathematics – 127 CreditsBSc in Microbiology – 136 CreditsBSc in Biotechnology – 136 CreditsBachelor of Pharmacy (Hons) - 164 CreditsQ. What Postgraduate programs does BRAC University offer? How many credits do I have to complete per program?

The available Postgraduate programs with credits are as follows:ECD – 36 creditsEMBA – 45 creditsMA ENG – 36 creditsMA TESOL – 39/From non-English graduate 45 creditsMBA – 60 creditsMSAE - 36 creditsMDS – 45 creditsMED – 36 creditsMPH – 51 creditsMPSM – 48 creditsMS BIO – 36 creditsMS EEE – 36 creditsMSC CSE – 36 creditsMDM – 36 credits

MAGD - 36 credits

The available Postgraduate programs with credits are as follows:ECD – 36 creditsEMBA – 45 creditsMA ENG – 36 creditsMA TESOL – 39/From non-English graduate 45 creditsMBA – 60 creditsMSAE - 36 creditsMDS – 45 creditsMED – 36 creditsMPH – 51 credits (Programs>Postgraduate)MPSM – 48 creditsMS BIO – 36 creditsMS EEE – 36 creditsMSC CSE – 36 creditsMDM – 36 credits

MAGD - 36 credits

BRACU website to have precise information about admission procedures.

Applicants can apply only online for admission (admissions.bracu.ac.bd).

International Applicants must contact the and can apply only online for admission (admissions.bracu.ac.bd). The application process is the same for all applicants

Candidates for EEE, ECE, CSE, APE and Physics must have minimum B grade in Physics & Mathematics in HSC/Equivalent and with minimum C grade in Physics & Mathematics at 'A'-Levels.

Candidates for BSc in Electronic & Communication Engineer and Electrical & Electronic Engineering having Physics and Mathematics but not Chemistry at HSC or 'A'-Levels must take a remedial course on Chemistry in addition to the courses required for the program.

Candidates for BSc in Computer Science and BSc in Mathematics must have minimum B grade in Mathematics at HSC or minimum C grade in Mathematics in 'A'-Levels.

Candidates for BSc in Biotechnology and Microbiology should have minimum C grade in Biology and Chemistry at HSC/ A-Levels/Equivalent. Candidates not having Mathematics at HSC or A-Levels must take a remedial course on Mathematics in addition to the courses required for the program.

Our requirement for undergraduate programs is to have a minimum GPA of 2.5 in ‘O’ Level and ‘A’ Level separately. For ‘O’ level best 5 subjects and ‘A’ level best 2 subjects will be counted. The grade will be calculated according to BRAC University scale: A=5, B=4, C=3 and D=2. Subjects with E Grade will not be considered. It is to be noted that applicants must fulfill subject-wise requirements.

You can check the below box for your help:

O/A Levels

Example (Best 5 for 'O' Level)

Grade Letter

As per following Scale

Total

Cumulative Grade

A

5

17

17/5=3.4

B

4

C

3

D

2

C

3

Applicants can check their application status online any time at admissions.bracu.ac.bd

Applicants are required to upload their scanned recent formal passport size photograph taken within last three months.

If applicants make any mistake during filling up their application form and submit it with that error, they are required to send an email immediately at to inform the authority. But till submission you can edit the form any time.

Local Applicants need to pay the non-refundable application fee Tk. 1500. For international applicants, the non-refundable application fee is US$ 20.

Applicants can pay their admission application fee through-

All candidates will have to sit for a written admission test and an interview in the manner as described:

Program

Test section

BSS in Anthropology, BA in English and LLB (Hons)

English

BBA and BSS in Economics

English and Mathematics

Bachelor of Pharmacy, BSc in Microbiology and BSc in Biotechnology

English, Mathematics and Biology & Chemistry

BSc in Computer Science, BSc in Computer Science and Engineering, BSc in Electronic and Communication Engineering, BSc in Electrical and Electronic Engineering, BSc in Physics, BSc in Applied Physics & Electronics, BSc in Mathematics

English, Mathematics,

Higher Mathematics and Physics

Bachelor of Architecture

English, Mathematics and Drawing

In order to qualify, candidates must pass each section separately. Candidates who qualified in pre-university on the basis of written tests, will be called for an interview for final selection. In order to qualify, candidates must pass each section separately. Those who qualify in the written test will be called for interview for final selection.

For each postgraduate program, except EMBA and MEd, the admission test consists of a written test and an interview. Candidates for EMBA and MEd are required to qualify only in the interview.

For getting admission into BRACU, International applicants must appear for written test and interview on the date and time mentioned in their admit cards in the designated exam site in their respective home countries.

Those who have a minimum of 3-years of professional work experience (After Graduation) can apply for the EMBA Program.

After the deadline of submitting the Admission Form, candidates will be notified through SMS to collect the admit card. They will be required to login to their account to collect the Admit Card. There will be an option named 'Ädmit Card' at the left side. They are required to download the Admit Card from that option and print it out in an 80 gm A4 size white page. Applicants must bring this printed Admit Card to the exam hall; otherwise, they will not be allowed to sit for the exam.

After the publication of the admission test result, applicants are required to check the result on the BRACU website.

Regarding all information about Financial Aid, please contact:

BRACU (Phone: +88 09638464646, Call Center Ext: 1001 or email at

(BRACU Scholarship Office)

Transfer of credits from an educational institution with a system similar to BRAC University may be considered after admission. Such candidates will have to apply with required documents and are subject to credit transfer rules of BRAC University.

We do not offer online courses or degree programs.

Many students willing to join the university might not have the required standard of proficiency in English language. If the students fail to attain the minimum standard of English proficiency required by the university, they may be asked to attend remedial English courses before the admission to the university. At the end of this course, they will have to take an English proficiency test and if qualified be admitted to BRACU.

A Pre-University course is for those students who proved themselves competent in their respective discipline during the admission test except in English. This is a twelve week long intensive program which assists students to become active and independent learners, critical thinkers, responsible and thoughtful adults, and most importantly successful university students. Many of the students who come from the mainstream Bangla medium background enjoy this intellectually challenging and welcoming atmosphere.

One can enroll as a visiting or guest student in BRACU. Visiting or guest students are required to submit their applications along with the relevant transcripts and no objection letters (NOC) from her/his University. Please contact for more details.

All students are required to attend a Residential Semester at the Savar Campus in the 3rd semester. The cost is Tk. 80,000 for the 4-month long semester. (*May increase with a notice before the semester)

Candidate must submit copies of your previous academic records which have to be verified from their respective institutions or attested from your respective Embassy/High Commission/Foreign Ministry and equivalence certificate from University Grants Commission (UGC) or Ministry of Education, Government of The People's Republic of Bangladesh before the end of the first semester.

There is no evening class for undergraduate programs.

All the classes of Post graduate programs are held in the evening. There is a Weekend

Program for EMBA and MSc CSE students.

Freshers can change the department before the class starts. They need to contact the Admissions Office.

Accommodation facilities are provided only to female student's subject to availability of seats. For details, please contact

Yes, BRACU () will assist in applying for the visa as needed by a foreign student.

For more queries, please contact the Inquiry, Registration & Admissions Office, BRACU (Phone: +88 09638464646, Call Center Ext: 1001. You can also send e-mail at .

National and International conferences, hosted by Brac University.

National and International conferences, hosted by Brac University.

ICBM 2019

ICEPE 2019

ICBM 2017

International conference on "Redrawing Gender Boundaries in Literary Terrains"

Department of English and Humanities organises International Conference on First Language Acquisition

Inter-University Conference on New Directions in Literary Research

19th International Mathematics Conference, Dhaka, 18-20 December 2015

Brac University Hosts 7th ICCIT

Math Olympiad in BracU

8th National Undergraduate Mathematics Olympiad- 2016

To be the leading global public health institute for the world's critical health challenges affecting disadvantaged communities.

To create innovative public health leaders and solutions through cutting-edge, experiential Education, Training, Research and Advocacy.

BRAC James P Grant School of Public Health (BRAC JPGSPH) was founded in 2004 to address the unmet public health challenges of the Developing World. The School was co-founded by BRAC, icddr,b and Brac University in recognition of the fact that the public health needs of a Developing Country like Bangladesh requires community-based immersive teaching and learning to provide locally innovated research with sustainable solutions. The School is named after James P Grant, former executive director of UNICEF, who dedicated his whole life to the cause of international development.

The School draws from the remarkable public health achievements in Bangladesh; BRAC Health's nationwide services;the innovative health research of icddr,b; and BRAC International's programmes in South Asia and Africa. The School offers unparalleled real-life, community-centric teaching and learning experiences on critical and emerging Global Public Health challenges.

BRAC JPGSPH employs an interdisciplinary integration of Education, Training, Research and Advocacy to effectively address diverse health realities affecting disadvantaged communities.

The School offers a 1-year intensive, community-immersive, transformative Master of Public Health (MPH) programme. Since its inception, 492 students from a diverse academic and professional background representing 30 countries in South Asia, Southeast Asia, Africa, Australia, North and South Americas, and Europe have graduated from JPGSPH. Graduating students have moved on to work for their respective governments, national and international NGOs and with various donor and UN agencies. Additionally, universities and research organisations have also employed a large number of our Masters of Public Health graduates. The School also boasts 21 national and international faculties who are leaders in their fields of expertise. Read more about our MPH programme here.

The School offers training on various aspects of public health, gender, sexual and reproductive health, and research under the Centre for Professional Skills Development in Public Health (CPSD). Since its inception, the School has offered over 100 short courses and seminars, and trained over 5300 public health and development professionals.

To date, the School has conducted over 90 research projects under its 5 Centres of Excellence with over 75 local and global partners. Over 550 articles have been published in journals, books and chapters since the School’s inception. Between 2016-2018 the BRAC JPGSPH has conducted 65 research projects and published 87 journals, articles and papers. Learn more about our 5 Centres of Excellence and our research here.

Our Centres of Excellence are dedicated towards utilizing research evidence towards advocacy platforms and policy. To date the School has conducted over 200 advocacy based events to ensure that communities, practitioners, policymakers, government and international institutions learn from and can translate research findings to improve public health outcomes for disadvantaged groups.

News Archive >>

News

Professor Sabina Faiz Rashid appointed as a member of the Research Advisory Group of FCDO, UK

Regional Symposium on Population Health Informatics Hosted by BRAC JPGSPH and CUNY SPH

Professor Syed Masud Ahmed Ranked #2 among Bangladeshi Top Scientists in the Area of Social Sciences and Humanities in 2023 by Research.com

BRAC James P Grant School of Public Health and University of Groningen holds conference on improving SDGs by enhancing women’s empowerment

Monthly E-Newsletter

May-June-July 2019

March-April 2019

January-February 2019

October-November-December

The School of Data & Sciences is composed of the Department of Computer Science and Engineering (CSE) and the Mathematics & Natural Sciences Department (MNS). It is the largest school in the university. The school's focus is core computer science and engineering and fundamental sciences. Using the combined expertise of both the CSE and MNS departments it aims to be a leader in data science and related areas.

Dr. Mahbubul H. Siddiqee is a life scientist primarily focused on generating research-based evidence that would be helpful to implement effective intervention strategies and to adopt/improve health related policies. He is the Chair of the Institutional Review Board (IRB) at BRAC University, Bangladesh.

Being primarily affiliated with BracU since 2010, Dr. Siddiqee has been working closely with a number of other research-focused government and non-government organizations both nationally and internationally at various capacities. He led the Microbiology program at Brac University from January 2020 to October 2022. During the COVID-19 pandemic, he has played a pivotal an important role in the mass communication media to disseminate health awareness among general population. He served as the Chair of the Biomedical Research Board at Brac University from January 2022 until the end of March 2023. He also played a crucial role as the Interim Chair to establish Research Metrics Committee (RMC) at Brac University. He also served as the BracU Ambassador to IIT Madras, India. Beyond BracU, Dr. Siddiqee is the honorary

Director of Research at Red and White Innovations – an organization that works on conducting research and building relevant capacities among university level students and academics.

His research interests encompass various aspects of epidemiology, public health microbiology, clinical science, climate and environmental health, etc. Key elements of his research endeavors include monitoring, identifying, and characterizing atypical, and often overlooked, risk factors in the realm of human health.

Dr. Mahbub Siddiqee received his PhD in public health microbiology from Monash University, Australia. Some of his ongoing research projects are focusing on the rise of antimicrobial resistance (AMR) among the human pathogenic bacteria including extended-spectrum beta-lactamase (ESBL) producing E. coli, and carbapenem-resistant Enterobacteriaceae (CRE) organisms. Among specific research aspects of combatting the rise of AMR, Dr. Siddiqee is interested in investigating the atypical (and somewhat overlooked) risk-factors (both natural and anthropogenic) that might be associated with higher rates of mutations or transmissions of resistance properties among various bacteria. In the last few years, Dr. Siddiqee has published several peer-reviewed scientific articles in many high-impact refereed journals.

With over 14 years of experience of working mostly in the resource-limited settings, and with a rich publication record, Dr. Siddiqee is a uniquely skilled health system analyst with an ability to identify gaps in existing practices, and to find out effective strategies to maximize the outputs of the under-performing systems.

As a prominent public health commentator, Dr. Siddiqee is a frequent contributor on national media – both print and electronic. He is a vocal proponent of evidence-based healthcare practices, and of integrating the life scientists who are working in academia to the mainstream healthcare management system in Bangladesh. As a recognition of his contributions, Dr. Siddiqee was recently nominated by the Bangladesh Society of Microbiologists (BSM) as one of the eight Young and Emerging Scientists (YES) from

Bangladesh.

Level: 5, Room: 5B12 Kha-224 Merul BaddaDhaka 1212. Bangladesh

● PhD in Public Health Microbiology with concentration on epidemiology and bacterial virulence (2019), MONASH University, Melbourne, Australia; recipient of the Australian Postgraduate Award (industry linkage).● Master of Science in Environmental Microbiology (2008-2009; Fist Class 3rd; recipient of the Outstanding Academic Performance award), University of Dhaka, Bangladesh

● Bachelor of Science (Hons.) in Microbiology (2002-2007; First Class 5th), University of Dhaka, Bangladesh

JournalsSiddiqee MH, Bhattacharjee B, Hasan M et al. (2023). Risk perception of sun exposure and knowledge of vitamin D among the healthcare providers in a high-risk country: A cross-sectional study. BMC Med Educ 23, 46 (2023). https://doi.org/10.1186/s12909-023-04001-0 (Q1, Springer Nature)Hossain MS, Siam MHB, Hasan MN, Jahan R, Siddiqee MH. (2022) Knowledge, attitude, and practice towards blood donation among residential students and teachers of religious institutions in Bangladesh – A cross-sectional study. Heliyon, 2022, 8(10), e10792 https://doi.org/10.1016/j.heliyon.2022.e10792 (Q1; Elsevier)

Siddiqee, M. H., Bhattacharjee, B., Siddiqi, U. R., & Rahman, M. M. (2022). High burden of hypovitaminosis D among the children and adolescents in South Asia: A systematic review and meta-analysis. Journal of Health, Population and Nutrition, 41(1). https://doi.org/10.1186/s41043-022-00287-w (Q1; Springer Nature)

Siddiqee, M. H., Bhattacharjee, B., Siddiqee, U. R., & Rahman, M. M. (2021). High prevalence of vitamin D insufficiency among South Asian pregnant women: A systematic review and meta-analysis. British Journal of Nutrition. https://doi.org/10.1017/S0007114521004360 (Q2; Cambridge University Press)

Siddiqee, M. H., Bhattacharjee, B., Siddiqi, U. R., & MeshbahurRahman, M. (2021). High prevalence of vitamin D deficiency among the South Asian adults: a systematic review

and meta-analysis. BMC Public Health, 21(1). https://doi.org/10.1186/s12889-021-11888-1 (Q1; Springer Nature)

Rahman, M. M., Bhattacharjee, B., Farhana, Z., Hamiduzzaman, M., Chowdhury, M. A. B., Hossain, M. S., … Uddin, J. (2021). Age-related risk factors and severity of SARS-CoV-2 infection: A systematic review and meta-analysis. Journal of Preventive Medicine and Hygiene, 62(2), 329–371. https://doi.org/10.15167/2421-4248/jpmh2021.62.2.1946 (Q3 Scopus; Italy)

Islam, R., Solaiman, A. H. M., Kabir, M. H., Arefin, S. M. A., Azad, M. O. K., Siddiqee, M. H., … Naznin, M. T. (2021). Evaluation of lettuce growth, yield, and economic viability grown vertically on unutilized building wall in Dhaka city. Frontiers in Sustainable Cities, 3. https://doi.org/10.3389/frsc.2021.582431 (Scopus; Frontiers)

Hossain, M. B., Khan, J. R., Adhikary, A. C., Anwar, A. H. M. M., Raheem, E., Siddiqee, M. H., & Hossain, M. S. (2021). Association between childhood overweight/obesity and urbanization in developing countries: Evidence from Bangladesh. Journal of Public Health (Germany). https://doi.org/10.1007/s10389-021-01560-8 (Q3; Springer Nature)

Hossain, M. S., Raheem, E., & Siddiqee, M. H. (2020). The forgotten people with thalassemia in the time of COVID-19: South Asian perspective. Orphanet Journal of Rare Diseases, 15(1). https://doi.org/10.1186/s13023-020-01543-0 (Q1; Springer Nature)

Hossain, M. S., Ferdous, S., Raheem, E., & Siddiqee, M. H. (2020). The double burden of malnutrition—further perspective. The Lancet, 396(10254), 813–814. https://doi.org/10.1016/S0140-6736(20)31359-3 (Q1; Elsevier)

Hossain, M. S., Siddiqee, M. H., Siddiqi, U. R., Raheem, E., Akter, R., & Hu, W. (2020). Dengue in a crowded megacity: Lessons learnt from 2019 outbreak in Dhaka, Bangladesh. PLoS Neglected Tropical Diseases, 14(8), 1–5. https://doi.org/10.1371/journal.pntd.0008349 (Q1; PLOS USA)

Hossain, M. S., Ferdous, S., & Siddiqee, M. H. (2020). Mass panic during Covid-19 outbreak- A perspective from Bangladesh as a high-risk country. Journal of Biomedical

Analytics, 3(2), 1–3. https://doi.org/10.30577/jba.v3i2.40 (CrossRef)

Hossain, M. S., Hasan, M. M., Raheem, E., Islam, M. S., Al Mosabbir, A., Petrou, M., … Siddiqee, M. H. (2020). Lack of knowledge and misperceptions about thalassaemia among college students in Bangladesh: A cross-sectional baseline study. Orphanet Journal of Rare Diseases, 15(1). https://doi.org/10.1186/s13023-020-1323-y (Q1; Springer Nature)

Siddiqee, M. H., Henry, R., Deletic, A., Bulach, D. M., Coleman, R. A., & McCarthy, D. T. (2020). Salmonella from a microtidal estuary are capable of invading human intestinal cell lines. Microbial Ecology, 79(2), 259–270. https://doi.org/10.1007/s00248-019-01419-2 (Q1; Springer)

Hossain, M. S., Siddiqee, M. H., Ferdous, S., Faruki, M., Jahan, R., Shahik, S. M., … Okely, A. D. (2019). Is childhood overweight/obesity perceived as a health problem by mothers of preschool aged children in Bangladesh? A community level cross-sectional study. International Journal of Environmental Research and Public Health, 16(2). https://doi.org/10.3390/ijerph16020202 (Q1; MDPI, Switzerland)

Siddiqee, M. H., Henry, R., Coleman, R. A., Deletic, A., & McCarthy, D. T. (2019). Campylobacter in an urban estuary: Public health insights from occurrence, hela cytotoxicity, and Caco-2 attachment cum invasion. Microbes and Environments, 34(4), 436–445. https://doi.org/10.1264/jsme2.ME19088 (Q1; Japanese Society of Microbiology)

Siddiqee, M. H., Henry, R., Coulthard, R., Schang, C., Williamson, R., Coleman, R., … McCarthy, D. (2018). Salmonella enterica serovar typhimurium and escherichia coli survival in estuarine bank sediments. International Journal of Environmental Research and Public Health, 15(11). https://doi.org/10.3390/ijerph15112597 (Q1; MDPI, Switzerland)

Siddiqee, M. H., Islam, M. S., & Rahman, M. M. (2013). Assessment of pollution caused by tannery-waste and its impact on aquatic bacterial community in Hajaribag, Dhaka.

Stamford Journal of Microbiology, 2(1), 20–23. https://doi.org/10.3329/sjm.v2i1.15208 (CrossRef)

Siddiqee, M. H., Sarker, H., & Shurovi, K. (2013). Assessment of probiotic application of lactic acid bacteria (LAB) isolated from different food items. Stamford Journal of Microbiology, 2(1), 10–14. https://doi.org/10.3329/sjm.v2i1.15206 (CrossRef)

Newspaper Articles

Siddiqee, M. H., & Shojon, M. (2020, July 28). Simple ways to alleviate antibiotic resistance amidst Covid-19. The Business Standard. Retrieved from https://www.tbsnews.net/thoughts/simple-ways-alleviate-antibiotic-resist...

Rafiqullah, I., & Siddiqee, M. H. (2020, July 14). COVID-19 is aggravating antimicrobial resistance. The Bangladesh Today. Retrieved from https://thebangladeshtoday.com/?p=21868

Siddiqee, M. H. (2020). Six critical points to prioritize during the covid pandemic. Amader Somoy.

Siddiqee, M. H. (2020, July 2). Covid vaccine: how far is it? Amader Somoy.

Siddiqee, M. H. (2020, July 2). □□□□□□□ □□□□□□□□ □□ □□□□□□□□□□□□ □□□□□□□ □□□? (Does ge long-term protection?). Daily Bonikbarta. Retrieved from https://bonikbarta.net/home/news_description/234309/□□□□□□□-□□□□□□□□-□□-□□□□□□□□%E0

Rafiqullah, I., & Siddiqee, M. H. (2020, June 25). □□□□□□□ □□□□□□□□□□□□□□□□□□□□□ □□□□□□□ □□□ during covid pandemic could have dire consequences). Daily Bonikbarta. Retrieved from https://bonikbarta.net/home/news_description/233683

Siddiqee, M. H. (2020, June 27). How likely is it to catch covid second time? Amader Somoy.

Siddiqee, M. H. (2020, June 29). The ’if’s and ’but’s regarding covid vaccine. Amader Somoy.

Siddiqee, M. H. (2020, June 26). □□□□□ □□□□□□□□□□□□□□ □□□□ □ □□□□□ (The health hazards masks). Dhaka Times. Retrieved from https://www.dhakatimes24.com/2020/06/26/172605/[illegible]-[illegible]-[illegible]-[illegible]-[illegible]

Siddiqee, M. H. (2020, June 18). [illegible] mandatory and a few suggestions). Kaler Kontho. Retrieved from https://www.kalerkantho.com/online/miscellaneous/2020/06/18/924358

Siddiqee, M. H. (2020, June 18). Which masks you should wear to stay protected from covid? Amader Somoy.

Siddiqee, M. H. (2020, June 13). [illegible] (Using masks risks). Dhaka Times. Retrieved from https://www.dhakatimes24.com/2020/06/13/170691/[illegible]-[illegible]-[illegible]-[illegible]%E0%A7%25

Siddiqee, M. H. (2020, June 9). [illegible] RT-PCR tests during Covid and how to do that). Dhaka Times. Retrieved from https://www.dhakatimes24.com/2020/06/09/169957

● Associate Professor (Since July 2022), Microbiology Program, Department of Mathematics and Natural Sciences (MNS), School of Data and Sciences (SDS), BRAC University, Bangladesh●Assistant Professor (March 2021-at present), Microbiology Program, Department of Mathematics and Natural Sciences (MNS), BRAC University, Bangladesh● Chair, Biomedical Research Board (IRB #1; Since March 2022), BRAC University, Bangladesh● Program Coordinator (January 2020 - at present), Microbiology program, BRAC University, Bangladesh● Member, Institutional Review Board (IRB) Core Committee, BRAC University, Bangladesh● Member, Self-Assessment Committee (SAC) (June 2017- at present), Internal Quality Assurance Cell (IQAC), Microbiology Program, BRAC University, Bangladesh● Honorary Senior Research Fellow (June 2020 - October, 2020), Biomedical Research Foundation, Bangladesh● Senior Lecturer (July 2018- February 2021), Dept. of Mathematics and Natural Sciences (MNS), BRAC University, Bangladesh● Research Fellow (September 2017- May 2020), Biomedical Research Foundation, Bangladesh● Lecturer (April 2010- June 2018), Microbiology Program, Department of Mathematics and Natural Sciences, BRAC University● Lecturer (April

2009- April 2010), Department of Microbiology, Stamford University Bangladesh● Lab. In-charge of the Life Science laboratories (Microbiology and Biotechnology); June 2017 to January 2019

● In-charge of requisitions for the Life Science laboratories (Microbiology and Biotechnology); June 2017 to January 2019

● Member, Syllabus Moderation Committee, Microbiology Program, BRAC University

● Environmental Microbiology● Analytical Microbiology● Microbial Biotechnology● Microbial Genetic Engineering● Introduction to Molecular Biology● Basic Techniques in Microbiology● Microbial Metabolism● General Microbiology

• Outstanding Academic Performance Award, 2007; Department of Microbiology, University of Dhaka

• Australian Postgraduate Award (APA), 2012-2016

● Member, Institutional Review Board (IRB) Core Committee, BRAC University, Bangladesh● Member, Self-Assessment Committee (SAC) (June 2017- at present), Internal Quality Assurance Cell (IQAC), Microbiology Program, BRAC University, Bangladesh● Member, Syllabus Moderation Committee, Microbiology Program, BRAC University

Mr. Mahbubul H. Siddiqee is specialized in the area of environmental health. The key research interest of Mr. Siddiqee include (but not limited to)- 1. monitoring, prediction, and communication of health risks from microbial pathogens in surface- and drinking water systems; 2. spread of antibiotic resistance in nature via horizontal gene transfer mechanisms (plasmids, transposons, integrons, and gene cassettes); 3. control of human-pathogens (including multidrug-resistant pathogens) using bacteriophages as bio-control agents; and 4. identification and characterization of risk-factors that are directly and indirectly associated with public health.

More about Dr. Siddiqee’s recent research activities can be found at the ‘research’ subsection of his ResearchGate profile-https://www.researchgate.net/profile/Mahbubul-Siddiqee/research

Mahbubul Hasan Siddiqee, PhD

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Most. Salma Yeasmin Jannaty completed her school and college from Ahmad Uddin Shah Shishu Niketon School and College, Gaibandha. She finished her B.Sc. and MS From University of Dhaka.

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• She worked as a part-time faculty at Ahsanullah University of Science and Technology during semester Fall-2021.• She conducted her thesis work under the supervision of Dr. Md. Jamil Hasan Karami, Professor, Department of Statistics, University of Dhaka on the topic 'Modeling of Velocity Distribution of Motor Nerves Using Compound Action Potential'.• She conducted a research project titled 'A Study on Empowerment of Educated Women in Bangladesh' jointly as per the requirement in Honors Final Year, and was supervised by Muhammad Ahsan Uddin, Professor, Department of Statistics, University of Dhaka.

• Elements of Statistics and Probability• Introduction of Statistics

• Received National Science and Technology Fellowship awarded by the Ministry of

Science and Technology of the People’s Republic of Bangladesh for MS thesis for the Fiscal year 2020-21.• Received scholarship from Dhaka University for academic performance in BSc Honours Examination 2018.

• Member of Bangladesh Physical Society

• Biostatistics• Biomedical Engineering• Statistical modeling etc.

Most. Salma Yeasmin Jannaty

News & Events

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The Academic Council recommends the educational policies of the University and also determines the curricula and courses that can help achieve high educational standards. The council is currently composed of the following academics and professionals:

Chairperson

Members

Ms. Rasheda K. ChoudhuryFormer Advisor, Caretaker Government of Bangladesh andExecutive Director, Campaign for Popular Education (CAMPE)

Ms. Sadaf Saaz SiddiqiDirector & Producer, Dhaka Lit FestCulture & Curator Activist and Entrepreneur and an Alumna of BRAC University

Professor Sultan Hafeez Rahman, PhDProfessorial Fellow and Former Executive Director, BRAC Institute of Governance and DevelopmentBRAC University

Dr. Laura Reichenbach, PhDDean, James P Grant School of Public Health BRAC University

Professor K. Shamsuddin MahmoodDean, School of LawBrac University

Professor Fuad Hassan Mallick, PhDDean, School of Architecture and DesignBrac University

Professor Arshad Mahmud Chowdhury, PhDDean, BSRM School of EngineeringBrac University

Professor Mohammad Mujibul Haque, PhDProfessor and Associate Dean (Acting Dean), BRAC Business SchoolBRAC University

Professor Muhammad Wasiqur Rahman KhanProfessor and Chairperson , Department of Economics and Social SciencesBrac University

Professor Firdous Azim, PhDChairperson, Department of English and HumanitiesBrac University

Dr. Samia HuqProfessor (ESS) and Dean, General EducationBrac University

Professor Md. Mosaddequr Rahman, PhDChairperson, Department of Electrical and Electronic EngineeringBrac University

Professor Eva Rahman Kabir, PhD Dean, School of Pharmacy Brac University

Professor Mahbubul Alam Majumdar, PhD Dean, School of Data and Sciences

Brac University

Dr. Sadia Hamid KaziChairperson, Department of Computer Science and Engineering Brac University

Professor Zainab Faruqui Ali, PhDChairperson, Department of ArchitectureBrac University

Dr. Md. Firoze H. Haque, PhDChairperson and Associate Professor, Department of Mathematics and Natural SciencesBRAC University

Dr. Shaila SultanaDirector & Professor, BRAC Institute of Languages BRAC University

Dr. Imran MatinExecutive Director, Brac Institute of Governance and DevelopmentBrac University

Dr. Erum Mariam Director, Institute of Educational Development Brac University

Member Secretary

Dr. David DowlandRegistrar & Controller of Examinations (acting)Brac University

Md. Mehedi Hasan completed his M.Sc. and B. Sc. in Physics from Jahangirnagar University in 2015 and 2014 respectively. A thesis work on Plasmas Physics (submitted on August 2016) was done by him under the direct supervision of Prof. Dr. A. A. Mamun, Department of Physics, Jahangirnagar University and some of his research articles have been published in renowned peer reviewed international journals like Vacuum-Journal-Elsevier, Chaos: An Interdisciplinary Journal of Nonlinear Science and Physics of Plasmas. He has been working in the Department of Mathematics and Natural Sciences at Brac University since May 2017. His current research interests on the fully relativistic plasma and astrophysical plasma.

Level: 5Kha-224 Merul BaddaDhaka 1212. Bangladesh

Journals

Chowdhury, N. A., Mannan, A., Hasan, M. M., & Mamun, A. A. (2019). Modulational instability, Ion-acoustic envelope solitons, and rogue waves in four-component plasmas. Plasma Physics Reports, 45(5), 459–465. https://doi.org/10.1134/S1063780X19050027

Chowdhury, N. A., Hasan, M. M., Mannan, A., & Mamun, A. A. (2018). Nucleus-acoustic envelope solitons and their modulational instability in a degenerate quantum plasma system. Vacuum, 147, 31–37. https://doi.org/10.1016/j.vacuum.2017.10.004

Chowdhury, N. A., Mannan, A., Hasan, M. M., & Mamun, A. A. (2017). Heavy ion-acoustic rogue waves in electron-positron multi-ion plasmas. Chaos, 27(9). https://doi.org/10.1063/1.4985113

Hasan, M. M., Hossen, M. A., & Mamun, A. A. (2017). Acoustic solitary waves in a magnetized degenerate quantum plasma. Physics of Plasmas, 24(7). https://doi.org/10.1063/1.4993052

Hasan, M. M., Hossen, M. A., Rafat, A., & Mamun, A. A. (2016). Effect of Bohm quantum potential in the propagation of ion-acoustic waves in degenerate plasmas. Chinese Physics B, 25(10). https://doi.org/10.1088/1674-1056/25/10/105203

May 2017 – PresentLecturerDepartment of Mathematics and Natural Sciences (MNS)Brac UniversityDhaka, BangladeshOthers:

• Involve in research.• Member of the revised syllabus committee, question moderation committee and final exam scrutinizing committee• Attending seminar and workshop.• Invigilator in exams.

• PHY 102: Fundamentals of Physics• PHY 111: Principles of Physics I• PHY 112: Principles of Physics II• PHY 403: Plasma and Astrophysics• PHY 116: Physics Lab I

• National Science & Technology Fellowship (2015-2016) from Ministry of National Science & Technology, Bangladesh.• Regular merit based annual scholarships (2012-2016) from Jahangirnagar University (JU).

• Seminar on “Challenges of Protection of Civilians in Peace Operation Insight from Africa: What Lies Ahead” organized by Bangladesh Institute of Peace Support Operation Training and Center for United Nations Peacekeeping, India.• Professional Training on “Certificate in Higher Education Teaching Program” conducted by Professional Development Centre (PDC) of BRACU, 2017/18.• Workshop on “Developing Effective Assessment: Strategies for Success” organized by PDC.• Speaker on a seminar entitle “Effects of Quantum Mechanical Bohm Potential on Ion-Acoustic Waves in Degenerate Plasmas” organized by MNS department, BRAC University.• Workshop on “Student Engagement through Web-Tools and Digital Contents” organized by PDC, 2018.• Seminar on “THEORY OF KNOWLEDGE (TOK)” organized by PDC.

Md. Mehedi Hasan

News & Events

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EEE student selected for Huawei program in China

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Workshop on Architecture Model Making Held at BRAC University

Subarno Hossain Turja completed his high school education from the Aga Khan School, Dhaka. He went on to obtain his BSc. in Biomedical Engineering from Bucknell University in Pennsylvania, USA, followed up by his MSc. in Biomedical Engineering from Drexel University in Pennsylvania, USA.

Level: 4Kha-224 Merul BaddaDhaka 1212. Bangladesh

BCH 101, BIO 101, BTE 102, BTE 103, BTE 155

Subarno Hossain Turja

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Romana Siddique is a Lecturer of biotechnology at the Department of Mathematics and Natural Sciences .She did her bachelor and masters in Genetic Engineering and Bio-technology from University of Dhaka. During her study she was involved in the research arena of molecular biology. Before joining BRAC University she worked in BRAC as a young professional which also develops her projects coordinating capacity and leadership skills. She gained good command of molecular biology techniques as it was learned the techniques during her research work at Dhaka University & BRAC Plant Biotechnology Laboratory.

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i. Awarded Bachelor of Science (Honours) in Genetic Engineering and Biotechnology from University of Dhaka.

ii. Awarded Masters of Science in Genetic Engineering and Biotechnology from University of Dhaka.

Genetic Engineering & Biotechnology Alumni Association

GNOBB (Global Network of Bangladeshi Biotechnologist)

Molecular biology, Plant Biotechnology, Bioinformatics

Romana Siddique

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Brac University believes in a holistic approach towards education and grooming future leaders who will possess high moral values, have empathy, more informed, scientific, and socially responsible alongside their academic excellence. Brac University has established 27 clubs to nurture the students to be future leaders in alignment with the vision of the institution. To supervise and support these clubs' activities, BracU created the Office of Co-curricular Activities (OCA).

Office of Co-curricular Activities (OCA) is a hub to foster BracU students' complete learning experience and to groom up the students in achieving the distinctive qualities of a real-life champion.

Through the whole year, the clubs remain engaged in their regular activities. Alongside they always contribute to the university's central events and also work collaboratively with the various departments of the university. The clubs of BracU respectfully participate and observe all national events. OCA always ensures that the club events are designed integrating community services and social contributions towards ensuring sustainable future.

OCA believes in a holistic approach towards education and will work in grooming future leaders who will possess high moral, ethical and humane values, have empathy, will be socially responsible, committed to cultural and patriotic values; more informed and scientific alongside maintaining their academic excellence.

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News and Events

BRAC University's Ramadan 2024: A Campus United in Generosity and Community Service

BRAC University observes National Mourning Day

OCA holds “The Leadership Bootcamp 2023”

BRAC University celebrates Martyr’s day and International Mother Language Day

Join "Bongo Marathon" to celebrate National Children’s Day and Birth Anniversary of Father of the Nation, 2023

Celebrating International Mother Language Day 2023

Freshers' Orientation Fall 2022

Observing "Mujib Borsho" the 100th Birth Year Celebration of the Father of the Nation

The second day of 1st International Conference on Business and Management, September 22, 2017 at the Westin Hotel, Dhaka, Bangladesh, Industry Talk session was chaired by Prof. Dr. Mirza Azizul Islam, Former Adviser of Caretaker Government, People’s Republic of Bangladesh and Professor of BRAC Business School and Prof. Dr. Salehuddin Ahmed, Former Governor of Bangladesh bank (The Central bank of

Bangladesh) and Professor of BRAC Business School. In the introductory speech, Dr. Salehuddin Ahmed urged, 'It's the people behind everything and they should be trained and motivated for the society.' Dr. Mirza Azizul Islam also pointed out that the country lacked skilled manpower and the issue had to be considered with importance.

Mr. Syed Abdul Momen, Head of SME, BRAC Bank limited as the first speaker in the industry talk session. Taking the floor, Mr. Momen provided an outline of how the bank pioneered in providing dedicated services for Small and Medium Enterprises (SME) and promoted sustainable economic, social and environmental development being a member of The Global Alliance for Banking on Values. He shared some statistics with the audience to give an overview of the customer segmentation of BRAC Bank Limited. He informed the audience that 76 per cent of BRAC Bank's customers are from micro and cottage enterprises segment while the rest 24 per cent are from small enterprises. Now going into the further segmentation of SME customers' we get to see 75 per cent of small business loans the bank disbursed every month were in rural areas around the country while 25 per cent were in Dhaka and Chittagong districts.

Sharing the future plan of BRAC Bank Mr. Momen said currently they are concerned about looking for ways to stimulate and support SME across the country and in near future they are on the way to "Promote Digitalization". After that he gave a brief overview on how BRAC Bank had taken the services to the door steps of their customers so that they can avail the services in the shortest possible time. Finally, he concluded his speech emphasizing on conducting researches time to time to do the impact analysis.Second speaker of this session, Mr. Syed Waseque Md Ali, Managing Director, First Security Islami Bank (FSIB) graced the floor with his presence and shared his view on how following the suggestions given by policy makers and findings of several researches, after 18 years of operations FSIB changed its track from a conventional Bank to an Islami Bank in 2009. After that, he went on describing the scenario of Islamic Banking from the perspective of Bangladesh in terms of profit sharing, safekeeping, joint venture and cost plan. Later on he concluded his speech highlighting the contribution of FSBI to the overall Bangladeshi Banking sector.

Last but not the least, Mr. ASM Mainuddin Monem, the Deputy Managing Director of Abdul Monem Ltd., was Industry Talk speaker. Throughout his speech, he emphasized on research and developments to bring the best minds of Bangladesh in a funnel to flourish innovation. Moreover, Mr. Monem also shared his vision to develop an 'Innovation and Research Institute' on the 314-acre land provided by the government to establish economic zone.

To sum up, The Industry Talk Session of day two became successful in its purpose to edify audiences about different industries, their operational process, expansion strategies and future goals. Finally, the session ended with a Q&A session where the Vice Chancellor of BRAC University, Prof. Syed Saad Andaleeb, Ph.D. informed his audience that he had proposed the government to form a research council and suggested industries to provide funds in small amounts to develop a 'Brain Trust' in this regard. Among others, Prof. Dr. Iftekhar Ghani Chowdhury, Dean of BRAC Business School and General Chair of ICBM 2017 and Assoc. Prof. Dr. Md. Mamun Habib, Program Chair of ICBM 2017, were attended along with local and international dignitaries.

With immense dedication and commitment BRAC Business School (BBS), BRAC University, Bangladesh, has proudly established a new milestone by hosting the 1st International Conference on Business and Management (ICBM 2017). Sir Fazle Hasan Abed KCMG, the chief patron, inaugurated ICBM 2017 by his statement. Sir Abed KCMG, welcomed the initiative of organizing the conference and encouraged to continue this kind of initiatives in the future with accolade.

The Honorary General Chair of ICBM 2017, Prof. Syed Saad Andaleeb, PhD. , the General Chair Prof. Dr. Iftekhar Ghani Chowdhury, the Program Chair Assoc. Prof. Dr. Md. Mamun Habib, three Keynote speakers, Prof. Dr. Mark GOH [NUS, Singapore], Prof. Dr. Premkumar Rajagopal [MUST, Malaysia] and Dr. Pairach Piboonrungroj [Chang Mai

University, Thailand] graced the inaugural session on September 21, 2017 at the hotel Westin, Dhaka.

The Honorary General Chair, Prof. Syed Saad Andaleeb, PhD. talked about the importance of research for the improvisation of academic standard. He drew the significance of research work and added that Bangladesh should open more research-based universities for the development of tertiary education in the country. Prof. Andaleeb exemplified the perseverance of the faculty members of research universities in California to publish their work in reputed journals. He also proclaimed that for research supremacy Asian universities are becoming the next superpowers with Japan, South Korea, Taiwan, Singapore and Hong Kong already being in the competition. The General Chair, Prof. Dr. Iftekhar Ghani Chowdhury, referred to the pivotal role of such conferences in the development of quality education and thanked BRAC Business School for successfully organizing such an international conference with massive participation from local as well as international.

The Program Chair, Dr. Md. Mamun Habib, the initiator of the conference as well, addressed that how an international conference contributes to international integration and building strong networks. This conference is a parameter to measure the research quality and education standard of our country in the international arena, he added. Prof. Habib urged that the students would be motivated after attending this conference in order to pursue further research based study. In addition, he hoped that this will unlock a new era of research in Bangladesh.

Dr. Habib appreciated the dedication and hard work of the conference secretariat, Mr. Zaheed Husein Mohammad, Mr. N M Baki Billah, Mr. Saif Hossain and Mr. Shamim Ahmed. He also admired the organizing committee members, namely Mr. Hasan Maksud Chowdhury, Mr. Riyashad Ahmed, Dr. Suman Paul Chowdhury and so on. He also thanked the volunteers for their fervent pro-activeness and devotion to organize the conference successfully. Finally, Prof. Habib expressed his special gratitude to the sponsors, particularly platinum sponsor (BRAC Bank), gold sponsor (FSIB), silver sponsors (UCB, BBS Cables, Abdul Monem Group, Bkash), supporting sponsors (Islami Bank, Dhaka Bank, IDLC, Aarang) and hospitality partner, The Westin.

The Vice-Chancellor and Pro-VC of various universities as well as corporate, namely Ms. Zara Mahbub (V.P, BRAC Bank), graced the conference ceremony. Among them from BRAC University, Prof. Dr. Hafiz G A Siddiqi, Dr. Akbar Ali Khan, Dr. Mirza Azizul Islam and Dr. Salehuddin Ahmed made an immense contribution in the conference. Mr. Shib Narayan Kairy, Registrar (a.i.), Mr. Monojit Kumar Ojha, FCA, Director of Finance, Mr. Md. Nurul Islam Bulbul, Senior Assistant Director, Admin; Ms. Ismat Shereen, Head of RMO, Ms. Sophia Najma Chaudhury, Communication Dept. and others were attended. A good number of foreign delegates' participation from various countries, including India, Thailand, Singapore, Malaysia, Sweden, USA, China and so on, enlighten and signify the conference in an international manner. There are various international universities, namely University of Texas, Arlington (UTA),USA; University of Kuala Lumpur (UniKL), Malaysia University of Science and Technology (MUST), Chiang Mai University, Thailand; Universiti Utara Malaysia (UUM, Malaysia), extended their partnership with ICBM 2017.

Ms. Adiba Naoshin, BBS faculty member, played the role of Master of Ceremonies and welcomed all the guests, session chairs, presenters, participants and the keynote speakers. Being unable to attend physically, on behalf of Sir Fazle Hasan Abed KCMG, the inauguration statement was read by Shamma Nafisa, a BBS student of BRAC University, Bangladesh.

Structure of the Program

A physics undergraduate program has been designed including topics of current interest and applications. Once a student undergoes this course successfully he will be well equipped to face the challenges of life. The program of study includes courses for improving communication skills, strengthening mathematical background and acquainting the student with the socio-economic & historical background of

Bangladesh. With this background it should not also be a problem to find a suitable and satisfying job in various universities & R/D organizations in the country. The total credit requirements for the degree of Bachelor of Science in Physics is 132. Out of these 21 credits are for general education. Twenty one major area compulsory courses account for 63 credits. The students are required to complete 3 courses (4.5 credits) of Physics Lab and write a dissertation/report on a suitable thesis/project topic. The thesis/ project work spread over the last two semesters will have a total of 4.5 credits. The students will be required to complete 12 credits choosing from several elective courses in their major field and 27 credits from outside their major specialization. The students may also be required to take non-credit remedial courses in English.

List of Courses for Bachelor of Science in Physics

a. General Education: (21 credits)

1. PHY 110 Mechanics and Properties of Matter 2. CSE 110 Programming Language I3. DEV 101 Bangladesh Studies4. ENG 101 English Fundamentals5. ENG 102 English Composition I6. HUM 103 Ethics and Culture7. MAT 102 Introduction to Mathematics

b. Core Courses: (63 credits)

1. PHY 113 Waves, Oscillation and Acoustics 2. PHY 114 Thermal Physics and Radiation 3. PHY 115 Electricity and Magnetism 4. PHY 201 Solid State Physics 5. PHY 202 Optics 6. PHY 204 Classical Mechanics and Special Theory of Relativity 7. PHY 205 Statistical Mechanics 8. PHY 301 Classical Electrodynamics 9. PHY 302 Fluid Mechanics 10. PHY 303 Quantum Mechanics I11. PHY 304 Quantum Mechanics II 12. PHY 305 Quantum Mechanics III 13. PHY 306 Basic Electronics14. PHY 401 Particle and Reactor Physics 15. PHY 402 Atmospheric Physics 16. PHY 403 Cosmology & Astrophysics 17. MAT 105 Calculus 18. MAT 203 Matrices, Linear Algebra and Differential Equations 19. MAT 204 Complex Variables and Fourier Analysis 20. MAT 205 Introduction to Numerical Methods 21. STA 201 Elements of Statistics and Probability

c. Elective Courses: (12 credits)

1. PHY 308 Methods of Experimental Physics and Instrumentation2. PHY 309 Introduction to Materials Science3. PHY 310 Advanced Solid State Physics4. PHY 311 X-Rays5. PHY 312 Nuclear Physics II 6. PHY 404 Electronic Devices and Circuits 7. PHY 405 Mathematical Physics 8. PHY 406 Medical Physics and Instrumentation9. PHY 407 Mathematical Modeling in Physics10. PHY 408 Advanced Quantum Mechanics 11. PHY 409 Physics of Radiology 12. PHY 410 Laser Physics 13. PHY 411 Geophysics 14. PHY 412 Dynamical and Tropical Meteorology 15. PHY 413 General Theory of Relativity 16. PHY 414 Field Theory 17. PHY 415 Neutron Scattering 18. PHY 416 Radiation Biophysics 19. MAT 301 Group Theory20. MAT 303 Tensor Analysis

d. Lab: (4.5 credits)

1. PHY 116 PHY Lab I2. PHY 203 PHY Lab II3. PHY 307 PHY Lab III

e. Thesis/Project: (4.5 credits)

1. PHY 400 Thesis/Project

f. Courses Outside Major Specialization: (27 credits)*

1. ANT 101 Introduction to Anthropology2. ARC 292 Painting**3. ARC 293 Music Appreciation**4. BI0 101 Introduction to Biology5. CSE 101 Introduction to Computer Science 6. ECE 310 Introduction to Communication Engineering7. ENV 101 Introduction to Environmental Sciences8. ECO 103 Principles of Economics9. HUM 111 History of Science10. HUM 101 World Civilization and Culture 11. MGT 211 Principles of Management12. PHY 313 Physics for Development 13. HUM 102 Introduction to Philosophy14 POL 103 Introduction to Political Science15. POL 245 Women, Power and Politics16 PSY 101 Introduction to Psychology17. SOC 101 Introduction to Sociology18. SOC 401 Gender and Development

* Or any other BRACU course outside major specialization** 2 Credit courses

Double Major in Physics

To satisfy the needs of students studying in various disciplines of BRACU desirous of pursuing a double major degree is also offered by the MNS Department.

Students who want to do a double-major, one of them being physics, will have to complete a total of 66 credits, the break down of which is given in the following:

A. Core Courses

Theory Courses: (45 credits)

PHY 110 Mechanics and Properties Of Matter PHY 113 Waves, Oscillation and Acoustics PHY 114 Thermal Physics and Radiation PHY 115 Electricity and Magnetism PHY 201 Solid State Physics PHY 202 Optics PHY 204 Classical Mechanics and Special Theory of Relativity PHY 205 Statistical Mechanics PHY 301 Classical Electrodynamics PHY 303 Quantum Mechanics PHY 304 Atomic and Molecular Physics MAT 105 Calculus MAT 203 Matrices, Linear Algebra and Differential Equations MAT 204 Complex Variables and Fourier Analysis STA 201 Elements of Statistics and Probability

Lab Courses: (4.5 credits)

PHY 116 PHY Lab IPHY 203 PHY Lab IIPHY 307 PHY Lab III

Elective Courses: (12 credits)

PHY 302 Fluid Mechanics PHY 305 Nuclear Physics I PHY 306 Basic Electronics109PHY 308 Methods of Experimental Physics and InstrumentationPHY 401 Reactor Physics PHY 402 Atmospheric Physics PHY 403 Plasma and Astrophysics PHY 407 Mathematical Modeling in PhysicsPHY 410 Laser Physics PHY 411 Geophysics MAT 205 Introduction to Numerical Methods

Thesis: (4.5 credits)

The thesis work should be in line with one of the majors or on some interactive topic involving the two majors.

Minor in Physics

The MNS Department also offers the program of minor in physics for students doing a major in CS, CSE, ECE or any other relevant discipline. Such major-minor combination will stand the students in good stead in the job market.

The details of the Minor in Physics program are given below.

Total credit requirement: 27 credits

There will be seven compulsory courses, each of three credits, which are as follows:

PHY 111 Principles of Physics IPHY 112 Principles of Physics IIPHY 204 Classical Mechanics and Special Theory of RelativityPHY 205 Statistical MechanicsPHY 210 Quantum Physics of Atoms, Solids and NucleiPHY 301 Classical ElectrodynamicsPHY 305 Nuclear Physics I

Students may choose any two courses from the following list of elective courses each of three credits offered by the Department.

PHY 302 Fluid MechanicsPHY 308 Methods of Experimental Physics and InstrumentationPHY 309 Introduction to Materials SciencePHY 311 X-RaysPHY 313 Physics for DevelopmentPHY 409 Physics of RadiologyPHY 403 Plasma and Astrophysics or any other physics course with the permission of the Chairperson of the MNS Department.

Md. Mohtasir Billah Sheraj completed his school and college from Government Laboratory High School & Dhaka College respectively. He completed his B.S. from Department of Physics, University of Dhaka in 2018 & finished his M.S. from the same department. He has been working in the field of network theory & percolation theory for nearly two years under the supervision of Dr. Kamrul Hassan, Professor, Department of Physics, University of Dhaka.

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SSC from Government Laboratory High School- 2011HSC from Dhaka College-2013Bachelor of Science in Physics, from Department of Physics, University of Dhaka-2018Masters of Science in Physics, from Department of Physics, University of Dhaka-2020

Equilibrium & Non Equilibrium Statistical Physics, Stochastic Process, Critical Phenomena, Network Theory, Chaos Theory.

Md. Mohtasir Billah Sheraj

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Workshop on Architecture Model Making Held at BRAC University

Pages

Muhammad Lutfor Rahman completed his B.Sc (Honours) in Physics from the Shah Jalal University of Science and Technology (SUST), Sylhet, Bangladesh. He also completed his M. Phil. degree in Condensed Matter Physics from Bangladesh University of Engineering & Technology (BUET), Dhaka. Mr. Lutfor Rahman started his career as a teaching assistant at the Department of Mathematics and Natural Sciences of BRAC University. He was appointed as a lecturer of Physics at the same department in 2011. Before joining Brac University as a lecturer he worked as a Part-Time Lecturer at Ahsanullah University of Science & Technology (AUST). He was involved in several experimental condensed matter research projects and his current research focuses on electronic transport measurements and characterizations of organic and inorganic materials, both magnetic as well as non-magnetic.

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• M. Phil. in Physics (Condensed Matter) Bangladesh University of Engineering and Technology (BUET), Dhaka.• B.Sc. Honors in Physics [4 Year Integrated Course] Shah Jalal University of Science and Technology (SUST), Sylhet.

Selected Publications

Journals

Arifuzzaman, M., Hossen, M. B., Harun-Or-Rashid, M., & Rahman, M. L. (2021). Structural and magnetic properties of nanocrystalline Ni0.7-xCuxCd0.3Fe2O4 prepared through Sol-gel method. Materials Characterization, 171. https://doi.org/10.1016/j.matchar.2020.110810

Alam, F., Rahman, M. L., Das, B. C., & Hossain, A. K. M. A. (2020). Effect of Cu2+ on structural, elastic and magnetic properties of nanostructured Mn–Zn ferrite prepared by a sol-gel auto-combustion method. Physica B: Condensed Matter, 594, 412329. https://doi.org/10.1016/j.physb.2020.412329

Ali, M. M., Hadi, M. A., Rahman, M. L., Haque, F. H., Haider, A. F. M. Y., & Aftabuzzaman, M. (2020). DFT investigations into the physical properties of a MAB phase Cr4AlB4. Journal of Alloys and Compounds, 821, 153547. https://doi.org/10.1016/j.jallcom.2019.153547

Noor, L., Ahsan, G., Rahman, M. L., & Haque, F. (2019). Simulation of electronic transport properties of a metallic quantum dot in transistor geometry at varying temperatures. Bangladesh Journal of Physics, 25, 11–20.

Rahman, M. L., Khan, M. H. R., Mahmud, S. T., & Hossain, A. K. M. A. (2011). Sintering temperature dependent permeability of nanocrystalline. Journal of Bangladesh Academy of Sciences, 35(1), 67–75. https://doi.org/10.3329/jbas.v35i1.7972

Hossain, A. K. M. A., & Rahman, M. L. (2011). Enhancement of microstructure and initial permeability due to Cu substitution in Ni0.50−xCuxZn0.50Fe2O4 ferrites. Journal of Magnetism and Magnetic Materials, 323(15), 1954–1962. https://doi.org/10.1016/j.jmmm.2011.02.031

Rahman, M. L., Mahmud, S. T., & Hossain, A. K. M. A. (2010). X-ray studies and magnetic properties of Ni-Cu-Zn ferrite. BRAC University Journal, VII(1&2), 9–14. Retrieved from http://hdl.handle.net/10361/893

Rahman, M. L., Haque, Y., Das, S. K., Hossain, M. M., & Rashid, M. H. (2009). Study of Attractive Nonlinearity by 1-D Nonlinear Schrödinger Equation in the Bose-Einstein Condensate. Journal of Bangladesh Academy of Sciences, 33(1), 87–98. https://doi.org/10.3329/jbas.v33i1.2953

Conferences

Rahman, A. R., Rahman, M. L., & Haque, M. F. H. (2019). Measurement and analysis of temperature dependent resistivity of Ni0.30Cu0.20Zn0.50Fe2O4 sample using Lab VIEW. National Conference on Physics. Dhaka, Bangladesh: University of Dhaka.

Shara, J., Mridula, Z. T., Rahman, M. L., Haque, F., & Hossain, A. K. M. A. (2018). Influence of Mn substitution on structural and AC magnetic properties of (Ni0.4Cu0.15Zn0.45)1-xMnxFe2O4. International Conference on Physics. Dhaka, Bangladesh: University of Dhaka.

Haque, F., Muthuraman, H., Rahman, M. L., & Hoq, M. (2018). Electronic transport study of carbon nanotube and poly3-hexylthiophene composite in poly(methyl methacrylate) matrix. International Conference on Physics. Dhaka, Bangladesh: University of Dhaka.

Alam, F., Rahman, M. L., & Hossain, A. K. M. A. (2011). Impact of Cr3+ on the structural and magnetic properties of Mn-cr-Zn ferrite. In Proceeding of International Conference on Advances in Electrical Engineering (ICAEE), 2011 (pp. 280–284).

Rahman, M. L., Khan, M. H. R., Faruk, M. O., & Hossain, A. K. M. A. (2011). Structural and Magnetic Properties of Ni0.45−xCu(x+y)Zn0.55−yFe2O4. AIP Conference Proceedings, 1347(1), 210–213. https://doi.org/10.1063/1.3601819

• Advisor – Physics and Applied Physics & Electronics Programs• Co-advisor - BRAC University Natural Sciences Club (BUNSC)• Editorial Assistant - BRAC University Journal• Member – Interview Board for Undergraduate Physics program• Lab incharge - Undergraduate Physics and Applied Physics & Electronics Laboratories

• PHY 306: Basic Electronics• PHY 115: Electricity and Magnetism• PHY112: Principles of Physics II• PHY111: Principles of Physics I• PHY102: Fundamentals of

Physics• PHY101: Introduction to Physics• APE 302 Electronic Devices and Circuits II • APE 205 Electronic Devices and Circuits I• APE 204 Digital Logic Design• APE 203 Electrical Circuits II• APE 103 Electrical Circuits I

• M. L. Rahman, M. O. Faruk, A .K .M. Akther Hossain, T. Yanagida and T. Kawai, “Magnetic properties of Cu substituted Ni-Zn ferrites”, Presented at the International Conference held at Bangladesh University of Engineering and Technology (BUET), Dhaka-1000, Bangladesh, on March 3-7, 2010• M .O. Faruk, M. H. R. Khan, M. L. Rahman, and A. K. M. Akther Hossain, “Colossal Dielectric Constant of Zn substituted polycrystalline CaCu3Ti4O12” Presented at the International Conference held at Bangladesh University of Engineering and Technology (BUET), Dhaka-1000, Bangladesh, on March 3-7, 2010• M. L. Rahman, S .T. Mahmud, M. H. R. Khan, M. O. Faruk and A. K. M. Akther Hossain,“Characterization of Ni0.50-xCuxZn0.50Fe2O4 Ferrites Prepared by Combustion Technique”, Presented at the National Conference held at Computer Centre, Pre-engineering Building, Chittagong University of Engineering and Technology (CUET), Chittagong-4349, Bangladesh, on 8 November, 2009• M. H. R. Khan, M. L. Rahman, Farhad Alam, J. U. Khan, D. K. Saha and A. K. M. Akther Hossain,“Microstructural Dependence of the Magnetic Properties of Co-Zn Ferrites”, Abstracts of National Conference CUET,Chittagong-4349,Bangladesh,on 8 November, 2009, p8• M. O. Faruk, A. K .M. Akther Hossain, M. L. Rahman, M. Belal Hossen and M. A. Rashid, “Colossal Dielectric Constant of Polycrystalline CaCu3Ti4O12”, Poster presented at the National Conference, on November 8,2009 at Pre-engineering Building (1st Floor) Chittagong University of Engineering and technology(CUET),Chittagong-4349, Bangladesh• M. L. Rahman, A. K. M. Akther Hossain and S. T. Mahmud, “Frequency Dependent Permeability of Nanocrystalline Ni-Cu-Zn Ferrites”, Abstracts of International Physics Conference, BUET, Dhaka, Bangladesh, 15-17 May, 2009, p85• M. L. Rahman, A. K. M. Akther Hossain and S. T. Mahmud, “Preparation and Characterization of Nanocrystalline Ni-Cu-Zn Ferrites”, Poster presented at the International Physics Conference,on May 16 ,2009 at Seminar Room(1st Floor), Bangladesh University of Engineering and Technology (BUET), Dhaka-1000, Bangladesh

• Bangladesh Physical Society• National Oceanographic and Maritime Institute• Member, Alumni Association, Department of Physics, Bangladesh University of Engineering and Technology (BUET), Dhaka• Member, Alumni Association, Department of Physics, Shah Jalal University of Science and Technology (SUST), Sylhet• Member, Physics Society, Department of Physics, Shah Jalal University of Science and Technology (SUST), Sylhet

• Electronic transport measurement• Material characterization• Nonlinear Optics • Micro-device fabrication• Magnetism

Muhammad Lutfor Rahman

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Farzana Siddika completed her school and college from Viqarunnisa Noon School and College. She finished her BSc. And MSc. From the department of Geography and Environment, University of Dhaka in 2010 and 2012 respectively. She completed another joint MSc. Programme from the department of Geography, University of Bonn and United Nations University, Bonn, Germany in 2017.

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Book Chapter

Siddika, F., & Rafiq, N. (2018). Detecting landuse change in Dhaka city using GIS and RS techniques. In H. Khatun, N. Ahmad, A. Q. M. Mahbub, & H. Kabir (Eds.), Environment and sustainable development in Bangladesh: Geographical perspectives (pp. 65–76). Dhaka, Bangladesh: Department of Geography and Environment, University of Dhaka.

Working Paper

Siddika, F. (2018). Promoting sustainable urban development: An assessment of land cover change of Dhaka city. Dhaka, Bangladesh: BRAC Institute of Governance and Development (BIGD), BRAC University.

Elements of Environmental Science (ENV 103)Introduction to Economic Geography (GEO 101)

Environmental Science and Management, Climate Change, Ecosystem based Adaptation, Disaster Risk Management, Sustainable Environmental Development.

Farzana Siddika achieved around five years of research experience working in multiple organizations including national and abroad. She worked as Research Intern at the German Research Center for Geoscience (GFZ), Germany; Visiting researcher at the International Center for Climate Change and Development (ICCCAD), Research Associate at the BRAC Institute for Governance and Development (BIGD), BRAC University.

Farzana Siddika

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Md Hasanuzzaman joined as a lecturer in Microbiology program at the Department of Mathematics and Natural Sciences, Brac University, Dhaka on May 2020. Before joining in Brac University, he worked as Research Investigator & Laboratory Manager in Child Health Research Foundation, Bangladesh. He completed his BSc (Hon.) and MSc in Microbiology from University of Dhaka. After completion of his academic degree, he worked as Research Assistant in the same department. In January 2013, he joined in Child Health Research Foundation and continued his work there before he joined at Brac University.

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BSc (Hons.) in Microbiology & MSc in Microbiology.

Journals

Saha, S., Malaker, R., Sajib, M. S. I., Hasanuzzaman, M., Rahman, H., Ahmed, Z. B., … Saha, S. K. (2020). Complete genome Sequence of a Novel Coronavirus (SARS-CoV-2) isolate from Bangladesh. Microbiology Resource Announcements, 9(24). https://doi.org/10.1128/MRA.00568-20

Saha, S., Ramesh, A., Kalantar, K., Malaker, R., Hasanuzzaman, M., Khan, L. M., … Derisi, J. L. (2019). Unbiased metagenomic sequencing for pediatric meningitis in bangladesh reveals neuroinvasive chikungunya virus outbreak and other unrealized pathogens. MBio, 10(6). https://doi.org/10.1128/mBio.02877-19

Malaker, R., Saha, S., Hanif, M., Ahmed, A., Saha, S., Hasanuzzaman, M., … Saha, S. K. (2019). Invasive pneumococcal infections in children with nephrotic syndrome in Bangladesh. Pediatric Infectious Disease Journal, 38(8), 798–803. https://doi.org/10.1097/INF.0000000000002386

Lo, S. W., Gladstone, R. A., van Tonder, A. J., Lees, J. A., du Plessis, M., Benisty, R., … Consortium, T. G. P. S. (2019). Pneumococcal lineages associated with serotype replacement and antibiotic resistance in childhood invasive pneumococcal disease in the post-PCV13 era: an international whole-genome sequencing study. The Lancet Infectious Diseases, 19(7), 759–769. https://doi.org/10.1016/S1473-3099(19)30297-X

Saha, S. K., Schrag, S. J., El Arifeen, S., Mullany, L. C., Shahidul Islam, M., Shang, N., … Baqui, A. H. (2018). Causes and incidence of community-acquired serious infections among young children in south Asia (ANISA): an observational cohort study. The Lancet, 392(10142), 145–159. https://doi.org/10.1016/S0140-6736(18)31127-9

Hasanuzzaman, M., Malaker, R., Islam, M., Baqui, A. H., Darmstadt, G. L., Whitney, C. G., & Saha, S. K. (2017). Detection of macrolide resistance genes in culture-negative specimens from Bangladeshi children with invasive pneumococcal diseases. Journal of Global Antimicrobial Resistance, 8, 131–134. https://doi.org/10.1016/j.jgar.2016.11.009

Hossain, B., Islam, M. S., Rahman, A., Marzan, M., Rafiqullah, I., Connor, N. E., … Saha, S. K. (2016). Understanding bacterial isolates in blood culture and approaches used to define bacteria as contaminants: A literature review. Pediatric Infectious Disease Journal, 35(5), S45–S51. https://doi.org/10.1097/INF.0000000000001106

Saha, S. K., Hossain, B., Islam, M., Hasanuzzaman, M., Saha, S., Hasan, M., … Whitney, C. G. (2016). Epidemiology of invasive pneumococcal disease in Bangladeshi children before introduction of pneumococcal conjugate vaccine. Pediatric Infectious Disease Journal, 35(6), 655–661. https://doi.org/10.1097/INF.0000000000001037

• July 2012 to December 2012: Worked as a Research assistant at Department of Microbiology, University of Dhaka in a project entitled “Detection of White tail disease causing Macrobrachium rosenbergii nodavirus (MrNV) and its associate extra small virus (XSV) from Shrimps collected from costal area in Bangladesh”• January 2013 to April 2016: Worked as a as Research Officer (Molecular Microbiologist) in Child Health Research Foundation, Dept. of Microbiology, Dhaka Shishu Hospital.• May 2016 to 2019: Working as a Senior Research Officer (Laboratory Manger) in Child Health Research Foundation, Dept. of Microbiology, Dhaka Shishu Hospital.• January 2029 to till: Working as a Research Investigator (Laboratory Manger) in Child Health Research Foundation, Dept. of Microbiology, Dhaka Shishu Hospital.

• 4th investigator meeting of ANISA (Aetiology of Neonatal Infection in South Ssia) at 2013 held in Sandiego, California, USA.• 9th World Congress of WSPID at 2015 held in Rio de Janeiro, Brazil.• 10th ISPPD at 2016 held in Glasgow, Scotland.• 10th ISPPD at 2016 held in Glasgow, Scotland.• 11th ISPPD at 2018 held on Melbourne, Australia.

• Postgraduate Fellowship, 2010-2011. Awarded by National Science and Technology, Bangladesh.• Achieved Dean’s Award for extraordinary performence in BSc. (hons.)• Awarded travel grant in 9th WSPID, 10th ISPPD and 11th ISPPD conference for the submitted abstracts.

Dhaka University Microbiology Alumni Association (DUMAA)Member of American Society of Microbiology

Understanding the etiology causing invasive diseases, dissemination and molecular epidemiology of drug resistance genes, maternal & child health and vaccine impact studies

• 28th August-1st September 2013 on TLDA (Taqman array low density) and MagNa pure automated total nucleic acid extraction method jointly organized by CMC (Christian Medical College) and CDC.• 2nd -12th September 2013 on Molecular typing of Rota virus at Virology lab of Christian Medical College, Vellore, Tamil Nadu, India.• 7th -25th September 2015 on Real-time PCR at Oxford University Clinical Research Unit, Vietnam.• Attended the DBD International Molecular Diagnostics Course organized by

CDC and Dept. of Public Health of University of Minnesota, May 7-11, 2018 in Saint Paul, Minnesota, USA.• Attended the MCHD STATA LEARNING WORKSHOP organized by Maternal & Child Health Division of ICDDR’B from 16-20th September, 2018.

Md Hasanuzzaman

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

With the aim to generate awareness about mathematics education at the tertiary level, Department of Mathematics and Natural Sciences (MNS) of BRAC University organised the “Inter University Mathematics Olympiad 2011” on Friday, December 10, 2011, in collaboration with Bangladesh Mathematical Society (BMS).

A total of 146 talented students from 22 different institutions from all over the country took part in this national competition. Professor Md. Golam Samdani Fakir, Pro Vice Chancellor, BRAC University was present as the chief guest and Prof. Md. Abdus Sattar, President, Bangladesh Mathematical Society, as special guest at the inauguration ceremony. The event celebrated the mathematical and analytical skills of the young undergraduate students who participated in the Olympiad and showcased their talent. Sheikh Ziauddin, Department of EEE, BUET stood first in the competition. Forshad-al-Hossain and Emroze Khan from the same institution stood second and third, respectively. Prof. Jamilur Reza Choudhury, former Vice Chancellor of BRAC University, graced the closing and prize distribution ceremony as the chief guest. A number of acclaimed professors of mathematics from different institutions also added glimmer to the event with their presence.

Thank you for showing interest in Brac University (BracU). Please complete the contact details box on the right.

A member of BracU International Student Recruitment Team will assist you to make your admission process as simple and easy as possible.

Submission of online Applications for Fall 2024 Semester is going on for both Undergraduate and Postgraduate programs. Please visit the link and apply: https://admissions.bracu.ac.bd/academia/

Brac University believes in a holistic approach towards education and grooming future leaders who will possess high moral values, have empathy, more informed, scientific, and socially responsible alongside their academic excellence. Brac University has established 27 clubs to nurture the students to be future leaders in alignment with the vision of the institution.

Brac University believes in a holistic approach towards education and grooming future leaders who will possess high moral values, have empathy, more informed, scientific, and socially responsible alongside their academic excellence. Brac University has established 27 clubs to nurture the students to be future leaders in alignment with the vision of the institution.

Inquiry

Campus Life

The last session of Day 2 of the International Conference on Business and Management (ICBM 2019), was a unique Academia - Industry Discussion where prominent scholars and industry practitioners participated in a roundtable discussion to discuss the co-creation of knowledge. ICBM 2019 organized this platform for the two sides of the coin to meet and discuss how best to bridge this gap and prepare a new generation of workforce that is qualified to tackle the unique challenges of this digital age. Prof. Dr. Mohammad Mahboob Rahman, Dean, Brac Business School conducted the discussion as the session chair.

The participants from academia were Prof. Dr. Hui-Ming Wee, Chung Yuan Christian University, Taiwan; Prof. Dr. Tzong-Ru Lee, National Chung Hsing University, Taiwan; Prof. Dr. Anwar Hossain, Vice Chancellor, Northern University; Prof. Dr. H.M. Jahirul Haque, Vice Chancellor, University of Liberal Arts Bangladesh; Prof. Dr. Abdur Rab, Vice Chancellor, International University of Business and Agricultural Technology; Prof. Dr. Abdul Hannan Chowdhury, Vice Chancellor, Prime Asia University; Prof. Dr. Razamin Ramli, Universiti Utara Malaysia; Dr. Erne Suzila Kassim, Universiti Teknologi Mara, Malaysia; Dr. Rina Fitriana, Universitas Trisakti, Indonesia as well as Prof. Dr. Rahim B. Talukder, Prof. Dr. Akbar Ali Khan, Prof. Dr. Mirza Azizul Islam, Prof. Dr. Salehuddin Ahmed from Brac Business School, Brac University.

To represent the corporate side, Mr. M. Anis Ud Dowla, Chairman, ACI Group, Mr. Syed Abdul Monem, Head of SME banking, BRAC Bank Ltd, Mr. Mominul Islam, Managing Director and CEO, IPDC Finance Ltd, Mr. Reazul Chowdhury, Managing Director and CEO, Runner Automobile Ltd., Mr. A.S.M. Mainuddin Monem, Managing Director, Abdul Monem Ltd, Mr. Anis A. Khan, Managing Director and CEO Mutual Trust Bank, Mr. Abu Hossain Khan, Chairman and Managing Partner, A.H. Khan and Co., Mr. Ishtiaq Ahmed, Kazi Farms Ltd were present.

A wide range of topics were discussed over this lively session that resulted in many suggestions for both academia and industry.

Unique insights on what actually constitutes corporate social responsibility, which interpersonal skills employers want to see in graduates, how the co creation of knowledge can be done by both sides in this era of “Industry 4.0”, what each party expects from the other were discussed during the lively session. Aspects of knowledge synthesis and corporations funding for business curriculums to be revised and made more tailor made to industry needs were discussed, as well as the need for traditional theory so that students can get a more utilitarian approach to education. A notable point put forward was the lack of case studies on Bangladeshi firms and students requiring lessons on the local business environment. The discussion ended on a promising note with Mr. Anis Ud Daula offering to arrange and fund such meetings with academicians in the future.

Ms. Sanjeeda Nazneen has been appointed as a lecturer in MNS Department, Brac University since September 2010. Before joining here, she worked as lecturer in University of Liberal Arts Bangladesh, University of Information Technology and Sciences and Eastern University. She graduated in Mathematics from Department of Mathematics, University of Dhaka, Dhaka, Bangladesh. Also she did her post graduation in Pure Mathematics from there.

Level: 4, Room: 4G47Kha-224 Merul BaddaDhaka 1212. Bangladesh

Selected Publications

Journals

Nazneen, S. (2019). A five step iterative method for solving nonlinear equations. International Journal of Pure and Applied Mathematics, 119(14), 207–212. Retrieved from https://acadpubl.eu/hub/2018-119-14/articles/4/23.pdf

Nazneen, S. (2018). A new approach of Newton’s method using Simpson’s rule. Asian Journal of Applied Science and Technology, 2(1), 135–138.

♦Merit based University scholarship (both in Honors and in MS)

♦ Life member - Bangladesh Mathematical Society

♦ Numerical Analysis

Sanjeeda Nazneen

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Md Tawsif Ur Rashid embarked on his academic journey with a solid foundation, completing his O and A levels from Scholastica under the board of CIE (Cambridge International Examination) in 2014 and 2016, respectively. His quest for knowledge led him to pursue a BSc in Biotechnology at BRAC University, where he graduated in 2021 with the highest distinction, reflecting his exceptional academic prowess.Md Tawsif Ur Rashid's academic excellence and dedication to his field were further recognized when he became one of the very few students Internationally in 2021 to be honored with the prestigious Danish Government Scholarship by the University of Copenhagen for their Master’s Program. Undoubtedly, this accolade underscores his exceptional capabilities and potential in the realm of biotechnology.Continuing his pursuit of academic excellence, he completed his postgraduate studies, earning an MSc in Biotechnology from the University of Copenhagen in 2023, under full scholarship. Throughout his postgraduate journey, he exhibited a keen interest and aptitude in various specialized areas, including Cellular Biology, Membrane Trafficking, Plant Immunology, and Epigenetics.Furthermore, Mr. Rashid possesses a rich repertoire of laboratory skills, spanning from molecular detection techniques to Confocal Imaging and Cloning

Methods. His proficiency extends across various model organisms, including Pichia Pastoris, Plant and Mammalian Cells, and Xenopus Oocyte to list a few, demonstrating his versatility and comprehensive grasp of biotechnological methodologies.In addition, his research interests are diverse yet interconnected, encompassing a range of topics crucial to advancing our understanding of biotechnology. His master's thesis focused on characterizing Qa-SNARE proteins in the innate and adaptive immunity of Arabidopsis Thaliana, highlighting his commitment to unraveling complex biological mechanisms.As a faculty member, Md Tawsif Ur Rashid aims to inspire and mentor the next generation of biotechnologists, fostering a culture of curiosity, innovation, and excellence. His commitment to research, coupled with his strong academic background and practical skills, equips him to make meaningful contributions to both teaching and research endeavors.

Level: 4Kha-224 Merul BaddaDhaka 1212. Bangladesh

BTE405 – Advanced ImmunologyBTE309- Food BiotechnologyBTE303- Basic ImmunologyBTE302- Microbial WorldBTE204- Fundamentals of Genetic EngineeringBTE203- Introduction to Molecular BiologyBTE155- Introduction to Lab IBTE102- Microbial WorldBIO101- Introduction to Biology

- Danish Government Scholarship, 2021-2023 by University of Copenhagen for Masters in Biotechnology- Highest Distinction for BSc in Biotechnology by BRAC University- Vice-Chancellors List by BRAC University from 2018 Spring till 2020 Fall (9-times)- Performance Based Scholarship by BRAC University from 2018 till 2020 Fall- Daily Stars Award, 2015 for Outstanding O levels Result under Cambridge Internation Examination (CIE)

-BRAC University Biotechnology Alumni-Association- Former CMO, H3O Bangladesh

- Cellular and Molecular Biology- Cancer- Cell Signaling in Health & Disease

Md Tawsif Ur Rashid

News & Events

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BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Safeguarding is a process of ensuring all the stakeholders/members of the institution are protected from any sort of sexual harassment, bullying, ragging, humiliation, abuse, exploitation, and threat. Brac University believes all individuals have a right to learn and grow within a safe environment, and we are pledged to provide a safe and positive atmosphere to all the members of Brac University, including students, staff, visitors, partners, faculty members, volunteers, and those who come into contact with the University activities. We will not endure any sexual harassment, bullying, ragging, humiliation, abuse, and violence on any platform and we will take all anticipatory measures to keep our campus and community safe.

Brac University desires to ensure the highest possible standards to meet its social, ethical, and legitimate responsibilities to protect and safeguard the welfare of all its stakeholders/members. Our prime purpose is to safeguard vulnerable groups or individuals from abuse, exploitation, and threat. Safeguarding policy recognizes the welfare of the student as a paramount importance. We always try to provide a safe campus life for our students. We will take the necessary steps to provide a safe learning campus in line with our standard policy and procedure.

Safeguarding policy states the University’s position on preventing and reducing harm for all stakeholders/members of Brac University. This policy promotes and prioritizes the safety and wellbeing of everyone. This policy ensures that roles and responsibilities are made clear in respect of safeguarding affairs. Safeguarding policy offers assurance to all the members of Brac University that safeguarding incidents will be addressed effectively and promptly. Brac University authority pledges that they will continue to find gaps and flaws in the safeguarding policy and address them whenever they are found. They will bring innovative safeguarding solutions to upgrade the policy if needed.

Finally, Brac University is committed to providing a safe environment for everyone regardless of any kinds of challenges. Campus Safeguarding is our supreme priority, and we care about everyone's safety against harassment, bullying, ragging, intimidation, humiliation, abuse, and any kind of violence.

Brac University medical center provides medical services to students, faculty and staff. The center is well equipped to provided necessary medical advice and services to the university community. Another wing of the Emergency Medical Centre is located in the Residential Campus at Saver where professional, qualified doctors and nurses provide medical care.

Medical Center (Dhaka)

Medical Center (Savar)

Badda Campus:Kha 224 Bir Uttam Rafiqul Islam AvenueMerul BaddaDhaka 1212. Bangladesh Ext. 1701

Residential Campus (Savar Campus)Sulla Building, Ground Flooremail: medicalcenter[at]bracu.ac[dot]bd

Dr. Mohammad Rafiqul Islam is mainly specialized in data management, cleaning, analyzing, sampling size and sampling design .Expert in R, SPSS, and familiar with SQL, Python, STATA and other programs. Practical Worked experience as a data analyst and statistician in various projects (with project design, impact evaluation and monitoring).Deep knowledge in Macroeconomics data analysis and models building. Research interest in: Sample size and sampling Design, Time series Analysis, Temporal Disaggregation, Econometrics, Systems of National Accounts, Macroeconomics, Bangladesh economy, Sampling Techniques, Big data, Machine learning.. Language skill in English and Russian. He has been teaching since 2008 as well as in overseas country (Oman). Dr Islam worked as a Senior Researcher at the project “Digital RMG (Readymade Garments) Factory Mapping in Bangladesh (DRFM-B)” funded by C&A foundation under the Centre for Entrepreneurship Development (CED), BRAC University. Dr. Islam received my PhD degree in Macroeconomic Statistics from the Moscow State University of Economics, Statistics and Informatics, December 2006 under Russian government scholarship. He also completed Post-doctoral Research Fellowship in the department of statistical sciences in Universita Degli Studi Di Padova, Italy, under the Research Fellowship of Erasmus Mundus Mobility with Asia (EMMA) funded by the European Commission, October 2009. Besides, Dr. Islam has successfully completed many online courses offered by MIT, Harvard University, JOHNS HOPKINS UNIVERSITY', The University of Texas at Austin, The IE BUSINESS SCHOOL, Madrid, Spain, DataCamp and others. He has 25 publications in national and international referred Journals and one book (PhD dissertation was published by the “Economic library” (online publication).

Level: 4, Room: 4G51Kha-224 Merul BaddaDhaka 1212. Bangladesh

Time series-Temporal Disaggregation (Econometrics)A research report on topic “Evaluation of Different Temporal Disaggregation Techniques and an Application to Italian GDP”, under the research fellowship of Erasmus Mundus Mobility with Asia (EMMA) funded by the European Commission.Department of Statistical Sciences, Universita Degli Studi Di Padova (State Univesity)

Via C. Battisti 241, 35121 Padova,, 35121 Padova, Padova ,Italy

Statistics (Macroeconomic Statistics),

Topics of PhD thesis (dissertation): “Statistical Analysis for the Economic Development of Bangladesh on the basis of System National of Accounts (SNA)”

Moscow State University of Economics, Statistics and Informatics (State University)

MESI, House-7, Street- Nezenskaya, 119501 Moscow, Moscow, Russia

Statistics First class (9th)

Topics of Master’s thesis: “Categorical Analysis of Some Data on Diabetes Mellitus”

Jahangirnagar University, Savar, Dhaka, Bangladesh

Savar, Dhaka, Dhaka-1342, Bangladesh

Statistics, First class (7th),

Jahangirnagar University, Savar, Dhaka-1342, Bangladesh

Diploma and certificate courses

Certificate on online learning of The University of Texas at Austin (UTAustinX),

UT.7.20x: Foundations of Data Analysis - Part 2: Inferential Statistics (with R)

A course of study offered by UTAustinX, an online learning initiative of University of Texas System through edX

Certificate on online learning of The Osaka University, Japan

MED101x: Introduction to Applied Biostatistics

A course of study offered by OsakaUx, an online learning initiative of Osaka University through edX

Certificate on online learning of The University of Texas at Austin (UTAustinX),

UT.7.01x: “Foundations of Data Analysis(with R)” Offered by The University of Texas at Austin (UTAustinX), an online learning initiative of The University of Texas System through edx.

Certificate with DISTINCTION on online learning of The JOHNS HOPKINS UNIVERSIT.

“Regression Models (with R programming)”offered by DEPARTMENT OF BIOSTATISTICS, BLOOMBERG SCHOOL OF PUBLIC HEALTH, an online learning initiative through “coursera.org”.

Certificate with DISTINCTION on online learning of The IE BUSINESS SCHOOL.

“Understanding economic policymaking”ffered by The IE BUSINESS SCHOOL, Madrid, Spain an online learning initiative through “coursera.org”.

Certificate with DISTINCTION on online learning of The JOHNS HOPKINS UNIVERSITY'.

“Statistical Inference (with R programming)”offered by DEPARTMENT OF BIOSTATISTICS,BLOOMBERG SCHOOL OF PUBLIC HEALTH,The JOHNS HOPKINS UNIVERSITY'S, an online learning initiative through “courser.org”.

Certificate with DISTINCTION on online learning of The University of Melbourne.

“Principles of Macroeconomics” offered by The University of Melbourne, an online learning initiative through “coursera”.

Certificate on online learning of The Massachusetts Institute of Technology(MIT)

14.73x: “The Challenges of Global Poverty” offered” by The Massachusetts Institute of Technology(MIT), an online learning initiative of MIT through edx.

9. 15 October 2012 – 14 January 2013

Certificate on online learning of Harvard University

PH207x: “Health in Numbers: Qualitative Methods in Clinical Public Health Research” Offered by Harvard University, an online learning initiative of Harvard University through edx.

10. February 2007 - October 2007

Diploma on Russian for Professional Communication at excellent level Russian language

Moscow State University of Economics, Statistics and Informatics (State University), House-7, Street-Nezenskaya, 119501 Moscow, Moscow (Russia

Selected Publications

Journals

Islam, M. R. (2019). Economic growth rates and exports of Bangladesh: The Bengal tiger? South Asia Research, 39(3), 285–303. https://doi.org/10.1177/0262728019861760

Islam, M. R. (2018). Intensive analyses of the growth rates and the trends of import and its impact to the total economy of Bangladesh. Journal of Business Economics and Management, 1(1), 1–9.

Islam, M. R. (2018). The growth and structural changes of GDP of Bangladesh. International Journal of Entrepreneurship and Development Studies (IJEDS), 6(1), 37–48.

Islam, M. R. (2009). Evaluation of different temporal disaggregation techniques and an application to Italian GDP. BRAC University Journal, VI(2), 21–32. Retrieved from http://hdl.handle.net/10361/460

Jahan, R., Islam, M. R., & Begum, F. (2007). Interrelation among Risk factors of diabetes mellitus and application of log linear models. Jahangirnagar University Journal of Science, 30(2), 191–203.

Islam, M. R. (2006). The comparative analysis of the structure and dynamics of GDP of Bangladesh. Economic Science, 8(21), 36–42.

Conferences

Haque, M., & Islam, M. R. (2019). Analysis the export performance of readymade garments of Bangladesh. In The Proceedings of the 2nd International Conference on Business and Management (ICBM 2019) (pp. 48–51). Dhaka, Bangladesh: Brac University.

Haque, M., & Islam, M. R. (2019). Analysis the export performance of readymade garments of Bangladesh. In 2nd International Conference on Business and Management (ICBM 2019). Dhaka, Bangladesh: Brac University.

Islam, M. R. (2019). Analysis of growth and forecasting the Gross Domestic Product (GDP) of Bangladesh. In 2nd International Conference on Business and Management (ICBM 2019). Dhaka, Bangladesh: Brac University.

Islam, M. R. (2019). Analysis of growth and forecasting the Gross Domestic Product (GDP) of Bangladesh. In The Proceedings of the 2nd International Conference on Business and Management (ICBM 2019) (pp. 73–76). Dhaka, Bangladesh: Brac University.

Islam, M. R., Mahbub, D., & Proma, S. N. (2017). The trends of import and its contribution to the economy of Bangladesh. In 1st International Conference on Business and Management (ICBM 2017). Dhaka, Bangladesh: Brac University.

Islam, M. R., & Haque, M. (2017). The trends of Export and its Consequences to the GDP of Bangladesh. In 1st International Conference on Business and Management (ICBM 2017). Dhaka, Bangladesh: Brac University.

Islam, M. R., & Bhuiyan, R. R. (2017). Analysis of the total economy of Bangladesh in the context of GDP. In 1st International Conference on Business and Management (ICBM 2017). Dhaka, Bangladesh: Brac University.

Islam, M. R. (2014). R program for temporal disaggregation: Denton's method. In 2nd International Conference on Applied Information and Communication Technology (pp. 183–189). Muscat,Oman: Reed Elsevier India Private Limited.

Academic responsibilities:

- Departmental Quality Control Coordinator

-Member of Core Facilitation Team for Faculty Development Initiative at BRAC University

- Adviser of the Natural Science club

- Served as a Convener of the departmental examination result scrutinizing committee (in four semesters)

-Member of the departmental question moderation committee.

Involvement in the Project (Consultancy)

Thesis supervised

(BRACU):

Courses Taught (Others)

viii. Mohammad Rafiqul Islam (2019). Presented paper on “Analysis of Growth and Forecasting the Gross Domestic Product (GDP) of Bangladesh”. 2nd International Conference on Business and Management (ICBM 2019), ISBN 978-984-344-3540, April 25-27, 2019,Dhaka, Bangladesh, BRAC Business School (BBS), BRAC University.

vii. Mohaiminul Haque Mohammad Rafiqul Islam (2019). Presented paper on “Analysis the Export Performance of Readymade Garments of Bangladesh”. 2nd International Conference on Business and Management (ICBM 2019), ISBN 978-984-344-3540, April 25-27, 2019,Dhaka, Bangladesh, BRAC Business School (BBS), BRAC University.vi. Mohammad Rafiqul Islam (2018). Presented paper on “Foreign Trade of Bangladesh-In the context of the growth rates of Export and Import and impact to the country Economy” 5th Annual International Conference on Business, Law & Economics, 7-10 May 2018, Athens, Greece by ATHENS INSTITUTE FOR EDUCATION AND RESEARCH (ATINER).

v. Mohammad Rafiqul Islam, Rafiur Rahman Bhuiyan (2017). Presented paper on “Analysis of the Total Economy of Bangladesh in the Context of GDP”. 1st International Conference on Business and Management (ICBM 2017), ISBN 978-984-34-2360-3, September 21-22, 2017 in Dhaka, Bangladesh, BRAC Business School (BBS), BRAC University.

iv. Mohammad Rafiqul Islam, Daniel Mahbub, and Shahida Najnin Proma (2017). Presented paper on “The trends of Import and its contribution to the Economy of Bangladesh”. 1st International Conference on Business and Management (ICBM 2017), ISBN 978-984-34-2360-3, September 21-22, 2017 in Dhaka, Bangladesh, BRAC Business School (BBS), BRAC University.

iii. Mohammad Rafiqul Islam, Mohaiminul Haque (2017). Presented paper on “The trends of Export and its Consequences to the GDP of Bangladesh”.1st International

Conference on Business and Management (ICBM 2017), ISBN 978-984-34-2360-3, September 21-22, 2017 in Dhaka, Bangladesh, BRAC Business School (BBS), BRAC University.

ii. Mohammad Rafiqul Islam (2015). Presented paper on “Sample Size Determination and a case study related to Central Limit Theorem (CLT)”. 19th international Math Conference on 18-20 December 2015, BRAC University, Dhaka, Bangladesh.

i. Mohammad Rafiqul Islam (2014). R Presented paper on “Program for Temporal disaggregation: Denton’s Method”. Proceedings of "The 2nd International Conference on Applied Information and Communications Technology" - ICAICT 2014, April, Middle East College, Muscat, Oman.

Books (Module Notes)

Mohammad Rafiqul Islam, PhD

News & Events

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EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

The professors of SHSS prepare students to the challenges of the 21st century by developing ethical and analytical skills that are critical to autonomy, resilience, innovation, and recovery. As should be expected from a faculty trained in several of the best universities in the world, our connections to leading international institutions are extensive. We participate in several global initiatives that mobilize knowledge and society and reflect the values and commitments of BRAC to justice and empowerment.

The departments of our School offer:

For an extensive description of our degree programs and to meet our faculty, you may visit the webpages of the Departments of Economics and Social Sciences (ESS) and of English and Humanities (ENH):

International Scholarship Committee has revised the criteria for International Scholarship/Financial Aid as follows; effective from Summer 2023:

1. BRAC Scholarship (A Needs Based Scholarship)

100% tuition and other fees (semester fees, lab fees, library etc.) waiver with food and accommodations awarded to economically disadvantaged but meritorious students

Eligibility and other Requirements:

2. Merit Scholarship Based on BracU Academic Results

Eligibility and other Requirements:

The CGPA range and respective scholarship percentages are as follows:

CGPA

3.70 - 3.84

3.85 - 3.89

3.90 - 3.94

3.95 - 3.99

4.00

Percentage

10%

25%

50%

75%

100%

General Conditions applicable for Scholarship/ Financial Aid:

International Undergraduate Students:

A1. In the first semester, students may take two non-credit courses plus one credit course (optional) or one non-credit course plus two credit courses.

A2. After the first semester students are required to take minimum 12 credits in each semester in order to continue any type of scholarship.

A3. If a student fails (F) in any subject(s), s/he will not be considered for tuition fee waiver for two subsequent semesters except Merit Scholarship. However, s/he will be allowed to re-apply with proper documents in the third semester with required CGPA.

A4. Tuition fee waiver is not awarded on retake/repeat subject(s). The student is required to pay for the retake/repeat subject(s). For Merit Scholarship, if a student takes any retake or repeat courses in the previous semester or in the current semester then s/he will not be eligible for scholarship.

A5. Students having “Incomplete (I)”/Pending (P) grade in a semester will be enlisted for Scholarship/Financial Aid in their current semester but have to pay for all the courses until s/he obtains the required CGPA after “make-up” or grade submission. His/her waiver will be reimbursed/adjusted in the next semester.

A6. Students, suspended/penalized for disciplinary reasons, may not be eligible for scholarship/financial aid for the remaining academic period at Brac University (starting from the date such decision is made).

A7. If a student is eligible for more than one category of Scholarship/Financial Aid, only the higher one will be adjusted.

A8. Scholarship is not applicable on Non-Tuition Fees (i.e., Admission Fee, IT Facility Uses Fee, PHR Lab Fee, Studio Lab Fee, CSE Lab Fee, Residential Fee, Library Fee, Library Membership Fee, Student Activity Fee etc.) except BRAC Scholarship.

A9. Scholarship is awarded to the number of credit hours assigned for completion of graduation for each program. Waiver is not awarded for additional credit hours taken by a student for their double major and others.

A10. Scholarship Committee may change the policy and eligibility criteria from time to time based on the availability of funds. The Scholarship Committee will make its endeavors to give advance notice of changes.

International Postgraduate Students:

A1. The students are required to take a minimum of 9 credit hours in all the semesters. (This will be applicable to students admitted from Summer- 2023 semester onwards.)

A2. Scholarship/Financial Aid will be applicable after considering the tuition fee waiver of 10% - 40% which is based on the student's undergraduate results.

A3. If a student fails (F) in any subject(s), s/he will not be considered for tuition fee waiver for two subsequent semesters except Merit Scholarship. However, s/he will be allowed to re-apply with proper documents in the third semester with required CGPA.

A4. Tuition fee waiver is not awarded on retake/repeat subject(s). The student is required to pay for the retake/repeat subject(s). For Merit Scholarship, if a student takes any retake or repeat courses in the previous semester or in the current semester then s/he will not be eligible for scholarship.

A5. Students having "Incomplete (I)"/Pending (P) grade in a semester will be enlisted for Scholarship/Financial Aid in their current semester but have to pay for all the courses until s/he obtains the required CGPA after "make-up" or grade submission. His/her waiver will be reimbursed/adjusted in the next semester.

A6. Students, suspended/penalized for disciplinary reasons, may not be eligible for scholarship/financial aid for the remaining academic period at Brac University (starting from the date such decision is made).

A7. If a student is eligible for more than one category of Scholarship/Financial Aid, only the higher one will be adjusted.

A8. Scholarship is not applicable on Non-Tuition Fees (i.e., Admission Fee, IT Facility Uses Fee, PHR Lab Fee, Studio Lab Fee, CSE Lab Fee, Residential Fee, Library Fee, Library Membership Fee, Student Activity Fee and etc.) except BRAC Scholarship.

A9. Scholarship is awarded to the number of credit hours assigned for completion of graduation for each program. Waiver is not awarded for additional credit hours taken by a student for their double major and others.

A10. Scholarship Committee may change the policy and eligibility criteria from time to time based on the availability of funds. The Scholarship Committee will make its endeavors to give advance notice of changes.

The BRAC Business School (BBS) offers an undergraduate program: Bachelor of Business Administration (BBA) and two graduate programs: Master of Business Administration (MBA) and Executive Master of Business Administration (EMBA). Each program offers a unique set of specializations in a wide variety of business disciplines: Accounting, Finance, E-Business, Entrepreneurship, Human Resource Management, Information System, Management, Marketing, Operations and Supply Chain Management and others.

The School of Data & Sciences is composed of the Deparment of Computer Science and Engineering (CSE) and the Mathematics & Natural Sciences Department (MNS). It is the largest school in the university. The school's focus is core computer science and engineering and fundamental sciences. Using the combined expertise of both the CSE and MNS departments it aims to be a leader in data science and related areas.

The BSRM School of Engineering is currently composed of the Department of Electrical

& Electronic Engineering (EEE). It intends to be the globally-recognized eminent engineering and computing school through excellence in teaching, research, innovation and technological entrepreneurship. It is our mission to produce technically competent, socially responsible, best quality engineering graduates with effective communication skills, highest ethical standards and human values...Read More

The School of Law at Brac University is a gateway through which students are prepared for careers in law, in administrative services, the judiciary and in the development sector. The four-year undergraduate programme at the School of Law culminates into a Bachelor of Laws (LL.B.) degree for successful students. Although the primary emphasis of the programme is on law and the legal profession, given that law is intertwined with economics, development, business and sociology, it also prepares students who are inclined to seek professions in other disciplines.

School of Humanities and Social Sciences (SHSS) Founded in 2019, our School supervises our university's teaching and research activities in the liberal arts. The SHSS Departments of Economics and Social Sciences and of English and Humanities offer:

Pharmacy is a multidisciplinary subject. The good practice of pharmacy today is one of the most crucial components in providing proper health services to people worldwide. It is a profession of honor and high esteem that demands huge responsibilities. Established in 2010, at the Department of Pharmacy, Brac University, we have designed an undergraduate programme to produce competent graduates, who will be capable of meeting the challenges of a rapidly evolving healthcare sector.

The James P Grant School of Public Health (JPGSPH) at Brac University was established in 2004 as an international educational and research institution focusing on the integral areas of teaching, research, and services. The goal of the School is not only to impart knowledge but also to act as a centre of excellence in knowledge creation through research and training that connect with practice.The School's guiding vision -

“knowledge and know-how for health equity” - draws on the inspirational leadership of James P. Grant, former executive director of UNICEF, after whom the School is named. The three primary areas of activity in the School are related to education, research and advocacy.

Aside from the flagship educational programmes, i.e. Master of Public Health (MPH), JPGSPH also provides public health short courses for health professionals through the Continuing Education Programme (CEP). Additionally, JPGSPH possesses a burgeoning research portfolio conducting innovative and pioneering studies on public health issues funded by multiple international donors. JPGSPH also offers services in the form of training, advocacy workshops, and seminars with a special focus on the rights of the marginalized and vulnerable population of Bangladesh.

The School of Architecture and Design(SoAD) is Brac University’s new venture. This follows from the tradition set by the Department of Architecture, which has established a name for itself in high quality design education. The School will be comprehensive in its scope to include design disciplines that has relevance to the local context as well as international trends.

Brac Institute of Languages (BIL) of BRAC University is working with the goal to assist students in developing the language skills they need to become successful in graduate schools and in their professional lives. BIL focuses on implementing student-centered and creative language teaching techniques by developing modules based on students’ proficiency level. This not only helps avoid having mixed ability classes, but also ensures small classes with excellent interaction between teachers and learners.

BIL offers well-designed, comprehensive courses ranging from the pre-intermediate to the upper-intermediate level in order to facilitate the English language requirements of the university students. Besides English, BIL currently offers the modern languages Bangla, German, Chinese, French, Arabic and Spanish to both its students and other professionals.

Additionally, BIL offers Pre-University, which is a unique course designed for those students who proved themselves competent in their respective disciplines during the admission test except in English. Besides, MA in TESOL, the flagship programme of BIL, is creating efficient English language teachers and leaders from around the country with its well organized curriculum, and innovative and practice-oriented teaching approach.

Since its inception in 2004, BRAC Institute of Educational Development (BRAC IED) has been an active advocate and proponent in playing a key role in the education sector worldwide. From professional development to capacity building and research, it aims to redefine, reframe and re- envision education to share the voice of South Asia and Africa. The Institute wants to tap into emerging areas of development by pioneering in Early Childhood Development (ECD), psychosocial wellbeing, model development, mental health, Primary and Secondary education, Higher education and curriculum.

Brac Institute of Educational Development (BRAC IED) runs two academic programmes, the Masters of Education (MEd)/ Post-Graduate Diploma (PGD) in Educational Leadership, Planning, and Management as well as the Masters in Science (MSc)/PGD in Early Childhood Development. Besides PGD/ MEd/MSc, BRAC IED has also designed a number of online short courses, as an initiative to bring innovative solutions in education and well-being.

Brac Institute of Governance and Development (BIGD) is a resource center that promotes research and develops knowledge on practical solutions to issues of poverty, inequity and social injustice. At the core of BIGD’s work are several academic courses, which include both classroom teaching and fieldwork experience:

Brac University’s General Education program provides a common education foundation to all students before they take courses specific to their major and minor fields of study. The program promotes the core aims of the liberal arts curricula: to develop well-rounded individuals who have a broad range of knowledge on a variety of subjects and a mastery of skills that can be used across disciplines. Students graduate from the program with the ability to think critically, learn creatively and communicate skilfully.

Brac Business School

School of Data and Sciences

BSRM School of Engineering

School of Law

School of Humanities and Social Sciences

School of Pharmacy

James P Grant School of Public Health

School of Architecture and Design(SoAD)

Brac Institute of Languages

Brac Institute of Educational Development

Brac Institute of Governance and Development

School of General Education

Iftekhar Bin Naser joined the Department of Mathematics and Natural Sciences at Brac University as an Assistant Professor in May, 2018. Under the MEXT scholarship, he received his Ph.D. in Medical Sciences from Kumamoto University, Japan. He completed his Master's (M.Sc.) and Bachelor's (B.Sc.) in Biochemistry and Molecular Biology from the University of Dhaka. Immediately after obtaining his Master's degree, he joined icddr,b as Research Officer in Molecular Genetics Laboratory. After completing the Ph.D. he performed two postdoctoral training in the same University of Japan and in University of Connecticut of USA. Through his research inquisition, he has explored the fields of Neurobiology, Molecular Biology, and epidemiology of enteric infectious disease in Bangladesh.

Level: 4, Room: 4G40Kha-224 Merul BaddaDhaka 1212. Bangladesh

• Doctor of Philosophy, Graduate School of Medical Sciences, Kumamoto University, 2009• Master of Sciences, Department of Biochemistry and Molecular Biology, University of Dhaka, 2001• Bachelor of Sciences, Department of Biochemistry and Molecular Biology, University of Dhaka, 1999

Journals

Naser, I. B., Mozammel Hoque, M., Faruque, S. N., Kamruzzaman, M., Yamasaki, S., & Faruque, S. M. (2019). Vibrio cholerae strains with inactivated cqsS gene overproduce autoinducer-2 which enhances resuscitation of dormant environmental V. Cholerae. PLoS ONE, 14(10). https://doi.org/10.1371/journal.pone.0223226

Naser, I. B., Hoque, M. M., Nahid, M. A., Tareq, T. M., Rocky, M. K., & Faruque, S. M. (2017). Analysis of the CRISPR-Cas system in bacteriophages active on epidemic strains of Vibrio cholerae in Bangladesh. Scientific Reports, 7(1). https://doi.org/10.1038/s41598-017-14839-2

Naser, I. B., Hoque, M. M., Abdullah, A., Bari, S. M. N., Ghosh, A. N., & Faruque, S. M. (2017). Environmental bacteriophages active on biofilms and planktonic forms of toxigenic Vibrio cholerae: Potential relevance in cholera epidemiology. PLoS ONE, 12(7). https://doi.org/10.1371/journal.pone.0180838

Hoque, M. M., Naser, I. B., Bari, S. M. N., Zhu, J., Mekalanos, J. J., & Faruque, S. M. (2016). Quorum regulated resistance of vibrio cholerae against environmental bacteriophages. Scientific Reports, 6. https://doi.org/10.1038/srep37956

Shinmyo, Y., Riyadh, M. A., Ahmed, G., Naser, I. B., Hossain, M., Takebayashi, H., … Tanaka, H. (2015). Draxin from neocortical neurons controls the guidance of thalamocortical projections into the neocortex. Nature Communications, 6. https://doi.org/10.1038/ncomms10232

Hossain, M., Ahmed, G., Naser, I. B., Shinmyo, Y., Ito, A., Asrafuzzaman Riyadh, M., … Tanaka, H. (2013). The combinatorial guidance activities of draxin and Tsukushi are essential for forebrain commissure formation. Developmental Biology, 374(1), 58–70. https://doi.org/10.1016/j.ydbio.2012.11.029

Ahmed, G., Shinmyo, Y., Ohta, K., Islam, S. M., Hossain, M., Naser, I. B., … Tanaka, H. (2011). Draxin inhibits axonal outgrowth through the netrin receptor dcc. Journal of Neuroscience, 31(39), 14018–14023. https://doi.org/10.1523/JNEUROSCI.0943-11.2011

Ahmed, G., Shinmyo, Y., Naser, I. B., Hossain, M., Song, X., & Tanaka, H. (2010). Olfactory bulb axonal outgrowth is inhibited by draxin. Biochemical and Biophysical Research Communications, 398(4), 730–734. https://doi.org/10.1016/j.bbrc.2010.07.010

Zhang, S., Su, Y., Shinmyo, Y., Islam, S. M., Naser, I. B., Ahmed, G., … Tanaka, H. (2010). Draxin, a repulsive axon guidance protein, is involved in hippocampal development. Neuroscience Research, 66(1), 53–61. https://doi.org/10.1016/j.neures.2009.09.1710

Su, Y., Zhang, S., Islam, S. M., Shinmyo, Y., Naser, I. B., Ahmed, G., & Tanaka, H. (2010). Draxin is involved in the proper development of the dI3 interneuron in chick spinal cord. Developmental Dynamics, 239(6), 1654–1663. https://doi.org/10.1002/dvdy.22299

Su, Y., Naser, I. B., Islam, S. M., Zhang, S., Ahmed, G., Chen, S., … Tanaka, H. (2009). Draxin, an axon guidance protein, affects chick trunk neural crest migration. Development Growth and Differentiation, 51(9), 787–796. https://doi.org/10.1111/j.1440-169X.2009.01137.x

Naser, I. B., Su, Y., Islam, S. M., Shinmyo, Y., Zhang, S., Ahmed, G., … Tanaka, H. (2009). Analysis of a repulsive axon guidance molecule, draxin, on ventrally directed axon projection in chick early embryonic midbrain. Developmental Biology, 332(2), 351–359. https://doi.org/10.1016/j.ydbio.2009.06.004

Islam, S. M., Shinmyo, Y., Okafuji, T., Su, Y., Naser, I. B., Ahmed, G., … Tanaka, H. (2009). Draxin, a repulsive guidance protein for spinal cord and forebrain commissures. Science, 323(5912), 388–393. https://doi.org/10.1126/science.1165187

Faruque, S. M., Naser, I. B., Fujihara, K., Diraphat, P., Chowdhury, N., Kamruzzaman, M., … Mekalanos, J. J. (2005). Genomic sequence and receptor for the Vibrio cholerae phage KSF-1Φ: Evolutionary divergence among filamentous vibriophages mediating lateral gene transfer. Journal of Bacteriology, 187(12), 4095–4103. https://doi.org/10.1128/JB.187.12.4095-4103.2005

Faruque, S. M., Naser, I. B., Islam, M. J., Faruque, A. S. G., Ghosh, A. N., Nair, G. B., … Mekalanos, J. J. (2005). Seasonal epidemics of cholera inversely correlate with the prevalence of environmental cholera phages. Proceedings of the National Academy of Sciences of the United States of America, 102(5), 1702–1707. https://doi.org/10.1073/pnas.0408992102

Basic Biochemistry; Medical Biotechnology; Animal tissue culture and its application; Recombinant DNA Technology, DNA fingerprinting and Molecular Diagnostics; Advanced Molecular Biology

2014 Poster presentation in neuroscience meeting in USA2012 Poster presentation in Cold spring harbor Neurobiology meeting in USA2011 Poster presentation in neuroscience meeting in USA2011 Oral presentation at Global COE annual conference in Japan2010 Poster presentation in Japanese neuroscience meeting in Japan2010 Poster presentation in Dev. Neurobiology meeting in Japan2009 Poster presentation in Japanese neuroscience meeting in Japan2016 Training on next generation sequencing protocol of illumina platform in Singapore2016 Training on automated DNA extraction by Maxwell machine in Singapore2011 Training on Developmental neurobiology at OIST in Japan

Water Microbiology and ecology, Bacteriophage and bacteria interaction, Genomics of enteric bacterial pathogensEpidemiology, evolution of enteric pathogensQuorum sensing and its effect on microbial virulence factorsBiofilms and their biology

Iftekhar Bin Naser, PhD

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Minor in biotechnology for students doing major in other disciplines is also offered. The details of the minor in biotechnology program are given below:Core Courses (21 credits): BCH 101: Basic BiochemistryBCH 202: Enzyme and Enzyme kineticsBTE 101: Introduction to Biotechnology and Genetic EngineeringBTE 103: Plants and PeopleBTE 203: Introduction to Molecular BiologyBTE 204: Fundamentals of Genetic EngineeringBTE 304: Environmental BiotechnologyElective Courses (6 credits):BTE 415: Entrepreneurship in BiotechnologyBTE 417: Enzyme BiotechnologyBTE 315: Bioremediation and BiodeteriorationBTE 313: Biomass and BiofuelsBTE 416: Biochemical Engineering Principles**Each of the above courses has in-built lab exercises.

1. 49 Postdoctoral Fellowship at University of Luxembourg, Luxembourghttps://www.vacancyedu.com/49-postdoctoral-fellowship-at-university-of-luxembourg-luxembourg/

Application Deadline: 21 March, 2022

2. Commonwealth Scholarships 2022 for International Students: Apply Onlinehttps://fully-fundedscholarships.com/commonwealth-scholarships-2022-for-international-students-apply-online/Application Deadline: March 10, 2022

3. Griffith University International Student Excellence Scholarships in Australia, 2022https://scholarship-positions.com/griffith-university-international-student-excellenc

e-scholarships-in-australia/2019/02/15/?utm_campaign=subscribers-NaN&utm_medium=subscribers_push_notification&utm_source=subscribers

Application Deadline: June 03

4. International Stellar Scholarships at Alliance Manchester Business School, UKhttps://scholarship-positions.com/international-stellar-scholarships-at-alliance-manchester-business-school-uk/2021/01/08/?utm_campaign=subscribers-NaN&utm_medium=subscribers_push_notification&utm_source=subscribers

Application Deadline: August 31, 2022

5. Sussex Bangladesh Scholarship (2022)

https://www.sussex.ac.uk/study/fees-funding/masters-scholarships/view/1327-Sussex-Bangladesh-ScholarshipApplication Deadline: September 01, 2022

6. 57 Postdoctoral Fellowship at University of Oxford, Englandhttps://www.vacancyedu.com/57-postdoctoral-fellowship-at-university-of-oxford-england/

Application Deadline: August 31, 2022

Application Deadlines: Different deadlines (Details are in the website)

Application Deadlines: Details are in the website

Application Deadlines: Details are in the website

Application Deadlines: Details are in the website

Application Deadline: December 01, 2021

Application Deadline: September 15, 2021

Application Deadline: July 15, 2021

https://www.studyinternational.com/news/scholarships-in-australia/

16. Yunnan University admissions for international students Chinese Government Scholarship is now open for application!

Application Deadline: April 08, 2022

For any inquiries, please feel free to contact: Ms. Fu/ Ms. Chu through 86-871-65032910 or lxsk@ynu.edu.cn

Anika Ferdous joined the Mathematics Program of the Department of Mathematics and Natural Sciences of Brac University as a Lecturer in January 2019. Prior to that she was a Part-Time Faculty in United International University, Dhaka. She completed her school and college from Viqarunnisa Noon School and College. She finished her BSc. from Department of Mathematics, University of Dhaka and MSc. (Thesis) from Department of Applied Mathematics, University of Dhaka.

Level: 4, Room: 4G44Kha-224 Merul BaddaDhaka 1212. Bangladesh

• MAT 092 (Remedial Course in Mathematics)• MAT 101 (Fundamentals of Mathematics)

• Got Board Scholarship for Bsc. (Honors) result at 2018.• Got Board Scholarship for SSC result at 2012.Fellowship• Got “National Science and Technology (NST)” fellowship as a thesis researcher at 2017.

Fluid Dynamics, Mathematical models in Biology, Numerical Analysis.

Anika Ferdous

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

The Department of Electrical and Electronic Engineering started its journey in 2009 with the primary goal of producing the best quality engineers and professionals in the relevant fields, who are capable of working at the frontier of cutting-edge technology, research, creation of new ideas, and design and development of new products. A dynamic and vibrant department, the Department of Electrical and Electronic Engineering boasts of having outstanding and highly dedicated faculty members with vast experience in both industry and academia, and state-of-the-art laboratory and research facilities. The primary goal of the department is to produce the best engineers, teachers and professionals who are capable of working at the frontier of cutting-edge technology and are involved in research, creation of new ideas, and design and development of new products. At present, the department is the home of 1200+ undergraduate and graduate students under its three programs and so far produced 1000+ alumni, who are significantly contributing nationally and internationally in their chosen field of work.

The department is committed to providing students with a solid foundation of a wide range of EEE courses designed in accordance with the curricula of renowned electrical and electronic engineering schools in North America and Europe, in an intellectually stimulating outcome-based student-centered teaching-learning setting, so that they are not only capable of using state-of-the-art engineering tools to identify, analyze and solve real world problems in sustainable way, but also inspired to strive for excellence in their personal and professional life, play leadership role and uphold human values.

Alongside providing a strong foundation with the fundamentals, the course curricula, laboratory facilities, teaching and learning strategies, and assessment tools and techniques are constantly being adapted to the fast-evolving technology and the stakeholders needs, so as not only to enable the graduates to meet the challenges of the emerging technologies of the 21st century, but also to help accelerate the growth of the new technologies.

At present, the Department of EEE offers the following undergraduate and graduate programs:

A good portion of the faculty members hold PhDs and have conducted a substantial amount of research and gathered experience on teaching and working at related industries at home and abroad. Most of the department's faculty members hold master's degrees from globally reputed universities and they are actively working in pushing the boundaries in their own fields of research, working as a team with students and research assistants towards BracU's common goal of inspiring excellence. The faculty members are continuously publishing their works in reputed journals and conferences in this field. Recent publications of the faculty members include ACS Nano, IEEE Transactions on Electron Devices, IEEE GHTC, IEEE GLOBECOM, IEEE TENCON, etc.

Since its establishment, the EEE department has continued to make significant contributions to social and humanitarian development through technology and research. There are a number of research groups and centres operating under the umbrella of this department. The Control and Applications Research Centre (CARC) focuses on areas of systems and control engineering. Most of the CARC projects involve close collaboration with national industry leaders such as EnergyPac Bangladesh, Brac Solar and IDCOL and international institutes like IEEE SIGHT.

The students from this department have developed the first nanosatellite titled "Brac Onnesha" for Bangladesh in collaboration with the Kyushu Institute of Technology, Japan. Faculty members along with students of this department have developed an up-to-the-minute ground station to communicate with the already deployed "Brac Onnesha" from International Space Station. It was taken there on SpaceX's Falcon 9 rocket.

Furthermore, the Robotics Lab had participated in an annual "Lunabotics Mining Competition" for two years in a row. "Chondrobot-2", a BracU product, did exceptionally well during the second phase of the competition and was ranked as the best performing robot among all others from Asian universities. Currently, the lab is working on other robotic designs and developments focusing on various engineering challenges. A memorandum of understanding was signed between the EEE department and the Bangladesh Council of Scientific and Industrial Research to fabricate and characterize efficient and cost-effective solar energy systems. This giant leap is taking our students' electronics designing skills to the fabrication and testing level in our country's context.

Faculty and Research Staff

Arshad M. Chowdhury, PhD

Md. Mosaddequr Rahman, PhD

Shahidul Islam Khan, PhD

A.K.M Abdul Malek Azad, PhD

Abu Hamed M. Abdur Rahim, PhD

Touhidur Rahman, PhD

Mohammed Belal Hossain Bhuian, PhD

Rony Kumer Saha, PhD

Abu S.M. Mohsin, PhD

A.S. Nazmul Huda, PhD

Saifur Rahman Sabuj, PhD

Md. Rakibul Hasan

Mohaimenul Islam

Ms. Sanjida Hossain Sabah

Aldrin Nippon Bobby

Nahid Hossain Taz

Afrida Malik

Md. Nahid Haque Shazon

Taiyeb Hasan Sakib

Md. Mahmudul Islam

Md. Mehedi Hasan Shawon

Tasfin Mahmud

Raihana Shams Islam Antara

Abdulla Hil Kafi

Md. Ehsanul Karim

Mohammed Thushar Imran

News and Events

EEE student selected for Huawei program in China

BSRM School of Engineering Teams Excel in University Innovation Hub Program at BRAC University

EEE students win Sheikh Jamal Innovation Grant 2024

BRAC University student becomes “Ethics Champion”

Workshop on PCB Design & Prototyping

International Conference On Energy And Power Engineering (ICEPE) 2019, Call For Paper

Seminar on Nano Technology

Seminar on "BRAC ONNESHA: Beyond The Horizon"

MDS

MPH

MS in Biotechnology

If you do not have Acrobat Reader you can download it for free. Click here to download Acrobat Reader at Adobe website.

Flyers

Founded in 2004, the School of Law at BRAC University is a gateway through which students are prepared for careers in law, in administrative services, the judiciary and in the development sector. The four-year undergraduate programme at the School of Law culminates into a Bachelor of Laws (LL.B.) degree for successful students. Although the primary emphasis of the programme is on law and the legal profession, given that Law is also intertwined with economics, development, business and sociology, it also prepares students who are inclined to seek professions in other disciplines.

The faculties of the School of Law are handpicked for their academic excellence and individual expertise. They bring with them teaching-learning experiences from Universities in the United Kingdom, Australia, Russia, the United States of America and Bangladesh. The faculty has individual expertise in the areas of child rights, criminal law, consumer laws, gender studies, business laws, economics, international laws and human rights; all which are shared with the students in course work and in workshops and lecture programmes organised by the School.The School of Law is also noteworthy in the fact that it has both a Law Club and a Moot Court Society, where students are constantly engaged in preparing for moot court competitions, writing research papers and articles for the Law Club newsletter ‘Acumen’ and planning lecture programmes and social awareness activities. The mooters have been successful in several national and international competitions, including the prestigious ICRC Henry Dunant Moot Court Competitions. They are a force to be reckoned with in the national arenaRecognising BRAC’s background and the goals and commitments of BRAC University, the School of

Law endeavours to impart legal education to seek legal solutions that respect social, cultural and aesthetic needs of the people. To meet this goal, it strives to impart to its students not only the tenets of the law, but also legal philosophy, rights-based issues and a broader awareness of the society they live in. Graduates from the School of Law are now pursuing careers as lawyers, judges, corporate legal officers, development workers and academics.

Faculty and Research Staff

K. Shamsuddin Mahmood

Saira Rahman Khan, PhD

Md. Mostafa Hosain

Md. Saimum Reza Talukder

Farhaan Uddin Ahmed

Ms. Sazeeda Johora Thakur

Anusheh Shiham Ali

Upal Aditya Oikya

News and Events

Law Alumni Association of BRAC University Hosts Meeting at Supreme Court

School of Law Faculty Awarded APrIGF 2024 Fellowship

School of Law Alumnus gets fully-funded PhD offers from the UK and Australia

Report Launch: Social Media Platform Regulation - Key Trends in Sri Lanka, India, and Bangladesh

5th Intra University Moot Court Competition- 2024

Certificate Course on ‘Intellectual Property for Innovation, Creativity and Branding’

Women in Innovative Ecosystem: Promoting Inclusivity and Diversity

4th Intra University Moot Court Competition Organized by BRAC University Law Society (BULS)

BRAC University uses Social Media and Networking tools to connect with current students, prospective students, faculty, staff, alumni and the community. Social media sites use two-way communication to create networks among users. Whether they enable discussion, the posting and sharing of photos and videos, or creating or promoting events; like, share and subscribe and get connected with BRAC University.

The official Social Media Pages and Channels are below:

[Contact: ]

(a) Theoretical Courses

ANT 101 Introduction to Anthropology 3 credits

Humans in nature, human evolution, history of culture, rise of early civilizations in the old and new world, organizations of pre-industrial society environment, resources and their distribution; gender, kinship and descent, religion, economics, politics, survival of indigenous groups, forms of culture and society among contemporary peoples, Comparative study of traditional and changing Third World societies, impact of modern world on traditional societies, power and social order; custom and law, conflict and change, Cultural and ethnic diversity.

Suggested Readings

1. Cultural Anthropology, a Global Perspective: R Scupin2. Anthropology: The Exploration of Human Diversity: Conrad P Kottak 3. Anthropology: C Ember and M Ember4. Cultural Anthropology: W A Haviland5. Anthropology – Social and Cultural: Kedar Nath, Ram Nath 6. An Introduction to Anthropology: Victor Barnouw.

ARC 292 Painting 2 credits

Painting as a form of artistic and architectural expression. Introduction to various media in painting. Still life sketches and painting. Study of forms in painting. Landscapes and cityscapes. Colour pencils, crayons, pastels and watercolour. Mixed media. Computers in painting.

ARC 293 Music Appreciation 2 credits

Musical form. Ingredients of music: sound and time. Indian and Western music: melody and harmony. Foundations of sub-continental music: raga system. Presentation of vocal and instrumental music. Modern Bengali music and works of major composers and demonstrations. Western classical music and works of major composers. Music and its rhythm, composition etc.

BCH 101 Basic Biochemistry 3 credits

Concept of life and living processes – The identifying characteristics of a living matter; The Cell and its evolution; From molecules to the first cell; From prokaryotes to eukaryotes; structure and function of sub-cellular organelles; Brief treatment of mitosis and meiosis; Cell Membrane and its Organization- Elementary idea of cellular constituents: Nucleus, Mitochondria, Golgi bodies, Endoplasmic reticulum, Lysogymes and Microbodies; Bacterial and Plant Cell walls; Biomolecules-The small molecules of life – Sugars, organic acids, amino acids, fatty acids and nucleotides; Macromolecules of life: polysaccharides, fats, proteins and nucleic acids; General idea of primary, secondary, tertiary and quaternary structures of proteins; Nucleus and Heredity-Nuclear membrane; Nucleolus- Nuclear pores; Chromosomes; Packaging of DNA; DNA as Genetic material; DNA replication- basic concept; Transcription-from DNA to RNA; Translation-RNA to protein Ribosomes and protein synthesis; Mitochondria –The power house; Structure, organization and function; Elementary account of Glycolysis and Krebs cycle, Electron transport chain and role of mitochondria in the later process.Chloroplasts- Capturing energy from the sun; Structure, organization and function; Basic information on 'light' and 'dark' reactions of photosynthesis and participation of chloroplast in the process.

Suggested Readings:

1. Lehninger, Albert L. 1978. Biochemistry., M/s Worth Publishers Inc,, New York 2.

Lehninger, Albert L. 1978. Principles of Biochemistry., M/s Worth Publishers Inc,, New York 3. Matthews & Van Holde, 2nd Ed. Biochemistry, Benjamin Cummings Pub. Co.4. Stryer, L. 4th Ed. Biochemistry.5. Rawn, 1989. Biochemistry.6. Voet & Voet, 1991. Biochemistry

Course Prerequisite: None

BCH 102 Biophysical Chemistry 3 credits

Basic concepts- Moles, Molarity, Normality, Avogadro’s number, Molality; ionization of acids and bases- Bronsted- Lowry concept, Lewis concept, Arhenius concept, strength of acids, pH and pKa of solutions, buffer solutions and buffer capacity, Henderson-Hasselbalch equation, acid base indicators; titration, choice of suitable indicators; Physical properties of water, ionic products of water and pH scale;Thermodynamics- First law; definition; nature of heat and work, first law of thermodynamics, internal energy, enthalpy, molar heat capacities, isothermal and adiabatic expansion;; Second law of thermodynamics- statement of the second law, entropy changes, phase transition, reversibility and irreversibility; Free energy, variation with temperature and pressure, Gibbs-Helmholtz equation, Clausius-Clapeyron equation; application of thermodynamics in biotechnology, open system, high energy compoundsThermochemistry- Exothermic and endothermic reactions, standard enthalpy formation, thermochemical equatios. Reaction enthalpy; Chemical equilibrium- the nature of chemical equlibrium, law of mass action, equilibrium constant, relationship between ^G and Keq; Effect of temperature and pressure, Le Chattelier principle, equilibrium reaction involving protons, coupling of reactions; Chemical kinetics-Definition, reaction rate, rate laws of zero, first and second order reactions, molecularity of a reaction , pseudo first order reaction, half life; determination of order of a reaction, effect of temperature on reaction rates; Catalysis-Definition, types, characteristics of catalysts, activation energy of catalysis,Course Prerequisite: None

Suggested Readings

1. Bahl, B.S. & Tuli, G.D. & Bahl A., Essentials of Physical Chemistry, S Chand & Company, New Delhi, 2004-05 2. A Text Book of Physical Chemistry, P.W. Atkins3. T. Hossain, A Text Book of Heat

BCH 201 Human Physiology 3 credits

Digestion and digestive system: Mechanisms and control of the secretion: Composition of digestive juices; digestion and absorption of foodstuffs; Blood and circulatory system : Composition, formation, destruction and function of blood; blood coagulation; blood groups; tissue fluid; cardiovascular system; Respiratory system and respiratory stimulants : structures and functions of lungs, liver, kidney, pancreas, spleen and nervous system; Water and electrolytic balance; Lymphoid and lymphatic system; Endocrinology : functions, mechanisms and function of testis, ovary, uterus and placenta.

Course Prerequisite: BCH 101

Suggested Readings:

1. M. Grifiths. Introduction of Human Physiology.2. R.F. Schumddt and G. Thews. Human Physiology.3. G. Thews, F. Mustscheler & P. Vaupe. Human Anatomy, Physiology and Pathohysiology.

BCH 301 Basic Immunology 3 credits

History and introduction of immunology: History and development of immunology; introduction to immune system; basic concept of innate and adaptive immunity; cellular and humoral immunity; Cells involved in immune response: General features and functions of lymphoid cells; mononuclear phagocytes; antigen presenting cells; polymorphs; mast cells and platelets; Lymphoid system: Primary and secondary lymphoid tissue; primary lymphoid organs; secondary lymphoid organs and tissue; Innate immunity: Phagocytosis; process of phagocytosis; complement system; activation and biological function of complements; Immunoglobulins: basic structure and function of imunoglobulins; immunoglobulins classes and sub-classes; physiochemical properties; distribution and function of different classes and subclasses of immunoglobulins; memory B cell; genetic basis of antibody heterogenecity; antibody class switching; Antigens: general properties of antigens; antigenic determinants; haptens; Membrane receptor for antigens: B cell surface receptors for antigens; T cell receptors (TCR) major histocompatibiliy complex (MHC); antigens structure function of MHC class 1 and class 2 molecules; gene map of MHC antigens; processing and presentation of peptides by MHC molecules; antigen recognition; antigen-antibody interaction; forces of antigen-antibody binding; haplotype restriction of T cell reactivity; Inflammation: patterns of cell migration; and inflammation and their control; Lymphocyte activation: Interaction of T lymphocyte and APC ; signals for T cell activation; B cell response to thymus dependent and independent antigens; B cell activation by surface Ig and T cells; Immune regulation: regulation of immune response by antigen antibody presenting cells and lymphocytes; idiotypic regulation of immune response; Effector molecules: cytokines; origin, source and effector function; cytokine action and network interaction; Immunity to infection: immunity to extracellular and intracellular bacteria; bacterial survival strategies; immunity to viral infection; innate and specific immune response to viruses; strategies for evading immune defenses for viruses; immunity to parasitic infection; Immunological techniques: precipitation reaction; immunodiffusion; immuno-electroporesis; agglutination; co-agglutination and hemagglutination; complement fixation; direct and indirect immulo-fluroescence; immunoassay; immunoblotting; immunopreciption; florescence-activated cell sorter (FACS); Monoclonal antibiotics: production of hybridoma screening; cloning and large scale production of monoclonal antibodies. Courrse Prerequisite: BTE 207

Suggested Readings:

1. Immunology – M. Roitt et al.2. Essential Immunology – I.M. Roill et al.3. Advanced Immunology – D.K. Male et al.4. Text book of immunology – T.J. Barrett.5. Immunology'

An introduction – I.R. Tizard

BCH 300 Basic Immunology 3 credits

History and introduction of immunology: History and development of immunology; introduction to immune system; basic concept of innate and adaptive immunity; cellular and humoral immunity; Cells involved in immune response: General features and functions of lymphoid cells; mononuclear phagocytes; antigen presenting cells; polymorphs; mast cells and platelets; Lymphoid system: Primary and secondary lymphoid tissue; primary lymphoid organs; secondary lymphoid organs and tissue; Innate immunity: Phagocytosis; process of phagocytosis; complement system; activation and biological function of complements; Immunoglobulins: basic structure and function of imunoglobulins; immunoglobulins classes and sub-classes; physiochemical properties; distribution and function of different classes and subclasses of immunoglobulins; memory B cell; genetic basis of antibody heterogenecity; antibody class switching; Antigens: general properties of antigens; antigenic determinants; haptens; Membrane receptor for antigens: B cell surface receptors for antigens; T cell receptors (TCR) major histocompatibiliy complex (MHC); antigens structure function of MHC class 1 and class 2 molecules; gene map of MHC antigens; processing and presentation of peptides by MHC molecules; antigen recognition; antigen-antibody interaction; forces of antigen-antibody binding; haplotype restriction of T cell reactivity; Inflammation: patterns of cell migration; and inflammation and their control; Lymphocyte activation: Interaction of T lymphocyte and APC ; signals for T cell activation; B cell response to thymus dependent and independent antigens; B cell activation by surface Ig and T cells; Immune regulation: regulation of immune response by antigen antibody presenting cells and lymphocytes; idiotypic regulation of immune response; Effector molecules: cytokines; origin, source and effector function; cytokine action and network interaction; Immunity to infection: immunity to extracellular and intracellular bacteria; bacterial survival strategies; immunity to viral infection; innate and specific immune response to viruses; strategies for evading immune defenses for viruses; immunity to parasitic infection; Immunological techniques: precipitation reaction; immunodiffusion; immuno-electroporesis; agglutination; co-agglutination and hemagglutination; complement fixation; direct and indirect immulo-fluroescence; immunoassay; immunoblotting; immunopreciption; florescence-activated cell sorter (FACS); Monoclonal antibiotics: production of hybridoma screening; cloning and large scale production of monoclonal antibodies. Courrse Prerequisite: BTE 207

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1. Immunology – M. Roitt et al.2. Essential Immunology – I.M. Roill et al.3. Advanced Immunology – D.K. Male et al.4. Text book of immunology – T.J. Barrett.5. Immunology' An introduction – I.R. Tizard

BI0 101 Introduction to Biology 3 credits

An introduction to the cellular aspects of modern biology including the chemical basis of life, cell theory, energetics, genetics, development, physiology, behavior, homeostasis and diversity, and evolution and ecology. This course will explain the development of cell structure and function as a consequence of evolutionary process, and stress the dynamic property of living systems.

Suggested Readings

1. Biology: P.H. Raven and G.B. Johnson2. Biological Science: G. W. Stout and D. J. Taylor3. Advanced Biology: J. Simpkins and J. J. Williams4. Biology: A Fundamental Approach: M. B. Roberts

BTE 207 Introduction to Molecular Biology 3 credits

Mendelism : Mendel's experiments and his interpretation; the basic principles of dominance, segregation and independent assortment; misinterpretations of Mendelian principles; Chromosomal basis of inheritance : the chromosome theory of heredity : sex chromosomes and sex determination; sex-linked genes in human beings; variation in chromosome number and structure; Chemical nature of hereditary material :

experiments with bacteria and bacteriophage indicating DNA to the material of heredity; Chromosome structure in prokaryotes; the Watson and Crick model of DNA structure; alternate forms of the double value and hybridization kinetics.; Replication of DNA : Semi-conservative replication: experiments of Meselson and Stahl; DNA polymerases; proofreading activities of DNA poly-merases; the mechanism of DNA replication; circular DNA replication; Transcription in prokaryotes and eukaryotes : different types of RNA molecules; prokaryotic and eukaryotic RNA poylmerases; mechanisms of transcription in prokaryotes and eukaryotes: post-transcription modification of RNA; interrupted genes in eukaryotes; Mechanism of removal of intron sequences; Translation and the genetic code : polypeptides and proteins; synthesis of polypeptide chain; nonsense mutation and suppressor mutation; the genetic code; Wobble hypothesis; post-translation modification of protein.Course prerequisite: BCH 101,

Suggested Readings:

1. Genetics – M.W. Striekberger2. Molecular Biology of the Gene – Watson, Hopkins, Roberts, Sgteitz and Weiner3. Molecular Biology – David Freifelder4. Essential Genetics – Peter J. Russel5. Principles of Genetics – D.P. Snustad, M. J. Simmons & J.B. Jenkins.

BTE 206 Introduction to Molecular Biology 3 credits

Mendelism : Mendel's experiments and his interpretation; the basic principles of dominance, segregation and independent assortment; misinterpretations of Mendelian principles; Chromosomal basis of inheritance : the chromosome theory of heredity : sex chromosomes and sex determination; sex-linked genes in human beings; variation in chromosome number and structure; Chemical nature of hereditary material : experiments with bacteria and bacteriophage indicating DNA to the material of heredity; Chromosome structure in prokaryotes; the Watson and Crick model of DNA structure; alternate forms of the double value and hybridization kinetics.; Replication of DNA : Semi-conservative replication: experiments of Meselson and Stahl; DNA polymerases; proofreading activities of DNA poly-merases; the mechanism of DNA replication; circular DNA replication; Transcription in prokaryotes and eukaryotes : different types of RNA molecules; prokaryotic and eukaryotic RNA poylmerases; mechanisms of transcription in prokaryotes and eukaryotes: post-transcription modification of RNA; interrupted genes in eukaryotes; Mechanism of removal of intron sequences; Translation and the genetic code : polypeptides and proteins; synthesis of polypeptide chain; nonsense mutation and suppressor mutation; the genetic code; Wobble hypothesis; post-translation modification of protein.Course prerequisite: BCH 101,

Suggested Readings:

1. Genetics – M.W. Striekberger2. Molecular Biology of the Gene – Watson, Hopkins, Roberts, Sgteitz and Weiner3. Molecular Biology – David Freifelder4. Essential Genetics – Peter J. Russel5. Principles of Genetics – D.P. Snustad, M. J. Simmons & J.B. Jenkins.

BTE 307 Advanced Molecular Biology 3 Credits

Mutation: mutation rate, types of mutations, detection of mutations; mutagenic agents molecular basis of mutagenesis; mutation induced by physical and agents. Effects of mutation on genomes, multicellular organisms and microorganism; DNA repair mechanism: Nature of DNA damage; Light dependent repair; excision repair; mismatch repair; post-replication repair; SOS repair; Gene transmission in bacteria: mutant phenotypes in bacteria; basic test for transmformation; conjugation and transduction; transformation and gene mapping; conjugation and gene mapping; the evolutionary significance of sexuality in bacteria; Plasmids: basic features; size and copy number; conjugation and compatibility; plasmid classification; plasmids in organisms other than bacteria; Regulation of bacterial gene expression: Constitutive; inducible; and repressive gene expression; lactose operon in E. coli , induction and catabolic repression; tryptophane operand in E. coli , repression and attenuation; arabionse operon in E. coli ; positive and negative control; transcriptional; translational and post-translational regulatory mechanisms; Genetic-recombination: types of recombination; models of general recombination; molecular basis of homologous and non-homologous recombination; Transposable genetic elements: Transposable elements in prokaryotes; IS elements; composite transposons; Tn3 elements; the medical significance of bacterial transposons; transposable elements in eukaryotes; Mobile genetic elements in eukaryotes(jumping gene)-relevance to plants, Studied in maize; Generation of antibody diversity.Course prerequisite: BTE 207

Suggested Readings:

1. Principles of Genetics – E.J. Gardner, M.J. Simmons & D.P. Snustad2. Molecular Biology of the Gene – Watson, Hopkins, Roberts, Steitz & Weiner3. Gene VI – B. Lewin4. Principles of Genetics – D.P. Snustad, M.J. Simmons & J.B. Jenkins.

BTE 315 Bioremediation and Biodeterioration 3 credits

Biodeterioration of materials: Basic concepts, factors involved in biodeterioration; biodeterioration of leather, wool, fur, feather, stones, plastics and rubber; Control of biodeterioration – physical, chemical and biological methods; Biodegradation of recalcitrant industrial wastes: xenobiotic chemicals in the environment; biodegradable, persistent and recalcitrant wastes; structure-recalcitrance relationship, ring cleavage-Ortho and para cleavage; factors affecting microorganisms to degrade xenobiotics; Biodegradation and metabolism: removal of substituent groups and ring opening in model molecules: biodegradation of pesticides; choloroorganics; organic dyes; phenols and petroleum hydrocarbons; Enrichment and isolation of degradative microbes: recent approaches to enrich and isolate microbes having catabolic properties; Biotechnological aspects for effluent treatment: genetic manipulation, enzyme and specialized bacteria; biodegradability testing; monitoring of the bioremediation of xenobiotic pollutants; Biosensor: use and application of biosensor for detection of pollutants; Approaches to bioremediation: environmental modification for bioremediation; microbial seeding and bioengineering approaches to the bioremediation of pollutants; Biological control of insects and pests: biopesticides of microbial origin; viral bacterial, protozoan and fungal pesticides; Water treatment systems: coagulation and flocculation: sedimentation; filtration; disinfection use of ozone, UV and activated carbon; measurement of treatment efficiency; recent advances in biochemical, serological and molecular techniques for the detection of indicators and pathogenic microorganisms in surface, ground and potable waters; Toxicity testing in wastewater: impacts of toxicity on wastewater treatment; heavy metals; organic toxicants; enzymatic assays and microbial bioassays; Pollution control biotechnology: production of microbial seeds; use of bioaugmentation in waste treatment, use of enzymes and immobilized microbial cells; removal of metals by microbes.Course prerequisite: BTE304,/MIC203

Suggested Readings:1. Microbial Ecology – Atlas & Bartha2. Current perspective in Microbial Ecology - Klug & Reddy 3. Ecological Systems and the Environment – I. Foin4. Biotreatment Systems, volume-II – D.L. Wise5. Wastewater Microbiology – Gabriel Bitton

BTE 401 Bioinformatics 3 credits

Biological databases- Primary sources of sequence and structure data; secondary data bases; Sequence analysis for Molecular biology- primer selection, restriction mapping, protein sequence analysis; Sequence alignment- Scoring matrices- PAM and BLOSUM- Local and Global alignment concepts- dynamic programming methodology; Needlham Wunsch algorithm, Smith-Waterman algorithm; Statistics of alignment score; Multiple sequence alignment; Progressive alignment; Heuristic methods for data base searching- BLAST and FASTA; Phylogenic analysis- Evolutionery chnges in Nucleotide sequences,; Rates and pattern of nucleotide substirution; Methods for phylogenic estimation- Maximum parsimony, Distance Matrix Methods and Maximum Likelihood Methods; Functional Perl Programming for bioinformatics; String processing; Regular Expressions; Object oriented programming in Perl.Course prerequisite: CSE 101, BTE 203/BTE 207, BTE204/MIC401

Suggested Readings

1. Baxevanis, A.D. Quellette, B.F.F. Bioinformatics2. Mount, D.W. Bioinformatics: Sequence and Genome analysis3. Gaur, D. Li, W-H. Fundamentals of Molecular evolution4. Tisdall,J.D. Mastering Perl foe Bioinformatics5. Claverie, J.M Notredamame, C. 2003. Bioinformatics for Dummies

BTE 404 Bioprocess Technology 3 credits

Development of bioprocess technology- Methods of biocatalysis and biotransformations, Concepts and general features of biotransformations- Procedures, techniques and media for biotransformations; Reaction in solvent mixtures; Equipment, automation, standardization, quality control and quality assurance; Optimization procedures; Examples of bioconversion processes; Scale up of microbial processes- Criteria used for scale up; Important factors for development of microbial processes- physical, chemical process and sterilization factors; Processing of Food and Feed- Raw materials, microorganisms, processing; Food value and economic importance of fermented foods-idli, dahi, tofu, tempeh; Microbial up gradation of poultry and animal feed; Biocomposting processes- succession of microorganisms, bioelemental changes, Applications; Single step microbial processes of industrial importance- Steroid biotransformation, vinegar production etc.Course Prerequisites: BTE306/MIC308, BTE302/MIC304

Suggested Readings

1. Rehm, H-J. and Reed G. Biotechnology 2nd Edition. Volume 3. Bioprocessing2. Domain, A.L. and Davies, J.E. Manual of Industrial Microbiology and Biotechnology3. Steinkraus, K.H. Handbook of Indigenous Fermented Foods4. Chahal, D.S. Food, Feed and Fuel from Biomass5. Mizrahi, A. and van Wezel, AL. Advances in Biotechnological Processes

MIC 101 General Microbiology 3 credits

Development of microbiology: Discovery of microorganisms; biogenesis versus abiogenesis; fermentation process; germ theory of disease; Koch’s postulates; development of laboratory techniques; vaccination; Chemotherapy; Scope of microbiology: in human welfare, agriculture, industry, health and sanitation; environment and pollution control; Prokaryotic and eukaryotic cells-Morphological characterization and ultra structure of procaryotic and eucaryotic cells: functions of different subcelluar elements; distinctive characteristics of the major groups of microorganisms; significance of smallness; endosymbiotic hypothesis; Bacteria: size, shape and arrangements; characteristics of major groups of Gram-negative and Gram positive bacteria; Archaea : general characteristics; morphological and physiological diversity.; Viruses : discovery : general characteristics; morphology; chemical composition; classification and nomenclature of bacteriophages; bacteriophage life cycle : lytic and lysogenic life cycles; replication and one-step multiplication curve; viroids; prions; Fungi : general morphological characteristics; growth and reproduction; classification; importance in industry and natural process; Algae : general characteristics; classification; microscopic algae and their importance; Protozoa : general characteristics; classification of major groups; importance in natural process.

course Prerequisite: None

Suggested Readings

1. G. Schlegel et al. 1993. General Microbiology-II, -. 7th Ed. Cambridge University Press2. T.D. Brock et al. Biology of Microorganisms . 9th Ed. Prentice Hall International Inc.3. M. J. Pelczar, Jr; E.C.S. Chan & N.R. Krieg. Microbiology. 5th Ed. Tata Mcgraw Hill Publishing4. G.J. Tortora et al. Microbiology: An Introduction. 6th Ed. The Benjamin Cunnings Publishing Co. 5. A.J. Salle et al.Fundamental principles of Bacteriology.

MIC 102 Basic techniques in Microbiology 3 credits

Microscopes and Microscopy: light spectrum, resolving power and magnification power; microscopes: light and electron microscopes; Microscopy; bright-field, dark-field, fluorescence, phase-contrast, differential interference contrast, transmission electron, scanning electron, scanning tunneling and atomic force microscopy; Observation of microorganisms under microscope: wet-mount and hanging-drop technique: preparation of microorganisms for staining; chemical properties of stains; mechanisms of staining; chemical positive and negative staining, simple, differential and special staining technique; Cultivation of microorganisms : nutritional requirements; physical and gaseous requirements; media used for cultivation microorganisms: chemically defined media, complex media, anaerobic growth media, selective and differential media; enriched culture; anaerobic culture method; pure culture techniques; Characterization of microorganisms : morphological characteristics; nutritional and cultural characteristics metabolic characteristics; antigenic characteristics pathogenic characteristics; genetic characteristics; Culture preservation : Techniques for long-term and short-term preservation of microbial culture; Measurement of growth : direct measurement of microbial growth; estimating bacterial number by indirect methods; Control of microbial growth : principles of microbial control; the rate of microbial death; the action of microbial control agents; conditions influencing microbial control; physical and chemical methods of microbial control.Course Prerequisites: MIC 101/BTE 102

Suggested Readings:

1. G. Schlegel . General Microbiology –II2. T.D. Brock et al. Biology of Microorganisms 3. M.J. Pelczar, Jr.; E.C.S. Chan & N.K. Krieg. Microbiology 4. G.J. Tortora et al. Microbiology : An Introduction5. A.J. Salle . Fundamental Principles of Bacteriology

MIC 201 Microbial Chemistry 3 credits

Biomolecules and biopolymers : Properties and functions of the major and minor essential elements, water, carbohydrate, lipids, nucleic acids and proteins; Molecular architecture of microbial cells : chemical composition and function of cellular structures and organelles : capsule, flagella, pili, cell-wall, cytoplasmic membrane, pigments, ribosome, mitochondria, cytoplasmic inclusions and endospore; Antimicrobial agents : type, chemistry, mode of action, efficiency, and the antimicrobial resistance; study of penicillin, tetracycline, chloramphenicol, nyastatin, gentamicin and griseofulvin; effectiveness of antimcrobial agents : assay of antibiotics by chemical methods.Course Prerequisites: MIC 101/BTE 102, BCH101

Suggested Readings:

1. Gottschalk . Bacterial Metabolism.2. A. H. Rose . Chemical Microbiology –3. N.S Agorov. Antibiotics : a scientific approach.4. A.L. Lehninger. Biochemistry.

MIC 202 Microbial Metabolism 3 credits

Introduction to metabolism: Important differences and relationships between anabolic and catabolic mechanisms in life; Cell bioenergetics energy production, ATP generation by different process, free energy, energy coupling; Membrane transport system: active. passive, facilitative and group translocation; Carbohydrate catabolism/Aerobic metabolism processes: the Embden-Meyerhof-Parnas pathway; tricarboxvlic acid cycle; electron transport chain; oxidative and substrate level phosphorylation; Alternate pathways of glucose catabolism: hexose monophosphate pathway: Entner-Doudoroff pathway, glyoxylate cycle, methyl-glyoxal bypass. Inter linkages of pathways. Anapleuretic reactions; Pathways for utilization of sugars other than glucose: starch, cellulose, maltose, sucrose, lactose, sorbitol & mannitol; Catabolic activities of aerobic heterotrophs: growth with organic acids (beta-oxidation), amino acids, aromatic compounds, aliphatic hydrocarbons and CI compounds; Anaerobic metabolic processes: fermentation of ethanol, acetate-butyrate, acetone-butanol, lactate and methane. Methane fermentation.Course Prerequisites: MIC 101, BCH 101

Suggested Readings:

1. A.G. Moat & J.F. Foster. Microbial Physiology.2. Gottschalk . Bacteril Metabolism .3. M.J. Pelezar, Jr. ; E.C.S. Char & N.R. Krieg . Microbiology.4. Lehninger. Principles of Biochemistry.

MIC 203 Environmental Microbiology 3 credits

Biological interactions: Microbial interaction within a single microbial populations, positive and negative interaction, interaction between diverse microbial population. neutralism. commensalisms, synergism, mutualism, competition. ammensalism, parasitism, predation; microbe-plant interaction and microbe animal interaction; Techniques for the study of environmental microbes : sample collection, sample processing, detection of microbial populations. determination of microbial numbers. Determination of microbial biomass, and measurement of microbial metabolism; Microbiology of potable water: introduction to indicator organisms, water borne pathogens, isolation and identification of indictor bacteria and water borne pathogens.; Sanitation and public health microbiology with special reference to Bangladesh : water supply, the use of safe water, public tube well coverage, sanitation, disposal of human excreta and refuse; Microorganisms and some novel pollution problem : persistence and biomagnification of xenobiotic molecules; recalcitrant halocarbons, polychlorinated biphenyls (PCBS), alkyl benzyl sulfonates, synthetic polymer; Sewage treatment : primary secondary treatment acrobic and anaerobic and tertiary treatment.Course Prerequisites: MIC 102

Books Recommended :

1. Microbial Ecology: Fundamentals and Applications R.M. – Atlas and Bartha2. Microbial Ecology: A Conceptual Approach – J.M. Lynch and Poole3. Microbiology – M.J. Pelezar, Jr.; E.C.S. Chan & N.R. Krieg4. Microbiology: An Introduction – G.J. Tortora et al.5. Microbial Ecology: Organisms, Habitats and Activities – Heinz & Stolph.

MIC 204 Medical Microbiology 3 credits

Infection and infectious diseases: Concept of infection and infectious diseases; pathogenesis of infectious diseases; virulence (ID 50, LD50); Brief introduction to virulence factors: Adherence factors; invasion of host cells and tissues; toxins; enzymes; intracellular pathogenesis; antigenic heterogeneity; iron acquisition; Identification of microbes that cause disease: Koch’s postulates and their limitations; Host-Microbe interaction : normal resident micro flora of human body and their role;

initial colonization of a new born; introduction to resident flora of skin, mouth, upper-respiratory tract, intestinal tract, uro-genital tract, eye; Non-specific host defenses against microbial pathogen : primary defenses conferred by tissues and blood; Major reservoirs of microbial pathogens : acquisition of and mode of transmission of diseases; Epidemiology : study of infectious diseases in population; The progress of an infection : true and opportunistic pathogens; portal of entry; size of inoculums; stages in the courses of infections and diseases; mechanism of invasion and establishment of the pathogens; signs and symptoms of a disease; portal of exit; Nosocomial infection : hospital as a source; Brief introduction to the microbiology of major infectious diseases : skins : respiratory system ; nervous system : genito-urinary tract; gastro-intestinal tract; circulatory system.Course Prerequisites:MIC 101, BCH 201

Suggested Readings:

1. Review of Medical Microbiology – E. Jawetz, J.L. Melnick & E.A. Adelberg2. Essential Clinical Microbiology : An introductory Test – E.M. Cooke & G. L. Gibson.3. Manual of Clinical Microbiology – H. Lennette4. Modern Medical Microbiology – M.R. Chowdhury5. Medical Microbiology- J.P. Duguld, B.P. Marimian & R.H.A. Swain.6. Microbial Pathogenesis : A Molecular Approach – A. A. Salvers & D.D. Whitt.7. Medical Microbiology – Mims, Playfair, Roitt, Wakelin & Williams.8. Medical Microbiology – Robert F. Boyed & J. Joseph Marr.

MIC 301 Virology 3 credits

Introduction to virology: Brief history and development of virology; Nomenclature and classification of animal and plant viruses; Virus cultivation: cultivation quantification of plant, animal, and bacterial viruses, purification and identification of virus, one step growth curve, inclusion bodies; Virus replication: steps in virus replication, multiplication and gene expression of DNA and RNA viruses; Pathogenesis of viral diseases.; Bacteriophages: Overview of bacteriophages, genome organization and multiplication of RNA and DNA bacteriophages, temperate bacteriophages, lytic and lysogenic phage ; transposable phages; Prevention and treatment of viral infection: viral vaccines, interferons, induction and action of interferons, antiviral chemotherapy; Viriods and Proins: General properties and diseases caused by viriods and prions; Cell culture : primary, secondary and continuous animal cell cultures.Course prerequisite: MIC 206, MIC 300

Suggested Readings:

1. Microbiology : Concepts and Application – M.J. Pelezar, Jr.; E.C.S. Chan & N.R. Krieg2. Biology of Microorganisms – T.D. Brock et al.3. Virology, 3rd ed. Volume –I and II – Fields4. Fundamentals of Virology – Fields.

MIC 302 Food Microbiology 3 credits

Food and food borne microbes: Introduction to various types of food; food preparation and spoilage; food borne diseases; Factors affecting microbial growth in foods: Intrinsic and extrinsic parameters; Food preservations: General principles of food preservation; preservation by high temperature; low temperature; drying; using food additives; and radiation; Food spoilage and preservation: Cereal and cereal products; sugar and sugar products; vegetables and fruits; meat and meat products; fish and other sea-foods, poultry, milk and milk products; Canned foods; Preparation of fermented foods: Bakery products; products; cheese; yogurt and curd; vegetables products; cabbage; cucumbers; oriental fermented foods.Course prerequisite: MIC 202

Suggested Readings:1. Food Microbiology – W.C. Frazier & D.C. Westhoff2. Modern Food Microbiology, 3rd edn. – James M. Jay

MIC 303 Agriculture Microbiology 3 credits

Major groups of microorganisms in soil: Bacteria, fungi, actinomycetes, algae and virus; Role of microbes in soil fertility and plant nutrition: use of microbial metabolites as major nutrients the effect of growth regulators produced by microorganisms; the liberation of unavailable nutrients from soil organic matter and mineral; suppression of plant pathogens; the production of phytotoxic substances by saprophytes and parasites; the production of enzymes and competition of microorganisms with plants for essential nutrients; Biogeochemical cycling of nutrient elements: The carbon cycle; the hydrogen cycle; the oxygen cycle; the nitrogen cycle; the sulfur cycle; the phosphorus cycle; Microbial degradation of cellulose, hemicellulose, and lignin; Microbial biofertilizer and inoculation of techniques; Microbiological aspects of pesticides behavior in the environment: Purpose and types of uses of pesticides, pesticides in the microbial environment; pesticides in the soil and aquatic environment; effect of pesticides; persistence of pesticides; metabolism of pesticides by microorganism; Microbes as plant pathogens: the concept of diseases in plants, diagnosis and control of plant diseases.Course prerequisite: MIC 203

Suggested Readings:

1. An Introduction to Soil Microbiology – M.Alexander 2. Soil Microorganisms – T.R.G. Gray & S.T. Williams3. Soil Microorganisms and Plant Growth – N.S. Subba Rao.4. Plant Microbiology – R. Campbell5. Plant Diseases – R.S. Shing6. Plant Pathology – Arios7. Microbial Ecology: A Conceptual Approach – J. M. Lyncy & Poole8. Biological Indicators of Soil Health – C.E. Pankhurs, B.M. Doube & ViV.S.R. Gupta9. Pesticide Microbiology – I.R. Hill & S.J.L. Wright

MIC 304 Microbial Biotechnology 3 credits

Historical development of Microbilal Biotechnology- from ancient fermentation to modern microbial technology, scope and essential features of applied microbiology and Biotechnology; Microorganisms of industrial importance- yeasts, molds, bacteria and actinomycetes; Screening and selection of microorganisms for useful products, genetic manipulation of microorganisms for increased productivity; Kinetics of microbial growth and product formation; Microbiological production of foods, drinks, industrial chemicals and pharmaceuticals; major classes of microbial products and processes; Microbiological production of foods- SCP and MBP; baker’s yeast; food additives; Production of industrial chemicals, Detail of some fermentation processes- SCP, baker’s yeast, alcohol; Organic acids ;acetate; citrate; lactate and amino acids; Solvents; alcohol; butanol and acetone; Enzymes; Pharmaceuticals; Antibiotics; Steroids; Vaccines and Antibiotics; Immobilization of microbial cells and enzymes; Immobilized technology for production of industrially important chemicals; Role of microbes in dairy products formation.Course prerequisite: MIC 202, BTE 207

Suggested Readings:

1. Miller & Lidsky. Industrial Microbiology2. G. Reed.Prescott and Dunn's Industrial Microbiology 3. H.J.Rehm, G.Reed & H.Pape.Biotechnology(Vol-1):Microbial FundamentalsIndustrial 4. J. Riviere Applications of Microbiology 5. Smith, J.E. Biotechnology Principles6. Bu Lock, J. 1987 Basic Biotechnology 7. Primrose, Modern Biotechnology.8. Fogerty, 1983. Microbial Enzymes and Biotechnology9. Smith, J.E. Biotechnology Principles10. Brown, C.M., Campbell, I. And Priest, F.G. Introduction to Biotechnology

MIC 306 Pharmaceutical Microbiology 3 credits

Ecology of microorganisms as it affect the pharmaceutical industry: Atmosphere, water, raw material personnel, etc; Sterilization methods: Heat, radiation, gases and filtration system; sterilization kinetics; Microbial spoilage, deterioration and preservation of pharmaceutical products: mixtures, suspension, syrups sterile products, cosmetics and toiletry products; Determination of potency/ concentration of antibiotics and antimicrobial preservatives in pharmaceuticals of antibiotics or products; Microbiological tests: test for sterility, MIC and MBC; pyrogen and pyrogen tests; Aseptic techniques: design and maintenance of an aseptic unit laboratory/ processing area; Production of immunological products: vaccines, immumosera and human globulins, and their quality control; Public health aspect of microbial contamination and infection of fish: fish quality assurance research methodology.Course prerequisite: MIC 202

Suggested Readings:

1. Pharmaceutical Microbiology – W.B. Huge & A.D. Russel2. Dispensing for

Pharmaceutics, Students – Cooper & Gums3. Preservatives in the Food, Pharmaceutical and Environmental Industries – R.G. Board, M.C. Allwood & J.G. Banks.4. Essays in Applied Microbiology – J.R. Noris & M.H. Richmond

MIC 401 Microbial Genetic Engineering 3 credits

Purification of DNA : preparation of total cell DNA; preparation of plasmid DNA; preparation of bacteriophage DNA; Techniques of molecular genetics: production of recombinant DNA in vitro; amplification of recombinant DNA in cloning vector; construction of DNA, RNA and protein by blot techniques, amplification of DNA by PCR: in vitro sitespecific mutagenesis; DNA manipulative enzymes: restriction endonucleases and other nucleases; ligases; polymerases DNA modifying enzymes; topoisomerases; Cloning vectors: cloning vectors for prokaryotic organisms; bacteriophage M13, bacteriophage l plasmid pBR322, plsmid pBR325, pUC119, cosmids, phagemids, and charomid: cloning vectors for eukaryotic organisms: yeast episomal plasmid (2 um circle), cloning vectors for higher plants and mammalian cell; Ligation systems: bustend ligation; sicky-end ligation; putting sticky ends on to a blunt-ended molecule: homopolymer tailing, use of linkers and adaptors; Introduction of recombinant DNA into living cells: transformation of bacterial cells and selection of recombinants; introduction of phage DNA into bacterial cells and selection of recombinant phage; transformation of non-bacterial cells; Gene location and gene structure: locating the position of a cloned gene; chromosome walking; DNA sequencing; the Sanger-Coulson method and the Maxam-Gilbert method; restriction fragment length polymorphism (RFLP) analysis; Expression of cloned gene: requirements for gene expression: expression vectors; transcript of a cloned gene; regulation of gene expression; identifying and studying the translation product of cloned gene;Course prerequisite: MIC 310

Suggested Readings:

1. Principle of Gene Manipulation – R.W. Old & Primrose.2. Molecular Biology of the Gene – J. Watson3. Genetic Engineering – Kingsman & Kingsman4. Principles of

Genetics – D.P. Snustad, M.J. Simmon & J.B. Jenkins5. Gene Cloning: An Introduction – T.A. Brown6. Molecular Cloning : A Laboratory Manual, 2nd Ed. – J. Sambrook & T. Maniatis7. Principles of Gene manipulation: An Introduction – R.W. Old.8. Current Protocol in Molecular Biology – J.A. Smith & K. Struhl.

MIC 402 Analytical Microbiology 3 credits

Spectroscopic techniques: Visible, ultraviolet and infrared spectrophotometers; spectrofluorimetry luminomery; NMR and mass spectrometry; Centrifugation techniques : principle of sedimentation; Centrifuges and their use; density gradient centrifugation and ultracetrifuge; Chromatographic techniques: principle of chromatography; column, thin-layer and paper chromatography adsorption, gas-liquid, ion-exchange, exclusion, affinity and high performance liquid chromatography; Electrophoretic techniques: principle, factors affecting electrophoresis, low and high voltave electrophresis; gel electrophoresis; preparative electrophoresis; Protein characterization: determination of molecular weight, amino acid composition and number of subunit; protein sequencing; Biosensor: principle, transducers; biocomponent of biosensor; application of enzyme-based, cell-based and organelle-based biosensors; affinity binding assay; biological reactant pairs; application of immunosensor and receptor-based sensor; Radioisotope techniques: nature, detection and measurement of radioactivity; application of radioisotopes in the biological sciences; safety aspects of the use of radioisotopes, Microarray, SELEX processCourse prerequisite: BCH301/MIC300

Suggested Readings:1. Protein Purification – Scopes2. Comprehensive Biotechnology, vol. 2. – Murray Moo-Young.3. A Biologists Guide to Principles and techniques of practical Biochemistry, 3rd Ed. – K. Wilson & K.H. Goulding.4. An Introduction to Practical Biochemistry, 2nd Edn. – D.T. Plummer5. Basic biochemical methods, 2nd Edn. –R.R. Alexander & J.M. Griffiths

MIC 403 Microbiological Quality Control of Food, Fish and Beverages 3 credits

Introduction: importance of quality control of food, fish, beverage and mineral water; The organization of quality control: the principles, application, organization, problems and techniques of quality control; the future of quality control; Microbiological quality control: principle and pitfalls; fundamentals of microbiological quality control; chemical and microbiological quality for quality assurance, standards monitoring to assess compliance with good practices; Sanitation and inspection: sanitation and hygiene of processing plant, water in processing and cleaning, waste/effluent treatment packaging, equipment, handling; Quality assurance: sampling, testing panel-sensory assessments in quality control; hazard analyses and critical control point(HACCP) systems; identification of potential hazards; monitoring system for critical control point (CCP), corrective actions, verification; Food laws and regulation: national and international standards and guidelines.Course prerequisite: MIC 302, MIC 308

Suggested Readings:1. Microbiology of Frozen Foods – R.K. Robinson2. Food Microbiology, 4th edn. – W.C. Frazier3. Food Microbiology – M.R. Adams & M.O. Moss4. Manual for the isolation and identification of fish bacterial pathogens – G.N. Frerichs and S.D.5. Miller Modern food microbiology, 3rd edn. – James Jay6. Quality Control in the Food Industry vol. 1. – S.M. Herschdoerfer.

c. Elective Courses (12 Credits):

BCH 203 Enzymes and Enzyme Kinetics credits (3 BTE 203) 3 Credits

Brief history, classification of enzymes; Enzyme assay, enzyme activity, units; Enzyme as biocatalysts, catalytic power, specificity and regulatory propertiesFactors affecting enzyme reactions- pH, Temperature, Oxygen, Substrate concentration; Enzyme kinetics, Mono-substrate reactions: Michaelis- Menten equation and its linear transformations; Km and Vmax-definitions, determination, significance; Enzyme inhibition; Identification of functional group essential for catalysis with biological examples; Factors affecting the efficiency of enzymes as catalysts- Specificity; ES complex; Enzyme regulation, allosteric enzymes, Monod and Koshland models;

examples of regulatory enzymes- aspartate trans carbamylase; phosphorylase; pyruvate dehydrogenase; hexokinase; Mechanism of enzyme action.Course prerequisite: BCH 101

Suggested Readings

1. Boyer, 1970, The Enzymes2. Leninger, A.L. 1987, Biochemistry3. Voet and Voet, 1991, Biochemistry

BTE 312 Computer Application in Biotechnology 3 credits

Introduction- Types of Computers (Mainframe, Mini, Micro0- Hardware, Software- Storage and Memory devices(RAM, ROM, Discs, Tapes etc.)I/O devices (Monitors, keyboards, Printers, Plotters etc.) Directory concepts- File operations; Operative system- Overview- Booting- DOS Files, DOS internal and external commands- file management (creating, editing, deleting and copying); Word Processing- Introduction to textfiles- Word Processors- Word Star package- File operation and utilities under WS.; Data Concepts; Database structure and dscription- Bibliographic and non-bibliographic- Biotechnology databases(Medlin, Agric, EMBL, Genebank, NBRF); Information Retrieval- Information Sources/ Search aids9 Directories, Dictioneries, Glossary; Search operations(Boolian, Keyword, String, sequences); Information access- Online, CD-ROM, NICNET, E-mail; Software- Demo of some PC based Educational Software packages in Genetic Engineering; Presentation of results in text/graphic mode- Harvard Graphics package; Staistical Concepts- Mean, Mode, Median, SD- Probabilty Correlation, Trends, Theorems. Permutation and combination- distributions- Line, bar, pie, sigma plot and Harvard Graphics software package;Analysis of Biotechnological data in Agro base 4 and preparation of similarity indices, dendograms using NTSYS; Analysis of robustness of clusters using WINBOOT; DNA sequence data management; Database and homology search of DNA and protein sequences; Labeling amtorads and Polaroid photographs using Microsoft Excel and presentation of nucleic acid sequences using power point; Scanning od DNA, Protein and Isozyme profiles using computers and scanners;

Retrieving information using INTERNET and CD-ROM.Course prerequisite: CSE 101

Suggested Readings

1. Lesk, M. 2002 Introduction to Bioinformatics2. Brown, T.A. Gene Cloning and DNA Analysis3. Rehm H-J and Reed, G Genomics and Bioinformatics. 2nd Edition. Vol 5b4. Grfith H.G. Griffin, A.M. DNA Sequencin Protocols

BTE 313 Biomass and Bio-fuel 3 credits

Introduction: Importance of fuel energy; Fuels from nature Basic bioenergy interconversion; Formation of biomass & its conversion to fuel; Potential Biomass for Fuel Production: Types of natural biomasses; Land crops aquatic plants & waste materials; Production of desirable biomass, Advantages & problems in utilization of biomass for fuel generation, Pretreatment of biomass for fuel production; Bioconversion of Biomass to Methane: Biomass composition & methane production; Synthesis of methane under natural conditions Potential microbes involved in methane generation, Man-made processes; Methane form sanitary landfills sewage farm industrial wastes & energy crops; Reactor design; Utilization of the methane as fuel; Production of Fuel Ethanol from Biomass: Potential biomass and microorganisms for ethanol production; Problems in production of ethanol from agro-industrial wastes; Development of technology of fermentation; Ethanol production from molasses by yeast; Ethanol production from cellulose by rumen bacteria; Ethanol production by Zymomonas mobilis; Future prospects of the industrial alcohol; Production of Hydrogen from Biomass: Potential substrates and microorganisms; Natural biosynthesis of hydrogen under natural habitats; Cell-free system and combined system for production of hydrogen.Course prerequisite: BTE 304/MIC203

Suggested Readings:

1. Food Feed and Fuel from Biomass – Chahal D.S.2. Biotechnology: Principles and Applications – Higgins I.J., Best D.J. & Jones J.

BTE 317 Biostatistics 3 credits

Definition, uses, limitation and scope; Role statistics in research and experiment; Sampling techniques- Basic statistical principles and terminology- Population and parameters, Samples; Variables; Distribution; Statistical concepts relating to interpretation and decision; Descriptive statistics- Estimation of uncertainties; Calculation of mean, variance and standard deviation; Standard deviation of the mean; Confidence limit of the mean; Test of significance-Measurement of data- t test in paired and non-paired experiments; Selection of apprpriate method for calculating t; Confidence limit of a difference between means; Analysis of variance- Single characterization data; Analysis of variance- Single classification data with subgroups, Multiple classification data; Ducan’s multiple range test; Least significance difference test; Relationship between t and f test; Test of significance- Chi square test-1xn table, 2xn table; Use of chi square with occurrence and non-occurrence data; chi square analysis of 2x2 or four fold table; Alternate methods of calculating chi square; Tests of significance when cell frequencies are small; Statistical methods based on binomials; Relationship between variables- Correlation;Linear regression; Least square regression line; Test of linearity of a regression; Confidence limit of regression coefficient; Dosage response data- Estimation of 50% end point; Graphical approximation of ED50 value; Estimation of relative frequency.Course prerequisite: STA 201

Suggested Readings

1. Daniel, W.W. Biostatistics- A Foundation for Analysis. John Wiley and Sons2. Zaman, S.H. Simple Lesson from Biometry3. Jalil, M.A. Ferdous, . Basic Statistics- Methods and Applications4. Mian, M.A. Miyan, A. Introduction to Statistics5. Snedecor, G.W. and Cochran, W.G. Staistical Methods

BTE 403 DNA Fingerprinting and Molecular Diagnostics 3 credits

Introduction- Basic genetic principle; Variable Number of Tandem Repeats (VNTRs)/ Minisattellite sequences; Short Tandem Repeats (STRs)/ Microsattellite sequences; Hybridization based DNA fingerprinting (RFLP)- Radioactive method, Fluorescent method, Chemiluminescent method; PCR-based DNA fingerprinting; Single locus and multi-locus DNA fingerprinting; Isolation of DNA from whole blood, soft tissues, semen stains and swabs, bones, plant material; Polymorphism: Polymorphism of some genetic locus in relation to disease (HLA, Apo and ACE gene);Applications of DNA fingerprinting; Criminal investigation (personal identification); Immigration; Paternity dispute; Identification of missing children, bodies found in plane crash, road accidents etc.; Varietal identification of plantsMolecular Dignostics-Diagnosis of Cystic fibrosis by multiplex PCR; Clinical implications: Abnormal mucus clearance from the respiratory tract with frequent infections, pancreatic insufficiency, abnormal salt transport, infertility in males.; Detection of b-Thalassemia mutation using ARMS-PCR; Clinical implications: Anemia (red cell deficiency); Detection of Fragile X syndrome by FMR-1 gene trinucleotide repeat analysis; Clinical implications: Mental retardation, long faces, large ear, prominent jaws, post-pubertal macroorchidism.; Detection of Philadelphia chromosome [BCL-ABL t(9:22) translocation] by genomic southern hybridization; Clinical implications: Acute leukemia (ALL) and Chronic myelogenous leukemia(CML); Bone marrow engraftment: DNA analysis to distinguish patient and donor cells as different using hypervariable tandem repeat polymorphic DNA markers; Identification of bacterial species based on the sequences of their 16S ribosomal RNA genes; DNA Microarrays/ DNA Chips/Gene Chips- Basic concept; Design of a DNA Microarray; Applications of DNA Microarray technology : Disease diagnosis, Drug discovery etc.Course prerequisite: BTE 307/MIC310

Suggested Readings

1. Micklos, Davod A. and Freyer , Greg A. 1990. DNA Science, Cold Spring harbor Laboratory Press and Carolina Biological Supply company.

BUS 101 Introduction to Business 3 credits

Basic principles and practices of contemporary business and its history; Forms of business organization and ownership; Environment of an enterprise ; Organizing and managing the enterprise; Management of: HR, market productions and operations, finance; discuss a broad range of business situations where analysis and decision-making are required. Management tools and information systems; International and globalization; External environments of business; Future outlooks of business and business ethics.CHE 101 Introduction to Chemistry 3 credits

The course is designed to give an understanding of basics in chemistry. Topics include nature of atoms and molecules; valence and periodic tables, chemical bonds, aliphatic and aromatic hydrocarbons, optical isomerism, chemical reactions.

Recommended Books:

1. Introduction to Modern Inorganic Chemistry: S. Z. Haider2. Physical Chemistry: Haque & Nawab 3. Organic Chemistry: R. T. Morrison & R. N. Boyd4. General Chemistry: Raymond Chang

CSE 101 Introduction to Computer Science 3 credits

Introduction to the use of computer hardware and software as tools for solving problems. Automated input devices and output methods (including pre-printed stationery and turnaround documents) as part of the solution. Using personal computers as effective problem solving tools for the present and the future. Theory behind solving problems using common application software including word processing, spreadsheets, database management, and electronic communications. Problem solving using the Internet and the World Wide Web. Programming principles and use of macros to support the understanding of application software. The course includes a compulsory 3 hour laboratory work each week.

Recommended Books:

1. Computer Science – A modern introduction: Goldschlager and Lister2. Fundamentals of Computers: V Rajaraman 3. Work Out Computer Studies GCSE (Macmillan Work Out S.): Graham Taylor

CSE 110 Programming Language 3 credits

An introduction to the foundations of computation and purpose of mechanised computation, techniques of problem analysis and the development of algorithms and programs, principles of structured programming and corresponding algorithm design, Topics will include data structures, abstraction, recursion, iteration as well as the design and analysis of basic algorithms, (language C is primarily used), introduction to digital computers and programming algorithms and flow chart construction, information representation in digital computers, writing, debugging and running programs (including file handling) on various digital computers using C. Prerequisite CSE 101

Recommended Books:

1. Working with C: Y KanetkarSchaums Outline of Theory and Problems of Programming With C: Byron S. Gottfried

DEV 101 Bangladesh Studies 3 credits

Socio-economic profile of Bangladesh, agriculture, industry, service sector, demographic patterns, social aid and physical infrastructures. Social stratification and power, power structures, government and NGO activities in socio-economic development, national issues and policies and changing society of Bangladesh.

Recommended Books:

1. Bangladesh: National Cultures and Heritage: An Introductory Reader: A.F. Salahuddin Ahmed & Bazlul Mobin Chowdhury2. The History of Bengal (Vol.1 &Vol.2) : R.C. Majumdar3. Banglapedia, 2003: Asiatic Society of Bangladesh4. Bangladesh Arthaniti: Khan, Md. Shamsul Kabir5. Bangladesh on the Threshold of the Twenty-First Century, Asiatic Society of Bangladesh, 2002: A.M Chowdhury and Fakrul Alam 6. Poverty Reduction & Strategy: What, Why & for Whom in Asit Biswas et.al.(ed) Contemporary Issues in Development : M.M Akash 7. Bangladesh 2020: A long-run perspectives study: The World Bank

ECO 103 Principles of Economics 3 credits

A study of the fundamentals of micro and macroeconomics, nature and method of economics, individual markets, demand and supply, elasticity of demand and supply. Production and cost, market structures with special focus on perfect competition and monopoly, economic efficiency and market failure, determination of national income. The aggregate supply model, unemployment, inflation, unemployment-inflation trade-off, government budget and fiscal policy, money creation and monetary policy, business cycles, economic growth, theory of comparative advantage, free trade versus protection, balance of payments and exchange rate policies.

Recommended Books:

Economics: John Solman Principles of Macroeconomics: Robert H FrankModern Economic Theory: K.K. DewettInternational Economics: Appleyard & Field

ENG 091 Foundation Course (non-credit)

The English Foundation Course is designed to enable students to develop their competence in reading, writing, speaking, listening and grammar for academic purposes. The students will be encouraged to acquire skills and strategies for using language appropriately and effectively in various situations. The approach at all times will be communicative and interactive involving individual, pair and group work.

Recommended Books:1. J. C. Richards, J. Hull, and S. Proctor, “New Interchange: Student 's Book 3-A”, Cambridge University Press, 20022. J. Nadel, B. Johnson, and P. Langan, “Vocabulary Basics”, Townsend Press, 1998.3. A. Hogue, “First steps in Academic Writing”, Longman, 19964. K. Blanchard, C. Root, “Get Ready to Write”, Longman, 1998.

ENG 101 English Fundamentals 3 credits

Developing basic writing skills: mechanics, spelling, syntax, usage, grammar review, sentence and essay writing.

Recommended Books:1. Fundamentals of English: Jack C. Richards

ENG 102 Composition I 3 credits

The main focus of this course is writing. The course attempts to enhance students’ writing abilities through diverse writing skills and techniques. Students will be introduced to aspects of expository writing: personalized/ subjective and analytical/persuasive. In the first category, students will write essays expressing their subjective viewpoints. In the second category students will analyze issues objectively, sticking firmly to factual details. This course seeks also to develop students' analytical abilities so that they are able to produce works that are critical and thought provoking.

Recommended Books:

1. Composition I: The Pearl; John Steinbeck

ENV 101 Introduction to Environmental Science 3 credits

Fundamental concepts and scope of environmental science, Earth's atmosphere, hydrosphere, lithosphere and biosphere, men and nature, technology and population, ecological concepts and ecosystems, environmental quality and management, agriculture, water resources, fisheries, forestry and wildlife, energy and mineral energy sources; renewable and non renewable resources, environmental degradation; pollution and waste management, environmental impact analysis, remote sensing & environmental monitoring.

Recommended Books:

1. Something New Under the Sun: An Environmental History of the Twentieth-Century World: J. R. McNeil & Paul Kennedy Principles of Ecology: R Brewer3. Fundamentals of Ecology: E. P. Odum

HUM 101 World Civilization and Culture 3 credits

A brief view of the major civilizations and cultural aspects in different continents covering ancient, medieval and modern civilizations. Topics include renaissance, reformation, and the beginning of the modern world, scientific revolution, industrial revolution, the age of democratic revolutions, nineteenth century Europe, Asia-Pacific Region, Africa, World Wars, South Asia: colonization, decolonization and after; contemporary world: Cold War and after.

Recommended Books:

World Civilization: Bums & othersCivilization: T Walter Walbank and othersA History of World Civilization: J. E. SwainWestern Civilization: Robert E. Lerner & Standish Meachem

HUM 102 Introduction to Philosophy 3 credits

Philosophy: Concept of philosophy; science and philosophy; religion, literature and philosophy; sources of knowledge: empiricism, rationalism and criticism; concepts of value, ethics and sources of ethical standards.

Recommended Books:

1. A Modern Introduction to Philosophy: P Edwards and A. Pap 2. Philosophy: R. J. Hirst 3. Introduction to Modern Philosophy: C.E.M. Joad 4. An Introduction to Philosophical Analysis: J. Hospers5. An Outline of Philosophy: A. Matin6. Introduction of Philosophy: T. W. Patrick

HUM 103 Ethics and Culture 3 credits

This course introduces the students to principles and concepts of ethics and their application to our personal life. It establishes a basic understanding of social responsibility, relationship with social and cultural aspects, and eventually requires each student to develop a framework for making ethical decision in his work. Students learn a systematic approach to moral reasoning. It focuses on problems associated with moral conflicts, justice, the relationship between rightness and goodness, objective vs. subjective, moral judgment, moral truth and relativism. It also examines personal ethical perspectives as well as social cultural norms and values in relation to their use in our society. Topics include: truth telling and fairness, objectivity vs. subjectivity, privacy, confidentiality, bias, economic pressures and social responsibility, controversial and morally offensive content, exploitation, manipulation, special considerations (i.e. juveniles, courts) and professional and ethical work issues and decisions. On conclusion of the course, the students will be able to identify and discuss professional and ethical concerns, use moral reasoning skills to examine, analyze and resolve ethical dilemmas and distinguish differences and similarities among legal, ethical and moral perspectives.

Recommended Books:

1. Ethics, Culture and Psychiatry - International Perspectives: Ahmed Okasha, Julio Arboleda2. Ethics and HRD: A New Approach to Leading Responsible Organizations: Tim Hatcher3. The Ethical Challenge: How to Lead with Unyielding Integrity: Noel M. Tichy and Andrew R. 4. Dignity of Difference: How to Avoid the Clash of Civilizations: Jonathan Sacks 5. Culture and Ethics: Michel Labour, Charles Juwah, Nancy White and Sarah Tolley6. Culture Matters: How Values Shape Human Progress: Samuel P. Huntington

HUM 111 History of Science 3 credits

This course will present a general overview of the development of scientific knowledge from ancient to modern times. It will examine how our modern scientific worldview developed over the ages in the fields of astronomy, physics, biology, chemistry, medicine, geology and other science disciplines. Focus will be on significant discoveries, the major scientists responsible for these revolutions, and the interrelation between science and society over the centuries. The course will contain the following:Science & philosophy, development of science in the ancient times, Greek & Egyptian science, science in the Orient, medieval science, science in the Islamic world, Western renaissance & industrialization, evolutionary theory, science in the modern ages. Science & religion, nature of scientific truth, validation of scientific theories.

Recommended Books:

1. Reader’s Guide to the History of Science: A. Hessenbruch2. Scientific Laws, Principles, and Theories: a Reference Guide: Robert E Krebs3. The History of Science: an Annotated Bibliography: G MillerA Guide to the History of Science: a First Guide for the Study of the History of Science, with Introductory Essays on Science and Tradition: G. Sarton5. Knowledge & the World: Challenges Beyond the Science Wars: M. Cavrier, J. Roggenhofer, G. Kuppers & P. Blanclard6. The Forgotten Revolution: How Science was

Born in 300 BC and Why It Had to be Reborn: Lucio Russo7. Hitler’s Scientists: Science, War and the Devil’s Fact: J. Cornwell

MAT 101 Fundamentals of Mathematics 3 credits

Basic techniques of algebra, analytical geometry, graphing, and trigonometry. Prerequisites : None

MGT 211 Principles of Management 3 credits

Meaning and importance of management, evolution of management thoughts; managerial decision making; Environmental impact, corporate social responsibility, planning, setting objectives, implementing plans, organizing; organization design, managing change, directing, motivation, leadership, managing work groups, controlling: principles, process and problems and managers in changing environment.

Recommended Books:

1. Management: Stephen P. Robbins and Mary Coulter2. Management: James A. F. Stoner, Edward R. Freeman & Daniel R. Gilbert

MIC 307 Microbiology of Frozen Food and Fish 3 credits

Normal flora of fish: Factors affecting types and load of micro flora on freshly caught fish; ine; Contamination and spoilage of fish and frozen fish: factors affecting kind and rate of spoilage; microbial spoilage; chemical spoilage; autolytical spoilage. Control of spoilage; Isolation and identification of fish bacterial pathogens; Effects of freezing/thawing on foods: basic concepts of freezing and thawing; influence of frozen temperature and time on foods, thawing methods; Freezing preservation: Influence on food quality, physical and chemical reactions during freezer storage; Microbiology of frozen meat and meat products: Effect of freezing on microorganisms; structure and composition of meat; freezing temperature and changes induced by freezing of meat; spoilage micro flora; Microbiology of frozen fish and related products: freezing of fish; effects of freezing on fish microorganisms; chill storage and freezing on microbial growth and survival; Microbiology of frozen dairy products: Microbiology of raw and frozen milk; microbial types and load on butter, ice cream ad frozen cheese. Course prerequisite: MIC 302/BTE309

Suggested Readings:

1. M.R. Adams & M.O. Moss. Food Microbiology.2. G.N. Frerichs and S.D Manual for the isolation and identification of fish bacterial pathogens..3. Miller Modern food microbiology, 3rd edn. – James Jay

MIC 308 Fermentation Technology 3 credits

Introduction to fermentation processes: range of fermentation processes; chronological development of the fermentation industry; component parts of fermentation process; Fermetor/bioreactor: types, configuration, mixing and aeration; power requirements, impeller designs baffle and aeration; Inoculum preparation and inoculum development: development of inoculum for yeast processes; development of inocula for bacterial processes; development of inocula for fungal processes; Fermentation modeling: rate eaquations for cell growth, substrate utilization, product formation; transfer across phase boundaries; Mode of fermentations: fed-batch and continuous culture processes and their control; Sterilization of fermenters and liquid media: medium sterilization; the desing of batch sterilization processes; the design of batch sterilization processes; the design of continuous sterilization of feed and air; Instrumentation and control: control systems: manual, automatic, and combinations of methods of control; methods of control of process variables like temperature, pH, flow measurement, pressure measurement, pressure control, safety valves, agitation-shaft power, stirring rate, foam sensing and control weight, measurement and control of dissolved oxygen; exit-gas and analysis; redox, and carbon dioxide electrodes.Course prerequisite: MIC 202

Suggested Readings:

1. B. McNeil Harvey Fermentation: A Practical Approach. IRL Press Oxford 2. P.F. Stanbury & A. Whitaker . Principle of Fermentation Technology

MIC 309 Immuno-pathology and Vaccine Development 3 credits

Inactivation and activation of biologically active molecules: Mechanism of antibody mediated inactivation and activation (hormone, receptor, ligand); neutralizing antibodies - cause and effect, Protective functions of inactivation Antibodies; Cytotoxic and cytolytic reaction: Mechanism of cytolytic reactions; Immuno-haematologic disease; erythrocyte, leukocyte, platelets; Detection of circulating cytotoxic antibodies, Protective and pathologic effects in infections diseases; Granulomatous reactions: Nature of ganulomatous reactions; Granulomatous disease; Infectious diseases-bacterial and parasitic; Inflammation: Nonimmune and immune inflammation, Immune specific protection against infections, Interaction of immune mechanism in infectious disease; Evasion of immune defense mechanism; Antibody engineering: Antibody gene cloning; Recombinant antibody gene expression; Applications of engineered antibodies; Vaccination: Designing of vaccines, attenuated vaccine, conjugate vaccine, subunit vaccine, DNA based and other vaccines; Vaccine Strategy: Experimental vaccines for botulism, anthrax, malaria, pneumonia, cholera, typhoid, hepatitis and tumors.Course prerequisite: BCH301

Books Recommended:

1. Immunology: Immunopathology and Immunity – Sell S. ASM Press, USA2. Bacterial Pathogenesis: A molecular Approach – Salyers A.A & Whitt D.D., ASM Press, USA.3. Molecular Immunology – Hames B.D. & Glover D.M., IRL Press, USA.4. Immunology Today: Elsevier Trends Journals, UK

MIC 404 Molecular Virology and Oncology 3 Credits

Persistence of Viruses: Patterns of virus infections; Mechanisms of viral persistence; Persistence of HSV EPV & HIV in humans; Viruses of Special Interest: Dengue and Japanese encephalitis virus; Ebola virus infection; Severe acute respiratory syndrome (SARS) & SARS corona virus; Other important viruses of recent epidemics; Virus Evolution & Emerging Viruses: How do viruses evolve? Emerging viruses; Emergence of dengue virus infection in Bangladesh; Oncology: Introduction & general terminology of oncology; Cancer: Development of cancer; Spread of cancer; Molecular mechanisms of transformation by DNA and RNA viruses; Physical & chemical factors contribute to cancer development; Cancer therapy.Course prerequisite: MIC 301, BTE 207

Suggested Readings:

1. Virology – Fields2. Principles of Virology – Filed3. Scientific American, September 1996.

MIC 405 Drinking Water microbiology 3 credits

Water and Health- Sources of water- Surface, ground, marine, recycled; Drinking water- Treatment and Distribution; Waste Water- Collection, Treatment and discharge; Quality of human drinking water ;Microbes in human and domestic waste water ;Microbes as a part of freshwater and marine ecology- Distribution of microorganisms in the aquatic environment – Planktons, benthic microbes, mixed flora; Microorganisms common to different aquatic habitats ; Eutrophication - Great Lakes ; Biochemical Oxygen Demand (BOD) ; Oligotrophication: microbial growth is suppressed; Acid rain algae and bacteria don't grow ,nutrients not returned ; Drinking water quality - faecal contamination - Salmonella, Vibrio, Shigella, Staphylococcus, Streptococcus; Eschericia coli, lactose-fermenting coliform ; E. coli O157:H7 ; Ways to look for coliform in water; Qualitative and quantitative measurements of microbes in water; Total Coliform and Total fecal coliform count, Indicator organisms -Most Probable Number (MPN) -membrane filter analysis ;Quality to ensure safe drinking water- Sources of contamination of community water supply and other water reservoirs used for drinking water; Recommendations of WHO and other organization on safe drinking water; Water treatment before supplying for the community- Physical, chemical and biological aspects. Waterborne diseases-Amoebic dysentery; Entamoeba histolytica ; Cryptosporidiosis ; Giardiasis (Beaver fever) ;Water treatment - Flocculation: large particles settle out; filtration: sand, activated charcoal traps protozoan cysts ; Chlorination/ozone/u.v. kills bacteria, but can be toxic ;Household waste water (sewage) treatment-- primary and secondary sludge fermented anaerobically - lagoon system: settling of primary sludge; some aerobic digestion of effluent (oxidation pond) - septic tank: primary treatment system; effluent trickles out into the field ; Water and Human and Ecosystem Health.Course prerequisite: MIC 203

Suggested Readings1. Cliver, D.O., Newman, R.A., Pickford, R.D. and Berger, S. 1984. Drinking Water Microbiology. USEPA Publication, Washington2. Mitchel, R. 1974. Water Pollution Microbiology. John Wiley and Sons.3. Ford, T.E. 1996. A Global decline in Microbiological Quality of Water. American Academy of Sciences, Washington

MIC 406 Medical and Diagnostic Microbiology 3 credits

Laboratory diagnoses of infections agents: different types of and approaches to clinical sample collection, maintenance and laboratory management; Diagnostic studies: principles of diagnoses of bacterial, fungal, rickettsial, parasites, spirochetal, viral and mycoplasmal diseases; Diagnosis of sexually transmitted diseases; Immunodiagnostic studies: collection of serum; antibody titer (such as ASO, Widal); agglutination, double diffusion; counter immuno-electrophoresis and immuno-fluorescence; complement fixation test; fluorescent antibody test (FAT and IFA); radio immunoassay (RIA); enzyme immunoassay (EIA) and enzyme linked immunosorbent assay (ELISA; Polymerase chain reaction (PCR): detection of genes for toxins and virulenc; Vaccine approaches and immunization.Course prerequisite: BCH 301; MIC 300

Suggested Readings:

1. David H. Persing, Thomas, F. Smith-Fred & Teaver T.J. Wite.Diagnostic Molecular Microbiology: Principle and Applications2 –Hand Book of Serodiagnosis in Infections DiseasesA Manual of Laboratory and Diagnostic Tests – Frances Fischbach 2. Diagnostic immunology Laboratory Manual – Ronald, J. Harbeck & Petricia C. Giclas.

MIC 407 Bacterial Pathogenesis and Molecular Epidemiology 3 credits

Pathogenesis with Special Reference to Molecular Basis: Shigellosis; Cholera; Bacterial ulcer in human; Meningitis and epiglottis; Botulism; Gas gangrene, Anthrax; Toxic shock syndrome; Plague, Pelvic inflammatory disease, Haemophilus influenzae, Listeria monocytogense; Bacterial Resistance to Antibiotics: Mechanism of antibiotic resistance; Antibiotic tolerance; Transfer of resistance genes; Infection Control & Prevention: Epidemic versus endemic; Steps in epidemiologic evaluation; Role of the laboratory in epidemiologic evaluation; Potential problems related to laboratory activities in epidemiologic investigations; Epidemiologic Analysis: Criteria for evaluating typing system; Phenotypic techniques biotyping, antimicrobial susceptibility testing, serotyping, bacteriophage typing, MLEE; Genotypic techniques; plasmid analysis, REA of chromosomal DNA. Southern blot analysis of RFLPs, PFGE of chromosomal DNA, typing system applying PCR, PCR-based detection of restriction sites, & nucleotide sequence analysis; Molecular typing of specific organisms; Application of microbial typing system; Implementing a molecular epidemiology laboratory.Course prerequisite: MIC 204; BTE 207

Suggested Readings:

1. Bacterial Pathogenesis: A Molecular Approach – Salyers A.A.. & Whitt D.D.2. Medical microbiology – Baron S.3. Medical Microbiology – Mims & Playfair4. Principles of Bacterial Pathogenesis – Groisman E.5. Virulence Mechanisms of Bacterial Pathogenes – Brogden K.A. et al.6. Mechanism of Microbial Diseases Schaechter M & Engelberg N.C.7. Manual of Clinical Microbiology, 7th ed. – Murray P.R., Baron E.J., Pfaller M.A.3. Tenover F.C. & Yolken R.H.

MIC 408 Extremophiles 3 credits

Extreme Environment: Microbial adaptations to extreme environments; Extreme environment as a resource for novel microorganisms; Biodiversity: Biodiversity at the molecular level: the domains kingdoms and phyla of life, Microbiological perspective; The primary divisions of life; domain archaea; domain bacteria, domain eukarya; Microbial diversity; The Archaea: Phylogenetic overview; Kingdom Euryarchaca; Kingdom Crenarchaea; Hyperthermophilic archaea and microbial evolution; Different

Types of Extemophiles: The extreme thermophiles; The extreme acidophiles; The extreme alkalophiles; The extreme halophiles; The extreme barophiles; Cultivation of Extremophillic Microorganisms: Various strategies applied for cultivation of extremophiles; Extremozymes: Biocatalysts under extreme conditions; Extremophiles as a source of novel enzymes for industrial application; Screening strategies for novel enzymes; Heat-stable amylase and glucomylase; Thermostable cellulases; Thermostable xylanases; DNA processing enzymes in PCR. High temperature reverse transcription; Thermostable DNA ligase; Other thermoactive enzymes of biotechnological interest; Thermostable glucose isomerases; Thermostable alcohol dehydrogenases; Biochemical basis of heat stability. Course prerequisite: MIC 203, BTE 304

Suggested Readings:

1. Microbiology of Extreme environments – Edward C.2. Microbial Growth and Survival in the Extreme Environments – Brock T.D.3. Microbial Life in Extreme Environments - Kushner D.J.4. Biotechnology: A Multi-Volume Comprehensive Treatieses, vol. 10 (Special Processes). Rehm H & Reed G.5. Biology of Microorganisms – Brock T.D.6. Microbes in Action: Concept and Application in Microbial Ecology – Lynch J.M. & Hobble J. E.7. Biodiversity: Measurement and Estimation. – Hawksworth D.L.

PHY 101 Introduction to Physics 3 credits

The course is designed to give an understanding of basics in physics. Topics include Vector; Motion; Force; Energy; Pressure; Heat and temperature; Mechanics; Forces; Gravitation; Sound, Light, Electricity and Magnetism; Atomic and Nuclear Physics. Basics of Astronomy. 3 credits, Prerequisites : None

POL 103 Introduction to Political Science 3 Credits

A study of political systems and process with special reference to Bangladesh. Topics include nature and origin of state, sovereignty of state, forms of political units, liberty, law, process of politics, political structure, political ideas- democracy, socialism, nationalism, peoples' behavior in politics. Political system, process and problems of Bangladesh.

Recommended Books:

1. A History of Political Thought: Plato to Marx: Subrata Mukherjee & Sushila Ramaswamy 2. An Introduction to Political Science: Rand DyckA Social Political History of Bengal and the Birth of Bangladesh: Kamruddin Ahmed4. Radical Politics and the Emergence of Bangladesh: Talukder Manizzaman 5. India and Pakistan: A Political Analysis: Hugh Tinker6. Politics and Policy Making in Developing Countries: Perspective on the New Political Economy: Gerald M. Meier7. Political Culture, Political Parties and Democratic Transition in Bangladesh: Shamsul I. Khan, S. Aminul Islam & Imdadul Haque 8. Involvement in Bangladesh's Struggle for Freedom: T. Hossain9. History of Bangladesh 1704-1971: Political History: Sirajul Islam10. Conflict and Compromise: An Introduction to Political Science: H.R. Winter

PSY 101 Introduction to Psychology 3 Credits

The objective of this course is to provide knowledge about the basic concepts and principles of psychology pertaining to real-life problems. The course will familiarize students with the fundamental process that occur within organism-biological basis of behaviour, perception, motivation, emotion, learning, memory and forgetting and also to the social perspective-social perception and social forces that act upon the individual.

Recommended Books:

Introduction to Psychology: C.T. Morgan Introduction to Psychology: R.F. Crider Understanding Psychology: Robert Feldman

SOC 101 Introduction to Sociology 3 credits

Perspectives on society, culture, and social interaction, Topics include community, class, ethnicity, family, sex roles, and deviance. Social problems and sociological problems. Problems, theories, and the nature of sociological explanation. Explanation, evidence and objectivity. Sociology as a comparative study of social action and social systems. Some models of sociological thinking as applied to the study of the following: aspects of social ranking; forms of interpersonal and personal relationships; the changing nature of the relationship between economy and society; the sociology of development; the origins and spread of capitalism and socialism; ideology and belief systems; religion and society; rationality and non-rationality; conformity and deviance.

Recommended Books:

1. Sociology: Anthony Giddens 2. Sociology: Richard T. Schaefer 3. Sociology: Rao and C.N. Shankar 4. Sociology: Neil J. Smelser

SOC 401 Gender and Development 3 credits

Position & role of women in society, contemporary issues, analysis of various aspects of gender relations, gender discrimination, societal attitude, different forms of feminism, women in higher education, employment of women & discrimination, workplace harassment, contribution of women in development: world picture I position in Bangladesh. Prerequisite SOC 101

Recommended Books:

1. Women and Social Security: Progress Towards Equality of Treatment, 1990 Geneva International Labor Office: Anne- Marie Brocas, Anne-Marie Cailloux and Virgine Oget.2. Impact of Women in Development Projects on Women Status and Fertility in Bangladesh, 1993, Dhaka, Development Researchers and Associates: M. Kabir, Rokeya Khatun, Ishrat Ahmed.3. Integration of Women in Development: Why, When and How: Ester Boseup, Cristine Liljencrantz.4. Women in the Third World: Gender Issue in Rural and Urban areas: Hants5. Women, Man and Society: The Sociology of Gender: Allyn and Bacon, Claire M Renzetti, Daniel J Curran.6. Race, Ethnicity, Gender and Class: the Sociology of Group Conflict and Change, London: Joseph F Healey

STA 201 Elements of Statistics and Probability 3 credits

Frequency distribution, mean, median, mode and other measures of central tendency, standard deviation and other measures of dispersion, measure of skewness and Whisker-Box plot, correlation and regression analysis, elementary probability theory, conditional and joint probability, Bayes' theorem, discrete probability distributions, binomial, hypergeometric, Poisson, geometric and negative binomial distributions, continuous probability distributions, normal and exponential distributions, sampling distributions for relevant statistics (Normal, t, chi-square, and F distributions), central limit theorem, confidence intervals and hypothesis testing for parameter (mean and proportion) .

Recommended Books:

1. Probability and Random Processes: G.R. Grimmett and D.R. Stirzaker2. Elementary Probability Theory with Stochastic Processes: K.L. Chung

(b) Practical Courses 6 Credits

MIC 105 Microbiol Lab I

Microscopy-Use & function of microscopes, Observation of stained cell preparations, Observation of living bacterial cells, Observation of living yeasts & molds,Micrometry : measurement of microbial cell; Bacterial staining-Simple staining & negative staining., Gram staining, Acid-fast staining, Capsule staining, Spore staining, Flagella staining; Cultivation techniques-Media preparation & sterilization techniques, Culture transfer techniques, Techniques for isolation of pure cultures, Techniques for preservation and maintenance of pure cultures, Observation of cultural characteristics of bacteria on various media, Observation of cultural characteristics of yeast on various media; Determination of biomolecules-Preparation of different lab solutions (molar, molal, normal and buffers);Determination of citric acid by titrimetric method; Determination of antibiotic agents; Determinations protein; Determinations reducing sugar; Detection of cytoplasmic inclusions (PHB and volutin)

MIC 205 Microbiol Lab II

Growth measurement-Techniques of pipetting and dilution; Determination of quantitative viable cells by serial dilution technique (Spread plate & pour plate) and making a growth curve. Techniques of enumeration of microorganisms : improved Neubaur counting chamber and Miles and Misra technique. Turbidimetric estimation of bacterial growth; Environmental influences-Effect of temperature on growth, Effect of heat on vegetative cells and spores of bacteria and on spores of yeast and mold, Effect of osmotic pressure on growth, Effects of pH, energy and buffer on growth; Metabolic activities of microorganisms- Starch, lipid, casein & gelatin hydrolysis test, Carbohydrate (LDS) fermentation, MIU, KIA & IMVIC tests, Nitrate reduction, oxidase, catalase & litmus milk reaction tests, Antimicrobial sensitivity test of microorganisms (qualitative); Identification of unknown bacterial culture with the held of Bergey's Manual of Systematic Bacteriology; Basic Microbial Genetics- Protoplast fusion test; Detection of genetic material by staining; Test or enzyme induction; Isolation of drug resistant mutant; Medical (Microbiology - Microscopic study of Parasites, Microscopic study of the pathogenic microorganisms present in air, water & soil (Gram reaction, morphology, motility etc.), Microbial flora of throat and skin, Identification of human staphylococcal pathogen, Identification of human streptococcal pathogens; Human Physiology-Test in circulatory system: Total blood cell count, Differential count for WBC; Determination of total bilirubin, cholesterol and non esterified fatty acids, uric acid, glucose, etc. in blood; Gastro-enteric system ,Genitourinary tract system ;Respiratory tract system .

MIC 210 Microbiol Lab III

Virology-Cultivation and enumeration of bacteriophages; Isolation of bacteriophages from raw sewage; Detection of HBs Ag from patients serum by serological methods; Isolation of TMV virus and infecting plants; Microbial Nutrition and Metabolism-Relationship of free oxygen to microbial growth ; Anaerobic culture of bacteria; Degradation of polymer by exoenzymes; Actions of antiseptics, disinfectants, UV light and photo reactivation and antimetabolites.; Microbial Molecular Genetics-Isolation of plasmid and chromosomal DNA; Detection of DNA by agarose gel electrophoresis; Transformation of E. coli by plasmid; Study of gene expression in E. coli.; Medical Microbiology II- Isolation, identification and antibiotic sensitivity pattern of pathogenic microorganisms from clinical specimens-Stool, (b) Urine, (c) Pus, (d) Blood, (e) CSF and (f) Biopsy; Immunology-I Preparation of bacterial whole cell extract; Preparation of outer membrane protein; Immunization protocol for animals; Collection of serum and plasma; Separation of blood leucocytes; Test for cell viability; Phagocytosis by neutrophils; Agricultural Microbiology- Microbial population of soil, rhizosphere and rhizoplane; Denitrification and ammonification; Nitrogen fixation test; Identification of plant pathogens; Food Microbiology-Quantitative examination of bacteria in raw and pasteurized milk; Methylene blue reduction test; Microbiological analysis of fermented foods and nonfermented foods; Detecting Salmonella sp. on poultry; Industrial Microbiology- Production of microbial extracellular enzymes; Production of SCP; Production of antibiotics; Production of alcohol from molasses; Enzymology-Determination of enzyme activity (qualitative and quantitative); Determination of kinetic properties of an enzyme; Determination of activators and inhibitors of enzymes; Determination of molecular weight and substrate specificity of enzyme; Pharmaceutical Microbiology-Microbiological assay of pharmaceutical raw material; Microbiological assay of pharmaceutical solids ointments & oral liquids; Bioassay of potency of antibiotics; Sterilization and sterility test, pyrogen test; Microbiology of Frozen Fish and Food- Identification of microbial flora of frozen food and fish; Identification of different fish pathogens; Determination of microbial flora of frozen food; Identification of different pathogens in frozen foods

MIC 305 Microbiol Lab IV

Virology (II)-Detection of viral Ags/Abs from patients’ sera by immunological techniques; PCR Amplification of HBV core and surface gene; Detection of viral DNA by PCR amplification and dot-blot hybridization; Use of RPHA method for the detection of viral

Ag/Ab; Titration of virus using immunofluorescent microscope; Immunology(II)- Detection of antigen and antibody (a) by Gel Immunodiffusion technique; (b) by Radial Immunodiffusion technique; (c) by Crossed immunoelectrophoresis technique; SDS-PAGE and immunoblotting of bacterial proteins; Complement fixation tests; . HLA typing; Environmental pollution- Enrichment and isolation of biodegradative microbes from environment; Non-culturable state of microorganisms (detection by FA or Acridine orange DVC); Detection of indicators and pathogenic microbes in potable water; Water purification (viz, flocculation, chlorination, ozonation etc.) Food Microbiology (II)- Detection of B. cereus and S. aureus in fast foods; Detection of E. coli and Aeromonas hydrophila in salad dressings; Isolation of Aspergillus flavas from oil seeds; Detection of emolysin and phospholipase C (toxins) from B. cereus; Genetic Engineering- DNA digestion by restriction enzymes; Ligation of DNA to appropriate vector; Study genetic map; Fermentations Technology- Determination of specific growth rate (μ), Doubling time (td) and generation time; Dough fermentation by baker’s yeast for bread making; Production of acetic acid by Acetobacter aceti; Demonstration of fermentor; Yogurt production by lactic starter; Production of citric acid by A. niger; Microbial Biotechnology- Whole cell immobllization by Ca-alginate; Determination of specific growth rate substrate utilization constant and biomass in a steady state batch culture; Pesticide degradation: Biodegradation of holozenated pesticite by bacterial dehalogenases; Diagnostic Microbiology- Determination of blood grouping; Coagulation/agglutination/hemagglutination; Determination of anti-streptolysin-O (ASO) titre; VDRL test; ELISA; Diect fluorescent antibody (DFA) detection of microbial pathogens; Plasmid fingerprinting in clinical diagnosis; Gene detection and DNA-hybridization analysis in clinical diagnosis; Complement activation; Tuberculin test; Widal test; Determination of C reactive protein (CRP); Determination of plasma fibrinogen level; Determination of fibrin degradation product (FDP); Radio immuno detection of Immunoglobulins (RID); Analytical Microbiology- Thin-layer chromatographic separation of amino acids Separation of sugars by paper chromatography; Determination of organic carbon in soil and waste water ; Microbiology Quality Control- Test for microbiological quality of water and beverages: standard qualitative analysis of water MPN, and quantitative analysis of water by membrane filter method

(c) Microbiology Internship

MIC 400 Industrial/Research Organizations Attachment 3 credits

Each student, individually or in a group of three will be sent to different food, beverages, pharmaceutical and other industrial concerns having some microbiology related work depending on the availability of space in the concerned organization, for a specific period of time every week for 12 weeks. Each student will have to submit a training report at the end of the attachment duly endorsed by the authority of the respective organization for evaluation and grading by the course teacher.

(d) MIC 450 Thesis/ Project 4 credits

Each student will be required to carry out thesis/project work in the 7th and 8th semester in a chosen field in any of the following areas as decided by the supervisor.

1. Food Microbiology2. Microbial biotechnology3. Industrial Microbiology4. Molecular biology & genetics5. Environmental microbiology6. Clinical microbiology7. Immunology8. VirologyThe supervisor may be a faculty member of the BRAC University or any other suitable expert from other universities, R&D organizations, private food and industrial microbiology enterprises. On completion of the thesis work he/she will have to submit a dissertation and face a viva board for the defence.

Mumtarin Jannat Oishee has joined as a lecturer in the Department of Mathematics and Natural Sciences on October, 2021. Before joining BRAC University, she had been working as an Assistant Lecturer at the Department of Microbiology, Gono Bishwabidyalay since October 2019. She was also an active team member of Gonoshasthaya-RNA Biotech Limited in developing immunoassay for rapid detection of

SARS-CoV-2. She has obtained her MSc. and B.Sc. degree in Microbiology from Jahangirnagar University securing top position. She has completed her secondary and higher secondary level from Feni Girls' Cadet College.

Level:Kha-224 Merul BaddaDhaka 1212. Bangladesh

2016-2017: M.Sc. in Microbiology, Jahangirnagar University, Dhaka, Bangladesh

2013-2016: B.Sc. in Microbiology, Jahangirnagar University, Dhaka, Bangladesh

2012: HSC, Feni Girls' Cadet College

2010: SSC, Feni Girls' Cadet College

Journals

Jamiruddin, M. R., Haq, M. A., Tomizawa, K., Kobatake, E., Mie, M., Ahmed, S., Khandker, S. S., Ali, T., Jahan, N., Oishee, M. J., Khondoker, M. U., Sil, B. K., Haque, M., & Adnan, N. (2021). Longitudinal antibody dynamics against structural proteins of SARS-CoV-2 in three COVID-19 patients shows concurrent development of IgA, IgM, and IgG. Journal of Inflammation Research, 14, 2497–2506. https://doi.org/10.2147/JIR.S313188

Sil, B. K., Jamiruddin, M. R., Haq, M. A., Khondoker, M. U., Jahan, N., Khandker, S. S., Ali, T., Oishee, M. J., Kaitsuka, T., Mie, M., Tomizawa, K., Kobatake, E., Haque, M., & Adnan, N. (2021). Aunp coupled rapid flow-through dot-blot immuno-assay for enhanced detection of sars-cov-2 specific nucleocapsid and receptor binding domain igg. International Journal of Nanomedicine, 16, 4739–4753. https://doi.org/10.2147/IJN.S313140

Oishee, M. J., Ali, T., Jahan, N., Khandker, S. S., Haq, M. A., Khondoker, M. U., Sil, B. K., Lugova, H., Krishnapillai, A., Abubakar, A. R., Kumar, S., Haque, M., Jamiruddin, M. R., & Adnan, N. (2021). Covid-19 pandemic: Review of contemporary and forthcoming detection tools. Infection and Drug Resistance, 14, 1049–1082. https://doi.org/10.2147/IDR.S289629

Sil, B. K., Jahan, N., Haq, M. A., Oishee, M. J., Ali, T., Khandker, S. S., Kobatake, E., Mie, M., Khondoker, M. U., Jamiruddin, M. R., & Adnan, N. (2021). Development and

performance evaluation of a rapid in-house ELISA for retrospective serosurveillance of SARS-CoV-2. PLoS ONE, 16(2). https://doi.org/10.1371/journal.pone.0246346

Preprint

Sil, B. K., Adnan, N., Oishee, M. J., Ali, T., Jahan, N., Khandker, S. S., Kobatake, E., Mie, M., Khondoker, M. U., Haq, M. A., & Jamiruddin, M. R. (2020). Development and evaluation of two rapid indigenous IgG-ELISA immobilized with ACE-2 binding peptides for detection neutralizing antibodies against SARS-CoV-2. MedRxiv. https://doi.org/10.1101/2020.12.19.20248535

Microbial genetic Engineering, Advanced Molecular Biology, Introduction to Biology, Immunology

Development and performance evaluation of a rapid in-house ELISA for retrospective sero-surveillance of SARS-CoV2, Oishee MJ*, Jahan N, Ali T, Khandker SS, Khondoker M, Haq MA, Jamiruddin MR, Adnan N, and Sil BK. 34th Annual BSM Conference (international e-conference) (*Poster Presenter).

Accuracy of a rapid flow-through dot-blot assay for the detection of anti-SARS-CoV-2 IgG in comparison with ELISA. Jahan N, Khandker SS, Oishee MJ, Ali T, Khondoker MU, Haq MU, Jamiruddin MR, Sil BK, and Adnan N. 34th Annual BSM Conference (international e-conference).

Occurrence of resistance phenomenon against third-generation cephalosporin and quinolone antibiotics in uropathogenic Escherichia coli Isolates from Bangladesh, Rahman MH, Arefin MS, Oishee MJ, and Anjum H. at ASM Microbe 2020. Session AAR01: Epidemiology and Mechanisms of Resistance in the Health Care Setting.

Occurrence of Resistance Phenomenon against Third-Generation Cephalosporin and Quinolone Antibiotics in Uropathogenic Klebsiella spp. Isolates From Bangladesh, Anjum,H; Arefin, M.S; Akter,S; Islam, M.R; Oishee,M.J.; Rahman, H.R.*, 34th BSM (Bangladesh Society of Microbiologists) Annual Conference (International e-conference), 2020, ID: AMR03, Antibody Kinetics in Three COVID-19 Patients against Four Structural Proteins of SARS-CoV-2: A Longitudinal Case Study, Adnan,N; Jamiruddin MR, Haq,MA; Khandker,SS; Oishee, MJ; Jahan,N; Ali,T; Khondoker,MU; Sil, BK. 34th BSM (Bangladesh Society of Microbiologists) Annual Conference (International e-conference), 2020 ID: PDM03,

Development and Internal Evaluation of a Rapid SARS-Cov2 Antigen Detection Assay Khandker,SS; Oishee,MJ; Chaity,MA; Jahan,N; Ali,T; Jamiruddin,M; Uzzal,AK; Nazim,AQ; Khondoker,MU; Ahmed,S; Haq,MA; Sil,BK; Jamiruddin,MR; Adnan, N; 34th BSM (Bangladesh Society of Microbiologists) Annual Conference (International e-conference), 2020, ID: DIGN03

Occurrence of Quinolone resistance genes in multidrug resistance clinical E.coli , Arefin MS, Akter S, Islam MR, Biswas S, Oishee MJ, Nurjahan A and Rahman MH., 32nd BSM Annual Conference, April 6, 2019.

2018-2019: National Science and Technology Fellowship, Ministry of Science and Technology, Govt. of Bangladesh.

2017: Govt. Merit Scholarship, Directorate of Secondary and Higher Education, Govt. of Bangladesh.

2013-2016: Merit Scholarship, Jahangirnagar University, Dhaka, Bangladesh.

2013-2016: Education Board Scholarship for excellence in H.Sc. level, Cumilla, Bangladesh.

2011-2012: Education Board Scholarship for excellence in S.Sc. level, Cumilla, Bangladesh.

Rotaract Club of Jahangirnagar University.

Antimicrobial resistance, Environmental Microbiology, Immunology

Mumtrarin Jannat Oishee

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Fahmida Sultana completed her school from Govt. Model Girls' High School and College from Brahmanbaria Govt. College. She finished her B.Sc And M.Sc in Microbiology from Jahangirnagar University at 2015 and 2016 respectively.

Level: 4, Room: 4G35 Kha-224 Merul BaddaDhaka 1212. Bangladesh

Journals

Urmi, U. L., Nahar, S., Rana, M., Sultana, F., Jahan, N., Hossain, B., Alam, M. S., Mosaddek, A. S. M., McKimm, J., Rahman, N. A. A., Islam, S., & Haque, M. (2020). Genotypic to phenotypic resistance discrepancies identified involving β-lactamase genes, blakPC, blaIMP, blaNDM-1, and blaVIM in uropathogenic klebsiella pneumoniae. Infection and Drug Resistance, 13, 2863–2875. https://doi.org/10.2147/IDR.S262493

Islam, S., Urmi, U. L., Rana, M., Sultana, F., Jahan, N., Hossain, B., Iqbal, S., Hossain, M. M., Mosaddek, A. S. M., & Nahar, S. (2020). High abundance of the colistin resistance gene mcr-1 in chicken gut-bacteria in Bangladesh. Scientific Reports, 10(1). https://doi.org/10.1038/s41598-020-74402-4

Ali, T., Rumnaz, A., Urmi, U. L., Nahar, S., Rana, M., Sultana, F., Iqbal, S., Rahman, M. M., Rahman, N. A. A., Islam, S., & Haque, M. (2021). Type-2 diabetes mellitus individuals carry different periodontal bacteria. Pesquisa Brasileira Em Odontopediatria e Clinica Integrada, 21. https://doi.org/10.1590/pboci.2021.049

• March 2015- February 2016: Worked as “Research fellow” at Department of Microbiology, Jahangirnagar University in a project entitled “Screening of arsenite tolerant and arsenite oxidizing bacteria from arsenic contaminated soil and water from Dohar, Dhaka”• February 2017-February 2018: Worked as “Research fellow” in One Health Laboratory, Department of Microbiology, Jahangirnagar University• March 2018-June 2019: Worked as “Executive, Microbiology” in Square Pharmaceuticals Ltd.(Dhaka Unit)• May 2019-October 2021: Worked as a “Lecturer” at Department of Microbiology, Primeasia University

General Microbiology, Microbial Ecology, Environmental Microbiology, Medical Microbiology, Clinical Microbiology, Virology, Immunology, Microbial Biotechnology, Biochemistry, Basic Techniques in Microbiology, Immunology, Industrial Microbiology, Bioinformatics etc.

Conference Abstracts• Salequl Islam, Umme Laila Urmi, Masud Rana, Nusrat Jahan, Fahmida Sultana, Billal Hossain, Samiul Iqbal, and Shamsun Nahar. “Prophylactic Uses of Antimicrobials in Bangladeshi Poultry Drive Microbial Resistance”. American Society for Microbiology (ASM) Biothreat. Feb 12-14, 2018. Baltimore, MD, USA.

• Shamsun Nahar, Fahmida Sultana, Masud Rana, Nusrat Jahan, Billal Hossain, Samiul Iqbal , Azizur Rahman, Umme Laila Urmi,and Salequl Islam. “Prior Exposures of Antimicrobials Contribute Acquisition of Mobile Colistin Resistance (mcr-1) Gene into Bangladeshi Poultry Microbiota”. American Society for Microbiology (ASM) Microbes, 2018, Atlanta, USA.

• Salequl Islam, Umme Laila Urmi, Masud Rana, Fahmida Sultana, Nusrat Jahan, Billal Hossain, Samiul Iqbal, Abu Syed Md. Mosaddek and Shamsun Nahar. “Poultry chicken gut-bacteria carry high extent of colistin resistant mcr-1 gene in Bangladesh”. International Journal of Infectious Disease (ICID), December, 2020.

• E-Conference on current issues in Microbiology, Primeasia University, Dhaka, Bangladesh• Poster presentation on Asian Federation of Biotechnology (AFOB), Dhaka, Bangladesh.• In-Plant Training at Square Pharmaceuticals Ltd.(Dhaka Unit)

• 2017: Postgraduate Fellowship, awarded by National Science and Technology, Bangladesh• 2012-2015: Undergraduate Scholarship, Jahangirnagar University, Bangladesh• Board Scholarship for Secondary and Higher Secondary results• Merit Scholarship for Primary and Junior level

• Member of American Society of Microbiology (ASM)• Member of Jahangirnagar University Microbiology Alumni Association (JUMAA)• Member of Graduate Microbiologist Association (GMA)

Bioinformatics, Clinical and Environmental Microbiology, Drug Design and Development etc.

Fahmida Sultana

News & Events

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EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Mahmuda Kabir has joined as a Lecturer in Biotechnology Program of the Department of Mathematics and Natural Sciences, BRAC University since September 2022. She has completed both her BS(Hons.) and MS from the Department of Genetic Engineering and Biotechnology, University of Dhaka. Her research interest encompasses molecular oncology, precision medicine and identification of genetic diseases. She did her BS project on ‘The Role of Golgi Protein in Cancer.’ During her master’s degree, she continued working on oncology with the dissertation title ‘Expression Status of HER2 and its Oncogenic Alterations in Bangladeshi Breast Cancer Patients: A Demographic, in silico and ctDNA-based Study.’Mahmuda has always been actively involved in leadership and co-curricular activities. She believes in promoting and popularization of science and scientific literacy in Bengali among mass people. During Covid-19 she worked as a trainer on a project launched by the EMK center to “Prepare STEM Teachers for Post-Covid Pandemic Pedagogy”. She holds the position of the first female CEO of the Network of Young Biotechnologists of Bangladesh (NYBB) – a national platform that works to connect and bring inspiration and innovation to every Bangladeshi Biotechnologist of the world.

Level: 4Kha-224 Merul BaddaDhaka 1212. Bangladesh

• Master of Science (MS) in Genetic Engineering and Biotechnology, University of Dhaka (2021)• Bachelor of Science (BS) in Genetic Engineering and Biotechnology, University of Dhaka (2019)

NewsletterKabir, M. (2018). Network of Young Biotechnologists of Bangladesh Arranges the First Career Festival in Bangladesh for Youth in Biotechnology. South Asia Biosafety Program Newsletter, 18(3).

BookletKabir, M. (2020). Cancer Awareness and Management- Part-01. Cancer Care & Research Trust, Bangladesh (CCRTB) Booklet.

Magazine ArticlesKabir, M. (2017, December). □□□□ □□□□□ □□□□ □□□□□. BigganChinta, 17.K

August). □□□□□□□□□ □□□□□□□ □□□□□ □□□□□□□. BigganChinta, 17.Kabir, M. (2017, April). □□□□□□

• Lecturer, Department of Mathematics and Natural Sciences, BRAC University (September 2022-Present)

• Introduction to Biotechnology and Genetic Engineering• Metabolism• DNA Fingerprinting and Molecular Diagnostics

• Participated in Global Community Bio Summit 5.0 (2021) and 4.0 (2020)• Organized

NYBB 5th Young Biotechnologists Congress 2020• Participated in the International Symposium on Integrated Management and Treatment of Cancer (2020)• Participated in The International CRISPR and Gene Editing Symposium

• Kabi Sufia Kamal Hall Trust Fund Scholarship (Talent pool) 2020, University of Dhaka• Scholarship for Bachelor of Science degree, University of Dhaka (2019)• National Science and Technology (NST) Fellowship for MS research, Session 2020-21; Ministry of Science and Technology, Government of the People's Republic of Bangladesh

• General member, Federation of Asian Biotech Associations (FABA) Bangladesh Chapter

• Molecular oncology• Precision Medicine• Genetic diseases• Genomics• Molecular Diagnostics

Mahmuda Kabir

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

MNS Department started the undergraduate program of BS in Microbiology (136 credits) in Summer 2010. Microbiology, an integral part of molecular biology is emerging as a key biological science involving the study of cellular information. The mission of the undergraduate program in microbiology is to prepare the students for employment in their chosen areas and to provide them with rigorous training so that they can develop proper skills for a successful career in microbiology. A student has option for doing a double-major with microbiology as the first major. One can take a minor in microbiology as well.

News Archive >>

News

Biotechnology seminar hosts Bangladeshi scientist who developed high-yielding “Panchabrihi” rice

Orientation held for Summer 2024 biotechnology students

Nanobiotechnology seminar organized by BUSB and Biotechnology Program

The biotechnology program and BUSB organized the inaugural Biotechnology Workshop Series on “Understanding and Designing Biosensors”

Academic excellence is recognized in a number of ways at BRAC University , including through scholarships and awards.

Based on semester GPA , students are placed on Dean's List & VC's List, published during appropriate semester, is recognition of an undergraduate's outstanding academic performance that particular semester.To be eligible for Dean's List & VC's List certificate , students must meet the following requirements:

BU awards three types of academic honors at graduation . These Distinction honors, based on the cumulative grade point average (CGPA), are awarded in three grades:

[Amendments to the Criteria for the Gold Medal]

Chancellor's Gold Medal and Vice Chancellor's Gold Medal, based on the cumulative grade point average (CGPA), realize for academic excellence achieved over the duration of an entire undergraduate and postgraduate career. Candidates must have a minimum CGPA of 3.65 in both the cases.

Chancellor's Gold Medal : One candidate among all Undergraduate programs with highest CGPA and One candidate among all Postgraduate programs with highest CGPA is awarded with Chancellor's Gold Medal.Vice Chancellor's Gold Medal : One candidate from each Undergraduate programs with highest CGPA and One candidate from each Postgraduate programs with highest CGPA is awarded with Vice Chancellor's Gold Medal.

Office of Academic Advising (OAA) of Brac University links the students to all the facilities and resources that are available in the university such that the students can get started and continue in the path to success. In collaboration with students, faculties, administrative staffs, and parents, OAA nurtures an optimal university experience. In pursuit of doing so, the office members advise, plan, arrange tutorials, point them to appropriate teachers, tutors and services that are present to serve them. In summary, OAA strives to solve any problem affecting their academic performance, staying within the boundaries of the rules and regulation of the university.

Also assist any interim students who approaches us for Assistance or Advise.

Our vision is to partner with the BracU students, in educational growth and development, by creating meaningful educational plans which is compatible with their life goals.

Email: bracu-oaa[at]bracu.ac[dot]bdWebsite: www.bracu.ac.bd

News Archive >>

News

BRAC University alumnus receives partial scholarship at Texas State University

First Year Advising Team (FYAT), Summer 2020

FYAT Groups for Freshmen Students

OAA meets parents of Spring 2019 students

Career at BracU

Candidates who have finished Bachelor Degree from abroad have to submit verified/attested copies of previous academic documents from their institute/Foreign Ministry and Equivalence Certificate from University Grants Commission (UGC), Dhaka.

The EMBA Program at BBS is a part-time, skill-based program requiring 45 credits expected to be completed within two years. Depending on the course load of an individual student, the duration may vary. A student may be eligible for a waiver of up to 12 credits depending on his/her academic background.

The classes are conducted on Thursday and Saturday (6.30 pm – 9.30 pm) and Friday (9.30 am – 9.30 pm).

Objectives

The EMBA Program aims at empowering corporate executives by enabling them to skillfully use theoretical knowledge in practical situations and take effective decisions.

Eligibility

Minimum 15 years of schooling with a Bachelor Degree from any discipline and 3 years of professional / executive experience after graduation or 14 years of schooling with 6 or more years of corporate experience having at least 2nd class or CGPA 2.5 out of 4.0 or equivalent (from any government authorized university).

Minimum GPA of 2.5 or 2nd Class in SSC/Equivalent and HSC/Equivalent separately (including additional subject).

Minimum GPA of 2.5 in O-Levels in five subjects and A-Levels in two subjects according to the scale (A=5, B=4, C=3, D=2 and E=1). Only one E is acceptable.

Areas of Concentration

Students having a minimum of second class /2.50 CGPA (out of 4) in any one of the following degree can apply:

Applicant’s proficiency in English will be assessed through a placement and interview.

Postgraduate Programs in Disaster Management (PPDM) is a 36 credit Master’s Degree program including 6 credit for dissertation. Full time student can complete the course in

4 consecutive semesters.

All classes are held in the evening: 6:30pm-9:30pm, Friday to Tuesday.

Eligibility

Admission Test

All eligible applicants, unless exempted*, are required to sit for a written admission test. Those who qualify in the written test will be called for interview for final selection.

*Eligibility for exemption from the written test:

(a) BracU graduates having CGPA 3.0 or above and

(b) Applicants with the GRE/GMAT scores as specified below:

- Old GRE score of 450 verbal/550 analytical/650 quantitative and
- Revised GRE score of 300 total for combined verbal and quantitative scores, or
- GMAT total combined score of 560.

[GRE/GMAT scores are considered valid for 5 (five) years after the test date]

Interview is mandatory for all qualified applicants

The Master of Science/Post Graduate Diploma in Early Childhood Development (ECD) is a pioneering programme, the first of its kind in Bangladesh. The programme offers a detailed pedagogy on Early Childhood Development (ECD), the formation & growth during the early years. The course will enhance the learning and knowledge of all who are interested in working in the field of ECD from both private and public sectors. The programme is designed with a unique combination of classroom studies and field research which caters to those working on early childhood development to further their understanding of major issues. These issues encompass play, parenting, school readiness, program design and advocacy. The programme is led by a team of highly qualified faculty members comprising of researchers, curriculum developers, ECD specialists, programme specialists and resource professionals of national and global repute. It comprises of (9-courses/27-credits) to complete PGD in ECD, & (9-courses plus a 9-credit 'dissertation'/total 36-credits) to complete MSc. in ECD. The program is designed and offered in a modular system of courses, and it takes 18-months to complete ‘PGD’, and, 24-months to complete ‘MSc’.

All classes are held in the evenings at BRACU.

All eligible applicants, except the BracU graduates having the CGPA 3.0 or above, are required to sit for the written admission test. Admission Test will will be conducted in 'two' parts: 1-hr 'written test', and, a short 'interview' to identify the applicant’s proficiency level in English language. Both the 'written test' and the 'interview' will be held on the same day in a short (15/30-minute) break. Interview is, however, mandatory for all applicants.

Click to know more

BRAC IED’s Master of Education in Educational Leadership and School Improvement programme is designed to develop the skills and capacities of students as education leaders and managers of tomorrow. The programme works towards enhancing the capacity of individuals who are interested in working in the multifaceted field of education. We focus on developing leadership, management, analytical and problem-solving skills, introducing participants to the world of academia and research. The programme is led by a team of highly qualified faculty members comprising of researchers, curriculum developers, educationists and resource professionals of national and global repute. It comprises of (10-courses/27-credits) to complete PGD in Educational Leadership and School Improvement, & (10-courses plus a 9-credit 'dissertation'/total 36-credits) to complete MEd. in Educational Leadership and School Improvement.

All classes are held in the evenings at BRACU.

There is no written admission test for this programme. All the eligible candidates would need to face an interview in order to get selected for the programme.

Click to know more

Masters in Procurement and Supply Management (MPSM) is a unique program – consisting of 48 credits - offered by the Brac Institute of Governance and Development (BIGD), Brac University. Nowadays, procurement is a key function for any organization, either public or private. With this context in mind, BIGD is offering the program to sensitize and build up the capacity of officials of public, non-government, and private sectors on procurement, purchasing and supply chain management issues.

Class time: Thursday - 6.00 pm -9.00 pm & Friday - 3.00 pm-9.00 pm

Eligibility

Admission Test

All eligible applicants, unless exempted*, are required to sit for a written admission test. Those who qualify in the written test will be called for interview for final selection.

*Eligibility for exemption from the written test:

(a) BracU graduates having CGPA 3.0 or above and

(b) Applicants with the GRE/GMAT scores as specified below:

- Old GRE score of 450 verbal/550 analytical/650 quantitative and
- Revised GRE score of 300 total for combined verbal and quantitative scores, or
- GMAT total combined score of 560.

GRE/GMAT scores are considered valid for 5 (five) years after the test date.

Interview is mandatory for all qualified applicants

Brac University MA in English program has two concentrations: a) Literature and b) ELT and Applied Linguistics. The program is based on the conviction that students will benefit from an in-depth study of many aspects of English language and literature. Required credit for MA in English is 36.

All classes are held in the evening, 6:30pm - 9:30pm, Sunday to Wednesday, 6.00pm – 9.00pm Thursday

Eligibility

Admission Test

All eligible applicants, unless exempted*, are required to sit for a written admission test. Those who qualify in the written test will be called for interview for final selection.

*Eligibility for exemption from the written test:

(a) BracU graduates having CGPA 3.0 or above and

(b) Applicants with the GRE/GMAT scores as specified below:

- Old GRE score of 450 verbal/550 analytical/650 quantitative and

- Revised GRE score of 300 total for combined verbal and quantitative scores, or

- GMAT total combined score of 560.

- Test of English as a Foreign Language (TOEFL): 580 (Paper-based test) or 240 (Computer based test) or 94 (Internet based test - iBT)

- International English Language Testing System (IELTS): Overall band score of 7.0, with sub scores of 6.0 each

GRE/GMAT scores are considered valid for 5 (five) years and IELTS/TOEFL for 2 (two) years after the test date.

Interview is mandatory for all qualified applicants

MA in TESOL is a full-time program for current English teachers or anyone interested to pursue a career as an English teacher. This program focuses on training teachers of English to acquire the knowledge and competence for upgrading their professional skills so that they can efficiently teach English to both adult and young learners.

-Candidates with a 4-year Bachelor degree in English/Applied Linguistics/Linguistics/Education will have to do 39 credits.

-Candidates with a 4-year Bachelor degree in any other subjects of Humanities/Social Sciences/Liberal Arts/Business Studies will have to do 45 credits.

-Candidates with a 3-year BA degree having English/Applied

Linguistics/Linguistics/Education with 2-years of teaching experience will have to do 45 credits.

All classes are held in the afternoon-evening hours, i.e. 3:00 pm - 8:00 pm, Thursday to Saturday. However, the timings are flexible according to learners' situation.

ELIGIBILITY

ADMISSION TEST

All eligible applicants, unless exempted*, are required to sit for a written admission test. Those who qualify in the written test will be called for interview for final selection.

*Eligibility for exemption from the written test:

(a) BracU graduates having CGPA 3.0 or above.

(b) Applicants will get exemption from the English Composition part of the written test if they have the following TOEFL/IELTS scores:

- Test of English as a Foreign Language (TOEFL): 580 (Paper-based test) or 240 (Computer based test) or 94 (Internet based test - iBT)
- International English Language Testing System (IELTS): Overall band score of 7.0, with a minimum score of 6.0 in each module.

[TOEFL/IELTS scores are considered valid for 2 (two) years after the test date]

Interview is mandatory for all qualified applicants

The Master of Business Administration (MBA) of the BRAC Business School is a skill-based, 60 credit (19 courses and an internship), two and a half years of full time Program. A student may, however, enroll as a part time student. The duration of the part time program depends on the chosen course loads.

All classes are held in the evening, 6:30pm-9:30pm, Saturday to Thursday.

Eligibility

Minimum Bachelor Degree (15 years of education or 14 years of education with 8 or more years of corporate experience) having at least 2nd Class/Division or CGPA 2.5 out of 4.0 or equivalent (from any government authorized university).

Admission Test

All eligible applicants, unless exempted*, are required to sit for a written admission test. Those who qualify in the written test will be called for interview for final selection.

*Eligibility for exemption from the written test:

(a) BRACU graduates having CGPA 3.0 or above and

(b) Applicants with the GRE/GMAT scores as specified below:

- Old GRE score of 450 verbal/550 analytical/650 quantitative and
- Revised GRE score of 300 total for combined verbal and quantitative scores,
- GMAT total combined score of 560.

GRE/GMAT scores are considered valid for 5 (five) years after the test date.

Interview is mandatory for all qualified applicants

Eligibility Criteria

Minimum 15 years of schooling and 2 years of professional/executive experience or 14 years of schooling with 8 or more years of corporate experience having at least 2nd class or 2.50 GPA (or equivalent) at every level of education.

The Master of Development Studies (MDS) is a 45-credit Masters program. The MDS program designed to provide an in-depth, inter-disciplinary understanding of development concepts and issues. The program provides a wide range of elective courses so that students can pursue their studies according to their academic interests and career objectives.

All classes are held in the evening, 6:00pm-9:00pm, Saturday to Thursday

Eligibility

Admission Test

All eligible applicants, unless exempted*, are required to sit for a written admission test. Those who qualify in the written test will be called for interview for final selection.

*Eligibility for exemption from the written test:

(a) BracU graduates having CGPA 3.0 or above and

(b) Applicants with the GRE/GMAT scores as specified below:

- Old GRE score of 450 verbal/550 analytical/650 quantitative and
- Revised GRE score of 300 total for combined verbal and quantitative scores, or
- GMAT total combined score of 560.

GRE/GMAT scores are considered valid for 5 (five) years after the test date.

Interview is mandatory for all qualified applicants

ADMISSION CRITERIA

APPLICATION DEADLINEInternational Application: 15th September, 2020National Application: 31st October, 2020

INQUIRIEStahsin.hossain[at]bracu.ac[dot]bd

The “MS in Biotechnology” curriculum at BracU is aimed to blend a balance between theory and practice. The curriculum at BracU is comprised of a total of 36 credits. There are core theory courses and a wide variety of elective courses and a compulsory research project (thesis) on an indigenous biotech-related problem. The thesis is obligatory to acquaint a student with the process of how biotechnological research is carried out starting from the planning stage of the research project to its execution. This will not only give an in-depth understanding of the research topic but also a hands-on experience of conducting a research study.

All classes are held in the evening, 6:30pm-9:30pm, Sunday to Thursday. In special cases classes will be taken on weekends.

Eligibility

Admission Test

All eligible applicants, unless exempted*, are required to sit for a written admission test. Those who qualify in the written test will be called for interview for final selection.

*Eligibility for exemption from the written test:

(a) BracU graduates having CGPA 3.0 or above

(b) Applicants of MS BIO will get exemption from the English Composition part of the written test if they have the following TOEFL/IELTS scores:

(Note that the applicants must appear in the program-wise sections of the written test)

[TOEFL/IELTS scores are considered valid for 2 (two) years after the test date]

Interview is mandatory for all qualified applicants

The objective of the degree is to produce a well-informed and well-balanced graduate who can use Computer Science and Engineering tools to solve real world problems. At Masters level, they would be trained in research methodology, so that they can contribute towards creation of new knowledge.

- M.Sc. in Computer Science and Engineering Courses =18 credits and Thesis = 18 credits- M.Engg. in Computer Science and Engineering Courses = 30 credits and Project = 6 credits

All classes are held during the weekends between 3:00pm-9:00pm.

Eligibility

Admission Test

All eligible applicants, unless exempted*, are required to sit for a written admission test. Those who qualify in the written test will be called for interview for final selection.

*Eligibility for exemption from the written test:

(a) BracU graduates having CGPA 3.0 or above

(b) Applicants with the GRE/GMAT scores as specified below:

(c) Applicants of MSc/Engg CSE will get exemption from the English Composition part of the written test if they have the following TOEFL/IELTS scores:

[GRE/GMAT scores are considered valid for 5 (five) years and TOEFL/IELTS scores are considered valid for 2 (two) years after the test date]

Interview is mandatory for all qualified applicants

The MSc/MEngg in EEE program provides students with broad, in-depth and up-to-date engineering knowledge and skills in three academic tracks of electrical engineering, namely, electronics, telecommunication, and power system. Students can pursue their Master’s degree with concentration/major in one of these academic tracks.

There are two options under which the Master’s degree can be pursued: the thesis option and the non-thesis option. In the thesis option, students are required to carry out original research work as a part of their degree requirement in addition to doing the required number of course works and the degree awarded is Master of Science (M.Sc.) in EEE. In the non-thesis option, students are required to carry out an independent study of an engineering topic of practical significance, in addition to doing the required number of course works and the degree awarded is Master of Engineering (M.Engg) in EEE.

Each student is required to successfully complete a minimum of 36 credit hours with a minimum CGPA of 2.65 out of 4.0 to graduate. Apart from the regular subject-related courses to fulfill the requirement for coursework, a student may also be required to take remedial and supplementary non-credit courses to improve study skills, presentation and communication skills, as determined by the admission committee.

All classes are held in the evening: Sunday to Thursday 6:00pm-9:00pm.

Eligibility:

A candidate must have a B.Sc. degree in any of the following:

I. Electrical and Electronic Engineering (EEE)II. Electrical Engineering (EE)III. Electronic and Telecommunication Engineering (ETE)IV. Electronic and Communication Engineering (ECE)V. Or, an equivalent degree in related areas

Admission Test

All eligible applicants, unless exempted*, are required to sit for a written admission test. Those who qualify in the written test will be called for interview for final selection.

*Eligibility for exemption from the written test:

(a)BracU graduates having CGPA 3.0 or above

(b) Applicants with the GRE/GMAT scores as specified below:

(c) Applicants of MSc/Engg EEE will get exemption from the English Composition part of the written test if they have the following TOEFL/IELTS scores:

[GRE/GMAT scores are considered valid for 5 (five) years and TOEFL/IELTS scores are considered valid for 2 (two) years after the test date]

Interview is mandatory for all qualified applicants

Bangladesh is an emerging economy where public policy has to be broadened up and it has to be everybody's business. It is not only about policy making but also how citizens engage with that policy making, how policies are actually being implemented on the ground and how it impacts people- these are all central to good policy making that can help governance work for development.

The Master of Arts in Governance and Development (MAGD) program, offered by BIGD at Brac University is a nationally and internationally acclaimed program that has been meticulously designed for government and non-profit officials, researchers, development practitioners, and young professionals who want to make public policy work for development.

Since its beginning, the MAGD program has been playing a significant role in nurturing promising policymakers, enabling them to take on multifaceted, complex governance and development challenges with confidence. The MAGD program is very important for BIGD because it is central to BIGD’s mission. BIGD wants to connect governance and development and wants to ensure that governance works for development and the way governance works for development is through public policy so from that point of view this is central to the mission of BIGD and the overall vision of ‘knowledge for a better world.’

To achieve its aim of developing highly effective policymakers and professionals in governance and development, the program offers a wide range of expertise among faculty, cross-learning opportunities with development programs and connecting with a strong alumni network. Access to digital content and an understanding of the role of the latest technologies in public policy analysis will ensure that students are well-equipped to be confident benefactors of the fourth industrial revolution.

The MAGD program requires 36 credits and can be completed within 2 years under an open-credit system. MAGD will offer a unique blend of teaching by globally acclaimed academics and the insights of locally rooted practitioners to build a strong theoretical foundation on governance and development and the capabilities to successfully implement learned theories into practice. The redesigned MAGD will be offered from Fall 2023 as a non-residential blended/hybrid program to offer the necessary flexibility to working students.

MAGD aims to enable policymakers and professionals to successfully address the 21st-century challenges and meaningfully contribute to the sustainable, inclusive development of their country. In order to achieve this, The curriculum has been designed in such a way that will cultivate critical thinking among the students and develop their soft and hard skills to turn them into strategic, agile, efficient, and responsive actors of governance and development. Therefore, MAGD graduates have an additional edge over their peers, giving them an advantage in terms of career progression and job postings.

Please visit BIGD website for details

The Master of Science in Applied Economics (MSAE) degree is designed to create highly competent economics professionals with capability to serve in the private and public sectors. It is a 36-credit program comprising core courses, primary concentration courses, free electives and a thesis.

All classes are held in the evening, 6:30 pm-9:30 pm, on weekdays.

Eligibility

Admission Test

All eligible applicants, unless exempted*, are required to sit for a written admission test. Those who qualify in the written test will be called for interview for final selection.

*Eligibility for exemption from the written test: (a) BracU graduates having CGPA 3.0 or above and (b) applicants with the GRE/GMAT scores as specified below:

- Old GRE score of 450 verbal/550 analytical/650 quantitative and
- Revised GRE score of 300 total for combined verbal and quantitative scores, or
- GMAT total combined score of 560.

GRE/GMAT scores are considered valid for 5 (five) years after the test date.

Interview is mandatory for all qualified applicants

Executive MBA (EMBA)

Master in Computer Applications (MCA) - (Discontinued)

Master in Disaster Management (MDM)

MSc. & PGD in Early Childhood Development

M.Ed and Post Graduate Diploma in Educational Leadership & School Improvement

Masters in Procurement and Supply Management (MPSM)

Master of Arts in English

MA in TESOL (Teaching English to Speakers of Other Languages)

Master of Business Administration (MBA)

Master of Bank Management (MBM)

Master of Development Studies (MDS)

Master of Public Health (MPH)

Master of Science in Biotechnology

M.Sc/M.Engg in Computer Science and Engineering

Masters in Development Management & Practice (MDMP)

M.Sc/M.Engg in Electrical and Electronic Engineering

MA in Governance and Development (MAGD)

Master of Science in Applied Economics (MSAE)

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Services and Resources

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News

BRACU library staff gets fund from Open Society University Network

BRAC University Ayesha Abed Library, in collaboration with the Open Society Network (OSUN), celebrates Open Access Week 2023

Author Workshop on “Understanding research metrics and finding relevant journals to read and publish in”

Brac University received 2019 Global Innovation Award

Dr Mahdi Moosa is a biophysicist and bioinformatician who has extensive research experience with disordered proteins and their physicochemical properties. He obtained his BSc (Hons.) and MS from the University of Dhaka, and PhD from the Scripps Research Institute, La Jolla, California. He got his postdoctoral trainings at the Baylor College Medicine (Texas) and the University at Buffalo (New York). Currently, he is an Assistant Professor at the Department of Mathematics & Natural Sciences (Biotechnology Program).

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• PhD in Biological Sciences, the Scripps Research Institute, California, USA• MS in Genetic Engineering & Biotechnology, University of Dhaka, Dhaka, Bangladesh• BSc (Hons) in Genetic Engineering & Biotechnology, University of Dhaka, Dhaka, Bangladesh

Book Chapters

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Mahmood, N., Moosa, M. M., Matin, S. A., & Khan, H. (2012). Members of ectocarpus siliculosus F-box family are subjected to differential selective forces. Interdisciplinary Bio Central, 4(1), 1–7. https://doi.org/10.4051/ibc.2012.4.1.0001

Maruf, A. A., Moosa, M. M., Rashid, S. M. M., Khan, H., & Yeasmin, S. (2012). Culture dependent molecular analysis of bacterial community of Hazaribagh tannery exposed area in Bangladesh MATeRIAlS And MeThodS. Agriculture, Food and Analytical Bacteriology, 2(2).

Moosa, M. M., Ayub, M. I., Bashar, A. E., Sarwardi, G., Khan, W., Khan, H., & Yeasmin, S. (2011). Combination of two rare mutations causes β-thalassaemia in a Bangladeshi patient. Genetics and Molecular Biology, 34(3), 406–409. https://doi.org/10.1590/S1415-47572011005000026

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Samira, R., Moosa, M. M., Alam, M. M., Keka, S. I., & Khan, H. (2010). In silico analysis of jute SSR library and experimental verification of assembly. Plant OMICS, 3(2), 57–65.

Ahmed, S., Nabi, M. Z., Alam, M. M., Islam, M. S., Samira, R., Moosa, M. M., & Khan, H. (2009). A computational and experimental approach for developing jute ESTs from genomic clones. Australian Journal of Crop Science, 3(6), 322–328.

Minhaz Ud-Dean, S. M., & Moosa, M. M. (2009). Sequence specific dihedral angle distribution: Application in protein structure prediction and evaluation. Plant Tissue

Culture and Biotechnology, 19(2), 217–226. https://doi.org/10.3329/ptcb.v19i2.5439

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o 2023- till date: Assistant Professor, BRAC University, Dhaka, Bangladesho 2019-2022: Postdoctoral Associate, University at Buffalo, New York, USAo 2017-2019: Postdoctoral Associate, Baylor College of Medicine, Texas, USAo 2010-2011: Research Associate, Department of Biochemistry & Molecular Biology, University of Dhaka, Dhaka, Bangladesh

o Structural & Functional Genomics Studieso Bioinformaticso Human Genomeo Advanced Molecular Biology

• Moosa, MM, Tsoi, PS, Choi, KJ, Ferreon, ACM, Ferreon, JC. Deciphering complexities in Sox2-DNA interactions using single-molecule spectroscopy (2018). 2018 BMB/Pharm Research Conference, Baylor College of Medicine, The Tremont House, Galveston, Texas.• Banerjee, PR, Moosa, MM, Deniz, AA. Single-Molecule Spectroscopy Reveals the Role of Two-Dimensional Crowding in Disordered Protein Folding (2017). Platform on Intrinsically Disordered Proteins (IDP) and Aggregates II in 61st Annual Biophysical Society Meeting, New Orleans, Louisiana.• Moosa, MM, Banerjee, PR, Deniz AA. Chaperone Hsp27 uncovers a hidden membrane-bound state of α-synuclein (2016). Symposium on Theory, Experiment and Computations: A Synergistic Approach to Research in the 97th Annual Meeting of the Pacific Division of the American Association for the Advancement of Science, San Diego, California.

o NIH T32 Interdisciplinary Translational Cancer Nanotechnology Fellowship (declined), MD Anderson Cancer Center- Rice University Joint Program (2017-2019)o The Joshi Memorial Prize for Outstanding Research Talk, 2018 BMB/Pharm Research Conference, Baylor College of Medicine, Texas (2018).o Dean’s Award, Faculty of Life Sciences, University of Dhaka, Bangladesh (2011)o National Science, Information &

Communication Technology Fellowship, Ministry of Science, Information & Communication Technology, Bangladesh (2009-2010)o SUMITOMO Corporation Scholarship, University of Dhaka, Bangladesh (2009)o Japan International Co-operation Foundation Scholarship, University of Dhaka, Bangladesh (2004)

o Biophysical Societyo American Association of Advancement of Sciences (AAAS)o American Chemical Society (ACS)o International Society for Computational Biology (ISCB)

o Protein structure & dynamicso Biomolecular assemblies/aggregateso Genomic variationso Bioinformatics

Mahdi Moosa, PhD

News & Events

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EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Science Exhibition 2019: an inter-university and college scientific project exhibition program organized by Department of Mathematics and Natural Sciences in the university premises to motivate students to involve themselves more into learning about basic sciences and their applications. The event is going to take place on Sunday, February 17, 2019, at BRAC University Auditorium (UB10005).

Science Exhibition - 2019

Dr. Syed Hasibul Hassan Chowdhury obtained his PhD degree in mathematics (specialising in mathematical physics) in 2013 from Concordia University, Montreal, Canada. Earlier, he did his B. Sc and M. Sc in Physics from the Physics department of Dhaka University. Soon after his PhD defence, he was employed in the same university as a research assistant professor. He stayed there for around a year and moved to the Chern Institute of Mathematics (CIM), Nankai University, China with a postdoctoral research fellowship. After the completion of his postdoctoral fellowship at CIM, he moved to Institute of Mathematical Research (INSPEM) at University Putra Malaysia (UPM) with a second postdoctoral fellowship. After half a year on the 7th March, 2017 he came to Bangladesh to join the physics department of Dhaka University. He was granted extraordinary leave from Dhaka University to serve the Department of Mathematics and Natural Sciences of Brac University from 28th March, 2020 as an associate professor.

Level: 4, Room: 4G39Kha-224 Merul BaddaDhaka 1212. Bangladesh

Name of the exam

Institute name

Passing year

Result

H. Sc

Notre Dame College

1995

First Division

B. Sc in Physics

Dhaka University

1998

First Class

M. Sc in Physics

Dhaka University

1999

First Class

PhD in Mathematics

Concordia University

2013

CGPA 4.1 out of 4.3

Journals

Jim, M. R., & Chowdhury, S. H. H. (2024). Supersymmetric quantum mechanics on a noncommutative plane through the lens of deformation quantization. Annals of Physics, 467, 169718. https://doi.org/10.1016/j.aop.2024.169718

Chowdhury, S. H. H., & Chowdhury, T. A. (2023). On a charged spinless point particle minimally coupled to a constant magnetic field in a noncommutative plane. Annals of Physics, 453. https://doi.org/10.1016/j.aop.2023.169308

Chowdhury, S. H. H., Chowdhury, T. A., Nasri, S., Nazim, O. I., & Saad, S. (2023). Quantum simulation of quantum mechanical system with spatial noncommutativity. International Journal of Quantum Information. https://doi.org/10.1142/S0219749923500284

Chowdhury, S. H. H., Chowdhury, T. A., & Duha, M. A. U. (2021). Gauge invariant energy spectra in 2-dimensional noncommutative quantum mechanics. Annals of Physics, 430. https://doi.org/10.1016/j.aop.2021.168505

Chowdhury, S. H. H., & Zainuddin, H. (2017). Wigner functions for gauge equivalence classes of unitary irreducible representations of noncommutative quantum mechanics. European Physical Journal: Special Topics, 226(10), 2359–2374. https://doi.org/10.1140/epjst/e2017-70066-8

Chowdhury, S. H. H., (2017). On the plethora of representations arising in noncommutative quantum mechanics and an explicit construction of noncommutative 4-tori. Journal of Mathematical Physics, 58(6). https://doi.org/10.1063/1.4985152

Chowdhury, S. H. H., Ali, S. T., & Engliš, M. (2017). Noncommutative coherent states and related aspects of Berezin-Toeplitz quantization. Journal of Physics A: Mathematical and Theoretical, 50(19). https://doi.org/10.1088/1751-8121/aa66a6

Jiang, J.-J., & Chowdhury, S. H. H. (2016). Deformation of noncommutative quantum mechanics. Journal of Mathematical Physics, 57(9). https://doi.org/10.1063/1.4963142

Chowdhury, S. H. H. (2016). On Goldman bracket for G 2 gauge group. Journal of High Energy Physics, 2016(2), 1–33. https://doi.org/10.1007/JHEP02(2016)001

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Chowdhury, S. H. H., & Ali, S. T. (2014). Triply extended group of translations of ℝ4 as defining group of NCQM: Relation to various gauges. Journal of Physics A: Mathematical and Theoretical, 47(8). https://doi.org/10.1088/1751-8113/47/8/085301

Ali, S. T., & Chowdhury, S. H. H. (2014). On the group theory of 2-dimensional noncommutative quantum mechanics. Malaysian Journal of Mathematical Sciences, 8(S), 1–14.

Chowdhury, S. H. H., & Ali, S. T. (2013). The symmetry groups of noncommutative quantum mechanics and coherent state quantization. Journal of Mathematical Physics, 54(3). https://doi.org/10.1063/1.4793992

Chowdhury, S. H. H., & Ali, S. T. (2011). All the groups of signal analysis from the (1+1)-affine Galilei group. Journal of Mathematical Physics, 52(10). https://doi.org/10.1063/1.3652697

Chowdhury, S. H. H., & Ali, S. T. (2013). Coadjoint orbits and wigner functions of (1+1)-extended affine Galilei group and Galilei group. Bollettino Di Matematica Pura Ed Applicata (BmPa), VI.

Calculus and analytic geometry, Linear algebra, Discrete Mathematics, Complex analysis, Topology, Mathematical Methods of Physics, General relativity and Cosmology, Mechanics and waves, Electricity and Magnetism, Physics I, Physics II.

i) Construction of Wigner functions from gauge equivalence classes of unitary irreducible representations of noncommutative quantum mechanics (NCQM), S. N. Bose National Centre for Basic Sciences, Kolkata, India, 28th November, 2018.ii) On Goldman bracket for G2 gauge group, IISER, Kolkata, India, 12th April, 2018.iii) On the Plethora of Representations Arising in Noncommutative Quantum Mechanicsand An Explicit Construction of Noncommutative 4-tori, ISI, Kolkata, India, 10th April, 2018.iv) On the Wigner functions of Noncommutative Quantum Mechanics: a group representation based construction, Xuefu Rd, Mengzi, Honghe, Yunnan, P. R. China, September 1-3, 2016.v) Deformation quantization of Noncommutative Quantum Mechanics, XXIVth International Conference on Integrable Systems and Quantum symmetries (ISQS-24), Czech Technical University Prague, June 13-17, 2016.vi) From Noncommutative Quantum Mechanics to Noncommutative Geometry, 8th Expository Quantum Lecture Series (EQuaLS8), Institute of Mathematical Research, Universiti Putra Malaysia (INSPEM, UPM), January 18-22, 2016.vii) On Alternative derivation of Goldman bracket, Workshop on Integrable systems and moduli spaces, Banff International Research Station, August 25-30, 2013.viii) On the symmetry groups of Noncommutative Quantum Mechanics, XXXI Workshop on Geometric Methods in Physics, Bialowieza, Poland, June 24-30, 2012.

• Research grant for International Young Scientists by NSFC (National Natural Science Foundation of China) in the year of 2015.• Best student prize for 2012-13 awarded by CRM Mathematical Physics laboratory.• Concordia Accelerator Award in 2013.• Concordia University International Fee Remission Award, 2011-12.• ISM (Institut des sciences Mathématiques) Doctoral Scholarship, 2009-2013.

Representation theory of locally compact and nilpotent Lie groups, Symplectic geometry, Chern-Simons theory, Connes-Kreimer theory in renormalization of QFT, Deformation quantization, Quantum groups, Noncommutative geometry.

Syed Hasibul Hassan Chowdhury, PhD

News & Events

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EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

He has completed both his B.Sc. (Hons.) in Mathematics in 2015 and M.Sc. in Applied Mathematics in 2017 from the Mathematics Discipline, Khulna University, Khulna, Bangladesh. His teaching career started with working in BRAC University under the Department of Mathematics and Natural Sciences as a Course Contract Faculty and taught the course MAT092 and MAT110 for the Spring semester 2018. Then he joined in American International University-Bangladesh (AIUB) as a Lecturer under the Department of Mathematics and worked there successfully from May 13, 2018 to September 1, 2018. After working four months in AIUB, he again joined in BRAC University on September 3, 2018 as a Lecturer under the Department of Mathematics and Natural Sciences.

Level:Kha-224 Merul BaddaDhaka 1212. Bangladesh

• Master of Science in Applied MathematicsMathematics DisciplineKhulna UniversityMedium of instruction - EnglishSession : 2014 - 2015Year of passing - 2017CGPA - 3.98 (in a scale of 4.00) equivalent to 97.40 % of total marks

• Bachelor of Science (Honours) in MathematicsMathematics DisciplineKhulna UniversityMedium of instruction - EnglishSession : 2010 - 2011Year of passing - 2015CGPA - 3.81 (in a scale of 4.00) equivalent to 83.80 % of total marks

• Higher Secondary CertificateB.A.F. Shaheen College, Tejgaon, DhakaDhaka BoardSession : 2008 – 2009Year of passing - 2010GPA - 5.00 (in a scale of 5.00)

• Secondary School CertificatePanchagarh B.P. Govt. High School, PanchagarhRajshahi BoardSession : 2006 – 2007Year of passing - 2008GPA - 5.00 (in a scale of 5.00)

• Currently working in BRAC University as a Lecturer under the Department of Mathematics and Natural Sciences since 3rd September, 2018.• Worked in American International University-Bangladesh (AIUB) as a Lecturer under the Department of Mathematics from 13th May, 2018 to 1st September, 2018.• Worked as a Course Contract Faculty in BRAC University under the Department of Mathematics and Natural Sciences from 9th January, 2018 to 30th April, 2018 (Spring semester - 2018).

MAT 092, MAT 101, MAT 105, MAT 110, MAT 120

• “Mathematical Analysis of Dynamic Model of Drug Addiction in Bangladesh”, A poster presented to the International Conference on Advances in Computational Mathematics (ICACM 2017) held on May 27 - 28, 2017 at University of Dhaka, Bangladesh.• “Optimal Control Applied to the Dynamic Model of Drug Addiction in Bangladesh”, A poster presented to the 20th International Mathematics Conference 2017 held on Dec 8-10, 2017 at University of Dhaka, Bangladesh.• “Mathematical Analysis and Control of Drug Addiction and its Harmful Consequences” an oral presentation in the 1st International Conference on Industrial and Mechanical Engineering and Operations Management (IMEOM) held on Dec 23 - 24, 2017 at Institution of Engineers Bangladesh, Dhaka, Bangladesh.

• A F Mujibur Rahman Gold Medal Award for Excellence in Mathematics (First Class First in Master of Science (M.Sc.) program)• National Science and Technology (NST) Fellowship for M.Sc. Thesis from Ministry of Science and Technology of Government Republic of Bangladesh, 2017.• Obtained the second place for best poster presentation in the International Conference on Advances in Computational Mathematics (ICACM 2017) held on May 27 - 28, 2017 at University of Dhaka, Bangladesh.• Best presentation award in 1st International Conference on Industrial and Mechanical Engineering and Operations Management (IMEOM) held on Dec 23 - 24, 2017 at Institution of Engineers Bangladesh, Dhaka, Bangladesh.• Merit scholarship from Khulna University.• Rajshahi board scholarship for outstanding performance in S.S.C. examination in 2008.

President, IEOM Society Bangladesh, Student Chapter, Khulna University.

Mathematical Modeling, Optimal Control Theory, Wavelet analysis, Numerical Analysis and Fluid Dynamics.

Md. Azmir Ibne Islam

News & Events

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As a leading private university in the country and one that strives to create educated youth ready to face the challenges of a highly competitive world, the University expects its faculty, students and the staff to display a positive image of the University in their dress and attire at all times. All are expected to come to the university wearing clean, decent dress that goes in conformity with its academic and cultural ethos. While a list of what one is expected to wear will be too exhaustive to be detailed, the University stipulates following dress items that are NOT permitted any time in the Campus:

a. Face mask or hood of any kind that makes the individual unidentified.b. Shorts of any kind, including three-quarter.c. Mini or midi skirts.d. Bathroom slippers (except for cleaning staff).e. A dress item that contains offensive slogan, picture or sign.

This does not bar any Department/Institute/School to prescribe any dress or attire that is considered necessary in the execution of its academic program.

BRAC University's New Campus is located at KHA 224, Progati Sarani, Merul Badda, Dhaka 1212. It's conveniently located near Hatirjhil, Rampura, Aftabnagar, Banasree, Niketon, Mohakhali, Gulshan, Banani, and Baridhara Residential Areas. The campus are covers a total land area of 5 acres, with the building spanning 1,700,000 square feet in total. WOHA Designs Pte Ltd., Singapore based world-renowned architectural firm has designed this project.

Salient Features The Campus is a 13-story building with three basements, designed to accommodate around 12,000-15,000 students. It's an environmentally conscious structure, allowing natural light and air to flow freely. Nearly half of the land is dedicated to the building itself, while the remainder is utilized for a water body, lake, plantation, and other features aimed at promoting biodiversity.

Sun-drenched CampusThe design of the building largely utilizes natural sunlight, thereby reducing the need for artificial lighting.

Draped in GreenLush greenery, both inside and outside, provides shade, temperature control, improved air quality, and aesthetics.

Nourished with RainwaterHarvested rainwater and a cleansing biotope meet 20% of the campus's water demand, especially for waterbodies, vegetation, toilet flushing, and firefighting.

Powered by Renewable EnergyA quarter of the campus's energy demand is met by solar panels on the roof, highlighting the commitment to sustainability.

Rooftop Green LawnThe greenery, sand, and gravel on the rooftop have therapeutic effects, promoting wellness.

Jogging TrackThe rooftop jogging track is a great place to get some exercise while enjoying a beautiful view.

Ayesha Abed LibraryThe library offers over 45,000 books along with AR/VR facilities for an immersive learning experience.

Library Study Places

Library staircase

Learning FacilitiesThe campus offers state-of-the-art learning facilities, including 123 classrooms, 5 lecture theatres, and 105 laboratories.

Alumni LoungeThe alumni lounge at BRAC University's new campus offers a welcoming space for graduates to reconnect, network, and reminisce.

Multipurpose HallA versatile space for hosting various events, including fairs, conferences, cultural events, and sports.

The CafeteriaThe cafeteria is a welcoming space for socializing and enjoying a meal.

Centre for Climate Change and Environmental Research (C3ER), BRAC University, was established in 2011. C3ER’s Vision is to craft sustainable & adaptive solutions for society, and its targets are to be established as a cutting-edge research and academic institute in Bangladesh working in the field of climate physical science and natural resources management. C3ER has extensive expertise in climate change adaptation, mitigation, loss and damage, natural resource management, knowledge sharing, capacity building, Remote Sensing, GIS and databases, policy advocacy, and governance, etc. Since its establishment, C3ER’s primary goal as an academic research institute is to generate new knowledge, share it with other stakeholders, and influence policymakers. C3ER adheres to the principle of “Leave No One Behind.”

Along with research activities, C3ER has expertise in formulating national policies and plans. It has been involved in the formulation process of several policy documents, including the National Adaptation Plan (NAP), 2023-2030; the updating of National Determined Contribution (NDC), 2021; Bangladesh Climate Change Strategy and Action Plan (BCCSAP), 2009; Bangladesh Climate Change Gender Action Plan (BCCGAP), 2022 and Sector Action Plan on “Environment & Climate Change”, 2022, etc.

In addition to this, C3ER has a capacity-building unit from where it provides capacity-building training on Climate change vulnerability, Disaster Management, Gender Mainstreaming Guidelines, Access to Green Climate Fund (GCF), Climate Finance, and Proposal Development Process, etc., and so far, the Centre has provided numerous trainings to the government officials, NGO, INGOs, youth, children, practitioners, researchers, and academician, etc.

For details please visit: https://c3er.bracu.ac.bd

The C3ER focus on:

Latest News and Events

C3ER, BRAC University hosted National Symposium to mark World Environment Day 2024

Fostering Bangladesh's Climate Resilience: Masterclass on Climate Resilient Infrastructure PPPs in Bangladesh

C3ER and Student Life of BRAC University host “Net-Zero Festival” with Korea’s DAEJAYON

Erratic weather alarming sign of climate change: Prof Ainun Nishat

Mr. Mohammad Maruf Ahmed was born in Dhaka, the capital of Bangladesh in 1978 and was brought up in a middle class Bangladeshi family. He is the eldest of the two sons of his parents and his career started from Dhaka. From his very childhood he used to take a lot of interest in confronting the challenging issues. He tries to keep pace with the changes and adjust himself accordingly. Throughout his entire educational career from the high school to the university levels apart from academic studies he used to take part in extracurricular activities like games and sports. He also participated in organizing events for the institutions where he studied. He was a sportsman and learnt how to prepare for, compete in and encounter adverse and competitive situations. He is now working as an Assistant Professor in Mathematics at BRAC University in the Department of Mathematics and Natural Sciences. As a teacher, he always tries to help his students in their learning process in the best possible way that he could. As a faculty member of this university, he also has to carry out the duties and responsibilities in addition to delivering lectures. He works as a member of a number of committees, formed for smooth operation of the activities concerned with the department as well as the university.

Level: 4, Room: 4G49Kha-224 Merul BaddaDhaka 1212. Bangladesh

Journals

Ahmed, M. M. (2017). Derivations of generalized inverse using contour integration and interpolation polynomiyal. BRAC University Journal, 4(1), 97–102. Retrieved from http://hdl.handle.net/10361/365

Ahmed, K. M., Ahmed, M., Khatun, G., Begum, F. A., Shahabuddiun, M., Das, T. K., & Saha, S. K. (2011). Study on cohomology of De Witt suppermanifolds. Jahangirnagar University Journal of Science, 34(2), 89–96.

Ahmed, M. M. (2006). Unified approach of generalized inverse and its applications.

BRAC University Journal, 3(2), 41–45. Retrieved from http://hdl.handle.net/10361/570

Conferences

Ahmed, M. M. (2012). Assorted representations of generalized inverse with numerical solutions. In International Workshop and Conference on Combinatorial Matrix Theory and Generalized Inverses of Matrices. Karnataka, India: Manipal University.

Ahmed, M. M. (2011). Role of visualization in learning and teaching mathematics for children. In International 17th Mathematics Conference. Dhaka: Jahangirnagar University.

♦ MAT 101: Fundamentals of Mathematics♦ MAT 102: Introduction to Mathematics♦ MAT 110: Differential Calculus and Coordinate Geometry♦ MAT 120: Integral Calculus and Differential Equation ♦ MAT 104: Mathematics♦ MAT 216: Linear Algebra and Fourier Analysis♦ MAT 215: Complex Variable and Laplace Transform♦ MAT 103: Basic Concepts in Mathematics♦ MAT 221 : Real Analysis I♦ MAT 111: Principles of Mathematics♦ MAT 122: Analytic and Vector Geometry♦ MAT 091: Basic course in Mathematics (MBA Program)

1. Participated as one of the speakers in “International Workshop and Conference on Combinatorial Matrix Theory and Generalized Inverses of Matrices” held in 2-11 January, 2012 at Manipal University, Karnataka, India. The title of my presentation was “Assorted Representations of Generalized Inverse with Numerical Solutions “2. Participated as one of the speakers in “ International 17th Mathematics Conference” held in 22- 24 December 2011 at Jahangirnagar University, Savar, Bangladesh. The title of my presentation was “ROLE OF VISUALIZATION IN LEARNING AND TEACHING MATHEMATICS FOR CHILDREN”. The conference was organized by Bangladesh Mathematical Society. 3. Participated as one of the speakers in “ Fifteenth International Mathematics Conference” held in 29-31, December 2007 at University of Dhaka, Dhaka, Bangladesh. The title of my presentation was “ Unified Approach of Generalized Inverse With Its Derivations, Representations and Applications”. The conference was organized

by Bangladesh Mathematical Society.

♦Merit based School scholarship♦Merit based University scholarship (both in Honors and in MS)

♦ Life member - Bangladesh Mathematical Society.♦ Life member – National Oceanographic And Maritime Institute (NOAMI), Dhaka, Bangladesh

♦ Oceanography♦ Dynamical System♦ Linear Algebra♦ Mathematical modeling in Biology

Maruf Ahmed

News & Events

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On 24 March 2016, the Department of English and Humanities organised an international conference on Language Acquisition, Impairment, and Intervention. Using an interdisciplinary approach, the conference aimed to bring together the practitioners working in related areas to have a consolidated understanding of the current practices, and the limitations in the context of Bangladesh with regard to language development and difficulties of Bangla-speaking children. The conference showcased invited papers in the areas of linguistics, education, neuroscience, and speech and language therapy.

The key speakers of the conference were Prof. Thomas Klee and Prof. Stephanie Stokes who are currently affiliated with the Division of Speech and Hearing Sciences, University of Hong Kong. Prof. Klee and Prof. Stokes previously engaged themselves in teaching and research, and in establishing child language centres at several universities around the world some of which included the Newcastle University, UK and the University of Canterbury, New Zealand. The speakers discussed the issues associated with the early identification of children with language difficulties, and how assessments were conducted crosslinguistically to determine children’s language development. The other invited speakers of the conference were the research groups working in Bangladesh who are involved in the assessment of children’s first language, and in the intervention process of children with language difficulties. The papers from Bangladesh included the research of the relevant teams from the ICDDR,B, Bangladesh Protibondhi Foundation, Dhaka Shishu Hospital, and the Institute of Education and Research, Dhaka University. The conference also proudly presented the research of two ENH students who worked on typical and atypical development of Bangla-speaking children.

The conference was well-attended with a large participation from linguists, speech and language therapists, school administrators, child caregivers, and students. The event was particularly significant to people working in related disciplines because platforms to explore these issues are extremely limited in Bangladesh. In addition, with regard to child language research, the resources available for Bangla are inadequate. A range of stardardised tests assessing various developmental abilities are still underway for Bangla, which makes it difficult to diagnose children accurately. There is a strong realisation among the researchers that a significant amount of language-specific research needed to be conducted before we could establish a comprehensive assessment and intervention framework to help Bangla-speaking children with a range of language difficulties. To this end, the conference opened possibilities for consolidating the existing work, and extending them to establish better resources for Bangla-speaking contexts.

By connecting to the wider community engaged in child language research, the Department of English and Humanities hopes to build meaningful collaborations with research organisations in Bangladesh as well as abroad, and inspire more students to pursue language acquisition research with Bangla-speaking children.

Keynote speaker Prof. Thomas Klee and Prof. Stephanie Stokes are delivering their lectures.

The department, currently chaired by Professor Zainab Faruqui Ali, PhD is guided by the following mission statement:The Department of Architecture prepares tomorrow’s professionals and critical thinkers driven by the development ethos of responsive, resilient, and inclusive built environment. With this mission, the Department pursues multidisciplinary knowledge and skills aimed at cultivating comprehensive aesthetic, technological, socioeconomic, and global understanding to meet environmental and developmental challenges. Its pedagogy aspires to empower students to realize their full potential for serving local and global communities in need of improved conditions.

The department, currently chaired by Professor Zainab Faruqui Ali, PhD is guided by the following mission statement:

The Department of Architecture prepares tomorrow’s professionals and critical thinkers driven by the development ethos of responsive, resilient, and inclusive built environment. With this mission, the Department pursues multidisciplinary knowledge and skills aimed at cultivating comprehensive aesthetic, technological, socioeconomic, and global understanding to meet environmental and developmental challenges. Its pedagogy aspires to empower students to realize their full potential for serving local and global communities in need of improved conditions.

Zero Carbon Prefabricated Bamboo Buildings

ArchKIDS

Angan Lecture Series

SoAD Alumni Highlights

Faculty and Research staff

Zainab Faruqui Ali, PhD

Iftekhar Ahmed, PhD

Mohammad Faruk, PhD

Mohammad Habib Reza, PhD

Md Rashed Bhuyan, PhD

Sajid Bin Doza, PhD

Sheikh Rubaiya Sultana

Dilruba Ferdous Shuvra

Saiqa Iqbal Meghna

Imon Chowdhooree

Abul Fazal Mahmudun Nobi

Tanjina Khan

Shams Mansoor Ghani

S M Kaikobad

Tasfin Aziz

Md. Samiur Rahman Bhuiyan

Mohammad Zillur Rahman

Shayeeka Binte Alam

Tanveer Ahamed Bin Ali Naser

Emmat Ara Khanam Ema

Mahia Mustary Nushin

Mahmudul Islam Chowdhury

Fardin Rahman

News and Events

Workshop on Architecture Model Making Held at BRAC University

BRAC University architecture team recognized at international design competition

Students of The Department Architecture Excel at AIUB Architecture Design Charrette

Professor Fuad Hassan Mallick and Professor Zainab Faruqui Ali Speak at Bengal Institute’s Lecture Series on Exploring Muzharul Islam

Exploring New Contextualism 2.0: An Approach to Architecture and Urban Design

Introduction to Virtual Reality in Architecture

Exploring New Contextualism: An Approach to Architecture and Urban Design

Remembering Professor Shaheda Rahman

For official statements and responses from Brac University please contact communications[at]bracu.ac[dot]bd.

Dr David Dowland [Details]Registrar & Controller of Examinations (acting)

Heads of the Departments/Schools/Institutes/Centre

Kashmery Khan is a dedicated biotechnologist who holds a faculty member position in the mathematics and natural sciences department at BRAC University. She earned her bachelor's degree in biotechnology and genetic engineering from Khulna University. She completed her postgraduation with merit in biotechnology at BRAC University by achieving the highest distinction. Her dynamic and passionate nature in research convinced her to join a unique master's program in molecular biology at the University of Hamburg. In recognition of her exclusive research contribution, she has been awarded prestigious Merit scholarship by the Ministry of Science, Germany. Her strong academic record and extensive research experiences prepare her to be a skillful academician with multidimensional thoughts.

Level: 4, Room: 4G45Kha-224 Merul BaddaDhaka 1212. Bangladesh

2018 - 2021

Master in molecular plant science, Universität Hamburg

Thesis title: Elucidating the role of a novel peroxisome-targeted Arabidopsis homolog of NDR1 in abiotic stress responses by reverse genetics and recombinant protein analyses

2012 - 2015

Master in biotechnology, BRAC University

Thesis title: Establishment of transformation protocol and regeneration potential comparison following introduction of antiporter gene in peanut

2007 - 2011

Bachelor in biotechnology and genetic engineering, Khulna University

Thesis title: Identification of potential reservoir host of Avian Influenza virus in Bangladesh

Journals

Rakib-Uz-Zaman, S. M., Iqbal, A., Mowna, S. A., Khanom, M. G., Al Amin, M. M., & Khan, K. (2020). Ethnobotanical study and phytochemical profiling of Heptapleurum hypoleucum leaf extract and evaluation of its antimicrobial activities against diarrhea-causing bacteria. Journal of Genetic Engineering and Biotechnology, 18(1). https://doi.org/10.1186/s43141-020-00030-0

July 2016 – Present (study leave)

Lecturer of biotechnology, BRAC University

Taught subjects: industrial biotechnology, bioprocess technology, virology oncology, Immunnology, DNA fingerprinting, GMOs and biosafety

January 2013 - July, 2016

Teaching assistant, biotechnology program, BRAC university

Key responsibilities: assistance in conducting senior level laboratory classes and training up students about laboratory techniques and maintaining biosafety rules and regulations

January 2012 - December, 2013

Part-time lecturer, department of Pharmacy, Dhaka international university

Taught subjects: Biochemistry, physiology and nutraceuticals

Syeda Fahria Hoque Mimmi, Kashmery Khan and Zubaida Marufee Islam, 2017, Prospects of Milk Protein as a Potential Alternative of Natural Antibiotic, International conference on Biotechnology in Health and Agriculture, December 29-30, Dhaka, Bangladesh.

Kashmery khan and Aparna Islam, 2017, How Our Young Generation Perceive GM Food Crop, Proceeding of the 5th South Asian Biosafety Conference, September 11, Bangalore, India

Z. M. Islam, K. Khan, M. Kibtia, S. Tabrejee, F. N.Rashid, 2017, Determination of antibacterial activity of Pimpinella anisum against commonly found pathogens in Bangladesh, 30th annual conference of Bangladesh society of microbiology, Dhaka.

K. Khan and A. Islam, 2017, Establishment of Transient Transformation Protocol Along with Comparison of Regeneration Potential Following Introduction of Antiporter Genes in Peanut (Arachis hypogaea L.), Proceeding of the 3rd International Conference on Multidisciplinary Innovation In Business Engineering Science & Technology, Scopus indexing conference proceedings, 5-6 April, Bangkok, Thailand.

R. T. Rudhy, K. Khan and Z. M. Islam, 2017, Isolation of Pathogenic Organisms from Meat, Proceeding of the 3rd International Conference on Multidisciplinary Innovation In Business Engineering Science & Technology, 5-6 April, Bangkok, Thailand.

K. Khan, Z. M. Islam, M. Giasuddin and M. Kibtia. 2017. Identification of Potential Reservoir Host of Avian Influenza Virus from Two Selected Districts in Bangladesh. Proceeding of the 1st International Congress by GCSTMR, Dhaka, Bangladesh.

A. Islam, S. Chowdhury, K. Khan; S. Rafiq; and I. M. Chowdhury. 2014. Preparing for transgenesis in a cleisotogamous crop: peanut (Arachis hypogaea L.). Proceeding of the

2nd Annual South Asia Biosafety Conference, Colombo, Sri Lanka, 15-16 September, 2014.

K. Khan, S. Rafiq, I.M. Chowdhury and A. Islam. 2014. Analysis of factors influencing successful Agrobacterium-mediated genetic transformation in three peanut (Arachis hypogaea L.) varieties from Bangladesh. Proceeding of the 7th International Plant Tissue Culture & Biotechnology Conference, Dhaka, Bangladesh, and 1-3 March, 2014.

N. Jahan, S. Chowdhury, K. Khan and A. Islam. 2015. Transformation of peanut (Arachis hypogaea L.) with salt tolerant gene (OsNHX1) following optimization of transformation parameters. Proceeding of the Annual Plant Tissue Culture & Biotechnology Conference, Cox's Bazar, Bangladesh, 19 November, 2015.

S. Chowdhury, K. Khan and A. Islam. 2013. Determination of an efficient regeneration protocol for peanut (Arachis hypogaea L.). Proceeding of the Annual Botanical Conference 2012, Chittagong, 5 January 2013.

• Merit Scholarship awardee, University of Hamburg, Ministry of Science, Germany• Vice Chancellor’s gold medal achievement for the highest distinction in master’s program at BRAC University.• NST research fellowship from ministry of science and technology, Government of the people’s republic of Bangladesh in the year 2013-14.• 100% Tuition Fee waiver at BRAC University for scoring 4 out of 4 in each semester of master’s program• Merit-based Scholarship, Khulna University, Bangladesh (Excellent performance in bachelor study)

Recombinant protein production, molecular cloning, plant tissue culture, Arabidopsis phenotyping, fluorescence microscopy, chromatographic techniques, enzymatic assays, antibacterial and phytochemical screening

Kashmery Khan

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get

Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

The I.T Department plays a pivotal role in the management of automation related activities of BRAC University. Many student information related activities including automated admissions management, student pre-advising related activities, online payments through selected banks, finance and accounts management, HR management activities, procurement related activities are automated for the convenience of both students and administration.

The I.T Department is staffed by tech-savvy personnel who provide prompt service to students, faculty members and staff of BRAC University who use computers. Also the department is continuously striving to improve the services by introducing and developing new innovative ideas.

The entire I.T activity is centered among a data centre with a host of servers, switches and other connectivity devices, which effectively connects the eight campus buildings using Fibre Optic cables, spread around the premises. There is a NOC manned by skilled technical personnel who are continuously monitoring the network, bandwidth availability, system functions, mail services and other IT communications related activities.

BRAC University is one of the very few private universities to have obtained connectivity with BdREN which in future shall be connected to TEIN (Trans Eurasian Information Network) to facilitate research collaborations.

BRAC University has a website available at www.bracu.ac.bd which has a very modern look and helps in disseminating various information related to the University.

A new cloud based DR site for hosting crucial operational data and application software has already been commissioned, which shall provide data security and continued operation using geographically separated DR site premises in case disaster strikes the primary site.

Network account management for faculty, staff and students

USIS, BRACU Mobile apps, Finance and Accounts management software support, HR Management software support, BRACU web portal, domain control, email services etc.

Multimedia Projector and projection screen support, Sound system equipment, Video conferencing

Classroom and lab software installation support, Classroom information and system support

Computer & other IT equipment installation and commissioning, decommissioning, computer hardware and software provisioning, Computer maintenance support, Standard software licenses, Network Printing

Email service - faculty & staff, alumni, students, TSR, Mailing lists/group

IP phone service through LANBRACU wireless networkNetwork connectivity & support

Bandwidth management from ISP, monitoring and maintenance.

Classroom and lab software installationClassroom information and supportIT trainingLive event support

About the Department

Following are services and supports provided by I.T

Download MP3

Dr. M. Mahboob Hossain joined the Microbiology Program of the Department of Mathematics and Natural Sciences of BRAC University as Associate Professor in July 2010. Prior to that he was a full-time faculty in the Department of Pharmacy, Northern

University Bangladesh for six months and in the Department of Pharmacy of University of Asia Pacific for 10 years. Before joining University of Asia Pacific, Mahboob worked as an Assistant Professor of Microbiology at Gonobishyabidyalay, Savar, Dhaka for about one year and six months. He also worked as a Research Fellow of Bangladesh Centre for Advanced Studies for one year. After completion of Masters Mahboob Hossain worked as an Assistant Microbiologist at the Organon Bangladesh Ltd. (a multinational pharmaceutical company of the Netherlands origin) for three and half years. He completed his Ph.D. from Shinshu University School of Medicine, Japan in 1996.

Level: 5, Room: 5B13Kha-224 Merul BaddaDhaka 1212. Bangladesh

• Ph.D., Shinshu University School of Medicine, Japan, 1996• M.Sc., Microbiology, University of Dhaka, 1983• B.Sc. (Hons.), Botany, University of Dhaka, 1982

Journals

Moin, A. T., Rani, N. A., Patil, R. B., Robin, T. B., Ullah, M. A., Rahim, Z., Rahman, M. F., Zubair, T., Hossain, M., Mollah, A. K. M. M., Absar, N., Hossain, M. M., Manchur, M. A., & Islam, N. N. (2024). In-silico formulation of a next-generation polyvalent vaccine against multiple strains of monkeypox virus and other related poxviruses. PLOS ONE, 19(5). https://doi.org/10.1371/journal.pone.0300778

Zaman, I., Turjya, R. R., Shakil, M. S., Al Shahariar, M., Emu, M. R. R. H., Ahmed, A., & Hossain, M. M. (2024). Biodegradation of polyethylene and polystyrene by Zophobas atratus larvae from Bangladeshi source and isolation of two plastic-degrading gut bacteria. Environmental Pollution, 345, 123446. https://doi.org/10.1016/j.envpol.2024.123446

Ahmed, N., Islam, M. A., Hossain, M. M., & Kabir, Y. (2024). XRCC1 and XPD polymorphisms: Clinical outcomes and risk of prostate cancer in Bangladeshi population. Molecular Biology Reports, 51(1), 893. https://doi.org/10.1007/s11033-024-09707-y

Islam, K., Tehjib Meghla, Y., Akhtaruzzaman, M., Shamsur Rouf, A. S., Hossian, M. S., &

Hossain, M. M. (2024). Quantitative determination of artificial sweeteners and sucrose in energy drinks and mango juice available in Dhaka city. Italian Journal of Food Safety, 13(1). https://doi.org/10.4081/ijfs.2024.10914

Ahmed, A., Naboni, N. S., Nowshen, A., Sultana, S., Haque, F., & Hossain, M. M. (2022). A review on the causative agents, risk factors and management of ventilator-associated pneumonia: South Asian perspective. Journal of Medicine (Bangladesh), 23(2), 151–161. https://doi.org/10.3329/jom.v23i2.60633

Aftab, I., Ahmed, A., Mumu, S., & Hossain, M. M. (2022). Management strategy for control and prevention of SARS-CoV-2 infection in hospital settings - a brief review. IMC Journal of Medical Science, 1–7. https://doi.org/10.55010/imcjms.16.016

Islam, T., Nusrat, F., Islam, M. K., & Hossain, M. M. (2022). Anorectal malformation diverted with transverse colostomy associated with recurrent urinary tract infections. Journal of Pediatric Surgery Case Reports, 79. https://doi.org/10.1016/j.epsc.2022.102235

Hossain, M. A., Hossain, M. M., & Begum, N. (2022). Antimicrobial susceptibility patterns of bacterial isolates from routine clinical specimens of a tertiary hospital in Bangladesh. IMC Journal of Medical Science, 16(1), 1–9. https://doi.org/10.55010/imcjms.16.004

Ashreen, S., Ahmed, A., Hasan, N., Akhtar, W., & Hossain, M. M. (2020). Isolation and identification of gram-negative bacteria from oral cancer site infections and study of their antibiotic resistance pattern. Bangladesh Journal of Microbiology, 36(2), 85–90. https://doi.org/10.3329/bjm.v36i2.45533

Rahman, S. S., & Hossain, M. M. (2019). Gulshan Lake, Dhaka City, Bangladesh, an onset of continuous pollution and its environmental impact: a literature review. Sustainable Water Resources Management, 5(2), 767–777. https://doi.org/10.1007/s40899-018-0254-4

Tabrejee, S., & Hossain, M. M. (2019). In silico approach to design a B-cell epitope based vaccine target against yellow fever virus. Bangladesh Journal of Microbiology, 35(1). https://doi.org/10.3329/bjm.v35i1.39801

Khan, U., Afsana, S., Kibtia, M., Hossain, M., Choudhury, N., & Ahsan, C. R. (2019). Presence of blactx-m antibiotic resistance gene in lactobacillus spp. Isolated from hirschsprung diseased infants with stoma. Journal of Infection in Developing Countries, 13(5), 426–433. https://doi.org/10.3855/jidc.10968

Mumu, S. K., & Hossain, M. M. (2018). Antimicrobial activity of tea tree oil against pathogenic bacteria and comparison of its effectiveness with eucalyptus Oil, Lemongrass Oil and conventional antibiotics. American Journal of Microbiological Research, 6(3), 73–78. https://doi.org/10.12691/ajmr-6-3-2

Mahjabeen, F., Khan, S., Choudhury, N., Hossain, M. M., & Khan, T. (2018). Isolation of cellulolytic bacteria from soil, identification by 16S rRNA gene sequencing and characterization of cellulase. Bangladesh Journal of Microbiology, 33(1–2). https://doi.org/10.3329/bjm.v33i1.39598

Rahman, S. S., Promon, S. K., Kamal, W., Hossain, M. M., & Choudhury, N. (2018). Bioethanol production using vegetable peels medium and the effective role of cellulolytic bacterial (Bacillus subtilis) pre-treatment [version 2; referees: 2 approved]. F1000Research, 7. https://doi.org/10.12688/f1000research.13952.2

Amin, R., Rahman, S. S., Hossain, M., & Choudhury, N. (2018). Physicochemical and microbiological qualities' assessment of popular Bangladeshi mango fruit juice. Open Microbiology Journal, 12, 135–147. https://doi.org/10.2174/1874285801812010135

Nasir, A., Rahman, S. S., Hossain, M. M., & Choudhury, N. (2017). Isolation of saccharomyces cerevisiae from pineapple and orange and study of metal's effectiveness on ethanol production. European Journal of Microbiology & Immunology, 7(1), 76–91. https://doi.org/10.1556/1886.2016.00035

Aditi, F. Y., Rahman, S. S., & Hossain, M. M. (2017). A study on the microbiological status of mineral drinking water. Open Microbiology Journal, 11, 31–44. https://doi.org/10.2174/1874285801711010031

Islam, T., Choudury, N., Hossain, M. M., & Khan, T. (2017). Isolation of amylase producing bacteria from soil, identification by 16S RNA gene sequencing and characterization of amylase. Bangladesh Journal of Microbiology, 34(1). Retrieved from https://www.banglajol.info/index.php/BJM/article/view/39224

Hossain, M., Biswas, S., & Marufee, Z. (2016). Study of some commonly prescribed antibiotics against some infectious diseases in Bangladesh. International Journal of Pharmaceutics and Drug Analysis, 4(4), 165–171. Retrieved from http://www.ijpda.com/admin/uploads/0Fh2jq.pdf

Rahman, S., Hossain, M. M., & Choudhury, N. (2016). Effect of various parameters on the growth and ethanol production by yeasts isolated from natural sources. Bangladesh Journal of Microbiology, 30(1–2). https://doi.org/10.3329/bjm.v30i1-2.28453

Shafa, F., Shahriar, M., Md.Opo, F. A. D., Akhter, R., Hossain, M. M., & Choudhury, N. (2016). Characterization of phytoconstituents, in vitro antioxidant activity and pharmacological investigation of the root extract of typhonium trilobatum. International Journal of Pharmaceutical Sciences and Research, 7(4), 1694–1704.

Shihab-us-Sakib, K., Islam, M. A., Moniruzzaman, M., Hussein, A., Hossain, M. M., & Mazid, M. A. (2012). Binding of desloratadine and atenolol with bovine serum albumin and their in vitro interactions. Ars Pharmaceutica, 53(4), 21–27.

Akter, F., Hossain, M. M., Rahman, A., Shaha, M., & Amani, A. E. A. A. A. (2012). Antimicrobials resistance pattern of escherichia coli collected from various pathological specimens. Pakistan Journal of Biological Sciences, 15(22), 1080–1084. https://doi.org/10.3923/pjbs.2012.1080.1084

Shahriar, M., Mawla, S., Bhuiyan, M., & Hossain, M. (2012). Study of the effect of Sodium Dodecyl Sulfate (SDS) and acridine orange on the isolation of plasmid and antimicrobial resistance pattern of clinical isolates of Klebsiella Sp. Daffodil International University Journal of Science and Technology, 7(1). https://doi.org/10.3329/diujst.v7i1.9646

Shahriar, M., Halder, L., Nushrat, N., Hossain, M., Kabir, A. N. M., & Kabir, S. (2010). Comparison of bactericidal activity of serum collected from typhoid patients and normal human against Salmonella typhi at various incubation time. Dhaka University Journal of Pharmaceutical Sciences, 9(1), 65–67. https://doi.org/10.3329/dujps.v9i1.7436

Kabir, F., K, N., Sadat, A., Hossain, M., & Mazid, M. (2010). Interaction of palmitic acid with losartan potassium at the binding sites of bovine serum albumin. Ars Pharmaceutica, 511, 28–36.

Hossain, M. M., Jahan, F. N., Rahman, M. S., Rahman, M. M., Gibbons, S., Masud, M. M., Rashid., M. A. (2009). Diphenylpropanoids from quisqualis indica linn. and their anti-staphylococcal activity. Latin American Journal of Pharmacy, 28(2), 279–283. Retrieved from http://sedici.unlp.edu.ar/handle/10915/7758

Rabbi, F., Sultana, N., Rahman, T., Al-Emran, H. M., Uddin, M. N., Hossain, M., Ahsan, C. R. (2008). Analysis of immune responses and serological cross reactivities among Vibrio cholerae O1, Shigella flexneri 2a and Haemophilus influenzae b. Cellular & Molecular Immunology, 5(5), 393–396. https://doi.org/10.1038/cmi.2008.49

Jahan, F. N., Rahman, Mohammad S., Hossain, M., & Rashid, M. A. (2008). Antimicrobial activity and toxicity of Quisqualis indica. Oriental Pharmacy and Experimental Medicine, 8(1), 53–58. https://doi.org/10.3742/OPEM.2008.8.1.053

Patoare, Y., Hossain, M., Islam, M., Chowdhury, A., Parveen, S., Hossain, M., & Hasnat, A. (2007). Effect of calcium carbide on rat tissue. Dhaka University Journal of Pharmaceutical Sciences, 6(2), 93–98. https://doi.org/10.3329/dujps.v6i2.682

Hossain, M., Ara, S., & Rahman, M. Z. (2004). Quantitative examination of aerobic bacteria and fungi in locally available antacid suspension and possible contamination by specified bacteria. Pakistan Journal of Biological Sciences, 7(11), 2014–2017. https://doi.org/10.3923/pjbs.2004.2014.2017

Hossain, M. M., Rahman, M., Talukder, M. A. S., Shahriar, M., & Zakaria, M. M. (2002). Prevalence of β-lactamse producing methicillin resistant staphylococcus aureus and antimicrobila sensitivity pattern. Bangladesh Pharmaceuticals Journal, 12(2), 1–5.

Okubo, Y., Hossain, M., Horie, S., Momose, T., Suzuki, J., Isobe, M., & Sekiguchi, M. (1997). Effect of herbal medicine (Sho-seiryu-to and Ryo-kan-kyo-mi-shin-ge-nin-to) on human eosinophil biological function. Phytotherapy Research, 11(7), 485–489. https://doi.org/10.1002/(SICI)1099-1573(199711)11:7<485::AID-PTR133>3.0.CO;2-N

Horie, S., Okubo, Y., Hossain, M., Sato, E., Nomura, H., Koyama, S., Sekiguchi, M. (1997). Interleukin-13 but not interleukin-4 prolongs eosinophil survival and induces eosinophil chemotaxis. Internal Medicine, 36(3), 179–185. https://doi.org/10.2169/internalmedicine.36.179

Okubo, Y., Hossain, M., Horie, S., Momose, T., Suzuki, J., Isobe, M., & Sekiguchi, M. (1997). Inhibitory effects of theophylline and procaterol on eosinophil function. Internal Medicine, 36(4), 276–282. https://doi.org/10.2169/internalmedicine.36.276

Horie, S., Okubo, Y., Hossain, M., Momose, T., Suzuki, J., Isobe, M., & Sekiguchi, M. (1997). Intercellular adhesion molecule-1 on eosinophils is involved in eosinophil protein X release induced by cytokines. Immunology, 90(2), 301–307. https://doi.org/10.1046/j.1365-2567.1997.00146.x

Hossain, M., Okubo, Y., Horie, S., & Sekiguchi, M. (1996). Analysis of recombinant human tumour necrosis factor-alpha-induced CD4 expression on human eosinophils. Immunology, 88(2), 301–307. https://doi.org/10.1111/j.1365-2567.1996.tb00019.x

Hossain, M., Okubo, Y., Yamaguch, S., Hone, S., Itoh, S., Sugii, S., Sekiguchi, M. (1996). Immunological analysis of organized pneumonia with eosinophilic pleural effusion. International Archives of Allergy and Immunology, 111, 195–198. https://doi.org/10.1159/000237368

Okubo, Y., Hossain, M., Kai, R., Sato, E., Honda, T., Sekiguchi, M., Takatsu, K. (1995). Adhesion molecules on eosinophils in acute eosinophilic pneumonia. American Journal of Respiratory and Critical Care Medicine, 151(4), 1259–1262. https://doi.org/10.1164/ajrccm.151.4.7697263

Okubo, Y., Sato, E., Hossain, M., Ota, T., Yoshikawa, S., & Sekiguchi, M. (1995). Periodic angioedema with eosinophilia: Increased serum level of interleukin 5. Internal Medicine, 34(2), 108–111. https://doi.org/10.2169/internalmedicine.34.108

Yamaguchi, S., Okubo, Y., Hossain, M., Fujimoto, K., Honda, T., Kubo, K., Takatsu, K. (1995). IL-5 predominant in bronchoalveolar lavage fluid and peripheral blood in a patient with acute eosinophilic pneumonia. Internal Medicine, 34, 65–68. https://doi.org/10.2169/internalmedicine.34.65

Hossain, M., Okubo, Y., & Sekiguchi, M. (1994). Eosinophil viability-enhancing activity in mite-sensitive bronchial asthma. Internal Medicine, 33(9), 529–535. https://doi.org/10.2169/internalmedicine.33.529

Hossain, M., Okubo, Y., & Sekiguchi, M. (1994). Effects of various drugs (staurosporine, herbimycin A, ketotifen, theophylline, FK506 and cyclosporin A) on eosinophil viability. Japanese Journal of Allergeology, 43, 711–717.

Hossain, M. M., Choudhury, N., & Islam, M. S. (1990). Genetic improvement of industrial microorganisms: induction of further mutation in Aspergillus niger strain 14/20 for higher yields of citric acid. Bangladesh Journal of Microbiology, 7(1), 37–41.

July 2016 -Till Date : Professor, Microbiology Program, Department of Mathematics and Natural Sciences, BRAC University, 66 Mohakhali, Dhaka-1212.July, 2010 –June 2016 : Associate Professor, Microbiology Program, Department of Mathematics and Natural Sciences, BRAC University, 66 Mohakhali, Dhaka-1212.Jan, 2010 –June 2010 : Associate Professor, Department of Pharmacy, Northern University Bangladesh, 93, Kazi Nazrul Islam Avenue, Kawran Bazar, Tejgoan, Dhaka-1215; Since January 2010- till June 4, 20102005-2009 : Associate Professor, Department of Pharmacy, The University Asia Pacific, House # 73, Road# 5A, Dhanmondi, Dhaka-1209, Bangladesh; Since April 2005- till January 6, 20102004-2005 : Post doctoral fellow, Department of Biochemistry, Niigata University of Pharmacy and Applied Life Sciences, Niigata, JAPAN; Since April 2004 to March 2005 1999-2004 : Assistant Professor, Department of

Pharmacy, The University Asia Pacific, House # 73, Road# 5A, Dhanmondi, Dhaka-1209, Since October 1999 to March 2004 1998-1999 : Assistant Professor, Department of Microbiology, Gono-Bishwabiyalay (Gono University), Savar, Nayerhat, Dhaka-1350, Bangladesh; Since April 20, 1998 till September 1 st 1999.1997-1998 : Research Fellow, Bangladesh Center for Advanced Studies, House No. 620, Road No. 10 A, Dhanmondi, Dhaka-1209, Since April 1997 to March 1998. 1988-1991 : Assistant Microbiologist, Organon (Bangladesh) Limited (a multinational company of Netherland origin), 97-98, Tongi Industrial Area, Gazipur (Currently known as Nuvista), Bangladesh; Since July 1988 to December 1991. Of the 18 years of teaching experiences Mahboob Hossain worked as Assistant Professor for six years (excluding one year as a post-doctoral research fellow),nine years as Associate Professor and two years as Professor.

General Microbiology (MIC-101)Environmental Microbiology (MIC-203)Agricultural Microbiology (MIC-303)Food Microbiology (MIC-302)Microbial Fermentation (BTE-302)Introduction to Biology (BIO-101)Biophysical Chemistry (BCH-102)Basic Biochemistry (BCH-101)Human Physiology (BCH-201)Basic Immunology (BCH-301)Medical Microbiology (MIC-204)Inorganic Pharmacy I (PHR-101)Pharmaceutical Microbiology (MIC-306)Principles of Chemistry (CHE-110)Medical and Diagnostic Microbiology (MIC-406)Analytical Microbiology (MIC-402)Fermentation Technology (MIC308)Immunopathology and Vaccine Development (MIC309)Advanced Immunology (BTE-405)Bacterial Pathogenesis and Molecular Epidemiology (MIC-407)

Attended and published several abstracts and evaluated the posters presented in The Thirtieth Annual Conference of the Bangladesh Society of Microbiologist, held on 29 April, 2017 at Nawab Ali Chowdhury Senate Bhaban,University of Dhaka,Bangladesh. The theme of the conference was “Microbiology Education and Research for Sustainable Development."

Attended the “National Conference on Diagnosis of Genetic Disorders in Bangladesh: Present Situation and Future Directions” held on April 15, 2017 organized by ideSHi

Financial year

Title of the project

2016-2017

Production of industrially important cellulolytic enzymes from microbes isolated various natural sources and study of their activities.

2014-2015

Isolation of ethanol producing yeast strains from various natural sources and production of ethanol using molasses and kitchen wastes as substrates.

2010-2011

Study of the microbiological status of food and drinks available in mobile restaurants and footpath of Dhaka city.

2005-2006

Study of the Prevalence, Antimicrobial Resistance pattern and Plasmid Profile of Methicillin Resistant Staphylococcus aureus in the clinical samples

2002-2003

Antibacterial principles from medicinal plants of Bangladesh

• Executive Member, Bangladesh Society of Microbiologists• Editorial Board Member of the Bangladesh Journal of Microbiology (2016 to 2018)• Life member, Bangladesh Association for the Advancement of Science (BAAS)• Vice President, Graduate Microbiologists Association (GMA)• Executive Committee member, Bangladesh Environmental Movement (BAPA)• Life Member, Japanese Universities Alumni Association Bangladesh (JUAAB)• Life Member, Japanese Universities Alumni Association Bangladesh (JUAAB)

Currently working on the:• isolation and identification of microorganisms present in the air of Dhaka city• Antibiotic stewardship• Frequency and identification of single nucleotide polymorphism in normal person and• Effect of plant extract on cancer• Isolation of polythene degrading bacteria from soilMahboob Hossain has been supervising research projects of several students of both undergraduate and Masters level for the last 18 years.

Mahboob Hossain, PhD

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Ms. Nabila Khan is currently serving the Microbiology Program, Department of Mathematics & Natural Sciences (MNS) as a lecturer. She has had a long journey with the excellence of the Microbiology Program, Brac University starting from being a student, Student Tutor (ST), and Teaching Assistant (TA) to a Lecturer.

Level: 5Kha-224 Merul BaddaDhaka 1212. Bangladesh

Purkaystha A., Megha SS., Khan N., Mati MN., Ahmed A. Isolation, Identification and Efficacy Assessment of Bacteria Degrading Azo Dye & Reactive Dye from River Sludge Sample. In: proceedings of 36 th Annual Conference of the Bangladesh Society of Microbiologists; 2023 December 19-20; Sylhet Bangladesh. Abstract Id: PIB-7Siddiqua I., Tahmid MI., Khan N., Ahmed A. Isolation, Identification, and Characterization of

Keratinase Enzyme-Producing Bacteria, Collected from Tannery Soil. In: proceedings of 36 th Annual Conference of the Bangladesh Society of Microbiologists; 2023 December 19-20; Sylhet Bangladesh. Abstract Id: PIB-6Paul M., Tahiat T., Khan N., Tasnim S., Ahmed A., Haque F. Isolation of multidrug-resistant β-lactamase producing gram-negative bacteria and carbapenem-resistant gene in raw meat, fish and milk sample. [abstract]. In:proceedings of 35 th Annual Conference of the Bangladesh Society of Microbiologists; 2021 December 11; Dhaka Bangladesh. Abstract Id: PACS-14

Ms. Khan has been placed on the “VC’s List” (6 times) and “Dean’s List” (2 times) as a recognition of her undergraduate's outstanding academic performance in the particular semesters.

Investigating the complex dynamics of disease and antibiotic resistance patterns within microbial communities is something she is quite enthusiastic about. Her deep curiosity about the mechanisms underlying the emergence of antibiotic resistance and the related pathogenic processes is what drives her research interest. She is currently conducting her research on the virulence factor and antimicrobial resistance in gram-negative pathogenic bacteria.

Nabila Khan

News & Events

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BS in Biotechnology program (136 credits) started in Summer 2010. Biotechnology involves applications of scientific and engineering principles for processing of materials (organic and inorganic) using biological systems to provide goods & services. Biotechnology is a multi-disciplinary science involving use of different disciplines like biochemistry, microbiology, genetics, chemical and bio-chemical engineering.Rapid advances in the field of biotechnology and their impacts on the socio-economic development of a country calls for development of trained manpower in this discipline. The undergraduate program will provide students with fundamental knowledge, theoretical and practical skills in biotechnology with good foundation in related physical and biological sciences. A student is offered the option of doing double-major with biotechnology as the first major. One can also take biotechnology as a minor.

News Archive >>

News

Biotechnology seminar hosts Bangladeshi scientist who developed high-yielding “Panchabrihi” rice

Orientation held for Summer 2024 biotechnology students

Nanobiotechnology seminar organized by BUSB and Biotechnology Program

The biotechnology program and BUSB organized the inaugural Biotechnology Workshop Series on “Understanding and Designing Biosensors”

BRAC University is the first university from South Asia to be part of the Open Society University Network (OSUN), a global network of educational institutions across the world that are dedicated to advancing knowledge, promoting civic engagement, and expanding access to global higher education.

The Open Society University Network (OSUN) is a groundbreaking model of global higher education. It brings together universities and colleges from around the world to integrate learning and knowledge creation across disciplines and borders. OSUN focuses on the social sciences, humanities, sciences, and arts, both at the undergraduate and graduate levels.

OSUN aims to educate students to address tomorrow’s global challenges, fostering critical thinking and open intellectual inquiry to strengthen the foundations of open society amid the current authoritarian resurgence. OSUN counteracts intellectual monocultures and polarization by uniting institutions around the world in collaborative research projects and by encouraging students to examine issues from different perspectives and through reasoned arguments. In addition, OSUN addresses inequality by expanding educational access to neglected and minority populations, such as incarcerated persons, the Roma, and refugees.

As a member of OSUN, BRAC University gains access to a global network of like-minded institutions committed to academic excellence, social justice, and open societies. This network provides unique opportunities for collaboration, knowledge exchange, and joint research initiatives, empowering our faculty and students to contribute to pressing global challenges. The Following are some of the programs that are currently ongoing at the university with OSUN.

Mahfuza Haque Mahi is a Lecturer of Statistics at Brac University in the Department of Mathematics and Natural Sciences. Mahi completed her Masters and Bachelor of Science from the Department of Statistics, University of Dhaka in 2018 and 2017 respectively. While she was a student, she won several prestigious awards and scholarships from the government for her consistent efforts in academia. Now as an academician she is involved in several research based activities that will help her to learn and grow in the field of data sciences. Her research interest lies in bio-statistics, public health and econometrics.

Level:Kha-224 Merul BaddaDhaka 1212. Bangladesh

• University of Dhaka Masters of Science, 2018Major in StatisticsCGPA 3.93 (out of 4.00)

• University of Dhaka Bachelor of Science, 2017Major in StatisticsCGPA 3.92 (out of 4.00)

• Lecturer, Department of Mathematics and Natural Sciences, Brac University (May 2022 – Present)• Adjunct faculty, Department of Mathematics and Natural Sciences, Brac University (February 2021 - May 2022)

• Elements of Statistics and Probability• Linear Algebra and Fourier Analysis • Remedial Course in Mathematics

• Presented paper in the 2nd International Conference on Applied Statistics at University of Dhaka (December 27-29, 2019)

• Received Dean’s Award from Faculty of Science for outstanding accomplishment in B.S. Honors examination of 2017• Received National Science and Technology Fellowship awarded by the Ministry of Science and Technology of the People’s Republic of Bangladesh for MS thesis for the Fiscal year 2019-2020• Received Samsun Nahar Mahmud Foundation Merit Scholarship Award for academic performance in B.S. Honours Examination 2017• Received scholarship from Dhaka University for academic performance in B.S. Honours Examination 2017• Received S N Mitra scholarship from Department of Statistics, University of Dhaka in 2017• Received Board Merit Scholarship for SSC result in 2011.

• Life member, Alumni Association (DUSDAA), Department of Statistics, University of Dhaka.

• Biostatistics• Public Health• Evidence Based Medicine• Econometrics• Market Research

Mahfuza Haque Mahi

News & Events

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BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

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Dr. Firoze Haque received his PhD in physics from the University of Central Florida (USA) in 2011 for his investigations onsingle-electron transport spectroscopy of magnetic molecules and nanoparticles. He spent over a year as a postdoctoral associate in the department of physics at University of Central Florida before joining BRAC University as an assistant professor in 2013. Dr. Haque's current research focuses on electronic transport measurementsand characterizations of organic and inorganic materials, both magnetic as well as non-magnetic.

Level: 5, Room: 5B04Kha-224 Merul BaddaDhaka 1212. Bangladesh

Journals

Ali, M. M., Hadi, M. A., Rahman, M. L., Haque, F. H., Haider, A. F. M. Y., & Aftabuzzaman, M. (2020). DFT investigations into the physical properties of a MAB phase Cr4AlB4. Journal of Alloys and Compounds, 821, 153547. https://doi.org/10.1016/j.jallcom.2019.153547

Noor, L., Ahsan, G., Rahman, M. L., & Haque, F. (2019). Simulation of electronic properties of a quantum dot in transistor geometry at varying temperatures. Bangladesh Journal of Physics, 25(11).

Haque, F., del Barco, E., Fishman, R. S., & Miller, J. S. (2013). Low temperature hysteretic behavior of the interpenetrating 3-D network structured $[Ru_2(O_2CMe)_4]_3[Fe(CN)_6]$ magnet. Polyhedron, 64, 73–76. https://doi.org/10.1016/j.poly.2013.02.049

Khallaf, H., Chen, C.-T., Chang, L.-B., Lupan, O., Dutta, A., Heinrich, H., … Chow, L. (2012). Chemical bath deposition of SnO_2 and Cd_2SnO_4 thin films. Applied Surface Science, 258(16), 6069–6074. https://doi.org/10.1016/j.apsusc.2012.03.004

Haque, F., Langhirt, M., del Barco, E., Taguchi, T., & Christou, G. (2011). Magnetic field dependent transport through a Mn_4 single-molecule magnet. Journal of Applied Physics, 109(7). https://doi.org/10.1063/1.3560891

Chen, H., Chunder, A., Liu, X., Haque, F., Zou, J., Austin, L., … Huo, Q. (2011). A multifunctional gold nanoparticle/polyelectrolyte fibrous nanocomposite prepared from electrospinning process. Materials Express, 1(2), 154–159. https://doi.org/10.1166/mex.2011.1016

Mackowski, S., Gurung, T., Haque, F., Jackson, H. E., Smith, L. M., Schallenberg, T., … Molenkamp, L. W. (2005). Spatial diffusion of carriers in a quantum dot system grown by shadow mask controlled epitaxy. AIP Conference Proceedings, 772(1), 657–658. https://doi.org/10.1063/1.1994278

Haque, F. H., & Mofiz, U. A. (1999). Study of nonlinear mode in the rotating neutron star magnetosphere plasma. The Bangladesh Journal of Astronomical Research, 2(31).

Conferences

Rahman, A. R., Rahman, M. L., & Haque, F. (2019, February 7-9). Measurement and analysis of temperature dependent resistivity of $Ni_{0.30}Cu_{0.20}Zn_{0.50}Fe_2O_4$ sample using Lab VIEW. In National Conference on Physics – 2019. Dhaka, Bangladesh: Bangladesh Physical Society.

Haque, F., Muthuraman, H., Rahman, M. L., & Hoq, M. (2018, March 8-10). Electronic transport study of carbon nanotube and poly3-hexylthiophene composite in poly (methyl methacrylate) matrix. In International Conference on Physics -2018. Dhaka, Bangladesh: Bangladesh Physical Society.

Shara, J., Mridula, Z. T., Rahman, M. L., Haque, F., & Hossain, A. K. M. A. (2018, March 8-10). Influence of Mn substitution on structural and AC magnetic properties of (Ni0.4Cu0.15Zn0.45)1-xMnxFe2O4. In International Conference on Physics -2018. Dhaka, Bangladesh: Bangladesh Physical Society.

Haque, F., Langhirt, M., Barco, E. del, Taguchi, T., & Christou, G. (2010, November). Magnetic field dependent electronic transport through a Mn4 single-molecule magnet. In 55th Annual Conference on Magnetism and Magnetic Materials. Atlanta, USA.

Haque, F., Langhirt, M., Barco, E. del, Taguchi, T., & Christou, G. (2010, March). Magnetic field dependent electronic transport of Mn4 single-molecule magnet. In American Physical Society Meeting. Portland, USA.

Haque, F., Stokes, P., Zhai, L., & Khondaker, S. I. (2008, March). Temperature dependent charge transport properties of poly(3-hexylthiophene) block poly(styrene) copolymer field-effect transistor. In American Physical Society Meeting. New Orleans, USA.

Haque, F. H., Gurung, T., Mackowski, S., Jackson, H. E., Smith, L. M., Schallenberg, T., … Molenkamp, L. W. (2004, March). Spatial diffusion of carriers in a quantum dot system grown by shadow mask controlled epitaxy. In American Physical Society Meeting2. Montreal, Canada.

Haque, F. H., & Mofiz, U. A. (1999, January). Study of nonlinear mode in the rotating neutron star magnetosphere plasma. In Fifth Chittagong Conference on Mathematical Physics. Chittagong, Bangladesh: Research Center for Mathematical and Physical Sciences (RCMPS), University of Chittagong.

Haque, F. H., & Mofiz, U. A. (1997). Study of nonlinear mode in the rotating neutron star magnetosphere plasma. In International Symposium on Recent Advancement in Physics. Dhaka, Bangladesh: Bangladesh Physical Society.

Md. Firoze H. Haque, PhD

News & Events

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EEE student selected for Huawei program in China

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Workshop on Architecture Model Making Held at BRAC University

Welcome to the Office of the Registrar. This office is responsible for preparing academic scheduling, registration into courses, the creation and maintenance of student academic records, academic transcripts, course information, student status, enrollment verification and graduation.Brac University is committed to prioritizing our students and is customer focused. The mission of this office is to provide students with the best possible service and the most up-to-date and supportive information in an efficient and expeditious manner.

Dr. Mir Mozibor Rahman completed his B. Sc in Applied Mathematics and Informatics in 1994 and later on he completed an MBA in 1996 at Peoples' Friendship University of Russia (PFUR) Moscow. After that he received his Ph. D degree in Economics at the same institution in 2003 for his research on "Reforms and Development Problems of External Economic Links of Bangladesh." He joined the Mathematics Program in the Department of Mathematics and Natural Sciences at BRAC University as an Assistant Professor in Jan 2012. Prior to that he was a full-time faculty member and worked as an Assistant Professor in the Department of Economics at Bangladesh University for around 4 years. Moreover he held the post of the Head of the Department of Economics for last two years. Before joining the Bangladesh University He had a three years teaching experience as TA in the Faculty of Economics at Peoples' Friendship University of Russia, Moscow. Dr. Mir's current research area is "Mathematical models in Economics".

Level: 5, Room: 5B07Kha-224 Merul BaddaDhaka 1212. Bangladesh

• Ph. D (2003) in Economics, Moscow, Russia • MBA (1996), Moscow, Russia• B. Sc (1994) in Applied Mathematics and Informatics, Moscow, Russia

Conferences

Rahman, M. M. (2002). Implementation of structural adjustment programs: Impact on Bangladesh economy. In Materials of scientific conference of young Economists.

Rahman, M. M. (2001). Some features of the foreign trade relations of developing countries in the present context. In Materials of scientific conference of young Economists (p. 12). Moscow: RUDN.

Rahman, M. M. (1999). The basic directions of liberalization of external economic policies of developing countries (With special inquiry into Bangladesh. In Materials of scientific conference for post-graduate students of Economics (p. 12). Russia.

■ He has been involved in BRAC University as an Assistant Professor in the Department of Mathematics and Natural Sciences since January 2012.■ He has been serving as a visiting faculty of MBA Program at BRAC Business School since January 2012.■ He was involved in Bangladesh University as an Assistant Professor in the Department of Economics from April 2008 to January 2012.■ He held the post of the Head of the Department of Economics at Bangladesh University for last two years(2011-2012).■ He has a three years teaching experience as TA in the Faculty of Economics, Peoples' Friendship University of Russia, Moscow

• MAT101(Fundamentals of Mathematics)• MAT105(Calculus)• MAT110(Differential Calculus & Coordinate Geometry)• MAT120(Integral Calculus & Differential Equations)• MAT203(Matrices, Linear Algebra and Differential

Equations)• MAT204(Complex Variables and Fourier Analysis)• MAT215(Complex Variables & Laplace Transformations)• MAT216(Linear Algebra & Fourier Analysis)• MAT091(Basic Course in Mathematics)• MAT501(Mathematics for Decision Making)

The Ph. D thesis Paper was applauded greatly and recommended for publication as a research monograph by the Scientific Council of PFUR and was published with warm support and financial assistance of Russia-Bangladesh Chambers of Commerce and Industries (RBCCI).

Mathematical Economics, Mathematical models in Economics and Econometric modeling.

Mir Mozibor Rahman, PhD

News & Events

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Second day's Parallel Session of International Conference on Business and Management (ICBM 2017) Organized by BRAC Business School, BRAC University, Bangladesh was held on September 22, 2017 at the Westin hotel, Dhaka, Bangladesh. Just like the day one, second day also had prominent academicians as session chairs from local and international arena who have years of experiences in corporate and academic field. Therefore, participants got the chance to present their papers in 3 different slots on 9 distinct tracks. On this day, around 81 papers had been presented by Students, Corporate people and Academicians.

There were 24 chairs altogether in the Second day’s Parallel Session had and all of them played their roles to bring out the best from our presenters. Among all of them some of the names to be mentioned, including Dr. Nazrul Islam (Uttara University, Bangladesh), Dr. Akbar Ali Khan (Former Advisor of Caretaker Government, People’s Republic of Bangladesh and Professor of BRAC Business School), Dr. Naffisah Mohd Hassan (UiTM, Malaysia), Dr. Mahbubur Rahman (NSU, Bangladesh), Dr. Noor’ain Mohamad Yunus, (UiTM, Malaysia), Ms. Aleya Ikbal (VP, Optima Project Management LLC, USA), Prof. Dr. Nurur Rahman (Presidency University, Bangladesh), Dr. Erne Suzila Kassim (UiTM, Malaysia), Prof. Dr. Farid Sobhani (Daffodil International University, Bangladesh), Mr. Manzur H. Khan (AIUB, Bangladesh), Prof. Dr. Sarwar Uddin Ahmed (IUB, Bangladesh), Mr. Monojit Ojha (BRAC University, Bangladesh), Prof. Dr. Nazrul Islam (Uttara University, Bangladesh), Prof. Ramayah Thurasamy (USM, Malaysia), Ms. Shakila Yasmin (IBA, DU, Bangladesh); Prof. Dr. Mohammad Iqbal (SUST, Bangladesh); Mr. N.I.M Alamgir, Mr. Ivan Bari and Ms. Hasina Afroz (BRAC University, Bangladesh).

List of tracks on which papers were presented (along with the numbers):

Finally, the session came to an end with the handover of certificates and crest among ICBM 2017’s Paper Presenters and Session Chairs.

Centre for Peace and Justice (CPJ) is a multi-disciplinary academic institute, which promotes global peace and social justice through quality education, research, training and advocacy. CPJ is committed to identifying and promoting sustainable and inclusive solutions to a wide range of global concerns and issues, including fragility, conflict and violence. The centre was established in March 2017 and joined the impressive cluster of other institutes and centres of Brac University. Under the leadership of its Executive Director, Manzoor Hasan the centre has already achieved vast research, training and advocacy experiences in its three thematic area (1) Peace and Fragility (2) Social

Justice, and (3) Open Society. Read More

For details, please visit: CPJ Website

CPJ on Social Media

CPJ on Social Media

News and Events

In loving memory of Riedwan Habibur Rahman: An ever-smiling man with a heart of gold

Peace Cafe launched as a club at BRAC University

A delegation of CPJ and UNDP DKC Programme team visits Women Peace Café, JKKNIU

Centre for Peace and Justice and Student Life hold ‘Public Lecture on Independent Investigative Mechanism for Myanmar’

10th Asia Pro Bono Conference & Access to Justice Exchange | Saturday, 25 September 2021 | 09:00 to 11:00 [BST]

Celebrating International Day of Peace with Centre for Peace and Justice, Brac University

Worldwide Commission to Educate All Kids (Post-Pandemic)

Online Film Festival to Celebrate World Day of Social Justice

Tushar Ahmed Shishir joined the Department of Mathematics and Natural Sciences at

BRAC University as a lecturer in September 2022. He completed his Master of Science in biotechnology and Bachelor of Science in Pharmacy from BRAC University. In the immediate aftermath of receiving his Master's degree, Tushar worked for the United Nations Population Fund (UNFPA) as a Laboratory Consultant during the COVID-19 pandemic for two years. He is involved in computational genomics, evolutionary biology, genomic epidemiology, water microbiology and bacteriophage research.

Level: 4Kha-224 Merul BaddaDhaka 1212. Bangladesh

Book Chapter

Azam, M. N. K., Shishir, T. A., Khandker, A., & Hasan, M. N. (2022). Value-added products from industrial wastes of phytopharmaceutical industries. In H. Thatoi, S. Mohapatra, & S. K. Das (Eds.), Innovations in Fermentation and Phytopharmaceutical Technologies (pp. 457–489). Elsevier. https://doi.org/10.1016/B978-0-12-821877-8.00002-6

Journals

Shishir, T. A., Jannat, T., & Naser, I. B. (2022). Genomic surveillance unfolds the SARS-CoV-2 transmission and divergence dynamics in Bangladesh. Frontiers in Genetics, 13. https://doi.org/10.3389/fgene.2022.966939

Shishir, T. A., Jannat, T., & Naser, I. B. (2022). An in-silico study of the mutation-associated effects on the spike protein of SARS-CoV-2, Omicron variant. PLoS ONE, 17(4 April). https://doi.org/10.1371/journal.pone.0266844

Naser, I. B., Shishir, T. A., Faruque, S. N., Mozammel Hoque, M., Hasan, A., & Faruque, S. M. (2021). Environmental prevalence of toxigenic Vibrio cholerae O1 in Bangladesh coincides with V. cholerae non-O1 non-O139 genetic variants which overproduce autoinducer-2. PLoS ONE, 16(7 July). https://doi.org/10.1371/journal.pone.0254068

Shishir, T. A., Naser, I. B., & Faruque, S. M. (2021). In silico comparative genomics of SARS-CoV-2 to determine the source and diversity of the pathogen in Bangladesh. PLoS ONE, 16(1 January). https://doi.org/10.1371/journal.pone.0245584

Shishir, T. A., Mahbub, N., & Kamal, N. E. (2019). Review on bioremediation: A tool to resurrect the polluted rivers. Pollution, 5(3), 555–568. https://doi.org/10.22059/poll.2019.272339.558

Introduction to Biology, Biophysical Chemistry, Bioorganic Chemistry

Vice-chancellor gold medal, BRAC University

Computational biology, Genomics epidemiology, Evolutionary biology, Genomics pathogenic bacteria, Bacteriophage–bacteria interaction, Phage therapy and Water microbiology

Tushar Ahmed Shishir

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Masuda Akther is a lecturer in Microbiology program, Department of Mathematics and Natural Sciences, BRAC University. Before joining BRAC University, she worked as Senior Research Assistant in International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b) and was involved in various collaborative research projects related to epidemiological study of cholera outbreaks and development of effective mitigation strategies for the population of Bangladesh. She graduated from University of Dhaka in 2018 and completed MSc from the same institution in 2019. Her thesis was about development of a multiepitope recombinant vaccine candidate against Foot-and-Mouth Disease Virus Serotype A circulating in Bangladesh. Along with Virology, she is also passionate about Immunology, Molecular Biology and Bioinformatics.

Level:Kha-224 Merul BaddaDhaka 1212. Bangladesh

BSc. from Department of Microbiology, University of Dhaka in 2018MSc in Microbiology from the same institution in 2019

Journals

Hoque, M. N., Akter, S., Mishu, I. D., Islam, M. R., Rahman, M. S., Akhter, M., … Hossain, M. A. (2021). Microbial co-infections in COVID-19: Associated microbiota and underlying mechanisms of pathogenesis. Microbial Pathogenesis, 156. https://doi.org/10.1016/j.micpath.2021.104941

Islam, M. R., Hoque, M. N., Rahman, M. S., Alam, A. S. M. R. U., Akther, M., Puspo, J. A., … Hossain, M. A. (2020). Genome-wide analysis of SARS-CoV-2 virus strains circulating worldwide implicates heterogeneity. Scientific Reports, 10(1). https://doi.org/10.1038/s41598-020-70812-6

Rahman, M. S., Islam, M. R., Hoque, M. N., Alam, A. S. M. R. U., Akther, M., Puspo, J. A., … Hossain, M. A. (2020). Comprehensive annotations of the mutational spectra of SARS-CoV-2 spike protein: a fast and accurate pipeline. Transboundary and Emerging Diseases, 68(3), 1625–1638. https://doi.org/10.1111/tbed.13834

Senior Research Assistant International Centre for Diarrhoeal Disease Research, Bangladesh (icddr,b) (March 2021- September 2021)

Bioinformatics (BTE 401), Introduction to Chemistry (CHE 101), Microbiol Lab I (MIC 155), Virology (MIC 301)

I. Talent pool Scholarship from Directorate of Secondary and Higher Education in 2021 II. Dean's award by Faculty of Biological Science, University of Dhaka in 2018III. National Science and Technology (NST) Fellowship, 2018-19.IV. Scholarship from University of Dhaka for academic excellence in Bachelor of Science

Global outreach member of American Society for Microbiology

Masuda Akther

News & Events

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The Department of Computer Science and Engineering strives to inspire excellence in the creation, application and dissemination of knowledge in computer science and engineering through comprehensive educational programs; research in collaboration with industry, NGOs, and government; diffusion through scholarly publications, and service to the national and international community.

Faculty and Research Staff

Mahbubul Alam Majumdar, PhD

Sadia Hamid Kazi, PhD

Mohammad Kaykobad, PhD

Matin Saad Abdullah, PhD

Md. Khalilur Rhaman, PhD

Md. Golam Rabiul Alam, PhD

Amitabha Chakrabarty, PhD

Muhammad Iqbal Hossain, PhD

Md Sadek Ferdous, PhD

Muhammad Nur Yanhaona

Abu Mohammad Khan, PhD

S M Taiabul Haque, PhD

Md. Ashraful Alam, PhD

Farig Yousuf Sadeque

Jannatun Noor Mukta

Aniqua Nusrat Zereen, PhD

Mr. Annajiat Alim Rasel

Md. Tawhid Anwar

Rubayat Ahmed Khan

Md.Saiful Islam

Sifat E Jahan

Moin Mostakim

A.M. Esfar-E-Alam

Md. Imran Bin Azad

Mehnaz Seraj

Shakila Zaman

Mohammed Abid Abrar

Najeefa Nikhat Choudhury

Arif Shakil

Mahrin Tasfe

Narzu Tarannum

Faisal Bin Ashraf

Warida Rashid

Salman Sayeed Khan

Tanvir Rahman

Dewan Ziaul Karim

Farhana Shahid

Trisha Das

Shaily Roy

Fairoz Nower Khan

Mirza Farhan Bin Tarek

Shahnewaz Ahmed

Beig Rajibul Hasan

Mursalin Habib

Shadman Shahriar

Aminul Huq

Jumana

Nadia Rubaiyat

Rafeed Rahman

Saadat Hasan Khan

Rasif Ajwad

Md. Farhan Shadiq

Syeda Ramisa Fariha

Arnisha Khondaker

Nabuat Zaman Nahim

Ismail Hossain

Ramkrishna Saha

Md Tanzim Reza

Zahin Ahmed

Ankan Ghosh Dastider

MD. Fakhruddin Gazzali

Md. Asif Hossain Bhuiyan

Md. Saiful Bari Siddiqui

Mir Sayeed Mohammad

Shoaib Ahmed Dipu

Tasnim Ferdous

Md. Shahrar Fatemi

Md Faisal Ahmed

Md. Shahriar Rahman

Rabeeb Ibrat

A.T.M Mizanur Rahman

Abid Jahan Apon

Avijit Biswas

Nazia Afreen

Mirza MD. Tanjim Shorif Mugdho

Ahashan Habib Niloy

Ayon Sarker

Dibyo Fabian Dofadar

Afia Mubassira Islam

Tamkin Mahmud Tan

Himaddri Roy

Md. Aquib Azmain

Md. Sabbir Ahmed

Md. Ahasanul Alam

Priata Nowshin

Purbayan Das

Riad Ahmed

Shafqat Hasan

Syed Zamil Hasan Shoumo

Md. Tanvir Arafat

Zahin Wahab

Zarin Tasnim Biash

Meem Arafat Manab

Mostafijur Rahman Akhond

Sowmitra Das

Mohammad Zahidul Hasan Zadid

Tanvir Ahmed

Mehedi Hasan Himel

A. K. M. Naziul Haque

Zaber Mohammad

Farhan Feroz

Taufiqul Islam Khan

News and Events

BRACU Alumni-led Startup Shines in Shark Tank Bangladesh

BRAC University Hosts First Coding Competition for Non-CS Students

BUCC holds webinar on “Pathway to Research and Industry in the USA”

CSE Faculty wins Best Presentation Award at ICHMI 2024

BracU IEEE SB Webinar Series: Learn from the Xperts

Project Showcasing In ESSAB "Abishkarer Khoje"

Short Course on Internet of Things (IoT)

Robu Fest 2016

Undergraduate Study

Postgraduate Study

Diploma and Certificates

Dr. Zayed completed his school and college from Government Laboratory High School, Dhaka and Notre Dame College respectively. He finished his BSc. From Bangladesh University of Engineering and Technology (BUET) in 2007 and Perused Ph.D in Nanotechnology from National University of Singapore (NUS) in 2013.

Level: 5, Room: 5B10 Kha-224 Merul BaddaDhaka 1212. Bangladesh

Journals

Badruddoza, A. Z. M., Shawon, Z. B. Z., Rahman, M. T., Hao, K. W., Hidajat, K., & Uddin, M. S. (2013). Ionically modified magnetic nanomaterials for arsenic and chromium removal from water. Chemical Engineering Journal, 225, 607–615. https://doi.org/https://doi.org/10.1016/j.cej.2013.03.114

Badruddoza, A. Z. M., Shawon, Z. B. Z., Tay, W. J. D., Hidajat, K., & Uddin, M. S. (2013). Fe3O4/cyclodextrin polymer nanocomposites for selective heavy metals removal from industrial wastewater. Carbohydrate Polymers, 91(1), 322—332. https://doi.org/10.1016/j.carbpol.2012.08.030

Shawon, Z. B. Z. (2012). Synthesis and characterization of janus magnetic nanoparticles and its application as an adsorbent. Journal of Chemical Engineering, 27(1), 64–68. https://doi.org/10.3329/jce.v27i1.15861

Badruddoza, A., Shawon, Z. B. Z., Tay, D., Hidajat, K., & Uddin, M. S. (2012). Endocrine disrupters and toxic metal ions removal by carboxymethyl-?-cyclodextrin polymer grafted onto magnetic nanoadsorbents. Journal of Chemical Engineering, 27(1), 69–73. https://doi.org/10.3329/jce.v27i1.15862

Conferences

Shawon, Z. B. Z., Badruddoza, A. Z. M., Louis, S. W. M., Hidajat, K., & Uddin, M. S. (2011). Accelerated procedure for synthesizing janus magnetic nanoparticles. In International Conference on Chemical Engineering 2011 (pp. 132–136). Dhaka, Bangladesh: Bangladesh University of Engineering and Technology.

Badruddoza, A. Z. M., Shawon, Z. B. Z., Daniel, T. W. J., Hidajat, K., & Uddin, M. S.

(2011). Carboxymethyl-β-cyclodextrin polymer modified magnetic nanoadsorbents for removal of endocrine disrupter and toxic metal ions. In International Conference on Chemical Engineering 2011 (pp. 30–34). Dhaka, Bangladesh: Bangladesh University of Engineering and Technology.

CHE101 (introduction to chemistry-3 credit), CHE110 (Principles of Chemistry- 3 credit)

1. Zayed Bin Zakir Shawon, Abu Zayed Md. Badruddoza, Soh Wei Min Louis, Kus Hidajat, Mohammad Shahab Uddin. Accelerated procedure for synthesizing janus magnetic nanoparticles, International Conference on Chemical Engineering 2011, ICChE 2011, Pages: 132-136, December, Dhaka, Bangladesh.2. Abu Zayed Md. Badruddoza, Zayed Bin Zakir Shawon, Tay Wei Jin Daniel, Kus Hidajat, Mohammad Shahab Uddin. Carboxymethyl-β-cyclodextrin polymer modified magnetic nanoadsorbents for removal of endocrine disrupter and toxic metal ions, International Conference on Chemical Engineering 2011, ICChE 2011, Pages: 30-34, December, Dhaka, Bangladesh.

Chemical Engineering Alumni Merit Award-2004

Nanoadsorbent, Catalysis, fluid mechanics, Corrosion Engineering

Zayed Bin Zakir Shawon, PhD

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Entrepreneurship plays an essential role in ensuring a strong resource for economic development and it is the key to high growth of economy in a country. It is also considered as the root of our economy. With a growing economy like Bangladesh, having an increasing youth population, entrepreneurs will be a key driving force to meet the needs of the industry and economy. It has the ability to foster growth and alleviate poverty. One of the major problems of entrepreneurship development in our country is the absence of the entrepreneurial skills in majority of the entrepreneurs. In line with this, BRAC University, first time in Bangladesh, came-up with an idea to establish a centre for entrepreneurship development with a commitment to contribute to the development of entrepreneurship in Bangladesh.Centre for Entrepreneurship Development (CED) started its journey in April 2011 with the view to encourage Bangladeshi entrepreneurs and engender entrepreneurial knowledge and skill so that they can develop and grow their own businesses. CED provides a platform for new, small and medium – scale enterprise through skills acquisition activities for the development and management of the enterprise. To understand the enterprise better CED emphasizes on research that will contribute to entrepreneurial development in Bangladesh, advance education and skill acquisition along this line, and popularize the idea of entrepreneurship.

Mission of the CED

"Helping people help themselves" through innovation and entrepreneurship

Function of CED

News Archive >>

Newsletter Archive >>

News

BRAC Business School Hosts Closing Ceremony for ‘Uddami Ami’ Women Entrepreneurship Training Program

BBS Faculty Publishes Research Paper in International Journal of Hospitality Management

Dr. Mohammad Shahidul Islam Brings New Light to Hospitality Studies with Innovative Research Method Video Reflexive Ethnography (VRE)

BRAC University students get Inspiring Women Award

Newsletter

Brac University follows a model of higher education consisting of semesters, courses, credit hours, continuous evaluation and letter grading.

Keeping with its mission and goals, the University strives to ensure high academic standards by implementing well-designed curricula, carefully selecting high quality students and faculty, utilizing modern and effective instructional methods and aids and by continuous monitoring and rigorous evaluation of all the pertinent activities and systems.

There are three equal semesters-Spring, Fall, and Summer. The duration of each semester is 13 working weeks. An additional week for each semester is allocated for final exams. Usually the Spring semester commences in January, the Summer semester in May and the Fall semester commences in September. The Semester Calendar provides deadlines for registration add/drop courses, withdrawals etc. This has financial and academic implications.

Each course is identified by a two-part numbering system. The first part with three alphabets represents the subject area and the second part refers to the level of the course as follows:

The series 100, 200, 300 and 400 numbers are intended to indicate progressively more demanding content of the course and correspondingly increasing competence on the part of the students enrolled in the course. For example, ACT 301 Intermediate Accounting is a third year course and it is assumed that a student registering in this course has already attended one or more second year (200 levels) courses in Accounting. A student may register in this course before third year if he has already fulfilled the prerequisites for the course. The course may be taken after third year also when the student has fulfilled the prerequisites. Courses numbered 001-099 are remedial or non-credit courses.

Students enrolled in the undergraduate programs of BRAC University are classified as Freshman, Sophomore, Junior, Senior and Advanced Senior based on the number of credit hours earned towards a degree. The basis and classification are as follows:

A student who registers for 12 credits or more is considered to be a full time student. A full-time student taking 12 credits in each semester will be able to complete the program in less than four years. Fee waivers, scholarships and academic honors are considered for full time students only.

Maximum period to complete the undergraduate program is eight years from the date of first admission to the university and minimum period to complete the undergraduate program is three years.

Students take courses each semester and these courses have credits assigned to them and the credits are counted towards the degree. Credit hours for a course are assigned on the basis of a 13-week semester. One (1) credit hour means that the course meets for 50 minutes in a class each week; Three (3) credits mean that the class will meet twice a week for 80 minutes in each session. The tutorial/lab/workshop sessions meets for 100 - 150 minutes each week. Two (2) credit courses mean that the course meets twice every week for 50 minutes in each class.

A student should take 4 or more courses per semester and must take at least 9 credits per semester to maintain full time student status. Full-time students of Law and

Architecture are required to take a minimum of 12 credits per semester and full-time students of Pharmacy are required to take a minimum of 18 credits per semester. With the approval of the Dean or Chair of Department, a student may take fewer than 12 credits per semester (for Law and Architecture) and 18 credits per semester (for Pharmacy) subject to completing the programme of study within the maximum registration period; the student will then be classified as part-time for that semester only.

The maximum period to complete the undergraduate program is eight years from the date of first admission to the university and the minimum period to complete the undergraduate program is three years (with the exception of the Bachelor of Architecture degree). The maximum period to complete the Bachelor of Architecture degree is ten years from the date of first admission to the University; a student may not complete the Bachelor of Architecture degree in less than the five-year standard period.

In addition to that, the working group approved the following regulations on course load.

For freshman (1st Semester):

For students in 2nd Semester and onward

All Programs (except Architecture and Pharmacy)

i. Minimum Course Load: Nine (9) credits

ii. Maximum Course Load: Fifteen (15) credits

iii. However, in special cases a student may register for eighteen (18) credits or more (see below). The applicability of the exception will be allowed only if recommended by the respective academic department. The advisor of the student in the respective program may propose the additional credits as a special case and the student may then take approval from the Dean/ Chairperson. The special case then needs to be approved by the Registrar.

Architecture Program

i. Minimum Course Load: Nine (9) credits

ii. Maximum Course Load: Twenty-One (21) credits with studio course

Pharmacy Program

iii. Minimum Course Load: Eighteen (18) credits

iv. Maximum Course Load: Twenty-Four (24) credits

A student may be granted leave of absence or semester drop for a period of up to three consecutive semesters or one academic year, subject to the student meeting academic requirements. This arrangement does not apply to students who received academic probation or who have been expelled from the university on disciplinary grounds or excluded on academic grounds. The decision to grant leave of absence will rest with the Dean or Head of Department in consultation with the Registrar. A student granted leave of absence must register in the following semester immediately after the expiry of the leave period.

The Registrar may withdraw a student who does not register for two consecutive semesters without prior approval. The student will be entitled to apply for a transcript of any credits obtained subject to paying any outstanding fees or other dues to the university. The Registrar will inform the school/department of the withdrawal.

Any student so withdrawn must apply for readmission as a new student if s/he wishes to return to the university. The applicant will write to the Registrar who, together with the Chair of the Admissions Committee and the Dean or Chair of Department, will decide the form of any admissions assessment to be undertaken and any other conditions attached to readmission. The standard admissions fee will be payable. The student will resume the student number previously allocated to him or her. The student will receive any academic credit previously awarded provided s/he remains within the maximum registration period.

A student who has been expelled from the University for a disciplinary cause will not be re-admitted to the University except with the exceptional approval of the

Vice-Chancellor, who will specify any conditions for re-admission. Any re-admission will be reported to the Admissions Committee. The student will be entitled, on expulsion, to a transcript with details of any academic credit awarded, subject to paying any outstanding fees or other dues to the university.

A student who has been excluded from the University on account of academic failure may be re-admitted to the University only with exceptional approval from the Chair of the Admissions Committee, the Registrar and the Dean or the Chair of Department to which the applicant seeks admission. Any such re-admission will be reported to the Admissions Committee.

Good advising is critical for successful graduation. For most students, University will be the first time that they will be responsible for things such as selecting courses or choosing a major. An academic advisor a faculty member can help with these and other decisions; however, advising is a two-way street. Students and advisors share the responsibility for successful advising.

The Registrar may only allow the students to register beyond the published deadline after consultation with the Dean or Head of Department. The student will be subject to a late registration fee, to be determined from time to time by the university management. The Registrar may excuse the late registration fee on the grounds of mitigating circumstances that prevented the student from registering on time because of significant reasons beyond his or her control.

The Dean or Head of Department, in consultation with the Registrar, may exceptionally authorize a student registered during any time of his or her programme of study to take up to (6 credit) two Independent Study Courses in order to complete studies in due time, or for any other authorized purpose. An independent study course is a specially designed course not usually offered through the programme. The student will take the independent study course through private study and a series of tutorials subject to a course plan approved by the Dean or the Head or Department. The number of students on an independent study course will be limited to two. The Course Plan will include the following details.

Once approved by the Dean of Head of Department, the Course Plan will be sent to the Registrar/Examination Controller to be retained on record. The student must complete the course within the specified semester or will otherwise receive an F Grade unless the Dean or Head of Department and the Registrar agree that there are mitigating circumstances arising from significant factors beyond the student’s control. The Registry/Examinations Office will inform the Academic Council periodically of the numbers of Independent Study Courses authorized.

The performance of the students will be evaluated throughout the semester by class tests, quizzes, assignments and midterm exams. End of semester evaluation includes comprehensive final exams, term papers, project reports etc. Numerical scores earned by a students in tests, exams, assignments etc are cumulated and converted to letter grades at the end of the semester.

The university follows modern teaching methods including interactive Internet, simulation, lab work, case analysis and field study. A special feature of BRAC University teaching is the workshop/lab sessions designed to assist students in learning application of concepts and theories. The medium of instructions in Brac University is English.

The grades at the university will be indicated in the following manner:

97 - 100* = A+ (4.0) Exceptional90 - <97* = A (4.0) Excellent85 - <90 = A- (3.7) 80 - <85 = B+ (3.3) 75 - <80 = B (3.0) Good70 - <75 = B- (2.7) 65 - <70 = C+ (2.3) 60 - <65 = C (2.0) Fair57 - <60 = C- (1.7) 55 - <57 = D+ (1.3) 52 - <55 = D (1.0) Poor50 - <52 = D- (0.7) <50 = F (0.0) Failure

* Grading system introduced from Fall 2020 Semester.

P: Pass I: IncompleteW: Withdrawal

Pass/Fail Option: A course may be taken for a pass/fail grade, providing that the instructor approves the option and the student carries 12 credits for regular letter grades in that semester. A maximum of 16 credits may be taken for credit with the pass/fail grading option. No more than four credits may be taken with the pass/fail grading option in any one semester. Departments may not approve the pass/fail grading option for some courses counting towards the major.

Incomplete Grade: An Incomplete (I) grade is assigned only when a student has failed to complete one or more requirements of the course for an unavoidable reason/accidental circumstance and has applied for I grade. The students who are permitted to appear in make up examination(s) will be assigned an 'I' grade for that course and this grade will stay until the student appears in the make up examination at the first available opportunity; if s/he fails to appear in the make up examination the 'I' grade will automatically be converted to 'F' grade.

Withdrawal (W): is assigned to a student who withdraws from the course within the deadline for withdrawal with 'W' grade. A student who withdraws after this date will earn the grade based on his performance before his withdrawal. Exception to this rule may be made on medical ground and on terms and condition imposed by the University.

The Grade Point Average (GPA) is computed in the following manner:

Sum of (Grade points H Credits)GPA = ---------------------------------- Sum of Credits attempted

The Committee on Academic Standard administers the Grading Regulations and reviews course grades submitted by the Departments.

Students will be expected to maintain standards in their academic work. They should be taking the requisite number of courses and maintain satisfactory grades in these courses. In particular, students are expected to maintain a GPA of 1.5 (both semester and cumulative), otherwise the student will be put on probation for the following semester. If a student on probation fails to raise CGPA to 1.5 in two consecutive semesters s/he will be dismissed from the University. Students whose grade point average is below 1.0 in their first semester may be asked to withdraw from the university.Students will be allowed to continue with a CGPA of 1.50 up to sixth semester, (in case of Pharmacy students up to fourth semester). However, at the end of the sixth semester/fourth semester for Pharmacy, students must have a CGPA of 2.00 in order to continue his/her studies in the undergraduate programs.

Probation: Students are expected to maintain a minimum CGPA of 1.5, otherwise the student will be put on probation for the following semester. If a student on probation fails to raise CGPA to 1.5 in two consecutive semesters s/he will be dismissed from the University. Students whose grade point average is below 1.0 in their first semester may be asked to withdraw from the university. Students will be allowed to continue with a CGPA of 1.50 up to sixth semester, (in case of Pharmacy students up to fourth semester). However, at the end of the sixth semester/fourth semester for Pharmacy, students must have a CGPA of 2.00 in order to continue his/her studies in the undergraduate programs.

Probation policy would be effective to all students enrolled in Summer-2022 and onward as followed:

Each student will be required to maintain a minimum CGPA of 2.0 otherwise s/he will be placed on probation for the following semester. The student will then have up to 3 consecutive semesters to achieve a CGPA of 2.0, with two consecutive semesters for Pharmacy.

The student will not have the option to drop a probationary semester, except in case of serious illness or other major circumstances beyond his or her control, as accepted by the Head of Department in consultation with the Registrar, and with confirmation of any medical evidence by the university Medical Centre. In this case the semester will be dropped as approved above.

A student who fails to achieve a CGPA of 2.00 within 3 consecutive semesters (or 2 semesters for Pharmacy) will be considered for withdrawal from the university. The withdrawal will be subject to review by the Dean or the Head of Department in consultation with the Registrar. The Dean or the Head of Department may recommend to the Pro-Vice-Chancellor that the student should continue for a defined period with a further opportunity to reach a CGPA of 2.

Such a recommendation may be made on the basis that the student was affected by major, adverse circumstances beyond his/her control (e.g. serious illness); that the student had a borderline failure to meet the 2.0 CGPA standard, or on the basis of an assessment of the overall student profile (such as transcript GPA). The Registry will retain a record of decisions and their reasons and the Academic Council will receive periodic reports on the numbers of exceptions approved.

Retake: A student who receives an “F” grade in a course may retake a course twice. By exception, the Head of the Department or the Dean may justify and arrange with the Registrar to allow students (such as those on probation) to take three retakes in order to safeguard their progression.

The student will be allowed only one retake in the case of a non-credit course (091/092) category) taken in the first or second semester.

No further attempt will be allowed except with the permission of the Head of the Department in consultation with the Registrar. The decision to allow a further attempt may be made only on the basis that the student was affected by major adverse circumstances beyond his/her control (e.g. serious illness). The best of the grades received will be counted for calculation of the CGPA.

Repeat: The current regulations allow a student to repeat a course in order to improve the grade. A student may repeat a course once in order to improve the grade, however, s/he must repeat the course within 2 semesters of the initial enrollment on the course. There will not be any cap on the grade of the repeated course and the latest grade earned would be counted for the CGPA calculation.

This Retake and Repeat policy is effective for all students.

Audit is a registration status allowing students to attend a course without receiving credit. Both undergraduate and postgraduate students enrolled in BRAC University may audit courses. Graduates of Brac University or other universities acceptable to Brac University may enroll for "Audit" of courses. The performance of students auditing a course will not be evaluated or graded and they will receive a grade 'AU'.

Students and alumni of Brac University will have to pay 50% of tuition fees and other fees. All other students will have to pay full tuition and other fees.

Students currently enrolled in universities acceptable to Brac University may enroll as a credit student in at best 10 courses (30 credits) on payment of full tuition and other fees of the university. Candidates seeking admission in one or more audit/credit course(s) must apply in prescribed form and the applications will be considered as individual cases. The university reserves the right to accept or reject the applications.

As Brac University is based on the US University system, all undergraduate degrees are for about four years duration. For each degree at least 120 credits are required. Students are responsible for meeting degree requirements. Before selecting the courses in each semester students should consult their academic advisor. The university reserves the right to bring in change into programs and curricula without notice whenever circumstances warrant such changes.

A minimum of 120 credits for a bachelor's degree out of which at least 70 must be earned at Brac University. For students of Architecture Department at least 134 credits must be earned at Brac University.

Attending Residential semester is compulsory for all BracU students. Students must complete all course requirements for the degree including General education courses, non-major area courses, major area courses, elective courses, courses for double major or minor.

A student must complete the requisite number of credits of course work and meet other requirements depending on the program in which he/she is enrolled and must maintain a minimum CGPA of 2.00.

A student must have clearance from Brac University Accounts, Library and Registrar's Office.

Fulfillment of the above conditions does not necessarily mean that a degree will be conferred on the student. The University reserves the right to refuse the awarding of degree on disciplinary or similar grounds.

The curriculum for degree requirements of graduate programs vary depending upon the degree offered.

Transfer of credits from institutions having equivalent curriculum grading system and grading standard may be allowed for a maximum of 30 credits provided that the student has obtained at least B+ grade(s) in the course(s) eligible for transfer. The university will consider applications for transfer of credit on a case-by-case basis.

The university follows modern teaching methods including interactive Internet, simulation, lab work, case analysis and field study. A special feature of Brac University teaching is the workshop/lab sessions designed to assist students in learning application of concepts and theories. The medium of instructions in BRAC University is English.

The grades at the university will be indicated in the following manner:90 - 100 = A (4.0) Excellent85 - <90 = A- (3.7) --80 - <85 = B+ (3.3) --75 - <80 = B (3.0) Good70 - <75 = B- (2.7) --65 - <70 = C+ (2.3) --60 - <65 = C (2.0) Fair57 - <60 = C- (1.7) --55 - <57 = D+ (1.3) --52 - <55 = D (1.0) Poor50 - <52 = D- (0.7) --<50 -- -- = F (0.0) Failure

Grades without numerical value:P: Pass I: IncompleteW: Withdrawal

The Grade Point Average (GPA) is computed in the following manner: Sum of (Grade points H Credits)GPA =---------------------------------- Sum of Credits attempted

A course may be taken for a pass/fail grade providing that the instructor approves the option and the student carries 12 credits for regular letter grades in that semester. Within the total credits required for a degree, a maximum of 16 credits may be taken for credit with pass/fail grading option. No more than 4 credits may be taken with the pass/fail grading option in any one semester. Departments may not approve the pass/fail grading option for some courses counting towards the major.

Incomplete Grade: An Incomplete (I) grade is assigned only when a student has failed to complete one or more requirements of the course for extreme medical conditions OR death in the immediate family during the semester and has applied for I grade. The students who are permitted to appear in Make up examination(s) will be assigned an 'I' grade for that course and this grade will stay until the student appears in the make up examination at the first available opportunity; if s/he fails to appear in the make up examination the 'I' grade will automatically be converted to 'F' grade.

Withdrawal (W): is assigned to a student who withdraws from the course within the deadline for withdrawal with W grade. A student who withdraws after this date will earn the grade based on his performance before his withdrawal. Exception to this rule may be made on medical ground and on terms and condition imposed by the University.

Review ProcedureThe Committee on Academic Standard administers the grading regulationand reviews course grades submitted by Departments.

Academic Standing Students are expected to maintain a consistently high standard in their academic work. They should be taking the requisite number of courses and maintain satisfactory grades in these courses. In particular students are expected to maintain a CGPA of 2.50 (both semester and cumulative), otherwise they will be put on probation for the following semester. If a student fails to maintain a CGPA of 2.50 in the two consecutive semesters then the university will review the student's record and recommend further action which may include withdrawing from the university.

Requirements for the DegreeFor graduation, a student must complete the requisite number of credits of course work and meet other requirements depending on the program in which he/she is enrolled and must maintain a CGPA of 2.50. The University, however, reserves the right to refuse the awarding of degree on disciplinary or similar grounds.

Student AdvisingWhen students first join the university, they are assigned an advisor, a faculty member who helps them in choosing their courses for the first year. Later students are assigned an advisor who then guides the student in choosing the courses of his/her major. Students will develop the direction of their study in consultation with their advisor.

Remedial CoursesMany students joining the university would be coming from Bangla medium schools and therefore would have to adjust to English as the medium of instruction. They may be asked to attend Remedial English courses during or proceeding the semester in which they take regular courses. Students from non-science background or who are weak in Mathematics may be asked to attend a remedial course in Mathematics. The University may ask the students to attend other remedial courses if necessary.

Undergraduate Programs

Postgraduate Programs

Academia and Industry are two different worlds which operate on different pedestals. Both have different purposes and different ideologies. Nevertheless, the rapid pace of change in the outside environment is compelling these two different worlds to come together to address and solve some of the real-world challenges. Realizing the immense urgency of the situation, 1st International Conference on Business and Management (ICBM 2017) organized an Academia-Industry Discussion session on September 21 at the evening at The Westin Hotel, Dhaka, Bangladesh where professionals from both sectors participated to analyze the current scenario and find out probable ways to minimize the gap.

From the Industry arena, Mr. K. Joaddar, Deputy Managing Director & CFO, BRAC Bank, Mr. ASM Mainuddin Monem, Deputy Managing Director, Abdul Monem Group Ltd., Mr. Syed Mahbubur Rahman, Managing Director of Dhaka Bank Ltd. and Mr. Syed Ferdous Raihan Kirmany, Deputy General Manager-Operations of BBS Cables Ltd., participated in the session to highlight the current scenario of the job market, industry's expectations from graduates, universities role to prepare the future leaders and finally the overall hurdles industries face to screen out appropriate candidates for their particular industries. Moreover, to improve the current state and ease the screening process to filter out the quality graduates, they proposed to have collaboration with the universities.

Moreover, the Vice Chancellor of BRAC University, Prof. Syed Saad Andaleeb, Ph.D, pointed out businesses' lack of interest in investing in researches and how this phenomenon is widening the gap between Bangladesh and the advanced nations. On financing, he said the Massachusetts Institute of Technology has an annual budget of $3.5 billion, 90 percent of which comes from some form of research and the rest from tuition but unfortunately in Bangladesh business communities are not really interested in making investment for research in universities. Therefore, Prof Andaleeb called on the businesses to make direct investment in universities for research purposes. He also expressed his anxiety that unless research-based institutes are developed and nurtured, our nation will fall behind compared to developed world.From the International Academic field, Prof. Premkumar Rajagopal, Vice-Chancellor, Malaysia University of Science and Technology, Prof. Pairach Piboonrungroj, Associate Dean, Maritime Studies and Management, Chiang Mai University, Thailand and Prof. Mark GOH, Business School, National University of Singapore shared the challenges faced by them during the early stage to make research- based universities. However, they also shared several keys to surmount those obstacles.

Apart from them other prominent scholars like Dr. Mirza Azizul Islam, Former Adviser of Caretaker Government, People's Republic of Bangladesh and Professor of BRAC

Business School; Dr. Akbar Ali Khan, Former Adviser of Caretaker Government, People's Republic of Bangladesh and Professor of BRAC Business School; Dr. Salehuddin Ahmed, Former Governor of Bangladesh bank (The Central bank of Bangladesh) and Professor of BRAC Business School; Prof. Dr Abdul Hannan, Vice Chancellor, Primeasia University; Prof. ATM Nurul Amin, Chairperson of ESS, BRAC University; Assoc. Prof. Dr. Md. Mamun Habib of BRAC Business school and Program Chair of ICBM 2017; Dr. Erne Suzila Kassim and Dr. Siti Noorsuriani Ma'on from UiTM, Malaysia; Dr. Mahbubur Rahman (NSU, Bangladesh), Dr. Ireen Akhter (IBA, JU, Bangaladesh), were also present in the session where they urged the necessity to create strong affiliation with different industry professionals to meet the needs of the overall employment of the country.

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

ENH lecturer attains Fulbright teaching assistantship

BRAC University architecture team recognized at international design competition

Urgent Action Needed to Protect Our Community

Announcement regarding admission test

BRAC University's Centre for Entrepreneurship Development Wins Global Impact Award at ICSB World Congress

Law Alumni Association of BRAC University Hosts Meeting at Supreme Court

Students of The Department Architecture Excel at AIUB Architecture Design Charrette

Professor Fuad Hassan Mallick and Professor Zainab Faruqui Ali Speak at Bengal Institute’s Lecture Series on Exploring Muzharul Islam

BRACU Alumni-led Startup Shines in Shark Tank Bangladesh

BRAC University and United Group entertainment venture - Courtside Sign Memorandum of Understanding

Awareness sessions on “Campus Safeguarding” at BRAC University’s Residential Campus conducted by the Office of the Proctor

BRAC Business School Hosts Closing Ceremony for ‘Uddami Ami’ Women Entrepreneurship Training Program

Pages

Events

Dr. Fahim Kabir Monjurul Haque joined the Department of Mathematics and Natural Sciences (Microbiology Program) at Brac University as an Assistant Professor in January 2018. Before joining here he was working as an Assistant Professor at Wakayama Medical University, Japan. He received his PhD in Medical Sciences from Kumamoto

University, Japan in 2017 under the HIGO Program scholarship which was funded by MEXT. He completed his Master's (M.Sc.) and Bachelor's (B.Sc.) in Microbiology from University of Dhaka. Immediately after obtaining his Master's degree he joined icddr,b in Virology Laboratory. He worked there as a Senior Research Officer until he left for perusing his PhD in 2013. Throughout his research career he has worked on developmental biology and public health related projects, such as, molecular mechanism of kidney development using animal model (transgenic and conditional knockout mice), congenital kidney disease pathogenesis using patient specific iPS cell derived kidney organoids, urogenital track development, effect of diabetes & low-testosterone in adult penile tissue homeostasis, child pneumonia etiology, clinical drug trial and HIV surveillance.

Level: 5, Room: 5B11Kha-224 Merul BaddaDhaka 1212. Bangladesh

Journals

Islam, H., Masud, J., Islam, Y., Haque, F. (2022). An Update on Polycystic Ovary Syndrome (PCOS): a review of the current state of knowledge in diagnosis, genetic etiology, and emerging treatment options. Women's Health (In Press)

Ahmed, A., Naboni, N. S., Nowshen, A., Sultana, S., Haque, F., & Hossain, M. M. (2022). A review on the causative agents, risk factors and management of ventilator-associated pneumonia: South Asian perspective. Journal of Medicine (Bangladesh), 23(2), 151–161. https://doi.org/10.3329/jom.v23i2.60633

Disha, T., & Haque, F. (2022). Prevalence and risk factors of vulvovaginal candidosis during pregnancy: A review. Infectious Diseases in Obstetrics and Gynecology, 2022. https://doi.org/10.1155/2022/6195712

Khan, T. S., Shirin, T., Alam, A. N., Shakhider, M. A. H., & Haque, F. (2022). Follow-up of practiced treatment regimens and health conditions of patients following recovery from COVID-19 residing in Dhaka City: a survey-based, descriptive, cross-sectional study. IJID Regions, 3, 68–75. https://doi.org/10.1016/j.ijregi.2022.03.005

Brooks, W. A., Zaman, K., Goswami, D., Prosperi, C., Endtz, H. P., Hossain, L., Rahman, M., Ahmed, D., Rahman, M. Z., Banu, S., Shikder, A. U., Jahan, Y., Nahar, K., Chisti, M. J., Yunus, M., Khan, M. A., Matin, F. B., Mazumder, R., Shahriar Bin Elahi, M., Saifullah, M., ..., Haque, F., ... Deloria Knoll, M. (2021). The Etiology of Childhood Pneumonia in Bangladesh: Findings From the Pneumonia Etiology Research for Child Health (PERCH) Study. The Pediatric infectious disease journal, 40(9S), S79–S90. https://doi.org/10.1097/INF.0000000000002648

Tanigawa, S., Islam, M., Sharmin, S., Naganuma, H., Yoshimura, Y., Haque, F., Nishinakamura, R. (2018). Organoids from nephrotic disease-derived iPSCs identify impaired NEPHRIN localization and slit diaphragm formation in kidney podocytes. Stem Cell Reports, 11(3), 727–740. https://doi.org/https://doi.org/10.1016/j.stemcr.2018.08.003

Haque, F., Kaku, Y., Fujimura, S., Ohmori, T., Adelstein, R. S., & Nishinakamura, R. (2017). Non-muscle myosin II deletion in the developing kidney causes ureter-bladder misconnection and apical extrusion of the nephric duct lineage epithelia. Developmental Biology, 427(1), 121–130. https://doi.org/https://doi.org/10.1016/j.ydbio.2017.04.020

Kaku, Y., Taguchi, A., Tanigawa, S., Haque, F., Sakuma, T., Yamamoto, T., & Nishinakamura, R. (2017). PAX2 is dispensable for in vitro nephron formation from human induced pluripotent stem cells. Scientific Reports, 7(1), 4554. https://doi.org/10.1038/s41598-017-04813-3

Haque, F., & Akhter, M. (2014). Plasmid Profile Analysis of Multi Drug Resistant Escherichia coli and Acinetobacter from Clinical Samples. Bangladesh Journal of Medical Science. Volume 20 & 21, Number 01 & 02, March & September 2014 & 2015, pp. 67-73

Preprints

Kenneth, M., and Haque, F. (2022). Bioinformatics Analysis Reveals miR-98-5p as a

potential Inhibitor of Tumor cell proliferation and metastasis in Colorectal Cancer by targeting the FZD3 receptor of the Wnt signaling pathway, 23 June 2022, PREPRINT (Version 1) available at Research Square. https://doi.org/10.21203/rs.3.rs-1737439/v1

General Microbiology (MI101), Immunopathology and Vaccine Development (MIC309), Bacterial Pathogenesis and Molecular Epidemiology (MIC407), Enzymology (BTC517), Fermentation and Industrial Biotechnology (BTC504)

Raydah, A., Khan, T., Shafi, S., Haque, F. (2022, March 28 – April 1). Environmental Dissemination of New Delhi Metallo-β-Lactamase-1 (NDM-1) Producing Bacteria in Community Wastewaters of Dhaka, Bangladesh [Conference presentation abstract]. 13th Annual Global Health Conference (CUGH 2022). Consortium of Universities for Global Health. https://cugh.confex.com/cugh/2022/meetingapp.cgi/Paper/1720

Haque, F., kaku, Y., Fujimura, S., Ohmori, T., Nishinakamura, R. (2017, May 10-13). Non-muscle Myosin II Deletion in the Developing Kidney Causes Ureter-bladder Misconnection and Apical Extrusion of the Nephric Duct Lineage Epithelia [Conference presentation]. 50th annual meeting of the Japanese Society of Developmental Biologists. Tokyo, Japan. https://www2.jsdb.jp/kaisai/jsdb2017/endaidisplay.php?id=P111

HIGO program Scholarship (A program for leading graduate schools), Japan 2013

Japanese Society of Developmental Biologist (JSDB)Dhaka University Microbiology Alumni Association (DUMAA)

Fahim Kabir Monjurul Haque, PhD

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Fariha Nusrat completed her school and college from Rajuk Uttara Model College. She completed her BSc (Hons.) and MSc in Microbiology from University of Dhaka. Immediately after obtaining her Master’s degree, she joined as a lecturer in Microbiology program at the Department of Mathematics and Natural Sciences, Brac University, Dhaka on January 2020. During her thesis, she worked in enzyme production, protein biotechnology and gene cloning.

Level:Kha-224 Merul BaddaDhaka 1212. Bangladesh

Master of Sciences, Department of Microbiology, University of Dhaka, 2018Bachelor of Sciences, Department of Microbiology, University of Dhaka, 2017

January 2020 to till: Lecturer, Microbiology Program, Department of Mathematics and Natural Sciences

MIC 102 Basic Techniques in MicrobiologyMIC 201 Microbial ChemistryMIC 204 Medical MicrobiologyMIC 308 Fermentation TechnologyMIC 401 Microbial Genetic EngineeringMIC 402 Analytical MicrobiologyMIC 403 Quality Control of Food, Fish and Beverages

• Government scholarship for Bachelor of Science degree, University of Dhaka (2020)Ministry of Education, Bangladesh.• Best lightening oral award (2019)4th IPFS-ICBHA 2019-GNOBB Conference, Dhaka.• National Science and Technology Fellowship (2019)Ministry of Science and Technology, Bangladesh.• Dean’s Award (2018)Faculty of Biological Sciences, University of Dhaka.

Member of Bangladesh Society of MicrobiologyDhaka University Microbiology Alumni Association (DUMAA)

Protein Biotechnology

Fariha Nusrat

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

The BRAC Business School (BBS) offers an undergraduate program: Bachelor of Business Administration (BBA) and two graduate programs: Master of Business Administration (MBA) and Executive Master of Business Administration (EMBA). Each program offers a unique set of specializations in a wide variety of business disciplines: Accounting, Finance, E-Business, Entrepreneurship, Human Resource Management, Information System, Management, Marketing, Operations and Supply Chain Management and others.

To become a globally-recognized center of excellence in developing business leaders and knowledge dedicated to championing human values.

The mission of BBS is two-fold: • To offer cutting-edge business education for grooming competent and ethical future leaders who cater to the interests of all stakeholders, and • To advance knowledge - in the local and global contexts - promoting sustainable development, equity and universal well-being.

• Transparent and shared-governance• Ethical conduct• Student-centric, exciting and rewarding education• Teamwork and collaboration among all stakeholders• Innovation and continuous improvements• Life-long learning and professional development• Creation and dissemination of impactful knowledge• Diversity, global perspectives and multiculturalism• Championship of human values

BBS is currently working on a 5-year comprehensive strategic plan. In order to achieve the stated vision and missions, the following complementary and instrumental medium-to-long term goals are set and in-progress:

• Deliver undergraduate and graduate business programs (BBA, MBA, EMBA and others) with contemporary curricula and multi-disciplinary focus across all functional areas of business, economics, ethics, sustainability, corporate governance and cross-cultural dimensions.

• Develop graduates with superior critical thinking, data analysis, problem-solving and innovative decision-making abilities to meet the dynamic and changing demands of the local and global business industries and other organizations as leaders and entrepreneurs.

• Attain internationalization through providing opportunities to students for two-way exchange and dual-degree programs with foreign universities, and facilitating exchange and cooperation for faculty and staff members with the counterparts of those universities.

• Achieve validation of high-quality learning and teaching, scholarly contributions, and services to the world-of-practice and the society-at-large; and global recognition through meeting and maintaining the standards of top accreditation agencies of business education.

• Establish and maintain close collaborations with alumni, local and multinational businesses, professional associations, government and development entities and the community-at-large.

• Attract and retain an adequate pool of academically and professionally qualified faculty members with diverse nationalities and academic, professional and cultural backgrounds; and facilitate their continuous professional development including scholarly activities.

• Facilitate, promote and disseminate academic, practice-based and pedagogical scholarly activities contributing towards excellence in higher education and sustainable and equitable global development.

• Deliver exciting and rewarding educational experience to students utilizing student-centric and competent faculty and staff members and state-of-the art pedagogy and resources.

• Impact the local and global community through ensuring dedicated support and contributing towards the success of all stakeholders of BBS in promoting sustainable development, equity and universal well-being.

• Transit BBS to an autonomous, resourceful and self-sustained center of excellence in higher education and sustainable and equitable global development.

Vision

Mission

Values

Strategic Goals:

Academic Programs

Faculty and staff

Mohammad Mujibul Haque, PhD

Mirza Azizul Islam, PhD

Salehuddin Ahmed, PhD

Mohammad Abdul Hoque, PhD

Sebastian Groh, PhD

Abu Reza Mohmmad Islam, PhD

Suman Paul Chowdhury, PhD

Syed Mahbubur Rahman, PhD

Mohammad Rabiul Basher Rubel, PhD

Zaheed Halim, PhD

Mozammel Mia, PhD

Nitai Chandra Debnath, PhD, FCMA

Muhommad Azizur Rahman, PhD

Md. Mizanur Rahman, PhD

Shamim Ehsanul Haque

Riyashad Ahmed

Md. Hasan Maksud Chowdhury

Saif Hossain

Faruk Bhuiyan, PhD

Md. Kausar Alam, PhD

Afsana Akhtar

Tarnima Warda Andalib, PhD

Mohammad Shahidul Islam, PhD., MM.

Ekramul Islam, PhD

Md Shamimul Islam, PhD

Mohammad Enamul Hoque, DBA

Sayem Hossain, PhD

Syed Far Abid Hossain, PhD

M. Nazmul Islam, PhD

Najmul Hasan, PhD

Md Arif Hossain Mazumder, PhD

Sau K Leung, PhD

Abu Saad Md. Masnun Al Mahi, PhD

Sayla Sowat Siddiqui, PhD

H. M. ARIF, PhD

Yasmin Jamadar, PhD

Farhana Zahir, CFA, FRM

Nusrat Hafiz, PhD

Saad Md Maroof Hossain, PhD

Mr. Zaheed Husein Mohammad Al-Din

Fabiha Enam

Tania Akter

Ummul Wara Adrita

Rahma Akhter

Jubairul Islam Shaown

Tanzin Khan

Ahmed Abir Choudhury

Md. Shamim Ahmed

Mohammad Atiqul Basher

Shihab Kabir Shuvo

Feihan Ahsan

Md. Shezanur Rahman

Tausif Bari

Takmilla Tabassum

Noshin Anjum Chaiti

Mukit Anis

Safarath Ahmed

Fairuza Mehreen

Nadia Afroze Disha

Md Tanvir Hasan

News and Events

BRAC Business School Hosts Closing Ceremony for ‘Uddami Ami’ Women Entrepreneurship Training Program

BBS Faculty Publishes Research Paper in International Journal of Hospitality Management

Dr. Mohammad Shahidul Islam Brings New Light to Hospitality Studies with Innovative Research Method Video Reflexive Ethnography (VRE)

BRAC University students get Inspiring Women Award

1st International Case Conference on Business and Management 2023 (ICCBM 2023) on July 27, 2023

BBA Frehser's Connect Fall 2022

BBS Townhall Meeting

Invitation to attend the Speech titled “Trends in Higher Education" by Dr. Mario Hayek, Dean, College of Business, Texas A&M – Commerce on Sunday, April 10, 2022

Newsletter

BBS Newsletter, June 23-Dec 23.

BBS Newsletter, July 2022

BBS Newsletter, December 2021

BBS Newsletter, August 2021

Students belonging to various disciplines of BRACU, specially physics, CS, CSE, ECE, ESS or any other relevant discipline desirous of doing a double major in mathematics, will have to complete a total of 67 credits, the breakdown of which is given below:

A. Core Courses

Theory Courses (48 credits)

1. MAT 121: Basic Algebra2. MAT 122: Analytical and Vector Geometry3. MAT 123: Calculus I4. MAT 211: Calculus II5. MAT 212: Linear Algebra6. MAT 221: Real Analysis I7. MAT 222: Differential Equations I8. MAT 223: Numerical Analysis I9. MAT 311: Abstract Algebra10. MAT 312: Numerical Analysis II11. MAT 314: Complex Analysis12. MAT 321: Real Analysis II13. MAT 322: Differential Equations II14. MAT 323: Dynamical Systems15. MAT 324: Graph Theory16. MAT 325: Mathematical Methods

Lab Courses (4 credits)

* MAT250: Mathematics Lab I* MAT350: Mathematics Lab II* 2 credit courses

B. Elective Courses (12 credits)

1. MAT 124: Scientific Programming2. CSE 220 : Data Structures3. CSE 221 : Algorithms4. MAT 203: Matrices, Linear Algebra & Differential Equations5. MAT 313: Differential Geometry6. MAT 316: Operations Research I7. MAT 326: Hydrodynamics8.

MAT 411: Topology 9. MAT 415: Finite Element Methods10. MAT 416: Tensor Calculus11. MAT 421: Fluid Mechanics12. MAT 422: Theory of Numbers13. MAT 423: Mathematical Modeling14. MAT 424: Operations Research II15. MAT 425: Advanced Numerical Methods16. PHY 112 : Principles of Physics II17. PHY 204 : Classical Mechanics and Special Theory of Relativity18. PHY 205 : Statistical Mechanics19. PHY 303 : Quantum Mechanics I20. PHY 413 : General Theory of Relativity21. STA 301 : Modern Probability Theory & Stochastic Processes

or any other mathematics course with the permission of the Chairperson of the MNS Department.

Project/Thesis (3 credits)

1. MAT 400: Project/Thesis

The project/thesis work should be in line with one of the majors or on some interactive topic involving the two majors.

Dr. Munima Haque is currently serving as an Associate Professor in Biotechnology program, Department of Mathematics and Natural Sciences, BRAC University. Dr. Munima received her Ph.D. from University of Illinois Urbana-Champaign (UIUC), IL, USA and M.S. from Southern Illinois University (Carbondale, IL, USA). She has over 17 years of teaching experience in in USA and Bangladesh Universities. Dr. Haque’s postdoctoral research was in biomedical engineering (nanotechnology), and her PhD research works in Biological Engineering (nanoscale characterization of corn protein). Dr. Haque is skilled in Nanomaterials, Health Physics, Cancer Biology, Bio nanotechnology, Nuclear Radiation Physics, Radiation Biotechnology. She has ~ 42 publications (37 international, 5 national) in journals and conference proceedings. During her Postdoctoral Research at Micro and Nanotechnology Laboratory, UIUC, USA she worked on gene therapy, targeted drug delivery. Dr. Haque’s research also involves biological effects of ionizing and non-ionizing radiation on human and animal biosystems, radioactivity in air water soil ecosystems, radioactivity in nuclear medicine wastes, food irradiation biotechnology, Radioisotope waste management, micro & nanoplastic pollution in ecosystems characterization and bioremediations, Radon concentration monitoring in environment, Radioisotope accumulation and testing in nuclear power plants, toxic materials measurement and health risk assessment, Dosimetry for radiotherapy treatment, obstetrics and gynecology, pediatric risk factors and health issues, biomedical data sciences. She is life member of Bangladesh Medical Physics Society and Bangladesh Physical Society, members of SARI radiation and GNOBB.

Level: 4, Room: 4G43Kha-224 Merul BaddaDhaka 1212. Bangladesh

Institution Name

Name of Exam

Results

Dissertation/Thesis Title

University of Illinois at Urbana-Champaign (UIUC), Urbana, IL, USA

Ph. D. in Agricultural and Biological Engineering

Concentration: Biological Engineering and Biophysics

GPA: 3.86/4.0 Ph.D. program

A colloidal science approach to characterize the nanoscale aggregation behavior of corn protein zein

Southern Illinois University (SIU), Carbondale, IL, USA

M.S. in Physics Concentration: Nuclear Radiation Physics

GPA: 3.64/4.0

Surface Cluster Model for cluster and alpha radioactivity

Rajshahi University, Rajshahi, Bangladesh

M. Sc. In Physics Concentration: Nuclear Physics

First class 1st position (combined: Nuclear Physics & Solid State Physics Group)

A study of the elastic α–scattering from 32,34S and the molecular resonances

Rajshahi University, Rajshahi, Bangladesh

B. Sc. (honors) in Physics Minor: Mathematics and Statistics

First class 2nd position

Selected Publications

Book Chapters

Quamruzzaman, M., Haque, M., Haque, S., & Das, U. C. (2021). Electromagnetic radiation from cell phones used in Dhaka city. In Y.-D. Zhang, T. Senjyu, C. SO–IN, & A. Joshi (Eds.), Smart Trends in Computing and Communications: Proceedings of SmartCom 2020 (pp. 147–157). Springer Singapore. https://doi.org/10.1007/978-981-15-5224-3_14

Journals

Kabir, M. R., Haque, M., Roy, R., Setu, S. P., & Huda, M. M. (2020). Factors affecting the survival time of cancer patient: a hospital registry-based study. EurAsian Journal of BioSciences, 14(2), 7817–7824. https://www.proquest.com/openview/1be77b813f2ba9b1e2f2ef18865deaf0/1?pq-...

Haque, M. (2019). Nano Fabrics in the 21st century: A review. Asian Journal of Nanoscience and Materials, 2(2), 131–148. http://www.ajnanomat.com/article_81050.html

Haque, M., Quamruzzaman, M., & Haque, S. (2018). Survey of EMF emitted by domestic appliances in Dhaka. ARPN Journal of Engineering and Applied Sciences, 13(18), 4976–4986. http://www.arpnjournals.org/jeas/research_papers/rp_2018/jeas_0918_7293.pdf

Shaheed, S., Haque, M., & Haider, R. (2018). Conservative management of single fetal demise in twin pregnancy in Dhaka Bangladesh: A case report. Dhaka Central

International Medical College Journal, 5(2), 105–111. https://dcimch.com/wp-content/uploads/2020/09/july18vol5no2.pdf

Haque, M., & Ferdous, M. J. (2017). Transfer of natural radionuclides from soil to plants in savar Dhaka. Spanish Journal of Soil Science, 7(2), 133–145. https://doi.org/10.3232/SJSS.2017.V7.N2.05

Haque, M., & Quamruzzaman, M. (2017). Survey of EMF emitted from air conditioners and switchboards in electrical and electronic engineering laboratories of Southeast University Bangladesh. Malaysian Journal of Public Health Medicine, 17(2), 35–41.

Haque, M., & Quamruzzaman, M. (2016). Health effects of EMF emitted from cell phones used by students of Southeast University in Bangladesh. Southeast University Journal of Science and Engineering, 10(1), 28–38. https://seu.edu.bd/seujse/vol_10_no_1_jun_2016.php

Sharmin, A. Al, & Haque, M. (2016). Susceptible factors of type-2 diabetes in a population of Bangladesh. Universal Journal of Public Health, 4(1), 38–45. https://doi.org/10.13189/ujph.2016.040106

Quamruzzaman, M., Haque, M., Ahmed, F., & Zaman, M. S. (2015). Effects of Electromagnetic Fields (EMF) near high voltage transmission line: A case study. Bangladesh Journal of Medical Physics, 7(1 SE-Short Notes), 66–68. https://doi.org/10.3329/bjmp.v7i1.25261

Quamruzzaman, M., & Haque, M. (2014). Non ionizing radiation (NIR), its harmful effects especially from mobile/cell phone and towers. Southeast University Journal of Science and Engineering, 8(1–2), 34–39.

Kulsharova, G. K., Lee, M. B., Cheng, F., Haque, M., Choi, H., Kim, K., O'Brien, W. D., & Liu, G. L. (2013). In vitro and in vivo imaging of peptide-encapsulated polymer nanoparticles for cancer biomarker activated drug delivery. IEEE Transactions on Nanobioscience, 12(4), 304–310. https://doi.org/10.1109/TNB.2013.2274781

Quamruzzaman, M., & Haque, M. (2012). Survey studies on effects of electromagnetic fields on living system and environment in Bangladesh. Southeast University Journal of Science and Engineering, 6(6), 23–33.

Conferences

Quamruzzaman, M., & Haque, M. (2012, December 28). Epidemiological survey of people working in EMF field exposed to high frequency. Proceedings of the Global Engineering, Science and Technology Conference.

Haque, M., Reichstein, I., Hooshyar, M., & Malik, F. (2005). Energy-density functional Approach to Fission, Cluster, and Alpha Radioactivity. Proceedings of XXIX International Workshop in Condensed Matter Theories, International Institute for Advanced Studies, 21.

Technical Report

Improved radiation dosimetry/risk estimates to facilitate environmental management of plutonium-contaminated sites” Scott, Bobby R.; Tokarskaya, Z.B.; Zhuntova, G.V.; Belyaeva, Z.D.; Osovets, S.V.; Syrchikov, V.; Okladnikova, N.D.; Khokhryakov, V.F.; Haque, Munima. Office of Science (BER), U.S. Department of Energy (DOE) project final report, USA 14 December 2007. https://www.researchgate.net/publication/274078803_Improved_Radiation_DosimetryRisk_Estimates_to_Facilitate_Environmental_Management_of_Plutonium-Contaminated_Sites

Preprint

Mim, H.; Haque, M.; Promon, S.K. Neural Implants: A Review Of Current Trends And Future Perspectives. Preprints 2022, 2022020050 (doi: 10.20944/preprints202202.0050.v1).

Position

Institution Name

Department/Discipline

Duration

Associate Professor (Full-time)

BRAC University

Biotechnology Program, Department of Mathematics and Natural Sciences (MNS)

16th May 2022- present

Assistant Professor (Part-time)

BRAC University

Biotechnology Program, Department of MNS

July 2020-May 2022

Research Fellow (Honorary)

Relevant Science and Technology Society Bangladesh (RSTS)

October 2016-May 2022

Assistant Professor (Full time)

Southeast University

Department of Electrical and Electronic Engineering (EEE)

May 2010-Sept 2016

Postdoctoral Research Associate

University of Illinois at Urbana-Champaign, IL, USA

Micro and Nanotechnology Laboratory (MNTL)

January 2009-January 2010

Graduate Research and Teaching Assistant

University of Illinois at Urbana-Champaign, IL, USA

Department of Agricultural and Biological Engineering

January 2006-December 2008

Graduate Research Associate

Lovelace Biomedical Research Institute, Albuquerque, NM, USA

Computational Radiation Biology Group

Summer 2005

Graduate Research and Teaching Assistant

University of Illinois at Urbana-Champaign, IL, USA

Department of Nuclear, Plasma & Radiological Engineering (NPRE)

August 2004-December 2005

Graduate Teaching and Research Assistant

Southern Illinois University, Carbondale, IL, USA

Department of Physics

August 2001-July 2004

Lecturer (Full-time)

Khulna University of Science and Technology, Khulna, Bangladesh

Department of Physics

April 2001-August 2001

Lecturer (part-time)

Rajshahi University of Science and Technology, Rajshahi, Bangladesh

Department of Physics

February 2001– April 2001

University Administrative Services

1. Member, Self-Assessment Committee (PSAC), B.Sc. and M.S. in Biotechnology Programs, The Institutional Quality Assurance Cell (IQAC), Department of Mathematics & Natural Sciences (MNS), School of Data and Sciences (SDS), BRAC University, Dhaka, Bangladesh. March 2022-present

2. Chairperson (Acting), Department of EEE and department of ETE, Southeast University, Dhaka, Bangladesh July-August 2013. Responsible for day to day smoothly functioning of the two departments EEE and ETE. In addition, provide assistance with the EEE department faculty team in the academic preparation and proper functioning of the required theory courses, experimental laboratories, departmental infrastructure for the then upcoming UGC accreditation for EEE program. The B.Sc. in EEE program obtained UGC approval and accreditation soon after this.3. Member of the Internal Quality Assurance Cell (IQAC), Southeast University, Dhaka, Bangladesh May 2013-October 2014. Responsible with the 11 member IQAC team to prepare and implement modern methods of teaching, learning and examination guidelines in various disciplines all campus, arrange for feedback responses from students and faculty members for quality-related institution processes, all campus mid-term and final exam venue invigilation and provide feedback report, and prepare focused annual quality assurance report (AQARs) for submission to Vice Chancellor on the quality.4. Convener of the seminar/workshop organizing committee for the EEE, CSE and ETE departments, Southeast University, Dhaka, Bangladesh Summer 2011-Fall 2015. Responsible for organizing seminars, inviting noted academic personalities as a speaker, budget allocations, overall seminar setup and processing.5. Convener of the Admission test committee for the B.Sc. in EEE, CSE and ETE programs, Southeast University, March 2012-Spring 2016. In charge of question paper setting, invigilation of admission exams, grading exam scripts and finalizing results.6. Member, question moderation committee, depts. of EEE & ETE, Southeast University, Summer 2013-Spring 2016. Responsible with

the team for moderating questions of faculties for mid-terms and final examinations.7. Member of Institute of Research & Training (IRT), Southeast University, Summer 2014-Spring 2016.8. Member of the In-house research works on Open Credit System qualitative & quantitative analysis “Introduction of OCS system at Southeast University”.

TEACHING PREPARATION

1. Workshop for 21st century biology education. Participated in a one-week workshop on bridging mathematics, computing, and quantitative science with biology for undergraduate education organized by National Center for Supercomputing Applications, University of Illinois at Urbana-Champaign, IL, USA during July 22-28, 2007.2. Center for teaching excellence and CTEN workshops. Attended several workshops offered by center for teaching excellence (CTE) and the college teaching effectiveness network (CTEN), UIUC, USA related to effective teaching. Some of them are: Test construction workshop, CTEN teaching a course on your own, IEF and ICEF workshop, CTEN effective online teaching, CTE microteaching, CTE getting your syllabus ready for next semester, CTE clicker technology, TA orientation, International ITA orientation during the periods 2005-2009.3. ACES Teaching College Course. The Teaching College Course is a 10-week faculty development program that is designed to assist the University faculty members and selected merit graduate students at the College of Agricultural, Consumer and Environmental Sciences (ACES), University of Illinois at Urbana-Champaign, IL, USA in designing and delivering effective instruction during the period August- December 2006.4. Charm School, Spring 2009. The Technology Entrepreneur Center’s (TEC's) Spring Charm School included the following topics: academic business travel & international business etiquette, business communication & writing, the art of public speaking , business lunch etiquette , how to handle difficult conversations, and life plan. I-Hotel and Conference Center, Champaign,

IL, USA on Saturday, April 4, 2009.

RESEARCH TRAINING1. Participated in the Global Enterprise for Micro-Mechanics and Molecular Medicine (GEM4) summer school an extensive course based on lectures offered by top researchers from USA in this field and lab work in four different research labs on campus for cellular biology, molecular biology, nano fabrication, and enabling technology during June 8 - 19, 2009 at University of Illinois Urbana Champaign (UIUC), IL, USA.2. Participated in the Nano Bio-photonics Summer School: an extensive summer school based on Principles of Nano-Biophotonics, Technology and methods of investigation, Current research (e.g. biomolecular sensing, nanoprobes, nonlinear microscopy, nanoscopy, nanoplasmonics) by leading speakers from USA. It consisted of 10 days: 4 hours of lectures per day, 4 hours of laboratory and computing per day on during 1-12 June, 2009 at UIUC, USA.3. Participated in the Center for Cellular Mechanics summer hands-on course on cell mechano-sensitivity: an extensive course based on lectures offered by top researchers in this field and lab work in research labs on campus for cell mechanics and cellular nano-devices during July 30 - August 3, 2007 at UIUC, USA.4. Participated and presented in the Joint ICTP-KFAS Conference on Nanotechnology for Biological and Biomedical Applications: a 1 week conference by leading researchers from around the globe to share their work and to derive synergy during 10-14 October, 2011 at the Abdus Salam International Centre for Theoretical Physics (ICTP), Trieste, Italy.5. Taken lab courses MCB 253 (Experimental techniques in cellular biology) and MCB 301 (Experimental Microbiology) at UIUC, USA as a training for properly performing the experiments in research.6. Participated in the workshop for successful installation, calibration and running of a LINAC at the Delta clinic, Dhaka, Bangladesh led by a specialized team (N. Suntharalingam, Emeritus Professor, Thomas Jefferson University, USA; Prof. Foiz Khan, Dept. of Radiology, Univ. of Minnesota; Leroy J. Humphries, Founder & President, Radiation Measuring Instrumentation & Accessories for Radiation Oncology, Nashville, TN, USA) in Jan-Feb 2001. The LINAC accelerator is being used for treating cancer, and acid burns.7. Attended fifty-one international and national (in USA, Italy, Bangladesh, and online) conferences and workshops which enhanced my experience and exposure to cutting edge research in engineering and science related to my research field: biophysics, biotechnology, radiation health physics, nanotechnology, biological modeling, and biomedical imaging during 1996-2021.

RESEARCH SKILLS• Instruments: Scanning Electron Microscope (SEM), Phase contrast inverted microscope, fluorescent microscope, Particle sizer instruments (dynamic light scattering), Atomic Force Microscope (AFM), plasma devices, vacuum chambers, UV Spectrophotometer, ellipsometer, profilometer, ebeam/thermal evaporator, NanoTech atomic layer deposition, sputterer, mask aligner, electron microscopes, Fourier-transform Infrared Spectroscopy (FTIR), spinner, centrifuge instruments and ultrasonicators.• Biolab skills: bio-hood, mammalian cell culture, human cancer cells culture, experimental techniques in cellular biology and molecular biology, microscope observations, micro-well plate reader, class 100 and 1000 cleanroom use, biosafety level-2 bionanotechnology complex access for experimentations. Had access to user facility at HMNTL, the BioNanotechnology Laboratory (BNL), UIUC, USA which provided 24/7 access to a Biosafety Level 2 facility with all the tools to conduct research experiments at the interface of biology and micro and nanotechnology, also suitable for research with infectious agents or biological research classified at BSL2.Software proficiency: SPSS, R, MATLAB, Mathematica, Maple, LABVIEW.

Other Co-curricular Activities

Introduction to Biotechnology and Genetic EngineeringMedical BiotechnologyBiomedical

Engineering/ElectronicsFundamentals of Nuclear EngineeringHealth PhysicsQuantum MechanicsNuclear Technology and Security Electrical Microcomputer Control Systems and Engineering applicationsTechnical Systems Management coursesModern PhysicsOpticsMechanicsPhysics I: Mechanics, Heat & Thermodynamics, Waves & OscillationsPhysics II: Electricity & Magnetism, Optics, Modern Physics, Quantum MechanicsEnglish for EngineersPhysics in ArchitectureGeneral Physics

Teaching & supervision of Physics and Engineering undergraduate labs (also lab development including purchasing new equipment and setup new experiments)Astronomy (Astronomy observations) labs. This lab is part of the University core-curriculum courses. Astronomy lab consists of both indoor (computer simulation programs) and outdoor (Sky observation with the aid of telescope) activities.Electrical Microcomputer Control Systems LabInternshipGraduate and undergraduate thesis research and projects

1. “Problematic/Excessive phone, computer, and online use among Bangladeshi Nationals amid COVID-19: the role of psychological well-being and pandemic related factors” Munima Haque, Md. Kamrul Hossain, Quamruzzaman Mohammad, Md. Shohag Hossain. International Conference on Physics-2022 (ICP 2022) 19-21 May 2022. Bangladesh Physical Society, Atomic Energy Centre, Dhaka, Bangladesh. Oral presentation2. “Epidemiological Survey on the Adverse Effects of High Frequency Electromagnetic Radiation on Living Systems and Environment from Various Sources, specifically from Cellphone and Cell Towers” Munima Haque, M. Quamruzzaman, Md. Shohag Hossain, Muhammad Abdur Raihan. “International Conference on Electronics and Informatics 2021 (ICEI 2021) 27-28 November 2021. Bangladesh Electronics Society, Atomic Energy Centre, Dhaka, Bangladesh. Oral presentation3. “Health Effects of Electromagnetic Radiation from Cell Phone and Towers: It’s Present Scenario in Bangladesh” Dr. Munima Haque, Dr. M. Quamruzzaman, Md. Shohag Hossain. National Conference on Physics 6-7 August, 2021 Bangladesh Physical Society, Dhaka,

Bangladesh. Oral presentation4. “Effects of electromagnetic radiation from mobile phone and towers: It’s present status in Bangladesh” M. Quamruzzaman, S. Haque, M. Haque; National Conference on Physics 2019, Dhaka University, Dhaka, Bangladesh 07-09 February 2019. Oral presentation5. “Radiation from mobile phone and towers: Its effects on human and environment” M. Quamruzzaman, S. Haque, M. Haque; International Conference on Electronics and ICT, Atomic Energy Centre, Dhaka, 25-26 November 2018. Poster presentation6. “Targeted drug delivery for cancer cells using hybrid nanocarriers” M. Haque, G. L. Liu; International Conference of Genomics, Nanotechnology and Bioengineering 2017, North South University, Dhaka, 14-16 May, 2017. Poster presentation7. M. Haque, M. Quamruzzaman; International Conference on Physics, Bangladesh Physical Society, Atomic Energy Center, Dhaka, March 10-12, 2016. Oral presentation8. M. Haque, M. Quamruzzaman; 2nd International Conference on Medical Physics in Radiation Oncology and Imaging (ICMPROI-2014), Bangladesh Medical Physics Society, Bangabondhu Sheikh Mujib Medical University (BSMMU), Dhaka, Bangladesh, Aug. 20-22, 2014. Oral presentation9. “Non Ionizing Radiation (NIR), its harmful effects especially from mobile/cell phone and towers” M. Quamruzzaman, M.Haque; National Conference on Electronics for Digital Society, held on 26th June 2014, Bangladesh Electronics Society, Atomic Energy Center, Dhaka, Bangladesh. Oral presentation10. “Epidemiological survey of people working in EMF field” M.Haque, M. Quamruzzaman; BPS International Conference on Physics for Energy and Environment, held on 6-8 March 2014, Bangladesh Physical Society, Atomic Energy Center, Dhaka, Bangladesh. Oral presentation11. “Environmental changes due to technological development” M. Haque, M. Quamruzzaman; BAS-TWAS ROCASA Workshop on Initiatives in Science Education, Research and Capacity Building, held on 14-15 September 2013, Bangladesh Academy of Sciences, Agargaon, Dhaka, Bangladesh. Oral presentation12. “Targeted cancer therapy for breast and prostate cancer cells using hybrid nanocarriers” Munima Haque, G. Logan Liu; International Bose

Conference-2013, held on 4th February 2013, at Dhaka, Bangladesh. Oral presentation13. “Effects of electromagnetic field on environment and living systems” M. Quamruzzaman, Munima Haque; International Bose Conference-2013, held on 4th February 2013, at Dhaka, Bangladesh. Poster presentation14. “Epidemiological survey of people working in EMF field exposed to high frequency” M. Quamruzzaman, and M. Haque. Global International conference for engineering, science and technology, held on 28-29 December, 2012 at Dhaka, Bangladesh. 2012 Oral presentation15. “Effect of non-ionizing radiation on environment” Md. Quamruzzaman, Munima Haque; Bangladesh Physical Society the International Conference on Physics of Today, Dhaka, Bangladesh: March 15-17, 2012. Oral presenation16. “A statistical approach to validate causal relationship existent between various conventional risk factors responsible for occurrence of cancer” Munima Haque, Ms. Tanzilah Noor Shabnam, Dr. Habibullah Talukder, Dr. Md. Quamruzzaman, Dr. Mohammad Ali Asgar; Bangladesh Physical Society the International Conference on Physics of Today, Dhaka, Bangladesh: March 15-17, 2012. Poster presentation17. “Biomarker enzyme activated cancer drug delivery via hybrid nanocarriers” M. Haque, G. Logan Liu; Joint ICTP-KFAS Conference on Nanotechnology for Biological and Biomedical Applications, Trieste, Italy: October 10-14, 2011. Poster presentation18. Electromagnetic hyper sensitivity: a case study. M. Quamruzzaman and M. Haque. Bangladesh Electronic Society Conference, 19th May, 2010, AECD, Dhaka, Bangladesh. Oral and abstract19. “Enhanced diagnostic imaging and high precision gene therapy of mesothelioma using nontoxic nanocarriers” M. Haque, G. Logan Liu; Radiation Research Society (RRS) annual meeting, Savannah, Georgia, USA: October 3-7, 2009. Poster presentation20. “Targeted cancer drug delivery using hybrid nanocapsules” M. Haque, G. Logan Liu; Imaging at Illinois Workshop; Beckman Institute, UIUC, IL, USA: October 1, 2009. Poster presentation21. “Novel approach to cancer therapy: targeted drug delivery by hybrid nanocarriers” M. Haque; MNTL internal research symposium, MNTL, UIUC, IL, USA: September 19, 2009.

Oral presentation22. “Biomarker enzyme activated breast and prostate cancer drug delivery via hybrid nanocarriers” M. Haque, F. Cheng, K. Kim, H. Choi, G. L. Liu; Center for Agricultural and Pharmaceutical Nanotechnology (CAPN) workshop at Illinois; MNTL, UIUC, USA: August 31- September 1, 2009. Poster presentation23. “Global practices in C-14 monitoring in nuclear power plants” M. Haque, D. W. Miller; Health Physics Society 54th Annual meeting, Minneapolis, Minnesota: July 12-16, 2009. Oral presentation24. “Enhanced diagnostic imaging and high precision gene therapy of mesothelioma using nontoxic nanocarriers” M. Haque, G. Logan Liu; International Symposium on Malignant Mesothelioma: Omni Shoreham Hotel, in Washington, DC. June 25-27, 2009. Poster presentation25. “Biomarker enzyme activated cancer drug delivery via hybrid nanocapsules” M. Haque, F. Cheng, K. Kim, H. Choi, G. L. Liu; GEM4 summer school; MNTL, UIUC, USA: June 8-19, 2009. Poster and oral26. “Nanoscale aggregation of corn protein: A colloidal approach” M. Haque, K. D. Bhalerao, H. Nguyen; CNST Nanotechnology Workshop; MNTL and Beckman, UIUC, IL, USA. September 4-5, 2008. Poster presentation27. “Analysis of C-14 accumulation at nuclear facilities” M. Haque, D. Miller, Health Physics Society 52nd Annual Meeting, Portland, Oregon, USA: July 8-12, 2007. Oral presentation28. “Deducing the structure of corn prolamin: a thermodynamic approach” M. Haque, K. D. Bhalerao, H. Nguyen; 2007 ASABE Annual International Meeting, Minneapolis, Minnesota: June 17-20, 2007. Presentation29. “Biological nanotechnology applied to zein” M. Haque, K. D. Bhalerao, University of Illinois Urbana-Champaign (UIUC). Engineering Open House (EOH), Digital Computer Laboratory, UIUC, 2007. Poster30. “Nanomaterials from corn” M. Haque, G. Nistala, Q. Wang, G. W. Padua, K. D. Bhalerao; Corn Utilization and Technology Conference, Hyatt Regency Dallas at Reunion, Dallas, Texas, USA. June 5-7, 2006. Poster31. “Dynamics of non-linear soft x-ray emission from a plasma discharge-driven hydride target”: Miley, George H. Miley, Yang Yang, Michael Romer, Munima Haque, Ian Percel, Andrei Lipson, Heinz Hora; American Physical Society (APS), APS March Meeting, March 13-17, 2006.

Presentation32. “Basic research results that do not support the BEIR VII report conclusions regarding the linear-no-threshold risk hypothesis” Bobby R. Scott, Munima Haque, and Jenni Di Palma; 9th International Conference on Environmental Mutagens, San Francisco, USA. Sept.3-8, 2005. Poster

1. “Best Presentation Award” for the oral presentation for the research paper “Epidemiological Survey on the Adverse Effects of High Frequency Electromagnetic Radiation on Living Systems and Environment from Various Sources, specifically from Cellphone and Cell Towers” Munima Haque, M. Quamruzzaman, Md. Shohag Hossain, Muhammad Abdur Raihan. “International Conference on Electronics and Informatics 2021 (ICEI 2021)”, 27-28 November 2021. Bangladesh Electronics Society, Atomic Energy Centre, Dhaka, Bangladesh.2. “Graduate Teacher Certificate” award, 2009, Center for Teaching Excellence, University of Illinois at Urbana-Champaign (UIUC), USA.3. “Steve Harris Outstanding Treasurer Award” for the work performance as the Vice President (Finance) of International Student Council, SIUC in 2003: an award given juried over annually by the Student Development amongst over 410 Registered Student Organizations (RSOs) in SIUC, USA.4. “University Award” for obtaining highest grades in the M.Sc. results from Rajshahi University, Bangladesh.5. Scholarship from Rajshahi University for pursuing M.Sc. in Physics at Rajshahi University, Physics department, Bangladesh.6. Higher Secondary Board Scholarship for pursuing undergraduate studies (B.Sc. in Physics, Honors) in Physics Department, Rajshahi University, Bangladesh.

1. Life member: Bangladesh Medical Physics Society2. Life member: Bangladesh Physical Society3. Life member: Bangladesh Association of Women Scientists4. Life member: American Alumni Association, Bangladesh5. Member: Organization for Women in Science for the Developing World – OWSD6. Member: Scientists for Accurate Radiation Information (SARI)7. Member: American Association for the Advancement of Science8. Member: Global Network of Bangladeshi Biotechnologists GNOBB

A brief synopsis of Dr. Haque’s contributions and the future plans for research are depicted.a) Nanobiotechnology: Nano biomaterials/nanomaterials characterizations/properties and bio applications. Nanocarriers for targeted drug delivery: this project aims for using a novel biodegradable polymer drug nanocarrier that can deliver the medical drug near diseased cancer cells and the extracellular matrix.b) Radiation Biotechnology consists of biological effects of ionizing and non-ionizing radiation on human and animal biosystems, radioactivity in air water soil ecosystems, radioactivity in nuclear medicine wastes, food irradiation biotechnology.c) Environmental Biotechnology spans areas of works on Radioisotope waste management, micro & nanoplastic pollution in ecosystems characterization and bioremediations.d) Health Physics & Medical Physics spans Radon concentration monitoring in environment, Radioisotope accumulation and testing in nuclear power plants, toxic materials measurement and health risk assessment; Dosimetry for radiotherapy treatment.e) Biomedical Data Sciences research revolves around data from biological and medical sciences for computational and biostatistical analysis.

Thesis Students Supervised (B.Sc.)

Munima Haque, PhD

News & Events

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Workshop on Architecture Model Making Held at BRAC University

Nanoscale science is a fascinating world for the upcoming advanced technology-driven century. Keeping that in mind, BRAC University Society for Biotechnology (BUSB) along with the Biotechnology program, BRAC University had arranged the third episode of the Biotechnology Seminar Series on “Nanobiotechnology” last 28th March 2024. The purpose of this seminar was to open the magic door of nanobiotechnology to the participants. This seminar mainly focusing on basic aspects of Nanobiotechnology was attended by MNS Chairperson Professor Dr. Yusuf Haider, faculties and BRACU students from different majors such as Biotechnology, CSE, Microbiology, and Pharmacy.

The speaker Dr. Munima Haque was introduced by Assistant Professor Dr. Jebunnessa Chowdhury of the Biotechnology program. Dr. Munima Haque, Associate Professor of Biotechnology Program, conducted an extremely engaging session on Nanobiotechnology as her research areas involved Nanobiotechnology, Radiation Biotechnology, Environmental Biotechnology, Medical Physics, and Biomedical Data Science.

Dr. Munima completed B.Sc. (honors) and M.Sc. in Physics with minors in Mathematics and Statistics from Rajshahi University, Bangladesh. Then she did another M.S. in Nuclear Radiation Physics from Southern Illinois University (SIU), Carbondale, IL, USA. She obtained her Ph. D. in Agricultural and Biological Engineering from the University of Illinois at Urbana-Champaign (UIUC), Urbana, IL, USA with a thesis titled “A colloidal science approach to characterize the nanoscale aggregation behavior of corn protein zein”. Her postdoctoral research was in the Micro and Nanotechnology Laboratory, University of Illinois at Urbana Champaign, IL, USA in Biomedical Engineering on Targeted Drug delivery using hybrid nanocarriers for breast and prostate cancer detection and treatment in addition to research on gene therapy.

The seminar discueesd that the fusion of nanotechnology and biotechnological advancement unveiled current possibilities of learning to participants. Nanobiotechnology focuses on studying and applying biological and biochemical processes found in nature to create innovative devices such as biosensors for disease detection. Targeted drug delivery is such an approach to create medication for a patient in a way that enhances the concentration of the medication in specific parts of the body compared to others. Moreover, Nanopharmacology promises targeted treatments with minimized side effects by combining nanotechnology with pharmacology and enhancing drug delivery with precision and efficacy.

The discussion flowed from medical applications of nanoparticles to the textile industry, food preservation, the agricultural sector, nanobioelectronics, and many more. The interactive format of the seminar encouraged participants to perform critical analysis and integrate nanomaterials with biological systems.

The technical session enlightened students about the potential of nanobiotechnology and the importance of achieving skills and insights for future biotechnological endeavors.

News Archive >>

Nanobiotechnology seminar organized by BUSB and Biotechnology Program

Related News

Biotechnology seminar hosts Bangladeshi scientist who developed high-yielding “Panchabrihi” rice

Orientation held for Summer 2024 biotechnology students

The biotechnology program and BUSB organized the inaugural Biotechnology Workshop Series on “Understanding and Designing Biosensors”

Orientation held for Spring 2024 biotechnology students

BracU offers excellent facilities for International students to study in Bangladesh, delivering various programs at postgraduate level. Visit the link for detailed admission programs: https://www.bracu.ac.bd/academics/programs/postgraduate

Application deadlines:

BracU offers three semesters each academic year - Spring, Summer and Fall. International students can apply any time during the year.

Application for admission

A member of BracU International Student Recruitment Team will assist you to complete your admission application. We aim to make your admission process as simple and easy as possible. The application and admission instructions of BracU and further information are available at the BRACU website www.bracu.ac.bd

Minimum qualifications for applying

Students have to submit all the academic documents to International Office or can email at info.international-office[at]bracu.ac[dot]bd. Student will be informed about the eligibility after checking all these documents and may proceed for admission.

Documents Required

The academic and other related documents required for admission at BracU which are as follows:

Academic Qualifications for Postgraduate Admission

For detailed postgraduate admissions requirements, please visit the link: https://www.bracu.ac.bd/admissions/postgraduate

Admission Test:

All eligible candidates will have to sit for a written admission test and an interview.

Those who qualify in the written test will be called for interview for final selection.

Written Admission test for the international students will be taken according to current practice, which is through Brac international Offices. Where there is no Brac office, a partnership agreement with local academic institutions will enable the potential student to underrtake the admissions test.

Applicants of EMBA are not required to give written test but must pass a viva for admission.

Interview

All qualified applicants for written admission tests will have to give their viva virtually on specific date and time.

Admission test exemption:

Applicants will gain exemption from the written test if they attain the following GRE/GMAT/ TOEFL / IELTS scores:

Programs

GRE

GMAT

TOEFL

IELTS

MA ENG

(Literature/applied linguistics and elt)

Earlier/Old GRE: 450 verbal/550 analytical/650 quantitative Revised GRE: 300 total for combined Verbal and Quantitative scores

580 (Paper-based test) or 240 (Computer-based test), 94 (Internet-based test - iBT)

Overall band score of 7.0, with sub scores of 6.0 each

MA TESOL

Earlier/Old GRE: 450 verbal/550 analytical/650 quantitative Revised GRE: 300 total for combined Verbal and Quantitative scores

580 (Paper-based test) or 240 (Computer-based test), 94 (Internet-based test - iBT)

Overall band score of 7.0, with sub scores of 6.0 each

MBA

Earlier/Old GRE: 450 verbal/550 analytical/650 quantitative Revised GRE: 300 total for combined Verbal and Quantitative scores

560 total for combined scores

MDS/MPSM

Earlier/Old GRE: 450 verbal/550 analytical/650 quantitative Revised GRE: 300 total for combined Verbal and Quantitative scores

560 total for combined scores

MDM

Earlier/Old GRE: 450 verbal/550 analytical/650 quantitative Revised GRE: 300 total for combined Verbal and Quantitative scores

560 total for combined scores

ECD/MED

Earlier/Old GRE: 450 verbal/550 analytical/650 quantitative Revised GRE: 300 total for combined Verbal and Quantitative scores

560 total for combined scores

Overall band score of 6.5, with sub scores of 6.0 each

MS BIO

550 (Paper-based test) or 213 (Computer-based test), 80 (Internet-based test - iBT)

Overall band score of 6.5, with sub scores of 6.0 each

MS CSE

550 (Paper-based test) or 213 (Computer-based test), 80 (Internet-based test - iBT)

Overall band score of 6.5, with sub scores of 6.0 each

MS EEE

Verbal:

140 and above Quantitative: 150 and above

Analytical:

3 and above

550 (Paper-based test) or 213 (Computer-based test), 80 (Internet-based test - iBT)

Overall band score of 6.5, with sub scores of 6.0 each

[GRE & GMAT scores are considered valid for five years after the test date] [IELTS & TOEFL scores are considered valid for two years after the test date]

Thank you for showing interest in Brac University (BracU). Please complete the contact details box.

Inquiry

Sharmin Akter joined as a lecturer of Mathematics in MNS department at BRAC

University since January 2015. She has completed her school and college education from Singair, Manikganj. Then she has completed her Bachelor’s and Master’s degree in Mathematics from Jahangirnagar University, Savar, Dhaka-1342, Bangladesh. Later she did another Master’s in “Mathematics International” specialization with “Industrial Mathematics: PDE” in 2021 from Technical University of Kaiserslautern, Germany. During her Master’s study she had an opportunity to work as a research assistant from July 2019 to June 2020 in a prestigious research institution named “Fraunhofer-Institution for Techno and Economic Mathematics ITWM, Kaiserslautern, Germany. Her research interest lies in the field of Industrial Mathematics-PDE, Dynamical Systems and Computational Fluid Dynamics (CFD).

Level: 4, Room: 4G35Kha-224 Merul BaddaDhaka 1212. Bangladesh

• B.Sc. (4-year Hons.) in Mathematics, Jahangirnagar University, Bangladesh (Session: 2008-2009) (Result published in 2012)• MS (Thesis) in Mathematics, Jahangirnagar University, Savar, Dhaka-1342, Bangladesh (Session: 2012- 2013) (Result published in 2014)• M.Sc. (Thesis) in Mathematics International (Specialization: “Industrial Mathematics-PDE”), Scholarship: “Mathematics as a key technology scholarship”, Institution: Technical University of Kaiserslautern, 67663, Germany, Session: Winter 2018/2019- Winter 2020/2021 (Result published in 26 March 2021

Journals

Ahmmed, M. U., Akter, S., Akhter, S., & Kamrunnahar. (2018). Space-Time based RLC model for gas flow simulation. Jahangirnagar Journal of Mathematics & Mathematical Sciences (JJMMS), 31, 49–60.

• January 8, 2015 to Present, Lecturer, Institution: BRAC University, 66 Mohakhali, Dhaka-1212, Bangladesh

• July 15, 2019 to June 30, 2020, Part time job as a research assistant, Institution: Fraunhofer–Institution for Techno and Economic Mathematics ITWM, Fraunhofer- Platz 1, 67663, Kaiserslautern, Germany.

• October 1, 2018 to December 31, 2019, Hiwi job (part time job) as a research assistant, Institution: Technical University of Kaiserslautern, Germany.

• MAT101: Fundamentals of Mathematics• MAT103: Basic Concepts in Mathematics• MAT104: Mathematics• MAT110: Differential Calculus and Coordinate Geometry• MAT120: Integral Calculus and Differential equations• MAT124: FORTRAN Programming• MAT203: Matrices, Linear Algebra and Differential Equations• MAT211: Calculus II • MAT212: Linear Algebra• MAT215: Complex variables and Laplace Transformation• MAT216: Linear Algebra and Fourier Analysis• MAT091: Basic Course in Mathematics • MAT092: Remedial Course in Mathematics

• “Flow behaviors simulations by dint of lumped parameter model” Sharmin Akter, Mahtab U. Ahmmed Published in International Conference on Advances in Computational Mathematics (ICACM), May 27-28, 2017, University of Dhaka, Bangladesh.• Special delegate of One Young World (OYW)-2013 from Bangladesh organized in Johannesburg, South Africa from October 2-6

• “Mathematics as a key technology scholarship” by Department of Mathematics of TU Kaiserslautern from 1/10/2018 to 30/9/2019.• “City Scholarship for young women” by City Foundation through Grameen Shikhkha Scholarship Management Program from June 2013 to May 2014 for excellent performance in master’s level of University Study.• “Shirin Merali Scholarship” from Honor’s third year to fourth year of undergraduate studies• Ivo and Evelyn Duvall’s Scholarship in Higher Secondary level of studies• Small planet fund scholarship from 2004 -2006 in Secondary level of studies

• Member of Bangladesh Mathematical Society (BMS)

• Industrial Mathematics-PDE, Dynamical Systems and Computational Fluid Dynamics

Sharmin Akter

News & Events

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EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Muhammad Shahnoor Rahman completed his school from Dakhin Banasree Model High School and college from National Ideal College. He completed his Honors From University of Dhaka in 2016 and Masters from the same institute in 2017.

Level: Kha-224 Merul BaddaDhaka 1212. Bangladesh

a. BSc. in Physics -Graduated in 2016b. MSc. In Physics -Graduated in 2017

Journals

Rahman, M. S., & Hassan, M. K. (2019). Redefinition of site percolation in light of entropy and the second law of thermodynamics. Physical Review E, 100(6). https://doi.org/10.1103/PhysRevE.100.062109

PrePrint

Rahman, R., Nowrin, F., Rahman, M. S., Wattis, J. A. D., & Hassan, M. K. (2020). Stochastic fractal and Noether's theorem. ArXiv.Org. https://arxiv.org/abs/2010.07953

a. Vice Chancellor's Fellow on Brac University from January 2020 to September 2020b. Bose Research Fellow at Bose Center at University of Dhaka from October 2019 to February 2020

a. Principles of Physics I (Mechanics), II (Electromagnetism and Introduction to Modern Physics)b. Fundamentals of Mathematics

a. In SPARQ seminar on January 2020 at Brac University.

a. Non-Equilibrium Physicsb. High Energy Physicsc. Application of Artificial Intelligence and Machine learning on Physics.

Muhammad Shahnoor Rahman

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Science Exhibition 2019: an inter-university and college scientific project exhibition program organized by Department of Mathematics and Natural Sciences in the university premises to motivate students to involve themselves more into learning about basic sciences and their applications. The event is going to take place on Sunday, February 17, 2019, at BRAC University Auditorium (UB10005).

Science Exhibition - 2019

Dr. Jebunnesa Chowdhury is a researcher with prevalent experience on plant genetic modification and modern mutation breeding. Dr. Chowdhury is currently working as an Assistant Professor, Biotechnology Program in the Dept. of Mathematics and Natural Sciences, Brac University. She joined at Brac University as a lecturer at 2010. Before joining at BracU she worked as a Research Associate in a USDA funded project for the improvement of the grain legume in the Dept. of Botany, University of Dhaka. She also worked as a Research Student in the department of Biochemistry and Molecular

Biology, University of Dhaka. Dr. Chowdhury received two prestigious fellowships from the Government of Bangladesh under the Ministry of Science and Information technology. The first one is NST fellowship for her MS research and the other one is Bangabandhu fellowship for her Ph.D. research. She also received EMMA staff fellowship where she had the opportunity to work at Heidelberg University and Max Planck Institute of Plant Breeding, Cologne, Germany. She received hands on training from ICGEB India and ICGEB Italy, on Plant Biotechnology and risk assessment and biosafety. Dr. Chowdhury is a fellow of JSPS and member of many scientific communities of her discipline. Dr. Chowdhury is working for the agricultural development of Bangladesh. Her research is involved in food security and molecular plant biotechnology coupling with marker assisted mutation breeding.

Level: 4, Room: 4G41Kha-224 Merul BaddaDhaka 1212. Bangladesh

• MS in Biotechnology (Brac University, Bangladesh)• Masters in Botany (National University, Bangladesh)• B. Sc (Honors) in Botany (National University, Bangladesh)

Journal

Islam, A., Chowdhury, J., & Seraj, Z. (2010). Establishment of optimal conditions for an Agrobacterium-mediated transformation in four tomato (Lycopersicon esculentum Mill.) varieties grown in Bangladesh. Journal of Bangladesh Academy of Sciences, 34(2), 171–179. https://doi.org/10.3329/jbas.v34i2.6863

Conference

Chowdhury, J., & Islam, A. (2012). A comparative study on in-vitro regeneration frequency of four locally grown popular tomato (Lycopersicon esculentum Miller) varieties of Bangladesh. In 2nd Annual International Conference on Advances in Biotechnology (BioTech 2012) Proceedings (pp. 26–30). https://doi.org/10.5176/2251-2489_BioTech42

• Senior Lecturer Biotechnology Program• Advisor - Biotechnology Program• Co-Advisor - BRACU Natural Science Club• Member of the Editorial Board of the journal ‘ Crop and Pasture Science’. This Journal was formerly known as Australian Journal of Agricultural Research, Publisher Commonwealth Scientific and Industrial Research Organization (CSIRO).

i. Introduction to Biology, BIO 101ii. Plant Biotechnology, BTE 103iii. Biotech lab I, BTE 155iv. Bioorganic Chemistry, BTE 201v. Introduction to molecular biology, BTE 203vi. GMOs and Biosafety, BTE 402vii. Enzyme Biotechnology, BTE 417viii. Basic Biochemistry, BCH 101ix. Biophysical Chemistry, BCH 102x. Enzyme and Enzyme kinetics, BCH 202xi. Microbial Chemistry, MIC 201xii. Introduction to chemistry, CHE 101xiii. Principles of Chemistry, CHE 110

Presentations at Conferences, Workshops and Lectures I. “GM crops and Biosafety regulation in Bangladesh: Present status and future perspective”. Workshop on “Risk Assessment: The Role of Science in GMO Decision-Making” 30 June - 4 July 2014, in the International Centre for Genetic Engineering and Biotechnology (ICGEB), Trieste, Italy.II. “Comparison between the regeneration responses of five different explants of local mungbean (Vinga radiata (L.) Wilczek) varieties regarding ten different media.” In “Annual Plant Tissue Culture & Biotechnology Conference 2012” held on November 30 – 01 December, 2012, at Shajalal University of Science and Technology, Sylhet, Bangladesh.III. “A Comparative study on in-vitro regeneration frequency of four locally grown popular tomato (Lycopersicon esculentum Miller) varieties of Bangladesh” at “2nd Annual International Conference on Advances in Biotechnology (BioTech 2012)” held on 12-13th March 2012, Bangkok, Thailand.IV. "Development of Transgenic tomatoes" (MS thesis work) on a seminar organized by the pharmacy dept. Heidelberg University, Germany (September 10, 2011).   V. "Regeneration protocol for Bangladeshi tomato varieties and its use in German tomato varieties" on a seminar held on September 25, 2011, Max Planck Institute of Plant Breeding, Cologne, Germany . VI. “Comparison of regeneration frequency of popular locally grown tomato (Lycopersicon esculentum Miller) varieties with commercial Indian variety.” In “Annual Botanical

Conference 2009" held on January 9-10, 2010, Chittagong, Bangladesh.VII. "Establishment of in-vitro regeneration protocol in locally grown tomato (Lycopersicon esculentum Miller)." In "Young Scientists Congress- 2009" held on (29th December, 2009) Dhaka, Bangladesh. VIII. "Establishment of a simple efficient in vitro regeneration protocol for locally grown tomato (Lycopersicon esculentum Miller) cultivars of Bangladesh." in Fourth International Botanical conference 2008 held on (16-18th January, 2009) Dhaka, Bangladesh. Participation in Seminar/Conference/Workshops:I. Stakeholder Workshop on the Finalization of the Guidelines of Environmental Risk Assessment (ERA) of Genetically Engineered Crops held on 20 August 2014, organized by Department of Environment(DOE), & South Asia Biosafety Program (SABP) in BRAC Centre Inn, Dhaka, Bangladesh. II. Workshop on "Risk Assessment: The Role of Science in GMO Decision-Making" organized by the International Centre for Genetic Engineering and Biotechnology (ICGEB) on 30 June - 4 July 2014, Trieste, Italy.III. 7th International Plant Tissue Culture & Biotechnology Conference, held on 1st-3rd March, 2014, Dhaka, Bangladesh.IV. Seminar entitled "New opportunities in Rice Genetics" on 5th December, 2012 at BRAC University.V. "Annual Plant Tissue Culture & Biotechnology Conference 2012" held on November 30 – 01 December, 2012, Sylhet, Bangladesh.VI. Seminar on National Industrial Biotechnology Guideline (5th April, 2012) organized by Ministry of industries (MoInd), GoB and BRAC University (BRACU), BRACU.VII. 2nd Annual Conference on Advances in Biotechnology (BioTech 2012) 2012, held on 12-13th March, 2012, Bangkok, Thailand.VIII. "Development of Transgenic tomatoes" (MS thesis work) on a seminar (September 10, 2011) organized by the dept. of Pharmacy, Heidelberg University, Germany . IX. "Regeneration Protocol for Bangladeshi tomato varieties and its use in German tomato varieties" September 25, 2011, Max Planck Institute of Plant Breeding, Cologne, Germany . X. "5th International Botanical Conference, 2011", held on 9-11 December 2011, at the department of Botany, University of Dhaka, Bangladesh.XI. "Annual

Botanical Conference 2009” held on January 9-10, 2010, Chittagong, Bangladesh. XII. Theoretical and Practical Course on “Transgene Expression in plants” from International Center for Genetic Engineering and Biotechnology (ICGEB), New Delhi, India (03/11/2008 to 14/11/2008).XIII. Seminar on Bioinformatics prospects and opportunities for Bangladesh, held on 5th August 2008, organized by Bangladesh Academy of Sciences, on Center of Excellence, University of Dhaka.XIV. Plant Tissue Culture & Biotechnology Conference 2008, On the theme “Opportunities and Challenges of Agricultural Biotechnology in Developing Countries”, held in April 11-13, 2008 organized by Bangladesh Association for Plant Tissue Culture & Biotechnology, Department of Botany, University of Dhaka. Bangladesh.XV. Seminar and Workshop on “Use of NCBI Resources for Research Advancement in Bangladesh” on 29th May, 2007 Organized by Department of Biochemistry and Molecular Biology, Centre of Excellence and Institute of Information Technology, University of Dhaka, Bangladesh.XVI. Conference on “Promotion of Biotechnology in Bangladesh: National and International Perspectives” held on 6-8 April, 2007, Organized by BRAC University, Ministry of Science and Information & Communication Technology, Bangladesh (MoSICT), Dhaka University, Bangladesh Academy of Sciences etc.XVII. Plant Tissue Culture & Biotechnology Conference 2006, held on March 09-10, 2007 Organized by Bangladesh Association for Plant Tissue Culture & Biotechnology (BAPTC & B), Department of Botany, University of Dhaka, Bangladesh.

i. ERASMUS MUNDUS MOBILITY WITH ASIA PROGRAM (EMMA) Staff Fellowship : (EMMA staff mobility to University of Heidelberg, Germany) 7 September 2011 to 7 October 2011.ii. National Science and Information & Technology (NSICT), Fellowship : Provided by the Ministry of Science and Information & Technology Government of Bangladesh from 2008-2009 (For MS thesis).

• Life Member: Bangladesh Botanical Society (BBS), Department of Botany, University of Dhaka, Bangladesh.• Member: Bangladesh Association for Plant Tissue Culture and

Biotechnology (BAPTC&B), Department of Botany, University of Dhaka, Bangladesh.• Member: Young Biotechnologist of Bangladesh (YoungBB).

• Plant Biotechnology• Molecular Biology and Genetic Engineering• Basic Biochemistry and Secondary Metabolites• GMOs, Biosafety and Risk Assessment

Jebunnesa Chowdhury, PhD

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Shabeeb completed his BSc in Physics from University of Dhaka in 2018.Shabeeb's current research activities include investigations into effective field theory techniques; particularly in the context of interacting higher-spin fields.

Level: Kha-224 Merul BaddaDhaka 1212. Bangladesh

BSc. in Physics-Graduated in 2018

PHY110 Mechanics and Properties of MatterPHY112 Principles of Physics II

Theoretical physics; particularly high energy theory and cosmology

Muhammad Shabeeb Ameen

News & Events

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Workshop on Architecture Model Making Held at BRAC University

Durdana Mahin Priom is a newly appointed Lecturer in the Microbiology Program of the Department of Mathematics and Natural Sciences of BRAC University from January, 2024. Before joining BRAC University, she worked as a Research Fellow in Virology Laboratory, Infectious Diseases Division, icddr,b.Durdana Mahin Priom completed her school and college from Viqarunnisa Noon School and College. She graduated from Microbiology, University of Dhaka securing Third position, in 2020 and completed M.S., securing First position in Microbiology from the same institution in 2021. Her thesis was about a Comprehensive Study on Bacterial Strains Isolated from Buriganga River Sediment for the Biodegradation of Synthetic Dyes and Polycyclic Aromatic Hydrocarbons. Along with this, she was involved in multiple projects on assessing Prevalence of Multidrug Resistant Gram-Negative Clinical pathogens along with whole genome sequencing analysis and microRNA expression analysis of COVID-19 patients.From her early years to university, she has participated in co-curricular and extracurricular activities. As a dedicated member of the Bangladesh Society of Microbiologists, she organized numerous scientific conferences, competitions, and cultural events.Her goal is to inspire students while adding value to life science research through her innovative research ideas and engaging teaching style.

Level: Kha-224 Merul BaddaDhaka 1212. Bangladesh

Introduction to Chemistry (CHE101)Basic Biochemistry (BCH101)Food Microbiology

(MIC302)Microbial Lab III (MIC-355)

• Environmental Microbiology• Antimicrobial resistance profiling• Virology• Immunology• Molecular Biology• Bioinformatics (Metagenome, whole genome)

Durdana Mahin Priom

News & Events

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BRAC University Annual Reports are avilable in this section in pdf format for download.

Md. Salman Shakil completed his studies at Jahangirnagar University, Bangladesh where he earned his BSc and MSc degrees (Biochemistry and Molecular Biology). He pursued his research Master's in Pharmacology at the University of Otago, New Zealand. In October 2021 he joined Brac University as a lecturer in the Microbiology Program at the Department of Mathematics and Natural Sciences. Shakil's research uses a combination of approaches from materials science and engineering, pharmacology, cell signalling, and toxicology.

Level:Kha-224 Merul BaddaDhaka 1212. Bangladesh

BSc with Honours in Biochemistry and Molecular Biology, Jahangirnagar University MSc in Biochemistry and Molecular Biology, Jahangirnagar University MSc in Pharmacology, University of Otago

Journals

Mahmud, K. M., Hossain, M. M., Polash, S. A., Takikawa, M., Shakil, M. S., Uddin, M. F., Alam, M., Ali Khan Shawan, M. M., Saha, T., Takeoka, S., Hasan, M. A., & Sarker, S. R. (2022). Investigation of antimicrobial activity and biocompatibility of biogenic silver nanoparticles synthesized using syzigyum cymosum extract. ACS Omega, 7(31), 27216–27229. https://doi.org/10.1021/acsomega.2c01922

Halder, S. K., Ahmad, I., Shathi, J. F., Mim, M. M., Hassan, M. R., Jewel, M. J. I., Dey, P., Islam, M. S., Patel, H., Morshed, M. R., Shakil, M. S., & Hossen, M. S. (2022). A comprehensive study to unleash the putative inhibitors of serotype2 of dengue virus: Insights from an in silico structure-based drug discovery. Molecular Biotechnology. https://doi.org/10.1007/s12033-022-00582-1

Khandker, S. S., Alam, M., Uddin, F., Shapla, U. M., Lubna, N., Mazumder, T. A., Marzan, M., Mondal, M., Khalil, M. I., Karim, N., Shakil, M. S., & Hossen, M. S. (2022). Subchronic toxicity study of alternanthera philoxeroides in Swiss albino mice having antioxidant and anticoagulant activities. Journal of Toxicology, 2022. https://doi.org/10.1155/2022/8152820

Shawan, M. M. A. K., Sharma, A. R., Bhattacharya, M., Mallik, B., Akhter, F., Shakil, M. S., … Chakraborty, C. (2021). Designing an effective therapeutic siRNA to silence RdRp gene of SARS-CoV-2. Infection, Genetics and Evolution, 93. https://doi.org/10.1016/j.meegid.2021.104951

Shakil, M. S., Parveen, S., Rana, Z., Walsh, F., Movassaghi, S., Söhnel, T., … Hartinger, C. G. (2021). High antiproliferative activity of hydroxythiopyridones over hydroxypyridones and their organoruthenium complexes. Biomedicines, 9(2), 1–18. https://doi.org/10.3390/biomedicines9020123

Hasan, M. M., Uddin, M. F., Zabin, N., Shakil, M. S., Alam, M., Achal, F. J., … Morshed, M. M. (2021). Fabrication and characterization of chitosan-polyethylene glycol (Ch-Peg) based hydrogels and evaluation of theirpotency in rat skin wound model. International

Journal of Biomaterials, 2021. https://doi.org/10.1155/2021/4877344

Niloy, M. S., Hossain, M. M., Takikawa, M., Shakil, M. S., Polash, S. A., Mahmud, K. M., … Ranjan Sarker, S. (2020). Synthesis of biogenic silver nanoparticles using caesalpinia digyna and investigation of their antimicrobial activity and in vivo biocompatibility. ACS Applied Bio Materials, 3(11), 7722–7733. https://doi.org/10.1021/acsabm.0c00926

Shakil, M. S., Hasan, M. A., Uddin, M. F., Islam, A., Nahar, A., Das, H., … Hoque, S. M. (2020). In Vivo toxicity studies of chitosan-coated cobalt ferrite nanocomplex for its application as MRI contrast dye. ACS Applied Bio Materials. https://doi.org/10.1021/acsabm.0c01069

Ali, M. Y., Kabir, A., Khandker, S. S., Hossan, T., Shakil, M. S., Islam, M. A., … Khalil, M. I. (2019). Assessment of toxicity and therapeutic effects of goose bone in a rat model. Journal of Chemistry, 2019. https://doi.org/10.1155/2019/1943601

Review Articles

Shakil, M. S., Bhuiya, S. M., Morshed, R. M., Babu, G., Niloy, S. M., Hossen, S. M., & Islam, A. M. (2022). Cobalt ferrite nanoparticle’s safety in biomedical and agricultural applications: A review of recent progress. In Current Medicinal Chemistry (Vol. 30, pp. 1–20). https://doi.org/10.2174/0929867329666221007113951

Shakil, M. S., Niloy, M. S., Mahmud, K. M., Kamal, M. A., & Islam, M. A. (2022). Theranostic potentials of gold nanomaterials in hematological malignancies. Cancers, 14(13). https://doi.org/10.3390/cancers14133047

Shakil, M. S., Rana, Z., Hanif, M., & Rosengren, R. J. (2022). Key considerations when using the sulforhodamine B assay for screening novel anticancer agents. Anti-Cancer Drugs, 33(1), 6–10. https://doi.org/10.1097/CAD.0000000000001131

Rumon, M.M.H.; Akib, A.A.; Moniruzzaman, M.; Sultana, F.; Shakil, M. S.; Roy, C.K.; Recent advancements in self-healable polymeric materials: their applications and future perspectives as a biomedical material. Polymers 2022, 14, 4539.

Shakil, M. S., Mahmud, K. M., Sayem, M., Niloy, M. S., Halder, S. K., Hossen, M. S., … Hasan, M. A. (2021). Using chitosan or chitosan derivatives in cancer therapy. Polysaccharides, 2(4), 795–816. https://doi.org/10.3390/polysaccharides2040048

Mahmud, K. M., Niloy, M. S., Shakil, M. S., & Islam, M. A. (2021). Ruthenium complexes: An alternative to platinum drugs in colorectal cancer treatment. Pharmaceutics, 13(8). https://doi.org/10.3390/pharmaceutics13081295

Rayhan, M. A., Hossen, M. S., Niloy, M. S., Bhuiyan, M. H., Paul, S., & Shakil, M. S. (2021). Biopolymer and biomaterial conjugated iron oxide nanomaterials as prostate cancer theranostic agents: A comprehensive review. Symmetry, 13(6). https://doi.org/10.3390/sym13060974

Niloy, M. S., Shakil, M. S., Hossen, M. S., Alam, M., & Rosengren, R. J. (2021). Promise of gold nanomaterials as a lung cancer theranostic agent: A systematic review. International Nano Letters, 11(2), 93–111. https://doi.org/10.1007/s40089-021-00332-2

Niloy, M. S., Shakil, M. S., Alif, M. M. H., & Rosengren, R. J. (2021). Using natural compounds to target KRAS mutated non-small cell lung cancer. Current Medicinal Chemistry. https://doi.org/10.2174/0929867328666210301105856

Khandker, S. S., Shakil, M. S., & Hossen, M. S. (2020). Gold nanoparticles; potential nanotheranostic agent in breast cancer: A comprehensive review with systematic search strategy. Current Drug Metabolism, 21(8), 579–598. https://doi.org/10.2174/1389200221666200610173724

Shakil, M. S., Hasan, M. A., & Sarker, S. R. (2019). Iron oxide nanoparticles for breast cancer theranostics. Current Drug Metabolism, 20(6), 446–456. https://doi.org/10.2174/1389200220666181122105043

Conferences

Shakil, M. S.; Hoque, S. M., Nahar, A.; Uddin, M. F.; Hasan, M. A.; In vitro and in vivo toxicity studies of chitosan-coated cobalt ferrite nanocomplex for the therapeutic purpose of cancer. In: International Conference on Physics; 2018 Mar. 08-10; Dhaka,

Bangladesh. Abstract #NM19

Mahmud, K. M.; Hossain M. M.; Polash S. A.; Takikawa M.; Shakil M. S.; Uddin M. F.; Alam M.; Shawan M. M. A. K.; Saha T.; Takeoka S.; Hasan M. A.; Sarker S. R.; Investigation of antimicrobial activity and biocompatibility of biogenic silver nanoparticles synthesized using Syzygium cymosum extract. In: International Conference on Physics; 2022 May. 19-21; Dhaka, Bangladesh. Abstract #NM 03

October 2022 - Present, Academic Editor, Plos OneOctober 2021 – December 2022, Lecturer, Microbiology Program, Department of Mathematics and Natural Sciences, Brac University, Dhaka, BangladeshJune 2021 – October 2021, Lecturer, Department of Biochemistry and Molecular Biology, Primeasia University, Dhaka, BangladeshOctober 2020 – December 2020, Assistant Research Fellow, Department of Pharmacology and Toxicology, University of Otago, New ZealandFebruary 2020 – June 2020, Demonstrator, Department of Pharmacology and Toxicology, University of Otago, New ZealandOctober 2018 – June 2019, Quality Control Executive, Quality Assurance Department, Eskayef Pharmaceuticals Limited, BangladeshJanuary 2015 - September 2018, Research Assistant, Department of Biochemistry and Molecular Biology, Jahangirnagar University, Bangladesh

Basic Biochemistry, Biophysical Chemistry, Introduction to Chemistry, Principals of Chemistry

Shakil, M. S.; Hoque, S. M., Nahar, A.; Uddin, M. F.; Hasan, M. A.; In vitro and in vivo toxicity studies of chitosan-coated cobalt ferrite nanocomplex for the therapeutic purpose of cancer. In: International Conference on Physics; 2018 Mar. 08-10; Dhaka, Bangladesh. Abstract #NM19

Member of Alumni Association of University of Otago, 2020Member of Alumni Association of Jahangirnagar University, 2015Member of Graduate Biochemist Alumni Association of Bangladesh, 2015

Nanomedicine, Cancer Biology, Pharmacology, Drug Delivery, Antimicrobial Resistance

Md. Salman Shakil

News & Events

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- See more at: http://www.bracu.ac.bd/admissions/postgraduate-programs#sthash.roKby67c....• Master of Business Administration (MBA)• Masters in Development Management and Practice (MDMP)• Master of Development Studies (MDS)• Master of Computer Applications (MCA)• Master of Science in CSE • Master of Arts in Teaching English to Speakers of Other Languages (TESOL)

Below are sample questions (program wise):

The Academic Council of BRAC University in its 17th meeting held on August 24,2005 approved the programme of minor in physics for students doing a major in CS, CSE, ECE or any other relevant discipline. Such major-minor combination will stand the students in good stead in the job market.

The details of the Minor in Physics programme are given below.

Total credit requirement: 27 credits There will be seven compulsory courses, each of three credits, which are as follows:

PHY 111: Principles of Physics IPHY 112: Principles of Physics IIPHY 204: Classical Mechanics & Special Theory of RelativityPHY 205: Statistical MechanicsPHY 210: Quantum Physics of Atoms, Solids & NucleiPHY 301: Classical ElectrodynamicsPHY 305: Quantum Mechanics III

Students may choose any two courses from the following list of elective courses each of three credits offered by the Department.

PHY 302: Fluid MechanicsPHY 308: Methods of Experimental Physics & InstrumentationPHY 309: Introduction to Materials SciencePHY 311: X-RaysPHY 313: Physics for DevelopmentPHY 409: Physics of RadiologyPHY 403: Cosmology & Astrophysics

or any other physics course with the permission of the Chairperson of the MNS Department.

Aksa Hossain Nizhum attained her secondary and higher secondary education from Joypurhat Girls' Cadet College. She earned her Bachelor of Science (Honours) and Master of Science degrees in Microbiology from the Department of Microbiology, University of Dhaka. Throughout her academic journey, spanning from college to university days, she has been actively engaged in both co-curricular and extra-curricular pursuits. As a conscientious member and organizer within the Bangladesh Society of Microbiologists, she adeptly orchestrated numerous scientific conferences, competitions, and cultural affairs. Acknowledged for her interest in debate and public speaking competitions, she has achieved multiple recognitions in esteemed national debating competions, including televised debate competitions. Driven by an unwavering dedication to pedagogy and groundbreaking research, her aspirations revolve around nurturing students' intellectual curiosity and augmenting the significance of life science research through her innovative ideas and contributions.

Level: 5Kha-224 Merul BaddaDhaka 1212. Bangladesh

She has started her professional life as a lecturer at the Mathematics and Natural Sciences (MNS) department of BRAC University.

Microbial Metabolism (MIC202)Human Physiology (BCH201)Basic Biochemistry (BCH101)

• Antimicrobial Resistance • Bioinformatics (Metagenome)• Environmental Microbiology• Virology

Aksa Hossain Nizhum

News & Events

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EEE student selected for Huawei program in China

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Enrolling is the process of registering for the courses students intend to study for a particular semester. After admission, students will enroll for courses based on the courses offered by the university.

The university offers the following courses for the undergraduate students in addition to General Education (GenEd) courses.

On the basis of the admission test results, a student will have to take one or more of the following courses:

a. ENG 101 (Credit course)b. ENG 102 (Credit course)c. ENG 091 (Non credit course)d. MAT 091 (Non credit course)e. MAT 092 (Non credit course)

The exact number of course(s) a student is required to take will be shown against each name of the student in the test result.

Initial admission fee includes a fee one course. Any additional course up to a maximum of four (4) in total, will incur a fee of Tk. 22,500 (all program except LAW) for each subsequent course.

In case of LAW Tk. 24,000 will be incurred for each subsequent course.

If a freshman is prescribed two (02) non-credit courses, the third course for him/her is optional.

This unique programme is designed for those students who are seeking to perfect and polish their English language skills to become successful BRAC University students. This course is for those students who proved themselves competent in their respective discipline during the admission test except in English.

This twelve weeks long intensive programme which encourages holistic approach based on critical thinking and inquiry is designed by an American fellow Beth Trudell. The goal of this programme is to assist the students to become active and independent learners, critical thinkers, responsible and thoughtful adults and most importantly successful university students. Reading and writing classes in this programme provide opportunities to read, annotate and talk about a variety of texts. It introduces writing as a systematic and creative task. Listening and speaking classes prepare students for real-world speaking situations such as solving problems, exchanging information and describing their experiences. Critical thinking and enhancement classes add a new creative element. These multi skilled classes based on specific interest provide exposure and help the students to reason, converse and build analytical skills. The students who come from mainstream Bangla medium background enjoy this intellectually challenging and welcoming atmosphere.

This course is designed for the intermediate level students to improve all four language skills- listening, speaking, reading and writing. The course is divided into two parts, each dealing with one pair of skills- the speaking and listening is one pair and the reading and writing is the other. The students are introduced to strategies of each skill and are expected to use them in and beyond the course. They practice reading both fiction and non-fiction, writing in different genres, listening to tracks and presenting to audience.

The materials for the course are developed in line with the students' present knowledge and the level they are expected to achieve. Besides, active participation of the students in the class proceedings is encouraged and enforced to create better effect of learning.

This course is designed for the students at an upper intermediate language proficiency level. To sharpen all four language skills, the strand is parted into two pairs of skills: the speaking and listening pair and the reading and writing pair.

This course exposes students to a variety of academic and professional use of language. It enhances students' vocabulary range, grammatical practice, organizational capacity and analytical understanding. It focuses on writing academic papers, analyzing reading materials and grooming students' listening and speaking skills, further emphasizing on strategies of the language skills.

The material of the course is designed in such a way that students can get the utmost benefit to improve themselves. This course, like all of the BIL courses, is learner-centered, and encourages and enforces active student participation.

This course is designed for advanced level students and is therefore focused on developing their capacity of critical reading and academic writing skills. Once again divided in pairs of skills- speaking & listening and reading & writing- this is more professional and academic in design, and concentrates on academic writing, critical and analytical reading and formal presentation. The course is heavy founded on extracting and developing the students' critical reading and critical thinking abilities.

Once again leaning on learner-centered and task-based teaching methodologies, the course materials are developed to expose students to more academic and professional readings. The students are definitely encouraged and enforced to actively participate in the classroom activities.

Topics including sets, relations and functions, real and complex numbers system, exponents and radicals, algebraic expressions; quadratic and cubic equations, systems of linear equations, matrices and determinants with simple applications; binomial theorem, sequences, summation of series (arithmetic and geometric), permutations and combinations, elementary trigonometry; trigonometric, exponential and logarithmic functions; co-ordinate geometry; statics-composition and resolution of forces, equilibrium of concurrent forces; dynamics-speed and velocity, acceleration, equationsof motion. No credit.

The idea and methods of describing a set, finite and infinite sets, empty set, subset, universal set, union and intersection of sets, disjoint set, difference of sets, complement of set, power set, cartesian product of sets, real and complex numbers system, number line, interval notation, absolute value, polynomials, exponents and radicals, algebraic expressions; variable, constant, power, coefficient, factorization, highest common factor, least common multiple, quadratic and cubic equations, variation, relation, functions and their graphs, rational functions and their graphs, simple functions and their graphs, systems of linear equations, matrices and determinants with applications; binomial theorem, sequences, summation of series (arithmetic and geometric), permutations and combinations, elementary trigonometry; trigonometric, exponential and logarithmic functions; calculus (rate of change, derivative) Cartesian coordinate system; various form of linear equations, distance between two points, formulas of area and volume (rectangles, parallelogram, triangles, circles, cylinders, spheres), composition and resolution of forces, equilibrium of concurrent forces; dynamics, statics, speed and velocity, acceleration, equations of motion, Mean (arithmetic, geometry, harmonic), mode, median, sample, variable, data, frequency distribution, graphical presentation, measures of central tendency and measures of dispersion, introduction to probability.

Course details:

Mathematics will also be offered as a minor for students doing a major in physics, CS, CSE, ECE, ESS or any other relevant discipline. Such major-minor combination will stand the students in good stead in the job market. The details of the minor in mathematics programme are given below:

Total Credits Requirement: (27 credits)

There will be seven compulsory courses, each of three credits, which are as follows:

1. MAT 111: Principles of Mathematics2. MAT 211: Calculus II3. MAT 221: Real Analysis I4. MAT 223: Numerical Analysis I5. MAT 311: Abstract Algebra6. MAT 316: Operations Research I7. MAT 322: Differential Equations II

Students may choose any two courses from the following list of elective courses each of three credits offered by the MNS Department.

1. MAT 312: Numerical Analysis II2. MAT 321: Real Analysis II3. MAT 325: Mathematical Methods4. MAT 326: Hydrodynamics5. MAT 411: Topology6. MAT 415: Finite Element Methods7. MAT 416: Tensor Calculus8. MAT 421: Fluid Mechanics9. MAT 422: Theory of Numbers10. MAT 423: Mathematical Modeling11. MAT 424: Operations Research II12. MAT 425: Advanced Numerical Methods13. MAT 490: Special Topics in Mathematics14. STA 301: Modern Probability Theory & Stochastic Processes15. CSE 490: Special Topics (Quantum Computing)

or any other mathematics course with the permission of the Dean of School of Data and Sciences or the Chairperson of the MNS Department.

Mohammad Mastak Al Amin is working as Assistant Professor of Statistics at the Department of Mathematics and Natural Sciences, BRAC University. His latest research interest is ‘population ageing & social policy’. Prior to joining BRAC University he taught more than two years at Eastern University Bangladesh from 2010-2013. He received his Master degree major in Economic Demography from School of Economics and

Management, Lund University, Sweden. He worked as an Assistant Director, Office of Co-curricular Activities (OCA), BRAC University from 2015-2016. He had been working as a Research Associates and Statistician in a research project funded by World Bank and also incorporated in the Consultant and Expert panel in the ARK foundation Bangladesh. He is also involving with British Council Bangladesh since 2011. He has published seven scientific papers in international and national journals with good impact factors.

Level:Kha-224 Merul BaddaDhaka 1212. Bangladesh

Master of Actuarial Science, Department of Banking and Insurance, University of Dhaka, Bangladesh. (2019)Master of Science in Economic Demography, School of Economics and Management, Lund University, Sweden (2010)Bachelor (4 Years) of Science in Statistics, Department of Statistics, University of Dhaka, Bangladesh. (2008)

Selected Publications

Journals

Rakib-Uz-Zaman, S. M., Iqbal, A., Mowna, S. A., Khanom, M. G., Al Amin, M. M., & Khan, K. (2020). Ethnobotanical study and phytochemical profiling of Heptapleurum hypoleucum leaf extract and evaluation of its antimicrobial activities against diarrhea-causing bacteria. Journal of Genetic Engineering and Biotechnology, 18(1). https://doi.org/10.1186/s43141-020-00030-0

Amin, M. M., Sajib, A., & Alamgir, M. (2015). Investigating causal relations of economic variables: Capital, GDP, labour and population in Sweden 1870-2000. Dhaka University Journal of Science, 62(2), 81–86. https://doi.org/10.3329/dujs.v62i2.21970

Amin, M. M. Al, & Sajib, A. H. (2013). Acculturation preferences and major adverse factors that Bangladeshi and finnish migrants faced to acclimatize in Sweden. Jahangirnagar University Journal of Information Technology (JIT), 2(23–32).

Conferences

Amin, M. M. (2010). Factors behind internal migration and migrant’s livelihood aspects:

Dhaka City, Bangladesh. In Second Nordic Conference on South Asian Studies for Young Scholars. Falsterbo, Sweden.

Reports

Amin, M. M. (2010). Aspects at the back of internal migration and exploring the changes in migrant's livelihood after their arrival in Dhaka City, Bangladesh. Finland: Siirtolaisuusinstituutti,Institute of Migration.

• Assistant Professor, Brac University from July 01, 2017 - till date• Program Coordinator, Statistics and Environmental Sciences, Department of Mathematics and Natural Sciences, Brac University from September 01, 2019 - till date• Office of Co-curricular Activities (OCA), Brac University from May 2015 to February 2016• Guest Faculty, Master of Actuarial Science Program, Department of Banking and Insurance, University of Dhaka from July, 2019 – December 2019 • Senior Lecturer, Brac University from October 01, 2015 - June 30, 2017• Lecturer, Brac University from January 13, 2013-September 30,2015• Senior Lecturer, Eastern University Bangladesh from October 25, 2011- January 13, 2013• Lecturer Eastern University, Bangladesh from October 10, 2010- October 24, 2011• Part Time Faculty, Institute of Cost and Management Accountants of Bangladesh (ICMAB) from January 01, 2015 to till date.• Assistant Director, Office of Co-curricular Activities (OCA), BRAC University from May 2015 to February 2016• IELTS Clerical Marker, British Council Bangladesh from May 01, 2012 to June 30, 2014• Invigilator, British Council Bangladesh from May 01,2011to till date.

Graduate Courses: (Master of Actuarial Science, MBA, ICMAB)

Undergraduate Courses:

Amin, M. M. (2010). Factors behind internal migration and migrant's livelihood aspects: Dhaka City, Bangladesh. Second Nordic Conference on South Asian Studies for Young Scholars. Falsterbo, Sweden.

Involvement with Learned Societies:

• Life Member, Lund University Alumni Association, Sweden• Life Member, Alumni Association (DUSSDA), Department of Statistics, Biostatistics and Informatics, University of Dhaka. • Life Member, Registered Graduate, University of Dhaka.• Life Member and Executive committee member, Nordic Alumni Association of Bangladesh (NAAB).• Life Member, Notre Dame Alumni Association (NDAA)• Life Member, Mathematical Society, Bangladesh.• Life Member, National Oceanographic and Maritime Institute (NOAMI), Bangladesh

Organization Activity:• Convener, Final examination scrutinizing committee, Department of Mathematics & Natural Science, BRAC University, Bangladesh, 2017• Member of advisory panel, First Year Advising Team (FYAT), BRAC University, Bangladesh from 2014- till date• Convener, Accommodation and Transportation Sub-committee, 19th International Math Conference, BRAC University, Bangladesh, 18-20 December 2015• Member of Panel of Judges, Hult Prize, BRAC University Chapter, Bangladesh, 2015• Convener, Semester result summary committee, Department of Mathematics & Natural Science, BRAC University, Bangladesh from 2013-2016• Member, Admission Test Committee, Department of Mathematics & Natural Science, BRAC University, Bangladesh, 2015 (Summer Semester)• Member, Organizing Committee, Eastern University Math Olympiad, Dhaka, Bangladesh, 2011• Editor, Publication Committee, Eastern University Math Olympiad, Dhaka, Bangladesh, 2011• Member, Organizing Committee, Golden Jubilee celebrations of Notre Dame College, Dhaka, Bangladesh, 1999Others Involvement:• Adviser, Agamirbangladesh Newspaper• Adviser, A FREE SCHOOL, community non-profit organization

Workshop:1. Workshop on “Democracy as Equality: Why is it important for BRACU-Clubs” with all the club presidents/vice presidents, conducted by BRAC University Office of Co-Curricular Activities (OCA) and Teaching and Learning Centre (TLC), January 14, 2016 at BRAC University auditorium.2. IQAC-TLC workshop on

"Roles, Responsibilities and Ethical Principles of University Teachers", 11th-12th November, 2015 at BRAC University Auditorium.3. Workshop on “Leadership training for BRACU-Clubs”, May 09, 2015 at BRAC CDM Rajendrapur.4. Professional Training of TLC workshop on “Teaching and Learning for Teachers (TLT): Preparing well-grounded Educators”, Conducted by BRAC University, held on May 06-09, 2013.5. Professional Training on “Developing Skills in Research Presentation in an International Workshop”, conducted by Bangladesh Academy of Sciences (BAS), 28th-29th December 2011.6. Professional Training on “Teaching Methods” training conducted by Robert Vance Selby, University of Virginia (USA), held on 6th - 8th January 2011, Eastern University Bangladesh. 7. Workshop arranged by Resource Center, British council Bangladesh for developing skill for teaching methods, May, 2011.

Mohammad Mastak Al Amin

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

EMB101 : Emergence Of Bangladesh

HUM103: Ethics and Culture

ENG101: English Fundamentals

ENG102: Composition 1

FRN101: Basic French

BNG 103: Bangla Language and Literature

An orientation program for first-year undergraduate students of Summer 2024 semester was organized under the Biotechnology Program of the Department of Mathematics and Natural Sciences (MNS) in cooperation with BRAC University Society for Biotechnology (BUSB) at BRAC University on 8 June 2024.

The event started off with Dr Munima Haque, Director of the Biotechnology Program, welcoming the freshers and informing of different teaching, research and academic facilities offered under the program.

Joining virtually, Professor Mahbubul Alam Majumdar, Dean of the School of Data and Sciences, shared opportunities available under the program, ways to collaborate with other programs under the MNS department and how students should always reach out to faculty members for assistance.

A video was screened giving an introduction to the Biotechnology Program. Professor Dr. Aparna Islam, Assistant Professor Dr. Mahdi Moosa and Associate Professor Dr. Iftekhar Bin Naser shared insights from their own research careers.

An alumnus, Rezwanul Kabir, spoke of his time as a student. BUSB President Md. Nafin Rahman enlightened the newcomers about the diversified aspects of both the Biotechnology Program and the student club.

The event ended with a cultural program comprising delightful songs, poetry recitations, and dance performances.

News Archive >>

Orientation held for Summer 2024 biotechnology students

Related News

Biotechnology seminar hosts Bangladeshi scientist who developed high-yielding "Panchabrihi" rice

Nanobiotechnology seminar organized by BUSB and Biotechnology Program

The biotechnology program and BUSB organized the inaugural Biotechnology Workshop Series on "Understanding and Designing Biosensors"

Orientation held for Spring 2024 biotechnology students

The BSRM School of Engineering at BRAC University is a great platform for teaching, learning, research and innovation in engineering education around the region. It is our mission to produce technically competent, socially responsible, best quality engineers with effective communication skills, highest ethical standards and human values.

The BSRM School of Engineering provides knowledge and skill-based learning environment to our students. At present, the school offers two undergraduate and two graduate programs in Electrical and Electronic Engineering, and Electronics and Communication Engineering. All of our undergraduate programs are accredited by the Board of Accreditation for Engineering and Technical Education (BAETE). Our faculty members are actively pursuing interdisciplinary research in engineering and computing, power and energy, robotics and automation, solar and nano-photonics, electronics and communication technology and relentlessly engaging our students to solve some of the society's most pressing and unique challenges in the context of Bangladesh. The school developed the nation’s first Nano-satellite “BRAC Onnesha” that led Bangladesh into space-age.

About BSRM School of Engineering

Office of the Dean

Arshad M. Chowdhury, PhD

Baby Naznin is working as a permanent lecturer at the BRAC University, Dhaka at the Biotechnology Program. She joined the University as a faculty just after finishing her second master’s degree in September 2023. She completed Infectious Diseases and One Health master’s degree with the prestigious Erasmus Mundus Joint Master’s Scholarship. The master’s degree was from three different universities named Université de Tours, France, Universitat Autònoma de Barcelona, Spain, Medizinische Hochschule Hannover, Germany, and the internship at INRAE, Paris. During her internship she did work on the project “Impact of the environmental stress (water quality, pollutants) on fish health from intestinal health to modulation of fish responses to viral infections”. During the master’s program, she did a fully funded summer internship at IRCMS (International Research Centre for Medical Science), Kumamoto University, Japan. Before going for post-graduation, she served as a scientific officer for Covid-19 diagnosis during the pandemic at DNA Solutions Ltd., a renowned molecular diagnostic and research center in Dhaka. She completed her first master’s degree from the department of Biochemistry and Molecular Biology, University of Dhaka standing position 6th among all. She did the thesis project with National Science and Technology (NST) fellowship on “Genetic polymorphism of MDM2-SNP309, Myc8q24 and CYP2E1*5B as a biomarker for the risk of bladder cancer in the Bangladeshi population”. She did complete her Bachelor’s (Hons) degree from the same department from University of

Dhaka having the position 6th.

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Introduction to Biology (BIO-101)Basic Biochemistry (BCH-101)Biophysical chemistry (BCH-102)Basic immunology (BTE-303)

ORAL PRESENTATION

1. Scientific Conference on preventive oncology, 2020- “Polymorphism of XPC and MTHFR gene in the development of bladder Cancer in Bangladesh: A case control study”

2. 4th Young Scientist Congress, 2019- “The Role of Genetic Polymorphism of rs710521 located near TP63 and G4C14 to A4T14 of TP73 as a Biomarker for the Risk of Bladder Cancer in Bangladeshi Population”

CONFERENCE & LEADERSHIP

1. National Youth Leadership Summit-2019 arranged by Dhaka walking Club, 2019

2. A seminar on “Breast Cancer Prevention” organized by Cancer Awareness Foundation, 2019

3. Participant of Daffodil International MUN, 2018

4. Participant at English Olympiad 2018

5. Leadership Training on Inclusive English Society & Scribe Award, 2018

6. Attending World Health Day Rally, 2018

7. National Young Entrepreneurs’ Conference, 2017

8. National Conference on “Biochemistry and Molecular Biology for Life Sciences”, 2016

VOLUNTEERING

1. 14th Asia Pacific Biorisk Conference, 2019

2. The Biggest Birthday Festival-Fuchka Bilash by MASTUL Foundation, 2017

3. Shopnaloron Bangladesh, 2017

4. Souhardo, 2017

5. Ramadan Mubarak-3 hosted by Asha, 2017

I. Erasmus Mundus Joint Master’s Scholarship, 2021-2023II. Fully funded internship from IRCMS in Japan, 2022III. National Science and Technology Fellowship (NST) as 2018-2019 MS research studentIV. Dr. Syed Salehin Qadri Scholarship, 2017V. Professor Dr. Kamaluddin Ahmed Scholarship, 2016VI. Full free govt. scholarship in HSC- 2014VII. Full free govt. scholarship in SSC-2012

I. Secretary of Documentation at Dhaka University Model OIC ClubII. General Member of Dhaka University Carrier ClubIII. Permanent Volunteer at SouhardoIV. Executive member of PDF (Physically challenged Development Foundation), University of DhakaV. Researcher of Bioscience group at Dhaka University Research Society

I. VaccinologyII. ImmunotherapyIII. Cancer BiologyIV. Molecular GeneticsV. Microplastics in environmental pollution

Baby Naznin

News & Events

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Course Structure of Undergraduate B.Sc in Microbiology

A 4-year (12 semesters) Microbiology undergraduate programme has been designed including topics of recent trends in this field with emphasis on applications of microbiology in different areas of socio-economic activities. Once a student has completed this course he/she should be well equipped to fulfill the requirement for jobs in public sectors and private enterprises involved in different microbiology research, development and applications and to face the challenges of new developments in this field.

The programme of study includes courses for improving communication skills, strengthening mathematical background and acquainting the student with socio-economic and historical background of Bangladesh. With this basic background the student should face no problem to find a suitable job in different universities, research and development organizations and private enterprises.

The total credit requirements for the degree of Bachelor of Science in Microbiology are 136. Out of these 21 credits is on general education, which is compulsory as per requirement of the BRAC University. The departmental core courses account for 76 credits including departmental core courses (63 credits) practical (6 credits), Microbiology internship (3 credits) and thesis /project (4 credits). In addition a student has to undertake 12 credits on elective courses and 27 credits from outside the major specialization. For elective and outside major courses a student can choose from several courses outlined in the syllabus in consultation with the department. The student may also be required to take non-credit remedial course in English.

Courses for Bachelor of Science in Microbiology

(a) General education (21 Credits):

1. MAT 101 Fundamentals of Mathematics2. DEV 101 Bangladesh Studies3. ENG 091 Foundation Course (non credit)4. ENG 101 English Fundamentals 5. CSE 101 Introduction to Computer Science6. ENG 102 Composition 17. HUM 103 Ethics and Culture8. BIO 101 Introduction to Biology

(b) Departmental Core Courses (63 Credits):

1. MIC 101 General Microbiology2. BCH 101 Basic Biochemistry3. BCH 102 Biophysical

Chemistry4. MIC 102 Basic techniques in Microbiology 5. MIC 201 Microbial Chemistry6. BCH 201 Human Physiology7. MIC 202 Microbial metabolism8. MIC 203 Environmental Microbiology9. MIC 204 Medical Microbiology10. BTE 207 Introduction to Molecular Biology11. MIC 301 Virology12. BCH 301 Basic Immunology13. BTE 307 Advanced Molecular Biology14. MIC 302 Food Microbiology15. MIC 303 Agriculture Microbiology16. MIC 306 Pharmaceutical Microbiology17. MIC 308 Fermentation Technology18. BTE 401 Bioinformatics19. MIC 401 Microbial Genetic Engineering20. MIC 402 Analytical Microbiology21. MIC 403 Microbiological Quality Control of Food, Fish and Beverages

(c) Elective Courses (12 Credits):

1. MIC 304 Microbial Biotechnology2. MIC 307 Microbiology of Frozen Food and Fish3. BTE 315 Bioremediation and Biodeterioration 4. BTE 312 Computer Application in Biotechnology5. BTE 317 Biostatistics6. MIC 309 Immuno-pathology and Vaccine Development7. MIC 404 Molecular Virology and Oncology8. MIC 405 Drinking Water Microbiology9. MIC 406 Medical and Diagnostic Microbiology10. BCH 202 Enzyme and Enzyme Kinetics11. MIC 407 Bacterial Pathogenesis and Molecular Epidemiology12. BTE 313 Biomass and Bio-fuel13. MIC 408 Extremophiles14. BTE 403 DNA Fingerprinting and Molecular Diagnostics15. BTE 404 Bioprocess Technology

(d) Lab (6 Credits):

MIC 105 Microbiol Lab IMIC 205 Microbiol Lab IIMIC 207 Microbiol Lab IIIMIC 305 Microbiol Lab IV

(e) Microbiology Internship (3 credits):

MIC 400 Industrial/Research Organizations Attachment

(f) Thesis/ Project (4 Credits):

MIC 450 Microbiology Project

(g) Courses Outside Major Specialization (27 Credits):

1. ANT 101 Introduction to Anthropology2. ARC 292* Painting3. ARC 293* Music Appreciation4. MIC 316 Microbes and Human Civilization5. BTE 317 Biostatistics 6. BUS

101 Introduction to Business7. CHE 101 Introduction to Chemistry8. CSE 110 Programming Language9. ENV 101 Introduction to Environmental Science10. ECO 10 Principles of Economics11. HUM 101 World Civilizations and Culture12. HUM 102 Introduction to Philosophy13. HUM 111 History of Science14. MGT 211 Principles of Management15. PHY 101 Introduction to Physics16. POL 103 Introduction to Political Science17. PSY 101 Introduction to Psychology18. SOC 101 Introduction to Sociology 19. SOC 401 Gender and Development20. STA 201 Elements of Statistics and Probability

* 2 Credit Courses

V. Semester wise Distribution of Courses

1st Year

Semester Course No. Course Name CreditsMIC 101 General Microbiology 3BIO 101 Introduction to Biology 3First BCH 101 Basic Biochemistry 3DEV 101 Bangladesh Studies 3MIC 101 Microbiol. Lab 1 1.5ENG 101 English Fundamentals 3Sub Total 16.5 Credits

Semester Course No. Course Name CreditsMIC 102 Basic Techniques in Microbiology 3Second CSE 101 Introduction to Computer Science 3MAT 101 Fundamentals of Mathematics 3ENG 102 Composition I 3BCH 102 Biophysical Chemistry 3HUM 103 Ethics and Culture 3MIC 205 Microbial Lab II 1.5

Sub Total 19.5 Credits

Total Credits in First year = 16.5+19.5= 36

2nd Year

Semester Course No. Course Name CreditsMIC 201 Microbial Chemistry 3BCH 201 Human Physiology 3Third MIC 202 Microbial Metabolism 3Elective Course (c) 3Elective Course outside major(g) 3Sub Total 15 Credits

Semester Course No. Course Name MIC 203 Environmental Microbiology 3MIC 204 Medical Microbiology 3Fourth BTE 207 Introduction to Molecular Biology 3Elective

Courses outside Major 6MIC 207 Microbiol Lab III 1.5

Sub Total 16.5 CreditsTotal Credits in 2nd Year 15+16.5= 31.5 Credits

3rd Year

Semester Course No. Course Name CreditsMIC 301 Virology 3BCH 302 Basic Immunology 3Fifth BTE307 Advanced Molecular Biology 3MIC 302 Food Microbiology 3Elective Course outside Major 3MIC 305 Microbiol Lab IV 1.5

Sub Total 16.5 Credits

Semester Course No. Course Name CreditsMIC 303 Agriculture Microbiology 3MIC 304 Microbial Biotechnology 3Sixth MIC 306 Pharmaceutical Microbiology 3Elective Course (c) 3Elective Courses outside Major 6

Sub Total 18 Credits

Total Credits in Third Year 16.5+18= 34.5

4th Year

Semester Course No. Course Name CreditsBTE 401 Bioinformatics 3MIC 401 Microbial Genetic Engineering 3Seventh Elective Course (c) 3Elective Courses outside Major(g) 6

MIC 450 Microbiology Project/Thesis 2

Sub Total 17 Credits

Semester Course No. Course Name CreditsMIC 402 Analytical Microbiology 3MIC 403 Microbiological Quality Control of Food, Fish and Beverages 3MIC 400 Industrial Attachment 3Eight MIC 450 Microbiology Project/Thesis 2Elective Course(c) 3Elective Course outside Major 3

Sub Total 17 Credits

Total credits in Fourth Year 17+17= 34

Total Credits 1st Year+ 2nd Year+ 3rd Year+4th year= 36+31.5+34.5+34=136

Academic System & Evaluation Method

(a) Academic Standards

In keeping with the mission and goals in mind, the Undergraduate programme in

Microbiology will strive to ensure high academic standards by implementing well-designed curricula, carefully selecting high quality students and faculty, utilizing modern and effective instructional methods and aides, and by continuously monitoring and rigorously evaluating all the pertinent activities and systems. A special feature of teaching will be the tutorial/ lab/ workshop sessions designed to assist students in learning application of the concepts and theories.

(b) Courses and Credit Requirements

The Bachelor of Science programme in Microbiology will follow the model of higher education consisting of semesters, courses, credit hours, continuous evaluation and letter grading as in all other courses of BRAC University. There are two regular semesters: Fall and Spring, each with a duration of 14 weeks and a Summer semester of 9 weeks. The undergraduate curriculum consists of general education courses, major courses, elective courses and non-major area courses.

Credit hours for a course are assigned on the basis of a 14 – week semester. One (1) credit hour means that the course meets for 60 minutes in a class each week; 3 credits mean that it will meet three times every week.

(c) Examination, Evaluation and Grading

The grading process will undoubtedly be transparent. The performance of the students is evaluated throughout the semester through class tests, quizzes, assignments, and midterm exams. End of semester evaluation includes final examinations, term papers, project reports etc. Numerical scores earned by a student in tests, examinations, assignments etc. are cumulated and converted to letter grades.

(d) Distribution of Marks & GPA Computation

The distribution of marks for the performance evaluation is as follows:

i. Theory Courses

Section Marks %1. Quizzes/ Class Tests/ Assignments/ Participation/ Attendance 302. Mid Term Examination 203. Final Examination (comprehensive), Projects 50Total Marks

100

ii. Lab Courses

Section Marks %1. Reports/ Class Tests/ Participation / Viva-Voce 302. Lab Experiments 303. Final Examination 40Total Marks 100

iii. Thesis / Project

Thesis / Project work will be spread over two semesters (7th & 8th). The mark distribution for the Thesis / Project will be as follows:

Section Marks%1. Viva-Voce (End of the 7th semester) 152. Thesis / Project Work 553. Defence (End of 8th semester) 30Total Marks 100

Class attendance is compulsory for every student. 5% of total marks in every course is allocated for attendance in classes including tutorials and labs. The basis for awarding marks for attendance is as follows:

Attendance Marks90% and above 585% to less than 90% 480% to less than 85% 375% to less than 80% 270% to less than 75% 1Less than 70% 0

If a student does not attend a minimum of 70% of the total classes including tutorials and labs, s/he will not be allowed to take the final exam.

Marks earned by the students in Class Tests, Quizzes, Assignments, Participation, Attendance, Midterm Exam , Final Exam, Projects, Term Papers etc are to be cumulated and the total is to be graded as per the scale given below:

Letter Grade GP 90 - 100 = A (4.0) Excellent85 - < 90 = A- (3.7) 80 - < 85 = B+ (3.3) 75 - < 80 = B (3.0) Good70 - < 75 = B- (2.7) 65 - < 70 = C+ (2.3) 60 - < 65 = C (2.0) Fair57 - < 60 = C- (1.7) 55 - < 57 = D+ (1.3) 52 - < 55 = D (1.0) Poor50 - < 52 = D- (0.7) <50 = F (0.0) Failure

P: PassI: IncompleteW: Withdrawal R: Retaken

GPA Computation

The Grade Point Average (GPA) is computed in the following manner:GPA = Sum of (Grade points x Credits)/ Sum of Credits Attempted

The Board of Trustees is the highest policy making body of BracU. It is responsible for ensuring that the best educational and administrative standards are set and maintained at BracU. The current Board of Trustees consists of the following eminent personalities of Bangladesh:

University

“Celebrating International Women’s Day 2023: Women in Science” organized jointly by the Bangladesh Society of Microbiologists (BSM) and Microbiology Program of BRAC University. We are honoured to have Professor Dr Malabika Sarker from BRAC University and Professor Dr Farhana Karabi Biswas from North South University (NSU) as the invited speakers in this event.

The event will take place on March 12, 2023 (Sunday) from 3 pm to 4:30 pm at the UB2 auditorium at BRAC University.

Celebrating International Women’s Day 2023: Women in Science

Tushar Mitra completed his school and college from Pubali Jute Mills High School and Narsingdi Model College. He finished his BS from University of Dhaka in 2018 in Physics. He then completed his MS from University of Dhaka in Physics in 2019 in the Thesis group. He worked as a Vice Chancellor’s Fellow at BRAC University from 2021-2022.

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Journals

Mitra, T., & Hassan, M. K. (2022). A weighted planar stochastic lattice with scale-free, small-world and multifractal properties. Chaos, Solitons and Fractals, 154. https://doi.org/10.1016/j.chaos.2021.111656

Mitra, T., & Hassan, K. (2021). Multi-multifractality and dynamic scaling in stochastic porous lattice. European Physical Journal: Special Topics, 230(21–22), 3835–3844. https://doi.org/10.1140/epjs/s11734-021-00329-0

Mitra, T., Hossain, T., Banerjee, S., & Hassan, M. K. (2021). Similarity and self-similarity in random walk with fixed, random and shrinking steps. Chaos, Solitons and Fractals, 145. https://doi.org/10.1016/j.chaos.2021.110790

a. Principles of Physics Ib. Principles of Physics IIc. Mechanics and Properties of Matterd.

Electricity and Magnetism

Non-equilibrium Statistical Mechanics, Percolation theory, Network theory, Fractals and Multifractals.

Tushar Mitra

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Course Contents

a. Theoretical Courses:

ANT 101 Introduction to Anthropology 3 credits

Humans in nature, human evolution, history of culture, rise of early civilizations in the old and new world, organizations of pre-industrial society environment, resources and their distribution; gender, kinship and descent, religion, economics, politics, survival of indigenous groups, forms of culture and society among contemporary peoples, Comparative study of traditional and changing Third World societies, impact of modern world on traditional societies, power and social order; custom and law, conflict and change, Cultural and ethnic diversity.

Recommended Books:

1. Cultural Anthropology, a Global Perspective: R Scupin2. Anthropology: The

Exploration of Human Diversity: Conrad P Kottak3. Anthropology: C Ember and M Ember4. Cultural Anthropology: W A Haviland5. Anthropology - Social and Cultural: Kedar Nath, Ram Nath6. An Introduction to Anthropology: Victor Barnouw.

ARC 292 Painting 2 credits

Painting as a form of artistic and architectural expression. Introduction to various media in painting. Still life sketches and painting. Study of forms in painting. Landscapes and cityscapes. Colour pencils, crayons, pastels and watercolour. Mixed media. Computers in painting.

ARC 293 Music Appreciation 2 credits

Musical form. Ingredients of music: sound and time. Indian and Western music: melody and harmony. Foundations of sub-continental music: raga system. Presentation of vocal and instrumental music. Modern Bengali music and works of major composers and demonstrations. Western classical music and works of major composers. Music and its rhythm, composition etc.

BI0 101 Introduction to Biology 3 credits

An introduction to the cellular aspects of modern biology including the chemical basis of life, cell theory, energetics, genetics, development, physiology, behaviour, homeostasis and diversity, and evolution and ecology. This course will explain the development of cell 11 structure and function as a consequence of evolutionary process, and stress the dynamic property of living systems.

Recommended Books:

1. Biology: P.H. Raven and G.B. Johnson2. Biological Science: G. W. Stout and D. J. Taylor3. Advanced Biology: J. Simpkins and J. J. Williams4. Biology: A Fundamental Approach: M. B. Roberts

CHE 101 Introduction to Chemistry 3 credits

The course is designed to give an understanding of basics in chemistry. Topics include nature of atoms and molecules; valence and periodic tables, chemical bonds, aliphatic and aromatic hydrocarbons, optical isomerism, chemical reactions.

Recommended Books:

1. Introduction to Modern Inorganic Chemistry: S. Z. Haider2. Physical Chemistry: Haque & Nawab3. Organic Chemistry: R. T. Morrison & R. N. Boyd4. General Chemistry: Raymond Chang

CSE 110 Programming Language I 3 credits

An introduction to the foundations of computation and purpose of mechanised computation, techniques of problem analysis and the development of algorithms and programs, principles of structured programming and corresponding algorithm design, Topics will include data structures, abstraction, recursion, iteration as well as the design and analysis of basic algorithms, (language C is primarily used), introduction to digital computers and programming algorithms and flow chart construction, information representation in digital computers, writing, debugging and running programs (including file handling) on various digital computers using C. The course includes a compulsory 3 hours laboratory work each week.

Recommended Books:

1. Working with C: Y Kanetkar2. Schaums Outline of Theory and Problems of Programming With C: Byron S. Gottfried

CSE 220 Data Structures 3 credits

Introduction to widely used and effective methods of data organization, focusing on data structures, their algorithms and the performance of these algorithms. Concepts and examples, elementary data objects, elementary data structures, arrays, lists, stacks, queues, graphs, trees, compound structures, data abstraction and primitive operations on these structures. Memory management; sorting and searching; hash techniques. Introduction to the fundamental algorithms and data structures: recursion, backtrack search, lists, stacks, queues, trees, operation on sets, priority queues, graph dictionary. Introduction to the analysis of algorithms to process the basic structures. A brief introduction to database systems and the analysis of data structure performance and use in these systems.

Recommended books:

1. Data Structure and Program Design: Robert L Kruse; PrenticeHall. 2. Data Structures, Algorithms, and Software Principles in C: Thomas A Standish; PrenticeHall.3. Data Structures and Algorithm Analysis in C: Mark Allen Weiss; Addison Wesley; Longman Inc. 4. Data Structure: Edward M Reingold, Wilfred J Hansen; CBS Publishers & Distributors. 5. Data Structures in C++: Using the Standard Template Library; Timothy Budd; Addison Wesley Longman Inc.

CSE 221 Algorithms 3 credits

The study of efficient algorithms and effective algorithm design techniques. Techniques foranalysis of algorithms, Methods for the design of efficient algorithms: Divide and Conquer paradigm, Greedy method, Dynamic programming, Backtracking, Basic search and traversal techniques, Graph algorithms, Elementary parallel algorithms, Algebraic simplification and transformations, Lower bound theory, NP-hard and NP-complete problems. Techniques for the design and analysis of efficient algorithms, Emphasising methods useful in practice. Sorting; Data structures for sets: Heaps, Hashing; Graph algorithms: Shortest paths, Depth-first search, Network flow, Computational geometry; Integer arithmetic: gcd, primality; polynomial and matrix calculations; amortised analysis; Performance bounds, asymptotic and analysis, worst case and average case behaviour, correctness and complexity. Particular classes of algorithms such as sorting and searching are studied in detail. The course includes a compulsory 3 hour laboratory work alternate week.

Recommended books:

1. Introduction to Algorithms: Thomas H Cormen, Charles E Leiserson, Ronald L Rivest, Clifford Stein; McGrawHill. 2. Fundamentals of Computer Algorithms: Ellis Horowitz, Sartaj Sahni; Galgotia Publications.3. Fundamental Algorithms: The Art of Computer

Programming. vol. 1; Donald E Knuth.4. Seminumerical Algorithms: The Art of Computer Programming. vol. 2; Donald E Knuth.5. Sorting and Searching: The Art of Computer Programming. vol. 3; Donald E Knuth.6. Data Structures and Algorithm Analysis in C: Mark Allen Weiss; Addison Wesley Longman Inc.

DEV 101 Bangladesh Studies 3 credits

Socio-economic profile of Bangladesh, agriculture, industry, service sector, demographic patterns, social aid and physical infrastructures. Social stratification and power, power structures, government and NGO activities in socio-economic development, national issues and policies and changing society of Bangladesh.

Recommended Books:

1. Bangladesh, National Cultures and Heritage , An Introductory Reader: A.F. Salahuddin Ahmed & Bazlul Mobin Chowdhury2. The History of Bengal (Vol.1 &Vol.2) : R.C. Majumdar3. Banglapedia, 2003: Asiatic Society of Bangladesh4. Bangladesh Arthaniti: Khan, Md. Shamsul Kabir 5. Bangladesh on the Threshold of the Twenty First Century, Asiatic Society of Bangladesh, 2002: A.M Chowdhury and Fakrul Alam6. Poverty Reduction & Strategy, What, Why & for Whom , Asit Biswas et.al.(ed) Contemporary Issues in Development : M.M Akash7. Bangladesh 2020, A longrun perspectives study: The World Bank

ECO 103 Principles of Economics 3 credits

A study of the fundamentals of micro and macroeconomics, nature and method of economics, individual markets, demand and supply, elasticity of demand and supply. Production and cost, market structures with special focus on perfect competition and monopoly, economic efficiency and market failure, determination of national income. The aggregate supply model, unemployment, inflation, unemployment-inflation tradeoff, government budget and fiscal policy, money creation and monetary policy, business cycles, economic growth, theory of comparative advantage, free trade versus protection, balance of payments and exchange rate policies.

Recommended Books:

1. Economics: John Solman2. Principles of Macroeconomics: Robert H. Frank3. Modern Economic Theory: K.K. Dewett4. International Economics: D.R. Appleyard &A.J. Field

ENG 091 Foundation Course (Non-credit)

The English Foundation Course is designed to enable students to develop their competence in reading, writing, speaking, listening and grammar for academic purposes. The students will be encouraged to acquire skills and strategies for using language appropriately and effectively in various situations. The approach at all times will be communicative and interactive involving individual, pair and group work.

Recommended Books:

1. New Interchange, Student 's Book 3A, Cambridge University Press, 2002 : J. C. Richards, J. Hull, and S. Proctor.2. Vocabulary Basics, Townsend Press, 1998: J. Nadel, B. Johnson, and P. Langan.3. First steps in Academic Writing, Longman, 1996: A. Hogue4. Get Ready to Write, Longman, 1998: K. Blanchard, C. Root.

ENG 101 English Fundamentals 3 credits

Developing basic writing skills: mechanics, spelling, syntax, usage, grammar review, sentence and essay writing.

Recommended Books:

1. Fundamentals of English: Jack C. Richards

ENG 102 Composition I 3 credits

The main focus of this course is writing. The course attempts to enhance students' writing abilities through diverse writing skills and techniques. Students will be introduced to aspects of expository writing: personalized/ subjective and analytical/persuasive. In the first category, students will write essays expressing their subjective viewpoints. In the second category students will analyze issues objectively, sticking firmly to factual details. This course seeks also to develop students' analytical abilities so that they are able to produce works that are critical and thought provoking.

Recommended Books:

1. Composition I: The Pearl; John Steinbeck

ENV 101 Introduction to Environmental Science 3 credits

Fundamental concepts and scope of environmental science, Earth's atmosphere, hydrosphere, lithosphere and biosphere, men and nature, technology and population, ecological concepts and ecosystems, environmental quality and management, agriculture, water resources, fisheries, forestry and wildlife, energy and mineral energy sources; renewable and non renewable resources, environmental degradation; pollution and waste management, environmental impact analysis, remote sensing & environmental monitoring.

Recommended Books:

1. Something New Under the Sun: An Environmental History of the TwentiethCentury World: J. R. McNeil & Paul Kennedy 2. Principles of Ecology: R Brewer3. Fundamentals of Ecology: E. P. Odum

HUM 101 World Civilization and Culture 3 credits

A brief view of the major civilizations and cultural aspects in different continents covering ancient, medieval and modern civilizations. Topics include renaissance, reformation, and the beginning of the modern world, scientific revolution, industrial revolution, the age of democratic revolutions, nineteenth century Europe, Asia-Pacific Region, Africa, World Wars, South Asia: colonization, decolonization and after; contemporary world: Cold War and after.

Recommended Books:

1. World Civilization: Bums & others2. Civilization: T Walter Walbank and others3. A History of World Civilization: J. E. Swain4. Western Civilization: : Robert E. Lerner & Standish Meachem

HUM 102 Introduction to Philosophy 3 credits

Philosophy: Concept of philosophy; science and philosophy; religion, literature and philosophy; sources of knowledge: empiricism, rationalism and criticism; concepts of value, ethics and sources of ethical standards.

Recommended Books:

1. A Modern Introduction to Philosophy: P Edwards and A. Pap2. Philosophy: R. J. Hirst3. Introduction to Modern Philosophy: C.E.M. Joad4. An Introduction to Philosophical Analysis: J. Hospers5. An Outline of Philosophy: A. Matin6. Introduction of Philosophy: T. W. Patrick7. Living Issues in Philosophy : H.H. Titus

HUM 103 Ethics and Culture 3 credits

This course introduces the students to principles and concepts of ethics and their application to our personal life. It establishes a basic understanding of social responsibility, relationship with social and cultural aspects, and eventually requires each student to develop a framework for making ethical decision in his work. Students learn a systematic approach to moral reasoning. It focuses on problems associated with moral conflicts, justice, the relationship between rightness and goodness, objective vs. subjective, moral judgment, moral truth and relativism. It also examines personal ethical perspectives as well as social cultural norms and values in relation to their use in our society. Topics include: truth telling and fairness, objectivity vs. subjectivity, privacy, confidentiality, bias, economic pressures and social responsibility, controversial and morally offensive content, exploitation, manipulation, special considerations (i.e. juveniles, courts) and professional and ethical work issues and decisions. On conclusion of the course, the students will be able to identify and discuss professional and ethical concerns, use moral reasoning skills to examine, analyze and resolve ethical dilemmas and distinguish differences and similarities among legal, ethical and moral perspectives.

Recommended Books:

1. Ethics, Culture and Psychiatry - International Perspectives: Ahmed Okasha, Julio Arboleda2. Ethics and HRD: A New Approach to Leading Responsible Organizations: Tim

Hatcher3. The Ethical Challenge: How to Lead with Unyielding Integrity: Noel M. Tichy and Andrew 4. Dignity of Difference: How to Avoid the Clash of Civilizations: Jonathan Sacks5. Culture and Ethics: Michel Labour, Charles Juwah, Nancy White and Sarah Tolley6. Culture Matters: How Values Shape Human Progress: Samuel P. Huntington

HUM 111 History of Science 3 credits

This course will present a general overview of the development of scientific knowledge from ancient to modern times. It will examine how our modern scientific worldview developed over the ages in the fields of astronomy, physics, biology, chemistry, medicine, geology and other science disciplines. Focus will be on significant discoveries, the major scientists responsible for these revolutions, and the interrelation between science and society over the centuries. The course will contain the following: Science & philosophy, development of science in the ancient times, Greek & Egyptian science, science in the Orient, medieval science, science in the Islamic world, Western renaissance & industrialization, evolutionary theory, science in the modern ages. Science & religion, nature of scientific truth, validation of scientific theories.

Recommended Books:

1. Reader's Guide to the History of Science: A. Hessenbruch2. Scientific Laws, Principles, and Theories: a Reference Guide: Robert E Krebs3. The History of Science: an Annotated Bibliography: G Miller4. A Guide to the History of Science: a First Guide for the Study of the History of Science, with Introductory Essays on Science and Tradition: G. arton5. Knowledge & the World: Challenges Beyond the Science Wars: M. Cavrier, J. Roggenhofer, G. Kuppers & P. Blanclard 6. The Forgotten Revolution: How Science was Born in 300 BC and Why It Had to be Reborn: Lucio Russo7. Hitler's Scientists: Science, War and the Devil's Fact: J. Cornwell

MAT 111: Principles of Mathematics 3 credits

Sets: Elementary idea of Set, Set notations, Set operations: union, intersection, complement, difference; Set operations and Venn diagrams. Set of Natural numbers, Integers, Rational numbers, Irrational numbers and Real numbers alongwith their geometrical representation, Idea of Open and Closed interval,. Idea of absolute value of real number, Variables and Constants, Product of two sets: Idea of product of sets, Product set of real numbers and their geometric representation, Axioms of real number system and their application in solving algebraic equations. Equation and Inequality, Laws of inequality, Solution of equations and inequalities. Variable and Functions: Variable of a set, Functions of single variable, Polynomial, Graph of Polynomial functions in single variable. Exponential, Logarithmic, Trigonometric functions and their graphs, Permutation and Combination. Binomial theorem.

Recommended Books:

1. Set Theory: Seymour Lipschutz.2. Functions and Graphs : R. David Gustafson & Peter D. Frisk3. Algebra and Trigonometry : Lial & Miller.4. Calculus: Howard Anton.

MAT 121 Basic Algebra 3 credits

Elements of logic: Mathematical statements, Logical connectives, Conditional and biconditional statements, Truth tables and tautologies, Quantifications, Logical implication and equivalence, Deductive reasoning. The concept of sets: Sets and subsets, Set operations, Family of Sets. Relations and functions: Cartesian product of two sets, Relations, Order relation, Equivalence relations, Functions, Images and inverse images of sets, Injective, surjective, and bijective functions, Inverse functions. Real number system: Field and order properties, Natural numbers, integers and rational numbers. Absolute value, Basic inequalities. (Including inequalities of means, powers, Weierstrass, Cauchy) Complex number system: Field of complex numbers, De Moivre's theorem and its applications. Elementary number theory: Divisibility, Fundamental theorem of arithmetic, Congruence. Summation of finite series: Arithmetic-geometric series, Method of difference, Successive differences, Use of mathematical induction. Theory of equations: Synthetic division, Number of roots of polynomial equations, Relations between roots and coefficients, Multiplicity of roots, Symmetric functions of roots, Transformation of equations.

Recommended Books:

1. Set Theory : Seymour Lipschutz.2. Higher Algebra : Shahidullah and Bhattacharjee.3. Higher Algebra : Bernard and Child.4. Higher Algebra : Md. Abdur Rahaman.

MAT 122 Analytic and Vector Geometry 3 credits

Two dimensional geometryCoordinates: Cartesian and polar Coordinates, Transformations of coordinates and its applications. Reduction of second degree equations to standard forms, Pairs of straight lines, Circles, Identification of conics, Equations of conics in polar Coordinates.

Three dimensional geometryCoordinates in three dimensions, Direction cosines, and Direction ratios. Planes, straight lines, shortest distance, sphere, orthogonal projection and conicoids.

Vector geometryVectors in plane and space, Algebra of vectors, scalar and vector products, Triple scalar products, its applications to Geometry.

Recommended Books:

1. Coordinate Geometry of Two Dimension : S. Loney.2. Textbook on Coordinate Geometry : Rahman and Bhattacharjee.3. Vector Analysis : Spiegel, M.R. Spiegel.4. A Textbook on Coordinate Geometry and Vector Analysis: Khosh Mohammad.

MAT 123: Calculus I 3 credits

Differential Calculus: Real number system and its geometrical representation, real variable, function of single (real) variable, parametric equations, limit, continuity and differentiability, derivatives of different types of functions, geometrical significance of derivative, Rolle's Theorem, Mean Value Theorem, Taylor's Theorem; maxima, minima, point of inflexion, concavity and convexity, sketching of curves using concepts of calculus; Indeterminate Form, L'Hospital Rule, Successive Differentiation, Leibnitz's Theorem, tangent, normal and related formulas, curvature.

Integral Calculus: Indefinite integrals of different types of functions, various methods of integrations, definite integrals, Fundamental Theorems of Definite Integrals, properties of definite integrals, Reduction formulas, Arc Length, Area under the curves, Surface area and Volume of a 3-D objects. Improper Integrals and applications.

Prerequisite: MAT122

Recommended Books:

1. Calculus : Howard Anton.2. Calculus and Analytical Geometry : Thomas Finney.3. Calculus with Analytic Geometry : E.R. Swokowski.4. Advanced Calculus : M.R. Spiegel.

MAT 124 Scientific Programming 3 credits

Problem-solving techniques using computers: Flowcharts, Algorithms, Pseudocode. Programming in FORTRAN: Syntax and semantics, data types and structures, input/output, loops, decision statements, arrays, user-defined functions, subroutines and recursion. Computing using FORTRAN: Construction and implementation of FORTRAN programs for solving problems in mathematics and sciences.

Recommended Books:

1. FORTRAN Programming : Schaum's Outline Series.

MAT 211 Calculus II 3 creditsFunctions of several variables, concept of surface, sketching of , contour sketch for surface, limit and continuity, partial derivative and its geometrical significance, chain rule of partial differentiation, concept of gradient, divergence and curl, directional derivative and tangent plane, concept of differential and perfect differential, linear approximation and increment estimation, maxima, minima and saddle point, Lagrange multiplier, higher order derivatives, Taylor's theorem of function of several variables. Multiple integrals: Double integrals, Double integrals in Polar coordinates, Triple integrals, Triple integrals in Cylindrical and Spherical coordinates, Change of variables in Multiple integrals, Jacobian, Line integrals, Green's theorem, Surface integrals, Applications of Surface integrals, Divergence theorem, Stoke's theorem.

Prerequisite: MAT123.

Recommended Books:

1. Calculus : Howard Anton, Irl Bivens & Stephen Davis2. Calculus and Analytical Geometry :Thomas Finney.3. Calculus with Analytic Geometry : E.R. Swokowski.4. Advanced Calculus : M.R. Spiegel.

MAT 212 Linear Algebra 3credits

Introduction to matrix, different types of matrices, equivalent matrices, determinants, properties of determinants, minors, cofactors, evaluation of determinants, adjoint matrix, inverse matrix, method for finding inverse matrix, elementary row operations and echelon form of matrix, system of linear equations (homogeneous and non-homogeneous equations) and their solutions; Vector, vector spaces and subspaces, linear independence and dependence, basis and dimension, change of bases, rank and nullity, linear transformation, kernel and images of a linear transformation and their properties, eigenvalues and eigenvectors, diagonalization, Cayley Hamilton theorem,

Recommended Books:

1. Elementary Linear Algebra and Applications: Howard Anton .2. Linear Algebra : S. Lipschutz.3. Linear Algebra : B. Kolman.

MAT 221 Real analysis I 3 credits

Real number system: Completeness of real numbers, supremum and infimum principles and their consequences, Dedekind's theorems, Bolzano-Weierstrass theorem. Sequences of Real Numbers: Infinite sequence, Convergent sequences, Monotone sequences, subsequences, Cauchy sequence, Cauchy criteria for convergence of sequences. Infinite Series: Concept of sum and convergence, series of positive terms, alternating series, absolute and conditional convergence, test for convergence, Convergence of sequences and series of functions. Continuity and Limits: Properties of continuous functions, Extreme Value Theorem and Intermediate Value Theorem, Uniform continuity concepts, Limits, Heine-Borel theorem. Integration: Necessary and sufficient conditions for integrability, Darboux Sums and Riemann Sums, Improper integral and their tests for convergence.

Recommended Books:

1. Principle of Mathematical Analysis : W. Rudin.2. Real Analysis (3rd edition): H.L. Royden.3. Mathematical Analysis: Tom .M. Apostol.4. Real Analysis: A. Malik.

MAT 222 Differential Equations I 3 credits

Ordinary differential equations and their solutions: Classification of differential equations, Solutions, Implicit solutions, Singular solutions, Initial Value Problems, Boundary Value Problems, Basic existence and uniqueness theorems (statement and illustration only), Solution of first order equations: Separable equations, Linear equations, Exact equations, Special integrating factors, Substitutions and transformations, Modeling with first order differential equations, Model solutions and interpretation of results, Orthogonal and oblique trajectories, Solution of higher order linear equations: Linear differential operators, Basic theory of linear differential equations, Solution space of homogeneous linear equations, Fundamental solutions of homogeneous solutions, Reduction of orders, Homogeneous linear equations with constant coefficients, Non homogeneous equation, Method of undetermined coefficients, Variation of parameters, Euler Cauchy differential equation, Modeling with second order equations, Initial Value Problems and Boundary Value problems, reduction of order, Euler equation, generating functions, eigenvalue problems. Inhomogeneous linear difference equations (variation of parameters, reduction of order, Series solutions of second order linear equations: Taylor series solutions about an ordinary point. Frobenious series solutions about regular singular points. Series solutions of Legendre, Bessel, Laguerre and Hermite equations. Systems of linear first order differential equations: Elimination method. Matrix method for homogeneous linear systems with constant coefficients. Variation of parameters. Matrix exponential.

Prerequisite: MAT211

Recommended Books:

1. Introduction to Differential Equations : S.L. Ross.2. Introduction to Differential Equations and Applications : D.G. Zill.3. Differential Equations : Boyce & D'Prima.4. Mathematical Methods (Volume I) : Md. Abdur Rahaman.

MAT 223 Numerical Analysis I 3 credits

Solution of equation of single variable: Fixed point iteration, Bisection algorithm, Method of False position, Newton-Raphson's method, Error Analysis for Iterative methods, Accelerating limit of convergence. Interpolation: Interpolating polynomials for equispaced and nonequispaced nodes, Lagrange's polynomial, Newton-Gregory's Interpolating polynomials, curve fitting with Least Square method, Iterated interpolation, Extrapolation. Differentiation and Integration: Numerical differentiation, Single point and (n+1)-point formulae of differentiation, Richardson's extrapolation, Numerical Integration, Gaussian quadrature formula, Trapezoidal, Simpson's, Weddle's Rules.

Prerequisite: MAT124

Recommended Books:

1. Numerical Analysis : R.L. Burden & J.D. Faires.2. Applied Numerical Analysis : C.I. Gerald & P.O. Wheatly.3. Introduction to Numerical Methods : S. Sastry.

MAT 311 Abstract Algebra 3 credits

Equivalence relation and residue classes modulo n. Groups and subgroups, Cyclic groups, Symmetric groups. Cosets and Lagrange's Theorem: Normal subgroups, Quotient groups, Permutation groups, Homomorphism, Isomorphism and Automorphism of groups with related theorems and problems, Cayley's Theorem, centralizer and normalizer of an element/ subset in a group. Rings, Ideals, and Quotient Rings, Prime and Maximal Ideals. Integral Domain, Field of fractions. Principal Ideal Domain, Euclidian Domain, Unique Factorization Domain.Polynomial Rings, Primitive polynomials, Gauss Theorem, Eisenstein's criterion for irreducibility. Prime Fields, characteristic of Fields.

Prerequisite: MAT121

Recommended Books:

1. Topics in Algebra: Herstein.2. First Course in Abstract Algebra: J.B. Frayleigh.3. A First Course in Abstract Algebra: H. Paley & P.M. Weicheel.4. Modern Algebra : R.S. Aggarwal.

MAT 312 Numerical Analysis II 3 credits

Part A: TheorySolutions of linear system of equations: Gaussian Elimination method with pivoting, Matrix inversion, Direct factorization of matrices, Iterative Techniques for solving linear system of equations: Jacobi's and Gauss-Seidel Method. Solution of tridiagonal system, Eigen values and Eigen vectors (Power Method). Numerical solution of Nonlinear system: Fixed point for functions of several variables, Newton's method, Quasi-Newton's method. Initial Value Problem for ODE: Euler's method, Higher order Taylor's method, Runge-Kutta methods, Multistep methods, Variable Stepsize Multistep methods. Boundary Value Problem: Linear Shooting method, Shooting method for non linear BVP. Boundary Value problem involving elliptic, parabolic and hyperbolic equations, explicit and implicit Finite Difference method.Part B: Numerical Analysis LabConstruction and implementation of FORTRAN / C, C++ programs of techniques in Numerical Analysis. There will be at least 15 lab assignments.

Prerequisite: MAT 223

Recommended Books:

1. Numerical Analysis, Brooks/cole : R.L. Burden & J.D. Faires.2. Applied Numerical Analysis : C.I. Gerald & P.O. Wheatly.3. Numerical Recipes in FORTRAN : W.H. Press, S.A. Teukolsky, W.T. Vetterling & B.P. Flannery.4. Numerical Solution of Partial Differential Equations: G.D. Smith.

MAT 313 Differential Geometry 3 credits

Curves in space: Vector functions of one variable, space curves, unit tangent to a space curve, equation of a tangent line to a curve, Osculating plane (or plane of curvature), vector function of two variables, tangent and normal plane for the surface , Principal normal, binormal and fundamental planes, curvature and torsion, Serret Frenet's formulae, theorems on curvature and torsion, Helices and its properties, Circular helix. Spherical indicatrik, Curvature and torsion. Curvature and torsion for spherical indicatrices. Involute and Evolute of a given curve, Bertrand curves. Surface: Curvilinear coordinates, parametric curves, Metric (first fundamental form), geometrical interpretation of metric, relation between coefficients E, F, G. properties of metric, angle between parametric curves, elements of area, second fundamental form. Derivatives of surface normal M (Weingarten equations), Third fundamental form, Principal sections, Principal sections, direction and curvature, first curvature, mean curvature, Gaussian curvature, normal curvature, lines of curvature, centre of curvature, Rodrigues's formula, condition for parametric curves to be line of curvature, Euler's Theorem, Elliptic, hyperbolic and parabolic points, Dupin Indicatrix.

Recommended Books:

1. Vector Analysis: Spiegel2. Differential Geometry: Henderson3. Foundations of Differential Geometry: S. Kobayashi, Katsumi Nomizu4. Differential Geometry : Lipschutz , McGraw Hill

MAT 314 Complex Analysis 3 credits

Introduction to complex numbers and their properties, complex functions, limits and continuity of complex functions, Analytic functions, Cauchy Riemann equations, harmonic functions, Rational functions, Exponential functions, Trigonometric functions, Logarithmic functions, Hyperbolic functions. Contour integration: Cauchy's Theorem, Simply and Multiply connected domain, Cauchy integral formula, Morera's theorem, Liouville's theorem. Convergent series of analytic functions, Laurent and Taylor series, Zeroes , Singularities and Poles, residues, Cauchy's Residue theorem and its applications, Conformal Mapping.

Prerequisite: MAT123

Recommended Books:

1. Complex Analysis and Applications: Churchill & Brown.2. Complex Analysis : M.R. Spiegel.

MAT 315 History of Mathematics 3 credits

A Survey of the development of mathematics beginning with the history of numeration and continuing through the development of the calculus. The study of selected topics from each field is extended to the 20th century. Biographical and historical aspects will be reinforced with studies of procedures and techniques of earlier mathematical cultures.

Recommended books:

1. A History of Mathematics; CB Boyer, UC Merzbac; Wiley.2. The History of Mathematics; VJ Katz; AddisonWesley.3. A Short Account of the History of Mathematics; WWR Bell; Macmillan.

MAT 316 Operations Research I 3 credits

Convex sets and related theorems, Introduction to linear programming, Formulation of linear programming problems, Graphical solutions, Simplex method, Duality of linear programming and related theorems, Sensitivity. Unconstrained optimization: Newton's method, Trust region algorithms, Least Squares and zero finding. Constrained optimization: linear/nonlinearequality/inequality constraints, Duality, working set methods. Linear programming: Simplex method, primal dual interior point methods.

Recommended Books:

1. Operations Research: An Introduction ,4th Edition by Hamdy A. Taha2. Convex Optimization: Stephen Boyd & Lieven Vandenberghe3. Schaum's Outline of Operations Research by Richard Bronson.

MAT 321 Real Analysis II 3 credits

Metric spaces: definition and some examples, open sets, closed sets, Convergence, Completeness, Baire's theorem. Connected set: Compact sets, locally compact sets and related theorems, connected sets, locally connected sets, continuity and compactness.

Sequence in metric space: Convergent and Cauchy sequence, Completeness, Banach Fixed Point theorem with applications, sequence and series of functions, pointwise and uniform convergence, differentiation and integration of series. Continuous function on metric space: Boundedness, Intermediate Value Theorem, uniform continuity. Differentiation in N : Jacobian , implicit and inverse function theorems. Integration in N : contents and integrals, Fubini's theorem, change of variables.

Prerequisite: MAT221

Recommended Books:

1. Principle of Mathematical Analysis: W. Rudin.2. Real Analysis (3rd edition) : H.L. Royden.3. Mathematical Analysis: Tom .M. Apostol.4. Real Analysis: A. Malik.5. Mathematical Analysis: K.G. Binmore.

MAT 322 Differential Equations II 3 credits

ODE: Existence and uniqueness theory: Fundamental existence and uniqueness theorem. Dependence of solutions on initial conditions and equation parameters. Existence and uniqueness theorems for systems of equations and higher order equations. Eigen value problems and Strum-Liouville boundary value problems: Regular Strum-Liouville boundary value problems. Solution by eigenfunction expansion. Green's functions. Singular Strum-Liouville boundary value problems. Oscillation and comparison theory. Nonlinear differential equations: Phase plane, paths and critical points. Critical points and paths of linear systems.

PDE: First order equations: complete integral, General solution. Cauchy problems. Method of characteristics for linear and quasilinear equations. Charpit's method for finding complete integrals. Methods for finding general solutions. Second order equations: Classifications, Reduction to canonical forms. Boundary value problems related to linear equations. Applications of Fourier methods (Coordinates systems and separability. Homogeneous equations.) Boundary value problems involving special functions. Transformation methods for boundary value problems, Applications of the

Laplace transform. Application of Fourier sine and cosine transforms.

Prerequisite: MAT222.

Recommended Books:

1. Introduction to Differential Equations: L. Ross2. Partial Differential Equation, SpringerVerlag: F. John:3. Differential Equations: Boyce & D'Prima4. Mathematical Methods : B.D. Gupta.5. Differential Equations, McGrawHill: Sneddon.6. Introduction to Partial Differential Equations: K. Sankara Rao.7. Mathematical Methods (Volume II) Md. Abdur Rahaman

MAT 323 Dynamical Systems 3 credits

Statics: Fundamental concept and principle of Mechanics. Statics of Particles: Review of vectors, vector addition of forces, resultant of several concurrent forces, resolution of forces into components, equilibrium of particles in a plane and in space. Rigid bodies: momentum of a force and a couple, Varignon's theorem, equivalent system of forces and vectors, reduction of system of forces. Equilibrium of rigid bodies: reactions at supports and connections of rigid bodies in two dimensions. Centroid and center of gravity: CG of two and three dimensional bodies, centroids of areas, lines and volumes. Moment of inertia, moments and products of inertia, radius of gyration, parallel axis theorem, principal axis and principal moments of inertia.

Dynamics: Kinematics of particles: rectilinear and curvilinear motion of particles. Kinematics of particles: Newton's second law of motion, linear and angular momentum of a particle, conservation of energy and momentum, principle of work and energy and their applications, motion under a central force and conservative central force, impulsive motion. System of particles: Newton's law, effective forces, linear and angular momentum, conservation of momentum and energy, work energy principle. Kinematics of rigid bodies: translation, rotation, and plane motion relative to rotating frame, Coriolis force. Plane motion of a rigid body: motion in two dimensions, Euler's equation of motion about a fixed point.

Prerequisite: MAT122.

Recommended Books:

1. Vector Mechanics for Engineers: F.P. Beer & E.R. Johnston.2. Principle of Mechanics : Soynge & Griffiths.

MAT 324 Graph Theory 3 credits

Number System: Numbers with different bases, their conversion and arithmetic operations, normalized scientific notations. Logic: Introduction to logic, logical operations, application of logic to sets. Mathematical Reasoning: Methods of proof, Mathematical induction, recurrence relations, generating functions. Boolean Algebra: Ordered sets, lattices, Boolean algebra and operations, Boolean expressions, logic gates, minimization of Boolean expressions, Karnaugh maps, Karnaugh map algorithm. Graphs: introduction and definitions, representing graphs, graph isomorphism, connected graph, planar graph, path and circuit, shortest path algorithm, Eulerian path, Euler's theorem, Seven Bridges of Problem, graph coloring. Application of graph: tree, tree reversal, trees and sorting, spanning trees, minimum spanning trees: related algorithms. Search trees: binary search tree, leaves on a rooted tree, spanning trees and the MST problems, network and flows, the max flow and the min-cut theorem. Binary trees.

Recommended Books:

1. Discrete Mathematics & its Applications: K.H. Rosen.2. Discrete Mathematics, Oxford : N.L. Biggs.3. Graph Theory, Coding Theory & Block Design, Cambridge: P.J. Cameron & J.H. Van Kaint.4. Discrete and Combinational Mathematics : Ralph P. Grimaldi.

MAT 325 Mathematical Methods 3 credits

Series solution: singularity of a rational function, series solution of linear differential equations at nonsingular and regular singular points. Fourier series: Introduction to orthogonal functions and Integral transform, Fourier integral, Fourier transform and their applications. Laplace transformation method: Definition, existence and properties of Laplace transform, Inverse Laplace transform, Transforms of discontinuous and periodic functions. Convolution. Impulses and Dirac delta function. Solving initial value problems. Solving linear systems, Harmonic functions: Laplace equation in different coordinates and its applications. Special functions: Legendre functions of first and second kinds, Hermite polynomials, generating function, Hypergeometric functions, Laguerre functions, Bessel function and their properties.

Prerequisite: MAT 322

Recommended Books:

1. Fourier Series and Boundary Value Problem : Churchill & Brown.2. Introduction to Differential Equations, Eddison Wesley: L. Ross.3. Mathematical Methods: B.D. Gupta.4. Mathematical Methods : Abdur Rahman.5. Laplace Transform : M.R. Spiegel.

MAT 326 Hydrodynamics 3 credits

Preliminaries: Concept of viscosity; Inviscid fluid; stream line, path line and streak lines; steady and unsteady motion. Equation of motion: Equation of continuity; Euler's equation of motion, conservative forces, Bernoulli's equation; circulation and Kelvin's circulation theorem; vorticity, irrotational and rotational motion, velocity potential; energy equation, Kelvin's minimum energy theorem. Two dimensional motion: vorticity, stream function and velocity potential function, streaming motion, complex potential and complex velocity, stagnation points, motion past a circular cylinder, circle theorem, motion past a cylinder, Joukowaski transformation, Blasius theorem; two dimensional source, sink and doublets, source and sink in a stream, the method of image. Vortex motion: vortex line, tube and filament, rectilinear and circular vortices, kinetic energy of system of vortices, vortex sheet, Karman's vortex street.

Recommended Books:

1. Fluid Dynamics : F. Chorlton.2. Theoretical Hydrodynamics: L.M. Milne Thompson.3. Hydrodynamics : H. Lamb.4. Hydrodynamics : P.P. Gupta.

MAT 411 Topology 3 credits

Metric Spaces: Definition and some examples. Open sets. Closed sets. Convergence. Completeness. Baire's theorem. Continuous mappings. Spaces of continuous functions. Euclidean and unitary spaces. Topological Spaces: Definition and some examples. Elementary concepts. Open bases and open subbases. Weak topologies. Function algebras. Compactness: Compact spaces. Product spaces. Tychonoff's Theorem. Locally compact spaces. Compactness for metric spaces. Separation: T1-spaces and Hausdorff spaces. Completely regular spaces and normal spaces. Urysohn's lemma. Connectedness: Connected spaces. Locally connected spaces. Pathwise connectedness. Banach Spaces: Definition and some examples. Continuous linear transformations. Hahn-Banach theorem. Natural embedding. Open mapping theorem. Conjugate of an operator. Hilbert Spaces: Definition and some simple properties. Orthogonal complements. Orthogonal sets. Conjugate spaces. Adjoint and self-adjoint operators. Fixed point theory: Banach contraction principle. Schauder Principle. Applications.

Prerequisite: MAT 221

Recommended Books:

1. Introduction to Topology & Modern Analysis: G.I. Simmons.2. General Topology : S. Lipschutz.3. General Topology : J.K. Killey.

MAT 415 Finite Element Methods 3 creditsBasic concept of finite element method, approximate solution of BVP, direct approach to Finite Element Methods. Galerkin's weighted residual method for one-dimensional BVP, the modified Galerkin's technique. Shape functions for one-dimensional elements. Division of a region into elements, linear and quadratic elements, numerical integration over elements. Finite element solution of one dimensional BVPs. Finite Element approximations of line and double integrals: line integral using quadratic elements, double integrals using triangular and quadrilateral elements, double integrals using curved elements. Finite Element solution of two-dimensional BVP: Galerkin formulation, matrix formulation for 2-D finite elements. Three-nodded triangular elements. Variational formulation of BVP: construction of variational functions, the Ritz method and finite elements, matrix formulation of the Ritz procedure, solution of two-dimensional problems. Pre-processing and solution assembly: mesh generation in one and two dimensions, techniques of assembly and solutions.

Prerequisite: MAT312.

Recommended Books:

1. Finite Element Analysis for Undergraduates : E.Akin.2. The finite Element Method: Principles and Applications, P.E. Lewis and J.P Ward. 3. The Finite Element Method in Engineering : S.S.Rao.4. Application and implimentation of Finite Element Methods, J.E.Akin.

MAT 416 Tensor Calculus 3 credits

Tensor: Coordinates, Vectors and tensors: Curvilinear coordinates, Kronecker delta, summation convention, space of N dimensions, Euclidean and Riemannian space, coordinate transformation, Contravariant and covariant vectors, the tensor concept, symmetric and skewsymmetric tensor. Riemannian metric and metric tensors: Basis and reciprocal basis vectors, Euclidean metric in three dimensions, reciprocal or conjugate tensors, Conjugate metric tensor, associated vectors and tensor's length and angle between two vector's, The Christoffel symbols. Covariant Differentiation of Tensors and applications: Covariant derivatives and its higher rank tensor and covariant curvature tensor.

Prerequisite: MAT313

Recommended Books:

1. Introduction to Tensor Calculus and Continuum Mechanics: J.H. Heinbockel2. Tensor Analysis : Richard L Bishop3. A Brief on Tensor Analysis : James G. Simmonds4. Tensor Analysis : J.L. Synge5. Tensor Analysis : Md. Abdur Rahaman.

MAT 421 Fluid Mechanics 3 credits

Preliminaries: Real and ideal fluids, Viscosity, Reynolds number, laminar and turbulent flows, boundary layers. Stress and rate of strain, General stress state of deformable bodies, General state of deformation of flowing fluid, Relation between stress and rate of deformation in general orthogonal coordinates. Equations of motion: Thermodynamic equation of state, Equation of continuity, Navier-Stokes equations, Energy equation, Equations of motion in different coordinate system. Exact solution of Navier-Stokes equations, Steady plane flow, Couette-Poiseuille flow, Plane stagnation-point flow, flow past parabolic body and circular cylinder, Steady axisymmetric flow, Circular pipe flow (Hagen-Poiseuille flow), Flow between two concentric rotating cylinders, Flow at a rotating disc, Unsteady plane flow, First Stokes problem, Second Stokes problem, Startup of Couette flow, Unsteady plane stagnation-point flow. Similarity analysis: Reynolds law of similarity, Dimensional analysis and theorem, Important non-dimensional quantities. Very slow motion: Equations of slow motion, Motion of a sphere in a viscous fluid, Theory of lubrication. Laminar boundary layer: Introduction to boundary layer, boundary layer equations in two dimensions, Dimensional representation of boundary layer equations, Displacement thickness, Friction drag, Flat plate boundary layer, Momentum thickness, Energy thickness, Similar solutions of boundary layer equations: Derivation of ODE, Wedge flow, Flow in a convergent channel, Integral relations of the boundary layer: Momentum-Integral equation, Energy-Integral equation.

Prerequisite: MAT 326

Recommended Books:

1. Boundary Layer Theory : Schilisting2. Introduction to Fluid Dynamics : G. Taylor3. Fluid Mechanics : F. Chorlton4. Fluid Mechanics : Raisinghania

MAT 422 Theory of Numbers 3 credits

Arithmetic in U. Euclidean algorithm. Continued fractions. The ring U and its group of units. Chinese Remainder Theorem. Linear Diophantine equations. Arithmetical functions. Dirichlet convolution. Multiplicative function. Representation by sum of two and four squares. Arithmetic of quadratic fields. Euclidean quadratic Fields.

Recommended Books:

1. Theory of Numbers: G.H. Hardy & E.M. Wright.2. Theory of Numbers : Niven & Zucherman.3. Theory of Numbers: J. Hunter.4. Theory of Numbers : Apostol.

MAT 423 Mathematical Modeling 3 credits

Modeling in BiologyContinuous population models for single species: Continuous growth models, Delay models, Periodic fluctuations, Harvesting models. Discrete population models for single species: Simple models, Cobwebbing, Discrete logistic models, Stability, Periodic fluctuations and Bifurcations, Discrete Delay models, Continuous models for interacting populations: Predator-prey models, Lotka-Volterra systems, Complexity and stability, Periodic behaviour, Competition Models, Mutualism. Discrete growth models for interacting populations: Predator-prey models. Epidemic models and dynamics of infectious diseases: Simple epidemic models and practical applications.

Modeling in EconomicsTheory of the household: Preference and indifference relations, Utility function, Order conditions of optimization, Stutsky equation, Demand functions, Revealed Preference hypothesis, Von Neumann-Morgenstern utility. Theory of the firm: Production function, Laws of production and scale, Optimizing behaviour, Cost curves and cost functions, Constrained output maximization. Theory of factor demand: Optimal input mix, Factor demand and supply curves, Elasticity of derived demand. Market structures and equilibrium: Market Economy and equilibrium, Stability of equilibrium, Dynamic stability.

Prerequisite: MAT 322.

Recommended Books:

1. The Nature of Mathematical Modeling, Cambridge University Press: Gerschenfeld.2. An Introduction to Mathematical Modeling, Dover Publication: F.A. Bender.3. A First Course in Mathematical Modeling, Brooks/cole: F.R. Giordano & M.D. Weir.4. Guide to Mathematical Modeling, MacMillan: D. Edwards & M. hamson.

MAT424 Operations Research II 3 credits

Transportation and Assignment Problem: Introduction and formulation; relationship with linear programming. Network models: shortest route problems, minimal spanning, maximal-flow problem. Sequencing problem: Minimax-maximin strategies, mixed strategies, expected pay-off, solution of and games, games by linear programming and Brown's algorithm. Dynamic programming: Investment problem, Production Scheduling problem, Stagecoach problem, Equipment replacement problem. Non-linear programming: Introduction, unconstrained problem, Lagrange method for equality constraint problem, Kuhn-Tucker method for inequality constraint problem. Quadratic programming.

Prerequisite: MAT 316

Recommended Books:

1. Operations Research: An Introduction ,4th Edition by Hamdy A. Taha2. Convex Optimization: Stephen Boyd & Lieven Vandenberghe3. Schaum's Outline of Operations Research by Richard Bronson

MAT 425 Advanced Numerical Methods 3 credits

Non-iterative (Newton's, steepest descent) methods for solution of a system of equation(linear and non-linear). Approximation theory: discrete least square applications, Chebyshev polynomial applications, rational function approximation, trigonometric polynomial approximation, Fast Fourier Transform. Approximating Eigenvalues: Honseholder's method, QR algorithm. BVP involving ODE: shooting method for linear and nonlinear problems, finite difference method for linear and nonlinear problems. PDE: Finite difference methods for elliptic, parabolic and hyperbolic problems.

Prerequisite: MAT 312

Recommended Books:

1. Advanced Methods for Scientists and Engineers, Mc.GrawHill Int: C.M. Bender and

S.A. Orszag.2. Asymptotic Expansions for Ordinary Differential Equations Wiley, New York, 1965.; W.A. Wasow.3. Introduction to Nonlinear differential and Integral Equations Dover Publications: H.T Davis.

MGT 211 Principles of Management 3 credits

Meaning and importance of management, evolution of management thoughts; managerial decision making; Environmental impact, corporate social responsibility, planning, setting objectives, implementing plans, organizing; organization design, managing change, directing, motivation, leadership, managing work groups, controlling: principles, process and problems and managers in changing environment.

Recommended Books:

1. Management: Stephen P. Robbins and Mary Coulter2. Management: James A. F. Stoner, Edward R. Freeman & Daniel R. Gilbert

PHY 111 Principles of Physics I 3 Credits

Vectors and scalars, unit vector, scalar and vector products, static equilibrium, Newton's Laws of motion, principles of conservation of linear momentum and energy, friction, elastic and inelastic collisions, projectile motion, uniform circular motion, centripetal force, simple harmonic motion, rotation of rigid bodies, angular momentum, torque, moment of inertia and examples, Newton's Law of gravitation, gravitational field, potential and potential energy. Structure of matter, stresses and strains, Moduli of elasticity Poison's ratio, relations between elastic constants, work done in deforming a body, bending of beams, fluid motion and viscosity, Bernoulli's Theorem, Stokes' Law, surface tension and surface energy, pressure across a liquid surface, capillarity. Temperature and Zeroth Law of thermodynamics, temperature scales, isotherms, heat capacity and specific heat, Newton's Law of cooling, thermal expansion, First Law of thermodynamics, change of state, Second Law of thermodynamics, Carnot cycle, efficiency, kinetic theory of gases, heat transfer. Waves & their propagation, differential equation of wave motion, stationary waves, vibration in strings & columns, sound wave

& its velocity, Doppler effect, beats, intensity & loudness, ultrasonics and its practical applications. Huygens' principle, electromagnetic waves, velocity of light, reflection, refraction, lenses, interference, diffraction, polarization.

Recommended Books:

1. Fundamentals of Physics: D. Halliday, R. Resnick & J. Walker2. University Physics: Francis W. Sears, Mark W. Zemansky, Hugh D. Young 3. Theory & Problems of Vector Analysis: Schaum's Outlines4. Outlines of Physics Vol.1: Dr. Giasuddin Ahmad5. Properties of Matter: Brij Lai and N. Subrahmanym6. Heat and Thermodynamics: Brij Lai and N. Subrahrnanyam7. A Textbook of Sound: N. Subrahmanyam and Brij Lal8. A Textbook of Optics : Brij Lal and N. Subrahrnanyam

PHY 112 Principles of Physics II 3 Credits

Electric charge, Coulomb's Law, electric field & flux density, Gauss's Law, electric potential, capacitors, steady current, Ohm's law, Kirchhoff's Laws. Magnetic field, BiotSavart Law, Ampere's Law, electromagnetic induction, Faraday's Law, Lenz's Law, self inductance and mutual inductance, alternating current, magnetic properties of matter, diamagnetism, paramagnetism and ferromagnetism. Maxwell's equations of electromagnetic waves, transmission along wave guides. Special theory of relativity, length contraction and time dilation, massenergy relation. Quantum theory, Photoelectric effect, xrays, Compton effect, dual nature of matter and radiation, Heisenberg uncertainty principle. Atomic model, Bohr's postulates, electron orbits and electron energy, Rutherford nuclear model, isotopes, isobars and isotones, radioactive decay, halflife, alpha, beta and gamma rays, nuclear binding energy, fission and fusion.Fundamentals of solid state physics, lasers, holography.

Recommended Books:

1. Fundamental of Physics: D. Halliday, R. Resnick and J. Walker2. University Physics: Francis W. Sears, Mark W. Zemansky, Hugh D. Young3. Electricity and Magnetism with Electronics: K.K. Tewari4. B.Sc. Physics Volume 1: C.L. Arora5. Perspectives of Modern

Physics: A. Beiser

PHY 204 Classical Mechanics and Special Theory of Relativity 3 credits

Classical Mechanics: Newtonian equations of motion, conservation laws of a system of particles, variable mass, generalized coordinates, generalized force, D' Alembert's Principle, variational method, EulerLagrange equations of motion, Hamilton's principles, twobody central force problem, elliptic orbit, scattering in a central field, Rutherford formula, kinematics of rigid body motion, Euler angles, rotating coordinates, Coriolis force, wind motion, principal axis transformation, top motion, principle of least action, Hamiltonian equations of motion, small oscillations, normal coordinates, normal modes. Special Theory of Relativity: Galilean relativity, MichelsonMorley experiment, postulates of special theory of relativity, Lorentz transformation, length contraction, time dilation, twin paradox, variation of mass, relativistic kinematics, mass energy relation.

Recommended Books:

1. Classical Mechanics: H. Goldstein2. Special Relativity: A.P. French3. Perspectives of Modern Physics: A. Beiser4. Special Relativity from Einstein to Strings: Patricia Schwarz5. An Introduction to Special & General Relativity: Hans Stephain

PHY 205 Statistical Mechanics 3 Credits

Statistical Mechanics: Phase space, concept of state and ensemble, microcanonical, canonical and grand canonical ensembles, Boltzmann probability distribution, Maxwell velocity distribution, derivation of Bose-Einstein and Fermi-Dirac statistics, ideal Fermi gas, degenerate Fermi system, equation of state of ideal gases, ideal Bose gas, application of Statistical mechanics in various fields in physical, biological, social sciences, economics, finance and in engineering & ICT.

Recommended Books:

1. Fundamentals of Statistical and Thermal Physics: F. Reif2. Statistical Mechanics: Kerson Huang3. Statistical Physics: F. Mandl4. Elementary Statistical Physics: C. Kittel

PHY 303 Quantum Mechanics I 3 Credits

Breakdown of classical physics, quantum nature of radiation, Planck's Law, photoelectric effect, Einstein's photon concept and explanation of photoelectric effect, Compton Effect, de Broglie wave, wave particle duality, electron diffraction, DavissonGermer experiment, emergence of quantum mechanics, Schrodinger equation, basic postulates of quantum mechanics, physical interpretation of wave function, wave packets, Heisenberg's uncertainty principle, linear operators, Hermitian operators, eigenvalue equation, onedimensional potential problem, harmonic oscillator, orbital angular momentum, rotation operator, spherical harmonics, spin angular momentum, spherically symmetric potential, solution of the Schrodinger equation for hydrogen atom, matrix formulation of quantum mechanics.

Recommended Books:

1. Quantum Mechanics: John L. Powell and B. Craseman2. Quantum Mechanics: L.I. Schiff3. Quantum Mechanics: E. Merzbacher4. Quantum Mechanics: A.M. Harun ar Rashid5. Introduction to Quantum Mechanics: P.T. Matthews6. Modern Quantum Mechanics: J.J. Sakurai7. Quantum Mechanics: D.R. Bes

PHY 312 Physics for Development 3 Credits

Twenty first century development issues, physics and break through technologies, ICT, fibre optics, quantum information theory, physics in genetics engineering and molecular biology, physics and health issues, bio and medical physics, materials science and physics, high temperature superconducting materials, space physics, microgravity experiments, onophysics,physics principles applied in sociology.

Recommended Books:

1. Oxford Companion to the History of Modern Science: J.L. Heilbron.2. The Evolution of Technology (Cambridge Studies in the History of Science): George Basalla & Owen Hannaway3. Technology in World Civilization: A ThousandYear History: Arnold Pacey4. Zoological Physics: B.K. Ahlborn5. New Directions in Statistical Physics: Econophysics, Bioformatics & Pattern Reorganization: L.T. Willie6. Knowledge & the World: Challenges

Beyond the Science Wars: M. Cavrier, J. Roggenhofer, G. Kuppers & P. Blanchard

PHY 413 General Theory of Relativity 3 Credits

Gravitation, Lagrangian Einstein equations, approximation of weak field and Hilbert's auxiliary conditions, comparison of corresponding relations with those of Newton's theory of gravitation, source of gravitation field, Schwarzschild's solution in isotropic and other coordinate systems, analogy between gravitation and electromagnetism, motion of test mass and geodetic lines, motion in Schwarszchild's field, equations of motion in general relativistic mechanics as a consequence of Einstein's equation of gravitational field, gravitational waves in weak field approximation, problem of energy transfer, exact wave solutions in the case of gravitational field, waves of matrices or wave of curvature, locally plane gravitational waves, Weber's and Braginski's experiments, prospects of future gravitational experiments.

Recommended Books:

1. A First Course in General Relativity: Bernard F. Schutz2. Gravitation and Cosmology: Steven Weinberg3. General Relativity: Robert Wald4. Relativity: Foster5. Relativity & Cosmology: J.N. Islam6. An Introduction to Special & General Relativity: Hans Stephain

POL 103 Introduction to Political Science 3 Credits

A study of political systems and process with special reference to Bangladesh. Topics include nature and origin of state, sovereignty of state, forms of political units, liberty, law, process of politics, political structure, political ideas democracy, socialism, nationalism, peoples' behaviour in politics. Political system, process and problems of Bangladesh.

Recommended Books:

1. A History of Political Thought: Plato to Marx: Subrata Mukherjee & Sushila Ramaswamy2. An Introduction to Political Science: Rand Dyck3. A Social Political History of Bengal and the Birth of Bangladesh: Kamruddin Ahmed4. Radical Politics and the Emergence of Bangladesh: Talukder Manizzaman5. India and Pakistan: A Political

Analysis: Hugh Tinker6. Politics and Policy Making in Developing Countries: Perspective on the New Political Economy: Gerald M. Meier7. Political Culture, Political Parties and Democratic Transition in Bangladesh: Shamsul I. Khan, S. Aminul Islam & Imdadul Haque8. Involvement in Bangladesh's Struggle for Freedom: T. Hossain9. History of Bangladesh 17041971: Political History: Sirajul Islam10. Conflict and Compromise: An Introduction to Political Science: H.R. Winter

POL 245 Women, Power & Politics 3 Credits

A critical examination of the impact of gender on forms and distributions of power and politics, with primary reference to the experience of Women in South Asia. Three major concerns will be addressed. First, what do we: mean by "sex", "gender". "women", "power" and "politics"? Second, how do issues of class and race/ethnicity inform our understanding of women and politics? Third, what is the relationship between women and the state? How can women organise collectively to challenge state policies, how does the state respond to organised women?

Prerequisite: POL 103

Recommended Books:

1. The Elusive Agenda: Mainstreaming Women in Development, UPL: Raunaq Jahan.2. Persistent Inequalities: Women and World Development, 1990: Frene Tinker3. Women Leaders in Development Organizations and Institutions: Sayeda Rawsan Kadir.4. Population Policy and Women Rights: Transforming Reproductive Choice: Prayer, Ruth DixonMueller.5. Women in Politics, 1994, Dhaka Women for Women: Najma Chowdhury, Hamida A Begum, Mahmuda Islam and Nazmunnessa Mahtab.

PSY 101 Introduction to Psychology 3 Credits

The objective of this course is to provide knowledge about the basic concepts and principles of psychology pertaining to reallife problems. The course will familiarize students with the fundamental process that occur within organismbiological basis of behaviour, perception, motivation, emotion, learning, memory and forgetting and also to the social perspectivesocial perception and social forces that act upon the individual.

Recommended Books:

1. Introduction to Psychology: C.T. Morgan2. Introduction to Psychology: R.F. Crider3. Understanding Psychology: Robert Feldman

SOC 101 Introduction to Sociology 3 credits

Perspectives on society, culture, and social interaction, Topics include community, class, ethnicity, family, sex roles, and deviance. Social problems and sociological problems. Problems, theories, and the nature of sociological explanation. Explanation, evidence and objectivity. Sociology as a comparative study of social action and social systems. Some models of sociological thinking as applied to the study of the following: aspects of social ranking; forms of interpersonal and personal relationships; the changing nature of the relationship between economy and society; the sociology of development; the origins and spread of capitalism and socialism; ideology and belief systems; religion and society; rationality and nonrationality; conformity and deviance.

Recommended Books:

1. Sociology: Anthony Giddens2. Sociology: Richard T. Schaefer3. Sociology: Rao and C.N. Shankar4. Sociology: Neil J. Smelser

SOC 401 Gender and Development 3 credits

Position & role of women in society, contemporary issues, analysis of various aspects of gender relations, gender discrimination, societal attitude, different forms of feminism, women in higher education, employment of women & discrimination, workplace harassment, contribution of women in development: world picture I position in Bangladesh.

Prerequisite: SOC 101

Recommended Books:

1. Women and Social Security: Progress Towards Equality of Treatment, 1990 Geneva International Labor Office: Anne Marie Brocas, AnneMarie Cailloux and Virgine Oget.2.

Impact of Women in Development Projects on Women Status and Fertility in Bangladesh, 1993, Dhaka, Development Researchers and Associates: M. Kabir, Rokeya Khatun, Ishrat Ahmed.3. Integration of Women in Development: Why, When and How: Ester Boseup, Cristine Liljencrantz.4. Women in the Third World: Gender Issue in Rural and Urban areas: Hants5. Women, Man and Society: The Sociology of Gender: Allyn and Bacon, Claire M Renzetti, Daniel J Curran.6. Race, Ethnicity, Gender and Class: the Sociology of Group Conflict and Change, London: Joseph F Healey

STA 201 Elements of Statistics and Probability 3 credits

Frequency distribution, mean, median, mode and other measures of central tendency, standard deviation and other measures of dispersion, moments, skewness and kurtosis, elementary probability theory and discontinuous probability distribution, binomial, Poisson and negative binomial distribution, continuous probability distributions, normal and exponential, characteristics of distributions, hypothesis testing and regression analysis, basic concepts and applications of probability theory and statistics, chisquared test.

Recommended Books:

1. Probability and Random Processes: G.R. Grimmett and D.R. Stirzaker2. Elementary Probability Theory with Stochastic Processes: K.L. Chung

STA 301 Modern Probability Theory & Stochastic Processes 3 Credits

Stochastic Processes: Definition of different types of stochastic processes, recurrent events, renewal equation, delayed recurrent events, number of occurrence of a recurrent event. Markov Chain: Transition matrix, higher transition probabilities, classification of sets and chains, ergodic properties. Finite Markov Chain: General theory of random walk with reflecting barriers, transient states, absorption probabilities, application of recurrence time, gambler's ruin problem. Homogeneous Markov Processes: Poisson process, simple birth process, simple death process, simple birth death process, general birth process, effect of immigration, nonhomogeneous birth death process, Queuing theory. Modern Probability Theory: Probability of a set function, Borel field and extension of probability measure, probability measure notion of random variables, probability space, distribution functions, expectation and moments.

Prerequisite: STA201

Recommended Books:

1. Probability and Random Processes by G.R. Grimmett and D.R. Stirzaker.2. Elementary Probability Theory with Stochastic Processes by K.L. Chung.3. The Theory of Stochastic Processes by D.R.Cox and W.Miller4. Introduction to Probability Models by S.M. Ross.5. Stochastic Processes by S.Ross6. Elements of Applied Stochastic Processes by U.N. Bhat.

b. Lab Courses:

MAT 250 Mathematics Lab I 2 Credits

Introduction to the computer algebra package MATHEMATICA/Matlab. Evaluation and graphical representation of function. Solution of linear and nonlinear equations by using False- Position, Bisection, Newton Raphson methods. Solution of system of linear equations by using Gaussian Elimination method. Interpolation and extrapolation. Numerical differentiations. Curve fitting. Trapezoidal and Simpson's rules for numerical integration. Problem solving in concurrent courses (e.g. Calculus, Linear Algebra and Geometry), using FORTRAN, MATHEMATICA and Matlab. Lab Assignments: There shall be at least 15 lab assignments

Recommended Books:

1. FORTRAN Programming: Schaum outline series.2. Mathematica : Stephen Wolfram.3. Matlab for engineering applications: W.J. PalmIII.

MAT 350 Mathematics Lab II 2 Credits

Solution of initial value or boundary value problems for Ordinary differential equations. Solution of initial value or boundary value problems for partial differential equations. Problem solving in concurrent courses (e.g; Calculus, Advanced linear algebra problems, Differential Equations and Numerical Analysis, Complex Analysis, Linear Programming, Numerical Analysis and Applied Mathematics) using FORTRAN and MATHEMATICA/Matlab. Lab Assignments: There shall be at least 15 lab assignments.

Recommended Books:

1. FORTRAN Programming: Schaum outline series.2. Mathematica : Stephen Wolfram.3. Matlab for engineering applications: W.J. PalmIII.

c. Dissertation:

MAT 400: Project /Thesis 3 Credits

A student is required to carry out project / thesis work in the last two semesters in her/his chosen field. There will be a supervisor who will either be a BRAC University faculty or any other suitable expert from universities and R/D organizations of the country to guide the project / thesis work .On completion of study and research s/he will have to submit the dissertation report and to face a viva board for the defence of the dissertation.

Mathematics Courses for Students of Other Disciplines

Mathematics & Natural Science Department of BRAC University offers the following mathematics courses to undergraduate programmes of Computer Science and Engineering, Computer Science, Electronics and Communication Engineering, Architecture, Bachelor of Business Administration, Economics English and Physics.

MAT 091 Basic Course in Mathematics (Non-credit)

Topics including sets, relations and functions, real and complex numbers system, exponents and radicals, algebraic expressions; quadratic and cubic equations, systems of linear equations, matrices and determinants with simple applications; binomial theorem, sequences, summation of series (arithmetic and geometric), permutations and combinations, elementary trigonometry; trigonometric, exponential and logarithmic functions; coordinate geometry; statics - composition and resolution of forces, equilibrium of concurrent forces; dynamics - speed and velocity, acceleration, equations of motion.

Recommended Books:

1. Algebra and Trigonometry: Lial and C.D. Miller2. College Algebra: Michael Sullivan

MAT101 Fundamentals of Mathematics 3 credits

Basic techniques of algebra, Sets and their applications, Polynomials, Inequalities, Linear inequalities , Exponential and Logarithmic functions, Trigonometric Functions, Function , Domain & Range of function, Sketching of functions, Matrix, Properties of Matrix, System of Linear Equations, Matrix solution of system of linear equations, Inverse matrix, determinant, Cramer's rule, Limit and Continuity, Elementary differentiation and application, Integration.

Recommended Books

1. Algebra and Trigonometry (6th edition) : Margaret L.Lial, Charles D. Miller, David I. Scneider.2. College Algebra (5th edition) : Raymond A. Barnett, Michael R. Ziegler3. College Algebra(5th edition) : Michael Sullivan

MAT 102 Introduction to Mathematics 3 credits

Factorisation, Synthetic Division, Zeros (Roots) of Polynomials, Relation between Roots and Coefficients, Nature of Roots (Descarte's Rule of signs); Complex Number System, Graphical representation of Complex Numbers (Argand Diagram), Polar form of Complex Numbers; Conic Sections, Parabola, Circle, Ellipse, Hyperbola, Transformation of Coordinates and Applications; Exponential Growth & Decay. Applications; Mathematical Induction; Determinants, Fundamental Properties of Determinants, Minors and Cofactors, Application of Determinants to solve System of Linear Equations (Cramer's, Rule); Introduction to Matrix Algebra, Matrix Multiplication, Augmented Matrix, Adjoint Matrix, Inverse Matrix, Application of Matrices-solution of System of Linear Equations (homogeneous & non-homogeneous), Consistency of System of Equations.

Recommended Books:

1. Algebra and Trigonometry: Lial and C.D. Miller2. College Algebra: Michael Sullivan

MAT103 Basic Concepts in Mathematics 3 credits

The real numbers, Absolute value of real numbers, Exponents, Polynomials, Basic operations and Factoring of polynomials, Rational expressions, Radicals. Linear Equations, Solutions, graphs and application, Quadratic Equations, Solutions, graphs and applications, Variation, Linear inequalities. Exponential and Logarithmic functions, Exponential growth and decay, Ratios, proportions, percent, application of simple and compound interest. Trigonometric Functions, the sine and cosine functions, Cartesian coordinate systems, Graphing, relations, Equations of a straight line, its slope, Equations of a circle, System of Linear Equations, Matrix. Population, Sample, Variable, Raw data, Frequency distribution table, Graphical presentation, Measures of central tendency and measures of dispersion.

Recommended Books

1. Algebra and Trigonometry (6th edition) - Margaret L.Lial, Charles D. Miller, David I. Scneider.2. College Algebra (5th edition) - Raymond A. Barnett, Michael R. Ziegler3. College Algebra(5th edition)- Michael Sullivan4. Statistics for management(7th edition)- Richard I. Levin, David S. Rubin

MAT 104 Mathematics 2 credits

Calculus: Function, definition of limit, continuity and differentiability, successive and partial differentiation, maxima and minima, integration, integration by parts, standard integrals, definite integrals.

Solid Geometry: Plane coordinates geometry, System of co-ordinates, Transformation of co-ordinates, projection, Straight lines, Pair of straight lines, Distance between two points, Sphere, Conicoids, Ellipsoid, Paraboloid.

Recommended Books

1. A Text Book on Co-ordinate Geometry and Vector Analysis : Khosh Mohammad(K.M)2. A Text Book on Co-ordinate Geometry and Vector Analysis : Rahman and Bhattacharjee

(R.B)3. Calculus (7th Edition) : Howard Anton (H.A)4. Calculus : Thomas and Finney.

MAT 105 Calculus 3 credits

Differential Calculus: Limits, continuity and differentiability, differentiation, Taylor's, Maclaurine's & Euler's theorems, indeterminate forms, tangent and normal, sub tangent and subnormal, maxima and minima, radius of curvature & their applications, introduction to calculus of function of several variables, Taylor's theorem, maxima and minima for function of several variables. Transformation of coordinates & rotation of axes, conic sections.

Integral Calculus: Definition of integration, techniques of integration for definite & indefinite integrals, improper integrals, area, volume and surface integration, arc length and their applications, multiple integrals, Jacobian, line integrals, divergence theorem and Stokes' theorem, beta function and gamma function.

Recommended Books:

1. Calculus: Howard Anton2. Calculus with analytic Geometry: E. W. Swokowski

MAT 110 Differential Calculus and Coordinate Geometry 3 credits

Differential Calculus: Limits. Continuity and differentiability. Successive differentiation of various types of functions. Liebnitz's Theorem. Rolle's theorem. Mean Value theorem. Taylor's theorem in finite and infinite forms. Maclaurine's theorem in finite and infinite forms. Lagrange's form of remainders. Expansion of functions. Evaluation of indeterminate forms by L'Hospitals rule. Partial differentiation. Euler's theorem. Tangent and normal. Subtangent and subnormal in Cartesian and polar coordinates. Determination of maximum and minimum values of functions and points of inflexion. Application. Curvature. Radius of curvature. Centre of curvature.

Coordinate Geometry: Change of axes. Transformation of coordinates. Simplification of equation of curves. Pair of straight lines. Conditions under which general equations of the second degree may represent a pair of straight lines. Homogeneous equations of the second degree. Angle between the pair of lines. Pair of lines joining the origin to the point of intersection of two given curves. System of circles; orthogonal circles. Radical axes, radical centre, properties of radical axes, coaxial circles and limiting points. Equations of ellipse and hyperbola in Cartesian and polar coordinates. Tangent and normal. Pair of tangent. Chord of contact. Chord in terms of its middle points, parametric coordinates. Diameters. Conjugate diameters and their properties. Director circles and asymptotes. 3 credits. [Students will be expected to attend a 3hour tutorial class, once each week and submit tutorial worksheets.]

Recommended books:

1. A Text Book on Coordinate geometry and Vector Analysis : Kosh Mohammad.2. The elements of Coordinate geometry : S.L. Loney.3. Calculus (7th Edition) : Howard Anton.

MAT 120 Mathematics II: Integral Calculus and Differential Equations 3 credits

Integral Calculus: Definitions of integration. Integration by the method of substitution. Integration by parts. Standard integrals. Integration by method of successive reduction. Definite integrals, its properties and use in summing series. Walli's formula. Improper integrals. Beta function and Gamma function. Area under a plane curve in cartesian and polar coordinates. Area of the region enclosed by two curves in cartesian and polar coordinates. Trapezoidal rule. Simpson's rule. Arc lengths of curves in cartesian and polar coordinates, parametric and pedal equations. Intrinsic equations. Volumes of solids of revolution. Volume of hollow solids of revolutions by shell method. Area of surface of revolution.

Ordinary Differential Equations: Degree of order of ordinary differential equations. Formation of differential equations. Solution of first order differential equations by various methods. Solutions of general linear equations of second and higher order with constant coefficients. Solution of homogeneous linear equations. Applications. Solution of differential equations of the higher order when the dependent and independent variables are absent. Solution of differential equations by the method based on the factorization of the operators. (The course also includes some lab work)

Prerequisites: MAT 110.

Recommended books:

1. Calculus with analytic geometry (7th edition) : Howard Anton2. A First course in Differential Equations With Modeling Applications (7th edition) : Dennis G. Zill3. Mathematical Methods : Md. Abdur Rahman.

MAT 203 Matrices, Linear Algebra & Differential Equations 3 credits

Matrices: Types of matrices, algebraic operation on matrices, determinants, adjoint & inverse matrix, orthogonality & diagonalization of matrix.

Linear Algebra: System of linear equations, vector space; 2D-space, 3D-space, Euclidean nD-space, sub space, linear dependence/independence, basis and dimension, row space, column space, rank and nullity, linear transformation, eigen value and eigen vector, matrix diagonalization, application of linear algebra.

Ordinary Differential Equations: Introduction to differential equations, first order differential equations and applications, higher order differential equations and applications, series solutions of linear equations, systems of linear first order differentialequations.

Prerequisite: MAT 105

Recommended Books:

1. Elementary Linear Algebra : Howard Anton2. Introductory Linear Algebra with Application : Bernard Kolman3. First Course in Linear Algebra : P.B. Bhattacharya & S.K. Jain4. A First Course in Differential Equations : D.G. Zill5. Introduction to Differential Equations : L. Ross

MAT 204 Complex Variables & Fourier Analysis 3 credits

Complex Variables: Complex number systems, general functions of a complex variable, limits and continuity of a function of complex variables and related theorems, complex differentiation and CauchyRiemann equations, mapping by elementary functions, line integral of a complex function. Cauchy's integral theorem, Cauchy's integral formula, Liouville's theorem, Taylor's and Laurent's theorem, singular points, residue, Cauchy's residue theorem, evaluation of residues, contour integration and conformal mapping.

Fourier analysis: Real and complex form, finite Fourier transform, Fourier integrals, Fourier transforms and their use in solving boundary value problems.

Prerequisite: MAT 105

Recommended Books:

1. Complex Variable and Applications: James W Brown and Ruel V Churchill2. Complex Variables: M R Speigel

MAT 205 Introduction to Numerical Methods 3 credits

Computer arithmetic: floating point representation of numbers, arithmetic operations with normalized floating point numbers; iterative methods, different iterative methods for finding the roots of an equation and implementation of them in computer programmings, solution of simultaneous algebraic equations by various methods, solution of tridiagonal system of equations, interpolation for equispaced and nonequispaced nodes, least square approximation of functions, curve fitting, Taylor series representation, Chebyshev series, numerical differentiation and integration and numerical solution of ordinary differential equations & partial differential equations.

Prerequisite: MAT 203

Recommended Books:

1. Numerical Analysis: R.L. Burden and J.D. Faires 2. Introductory Methods of Numerical Analysis: S.S. Sastry3. Numerical Recipes: W.H. Press, S.A. Teukolsky, W.T. Vetterling & B.P. Flannery.

MAT 215 Mathematics III: Complex Variables & Laplace Transformations 3 credits

Complex Variables: Complex number systems. General functions of a complex variable. Limits and continuity of a function of complex variables and related theorems. Complex differentiation and CouchyRiemann equations. Mapping by elementary functions. Line integral of a complex function. Cauchy's integral theorem. Cauchy's integral formula.

Liouville's theorem. Taylor's and Laurent's theorem. Singular points. Residue. Cauchy's residue theorem. Evaluation of residues. Contour integration. And conformal mapping.

Laplace Transformations: Definition. Laplace transformations of some elementary functions. Sufficient conditions for existence of Laplace transforms. Inverse Laplace transforms. Laplace transforms of derivatives. The unit step function. Periodic function. Some special theorems on Laplace transforms. Partial fractions. Solutions of differential equations by Laplace transforms. Evaluation of improper integrals.

Prerequisites: MAT120.

Recommended books:

1. Complex Variable and Applications; Author: James W Brown and Ruel V. Churchill; 6th Edition2. Complex Analysis, Cambridge University Press: Stewart and D. Tall.3. Complex Variables; M R Spiegel; Schuam's Outline Series4. A First Course in Differential Equations; Author: Dennis G. Zill; 7th Edition5. Laplace Transform; Author: M R Spiegel; Schuam's Outline Series

MAT 216 Mathematics IV: Linear Algebra & Fourier Analysis 3credits

Matrices: Definition of matrix. Different types of matrices. Algebra of matrices. Adjoint and inverse of a matrix. Elementary transformations of matrices. Rank and Nullity. Normal and canonical forms. Solution of linear equations. Matrix polynomials. Eigenvalues and Eigenvectors. Vectors: Scalars and vectors, equality of vectors. Addition and subtraction of vectors. Multiplication of vectors by scalars. Scalar and vector product of two vectors and their geometrical interpretation. Triple products and multiple products. Linear dependence and independence of vectors. Differentiation and integration of vectors together with elementary applications. Definition of line, surface and volume integrals. Gradient, divergence and curl of point functions. Various formulae. Gauss's theorem, Stroke's theorem, Green's theorem. Fourier Analysis: Real and complex form. Finite transform. Fourier integral. Fourier transforms and their uses in solving boundary value problems.

Prerequisite: MAT 120

Recommended books:

1. Elementary Linear Algebra (8th ed.) by Howard Anton and Chris Rorres.2. Calculus (7th ed.) by Howard Anton.3. Introductory linear algebra with applications. (7th ed.) by Bernard Kolman.4. First course in linera algebra by PB Bhattachary, S K Jain & S R Nagpaul.5. Mathematical methods Volume2 by Md. Abdur Rahman.

MAT 301 Group Theory 3 Credits

Definition and various examples of groups, subgroups, cosets, normal subgroups, quotient (factor) groups, permutation groups, cyclic groups, generator of a cyclic group, centre of a group, Abelian group, normalizer and centralizer of an element/ subset of a group and its application to physics, group homomorphism, isomorphism and automorphism and related theorems, symmetry groups, SU (3), SU (6), application of group theory in solid state physics & elementary particles.

Recommended Books:

1. Group Theory: Hammermesh2. Abstract Algebra: P.B Bhattacharya & S.K. Jain & S.R. Nagpaul3. A First Course in Abstract Algebra: J.B. Fraleigh4. Modern Algebra: A.K. Agarwal

MAT 303 Tensor Analysis 3Credits

Definition of tensor, tensor density, affine tensor and geometrical object, properties of tensor symmetry, criteria of tensor properties, metric tensor, Kronecker symbol and LeviCivita's symbol, determinant of metric tensor, connection between metric tensor and Dirac's matrices in the Sommerfeld representation, evolution of square root from four dimensional interval in matrix sense, transformation properties of vector partial derivatives by coordinates, connection coefficients and covariant derivatives, Christoffel's symbols, geodetic lines (geodetics) as a generalization of notion of straight line, variation principle for geodetics, parallel transport, connection between geodetics and covariant differentiation, transport along closed line curvature tensor of the 4th rank, curvature tensor of the 2 D rank, scalar curvature, equations of geodetic deviation, curvature expression in terms of Dirac's matrices, Bianchi's identity, Einstein's conservative tensor, integral operations and corresponding theorems.

Recommended Books:

1. Introduction to Tensor Calculus and Continuum Mechanics: J.H.Heinbockel2. Vector Analysis: Lipschutz, Schaum's Outline Series3. Tensor Analysis: Md. Abdur Rahaman.

Tuition fees for international students will cover:

For detailed information on Fee structure of International students, please visit:

https://www.bracu.ac.bd/admissions/tuition-and-fees

Accommodation, food and transportation Cost

Approximately US$ 400 per month will be charged for accommodation and food.

* These fees may change from time to time according to the decision of the BracU authority

Tasnia Islam is a lecturer of the Microbiology Program at the Department of Mathematics and Natural Sciences, Brac University, Dhaka. She has graduated in Microbiology from Brac University in 2016 with highest distinction. Her undergraduate thesis was based on isolation of amylase producing bacteria from soil and identification using 16S rRNA gene sequencing technique. She went on to pursue her Masters on Infection Biology at the University of Glasgow, UK, after receiving the ‘University of Glasgow Chancellors Award’ and successfully graduated in 2018 with distinction. She carried out her Master’s thesis on inspecting the effect of surface layer in bacterial characteristics of Clostridium difficile and its adhesion on mammalian tissue cultures. Her research interest lies in virology, bacteriology and microbial pathogenesis.

Level:Kha-224 Merul BaddaDhaka 1212. Bangladesh

BSc in Microbiology, Department of Mathematics and Natural Sciences, Brac University in 2016

Masters in Infection Biology from the University of Glasgow, UK in 2018.

Journals

Islam, T., Choudury, N., Hossain, M. M., & Khan, T. T. (2018). Isolation of amylase producing bacteria from soil, identification by 16S rRNA gene sequencing and characterization of amylase. Bangladesh Journal of Microbiology, 34(1), 1–6. Retrieved from https://www.banglajol.info/index.php/BJM/article/view/39224

Lecturer, Microbiology Program, Department of Mathematics and Natural Sciences, Brac University.

Introduction to Biology (BIO101), Environmental Microbiology (MIC203), Introduction to Molecular Biology (MIC206), Food Microbiology (MIC302), Analytical Microbiology (MIC402), Diagnostic Microbiology (MIC406)

Islam T, Choudury N, Hossain MM and Khan TT. (2016). Screening of Amylase Producing Bacteria from Soil Followed by Identification via 16S rRNA Gene Sequencing and Characterization of the Crude Amylase, 1st International GCSTMR Congress 2017 By Global Circle for Scientific, Technological and Management Research (GCSTMR), Dhaka, Bangladesh, 2017.

1. University of Glasgow Chancellors Award (2017)2. Brac University Vice Chancellor’s and Dean’s list of meritorious students (2012-2015)

Bangladesh Society of Microbiology

Virology, bacteriology and microbial pathogenesis.

Tasnia Islam

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Minor in microbiology for students doing major in other disciplines is also offered. The details of the minor in microbiology program are given below:

Core Courses (18 Credits):

MIC 101 General MicrobiologyBCH 101 Basic BiochemistryMIC 102 Basic techniques in Microbiology BCH 201 Human PhysiologyMIC 201 Microbial ChemistryMIC 202 Microbial Metabolism

Elective Courses (6 Credits):

MIC 203 Environmental MicrobiologyMIC 204 Medical MicrobiologyBTE 207 Introduction to Molecular BiologyMIC 302 Food MicrobiologyMIC 304 Microbial BiotechnologyMIC 306 Pharmaceutical MicrobiologyBTE 401 Bioinformatics

Lab (3 Credits):

MIC 105 Microbiol Lab IMIC 205 Microbiol Lab II

The postgraduate program of MS in Biotechnology (36 credits) started in Fall 2007. The course is so designed for creating highly competent biotechnologists to serve in the universities and other academic institutions, R&D organizations and biotech-related industries both in the public and private sectors of the country. With this academic background the graduates will be able to pursue higher studies leading to Ph.D. both at home and abroad.For the fulfillment of the MS degree requirement a student needs to complete 36 credits out of which 6 credits are assigned to Research Project/Thesis.So far, thirty six students have graduated with MS degree in Biotechnology and at present we have a total of seventy students pursuing the degree, who are at different stages (semesters) of their study.MNS Department has active collaboration with the University of Dhaka, ICDDR,B, BRAC ARDC, BIRDEM with excellent research and training facilities for research and project work of our MS Biotech students.

News Archive >>

News

Biotechnology seminar hosts Bangladeshi scientist who developed high-yielding "Panchabrihi" rice

Orientation held for Summer 2024 biotechnology students

Nanobiotechnology seminar organized by BUSB and Biotechnology Program

The biotechnology program and BUSB organized the inaugural Biotechnology Workshop Series on "Understanding and Designing Biosensors"

Links

Notices and Circulars

Anti Sexual Harassment Policy

Anti-Ragging and Anti Bullying Policy

Student Code of Conduct

Basic First Aid Handbook

Complaint Form

Format/Guidelines for first information report FIR

Format/Guidelines for General Diary

USEFUL LINKS

DMP Website

Cyber Crime Unit, DMP website

BLAST Website

BNWLA Website

Nari O Shishu Nirjaton Domon Ain 2000 (Amended 2003)

Pornography Act

Digital Security Act 2018

HRLS/BRAC web page

A world where all children grow up with dignity.

Create innovative solutions for children to become engaged, empathetic and empowered individuals.

Since its inception, BRAC Institute of Educational Development (BRAC IED) has been an active advocate and proponent of contributing to the development of education from a holistic perspective through professional development, capacity building, skills development, training and research. The Institute aims to tap into emerging areas of development by exploring areas including early years' learning & early childhood development, play-based learning methodology, psychosocial wellbeing of stakeholders from diverse socio-economic backgrounds, adolescent mental health, Primary and Secondary education and curriculum & textbook design.

News Archive >>

News

Prof. Manzoor Ahmed invited to become associate editor of International Journal of Educational Development by Elsevier

BRAC IED, BRAC University organizes a seminar on “World Suicide Prevention Day”

BRAC University alumnus chosen for Chevening scholarship

“Show You Care”: Public Speaking Event and Self Care Workshop

Students doing BS in microbiology can do a second major in biotechnology. For doing the double major in microbiology and biotechnology one has to complete extra 15 credits so that as a whole one has to complete a total of 151 credits. The required courses are given below.

Core Courses (39 credits):

BCH 101 Basic Biochemistry*BCH 102 Biophysical Chemistry*BCH 201 Human

Physiology*BCH 202   Enzyme and Enzyme KineticsBTE 203   Introduction to Molecular Biology (Eq to BTE 207   Introduction to Molecular Biology)BTE 303   Basic Immunology (Eq to BCH 301   Basic Immunology)BTE 307   Advanced Molecular Biology (Eq to MIC310   Advanced   molecular   Biology)BTE   401   Bioinformatics***BTE   204   Fundamentals of Genetic EngineeringBTE 402   GMOs and Biosafety**BTE 308   Plant Biotechnology and Genetic EngineeringBTE 403   DNA Fingerprinting and Molecular Diagnostics*BTE 404   Bioprocess Technology*

** Students from microbiology background may be exempted to do prerequisite BTE 101 (Introduction to Biotechnology and Genetic Engineering) for relevant courses.

Elective courses (12 credits)

BTE 411   Molecular Virology and Oncology (Eq to MIC 301Virology)BTE 313   Biomass and Biofuel*BTE 317   Biostatistics*BTE 312   Computer Application in Biotechnology*BTE   315   Bioremediation   and   Biodeterioration*BTE   415   Entrepreneurship in BiotechnologyBTE 417   Enzyme Biotechnology

* refers to courses common for undergraduate microbiology and biotechnology programmes.

'Eq' refers to equivalent courses for undergraduate microbiology and biotechnology programmes.

Internship (3 credits)

The internship should be in line with one of the majors or on some interactive topic involving the two majors.

Thesis (4.5 credits)The thesis work should be in line with one of the majors or on some interactive topic involving the two majors.Lab (6 credits)As students who are doing major in microbiology are going through 4 different lab courses that include most of the major practical oriented problems of biotechnology so students are exempted from doing any extra lab course in biotechnology.

Prof. Dr. Razamin Ramli of Universiti Utara Malyasia was the keynote speaker for the second keynote of Day 2 of ICBM 2019. The session was chaired by Prof. Dr. Tzong-Ru Lee, Department of Marketing, National Chung Hsing University, Taiwan and Prof. Imran Rahman of University of Liberal Arts Bangladesh (ULAB). Prof. Dr. Salehuddin Ahmed was the special guest speaker and two industry talks were delivered by Mr. A. S. M. Mainuddin Monem, Deputy Managing Director, Abdul Monem Limited and Mr. Syed Abdul Monem, Head of SME Banking, Brac Bank Limited.

Prof. Dr. Ramli, whose research focuses on social and practical problems, presented a paper titled “Malaysian harmony Index Based on Culture and Identity of the Nation”. The main objective of Prof. Dr. Ramli’s research was to develop a scoring model based on shared national culture and identity which is then translated into a harmony index in order to measure the level of harmony in Malaysia. Based on 11 cultural criteria and seven criteria related to identity, she along with her a group of researchers were able to establish a novel framework for a Harmony Index. Based on the research, the 4 most important factors for harmony and coexistence were tolerance, mutual understanding, communication and language.

Prof. Dr. Salehuddin Ahmed lauded Bangladesh’s remarkable economic performance in recent years but highlighted some of the many issues that still plague us. Some of the major flaws that Dr. Ahmed honed in on were inequality of income, poor distribution of wealth and concentration of wealth in the hands of a few. We have focused entirely on growth but not on distribution and we need to shift the focus on social development as well, he said. Prof. Ahmed also spoke about the role of the government and emphasized that economic decisions should only be taken by financial experts and based on economic considerations and ought to be free of political influence.

Mr. Syed Abdul Monem extensively detailed the small and medium enterprise (SME) portfolio of Brac Bank and discussed how aiding the least privileged and the disenfranchised was a fundamental motivation in the formation of BRAC Bank. In the course of his speech Mr. Monem shared some statistics with the audience - almost 99%

of our population has access to electricity, mobile penetration rate is almost at 82% and almost 50% of our rural population has access to the internet. Brac Bank plans to take advantage of these numbers and extend their banking services to the masses. Mr. Monem also spoke about BRAC Bank's plans for the expansion of agent banking to be able to serve more people and help SMEs grow with collateral free loans. As for the future, he said that their next plan is to go digital and in order to do so he implored the younger generation to step up and work alongside them to build their digital capabilities.

The last speaker of that session, A. S. M. Mainuddin Monem mainly focused on the grave importance of developing, nurturing and managing human resources. Mr. Monem lamented the fact that in Bangladesh there is a huge disconnect between what is taught in class and what is required in the field. The curriculum needs to be redesigned to address the lack of practical and technical approach to innovation and education has to be more pragmatic and application based, he said.

Double Major

A number of students studying in various disciplines have shown their interest in pursuing a double major curriculum if offered by the university. Discussions were held on the modus operandi for introducing a double-major option for BU.

At the MNS Department, we are at present offering Bachelor of Science programme in physics and also minor in physics programme.

Students who want to do a double-major, one of them being physics, will have to complete a total of 66 credits, the break down of which is given in the following:

A. Core Courses

a. Theory Courses: (45 credits)

1. PHY 110 Mechanics and Properties Of Matter 2. PHY 113 Waves, Oscillation &Acoustics 3. PHY 114 Thermal Physics & Radiation 4. PHY 115 Electricity and Magnetism 5. PHY 201 Solid State Physics 6. PHY 202 Optics 7. PHY 204 Classical

Mechanics and Special Theory of Relativity 8. PHY 205 Statistical Mechanics 9. PHY 301 Classical Electrodynamics 10. PHY 303 Quantum Mechanics I11. PHY 304 Quantum Mechanics II 12. MAT 105 Calculus 13. MAT 203 Matrices, Linear Algebra & Differential Equations 14. MAT 204 Complex Variables & Fourier Analysis 15. STA 201 Elements of Statistics and Probability

b. Lab Courses: (4.5 credits)

1. PHY 116 PHY Lab I2. PHY 203 PHY Lab II3. PHY 307 PHY Lab III

B. Elective Courses: (12 credits)

1. PHY 302 Fluid Mechanics 2. PHY 305 Quantum Mechanics III3. PHY 306 Basic Electronics4. PHY 308 Methods of Experimental Physics & Instrumentation5. PHY 401 Particle and Reactor Physics 6. PHY 402 Atmospheric Physics 7. PHY 403 Cosmology & Astrophysics 8. PHY 407 Mathematical Modelling in Physics9. PHY 410 Laser Physics 10. PHY 411 Geophysics 11. MAT 205 Introduction to Numerical Methods

Thesis: (4.5 credtis)

The thesis work should be in line with one of the majors or on some interactive topic involving the two majors.

(Effective from Summer 2017)

Graduates from Public University and UGC approved Universities will get 20% - 40% waiver on their tuition fees based on their undergraduate results (Except MBA and EMBA programs).

Semester Fee:

Discounted fee – BDT 4,000/- per semesterPlease visit Freshman fee structure for all fees details.

Sl.

Scholarship Particulars

Existing

Effective from Fall '22

1.

Merit Based Tuition Fee Waiver for the Graduated Students of BRACU

40% (Applicable to all the graduated students of BRACU)

25% (Only to the eligible BRACU graduates with a minimum CGPA of 3.00 and above)

2.

Merit Based Tuition Fee Waiver for the Graduated Students of other Universities

Up to 40%

Up to 20%CGPA 3.00 - 3.49 : 10% waiverCGPA 3.50 - 4.00 : 20% waiver

3.

Institute/Corporate Discount

10% – 20%

10 – 20% (no change)

4.

Tuition fees waiver for the full-time confirmed employees of BRAC Enterprises

40%

25%

5.

Tuition Fee Waiver for the Students from Ethnic Background (New)2

-

25%

6.

Tuition Fee Waiver for the Female Students & 3rd Gender (New)

-

10%

*Graduate Students of other Universities

CGPA

Tuition Fee Waiver

3:00 - 3.49

10%

3.50 - 4.00

20%

Note:

1If a student is eligible for more than one tuition fee waiver, he/she will be awarded only the highest applicable scholarship.

2Students from Indigenous (Tribal/Ethnic) communities of Bangladesh include: Chakma Tribe, Marma Tribe, Tripura Tribe, Santals, Tanchangya Tribe, Khumi Tribe, Mro Tribe, Lushai Tribe, The Jaintia, Garos and Khasi Tribe, Khiang Tribe, Bawm and Pankhu Tribe, Chak Tribe, The Mros, Keot(Kaibarta), “Pangal or Pangan” (The Muslim tribes), Pankho, Khumi, Kokborok, Sadri, and Chak Tribe.

A minimum of 2 courses per semester is required in order to avail the scholarship.

Students having tuition fee waiver may also apply for other scholarships, if eligible.

For further query please contact: scholarship[at]bracu.ac[dot]bd

The speaker for the third Keynote Presentation was Prof. Dr. Hui-Ming Wee, Chung Yuan Christian University, Prof. Dr. Mirza Azizul Islam, Brac Business School delivered the invited talk. Mr. Reazul Chowdhury, Managing Director and CEO, Runner Automobiles Limited and Mr. Mominul Islam, Managing Director & CEO, IPDC Finance Limited presented the industry talk at that session. Prof. Dr. Akbar Ali Khan and Prof. Dr. Rahim B. Talukdar, Brac Business School, chaired the session.

Prof. Dr. Hui-Ming Wee’s speech focused on factors that drive logistic innovation and the many ways in which big data has transformed logistics. According to Prof. Dr. Wee, there has been a rapid growth in technological progress and computing power and by 2020 there will be 24 billion connected devices around the world. This interconnectedness and the data that can be gleaned from it can help us compute and solve many problems. He concluded his speech by highlighting that big data analytics enables the conversion of information into an unprecedented amount of business intelligence, the applications of which are manifold.

Mr. Reazul Chowdhury began his speech by sharing some of his personal and professional experiences with the conference. He outlined 2 important factors for success - strategy and more importantly, execution and for successful execution of any business's plans the company needs a talented workforce. He stressed the importance of certain skills and qualities other than pedagogical knowledge for success. These skills include a desire to learn, a strong ethical and moral background and creativity. In the same vein he said, "In my view, a university degree provides an entry pass to the interview, nothing more than that". He further asked teachers to support students and graduates to realize that a career is something that needs time and patience and constantly switching jobs due to the seduction of a higher salary is overall detrimental to building a career.

Prof. Dr. Mirza Azizul Islam shifted the focus of the discussion from microanalysis to a macro perspective. His topic was "Macroeconomic Performance of Bangladesh: Achievements and Challenges" and he focused on two critical areas - economic growth and poverty alleviation. Prof. Dr. Azizul noted that Bangladesh has achieved considerable success in addressing both these issues. Our GDP growth has doubled since the early 1980's and currently stands at over 15%. He detailed a few underlying factors responsible for a country's economic growth. As for poverty alleviation, at the onset of the 21st century about 50% of our population was under the poverty line but currently that number stands at 22%. Prof. Dr. Azizul also discussed in detail the minutia of the many problems that are an obstruction to our future growth and economic performance. He suggested certain steps such as increasing allocation of funds on social safety net programs, increase expenditure on education and ensuring effective use of resources to ensure steady economic growth.

The topic of Mr. Mominul Islam's presentation was "Employability Skills in the Era of the Fourth Industrial Revolution". He condemned the culture of rote memorization in our schools and educational institutes, which stymies creativity, discourages students to think outside the box and does not equip them for collaborative tasks. He emphasized the need to develop crucial and sought after skills such as complex problem solving, critical thinking, creativity and people management for better employability. Mr. Islam further elaborated three simple policies that encapsulate the essence of good leadership into 3 P's - Perseverance, passion for learning and sense of purpose. In his closing remarks, he mentioned that the world does not need employees but entrepreneurs and people who think like future CEOs.

The student may transfer with the approval of the Deans or Heads of the student's current and proposed departments. The student must meet any pre-requisite and departmental requirements for the transfer as per his/her advisor's instructions. The student must pay the required transfer fee will be as determined from time to time by the university management. Students of Pharmacy and Law will not be allowed to transfer except with the special permission respectively of the Head of Pharmacy or the Dean of Law in consultation with the Registrar. Students beyond 60 credits will not be allowed to transfer except with special permission by the respective Heads of the Departments or Deans in consultation with the Registrar.

A student interested in transfer shall collect the Inter Department Transfer Form, available at Office of the Controller of Examination, information Desk, or at our webpage, attach copies of Grade Sheets of all semesters completed and submit a list of courses s/he wants to transfer on the reverse side of the form and follow the instruction below:

After the transfer is accepted,

a) The grade sheet /transcript of the student will show grades all the courses taken by the student.b) Grades of all required general education courses and of the required courses of the new major must be transferred. c) After transfer CGPA will be computed on the basis of courses transferred and taken thereafter.d) For graduation, grades of all courses required for the program and the degree will be included to compute the CGPA.

Since its inception, Brac University has conducted a series of cross-sectoral research on climate change and disaster management in direct collaboration with BRAC. To coordinate and manage these different activities, the Syndicate and the Board of Trustees of BRAC University have accorded for establishment of a research center titled “Centre for Climate Change and Environmental Research (C3ER)". The Centre establishes a synergy between BracU and BRAC in the field of climate change and other environmental issues.

Centre for Entrepreneurship Development (CED) started its journey in April 2011 with the view to encourage Bangladeshi entrepreneurs and engender entrepreneurial knowledge and skill so that they can develop and grow their own businesses.

The Centre for Peace and Justice was recently established as multi-disciplinary post-graduate institute within Brac University. The mission of the Centre is ‘to promote global peace and social justice through the means of education and training, research and advocacy’ and with the vision of ‘a just, peaceful and inclusive society’

Centre for Inclusive Architecture and Urbanism (Ci+AU) at Brac University is multidisciplinary research institute and offers research-based consultancy on urban issues – the exploration of conditions, processes, and policies that affect the livability, resilience, and sustainability of the built environment. The Centre's mission is to create a dynamic culture of architecture and urbanism that is inclusive, empathetic, environmentally conscious, and humanistic.

Counseling and Wellness Centre is to support an environment that fosters personal growth, development, and psychological well-being of students, faculty members, and staff through direct counseling service, education, and prevention

News Archive >>

News

BRAC University's Centre for Entrepreneurship Development Wins Global Impact Award at ICSB World Congress

C3ER, BRAC University hosted National Symposium to mark World Environment Day 2024

Fostering Bangladesh's Climate Resilience: Masterclass on Climate Resilient Infrastructure PPPs in Bangladesh

C3ER and Student Life of BRAC University host “Net-Zero Festival” with Korea’s DAEJAYON

Over the years, BRAC University has partnered with the following reputed academic and research institutions around the world to enhance our own educational experience by learning from others.

Faculty Name

Working Relationship with Other National, Regional and International Institutions

Prof. ATM Nurul Amin

· Asian Institute of Technology (AIT), Bangkok as its Professor Emeritus

· United Nations Centre for Regional Development, Nagoya, Japan as Guest Editor of its journal, Regional Development Dialogue (RDD), Vol 36, Issue Regional Development in the Context of 2030 Agenda for Sustainable Development and as an of Expert Member of its 3R international team.

· Centre of Development and Employment Research (CDER) as Research Fellow and its BoT Member.

· Asian Disaster Preparedness Centre (ADPC) and Rajshahi Development Authority (RDA) for contributing to, as an urban and regional economist, prepare the Master Plan of Rajshahi Metropolitan Development Plan.

· Contributing to Local Economic Development (LED) Strategy Development Project of

Shibgonj/Rajsjahi and Jashore, swisscontact/SDC.

Prof. Shahidur Rahman

· Copenhagen Business School Tufts University and Danish Ethical Trading Initiative , funded by the Danish Ministry of Foreign Affairs.

· Royal Holloway, University of London and Aconsa AB, a Swedish Consultant firm.

· BIGD, LSE,University of New South Wales, The Freie University, Volkswagen Foundation

· Ministry of Labour and Employment

· Center for Entrepreneurs and development (CED)

· University of Gothenburg

· University of Essex

· C&A Foundation

· Zurich University of Applied Science

· Department of Inspection for Factories and Establishment (DIFE).

Prof. Farzana Munshi

· RAND Corporation, USA

· ILO

· Development Economics Unit, University of Gothenburg

· World Bank

· Bangladesh Employers Federation

· Bangladesh Institute of Development Studies (BIDS),

· Planning Commission of Bangladesh.

Dr. Samia Huq

· UNWOMEN

· Chowdhory Center for Bangladesh Studies at UC Berkley

· Musawah: Equality in the Muslim family Malaysia

· Princeton University, University Center for Human Values

· University of Connecticut

· International editorial board of contemporary South Asia Taylor n Francis.

· Georgetown University, Berkley Center for Religion, Peace, World Affairs

· Network for Religious and Traditional Peacemakers

· Keough School of Global Affairs, University of Notre Dame, USA

Dr. Seuty Sabur

· Oxford University Press, Bibliography

· Center for Bangladesh Studies at UC Berkley

· National Women’s Studies Association’s

· Department of Women & Gender Studies, College of Liberal arts at the UMass Boston

· Cardiff university

· IIT Delhi

· Singapore National University

· Inter-Asia Cultural Studies Society

· Bangladesh Mahila Parishad

· Bangladesh Garments Shanghati

· Dhaka University

Dr. Rubana Ahmed

· Beautiful Mind (School for autistic children)

· UCEP

Mr. Tanvir Shobhan

· Queens University, Belfast, UK

· Swiss Development Corporation

· Economic Research Group (ERC)

· BIGD

Mr. Shehzad M Arifeen

· Inter-Asia Cultural Studies Society

Ms. Nazrina Haque

· ADB

1. Eva R. Kabir, Ph.D. - University of North Dakota (research collaboration)

Academic and Research Affiliation of CPJ

Asia Pacific Partnership for Atrocity Prevention, University of Queensland, Australia

Begum Rokeya University, Rangpur, Bangladesh

Berkley Center for Religion, Peace, and World Affairs, Georgetown University, Washington, USA

Centre for Genocide Studies, University of Dhaka, Bangladesh

Central European University, Budapest, Hungary

Centre for Asia Pacific Refugee Studies, University of Auckland, New Zealand

International Institute of Social Studies, Erasmus University Rotterdam, Netherlands

International Development Research Centre, Canada

Jatiya Kabi Kazi Nazrul Islam University, Mymensingh, Bangladesh

Lee Kuan Yew School of Public Policy, National University of Singapore, Singapore

National Law University, Odisha, India

North South University, Bangladesh

Peace Research Institute Oslo, Norway

University of New South Wales, Australia

Working Relationship with National, Regional and International Institutions

ActionAid Bangladesh

Ain o Salish Kendro, Bangladesh

Amala, England (Old Sky School)

Alliance for Social Dialogue, Nepal

Amnesty International

Asia Pacific Refugee Rights Network

Asia Justice Coalition

Bangladesh Legal Aid and Service Trust, Bangladesh

Bangladesh National Women Lawyers Association, Bangladesh

BRAC, Bangladesh

Centre for Social Justice, Ahmedabad, India

Council of Minorities, Bangladesh

Department for International Development

Friendship, Bangladesh

International Commission of Jurists

LAND Network, Sirajgonj, Bangladesh

LightCastle Partners, Bangladesh

Madaripur Legal Aid Association, Bangladesh

Management System International

Move Foundation, Bangladesh

MUKTI, Kushtia, Bangladesh

Nagorik Uddyog, Bangladesh

Naripokkho, Bangladesh

Open Society Foundation, Hungary

Open Society Justice Initiative, Hungary

Save and Serve Foundation, Chattogram, Bangladesh

The Asia Foundation

UN Women, Bangladesh

United Nations Development Programme

United Nations Office for Project Services

United States Agency for International Development

BRAC Business School

James P Grant School of Public Health

Department of Architecture

Postgraduate Programs in Disaster Management

Department of Mathematics and Natural Sciences

School of Humanities and Social Sciences

Department of Economics and Social Sciences, School of Humanities and Social Sciences

Department of English & Humanities, School of Humanities and Social Sciences

Department of Pharmacy

BRAC Development Institute

Institute of Educational Development

BRAC Institute of Languages

Centre for Peace and Justice (CPJ)

Tokee Mohammed Tareq started as a lecturer in MNS department, biotechnology program from fall, 2018. He completed his Bachelor of Science and Master of Science degree from Biochemistry and Molecular Biology, University of Dhaka at 2010-11 and 2014-15 session, respectively.

Level:Kha-224 Merul BaddaDhaka 1212. Bangladesh

BS in Biochemistry and Molecular Biology, 2014MS in Biochemistry and Molecular Biology, 2015

Journals

Akter, M., Brown, N., Clokie, M., Yeasmin, M., Tareq, T. M., Baddam, R., Talukder, K. A. (2019). Prevalence of Shigella boydii in Bangladesh: Isolation and characterization of a rare phage MK-13 that can robustly identify shigellosis caused by Shigella boydii type 1. Frontiers in Microbiology. https://doi.org/10.3389/fmicb.2019.02461

Rahman, M. S., Tareq, T. M., Sarker, P. K., Rashid, E. S. M. H., Yasmeen, R., Ali, M. A., Shimono, H. (2019). Genetic variation of phenotypic plasticity in Bangladesh rice germplasm. Field Crops Research, 243, 107618. https://doi.org/10.1016/j.fcr.2019.107618

Tareq, T. M., Rahman, M. S., Jewel, N. A., Islam, T., Shimono, H., & Seraj, Z. I. (2018). Relative Response of Indigenous rice genotypes to low versus normal planting density for determination of differential phenotypic plasticity in traits related to grain yield. Plant Tissue Culture and Biotechnology, 28(1), 109–124. https://doi.org/10.3329/ptcb.v28i1.37203

Naser, I. B., Hoque, M. M., Nahid, M. A., Tareq, T. M., Rocky, M. K., & Faruque, S. M. (2017). Analysis of the CRISPR-Cas system in bacteriophages active on epidemic strains of Vibrio cholerae in Bangladesh. Scientific Reports, 7(1). https://doi.org/10.1038/s41598-017-14839-2

Research Associate, Plant Biotechnology Laboratory, University of DhakaResearch Officer, Laboratory Science and Services Division, International Centre for Diarrhoeal Disease Research, Bangladesh

Introduction to BiologyBasic BiochemistryEnzymes and Enzyme KineticsMicrobial BiotechnologyGMOs and BiosafetyGenomics and Proteomics

Tokee Md Tareq, Nurnabi Azad Jewel, Taslima Haque, Zeba Islam SerajIn silico identification and validation of polymorphic SSR markers from assembled IR29 and Horkuch genome. 6th International Conference on Bioinformatics and Bioinformatics for Agriculture, Health and Environment; (January 20-23, 2017)

Tokee Md Tareq, Taslima Haque, Nurnabi Azad Jewel, Zeba I. SerajPopulation stratification of Bangladeshi rice landraces and comparative genetic architecture of the groups. International Conference on Genomics Nanotech & Bioengineering; (ICGNB 2017)

National Science and Technology fellowship from Ministry of Science and Technology, Bangladesh for outstanding research activity in MS thesis.

Computational biology, Comparative genomics, NGS data analysis, Phage-bacteria interaction, Phage therapy, Plant plasticity, Genotype-environment interaction of plant physiology.

Tokee Mohammad Tareq

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Nowrin Hossain is a passionate and enthusiastic lecturer who has joined Microbiology Program at the Department of Mathematics and Natural Sciences, BRAC University. With a Bachelor's degree in Microbiology from the Department of Mathematics and Natural Sciences, BRAC University, and a Master's degree in Microbiology from Department of Microbiology, Jagannath University, she developed a profound interest in the fascinating world of microorganisms. Her thesis focused on “Drinking Water Safety” and delved into the assessment of the prevalence and persistence of diarrheagenic Escherichia coli strains and pathogenic Salmonella spp., in tap and drinking water samples. This research unveiled significant public health hazards in a megacity, Dhaka.

Prior to joining as a Lecturer, she has experience of working as a Teaching Assistant in Microbiology Laboratory as well as a Part Time Lecturer in Microbiology Program at BRAC University. Along with this, she was involved as a Research Assistant at Osaka University for a project titled “Genomic characterization of carbapenemase-producing Enterobacteriaceae from Dhaka food markets unveiled the spread of high-risk antimicrobial resistant clones and plasmids co-carrying blaNDM and mcr1.1”

She holds an interest in Environmental Microbiology, Molecular Biology and Bioinformatics. She believes that practical knowledge, critical thinking, and a deep appreciation for microorganisms are essential for success in this field. Therefore, she is excited to bring her real-world experience and passion for microbiology to the classroom, where she will encourage students to explore this intricate and vital realm of science.

Level: 5Kha-224 Merul BaddaDhaka 1212. Bangladesh

Nowrin Hossain

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Department of Architecture

Department of Computer Science and Engineering

Department of Economics and Social Sciences

Department of Electrical and Electronic Engineering

Department of English and Humanities

Department of Mathematics and Natural Sciences

School of Pharmacy

News and Events

EEE student selected for Huawei program in China

Workshop on Architecture Model Making Held at BRAC University

ENH lecturer attains Fulbright teaching assistantship

BRAC University architecture team recognized at international design competition

5th Intra University Moot Court Competition- 2024

Exploring New Contextualism 2.0: An Approach to Architecture and Urban Design

Certificate Course on ‘Intellectual Property for Innovation, Creativity and Branding’

Introduction to Virtual Reality in Architecture

On Friday, April 26th, twelve parallel sessions of ICBM-2019 were held at The Westin, Dhaka. There were three different slots for the parallel sessions 1, 2 and 3 with paper presentations by the participants taking place in four rooms A, B, C, and D simultaneously during each slot. Each parallel session concentrated on a specific subject of business and management and had been named accordingly.

In total 79 papers were presented on the first day of the parallel session which saw prestigious session chairs from home and abroad, such as Prof. Dr. H.M. Jahirul Haque and Prof. Dr. C. V. Gopinath in 1A: (Theme: Education), Prof. Dr. Md. Azizul Baten and Tiena Gustina Amran in 1B: (Theme: Operations and Supply Chain Management), Prof. Dr. Hasnan Ahmed and Dr. Naffisah MD Hassan in 1C: (Theme: Accounting, Banking and

Finance), Prof. Dr. Golam Samdani Fakir and Assoc. Prof. S.A.M. Manzur H Khan in 1D: (Theme: Accounting, Banking and Finance).

Parallel Session 2 had Prof. Dr. Razamin Ramli and Mr. Mohammad Ismail Majumder as chairs in 2A: (Theme: Operations and Supply Chain Management), Prof. Dr. Mohammad Iqbal and Dr. Rina Fitriani in 2B: (Theme: Operations and Supply Chain Management), Prof. Dr. Arabinda Saha and Dr. Mohamad Safiai Saad in 2C: (Theme: Accounting Banking and Finance) and Prof. Dr. Akmal S. Hyder and Assoc. Prof. Dr. Mohammad Basir B Saud in 2D: (Theme: Marketing Management), Parallel Session 3A: (Theme: Economics) was chaired by Prof. Dr. ATM Nurul Amin, Parallel Session 3B: (Theme: Entrepreneurship) by Prof. Dr. Abdul Hannan Chowdhury and Dr. Erne Suzila Kassim, Parallel Session 3C: (Theme: Human Resource Management) by Prof. Dr. Abdur Rab and Assoc. Prof. Dr. Ireen Akhter, Parallel Session 3D: Human Resource Management by Prof. Dr. Yousuf Mahbubul Islam and Prof. Dr. B. Chandrachoodan Nair.

On the first day of the parallel session seven (7) papers on Education were presented, six (6) on Entrepreneurship, fourteen (14) on Human Resource Management, twenty one (21) on Operations and Supply Chain Management, eighteen (18) on Accounting, Banking and Finance, seven (7) on Marketing Management and six (6) on Economics.

Each session ended with some valuable feedback from the session chairs followed by the distribution of certificates for the participants.

Local and international participants particularly from Canada, Sweden, Singapore, Malaysia, Thailand, Taiwan, Indonesia, India, Nigeria, China, UK, presented their papers in various sessions.

Brac University (BracU) always pays close attention to convenient approach in terms of Admission Procedures. The entire admission process will be online and applicants can apply from the convenience of their home. The entire process of admission goes through an intensely selective method where applicants are asked to follow each phase precisely within the given framework of instructions. To get admitted into BracU, applicants must qualify in the admission test (written and an interview) after submitting their online application form.

As the admission test is a highly competitive one, applicants are required to perform satisfactorily in the examination in order to secure their seats.

Brac University excels at educating undergraduate students! Get yourself admitted to the Nation’s highest ranking private university as ranked in the QS Asian University rankings 2018 and build yourself a successful career!!Apply Now Admission RequirementsFreshman Course Offerings

Brac University excels at educating undergraduate students! Get yourself admitted to the Nation’s highest ranking private university as ranked in the QS Asian University rankings 2018 and build yourself a successful career!!

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Freshman Course Offerings

Get yourself ready!Find answer of your queries and admission test sample questions for various programs.Sample Questions FAQsAdmission Flyer Freshman Fee Structure

Find answer of your queries and admission test sample questions for various programs.

Sample Questions FAQs

Admission Flyer Freshman Fee Structure

The advantage of Postgraduate study is that it allows students to enter employment with additional skills and knowledge. One of the benefits of a Postgraduate research degree and the purpose of research is the output of original contributions to knowledge. The degree can demonstrate capability and tenacity to undertake an extended piece of investigative work.Apply Now Admission RequirementsWaiver for Postgraduate Programs

The advantage of Postgraduate study is that it allows students to enter employment with additional skills and knowledge. One of the benefits of a Postgraduate research degree and the purpose of research is the output of original contributions to knowledge.

The degree can demonstrate capability and tenacity to undertake an extended piece of investigative work.

Apply Now   Admission Requirements

Waiver for Postgraduate Programs

Scholarship & Financial AidEach semester, Brac University is proud to honor academically talented and exceptionally skilled students with a variety of scholarships and awards. The university annually awards more than 100 million takas as scholarships to both undergraduate and postgraduate students.Know More

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Know More

Brac University personifies a broad portrayal of states and cultures from all over the world in a single platform. International Students are given special attention and care to make the foreign students feel at home. Our stature regarding International Students comes firstly from an excellent broad-based education and the factual commitment we provide to every pupil.

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Join the “Highest ranked among private universities in Bangladesh - QS Asian University rankings 2018.”

Undergraduate Admissions

Get yourself ready!

Postgraduate Admission

Scholarship & Financial Aid

International Students’ Admissions

Course Structure of Undergraduate Mathematics Programme

Mathematics undergraduate programme has been designed including topics of current interest and applications. Once a student undergoes this course successfully he will be well equipped to face the challenges of life. The programme of study includes courses for improving communication skills, strengthening basic science and computational mathematics background and acquainting the student with the socio-economic & historical background of Bangladesh. With this background it should not also be a problem to find a suitable and satisfying job in both public and private sectors, various universities, financial institutions and software & R/D organizations in the country. The total credit requirement for the degree of Bachelor of Science in Mathematics is 127. Out of these 21 credits are for general education. Nineteen major area compulsory courses account for 57 credits. The students are required to complete 2 courses (4 credits) of Mathematics Lab and write a dissertation on a suitable project/thesis topic. The project/thesis work spread over the last two semesters will have a total of 3 credits. The students will be required to complete 15 credits choosing from several elective courses in their major field and 27 credits from outside their major specialization. The students may also be required to take noncredit remedial courses in English.

Courses for Bachelor of Science in Mathematics

The graduation programme for Mathematics will contain the courses as appended below . However, the courses listed below may change as the need arises in future.

a. General Education (21 credits)

1 BIO 101 Introduction to Biology2 CSE 110 Programming Language I3 DEV 101 Bangladesh Studies4 ENG 091 Foundation Course (Non-credit)5 ENG 101 English Fundamentals *6 ENG 102 Composition I *7 HUM 103 Ethics and Culture8 PHY 111

Principles of Physics I

b. Departmental Core Courses: (57 credits)

1 MAT 111 Principles of Mathematics2 MAT 121 Basic Algebra3 MAT 122 Analytic and Vector Geometry4 MAT 123 Calculus I5 MAT 124 Scientific Programming 6 MAT 211 Calculus II7 MAT 212 Linear Algebra8 MAT 221 Real Analysis I9 MAT 222 Differential Equations I10 MAT 223 Numerical Analysis I11 MAT 311 Abstract Algebra12 MAT 312 Numerical Analysis II13 MAT 314 Complex Analysis14 MAT 321 Real Analysis II15 MAT 322 Differential Equations II16 MAT 323 Dynamical Systems17 MAT 324 Graph Theory18 MAT 325 Mathematical Methods19 STA 201 Elements of Statistics and Probability

c. Elective Courses: (15 credits)

1 CSE 220 Data Structures2 CSE 221 Algorithms3 MAT 313 Differential Geometry4 MAT 316 Operations Research I5 MAT 326 Hydrodynamics6 MAT 411 Topology7 MAT 415 Finite Element Methods8 MAT 416 Tensor Calculus9 MAT 421 Fluid Mechanics10 MAT 422 Theory of Numbers11 MAT 423 Mathematical Modeling12 MAT 424 Operations Research II13 MAT 425 Advanced Numerical Methods14 PHY 112 Principles of Physics II15 PHY 204 Classical Mechanics and Special Theory of Relativity16 PHY 205 Statistical Mechanics17 PHY 303 Quantum Mechanics I18 PHY 413 General Theory of Relativity19 STA 301 Modern Probability Theory & Stochastic Processes

d. Lab: (4 credits)

1 MAT 250 Mathematics Lab I2 MAT 350 Mathematics Lab II

e. Project / Thesis: (3 credits)

1 MAT 400 Project / Thesis

f. Courses Outside Major Specialization: (27 credits)

1 ANT 101 Introduction to Anthropology**2 ARC 292* Painting**3 ARC 293* Music Appreciation4 CHE 101 Introduction to Chemistry5 ECO 103 Principles of Economics6 ENV 101 Introduction to Environmental Science7 HUM 101 World Civilization and

Culture8 HUM 102 Introduction to Philosophy9 HUM 111 History of Science10 MAT 315 History of Mathematics11 MGT 211 Principles of Management12 PHY 312 Physics for Development13 POL 103 Introduction to Political Science14 POL 245 Women, Power & Politics15 PSY 101 Introduction to Psychology16 SOC 101 Introduction to Sociology17 SOC 401 Gender and Development

or any other BRACU courses outside major specialization.* Other courses in English may substitute these courses depending on the level of the student's proficiency in English. ** 2 Credit Courses

Academic System & Evaluation Method

a. Academic Standards

In keeping with the mission and goals in mind, every effort will be made to ensure high academic standards by implementing well-designed curricula, carefully selecting high quality students and faculty, utilising modern and effective instructional methods and aides, and by continuously monitoring and rigorously evaluating all the pertinent activities and systems. A special feature of teaching will be the lab/workshop sessions designed to assist students in learning application of the concepts and theories.

b. Courses and Credit Requirements

The Bachelor of Science programme in Mathematics will follow the model of higher education consisting of semesters, courses, credit hours, continuous evaluation and letter grading as in all other courses of BRAC University. The undergraduate curriculum consists of general education courses, major courses, elective courses and non-major area courses.

c. Examination, Evaluation and Grading

The grading process will undoubtedly be transparent. The performance of the students is evaluated throughout the semester through class tests, quizzes, assignments, and midterm exams. End of semester evaluation includes final examinations, term papers, project reports etc. Numerical scores earned by a student in tests, examinations, assignments etc. are cumulated and converted to letter grades.

d. Distribution of Marks & GPA Computation

The distribution of marks for the performance evaluation is as follows:

i. Theory Courses

Section Marks %1. Quizzes/ Class Tests/ Assignments/ Participation/ Attendance 302. Mid Term Examination 203. Final Examination (comprehensive), Projects 50Total Marks 100

ii. Lab Courses

Section Marks%1. Reports/ Class Tests/ Participation / Viva-Voce 302. Lab Experiments 303. Final Examination 40Total Marks 100

iii. Project / Thesis

Project / Thesis work will be spread over the last two semesters of study. The mark distribution for the Project / Thesis will be as follows:

Section Marks %1. Viva-Voce 152. Thesis / Project Work 453. Defence 40Total Marks 100

Class attendance is compulsory for every student. 5% of total marks in every course is allocated for attendance in classes including tutorials and labs. The basis for awarding marks for attendance is as follows:

Attendance Marks90% and above 585% to less than 90% 480% to less than 85% 375% to less than 80% 270% to less than 75% 1Less than 70% 0

If a student does not attend a minimum of 70% of the total classes including tutorials and labs, s/he will not be allowed to take the final exam.

Marks earned by the students in Class Tests, Quizzes, Assignments, Participation, Attendance, Midterm Exam , Final Exam, Projects, Term Papers etc are to be cumulated and the total is to be graded as per the existing BRAC University rules and regulations :

GPA Computation

The Grade Point Average (GPA) is computed in the following manner:

$$GPA = \frac{\text{Sum of (Grade points Credits)}}{\text{Sum of Credits Attempted}}$$

BS in Physics program started in Fall, 2005. It may be mentioned that BRACU is the only private university in Bangladesh which offers a BS degree in physics. Physics is cutting across the edge of all aspects of engineering, information technology, biology, medicine, economics, finance, accounting and even sociology. A total of 132 credits is needed to complete the whole program. There also exist provisions for doing double-major in Physics with a second major in Computer Science/Computer Science and Engineering (CS/CSE), Electronics and Communication Engineering (ECE), Electrical and Electronic Engineering (EEE), Economics or any other relevant discipline and vice-versa. The Department also offers minor in Physics for students doing a major in CS/CSE, ECE, EEE or any other relevant discipline.

So far eight students have completed their BS degrees in physics. Some of them have gone abroad for higher studies and some are gainfully employed in Bangladesh. In addition to this a few students of other department did a second major in physics or had a minor in physics.

News Archive >>

News

Biotechnology seminar hosts Bangladeshi scientist who developed high-yielding “Panchabrihi” rice

Orientation held for Summer 2024 biotechnology students

Nanobiotechnology seminar organized by BUSB and Biotechnology Program

The biotechnology program and BUSB organized the inaugural Biotechnology Workshop Series on “Understanding and Designing Biosensors”

Mr. Mahfuzur Rahman joined as a Lecturer of Statistics at the Department of

Mathematics and Natural Sciences of Brac University in the Fall Semester, 2018. He has completed his B.S. (Hons) and M.S. degrees from the Department of Statistics of the University of Dhaka. Prior to joining Brac University he was a Lecturer of Statistics in Green University of Bangladesh. Before joining Green University of Bangladesh, Mr. Mahfuz worked as a Research Associate at HEQEP Sub-Project under the Department of Applied Statistics, East West University. He published several research articles in different prestigious journals. His research interests are in the areas of Biostatistics, Longitudinal data, Epidemiology and Public Health.

Level: 5, Room: 5B17Kha-224 Merul BaddaDhaka 1212. Bangladesh

• Master of Science (M.S.) in Statistics, Biostatistics and Informatics, Department of Statistics, University of Dhaka.• Bachelor of Science (B.S.) in Statistics, Biostatistics and Informatics, Department of Statistics, University of Dhaka.

Journals

Rahman, F. M. A., & Rahman, M. M. (2019). Analysis of Dhaka stock market data for explaining the stock market behavior: Before and after 2010 market fall. International Journal of Applied Mathematics and Statistics, 58(1), 109–126. Retrieved from http://www.ceser.in/ceserp/index.php/ijamas/article/view/5997

Jahan, F., Sinha, N. C., Rahman, M. M., Rahman, M. M., Mondal, M. S. H., & Islam, M. A. (2019). Comparison of missing value estimation techniques in rainfall data of Bangladesh. Theoretical and Applied Climatology, 136(3), 1115–1131. https://doi.org/10.1007/s00704-018-2537-y

Hosen, S., Jahan, S., Rahman, M. M., Siddiquee, M. A., & Shozib, H. B. (2016). Grain quality evaluation and comparative analysis of physicochemical properties of traditional cultivars and High Yielding (HYV) Aman rice varieties in Bangladesh. Biojournal of Science and Technology, 5(2), 2–3.

Conference Proceedings

Sultana, Z., Gupta, M., & Rahman, M. M. (2021, October 23-24). Time series analysis of

real GDP of four South Asian countries based on the Box Jenkins Model. International Conference on Innovation and Transformation for Development (ITD), 26.

Sultana, Z., Rahman, M. M., & Gupta, M. (2021, October 23-24). Financial literacy among university students in Bangladesh: An empirical study based on Dhaka city. International Conference on Innovation and Transformation for Development (ITD), 53.

• Senior Lecturer (Statistics), Brac University. July, 2022- Present• Lecturer (Statistics), Brac University. September, 2018- June 2022.• Lecturer (Statistics), Green University of Bangladesh. October, 2017- August, 2018.• Research Associate, HEQEP Sub-Project, Dept. of Applied Statistics, East West University. February, 2016- September, 2017.• Graduate Teaching Assistant (GTA), Dept. of Applied Statistics, East West University. September, 2015- January, 2016.

o Introduction to Statisticso Elements of Statistics & Probabilityo Modern Probability Theory & Stochastic Processeso Biostatisticso Business Statisticso Basic Mathematicso Business Mathematicso Introduction to SPSS Statistics

Published Abstracts in conferences:1. Jahan F, Sinha NC, Rahman MM, Haque MS, Rahman M, and Islam MA. Comparison of Missing Value Estimation Techniques in Rainfall Data of Bangladesh. International Conference on Analysis of Repeated Measures Data, November 25-26, 2016, page-30.2. Rahman FMA, Rahman MM. Cluster Analysis of Dhaka Stock Market Data for Explaining Role of Circuit Breaker on Performance. International Conference on Analysis of Repeated Measures Data, November 25-26, 2016, page-31.3. Rahman FMA, Rahman MM. Trend in Turnover, Trade Volume, and Market Return of Dhaka Stock Market. International Conference on Analysis of Repeated Measures Data, November 25-26, 2016, page-51.4. Jahan F, Sinha NC, Rahman MM, Haque MS, Rashid OR, and Islam MA. Probability Distribution and Trend of Seasonal Rainfall Data for Some Selected Stations in Bangladesh. International Conference on Analysis of Repeated Measures Data, November 25-26, 2016, page-56.

• Life Member, DExSA (Ex- Students Association of Dhanmondi Govt: Boys High

School)• Life Member, Alumni Association (DUSSDA), Department of Statistics, Biostatistics and Informatics, University of Dhaka. Organization Activity:• Organizing Committee Member, “International Conference on Analysis of Repeated Measures Data”, 2016, Department of Applied Statistics, East West University.

• Biostatistics, • Longitudinal Data, • Epidemiology or Public Health and • Econometrics.

• Webinar on “Modelling small area level data using Bayesian spatial models” October 27, 2021, Organized by QUT Centre for Data Science, Australia.• Workshop on “Capacity Building of Higher Education Teachers of Bangladesh on using and Creating the Open Education Resources” 29 -31 July, 2021, Organized by UGC & CEMCA.• The Statistics Workshop Series in collaboration with the Therapeutics Initiative (TI) Methods Speaker Series: “A Practical Introduction to Propensity Score Analysis using R” September 30, 2020, hosted by the Therapeutics Initiative, University of British Columbia.• Training on “DHS eLearning Survey Sampling Training Course - Summer 2020” July 13- August 31, 2020, Organized by Demographic Health Survey Program- A project of the USAID.• Workshop series on “Application of Statistical Analysis and Machine learning in Healthcare Industry” July 18- August 8, 2020, University of Dhaka.• Workshop on “Applying Propensity Score Approaches to Complex Survey Data Analyses” December 26, 2019, 2nd International Conference on Applied Statistics 2019.• Workshop on “Navigating the Data Science Landscape” December 26, 2019, 2nd International Conference on Applied Statistics 2019.• Training on “Research Proposal Writing” April 16, 2018 at Green University of Bangladesh.• Workshop on” Introduction to Robust Statistics”, July 22, 2017 at East West University.• Workshop on” Dimension Reduction and Classification Techniques for Big Data Analysis”, April 01, 2017 at East West University.• Workshop on “Applied Longitudinal Data Analysis”, November 24, 2016, International Conference on Analysis of Repeated Measures Data.• Workshop on “Missing Data in Longitudinal Studies”, November 24, 2016, International

Conference on Analysis of Repeated Measures Data.

Md. Mahfuzur Rahman

News & Events

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Workshop on Architecture Model Making Held at BRAC University

By donating to BracU Scholarship/Financial Aid program, you are empowering young leaders who will contribute to the communities in a way that is responsive to society's needs.

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"It was not possible for my parents to fund the cost of food and my accommodation. Moreover, I had a responsibility to finance my family partly as well. The reality would have been different without the scholarship as I would have had to get a full-time employment for my family."

- Mohsin applied and received the BRAC Scholarship in 2003.

“Without the scholarship, I never would have been able to attend a school like Brac University. Since I enrolled, I have maintained a perfect GPA. I'm involved in a lot of extracurricular activities, but my passion is biotechnology. I'm currently studying bacteria in the lab. It's not just about myself It's for the community, for the world, for my family. I believe if I can work hard enough, and if I'm supported enough, I can make a difference."

"I extend my sincere and heartfelt gratitude for the scholarship awarded to me. Engineering has been my dream from the age of eight up to date. Having narrowly missed the government scholarship [in Uganda], I had no more hopes left for higher education. Thanks to God who restored my hopes by bringing the Brac University opportunity to me."

PoliciesScholarship/Financial Aid will be awarded to all undergraduate and graduate students who meet the eligibility criteria*. All applications for scholarship/tuition waiver must be submitted within the deadline announced by the university.Only one of the waivers will be awarded to a particular student.All categories of tuition fee waiver will depend on the availability of funds.Students applying for renewal of their tuition fee waiver in any category will have to take minimum 12 (18 for pharmacy) credit course load.*The financial aid policy is subject to change as guided by exigencies.

Scholarship/Financial Aid will be awarded to all undergraduate and graduate students who meet the eligibility criteria*.

To keep pace with BRAC's focus on adolescents and youth as a special group of the country, Brac University (BracU) offers quality tertiary education, as well as financial support to meritorious students who come from disadvantaged communities. Brac University promotes the values of lifelong learning as part of its social responsibility initiative for the development of the country. Student recruitment, retention and graduation are key components of BracU's strategic plan. Hence, scholarship and tuition waivers are offered to attract students with high potential, to remove financial barriers and provide access to qualified students.

Each semester, Brac University is proud to honor academically talented and exceptionally skilled students with a variety of scholarships and awards. The university annually awards more than 100 million takas as scholarships to both undergraduate and postgraduate students.

Eligibility Criteria

The process is designed to attract a high quality student body, add diversity to the student population, remove financial barriers for qualified but financially disadvantaged students and assist in meeting institutional and regulatory compliance requirements. For any queries or for sending Scholarship application please email to: scholarship[at]bracu.ac[dot]bd

Read eligibility in detail (Undergraduate) Read eligibility in detail (Postgraduate)

Merit Based Scholarship Policy (Updated)

Application form for Local students (Undergraduate) Application form for Local students (Postgraduate)

Application form for Foreign students

20-40% Tuition fees waiver for postgraduate programsView More

View More

The impact and opportunity of the Brac Scholarship program during the last fourteen years has been substantial. Financial support under the program significantly improved students' chances of graduating from Brac University. Students also reflect the collective belief that higher education will lead to a stronger and more vibrant community and ultimately greater achievement in the job world. Year 2013 2014 2015 201620172018 Number of Awardees 570 597 609 690846864

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Policies

20-40% Tuition fees waiver for postgraduate programs

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BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

5th Intra University Moot Court Competition- 2024

Exploring New Contextualism 2.0: An Approach to Architecture and Urban Design

LSDB’s Regional Education Award 2023 (LREA-2023)

Certificate Course on ‘Intellectual Property for Innovation, Creativity and Branding’

The undergraduate and graduate study programs are offered by a highly qualified and dedicated faculty body within the department, contributed additionally by faculty and experts from other universities or institutions from home and abroad. More than half of the faculty members obtained their respective PhD degree from internationally reputed universities of the world, and have strong teaching, research and publication records. The department’s young bright teachers are dedicated to developing their academic careers and contributing to development of students’ all round quality and development. All faculty members are committed to close engagements with students to develop their intellectual curiosity and motivation for learning.

The undergraduate and graduate study programs are offered by a highly qualified and dedicated faculty body within the department, contributed additionally by faculty and experts from other universities or institutions from home and abroad. More than half of the faculty members obtained their respective PhD degree from internationally reputed universities of the world, and have strong teaching, research and publication records. The department’s young bright teachers are dedicated to developing their academic careers and contributing to development of students’ all round quality and development. All faculty members are committed to close engagements with students to develop their intellectual curiosity and motivation for learning.

Founded in 2003, the Department of Economics and Social Sciences (ESS) Brac University, has now expanded to 26 full time faculty members, 500 students and an alumni body of over 300 graduates. ESS seeks to make students understand the economic and social realities they live in, both in Bangladesh and the globalized world, and to develop critical insights and analysis to help them determine solutions. Along with emphasis on growth as well-rounded individuals with strong ethical values, ESS prepares students both for the job market and for graduate studies and research at home and abroad.

Other than the BracU-wide broad based education that imparts critical and global thinking, quantitative and communication skills, ESS offers its students a comprehensive, demanding and innovative curriculum for BSS in Economics. The major in Economics:

Curriculum Structure

The program requires 120 credit hours in total where:

Minor in Economics: For doing a minor in economics, students require to complete 24 credit hours from economics core courses and 3 credit hours from economics elective courses. In total 27 credit hours are required.

ESS has been offering a major in Anthropology since the Spring of 2013. The program is committed to rigorous teaching, writing and analytical skill-building on issues such as development, modernity, nationalism, sovereignty and culture. Close mentoring of students, both in the class room as well as in the field, ensure high-quality social science training, with prospects for advanced degrees and interesting careers in the development sectors and elsewhere. The Anthropology program thus offers the following:

Curriculum Structure

The program requires 120 credit hours in total where:

Minor in Anthropology: To be eligible for a minor in anthropology, students are required to complete a total of 27 credits where 12 credits must be in the core areas of

Anthropology.

Minor in Sociology: To be eligible for a minor in sociology, students are required to complete a total of 24 credits where 9 credit hours are required from core sociological theories. Among the rest of 15 credits, at least 9 credits are required from 300-level and 3 credits are from 400-level courses.

The Master of Science in Applied economics (MSAE) aims to creating highly competent economics professionals for serving in the private and public sectors of the country. The Master of Science in Applied Economics (MSAE) degree is designed to:

Curriculum Structure

The program requires 36 credit hours in total where:

BRAC UniversityKha 224 Bir Uttam Rafiqul Islam AvenueMerul BaddaDhaka 1212. Bangladesh

ESS:

ESS[at]bracu.ac[dot]bdAdmission: admissions[at]bracu.ac[dot]bd (Admission)General Information: info[at]bracu.ac[dot]bd FB: https://www.facebook.com/bracuess/

Faculty members, Economics and Social Sciences

Faculty and Research Staff

Wasiqur Rahman Khan, PhD

Farzana Munshi, PhD

Shahidur Rahman, PhD

Samia Huq, PhD

Syed A. Mamun, PhD, FCMA

Seuty Sabur, PhD

Salma Begum,PhD

Muhammad Shafiullah, PhD

Rubana Ahmed, PhD

A S M Shakil Haider, PhD

Moiyen Zalal Chowdhury, PhD

K.M. Arefin Kamal, PhD

Sarah Salahuddin, PhD

Meheri Tamanna

Muhammad Abdur Raqib

Tanvir Sobhan

Shehzad M Arifeen

Sifat Islam Ishty

Md. Aftab Alam

Rafiu Ibrahim

Ahsan Senan

Maliha Rahanaz

Nazrina Haque

Azraf Uddin Ahmad

Ishmam Al Quddus

Nazia Sharmin

Naima Zaman

Nazia Binte Mahmud

News and Events

Professor Farzana Munshi Publishes a paper titled "Does Access to Credit Reduce Child Labor: New Evidence from Bangladesh"

ESS alumna receives Don Lavoie Fellowship of George Mason University

Research report publication of ESS research team led by Dr Shahidur

ESS faculty participates in workshops organized by the Danish Embassy, RISE and GIZ

Technical Workshop on National Budget" by Dr. Zahid Hussain

“Launch Event of BRAC University-University of Kent Collaboration”

Brown Bag Seminar

Alaap2: Don’t you know you are toxic?

Dr. Hasibun Naher received her PhD from the School of Mathematical Sciences, Universiti Sains Malaysia (USM), Penang, Malaysia. She completed her B.Sc (Honours) as well as M. Sc in Mathematics from the Jahangirnagar University, Savar, Bangladesh and secured first class in both. In addition to her PhD period, she also taught several mathematics courses to the undergraduate students at School of Mathematical Sciences, Universiti Sains Malaysia (USM), from 2011 to 2013.

She is now working as an Associate Professor in mathematics at the Department of Mathematics and Natural Sciences, BRAC University (BU), Dhaka, Bangladesh. She joined BU in 2007 as a lecturer. Before joining BU she worked as a Lecturer in mathematics in different private universities of Bangladesh since 2001.

Her research background is quite commendable. She has published twenty five scientific papers in international journals with ISI (Thomson Reuters) having good Impact Factors, two conference proceedings with ISI and two books. She has been reviewing research papers of forty renowned journals and working as a member of editorial boards of one hundred fifty-five international journals of her field of expertise. She is also serving as an International Scientific Committee with several International

Conferences on Mathematics.

She is a life member of: NOAMI (National Oceanographic and Maritime Institute);

IAENG (International Association of Engineers) Membership Number: 126196 (Hong Kong), IAENG Society of Computer Science, IAENG Society of Electrical Engineering, IAENG Society of Industrial Engineering; MARQUIS Who’s Who in the World, Publications Number: 36725187, USA; South Asian Institute of Science and Engineering (SAISE): Membership Number: 20140702002; Organization for Women in Science for the Developing World (OWSD), Membership Number: 5165.

Level: 4, Room: 4G48Kha-224 Merul BaddaDhaka 1212. Bangladesh

Selected Publications

Books

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Awarded as the Best Researcher in BRAC University Faculty & Staff Award 2015, Bangladesh

Nominated as the Best Researcher in BRAC University Faculty & Staff Award 2014, Bangladesh

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Applied Mathematics, PDEs, ODEs, Mathematical Physics, Nonlinear Evolution Equations, Analytical Solutions, Travelling Wave Solutions, Analytical Methods, (G’/G)-Expansion Method and its various extensions, Exp-Function Method.

1. Applied Mathematics,2. Acta Physica Polonica A3. Abstract and Applied Analysis, Hindawi4. Mathematical Modelling and Analysis, Taylor & Francis5. Applied Mathematics and Computation (AMC), Elsevier6. Physical Review & Research International7. Journal of Scientific Research and Reports8. British Journal of Mathematics & Computer Science9. Journal of Modern Mathematics Frontier (JMMF)10. Walailak Journal of Science and Technology (WJST)11. Ain Shams Engineering Journal (ASEJ), Elsevier12. World Applied Sciences Journal13. Alexandria Engineering Journal (AEJ), Elsevier14. Journal of Egyptian Mathematical Society (JOEMS), Elsevier15. Results

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43. Journal of Basic and Applied Research International, International Knowledge Press44. Mathematical Methods in the Applied Sciences, John Wiley & Sons45. Zentralblatt MATH, NO.:15635

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Hasibun Naher, PhD

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

The 2nd International Conference on Business and Management (ICBM), 2019 saw the introduction of the PhD Colloquium & Paper Producing workshop for the conference participants which took place on the 25th of April at The Westin, Dhaka.

The speakers for the workshop were Prof. Dr. Tzong-Ru Lee from the Department of Marketing, National Chung Hsing University, Taiwan and Assoc. Prof. Dr. Md. Mamun Habib from Brac Business School, Brac University. This session was chaired by Prof. Dr. Mohammad Mahboob Rahman, Dean of Brac Business School, Brac University.

While assessing the paper submissions for ICBM 2019, the conference secretariat noticed that participants were unable to produce good quality papers that match international standards. The PhD Colloquium and Paper Producing workshop was arranged to ensure the submission of good quality papers in the future and to help the participants understand what mistakes they often make that degrades the paper’s quality.

The event kicked off at 9:30AM as the speakers and session chairs took their seats. Dr Mahboob then went on to state that the workshop was an initiative that would prove to be quite beneficial to the students who are involved in research and who plan on getting started for their doctoral. He then gave the floor to the speakers and the main session was underway.

At the very beginning of the session, Dr. Lee pointed out that when it comes to research, especially for a PhD, the research topic must be of an intriguing kind, a hot topic that will peak the interest of academicians everywhere. To this, Dr. Mamun added that our research must be able to bring benefit to the society in some form. Dr. Mamun also emphasised on the role a supervisor plays when it comes to getting your PhD. From the submission of the proposal to the final paper, one must co-ordinate with their respective supervisor.

The speakers then went on to present a comprehensive approach towards academic research which included selecting a topic as the main focus of research, suggesting different sites and journals for exploring papers related to the topic, where and how to submit a research paper, and a full on dissection of a well-structured research paper.

The entire workshop was comprised of 4 modules namely research-based ideas, research-publications, outlines of research publications and academic writing. These modules were presented in two sessions with a tea break in between.

After the speakers had finished taking questions from the audience at the end of the second session, the workshop came to a close and all the guests and participants were provided with lunch.

The workshop saw the participation of academicians from Bangladesh, India, Malaysia and Sweden. The research interests of these participants included advertising, consumer behaviour, micro finance in rich, developed countries, labour law, industrial law, supply change, health management and so on.

Md. Rafsanjany Jim completed his college from Govt. A. H. College, Bogura. He finished his BS. and MS. from University of Dhaka in 2018. Soon after his MS. he joined Brac University as a Vice-chancellor's fellow at the Department of Mathematics and Physics. Later he was appointed as a lecturer in the same department. He has a strong research interest in Noncommutative geometry and Operator theory.

Level: 4Kha-224 Merul BaddaDhaka 1212. Bangladesh

Differential Calculus and Coordinate geometry, Fundamentals of Mathematics (Introduction to Game Theory), Mathematical Methods, Linear Algebra, Calculus I, and Real Analysis.

Noncommutative Geometry, Operator Theory, Axiomatic Quantum Mechanics, Quantum field theory, and General Theory of Relativity.

Md. Rafsanjany Jim

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

1st International Conference on Business and Management (ICBM 2017), the two-day International Conference, Organized by BRAC Business School (BBS), BRAC University, Bangladesh, focused on the theme of 'Global Contemporary Practices in Business and Management'. The conference arena turned into a knowledgeable learning environment by the presence of prominent practitioners from different parts of the world such as

Singapore, Thailand, Malaysia, India, Sweden, China and USA. This event enhances its momentum through the Keynote Speech Session where academicians from different countries shared their thoughts and experiences on the conference theme.

Keynote Session 1 and 2

First keynote session of the day had been chaired by Dr. Mirza Azizul Islam, Former Adviser of Caretaker Government, People's Republic of Bangladesh and Professor of BRAC Business School, BRAC University. Prof. Dr. Mark GOH, National University of Singapore, was the keynote speaker for the session. Prof. GOH delivered his speech on 'Emerging Technologies, Global Contemporary Practices in Business, and Supply Chain Management: What Do They Have in Common?' He also discussed about how emerging technologies will change the contemporary business practices globally and how it will affect supply chain management. He highlighted on the imminent issues like artificial intelligence, big data, machine learning and technological advancement which has countless contributions to make our lives easier. He further emphasized on the fact that disruptive technology has changed our environment, society, economy and lives, especially the way we conduct our day to day activities get highly influenced by all those. According to his opinion, the world have already experienced major technological advances and will experience many more in near future, but the only thing that has been and will always remain constant is the status quo.

Prof. Dr. Syed Saad Andaleeb, Chancellor, BRAC University was the keynote speaker in the second session of the day. Dr. Salehuddin Ahmed, Former Governor of Bangladesh bank (The Central bank of Bangladesh) and Professor of BRAC Business School, BRAC University as the session chair. The topic of the keynote speech delivered by Prof. Andaleeb was 'Building a Research Culture'. To clarify the current world scenario he cited the example of California where the first tier of research universities have faculty members who are expected to just create knowledge and who pass sleepless nights to get their works into reputed journals at the year's end.

He further added that the national education policy for 2010 of Bangladesh espouses the need for quality research for finding solutions to problems of the state but there is a general lack of interest among teachers to conduct research. Moreover, referring to Asian Universities he said they are now being considered as the next higher education superpowers with Japan, South Korea, Taiwan, Singapore and Hong Kong already in the competition for research supremacy with policies designed to promote world-class universities, now it’s our turn to promote the research based culture to compete successfully with the developed nations of the world.

Keynote Session 3 and 4

Prof. Dr. Iftekhar Ghani Chowdhury, Dean, BRAC Business School, BRAC University chaired the third keynote session, where Prof. Dr. Prem Kumar Rajagopal, Vice-Chancellor of Malaysia University of Science and Technology (MUST), was the keynote speaker. Prof. Rajagopal delivered his speech on ‘Industry 4.0 and Supply Chain Responsiveness’. According to his opinion, industry and academia are not going hand in hand. The industry has already started to produce forth computing technologies where devices make decisions rather than sending it to any CPU, Artificial Intelligence (AI) is playing a big role in it, whereas academia seems to be lacking that buzz and only talking about matching with the fourth industrial revolution whilst doing nothing significant to keep pace with the change.

He then stressed on the fact that there is a wide disparity between the industry requirements and skill set offered by students, that’s why students are not being able to solve problems and communicate in accordance. He remarked ‘Smart Data’ and ‘Internet Industry’ as the key to “Industry 4.0”. His session made audience think to re-consider the overall operational process of academia and how it can be upgraded.

The fourth and the final keynote session had been chaired by Prof. Dr. Saiful Majid, Director of Institute of Business Administration (IBA), University of Dhaka. Dr. Pairach Piboonrungroj, Associate Dean, Maritime Studies and Management, Chiang Mai

University was the keynote speaker in that session. He talked about ‘Globalization or Delocalization? - A Supply Chain Perspective’ and demonstrated the mechanism of conflict between these two. The topic on which he focused on was quite a diverse one compared to rest of the keynote speakers.

Some countries have seen the peak of globalization whilst some suffered to a great deal, he added. In light of both situations, he presented illustration of Facebook-bridging globalization while country like China is staying out of it. Dr. Pairach concluded the session beautifully where he tried to analyze the overall situation through the lens of supply chain and showed how we are adapting to such changes gradually.

Available Program For Admission

1st International Conference on Business and Management (ICBM 2017) organized by BRAC Business School (BBS), BRAC University, under the theme of “Global Contemporary Practices in Business and Management” on September 21-22, 2017 at The Westin, Dhaka, Bangladesh.

1st International Conference on Business and Management (ICBM 2017) organized by BRAC Business School (BBS), BRAC University, under the theme of “Global Contemporary Practices in Business and Management” on September 21-22, 2017 at The Westin, Dhaka, Bangladesh.

BRAC Business School (BBS), BRAC University, Bangladesh organized its 1st International Conference on Business and Management (ICBM 2017) under the theme of “Global Contemporary Practices in Business and Management” on September 21-22, 2017 at The Westin, Dhaka, Bangladesh. Basically, this conference was one of the first major international initiatives in Bangladesh of recent times to develop an intellectual ambience with the presence of prominent practitioners coming from the field of both academia and corporate. The conference aimed to provide a forum for academics, researchers and practitioners to exchange ideas and to discuss about recent developments in the field of Business and Management.

To adorn the grace of the conference, ICBM 2017 organizing committee felt extremely proud to have the involvement of Sir Fazle Hasan Abed KCMG, the founder and chairman of BRAC as the Chief Patron. Other highlights of the conference were the presence of notable academicians from countries like Singapore, Thailand and Malaysia as Keynote Speakers, renowned national corporate personalities as Industry Talk specialists, Academia-Industry Discussion session and so on. Moreover, ICBM 2017 also efficaciously ensured the participation of scholars' from well-known universities such as those of Singapore, Thailand, Malaysia, China, India, Sweden, and USA. It saw over 150 academic papers being presented in 13 different tracks covering Business and Management issues in multiple parallel sessions and talks bringing together industry and academia.

Finally, the conference ended in a positive note through the distribution of crest and certificates as a token of appreciation and encouragement among speakers of Keynote Addresses, Industry Talk and Paper Presenters respectively and most significantly with a promise to be back once again with more invigorating highlights in 2nd ICBM 2019.

News and Events

Post Conference Event: ICBM 2017, Moving Forward…

ICBM 2017: Post Conference Meeting with Sir Abed

Update of BBS Conference (ICBM 2017)

5th Intra University Moot Court Competition- 2024

Exploring New Contextualism 2.0: An Approach to Architecture and Urban Design

LSDB's Regional Education Award 2023 (LREA-2023)

Certificate Course on 'Intellectual Property for Innovation, Creativity and Branding'

Moushumi Zahur is now working as Assistant Professor, Department of Mathematics and Natural Sciences, BRAC University, Dhaka, Bangladesh. She completed Master of Environmental Management (2014) from the University of Kitakyushu, Fukuoka, Japan. She completed Master of Urban and Regional Planning (2008) from Bangladesh University of Engineering and Technology. She completed her B.Sc. (Honors) and M. Sc (Thesis group) degrees in Geography and Environment from the University of Dhaka in the years 1998 (held on 2000) and 1999 (held on 2001) respectively. She published one research monograph and several research articles in different reputed journals and also participated at seminars & conferences in home and abroad.

Level: 4, Room: 4G46Kha-224 Merul BaddaDhaka 1212. Bangladesh

• Master of Environmental Management (MEM) 2014, Faculty of Environmental Engineering, University of Kitakyushu, Fukuoka, Japan.• Master of Urban and Regional Planning (MURP) 2008, Department of Urban and Regional Planning, Bangladesh University of Engineering and Technology• Master of Scince (M.Sc) 1999, Department of Geography and Environment, University of Dhaka• Bachelor of Scince (B.Sc) 1998, Department of Geography and Environment, University of Dhaka

Journals • Zahur, Moushumi (2007), "Solid Waste Management of Dhaka City: Public Private Community Partnership", BRAC UNIVERSITY Journal, Volume 4, No.2.• Zahur, Moushumi and Islam, Nazrul, (2005) "Chattagram Shaharer Abashik Bhumi Baboharer Bikash O prokriti"(in Bengali), Bhugal O Poribesh Journal, vol. 5, special series, page: 15-37.• Pavin, Gulshan Ara, Akbar, Marshia Tashmim & Zahur, Moushumi (2004), "NGOs in Bangladesh: Examination People's Perceptions", The Jahangirnagar Review, Part II: Social Science. Vol.XXVIII.Books (research monograph)• Zahur, Moushumi, (Nov, 2010) Private Apartment Housing: A Study on Affordability of the Middle Income People of Dhaka, Bangladesh, ISBN: 978-3-8433-5920-7, LAP LAMBERT Academic Publishing GmbH & Co. KG, Germany.

• Advisor - BRAC University Drama and Theatre Forum (BUDTF), April 2009 to February 2010.• Was in charge for the preparation and sending of the Departmental News for BRAC University Monthly Bulletin.• Member – Scrutinizing Committee of the Department of Mathematics & Natural Sciences (MNS).• Member - Moderation Committee of the Department of Mathematics & Natural Sciences (MNS).• Served as an invigilator in each of the undergraduate admission test of BRACU.• Regularly attended seminars organized by BRACU authority.• Worked as a volunteer in Sub Committee of BRACU Picnic 2009.• Responsible for coordinating and maintaining liaison between BRAC University Library and the Department of Mathematics of Natural Sciences.

• Introduction to Economic Geography (GEO101)• Elements of Environmental Sciences (ENV103)

International Conference Proceeding• Zahur, Moushumi and Otoma, Suehiro, “Informal Solid Waste Recovery and Recycling in Dhaka City: Context of Environmental and Socio-Economic Aspects”, 3rd International Conference on Solid Waste Management in Developing Countries, February 10-12, 2013, Khulna, Bangladesh.• Zahur, Moushumi, “Monitoring the Impacts of Climate Change: exploring framework in the context of Bangladesh”, International Training Workshop on Regional Climate Change and Its Impact Assessment organized by CAS-TWAS-WMO Forum on Climate Science. September 26 to 29, 2011, Beijing, China.• Zahur, Moushumi and Islam, Ishrat, “Gap Between Expectation & Reality: A Study on Housing Affordability by Middle-Income Group of Dhaka”, Pacific Regional Science Conference Organization, May15-17, 2008, Dhaka.Poster Presentation• Zahur, Moushumi and Otoma, Suehiro,“ Informal Waste Recycling Activities: A Case Study of Dhaka City, Bangladesh”, The 24th

Annual Conference of JSMCWM, Hokkaido University, Sapporo, Japan. November 2-4, 2013. (poster and short oral presentation)• Zahur, Moushumi and Otoma, Suehiro,“ Scavengers and Informal Recycling Activities in Dhaka, Bangladesh”, Annual Regional Meeting of JSMCWM, Kyushu Branch, Fukuoka University, Fukuoka, Japan. May 18, 2013.• Zahur, Moushumi and Otoma, Suehiro, “Environmental and Socio-Economic Aspects of Solid Waste Recovery and Recycling in Bangladesh: A Case Study of Dhaka City”, The 10th International Conference on Ecobalance-2012, November 20-23, Yokohama, Japan. Training and other programs• Participated in the International Training Workshop on Science Education for Sustainable Development held on Dhaka, Bangladesh from June 27-28, 2011.• Participated in the Young Scientists of Asia Conclave-2009 held on Bangalore, India from January 15 to January 17, 2009.• Participated in the 19th Asian International Seminar held in Kathmandu, Nepal from October 19–24, 2008. Participated in a group discussion about “How To Address Our Local Environment Degradation Issues” and gave a presentation on the same topic.• SAARC Exchange Program, India from December 1 to December 10, 2007 (selected by the Bangladesh University Grants Commission).• Participated in the 46th Annual General Meeting and Day-Long seminar on “YOUNG GEOGRAPHERS’ RESEARCH” organized by Bangladesh Geographical Society (BGS) held on 29th June, 2001 at Geography & Environment Department, University of Dhaka.• Participated in workshop on “Urbanization on Bangladesh” organized by Bangladesh Social and Economical Forum in association with Real Estate & Housing Association of Bangladesh (REHAB). Held on 3-5 May 2001.• Participated in Seminar on “Bangladesh on the Threshold of the twenty first century” organized by Asiatic Society of Bangladesh. Held on 27-28 April 2001.• Participated in seminar on “World Habitat Day” organized by RAJUK held on 3rd October, 2000.

• JICA Scholarship from April 2012 to March 2014• Dean’s honor (1998) from “The Faculty of Science”, University of Dhaka.• General Scholarship (Academic Year

1995-96 to 1998-1999), Board of Intermediate and Secondary Education, Dhaka, Bangladesh• General Scholarship (Academic Year 1993-94 to 1994-95), Board of Intermediate and Secondary Education, Dhaka, Bangladesh

• Alumni SUW program, University of Kitakyushu, Fukuoka Japan• JICA Student Association Alumni• Life member of Centre for urban Studies (CUS)• Life Member of National Oceanographic and Maritime Institute (NOAMI) and Centre for Urban Studies (CUS).• Member- Bangladesh Geographical Society (BGS)

• Waste management • Environmental Management• Disaster Management

Moushumi Zahur

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

BRAC University is the first university from South Asia to be part of the Open Society University Network (OSUN), a global network of educational institutions across the world that are dedicated to advancing knowledge, promoting civic engagement, and expanding access to global higher education.

The Open Society University Network (OSUN) is a groundbreaking model of global higher education. It brings together universities and colleges from around the world to integrate learning and knowledge creation across disciplines and borders. OSUN focuses on the social sciences, humanities, sciences, and arts, both at the undergraduate and graduate levels.

OSUN aims to educate students to address tomorrow’s global challenges, fostering critical thinking and open intellectual inquiry to strengthen the foundations of open society amid the current authoritarian resurgence. OSUN counteracts intellectual monocultures and polarization by uniting institutions around the world in collaborative research projects and by encouraging students to examine issues from different perspectives and through reasoned arguments. In addition, OSUN addresses inequality by expanding educational access to neglected and minority populations, such as incarcerated persons, the Roma, and refugees.

As a member of OSUN, BRAC University gains access to a global network of like-minded institutions committed to academic excellence, social justice, and open societies. This network provides unique opportunities for collaboration, knowledge exchange, and joint research initiatives, empowering our faculty and students to contribute to pressing global challenges. The Following are some of the programs that are currently ongoing at the university with OSUN.

There is a natural synergy between the multi-faceted areas of physics in general and electronics in particular. It is well nigh impossible to imagine any field or area where these two subjects have not had profound impacts. These ever-increasing technological aspects and applications have changed forever the life of an individual and the ambience and surroundings amidst which he lives. The total credit requirement for BS in APE degree is 130. Options for doing a double- major with another major or a minor in APE are also available.

News Archive >>

News

Biotechnology seminar hosts Bangladeshi scientist who developed high-yielding "Panchabrihi" rice

Orientation held for Summer 2024 biotechnology students

Nanobiotechnology seminar organized by BUSB and Biotechnology Program

The biotechnology program and BUSB organized the inaugural Biotechnology Workshop Series on "Understanding and Designing Biosensors"

Brochure/Flyer (Printable Versions)

Brochure/Flyer (Printable Versions)

There is a wide range of Undergraduate Programs that Brac University offers so that prospective students can make their degrees as exclusive as they may desire. At BracU, we are committed to assuring academic excellence and a dynamic learning environment in each of our Undergraduate Programs. Our superlative faculty is always ready to support students with the greatest assistance.

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Minimum Qualifications (except Bachelor of Pharmacy)

Particulars

Required Minimum GPA

SSC/Equivalent and HSC/Equivalent

(including additional subject)

3.5 in SSC and HSC separately

O-level with five subjects and A-level with two subjects as per the scale of A=5, B=4, C=3, D=2

2.5 in O-Level and A-Level separately

Candidates applying for EEE, ECE, CSE, APE and Physics

HSC/ Equivalent

“B” grade in Physics & Mathematics

A-levels

“C” grade in Physics & Mathematics

Candidates for BSc in Computer Science and BSc in Mathematics

HSC/ Equivalent

“B” grade in Mathematics

A-levels

“C” grade in Mathematics

Candidates for BSc in Electronic & Communication Engineering and Electrical & Electronic Engineering who had Physics and Mathematics but not Chemistry in HSC/A-level/Equivalent will have to take a remedial course on Chemistry in addition to the courses required for the program.

Candidates for BSc in Biotechnology and Microbiology should have minimum C grade in Biology and Chemistry in HSC/A-level/Equivalent. Candidates who did not have Mathematics in HSC/A-level/Equivalent must take a remedial course on Mathematics in addition to the courses required for the program.

Candidates who have qualified in HSC/A levels or recognized equivalent examination in current year or two previous years are eligible to apply.

“E” grades will not be considered.

(Public/ Private University B. Pharm Program Accreditation and Student Admission Guideline)

Candidates must have a minimum aggregate GPA of 8.00 out of 10 (on a scale of 10), with no grade lower than 3.5 in SSC/O Level/Equivalent and HSC/A Level/Equivalent. In addition, they must achieve a minimum GPA of 4.00 in Chemistry and Biology in both examinations. They also need a minimum GPA of 3.5 in Mathematics and Physics in both examinations (on a scale of 5 in HSC or equivalent). For “O” Level and "A" Level candidates, they must have a B grade in Chemistry, Biology, and a minimum C grade in Mathematics and Physics.

Candidates having no Mathematics or failed in the subject at the HSC/”A” Level or recognized equivalent exam may be admitted, but they need to take an extra 3 (three) credits course on Mathematics relevant to the B.Pharm. curriculum.

Candidates must pass HSC/ "A" level or recognized equivalent examinations in the current year or the previous year. For example, the 2024 January session candidate must pass their HSC/ "A" level or recognized equivalent examination in 2022 or 2023.

Foreign students must have passed 12 education years and got the same grades in recognized equivalent examinations from foreign recognized institutions and must get an equivalence certificate from the Pharmacy Council before admission. Recognized equivalents mean Education Board, Madrasah Board and Bangladesh Technical Education Board, etc.

Candidates completed IB Diploma with a minimum DP Score of 24 are eligible to apply. (before applying please contact through admissions[at]bracu.ac[dot]bd)

❖ Candidates completed 12 years of schooling from a reputed educational institution may also apply. (before applying please contact through admissions[at]bracu.ac[dot]bd)

❖ GED is not acceptable.

❖ Cases where HSC or equivalent is more than three years previous to date need to apply to the chair admissions committee for review. (please contact through admissions[at]bracu.ac[dot]bd)

❖ Candidates completed schooling outside Bangladesh will have to submit verified/attested copies of previous academic documents from their Institute/Foreign Ministry (of the country that they studied in) and equivalence certificates from the Board of Intermediate and Secondary Education, Dhaka.

All eligible candidates will have to sit for a written admission test on section/sections as shown in the table:

Program

Test section

BSS in Anthropology, BA in English and LLB (Hons)

English

BBA and BSS in Economics

English and Mathematics

Bachelor of Pharmacy, BSc in Microbiology and BSc in Biotechnology

English, Mathematics and Biology & Chemistry

BSc in Computer Science, BSc in Computer Science and Engineering, BSc in Electronic and Communication Engineering, BSc in Electrical and Electronic Engineering, BSc in Physics, BSc in Applied Physics & Electronics, BSc in Mathematics

English, Mathematics,

Higher Mathematics and Physics

Bachelor of Architecture

English, Mathematics and Drawing

Candidates who have qualified in HSC/A levels or recognized equivalent examination in current year or two previous year are eligible to apply (except Bachelor of Pharmacy). For Pharmacy, Candidate must pass HSC/ "A" level or recognized equivalent examination in current year or one previous year.

Transfer of credits from an educational institution following a system similar to BRAC University may be considered after passing the admission test. Such candidates will have to apply with required documents and decision on such cases will be determined by the credit transfer policy of BRAC University.

Visiting or guest students are required to submit their applications along with the relevant transcripts and no objection letter (NOC) from her/his university subject to approval from chair, admissions committee. Visiting students are not required to appear in the admission test. A visiting student will be able to take a maximum of 2 courses in a semester and a maximum of 8 courses over time. This is to ensure that a visiting student does not take too many courses and apply for a degree at BRACU.

Admission Requirements

M H M Mubassir is teaching at the BRAC University in Dhaka as a lecturer of the Biotechnology Program under the Department of Mathematics and Natural Sciences. Bestowed with prestigious Malaysia International Scholarship (MIS) he completed his Master of Philosophy (MPhil) under the Faculty of Biosciences and Medical Engineering, Universiti Teknologi Malaysia (UTM). Mubassir completed his graduation in Agriculture and Master's in Biotechnology from Bangladesh Agricultural University, Mymensingh.

Level: Kha-224 Merul BaddaDhaka 1212. Bangladesh

Master of Philosophy in Bioscience; Universiti Teknologi MalaysiaMaster of Science in Biotechnology; Bangladesh Agricultural UniversityBachelor of Science in Agriculture; Bangladesh Agricultural University

Journals

Mubassir, M. H. M. (2019). A synopsis of different plant LRR-RLKs structures and

functionality. American Journal of Biomedical Science & Research, 1(2), 84–86. https://doi.org/10.34297/AJBSR.2019.01.000517

Conferences

Jawad, T., & Mubassir, M. H. M. (2018). Bioinformatics approach of structural modelling and molecular dynamics simulation of pattern recognition receptor CORE. In 6th Annual South Asia Biosafety Conference 2018. Dhaka, Bangladesh.

Jawad, T., & Mubassir, M. H. M. (2018). Bioinformatics approach of structural modelling and molecular dynamics simulation of pathogen-associated molecular pattern CSP22. In National Biotechnology Fair. Dhaka, Bangladesh.

Mubassir, M. H. M. (2018). Molecular dynamics simulation of pathogen-associated molecular pattern flg22. In Second International Conference on Multidisciplinary Contemporary Research. Singapore.

Mubassir, M. H. M., Fathin, A., Naser, M., Abdul-Wahab, M. F., & Hamdan, S. (2017). Structural dynamics of endogenous peptide AtPep1. In ASIA International Multidisciplinary Conference (AIMC-2017). Malaysia.

Fathin, A., Mubassir, M. H. M., Naser, M., Abdul-Wahab, M. F., & Hamdan, S. (2017). In-Silico structural modeling of pathogen-associated molecular pattern Elf18. In ASIA International Multidisciplinary Conference (AIMC-2017). Malaysia.

BIO101: Introduction to Biology

BTE103: Plants and People

BCH101: Basic Biochemistry

BTE201: Bioorganic Chemistry

BTE203: Introduction to Molecular Biology

BTE258: Biotech Lab 3

BTE307: Advanced Molecular Biology

BTE308: Plant Biotechnology and Genetic engineering

BTE403: DNA Fingerprinting and Molecular Diagnostics

BTC509: Genomics (Bioinformatics)

CHE110: Principles of Chemistry

M H M Mubassir

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Curriculum in Details

a. General Education

BI0 101 Introduction to Biology 3 credits

An introduction to the cellular aspects of modern biology including the chemical basis of life, cell theory, energetic, genetics, development, physiology, behavior, homeostasis and diversity, and evolution and ecology. This course will explain the development of cell structure and function as a consequence of evolutionary process, and stress the dynamic property of living systems.

Suggested Readings

1. Biology: P.H. Raven and G.B. Johnson 2. Biological Science: G. W. Stout and D. J. Taylor 3. Advanced Biology: J. Simpkins and J. J. Williams 4. Biology: A Fundamental Approach: M. B. Roberts

CSE 101: Introduction to Computer Science 3 credits

Introduction to the use of computer hardware and software as tools for solving problems. Automated input devices and output methods (including pre-printed stationary and turnaround documents) as part of the solution. Using personal computers as effective problem solving tools for the present and the future. Theory behind solving problems using common application software including word processing, spreadsheets, database management, and electronic communications. Problem solving using the Internet and the World Wide Web. Programming principles and use of macros to support the understanding of application software. The course includes a compulsory 3 hour laboratory work each week.

Suggested Texts: L. Goldschlager and A. Lister, “Computer Science – A modern introduction”, 2nd ed., Prentice Hall, 1988.

DEV 101 Bangladesh Studies 3 credits

Socio-economic profile of Bangladesh, agriculture, industry, service sector, demographic patterns, social aid and physical infrastructures. Social stratification and power, power structures, government and NGO activities in socio-economic development, national issues and policies and changing society of Bangladesh.

Suggested Readings

1. Bangladesh: National Cultures and Heritage: An Introductory Reader: A.F. Salahuddin Ahmed & Bazlul Mobin Chowdhury 2. The History of Bengal (Vol.1 &Vol.2) : R.C. Majumdar 3. Banglapedia, 2003: Asiatic Society of Bangladesh 4. Bangladesh Arthaniti: Khan, Md. Shamsul Kabir 5. Bangladesh on the Threshold of the Twenty-First Century, Asiatic Society of Bangladesh, 2002: A.M Chowdhury and Fakrul Alam 6. Poverty Reduction & Strategy: What, Why & for Whom in Asit Biswas et.al.(ed) Contemporary Issues in Development : M.M Akash 7. Bangladesh 2020: A long-run perspectives study: The World Bank 4. ENG 091 Foundation Course (non-credit) The English Foundation Course is designed to enable students to develop their competence in reading, writing, speaking, listening and grammar for academic purposes. The students will be

encouraged to acquire skills and strategies for using language appropriately and effectively in various situations. The approach at all times will be communicative and interactive involving individual, pair and group work.

Suggested Readings 1. J. C. Richards, J. Hull, and S. Proctor, “New Interchange: Student 's Book 3-A”, Cambridge University Press, 2002 2. J. Nadel, B. Johnson, and P. Langan, “Vocabulary Basics”, Townsend Press, 1998. 3. Hogue, “First steps in Academic Writing”, Longman, 1996 4. K. Blanchard, C. Root, “Get Ready to Write”, Longman, 1998.

ENG 101 English Fundamentals 3 credits

Developing basic writing skills: mechanics, spelling, syntax, usage, grammar review, sentence and essay writing.

Suggested Readings

1. Fundamentals of English: Jack C. Richards

ENG 102 Composition I 3 credits

The main focus of this course is writing. The course attempts to enhance students’ writing abilities through diverse writing skills and techniques. Students will be introduced to aspects of expository writing: personalized/ subjective and analytical/persuasive. In the first category, students will write essays expressing their subjective viewpoints. In the second category students will analyze issues objectively, sticking firmly to factual details. This course seeks also to develop students’ analytical abilities so that they are able to produce works that are critical and thought provoking.

Suggested Reading

1. Composition I: The Pearl; John Steinbeck

HUM 103 Ethics and Culture 3 credits

This course introduces the students to principles and concepts of ethics and their application to our personal life. It establishes a basic understanding of social responsibility, relationship with social and cultural aspects, and eventually requires each student to develop a framework for making ethical decision in his work. Students learn a systematic approach to moral reasoning. It focuses on problems associated with moral conflicts, justice, the relationship between rightness and goodness, objective vs. subjective, moral judgment, moral truth and relativism. It also examines personal ethical perspectives as well as social cultural norms and values in relation to their use in our society. Topics include: truth telling and fairness, objectivity vs. subjectivity, privacy, confidentiality, bias, economic pressures and social responsibility, controversial and morally offensive content, exploitation, manipulation, special considerations (i.e. juveniles, courts) and professional and ethical work issues and decisions. On conclusion of the course, the students will be able to identify and discuss professional and ethical concerns, use moral reasoning skills to examine, analyze and resolve ethical dilemmas and distinguish differences and similarities among legal, ethical and moral perspectives.

Suggested Readings

1. Ethics, Culture and Psychiatry - International Perspectives: Ahmed Okasha, Julio Arboleda 2. Ethics and HRD: A New Approach to Leading Responsible Organizations: Tim Hatcher 3. The Ethical Challenge: How to Lead with Unyielding Integrity: Noel M. Tichy and Andrew R. 4. Dignity of Difference: How to Avoid the Clash of Civilizations: Jonathan Sacks 5. Culture and Ethics: Michel Labour, Charles Juwah, Nancy White and Sarah Tolley 6. Culture Matters: How Values Shape Human Progress: Samuel P. Huntington

MAT 101 Fundamentals of Mathematics 3 credits

Basic techniques of algebra, analytical geometry, graphing, and trigonometry.. Prerequisites: None

b. Departmental Core Courses: (63 credits)

BCH 101 Basic Biochemistry 3 credits

Concept of life and living processes – The identifying characteristics of a living matter;

The Cell and its evolution; From molecules to the first cell; From prokaryotes to eukaryotes; structure and function of sub-cellular organelles; Brief treatment of mitosis and meiosis; Cell Membrane and its Organization- Elementary idea of cellular constituents: Nucleus, Mitochondria, Golgi bodies, Endoplasmic reticulum, Lysogymes and Microbodies; Bacterial and Plant Cell walls; Biomolecules-The small molecules of life – Sugars, organic acids, amino acids and nucleotides; Macromolecules of life: polysaccharides, fats, proteins and nucleic acids; General idea of primary, secondary, tertiary and quaternary structures; Nucleus and Heredity-Nuclear membrane; Nucleolus- Nuclear pores; Chromosomes; Packaging of DNA; DNA as Genetic material; DNA replication- basic concept; Transcription-from DNA to RNA; Translation-RNA to protein Ribosomes and protein synthesis; Mitochondria –The power house; Structure, organization and function; Elementary account of Glycolysis and Krebs cycle and role of mitochondria in the laterprocess; Chloroplasts- Capturing energy from the sun; Structure, organization and function; Basic information on ‘light’ and ‘dark’ reactions of photosynthesis and participation of chloroplast in the process.

Suggested Readings:

1. Lehninger, Albert L. 1978. Biochemistry., M/s Worth Publishers Inc,, New York 2. Lehninger, Albert L. 1978. Principles of Biochemistry., M/s Worth Publishers Inc,, New York 3. Matthews & Van Holde, 2nd Ed. Biochemistry, Benjamin Cummings Pub. Co. 4. Stryer, L. 4th Ed. Biochemistry. 5. Rawn, 1989 . Biochemistry. 6. Voet & Voet, 1991. Biochemistry

BCH 102 Biophysical Chemistry 3 credits

Basic concepts- moles, Molarity, Normality, Avogadro’s number, Molality; ionization of acids and bases- Bronsted- Lowry concept, Lewis concept, Arhenius concept, strength of acids, pH and pKa of solutions, buffer solutions and buffer capacity, Henderson-Hasselbalch equation, acid base indicators; titration, choice of suitable indicators; Physical properties of water, ionic products of water and pH scale;

Thermodynamics- First law; definition; nature of heat and work, first law of thermodynamics, internal energy, enthalpy, molar heat capacities, isothermal and adiabatic expansion;; Second law of thermodynamics- staement o0f the second law, entropy changes, phase transition, reversabilty and irreversibility; Free energy, variation with temperature and pressure, Gibbs-Helmholtz equation, Clausius-Clapeyron equation; application of thermodynamics in biotechnology, open system, high energy compounds Thermochemistry- Exothermic and endothermic reactions, standard enthalpy formation, thermochemical equatios. Reaction enthalpy; Chemical equilibrium- the nature of chemical equlibrium, law of mass action, equilibrium constant, relationship between ^G and Keq; Effect of temperature and pressure, Le Chattelier principle, equilibrium reaction involving protons, coupling of reactions; Chemical kinetics-Definition, reaction rate, rate laws of zero, first and second order reactions, molecularity of a reaction , pseodo first order reaction, half life; determination of order of a reaction, effect of temperature on reaction rates; Catalysis-Definition,types, characteristics of catalysts, activation energy of catalysis,

Suggested Readings

1. Bahl, B.S. & Tuli, G.D. & Bahl A., Essentials of Physical Chemistry, S Chand & Company, New Delhi, 2004-05 2. A Text Book of Physical Chemistry, P.W. Atkins 3. T. Hossain, A Text Book of Heat

BCH 201 Human Physiology 3 credits

Blood cells, lymph, serum, plasma, origin of blood cells; Homeostasis, biochemistry of blood clotting, blood grouping; Circulatory system; Lymphatic and lymphoid system-lymph and lymph vessels, lymphatic circulation; structure and function of lymph nodes, spleen thymus and bone marrow in immunity Digestive system- digestive enzymes, digestion of food components and absorption of food components and absorption of digested products; Kidney- Excretory system, role of kidney in water, electrolyte and acid-base balance of the body; buffers of cells, regulation of extracellular fluid composition and formation and excretion of urine; Liver- Structural organization and function; Reproductive system- male reproductive system, spermatogenesis, hormonal factors that stimulate spermatogenesis. Regulation of male reproductive functions by various hormones; Female reproductive system-monthly ovarian and uterian cycle and function of gonadotrpic hormones; Preganacy, function of placenta, hormonal factors in pregnancy, lactation and manopause. Nervous system- Nerve cells, ionic basis of excitation and conduction, Synaptic transmission, the sense and sense receptors; Hypothalamus and temperature regulation.

Suggested Readings

1. Gyton, M.D. 1986, Text Book of Medical Physiology, W.B Saunders Company, London
2. Smith, E., Paterson, C.R., Scratecherd, T. and Read, N.W. 1988, Text Book of Physiology, Longman Group Ltd, Honkong

BCH 202 Enzymes and Enzyme Kinetics 3 credits

Brief history, classification of enzymes;Enzyme assay, enzyme activity, units; Enzyme as biocatalysts, catalytic power, specificity and regulatory properties Factors affecting enzyme reactions- Enzyme kinetics, Monosubstrate reactions: Michaelis- Menten equation and its linear transformations; Km and Vmax-definitions, determination, significance; Enzyme inhibition; Identification of functional gro up essential for catalysis with biological examples;Factors affecting the efficiency of enzymes as catalysts-Specificity; ES complex; Enzyme regulation, allosteric enzymes, Monod and Koshland models; examples of regulatory enzymes- aspartate trans carbamylase; phosphorylase; pyruvate dehydrogenase; hexokinase;Mechanism of enzyme action.

Suggested Readings

1. Boyer, 1970, The Enzymes 2. Leninger, A.L. 1987, Biochemistry 3. Voet and Voet, 1991, Biochemistry

BTE101 Introduction to Biotechnology and Genetic Engineering 3 credits

Basic concept about biotechnology and Genetic Engineering; Appreciation of how genetic engineering and biotechnology will influence agriculture, health, environment and society; A basic understanding of structure and function of cell, types of biomolecules and their function in cells; Basic knowledge of genes and how genes are expressed (DNA to RNA to protein); Basic understanding of important techniques of genetic engineering and biotechnology; Application of biotechnology in agriculture, medicine, industry and environment; Genetic engineering for better food; Genetic testing of diseases;, Gene therapy to treat diseases; DNA technology in criminal cases; Cloning; Prospects of biotechnology and genetic engineering; Biosafety and Bioethics; biodiversity.

(a Scope of Biotechnology for research (b) Agriculture related applications: Plant character amenable to change by biotechnology -- seed quality, photosynthesis, nitrogen fixation, herbicide resistance. (c) Medicine related applications: Commercial synthesis of hormones, vaccines etc., Gene therapy, Disease diagnosis. (d) Microbial applications: Large scale preparation of organic chemicals, biomining, microorganisms as feed of livestock. (e) Biotechnology in service of environment-related applications: Pollution control, waste disposal, biogas. (f) Livestock improvement: Dairy products, meat quality (g) Monoclonal antibodies and their applications.

Suggested Readings

1. Watson, J.D., Tooze, J. and Kurtz, D.T. 1983. Recombinant DNA: A Short Course, Scientific American Books, New York. 2. Drlica, K. 1984., Understanding DNA and Gene Cloning: A Guide for the Curious, John Wiely & Sons, New York. 3. Steven, P., 1984., Biotechnology – A New Industrial Revolution, George Braziller Inc. USA. 4. Antebi, E. and Fishlock, D. 1986. Biotechnology, The MIT Press, USA. 5. Marx, J.L., 1989. A Revolution in Biotechnology, Cambridge Univ. Press, UK. 6. Old, Principles of Gene Manipulation and Introduction to Genetic Engineering, 3rd Ed. 7. Koshland, Biotechnology. 8. Smith, 1996, Biotechnology. 9. Rehm, 1986. Biotechnology. 10. Brown, 1987. Introduction to Biotechnology. 11. Kingsman and Kingsman, 1988, Genetic Engineering.

BTE 102 Microbial World 3 credits

Historical development of microbiology from Antonie Van Keewenhi\ock to Alexander Faming Spontaneous generation theory; Involvement of microorganisms in fermentation; Development of microscopy and characteristic of different types of microscopes; Bacteria: Nutritional aspects of bacterial cultivation media; composition; types and sterilization, growth and reproduction, isolation of pure culture from natural sources, enumeration and preservation culture Fungi: Brief outline on growth and reproduction; importance in natural process Viruses: Classification with representative examples; TMV and l phage, lytic cycle and lysogeny Actinomycetes: Importance in industry and natural process Method of isolation and identification from different sources and environments; Protists, Eukaryotic and Prokaryotic microorganisms and their differentiation. Nutrient requirement of microorganisms; Carbon, nitrogen and mineral metabolism; Autotrophic, heterotrophic and chemilithotrophic microorganisms; Growth of microorgnisms, Generation time, Different phase of growth curve; Biosynthetic substances produce by microorganisms; importance of microorganisms in industrial process and agriculture. Natural resistance, pathogenecity and virulence, microbial toxins, transmission and prevention of common infectious diseases e.g. Cholera, Tuberculosis, Tetanus.

Suggested Readings

1. Brock, T.D. and Madigan, M.T. Biology of Microorganisms, Prentice Hall International, 1997. 2. Tortora, Funke and Case, Microbiology an Introduction, Addison Wesley longman Inc, 1997. 3. Atlas, R.M., Principles of Microbiology, 2nd Ed. Wm.C. Brown Publishers, 1997. 4. Pelczer & Reid, Microbiology.

BTE 103 Plants and People 3 credits

Knowledge about Basic Botany and pivotal role that plants play in the every day life of humans and also the role that man has to play in the health and survival of plants for his own survival and comfort; Energy, Plants and Environment; Photosynthesis and respiration, Mendelian Genetics, Origins of agriculture; Old fashioned and modern agriculture; Agriculture and environment; Cereal grains Forage grasses; Legumes; Fruits and Vegetables; Herbs and spices; Vegetable oils and waxes; Medicinal Plants- current revival of herbal medicines Wood Plants; Ornamental plants; Fibres; Plant t structure and reproduction; Plant breeding; Some novel aspects of plant life; Human nutrition and plants as food; Algae, Fungi and some Bacteria; Plant exploitation in world history; Ecology and succession; Terrestrial biomes; Biodiversity and Extinction; Concept of Plant Biotechnology; Importance of plants in culture, history and modern society; Management and protection of world natural resources;; Biodiversity- Threatened biodiversity, causes of loss of biodiversity, need and methods of biodiversity conservation; Pattern of Biodiversity in Bangladesh; Indigeous, exotic, common, rare, threatened and endangered species of Bangladesh; CBD and Caratagena Protocol

Suggested Readings

1. Estelle, L. Mcmahon, K. Plants and Society, 3rd Edition,2003 2. Ararwal, K.C. Biodiversity 3. Kumar, H.D. General ecology 4. Pandy, S.N. Text Book of Botany

BTE 201 Bioorganic Chemistry 3 credits

Chemical bonding- Covalent bond, ionic bond, hybrid orbital, polarity of bonds, electronegativity, dipole moment, potential curve, weak bonds, hydrogen bonds and hydrophilic interactions, intermolecular forces, boiling point, melting point, solubility; Methane- Structure, physical and chemical properties; Alkanes- Occurrence, structure, nomenclature, synthesis, physical and chemical properties, free radical substitutions, stability of free radicals, halogination; Alkenes and alkynes - Occurrence, structure, nomenclature, synthesis, physical and chemical properties, free radical substitutions, stability of free radicals, halogination; Alcohols, ethers, epoxides and diols- - Occurrence, structure, nomenclature, synthesis, physical and chemical properties and uses; Aldehydes and ketones- Nomenclature, synthesis, nucleophilic addition, elimination reaction, oxidation reduction of carbonyl compounds; enolization in biological system; Aldol condensation, benzoin condensation, Claison condensation; Stereochemistry- Sterioisomerization, anatiomers, polarimetry, optical isomers, racemic modification, meso compounds ;Aromaticity- Strucure of benzene, sources of aromatic hydrocarbons, industrially important aromatic compounds, chemistry of aromatic-aliphatic compounds; Carboxylic acids and their derivatives- nom3enclature, synthesis, classification, properties, reactions, uses, decarboxylation reaction; Nitro-compounds and amines- Occurrence, nomenclature, synthesis, classification, properties, reaction, uses diazonium compounds

Suggested Readings

1. Morison and Boyd Organic Chemistry 2. Bahl, B.S. and Bahl. A. Advanced Organic Chemistry 3. Atkins and Carey, 1991. Organic Chemistry: A Short Course. McGraw Hill Publishing Company 4. Eliel, 1962. Steriochemistry of Carbon Compounds

BTE 202 Metabolism 3 credits

General aspects of metabolism- Experimental approaches to the study of metabolism, metabolic and energy transfer pathways, a survey of intermediary metabolism; catabolism and anabolism;Glycolysis- The glycolytic pathway,Anaerobic and aerobic metabolism, regulation of glycolytic pathway, metabolism of disaccharides, pentoses and hexoses; Importance of anaerobic glycolysis, anaerobic glycolysis and tumour; anaerobic glycolysis and heart attack; Fructose intolerance, hypoglycemia;The TCA cycle- Overview, amphibolic nature of the cycle, anaplerotic reactions, regulation of TCA cycle Bioenergetics- high energy compounds, the ATP cycle, Central role of ATP in metabolism; Electron transport and oxidative phosphorylation, oxidation-reduction enzymes and electron transport, oxidative phosphorylation; Uncouplers and inhibitors of oxidative phosphorylation; Mitochondrial structure and compartmentalization of respiratory metabolism;Biosynthesis of carbohydrates- Gluconeogenesis and its regulation;Amino acid metabolism- Overview, general reactions of amino acids;

odegradation of aminpo acids; biosynthesis of amino acids; Inborn error of amino acid metabolism- phenylketonuria, alkaptonuria, hepatic coma, deficiencies of urea cycle enzymes;Nucleotide metabolism- Overview, metabolic function sof nucleotides, synthesis of purine and pyrimidine nucleotides, nucleotide degradation; Metabolic interrelationships- Overview, starve-fed cycle, metabolism in exercise; Brief introduction to lipid metabolism, fatty acid oxidation, ketone body formation and utilization of fatty acid; Fatty acid biosynthesis, regulation of fatty acid metabolism

Suggested Readings

1. Leninger, A.L. 1987, Biochemistry 2. Stryer, L., 1988, Biochemistry, W.H. Freeman and Company, New York

BTE 203 Introduction to Molecular Biology 3 credits

Nucleic acids- Occurrence, isolation, purification and molecular weight determination; Strucure of RNAs- primary, secondary and tertiary) and their sequences; Structure of DNAs-Different physcochemical properties such as Tm value, Cot value, hybridization kinetics, different conformations of DNA; DNA sequencing- Chemical, enzymatic and polymerase chain reaction methods of sequencing; Recombinant DNA- Plasmid and phage vector construction, isolation of recombinant DNA; restriction enzymes, DNA polymerase reaction and its applicationDNA replication- Mode of replication;Transcription-brief introduction, RNA polymerase, regulation of transcription; Translation- Genetic code, specificity, redundancy and wobble hypothesis; Protein synthesis- Ribosome as protein fctory, Initiation, elongation and termination of translation; Molecular biology and biotechnology.

Suggested Readings

1. Stryer, L., 1989 Molecular Design of Life, W.H. Freeman and Company, New York 2. Darnell, J. Lodish, H. and Baltimore, D. 1986. Molecular Cell Biology., Freeman and Company, New York. 3. Lehninger, A.L. Nelson and Cox, 1990. Principleas of Biochemistry

BTE 204 Fundamentals of Genetic Engineering 3 credits

Recombinant DNA Technology- Gene cloning concept and basic steps; Purification of total cell DNA; plasmid DNA and phage DNA; Techniques in molecular genetics- Production of recombinant DNA; DNA manipulative enzymes,; Cloning vectors and ligation system; Application of bacteria and viruses in genetic engineering; Molecular biology of E.coli and bacteriophages in the context of their uses in genetic engineering; Gene cloning- Restriction endonuclease, ligases and other enzymes useful in gene cloning, PCR technology for gene/DNA detection, DNA, usage of plasmid and phages ias vectors; Use of Agrobacterium for genetic engineering of plants; Gene libraries; Use of marker gens; Applications of genetic engineering in agriculture, medicine and environment.

Suggested Readings

1. Watson, J.D., Tooze, J, and Kurtz, D.T. 1983. Recombinant DNA a Short Course, Scientific American Books, New Ypork 2. Marx, J.L 1989. A Revolution in Biotechnology, Cambridge university Press 3. Old, Principles of Gene Manipulation and Introduction to Genetic Engineering, 3rd Edition.

BTE 302 Microbial Biotechnology 3 credits

Historical development from ancient fermentation to modern microbial technology, scope and essential features of applied microbiology and Biotechnology; Microorganisms of industrial importance- yeasts, molds, Bacteria and Actinomycetes; Screenung and selection of microorganisms for useful products, genetic manipulation of microorganisms for increased productivity; Kinetics of microbial growth and product formation; Microbiological production of foods, drinks, industrial chemicals and pharmaceuticals; Detail of some fermentation processes- SCP, baker’s yeast, alcohol, organic acids- vinegar, acetic acid, citric acid, amino-acid, and enzymes; Immobilization of microbial cells and enzymes; Immobilized technology for production of industrially important chemicals; Role of microbes in dairy products formation; Indigenous fermented foods

Suggested Readings

1. Smith, J.E. Biotechnology Principles 2. Reed. G. Prescott and Dunn's Industrial Microbiology 3. Bu Lock, J. 1987 Basic Biotechnology 4. Primrose, Modern Biotechnology. 5. Fogerty, 1983. Microbial Enzymes and Biotechnology 6. Smith, J.E. Biotechnology Principles 7. Higgins, D.J., Best and Jones, J. Biotechnology- Principles and Application 8. Brown, C.M., Campbell, I. And Priest, F.G. Introduction to Biotechnology 9. Laskin, 1985. Enzymes and immobilized cells in biotechnology 10. Davis, 1976. Single Cell protein.

BTE 303 Basic Immunology 3 credits

Molecular and cellular basis of immune system; Self verses non-self recognition; antigen recognition and the clonal selection theory of the immune response; The humoral immune response; antibodies- structure, diversity function and mechanism of action; The cell mediated immune response; recognition of antigen by T cells; antigen presentation; the major histocompatibility proteins; the role of cytokines and the regulation of immune responses The immune system in health and disease – response to infection, development of vaccines; the development of allergies; auto-immune disease; and immunodeficiency; transplantation immunology; manipulation of the immune system; Complements: Activities of complement proteins, activation of complement, classical pathway, regulation of classical pathway activation, alternative pathway activation and amplification loop, their regulation, membrane attack complex, biological effects of complement; The techniques of immunology and their application: Precipitation, agglutination, simple immunodiffusion, double immunodiffusion, immunoelectrophoresis; two-dimensional and counter immunoelectrophoresis, immunofluorescence, complement fixation, RIA, ELISA, Immunoblotting and immunoprecipitation.

Suggested Readings

1. Jains Kuby, Immunology., W.H. Freeman & Co. N.Y. 2. Alberts, B. Bray, D. Lewis, J., 1989. Molecular Biology of the Cell. Garland Publishing, Inc. New York. 3. Roitt, Brostoff, Nale, 1999. Immunology. 4. Roitt, 1998. Essential Immunology.

BTE 304 Environmental Biotechnology 3 credits

Microbial World; Microbial Metabolism and Growth; Role of Microorganisms in Biogeochemical Environment; Environmental effects of Biogeochemical Cycles; Biodegradation of recalcitrant industrial wastes- Xenobiotic chemicals in the environment; biodegradable and recalcitrant wastes; structure-recalcitrance relationship; Factors affecting biodegradation of xenobiotics; Hazardous waste treatment technologies- solid waste- Landfils and Composting; Liquid wastes- Primary, secondary and tertiary treatment processes; Genetic manipulation, enzymes and specialized bacteria; In situ analysis of microbial community and activity in bioremediation, DNA based method and RNA based methods Biodegradabilty testing and monitoringof the bioremediation of xenobiotic pollutants; Environmental manipulation for increased biodegradation; Bioengineering; Biological control of pests insects and pests- Biopesticides of microbial origin, Bacillus thuringiensis, Bacillus sphericus,etc.; Pathogens and Parasites in Water and W aste Water; Indicator Organisms and Disinfections;Water treatment; Use of commercial blend of microorganisms and enzymes in waste water treatment; Potential applications of recombinant DNA technology in waste treatment Waste Water Treatment; Stoichiometry and Kinetics; Biological Nutrient removal/Anaerobic Process. Biosensors- Recent development in biosensors in environment pollution detection

Suggested Readings

1. Environmental Microbiology- Maier, Pepper and Gerba, a Academic Press, 2000 2. Waste Water Microbiology, 2nd Edition- Gabriel Biton, a John Wiley and Sons, 1999,Biological Waste Water Treatment 2nd Edition- Grady 3. Atlas and Bartha, Microbial Ecology 4. Klug and Reddy, Current Perspectives in Environmental ecology 5.

Wise, D.L. Biotreatment Systems Volume II 6. Crawford. R.L. Crawford, D.L. Bioremediation Principles and applications 7. Waste Water Microbiology, 2nd Edition, Wiley GB 8. Maier, R.M., Pepper I.L.and Gerba, C.P., Environmental Microbiology

BTE 306 Industrial Biotechnology 3 credits

Inoculum preparation and development for industrial fermentation; Fermenter/bioreactor design- type, configuration, mixing and aeration, power input, impeller design, baffle and aeration and agitation; Fermentation kinetics- rate equations for cell growth, substrate utilization and product formation; Batch, fed batch, continuous culture of microorganisms; Upstream processing- media preparation, inoculum preparation, sterilization of bioreactor and liquid media, scale up problem; Monitoring of growth and product formation; Methods of process control- variable s-temperature, pH, Dissolved oxygen tension, pressure measurement and control, safety valves rate of stirring, foam sensing, control of dissolved oxygen and related sensors; Downstream processing- product separation and unit operations involved in industrial fermentation processes .

Suggested Readings

1. Mcneil, B. and Harvey Fermentation: A Practical Approach 2. Standbury , P.F. and Whitaker Principles of Fermentation technology 3. Mcneil, B. Harvey. Fermantation- a Practical approach 4. Standbury, P.F. and Whitaker, A. Principles of fermentation technology

BTE 307 Advanced Molecular Biology 3 credits

Chromosome structure: Basic chemical aspects – DNA, histones and non-histones; Basic structural aspects – the nucleosomes, euchromatin and heterochromatin; Organization of the genome in Eukaryotes: Gene and gene number; C-value paradox; Organization of replication; Gene amplification, Chromosomal redundency, Repetitive DNA and its relevance to plants and animals, inverted and tandem repeats; Regulation of gene expression: Transcription – multiple RNA polymerases, sigma like factors in eukaryotes;

Heterogeneous nuclear RNA; Messenger RNA -- structure and complexity; Interrupted genes and RNA splicing; Expression of specific genes; Genes for ribosomal RNA; Histone genes; Globin genes; Heat-shock genes; Leghaemoglobin genes; genes for storage proteins of legumes and cereals; possible role of middle middle repetitive DNA in control of gene expression; Britten –Davidson model; Regulation of gene expression in prokaryotes; Lac and Trp operon, inducible and repressible systems; positive and negative control; Brief introduction to the complexity of eukaryotic genetics; Mutation and repair of DNA: Type of mutaion; Spontaneous and Induced mutation; Physical and chemical mutagens; Molecular basis of mutaion; in vitro mutagenesis, site-directed mutagenesis; transposons and insertional elements; Repair mechanism in mutation, mutation rate and its measurement.

Suggested Readings

1. Lewin B, Genes VII, Oxford University Press, Sixth Edition 2. Darn ell, J., Lodesh, H. and Baltimore, D. 1986. Molecular Cell Biology., W.H. Freeman and Company, New York. 3. Alberts, B. Bray, D. Lewis, J., 1989. Molecular Biology of the Cell. Garland Publishing, Inc. New York. 4. Wolfe, S.L., Molecular and Cellular Biology, Wardsworth, Belmont, CA.

BTE 308 Plant Biotechnology and Genetic engineering 3 credits

Introductory history: Laboratory organization; Sterile techniques; Nutrition of plant cells; Media composition – solid and liquid; Tissue and organ culture; Establishment and maintenance of callus and suspension cultures; Cellular differentiation and regulation of morphogenesis; Somatic embryogenesis; Control of organogenesis and embryogenesis; Single cell methods; Cytology of callus, Tissue culture & genetic engineering; Tissue culture techniques: Haploid production - Androgenesis; Anther and microspore culture; Gynogenesis; Embryo culture and rescue in agricultural and horticultural corps; In vitro pollination and fertilization; Protoplast isolation; Culture – regeneration; Somatic hybrid-cybrids; Somatic embryogenesis and artificial seeds; In vitro selection of mutants –mutants for salts, disease, cold, drought, herbicide and other stress conditions; Plant micropropagation: Application of micropropagation in forestry and historical crops, Micrografting – in vitro clonal multiplication – Meristem culture and virus elimination; Shoot tip culture. Cryopreservation and germplasm conservation, In vitro conservation. Tissue culture applications: Improved crop varieties through somaclonal variation in in vitro cultures -- Causes- stability and utilization – genetic and epigenetic basis; Establishment of cell lines and evaluation; Secondary metabolite in cell culture; Application of tissue culture for crop improvement in agriculture, horticulture and forestry; Genetic engineering of plants: Methodology; Plant transformation with Ti plasmid of Agrobacterium tumifacians; Ti plasmid derived vector systems; Physical methods of transferring genes to plants – Microprojectile bombardment, Electroporation; Use of reporter genes in transformed plant cells; Manipulation of gene expression in plants; Production of marker free transgenic plants; Application of plant genetic engineering: Developing insect-resistance, disease-resistance and herbicide resistance in plants. Developing stress and senescence-tolerance in plants – oxidative, salt and submergance stress, fruit ripening. Genetic manipulation of flower pigmentation. Developing quality of seed storage. Provitamin A, iron proteins in rice. Modification of food plant taste and appearance, yield increase inplants, Wild plant relatives as a source of novel genes, Plants as bioreactor – antibodies, polymers, foreign proteins in seeds.

Suggested Readings

1. Walton, P.D., Principles and Practices in Plant Science, Prentice Hall 1988. 2. Bhowjwani, S.S., Plant Tissue Culture: Application and Limitations 1990. 3. Dixon, 1994. Plant Cell Culture: A practical approach. 4. Gresshoff, P.M. Plant Biotechnology and Delelopoment., SRC Series of Current Topics in Plant molecular Biology. 5. Anderson, L.A., Plant Cell Culture. Advances in Biochemical Engineering and 6. Biotechnology. 7. Watson, 1992. Recombinant DNA. 8. Portykus, 1995. Gene transfer to Plants. 9. 8 .Mantell and Smith, 1984. Plant Biotechnology. 10. Kosuge, 1983. Genetic Engineering

of Plants.

BTE 401 Bioinformatics 3 credits

Prokaryotic and eukaryotic genomes, structure, organization and function; Molecular evolution, Gene structure, genetic code and mutation; Biological databases- Primary sources of sequence and structure data; secondary data bases; Sequence analysis for Molecular biology- primer selection, restriction mapping, protein sequence analysis; Sequence alignment- Scoring matrices- PAM and BLOSUM- Local and Global alignment concepts- dynamic programming methodology; Needlham Wunsch algorithm, Smith-Waterman algorithm; Statistics of alignment score; Multiple sequence alignment; Progressive alignment; Heuristic methods for data base searching- BLAST and FASTA; Phylogenic analysis- Evolutionery chnges in Nucleotide sequences,; Rates and pattern of nucleotide substirution; Methods for phylogenic estimation- Maximum parsimony, Distance Matrix Methods and Maximum Likelihood Methods; Functional Perl Programming for bioinformatics; String processing; Regular Expressions; Object oriented programming in Perl.

Suggested Readings

1. Baxevanis, A.D. Quellette, B.F.F. Bioinformatics 2. Mount, D.W. Bioinformatics: Sequence and Genome analysis 3. Gaur, D. Li, W-H. Fundamentals of Molecular evolution 4. Tisdall,J.D. Mastering Perl foe Bioinformatics 5. Claverie, J.M Notredamame, C. 2003. Bioinformatics for Dummies

BTE 402 GMOs and Biosafety 3 credits

Topics of concern related to the environmental release of genetically modified organisms (GMOs); Risk for animal or human health – toxicity and food quality/safety, allergies, Pathogen drug resistance (antibiotic resistance); Risk for agriculture – weeds or superweeds, alteration of nutritional value (attractiveness of the organism to the pests), reduction of cultivars (increase of susceptibility) and loss of biodiversity; Risk of pollution with non-target organism – genetic pollution through pollen or seed disposal, horizontal gene transfer (transgene or promoter dispersion), transfer of foreign gene to microorganisms (DNA uptake), generation of new live viruses by recombination; Risk for the environment – Persistancy of gene or transgene or transgene products, resistance/tolerance of target organism or susceptibility of non-target organisms, increased use of chemicals in agriculture, unpredictable gene expression or transgene instability; General concerns – loss of familiarity, higher cost of agriculture, field trials not planned for risk assessment, ethical issues (labelling).; Genetically modified foods -- Benefits and Risks, Regulations and public acceptance Biosafety regulations to protect nature, growers and consumers interest and national interest.

Suggested Readings

1. Maurizio G. Paoletti and David Pimentel, Genetic Engineering in Agriculture and the Environemnt: Assessing risk and benefits. 2. Rissler, J. and Mellon, M., 1996. The Ecological Risks of Engineered Crops, Cambridge, USA: The MIT Press.

BTE 403 DNA Fingerprinting and Molecular Diagnostics 3 credits

Introduction; Basic genetic principle; Variable Number of Tandem Repeats (VNTRs)/ Minisattellite sequences; Short Tandem Repeats (STRs)/ Microsattellite sequences; Hybridization based DNA fingerprinting (RFLP)- Radioactive method, Fluorescent method, Chemiluminescent method; PCR-based DNA fingerprinting; Single locus and multi-locus DNA fingerprinting; Isolation of DNA from whole blood, soft tissues, semen stains and swabs, bones, plant material; Polymorphism: Polymorphism of some genetic locus in relation to disease (HLA, Apo and ACE gene);Applications of DNA fingerprinting; Criminal investigation (personal identification); Immigration; Paternity dispute; Identification of missing children, bodies found in plane crash, road accidents etc.; Varietal identification of plants Molecular Dignostics-Diagnosis of Cystic fibrosis by multiplex PCR; Clinical implications: Abnormal mucus clearance from the respiratory tract with frequent infections, pancreatic insufficiency, abnormal salt transport, infertility in males.; Detection of b-Thalassemia mutation using ARMS-PCR; Clinical implications: Anemia (red cell deficiency); Detection of Fragile X syndrome by FMR-1 gene trinucleotide repeat analysis; Clinical implications: Mental retardation, long faces, large ear, prominent jaws, post-pubertal macroorchidism.; Detection of Philadelphia chromosome [BCL-ABL t(9:22) translocation] by genomic southern hybridization; Clinical implications: Acute leukemia (ALL) and Chronic myelogenous leukemia(CML); Bone marrow engraftment: DNA analysis to distinguish patient and donor cells as different using hypervariable tandem repeat polymorphic DNA markers; Identification of bacterial species based on the sequences of their 16S ribosomal RNA genes; DNA Microarrays/ DNA Chips/Gene Chips- Basic concept; Design of a DNA Microarray; Applications of DNA Microarray technology : Disease diagnosis, Drug discovery etc.

Suggested Readings

1. Micklos, Davod A. and Freyer , Greg A. 1990. DNA Science, Cold Spring harbor Laboratory Press and Carolina Biological Supply company.

BTE 404 Bioprocess Technology 3 credits

Development of bioprocess technology- Methods of biocatalysis and biotransformations, Concepts and general features of biotransformations- Procedures, techniques and media for biotransformations; Reaction in solvent mixtures; Equipment, automation, standardization, quality control and quality assurance; Optimization procedures; Examples of bioconversion processes; Scale up of microbial processes- Criteria used for scale up; Important factors for development of microbial processes- physical, chemical process and sterilization factors; Processing of Food and Feed- Raw materials, microorganisms, processing; Food value and economic importance of fermented foods-idli, ogi, dahi, tofu, tempeh; Microbial upgradation of poultry and animal feed; Biocomposting processes- succession of microorganisms, bioelemental changes, Applications; Single step microbial processes of industrial importance- Steroid biotransformation, vinegar production etc.

Suggested Readings

1. Rehm, H-J. and Reed G. Biotechnology 2nd Edition. Volume 3. Bioprocessing 2. Domain, A.L. and Davies, J.E. Manual of Industrial Microbiology and Biotechnology 3. Steinkraus, K.H. Handbook of Indigenous Fermented Foods 4. Chahal, D.S. Food, Feed and Fuel from Biomass 5. Mizrahi, A. and Wezel, A.L. Advances in Biotechnological Processes

c. Elective Courses (12 credits)

BTE 309 Food Biotechnology 3 credits

Long chain of foods- from farm to fork; Biotechnology for improved food production with nutritional qualities; Impact of biotechnology on food production and protection; Food sources – Grains, animals, fish, vegetables, fruits and derived products Plants and Food biotechnology; Animals and food Biotechnology; Handling, storage, distribution and post harvest technologies for food preservation; High temperature, low temperature, and other physical methods of food reservation; Additives and chemical preservatives; Microbiological methods for food preservation, Quality control and quality assurance in food and beverage industries- GMP, GHP for QC and QA; Effects of processing technologies on physical, chemical, microbiological and nutritional qualities of foods; Improvement of food quality through application of modern biotechnology; Production of fermented foods- bakery products, dairy products, cheese, yugurt; Fermented vegetable products- cabbage, carrot, cucumbers; Lactic acid fermentation as a preservetion strategy for foods; Oriental fermented foods.

Suggested Readings

12. James M.J. Modern Food Microbiology 13. WHO Publication. 2005 Modern Food Biotechnology, human health and development

BTE 310 Medical Biotechnology 3 credits

Brief resume of Biochemistry and Cell Biology, Human Physiology; DNA based diagnostics for the identification of both genetic and infectious diseases; Genetic Counseling of patients at risk of acquiring genetic diseases and would be parents who are at a risk of giving birth to babies with genetic defects; Genome sequencing on disease causing microorganisms including viruses for better therapy and prognosis through appropriate drug targets and vaccines; Efficacy, product cost effectiveness of molecular medicines; Biotechnology for production pharmaceutical products including medicinal plants, vaccines, laboratory kits and tools; Stem Cell research and applications, Ethical issues in medical biotechnology

Suggested Readings

1. Albert Sasson, Medical Biotechnology: Achievements, Prospects And Perception, United Nations Pubns (November 30, 2005) 2. Bruce Alberts, Alexander Johnson, Julian Lewis, Martin Raff, Keith Roberts, Peter Walter Molecular Biology of the Cell, Fourth Edition, Garland Publishing; 4th edition (March, 2002) 3. Ian Dunham Genome Mapping and Sequencing, Horizon Scientific Pr (September, 2003)

BTE 311 Aquaculture and Marine Biotechnology 3 credits

Biotechnology in aquaculture- Fish, mollusks, crustaceans and aquatic plants; intervention in rearing process for enhancing growth, development and nutrition,; Improving health and well-being; Improving quality and value; Reproductive potential, disease resistance and ability to endure adverse environmental conditions; Fish health and nutrition;; Conservation and stock management; Diagnostics and disease control; Microbiology in aquaculture; Vaccine development against disease agents and parasites; Microinjection of growth hormone for accelerated growth; Transgenic fish; Use of DNA fingerprinting techniques for bloodstock management; DNA markers for analysis of genetic variation of species; conservation and stock management Novel marine organisms for natural products-Pharmaceuticals, Industrially important enzymes, biomonitors, biopesticides and other compounds with unique properties from marine algae, corals, sponges and tunicates; Seaweeds with antibacterial, anticoagulant, antithrombic properties for human use; Natural biosensors of marine organisms; Microbial consortia/symbiosis Genes with unique properties like salt tolerance, freeze tolerance and other stress tolerance from marine fishes; Aquatic bioremediation; Biofilms/biofouling ; Marine microbial and extremophile processes; Harmful algae; Genetic and ecological impacts of aquatic and marine biotechnology-Transgenic fish with modified traits;; Genetic impact of escaped organisms Inbreeding between transgenic and nontransgenic fish; Environmental safety issues.

Suggested Readings

1. Recent Advances in Marine Biotechnology: Aquaculture-Fishes (Recent Advances in Marine Biotechnology) by Milton Fingerman (Editor), Rachakonda Nagabhushanam (Editor) Publisher: Science Publishers Inc (February, 2000) 2. Biotechnology in aquaculture by Edward M Donaldson U.S. Congress, Office of Technology Assessment (1994) 3. Biotechnology and Genetics in Fisheries and Aquaculture (Hardcover) by A. R. Beaumont, K. Hoare, Blackwell Publishers (April 30, 2003)

BTE 312 Computer Applications in Biotechnology 3 credits

Introduction- Types of Computers(Main, Mini, Micro0- Hardware, Software- Storage and Memory devices(RAM, ROM, Discks, Tapes etc.)I/O devices (Monitors, keyboards, Printers, Plotters etc.) Directory concepts- File operations; Operative system- Overview- Booting- DOS Files, DOS internal and external commands- file management(creating, editing, deleting and copying); Word Processing- Introduction to textfiles- Word Processors- Word Star package- File operation and utilities under WS.; Data Concep Database structure and dscription- Bibliographic and non-bibliographic- Biotechnology databases(Medlin, Agric, EMBL, Genebank, NBRF) Information Retrieval- Information Sources/ Search aids9 Directories, Dictioneries, Glossary; Search operations(Boolian, Keyword, String, sequences); Information access- Online, CD-ROM, NICNET, E-mail Software- Demo of some PC based Educational Software packages in Genetic Engineering; Presentation of results in text/graphic mode- Harvard Graphics package;

Staistical Concepts- Mean, Mode, Median, SD- Probabilty Correlation, Trends, Theorems.

Permutation and combination- distributions- Line, bar, pie, sigma plot and Harvard Graphics software package; Analysis of Biotechnological data in Agro base 4 and preparation of similarity indices, dendograms using NT SYS; Analysis of robustness of clusters using WINBOOT; DNA sequence data management; Database and homology search of DNA and protein sequences; Labeling amtorads and Polaroid photographs using Microsoft Excel and presentation of nucleic acid sequences using power point; Scanning od DNA, Protein and Isozyme profiles using computers and scanners; Retrieving information using INTERNET and CD-ROM.

Suggested Readings

1. Lesk, M. 2002 Introduction to Bioinformatics 2. Brown, T.A. Gene Cloning and DNA Analysis 3. Rehm H-J and Reed, G Genomics and Bioinformatics. 2nd Edition. Vol 5b 4. Grfith H.G. Griffin, A.M. DNA Sequencin Protocols

BTE 313 Biomass and Biofuels 3 credits

Importance of fuel energy; Fuels from nature, Basic bioenergy interconversion; Formation of biomass and its conversion to fuel; Potential biomass for fuel p- Land crops, aquatic plants and waste materials; Production of desirable biomass; Advantages and problems in utilization of biomass for fuel production; Pretreatment of biomass for fuel production; Bioconversion of biomass to methane- synthesis of methane under natural conditions; Potential methanogenic microorganisms; Methane from sanitary landfills, sewage, farm, industrial wastes and energy crops; Bioreactor design; Utilization of methane as fuel; Fuel ethanol (gasohol) from biomass- Microorganisms for ethanol production; Production of ethanol from agroindustrial waste, associated fermentation technology; Ethanol fermentation from molasses by yeast; Ethanol fermentation by rumen bacteria; Ethanol production by Zymomonas mobilis; Cost-economics of industrial ethanol production from various raw materials; Production of hydrogen from biomass- Potential substrates and microorganisms; Biosynthesis of hydrogen under natural habitats; Cell free system and combined system for hydrogen production.

Suggested Readings

1. Chahal. D.S. Food, Feed and Fuel from Biomass 2. Higgins, I.J. Best D.J. and Jones, J. Biotechnology- Principles and applications

BTE 314 Livestock and Fisheries Biotechnology 3 credits

Livestock breeding-Techniques of modern biology such as molecular cloning of genes, gene transfer, genetic manipulation of animal; Artificial insemination; Super ovulation; embryo transfer Technology (ETT) for improved breed of livestock; Genetic manipulation of rumen microbes; Chemical and biological treatment of low quality animal feeds for improved nutritive value; Genetically engineered immunodiagnostic and immunoprophylactic agents as well as veterinary vaccines; New and emerging animal vaccines and new ways of using vaccines; Biotechnology for unprecedented opportunities for increasing animal productivity and for protecting the environment through reduced use of agro-chemicals; New animal products through biotechnology-new foods and food ingredients; Transgenic animals for adding extra value to food; Biotechnology options for livestock improvement in developing countries.

Suggested Readings

1. Murray Moo-Young, 1989. Animal Biotechnology: Comprehensive Biotechnology. Pergmon Press 2. Fao. 1989 Biotechnology for Livestock Production Kluwer Academic Publishers

BTE 315 Bioremediation and Biodeterioration 3 credits

Biodeterioration of materials- Basic concepts, factors involved in biodeterioration; Biodeterioration of leather, wool, fur, feather, stones, plastics and rubber; Microbial production of biodegradable plastics; Control of biodeterioration- physical, chemical and biological methods; Biodegradation of recalcitrant industrial wastes- xenobiotics in the environment; Biodegradable, persistent and recalcitrant wastes; Factors affecting microbial degradation of xenobiotics; Enrichment and isolation of microorganisms for biodegradation; Hazardous waste treatment technologies- physical, chemical and biological treatment; activated sludge and advanced treatment; Biological removal of nitrogen and phosphorus; Biotechnology of effluent treatment- genetic manipulation, enzymes and specialized bacteria, biodegradability testing, monitoring of the bioremediation of xenobiotic pollutants; Biosensors-use and application of biosensors for detection of pollutants; Biological control of insects and pests- biopesticides of microbial origin, viral, bacterial, protozoal and fungal pesticides; Water treat system- Coagulation and Flocculation, sedimentation; filtration; Disinfections using ozone, uv and activated carbon; Measurement of treatment efficiency; Advances in biochemical, serological and molecular techniques for detection of indicators and pathogens in surface, ground and potable water; Pollution control biotechnology- Production of microbial seeds; Use of bioaugmentation in waste treatment; Use of enzymes and immobilized cells; Removal of metals by microorganisms.

Suggested Readings

1. Atlas and Bartha. Microbial Ecology 3. Klug and reddy. Current Perspectives in Microbial ecology 4. Bitton, G. Waste Water Microbiology 5. Wise. Biotreatment System. Vol. II

BTE 405 Advanced Immunology 3 credits

Immunological tolerance- mechanisms of tolerance; thymic tolerance to self antigens; B cell tolerance artificially induced tolerance; Prophylaxis- antigens used as vaccines; Effectiveness and safety of vaccines; Modern approaches to vaccine development; Immunodeficiency- Primary immunodeficiencies; deficiencies of innate immunity; Primary b cell deficiency; Primary T cell deficiency; Combined immunodeficiency;Secondary immunodeficiency; Hypersensitivity- Hypersensitivity Type I, type II , Type III and Type IV reactions; Transplantation- Barriers of transplantation; Law of transplantation; Role of T Lymphocytes in rejection; Prevention of rejection; Tumour immunology- Surface markers of cancer cells; Lymphoproliferative disorders due to tumours; Cancer immunotherapy; Autoimmunity and auto immune diseases- Association of autoimmunity with diseases; Genetic factors in pathogenesis; aetiology and treatment of autoimmune diseases; Diagnostic and prognostic value of autoimmune diseases.

Suggested Readings

1. Roitt, I.M. and others. Immunology 2. Roitt and others. Essential Immunology 3. Male, D.K and others. Advanced Immunology 4. Barrett, T.J. Advanced Immunology 5. Tizard, I.R. Immunology- An Introduction

BTE 406. Genomics and Proteomics 3 credits

The structure, function and evolution of the human genome; Introduction to human genome project; Physical mapping of Genome sequence-acquisition and analysis; Strategies for large-scale sequencing projects ; Evolution and genome; Genomic identification; Genomic variation;Comparative homologies, evolutionary changes and Single Nucleotide Polymorphisms; Expression and analysis of expressed genes; Epigenetics-Control of gene expression and it affects proteins that are present in cell at any one time SNPs and pharmacogenomics; Human disease gene expression; Comparative genomics; Bioinformatics for the analysis of sequence data; approaches for determining gene expression patterns and functions; Post genomics. Proteomics- Basic issues and concepts;Protein structure, secondary structure and super-secondary structure. Mechanisms of protein folding, tertiary folds. Formation of oligomers; Protein sequence and alignment; Relationship between protein structure and function; ; Computer aided identification of antigen region; Important points in drug design Prions. Structure prediction and human proteomics. Mutant proteins. Use of computer simulations and knowledge-based methods in the design process. De-novo design; making use of databases of sequence and structure. Protein structure and drug discovery, Proteins in disease; Protein sequence alignment and data base searching; Public dimain resources in Biology; Sequence comparison- BLAST and beyond;

Suggested Readings 1. Campbell, M.A. and Heyer, L.J. 2002. Discovering Genomics, Proteomics and Bioinformatics , Benjamin Cummings, 2002. 2. Lesk, Arthur M, Introduction to Informatics, 2nd Edition, Oxford University Press, 2005 3. Lesk, Arthur M, Introduction to Protein Science, 2005.Oxford University Press 4. Fersht, A. Structure and Mechanism in Protein Science, W. H. Freeman (1999). 5. Carey P. R. (Ed.) Protein engineering and design, Academic Press (1996). 6. Strachan T. and Read A. P. Human Molecular Genetics, 2nd edition. Bios (1999) 7. Glick, Bernard R. and Pasternak J. J., Molecular Biotechnology: principles and applications of recombinant DNA, 2nd ed. ASM Press (1998)

BTE 407 Waste Biotechnology 3 credits

Waste water flows and characteristics; Basic waste water treatment processes-Preliminary and primary treatment, Secondary (Biological) treatment; tertiary and advanced treatment; Role of microorganisms in biogeochemical cycles for wastewater treatment; Microbial oxygen demand: self-purification; Biochemical oxygen demand; Microbial growth kinetics and its application to treatment processes; Biological Processes- Activated sludge; Fixed film reactors; Trickling filters; Rotating fixed film reactors; Stabilization ponds; Anaerobic unit processes; Other biological processes; sludge treatment and disposal; Public health- waterborne diseases; INDICATOR ORGANISMS; removal of pathogenic organisms; Biotechnology for waste water treatment and waste utilization; Special topics in waste btreatment; Visit to waste treatment plant(s).

Suggested Readings

1. Bitton, G. 1994. Wastewater Microbiology. Wiley-Liss Inc., New York 2.Gray, N.F. 1989. Biology of Wastewater Treatment. Oxford University Press, New York

BTE 408 Advanced Agricultural Biotechnology 3 credits

Mendelian genetics; Plant domestication; Traditional methods of crop improvement; Plant tissue culture; Transformation;Transgene analysis and expression; Insect resistance; abiotic stress; herbicide tolerance; virus resistance; value-added traits;; DNA technology; Genetic mapping; DNA fingerprinting; Marker assisted selection; similarities and differences between methods for traditional and non-traditional crop improvement; the techniques required for the development of improved cultivars through biotechnology; hybrid seed technology; hoticulture biotechnology- commercial production of horticultural, ornamental, timber and medicinal plants; Livestock and fisheries biotechnology; Transgenic animals and cloning; Immunotherapeutic drugs-edible vaccines, antibodies, interferon from plants; benefits and risks associated with biotechnology; the current impact of biotechnology on society;. bioethics related to agricultural biotechnology.

Suggested Readings

1. Dubey, R.C. 2006. A Text Book of Biotechnology. S. Chand and Company Ltd., New Delhi. 2. Slater, A. Plant Biotechnology: the Genetic Manipulation of Plants. Oxford University Press

BTE 409 Advanced Bioinformatics 3 credits

Genomics; Known Genomes; Genome variability and disease; Genome sequencing; Signal processing methods in sequence analysis; Fragment assembly- gene assembly; Gene identification; Machine learning methods: Hidden Markov model; Neural Networks and support vector machines; Genome annotation- Feature identification, annotation pipelines, data verification and gene ontology; Gene expression in prokaryotes and eukaryoutes; Methods of global expression analysis; Genome comparison- Suffix trees for whole genome sequence comparison; Bioinformatics and analysis of biological pathways, metabolic pathways and ecosystems.

Suggested readings

1. Durbin, R. et al. Biological sequence analysis: Probabilistic Models for Protein and Nucleic acid 2. Baxevanivanis et al., 2001.Bioinformatics: A Practical Guide to the Analysis of Genes and Proteins

BTE 410 Human Genome Project 3 credits

An overview of Human Genome Project-Brief history; From Genome to Proteome; Genome;sequencing; Employment of Restriction fragment length polymorphism; Yeast artificial Chromosome; Bacterial Artificial Chromosome; Bacterial and yeast genome; Parasite genome; Scientific and technical aspects of genome sequencing; Genome resource data bases; Maps and map construction; Access to data generated by genome projects and interpretation and application of genomic data; Gene hunting and comparative genomics; Newly emerging insights into evolution and evolutionary relationship between species; Scale and scope of HGP; Potential for new developments in medicine raised by HGP; Biomarkers for monitoring genetic diseases; Contribution of environmental and genetic susceptibility to the causes of cancer and cardiovascular diseases; Molecular epidemiological techniques to understand disease mechanisms at the molecular level’

Suggested Readings 1. Joel L. Davis Wiley Mapping the Code : The Human Genome Project and the Choices of Modern Science (Wiley Science Editions); 1 edition (February 27, 1991) 2. Walter Bodmer, Robin McKie, The Book of Man: The Human Genome Project and the Quest to Discover Our Genetic Heritage; Oxford University Press; Reprint edition 1997 3. Wiley-Interscience Genomics: The Science and Technology Behind the Human Genome Project; 1 edition (February 2, 1999)

BTE 411 Virology and Oncology 3 credits

Introduction to Virology- Brief history and development of virology, nature of virions-morphology, physical and chemical properties; replication; Pathogenesis of viral diseases; Bacteriophages- Overview, genome organization and multiplication; Prevention and treatment of viral diseases; Virions and Prions; Persistence of viruses-Patterns of viral infection; Mechanism of viral persistence; Persistence of HSV, EPV and HIV in humans; Viruses of special interest- Dengu and Japanese encephalitis virus; Ebola virus infection; Severe Acute Respiratory Syndroem(SARS) and SARS corponavirus; Other important viruses of recent epidemics; Virus evolutionnand emerging viruses; Oncology- Introduction and general terminology of oncology; Cancer- Development of cancer; Spread of cancer; Molecular mechanisms of transformation by DNA and RNA viruses; Physical and chemical factors contributing to cancer development; Cancer therapy.

Suggested Readings

1. .Fields. Virology 2. .Fields Principles of Virology 3. .Scientific American

BTE 412 IPR and Biotechnology 3 credits

Juripedential definition and concept of property; rights; duties and responsibilities; History and evolution of IPR- Patent; Design; Copyright; Distinction among various forms of IPR; Requrement of a patentable invention like novelty, inventive step and prior art, state-of-art; Rights/protection, infringement or violation, remedies against infringement- civil and criminal; WTO; TRIPS; Problems in Patenting living organisms; IPR and biodiversity; IPR and indigenous community knowledge; Plant breeders rights; IPR and farmer’s right; IPR as an instrument of piracy.

Suggested Readings

1. Intellectual Property Rights in Agricultural Biotechnology (Biotechnology in Agriculture Series, 28) (Hardcover) by Frederic H. Erbisch (Editor), Karim M. Maredia (Editor) CABI Publishing; 2nd edition (February, 2004) 2. Intellectual Property Rights, Trade and Biodiversity (Paperback) by Graham Dutfield Earthscan Publications (February 1, 2002) 3. Intellectual Property Rights in Biotechnology Worldwide (Hardcover) by Stephen A. Bent, Richard L. Schwaab, David G. Conlin, Donald Jeffrey Stockton Pr (December, 1987) 4. Intellectual Property Rights and the Life Science Industries: A 20th Century History (Globalization and Law) (Hardcover) by Graham Dutfield Ashgate Publishing (August, 2003) 5. Intellectual Property Rights In Frontier Industries: Software And Biotechnology (Paperback) American Enterprise Institute Press (March 30, 2005)

BTE 413 Bioethics 3 credits

Concept of bioethics; Ethical arguments for and against new genetics since early 70s; Historical context of current debate over recombinant DNA technology; Ethical aspects on research on topics that public perceives to be risky or morally objectionable; Concerns arising out of new genetics- Social concerns, privacy, confidentiality, psycological impacts; Ethical consiiderations in plant biotechnology-environmental risks, labeling, humanitarian issues in agricultural biotechnology; Ecosabotage; Animal biotechnology- Ethhical justifiability of using animals for medical experiments, use of genetic engineering to change animals nature to better suite human needs; Health, humanitarian and environmental issues concerning genetically modified foods; Golden rice; Commercial and ethical aspects of gene patenting; Genetic privacy and discriminations; Human genetics-Reproductive issues-informed consent for complex and potentially controversial reproductive procedures; Clinical issues-including education of doctors, patients and public on genetic capabilities, scientific limitations and social risks; Gene testing and gene therapy-conceptual and philosophical implications; Ethical, legal and social issues related to human genome project; Human stem cell research and human cloning-ethical aspects; Genetics in the courtroom.

Suggested Readings

1. Beachamp,T. and Bowie,N. 1989. Principles of Bioethics. Oxford University Press, N.Y. 2. Cummings, M.R. 2003 Human heredity- Principles and Issues, 6th Ed. Books/Cole Publishing Pacific Grove, C.A. 3. Reiss, M.J. Stranghan,R. 1996. Improving Nature. Cambridge University Press, N.Y. 4. Jackson D.A. Stich, S.P. 1997. The Recombinant DNA Debate. Engle wood Cliffs, N.Y.

BTE 414 Global Impact of Biotechnology 3 credits

Global governance of Biotechnology; Technology and Globalization-Linkage between technological advancement and global competitiveness; Technological factors shaping the market rise of biotechnology; Biotechnology and Development- Import og GM foods to Developing counties versus capacity building; Social, politicaland economic conditions in developing countries which influence food security, environment protection and life expectancy; Biotechnology and Institutional changes- Co-evolutionary dynamics in technological innovation and adjustment in existing institutional organization arrangement;; Technological competition- Biotechnology as a generic set of tools that enhances international competitiveness; Intellectual property-WTO, TRIPS and biotechnology;; Evolution of biotechnology industry-Unversity reseach as the basis for star-up business in biotechnology; integration of biotechnological practices to seed and agro-chemical industries; Consolidation of biotechnology activity in an small number of large farms- debate over control over food sectors; Industrial and environmental Biotechnology- Emerging opportunities; Biopharmaceutical technology- Global research and development effort on diseases of temperate region; Emergrnce of new biotechnology especially genomcs; New opportunities for developing countries; Food safety and international trade-debate over introduction of GM foods into the global economy; Precautionary principles and internationl trade; Ecological impact of Biotechnology; ; Science, technology and society.

Suggested Readings

1. Nye,J. Donahue.J. 2000 Governance in Globalizing World. Bookings Institution Press, Washington 2. Grace, E. 1997. Biotechnology Unzipped: Promises and Realities. Joseph Henry Press. Washington 3. Serageldin, I. 1999 Biotechnology and Food security in the 21st Century. In Science, 245, 387-389 4. Nelson, R. 1994. Economic Growth via Coevolution of Technology and institutions. In Leydesdorff, L and van den Basselaar(Eds). Evolutionary Economics and Chaos theory: New Directions in Technology Studies. Printer Publishers, London. 5. National Research Council. 2000. Genetically modified pest protected plants. National Academy Press, Washington

BTE 415 Entrepreneurship in Biotechnology 3 credits

Entrepreneur, Entrepreneurial attitudes, abilities and behaviours, Entrepreneurship: financing, market analysis and business models for commercialization , Social Entrepreneurship, developing an entrepreneurial culture; opportunity recognition and viability screening; first-mover advantages and disadvantages; risk recognition and risk reduction strategies; intellectual property protection, Biotechnology - industry overview and technologies, Industrial R&D and project management, Team work and alliances in R&D, Perspective on the biotechnology industry, Intellectual property – basics and business models, Product development and transition to business units, Suggested Readings

1. Grossmann, Martin Entrepreneurship in Biotechnology:Series: Contributions to Management Science 2003, XVI, 323 p. 95 illus., 2. Eric Grace Biotechnology unzipped: Promises and realities Joseph Henry Press. 1997. 3. Arthur Kornberg The golden helix Sausalito, CA: University Science Books. 1995. 4. Richard Oliver The coming biotech age: The business of biomaterials McGraw Hill. 2000. 5. Ruth Ellen Bulger The ethical dimensions of the biological sciences Cambridge University Press. 1993.

BTE 416 Biochemical Engineering Principles 3 credits

Structure of metabolism- Growth of an undifferentiated organism as a physicochemical process leading to quantification of growth processes; Cells as chemical plants, variation in types of cells and how it affects technological applications; Brief review of important biomolecules of industrial importance; Enzyme kinetics, Kinetics of microbial growth in batch, semi-batch and continuous culture Structure and function of a single cell, structure of metabolic processes; Energy metabolism balance; Small metabolite production, Macromolecule production; Coordination and control of cellular processes; Industrial bioprocesses- A review of bioprocess industries; Selection, screening and maintenance of commercial cultures; Optimization of bioprocesses- Batch and Continuous fermentations; Enzyme technology; Single Cell Protein; Biodereriration and microbial stability; Microbilal dynamics and Energetics; Quantification of biomass and growth processes; Quantification of product formation; Distributed, segregated unstructured and structured growth; Overall energetics of growth processes; The energetics of cell processes and prediction of yields and metabolic heat evolution; Bioreactor design- CSTR design equation; Scale-up issues- Mass tranfer and energy transfer; Sterilization; Recovery and purifucation – unit operation in product recovery; Common techniques extraction, precipitation, adsorption; Recovery and purification- Dialysis; Revesre osmosis; Ultrafiltration; Bioprocess economics (antibiotic and protein).

Suggested Readings

1. Mukhopadhyay, S.N. 2001. Process Biotechnology Fudamentals. Viva Books Private Ltd. 2. Shuler, M.L. and Kargi, F. 1991. Bioprocess Engineering: Basic Concepts, Prentice Hall 3. Bailey, J.E. and Ollis, D.F.1946 Biochemical Engineering Fudamentals, 2nd Edition, Mcgraw-Hill, New York 4. Blanch, H.W. and Clark, D.S. 1996. Biochemical engineering, Marcel Dekker 5. Humphrey and Davis. Biochemical engineering

BTE 417 Enzyme Biotechnology 3 credits

Fundamentals of enzyme kinetics; Mechanism of enzyme action; Effect of ph, temperature and pressure on enzyme action; Enzyme inhibition; Enzyme preparation and use-Plant, animal and microbial sources of enzymes; Screening for novel enzymes; Advantages of microbial production of enzymes; Production, separation and purification techniques for microbial enzymes, Production of immobilized enzymes- Methods of immobilzation; Large scale use of enzymes in solution; Enzymes in detergent industry; Enzymes in food industry; Enzymes in leather and wool industries; Starch hydrolysis; Production of glucose syrup; Lactase in dairy industry; Medical applications of enzymes; Enzyme reactors- Membrane reactor; Continuous flow reactors, Packed bed reactors; Immobilzed enzyme reactors Examples of use in amino acid , lactase antibiotic and other production processes; Enzymes in analysis- Biosensors; Immunosensors; Recent advances in enzyme technology- Enzyme reactions in biphasic liquid systems; Glycosidases used in synthetic reactions; Future prospects for enzyme technology-Enzyme engineering; Artificial enzymes; Coenzyme regenerating systems.

Suggested Readings

1. Chaplin, M. and Christol. 1990. Enzyme Technology. Cambridge University Press, New York. 2. Enzyme Technology (Biotechnology Series) (Paperback) by P. Gacesa, J. Hubble Open Univ Pr (June, 1987) 3. Biocatalysts and Enzyme Technology (Paperback) by Klaus Buchholz, Volker Kasche, Uwe Theo Bornscheuer John Wiley & Sons (April 22, 2005)

d. Lab: (6 credits)

BTE 106 Biotech Lab I

Practicals based non BTE 101, 102, 103 and 104

Use of analytical balance; Preparation of standard solutions; Standardization of acids and bases; Preparation of buffers and determination of pKa of acetic acid; Titration curves for acids and bases; Separation of amino acid mixtures by paper chromatography; estimation of ascorbic acid from biological samples; Use and function of microscopes; Observation of plant and microbial cells; Study of fresh water algae; Preparation of stains, dehydrating agents and clearing agents; Sectioning of roots, stems and leaves; Observation of living bacterial and yeast cells; Staining of bacteria; Cultivation of bacteria- Media preparation and sterilization; Culture technique; Isolation of single colony; Determination of Growth curve.

BTE 205 Biotech Lab II

Practicals based on BTE 201, 202, 203

Determination of Lambda-max and verification of Beer’s Lambert’s law; Qualitative and quantitative estimation of biomolecules- carbohydrates, proteins and lipids; Tests in circulatory system- Total blood cell count, Differential count for WBC; Blood pressure, Blood groups, ESR and hemoglobin; Experiments on enzyme isolation and enzyme assay- Determination of GOT and GPT activity; Activity of salivary amylase; alkaline phosphatase.

Biotech Lab III

Practicals based on BTE 204, 206 and 207

Study on solubility and precipitation - Effect of ionic strength on solubility of proteins; Determination of total globulin in serum by precipitation; Effect of pH and temperature on solubility of proteins; Detyermination of Isoelectric pH; Isolation of milk casein by precipitation and determination of its isoelectric point; Blood sugar under normal and diabetic conditions; Isolation of cholesterol from gallstones; Determination of blood cholesterol; Isolation of DNA and RNA from a samplei, Agarose gel electrophoresis of nucleic acids; Estimation ofquantity and molecular weight of DNA and RNA from a sample; Extraction of chloroplast and mitochondrial DNA from leaf samples;

BTE 208 Biotech Lab IV

Practicals based on BTE 301, 302 303 and 304

Bacterial conjugation and tranformation experimentsIsolation of plasmid DNA from E.coli; Transormation of E.coli with plasmid DNA; Isolation and assay of bacteriophage lambda; Amplification of DNA by polymerase chain reaction(PCR); Production of citric acid by Aspergillus niger; Dough fermentation by baker's yeast for bread making; Yogurt fermentation by lactic acid bacteria; Simple experiment on bioreacor design; Detection of antigen and antibody by immunodiffusion techniques; SDS-PAGE and immunoblotting of bacterial proteind; Complement fixation tests Detection of indicators and pathogenic organisms in potable water; Water purification by chlorination, flocculation and ozonization etc.;

e. Biotechnology Internship

BTE 400 Industrial Attachment 3 credits

Each student individually or in group of three will be sent to the different biotechnology industries in food, agriculture, industry and pharmaceuticals depending on availability of space in the concerned enterprise for a specific period of time every week for 12 weeks. Each student will have to submit a training report duly endorsed by the authority of the respective enterprise for evaluation and grading by the course teacher.

f. Thesis/Project

BTE 450 Biotech Project 3 credits

A student is required to carry out thesis/project work in the 7th and 8th semester in a chosen field decided by the supervisor. The supervisor may be a faculty member of the BRAC University or any other suitable expert from other universities, R& D organizations or biotech industries. On completion of the thesis work he/she will will have to submit a dissertation and face a viva board for the defence

f. Courses Outside Major Specialization

ANT 101 Introduction to Anthropology 3 credits

Humans in nature, human evolution, history of culture, rise of early civilizations in the old and new world, organizations of pre-industrial society environment, resources and their distribution; gender, kinship and descent, religion, economics, politics, survival of indigenous groups, forms of culture and society among contemporary peoples, Comparative study of traditional and changing Third World societies, impact of modern world on traditional societies, power and social order; custom and law, conflict and change, Cultural and ethnic diversity.

Suggested Readings

1. Cultural Anthropology, a Global Perspective: R Scupin 2. Anthropology: The Exploration of Human Diversity: Conrad P Kottak 3. Anthropology: C Ember and M Ember 4. Cultural Anthropology: W A Haviland 5. Anthropology – Social and Cultural: Kedar Nath, Ram Nath 6. An Introduction to Anthropology: Victor Barnouw.

ARC 292 Painting 2 credits

Painting as a form of artistic and architectural expression. Introduction to various media in painting. Still life sketches and painting. Study of forms in painting. Landscapes and cityscapes. Colour pencils, crayons, pastels and watercolour. Mixed media. Computers in painting.

ARC 293 Music Appreciation 2 credits

Musical form. Ingredients of music: sound and time. Indian and Western music: melody and harmony. Foundations of sub-continental music: raga system. Presentation of vocal and instrumental music. Modern Bengali music and works of major composers and demonstrations. Western classical music and works of major composers. Music and its rhythm, composition etc.

BTE 316 Social Impact of Biotechnology 3 credits

Basic concepts of Ecology- Atmosphere, ocean and World Biomass-deserts, grasslands, forests, tundra etc.; Dynamics of ecosystems and Biogeological cycles; Ecological succession; Flow of energy; Communities- Predator-prey interaction; Population Ecology; Environmental Biology and Man- Future of Biosphere; Global climate change- Changes in the atmosphere; Increase of carbon dioxide; Greenhouse effect; Ozone hole; EL Nino and oscillations in world’s climate; Desertification as affected by climate changes; Environment and organismal health- Pollution of air water, soil and their mitigation; Nuclear winter- Biological consequences; Biology and Future of man- New applications of biological sciences towards human welfare; Ethical and environmental issues concerning use of cloned gene in medicinal, agricultural and industrial fields; Effects of GMOs on biodiversity conservation; Inventory of world’s plants; Cryopreservation; Germplasm BankControl of human fertility- Ethical considerations; Invitro fertilization, Eugenics, Guarding the genetic quality of man; Future of Homo sapiens.

Suggested Readings

1.Handler, P. 1970. Biology and the Future of Man. Oxford University Press, U.k. 2.Watson, J.D. Tooze, J. 1941. DNA Story. W.H. Freeman and Company, N.Y. 3. Ricklef, R.E. 1990. Ecology. W.H. Freeman and Company, N.Y.

BTE 317 Biostatistics 3 credits

Definition, uses, limitation and scope; Role statistics in research and experiment; Sampling techniques- Basic statistical principles and terminology- Population and parameters, Samples; Variables; Distribution; Statistical concepts relating to interpretation and decision; Descriptive statistics- Estimation of uncertainties; Calculation of mean, variance and standard deviation; Standard deviation of the mean; Confidence limit of the mean; Test of significance-Measurement of data- t test in paired and non-paired experiments; Selection of apprpriate method for calculating t; Confidence limit of a difference between means; Analysis of variance- Single characterization data; Analysis of variance- Single classification data with subgroups, Multiple classification data; Ducan’s multiple range test; Least significance difference test; Relationship between t and f test; Test of significance- Chi square test-1xn table, 2xn table; Use of chi square with occurrence and non-occurrence data; chi square analysis of 2x2 or four fold table; Alternate methods of calculating chi square; Tests of significance when cell frequencies are small; Statistical methods based on binomials; Relationship between variables- Correlation;;Linear regression; Least square regression line; Test of linearity of a regression; Confidence limit of regression coefficient; Dosage response data- Estimation of 50% end point; Graphical approximation of ED50 value; Estimation of relative frequency.

Suggested Readings

1. Daniel, W.W. Biostatistics- A Foundation for Analysis. John Wiley and Sons 2. Zaman, S.H. Simple Lesson from Biometry 3. Jalil, M.A. Ferdous, . Basic Statistics- Methods and Applications 4. Mian, M.A. Miyan, A. Introduction to Statistics 5. Snedecor, G.W. and Cochran, W.G. Staistical Methods

BUS 101 Introduction to Business 3 credits

Basic principles and practices of contemporary business and its history; Forms of business organization and ownership; Environment of an enterprise ; Organizing and managing the enterprise; Management of: HR, market productions and operations, finance; discuss a broad range of business situations where analysis and decision-making are required. Management tools and information systems;

International and globalization; External environments of business; Future outlooks of business and business ethics. Prerequisites : None

CHE 101 Introduction to Chemistry 3 credits

The course is designed to give an understanding of basics in chemistry. Topics include nature of atoms and molecules; valence and periodic tables, chemical bonds, aliphatic and aromatic hydrocarbons, optical isomerism, chemical reactions.

Suggested Readings

1. Introduction to Modern Inorganic Chemistry: S. Z. Haider 2. Physical Chemistry: Haque & Nawab 3. Organic Chemistry: R. T. Morrison & R. N. Boyd 4. General Chemistry: Raymond Chang

CSE 110: Programming Language I 3 credits

An introduction to the foundations of computation and purpose of mechanised computation. Emphasis placed on techniques of problem analysis and the development of algorithms and programs. Principles of structured programming and corresponding algorithm design. Topics will include data structures, abstraction, recursion, iteration, as well as the design and analysis of basic algorithms. The language used is C or C++. Lecture, discussion, and laboratory. Students will be expected to do homework assignments in problem solving and program design as well as weekly laboratory assignments to reinforce the lecture material. Introduction to digital computers and programming algorithms and flow chart construction. Information representation in digital computers. Writing, debugging and running programs (including file handling) on various digital computers using C. The course includes a compulsory 3 hour laboratory work each week.

Suggested Texts: B. W. Kernighan and D. M. Ritchie, “The C Programming Language”, 2nd ed., Prentice-Hall, 1988.

ENV 103: Elements of Environmental Sciences 3 credits

Fundamental concepts and scope of environmental science, Earth's atmosphere, hydrosphere, lithosphere and biosphere, men and nature, technology and population, ecological concepts and ecosystems, environmental quality and management, agriculture, water resources, fisheries, forestry and wildlife, energy and mineral energy sources; renewable and non renewable resources, environmental degradation; pollution and waste management, environmental impact analysis, remote sensing and environmental monitoring.

Text and Reference Books:1. Goudie, A., "The Nature of the Environment", Blackwell Publishers, UK, 1993.2. Chhokar, Kiran B., Pandya, Mamata and Raghunathan, Meena, "Understanding Environment", Sage Publications, New Delhi , 2004.3. Pallister, John, "Environmental Management", Oxford University Press, UK 4. Morse, Stephen and Stocking Michael (ed.), "People and Environment", UCL Press limited, UK, 1995. 5. Spellman, Frank R., "The Science of Environmental Pollution", Technomic Publishing Company, USA, 1999.6. Manjunath, D.L., "Environmental Studies", 2007.

ECO 103 Principles of Economics 3 credits

A study of the fundamentals of micro and macroeconomics, nature and method of economics, individual markets, demand and supply, elasticity of demand and supply. Production and cost, market structures with special focus on perfect competition and monopoly, economic efficiency and market failure, determination of national income. The aggregate supply model, unemployment, inflation, unemployment-inflation trade-off, government budget and fiscal policy, money creation and monetary policy, business cycles, economic growth, theory of comparative advantage, free trade versus protection, balance of payments and exchange rate policies.

Suggested Readings

Economics: John Solman Principles of Macroeconomics: Robert H Frank Modern Economic Theory: K.K. Dewett International Economics: Appleyard & Field

HUM 101 World Civilization and Culture 3 credits

A brief view of the major civilizations and cultural aspects in different continents covering ancient, medieval and modern civilizations. Topics include renaissance, reformation, and the beginning of the modern world, scientific revolution, industrial revolution, the age of democratic revolutions, nineteenth century Europe, Asia-Pacific Region, Africa, World Wars, South Asia: colonization, decolonization and after; contemporary world: Cold War and after.

Suggested Readings

1. World Civilization: Bums & others 2. Civilization: T Walter Walbank and others 3. A History of World Civilization: J. E. Swain 4. Western Civilization: : Robert E. Lerner & Standish Meachem

HUM 102 Introduction to Philosophy 3 credits

Philosophy: Concept of philosophy; science and philosophy; religion, literature and philosophy; sources of knowledge: empiricism, rationalism and criticism; concepts of value, ethics and sources of ethical standards.

Suggested Readings

1. A Modern Introduction to Philosophy: P Edwards and A. Pap 2. Philosophy: R. J. Hirst 3. Introduction to Modern Philosophy: C.E.M. Joad 4. An Introduction to Philosophical Analysis: J. Hospers 5. An Outline of Philosophy: A. Matin 6. Introduction of Philosophy: T. W. Patrick 7. Living Issues in Philosophy : H.H. Titus

HUM 111 History of Science 3 credits

This course will present a general overview of the development of scientific knowledge from ancient to modern times. It will examine how our modern scientific worldview developed over the ages in the fields of astronomy, physics, biology, chemistry, medicine, geology and other science disciplines. Focus will be on significant discoveries, the major scientists responsible for these revolutions, and the interrelation between science and society over the centuries. The course will contain the following: Science & philosophy, development of science in the ancient times, Greek & Egyptian science, science in the Orient, medieval science, science in the Islamic world, Western renaissance & industrialization, evolutionary theory, science in the modern ages. Science & religion, nature of scientific truth, validation of scientific theories.

Suggested Readings

1. Reader’s Guide to the History of Science: A. Hessenbruch 2. Scientific Laws, Principles, and Theories: a Reference Guide: Robert E Krebs 3. The History of Science: an Annotated Bibliography: G Miller A Guide to the History of Science: a First Guide for the Study of the History of Science, with Introductory Essays on Science and Tradition: G. Sarton 5. Knowledge & the World: Challenges Beyond the Science Wars: M. Cavrier, J. Roggenhofer, G. Kuppers & P. Blanclard 6. The Forgotten Revolution: How Science was Born in 300 BC and Why It Had to be Reborn: Lucio Russo 7. Hitler’s Scientists: Science, War and the Devil’s Fact: J. Cornwell

MGT 211 Principles of Management 3 credits

Meaning and importance of management, evolution of management thoughts; managerial decision making; Environmental impact, corporate social responsibility, planning, setting objectives, implementing plans, organizing; organization design, managing change, directing, motivation, leadership, managing work groups, controlling: principles, process and problems and managers in changing environment.

Suggested Readings

1. Management: Stephen P. Robbins and Mary Coulter 2. Management: James A. F. Stoner, Edward R. Freeman & Daniel R. Gilbert

PHY 101 Introduction to Physics 3 credits

The course is designed to give an understanding of basics in physics. Topics include Vector; Motion; Force; Energy; Pressure; Heat and temperature; Mechanics; Forces; Gravitation; Sound, Light, Electricity and Magnetism; Atomic and Nuclear Physics. Basics of Astronomy. 3 credits, Prerequisites : None

POL 103 Introduction to Political Science 3 Credits

A study of political systems and process with special reference to Bangladesh. Topics include nature and origin of state, sovereignty of state, forms of political units, liberty, law, process of politics, political structure, political ideas- democracy, socialism, nationalism, peoples' behaviour in politics. Political system, process and problems of Bangladesh.

Suggested Readings

1. A History of Political Thought: Plato to Marx: Subrata Mukherjee & Sushila Ramaswamy 2. An Introduction to Political Science: Rand Dyck A Social Political History of Bengal and the Birth of Bangladesh: Kamruddin Ahmed 4. Radical Politics and the Emergence of Bangladesh: Talukder Manizzaman 5. India and Pakistan: A Political Analysis: Hugh Tinker 6. Politics and Policy Making in Developing Countries: Perspective on the New Political Economy: Gerald M. Meier 7. Political Culture, Political Parties and Democratic Transition in Bangladesh: Shamsul I. Khan, S. Aminul Islam & Imdadul Haque 8. Involvement in Bangladesh's Struggle for Freedom: T. Hossain 9. History of Bangladesh 1704-1971: Political History: Sirajul Islam 10. Conflict and Compromise: An Introduction to Political Science: H.R. Winter

PSY 101 Introduction to Psychology 3 Credits

The objective of this course is to provide knowledge about the basic concepts and principles of psychology pertaining to real-life problems. The course will familiarize students with the fundamental process that occur within organism-biological basis of behaviour, perception, motivation, emotion, learning, memory and forgetting and also to the social perspective-social perception and social forces that act upon the individual.

Suggested Readings

1. Introduction to Psychology: C.T. Morgan 2. Introduction to Psychology: R.F. Crider 3. Understanding Psychology: Robert Feldman

SOC 101 Introduction to Sociology 3 credits

Perspectives on society, culture, and social interaction, Topics include community, class, ethnicity, family, sex roles, and deviance. Social problems and sociological problems. Problems, theories, and the nature of sociological explanation. Explanation, evidence and objectivity. Sociology as a comparative study of social action and social systems. Some models of sociological thinking as applied to the study of the following: aspects of social ranking; forms of interpersonal and personal relationships; the changing nature of the relationship between economy and society; the sociology of development; the origins and spread of capitalism and socialism; ideology and belief systems; religion and society; rationality and non-rationality; conformity and deviance.

Suggested Readings

1. Sociology: Anthony Giddens 2. Sociology: Richard T. Schaefer 3. Sociology: Rao and C.N. Shankar 4. Sociology: Neil J. Smelser

SOC 401Gender and Development 3 credits

Position & role of women in society, contemporary issues, analysis of various aspects of gender relations, gender discrimination, societal attitude, different forms of feminism, women in higher education, employment of women & discrimination, workplace harassment, contribution of women in development: world picture I position in Bangladesh. Prerequisite SOC 101

Suggested Readings

1. Women and Social Security: Progress Towards Equality of Treatment, 1990 Geneva International Labor Office: Anne- Marie Brocas, Anne-Marie Cailloux and Virgine Oget. 2. Impact of Women in Development Projects on Women Status and Fertility in Bangladesh, 1993, Dhaka, Development Researchers and Associates: M. Kabir, Rokeya Khatun, Ishrat Ahmed. 3. Integration of Women in Development: Why, When and How: Ester Boseup, Cristine Liljencrantz. 4. Women in the Third World: Gender Issue in Rural and Urban areas: Hants 5. Women, Man and Society: The Sociology of Gender: Allyn and Bacon, Claire M Renzetti, Daniel J Curran. 6. Race, Ethnicity, Gender and Class: the Sociology of Group Conflict and Change, London: Joseph F Healey

STA 201 Elements of Statistics and Probability 3 credits

Frequency distribution, mean, median, mode and other measures of central tendency, standard deviation and other measures of dispersion, moments, skewness and kurtosis, elementary probability theory and discontinuous probability distribution, binomial, Poisson and negative binomial distribution, continuous probability distributions, normal and exponential, characteristics of distributions, hypothesis testing and regression analysis, basic concepts and applications of probability theory and statistics, chi-squared test.

Suggested Readings

1. Probability and Random Processes: G.R. Grimmett and D.R. Stirzaker 2. Elementary Probability Theory with Stochastic Processes: K.L. Chung

S M Rakib-Uz-Zaman joined the Biotechnology Program of the Department of Mathematics and Natural Sciences of BRAC University in September 2017. He completed his M.Sc. degree in Protein Science and Biotechnology under the faculty of Biochemistry and Molecular Medicine from the University of Oulu, Finland and Leibniz Institute on Aging-Fritz Lipmann Institute, Germany with the Erasmus+ Scholarship. He was awarded the Erasmus+ Fellowship to conduct his research work at Leibniz Institute. During his post-gradu study, he had completed theoretical courses from the University of Oulu, Finland and experimental parts from Leibniz Institute on Aging-FLI, Germany. The topic of his master’s thesis was “A Novel Method to Quantify DNA G-Quadruplex Structures in Human Cells”. He graduated from the Department of Genetic Engineering and Biotechnology from Shahjalal University of Science and Technology, Sylhet. He has extensive research experience in Cancer Biology and Molecular Medicine.

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• MSc in Protein Science and Biotechnology from the Faculty of Biochemistry and Molecular Medicine from University of Oulu, Finland and Leibniz Institute on Aging-FLI, Jena, Germany with Erasmus+ scholarship and Fellowship. Erasmus is the flagship and most prestigious scholarship program offered by EU.• MS in Biotechnology and Genetic Engineering from Shahjalal University of Science and Technology, Sylhet• BSc in Biotechnology and Genetic Engineering from Shahjalal University of Science and Technology, Sylhet• HSC from Shariatpur Govt. College, Shariatpur• SSC from Palong Tulashar Gurudas Govt. High School, Shariatpur

Journals

Khan, G., Akter, N., Uddin, M. D., & Rakib-Uz-Zaman, S. M. (2019). Influence of socio-demographic factors on the diarrheal disease management approaches taken by two distinct communities of Bangladesh. Journal of Advanced Biotechnology and Experimental Therapeutics, 2(2), 78–86. https://doi.org/10.5455/jabet.2019.d29

Hossain, M. U., Keya, C. A., Das, K. C., Hashem, A., Omar, T. M., Khan, M. A., Rakib-Uz-Zaman, S. M., Salimullah, M. (2018). An immunopharmacoinformatics approach in development of vaccine and drug candidates for West Nile Virus. Frontiers in Chemistry, 6(JUL). https://doi.org/10.3389/fchem.2018.00246

Hossain, M. U., Khan, M. A., Rakib-Uz-Zaman, S. M., Ali, M. T., Islam, M. S., Keya, C. A., & Salimullah, M. (2016). Treating diabetes mellitus: Pharmacophore based designing of potential drugs from Gymnema sylvestre against insulin receptor protein. BioMed Research International, 2016, 1–14. https://doi.org/10.1155/2016/3187647

Khan, M. A., Hossain, M. U., Rakib-Uz-Zaman, S. M., & Morshed, M. N. (2015). Epitope-based peptide vaccine design and target site depiction against Ebola viruses: an immunoinformatics study. Scandinavian Journal of Immunology, 82(1), 25–34. https://doi.org/10.1111/sji.12302

• Lecturer, Biotechnology Program, Department of Mathematics and Natural Sciences, BRAC University from September 2017 till date.• Erasmus+ Fellow, Leibniz Institute on Aging-FLI, Jena, Germany. Project Title: “Development of a novel approach to quantitatively visualize the DNA G-quadruplex structures in human cells” • Research assistant, Faculty of Biochemistry and Molecular Medicine, University of Oulu, Finland.

Project Title: “Investigating the role of alcohol and hypoxia in metabolism and cancer”.• Research Assistant, The World Academy of Sciences (TWAS), Italy. Project: Title: “Production, Partial Purification and Characterization of Bacterial Protease Produced with Municipal Solid Waste”.

1. Introduction to Biology (BIO101)2. Biophysical Chemistry (BCH102)3. Biotech Lab I (BTE155)4. Introduction to Molecular Biology (BTE203)5. Microbial Biotechnology (BTE302)6. Environmental Biotechnology(BTE304)7. Advanced Molecular Biology (BTE307)8. Enzyme Biotechnology (BTE417)9. Principles of Chemistry (CHE110)

• S M Rakib-Uz-Zaman (2017). A novel approach to quantitatively visualize the DNA G-quadruplex structures in human cells” at Leibniz Institute on Aging-FLI, Germany.• Zaman et al. (2018). Rapid Biological Synthesis of Silver Nanoparticles from Cymbopogon citratus Leaf Extract and Evaluation of its Antimicrobial Properties. National Institute of Biotechnology, Bangladesh.• Presentation on Production of Microbial Lipid for Biodiesel Synthesis and Industrially Important Enzymes by Using Biomass of Municipal Solid Waste. International Conference of Biotechnology in Health and Agriculture (ICBHA), Bangladesh• Conference Presentation on "Epitope-based peptide vaccine design and target site depiction against Ebola viruses: an immunoinformatics study", Malaysia. • Conference Presentation on "Occurrence and Consequences of Nipah Virus in Bangladesh". Third Congress of Young Biotechnologist of Bangladesh, Sylhet.• Conference Presentation on "Partial Purification and Characterization of Protease from Bacterial Isolates Using Municipal Solid Waste".

• Awarded Europe's most prestigious scholarship, the Erasmus+, to pursue the master's degree in Biochemistry and Molecular Medicine. Erasmus+ is the flagship educational program offered by the European Union. • Obtained first prize in scientific poster presentation on "Production of Microbial Lipid for Biodiesel Synthesis and Industrially Important Enzymes by Using Biomass of Municipal Solid Waste" in " International Conference of Biotechnology in Health and Agriculture (ICBHA), 2015". • 2nd position in both MSc and BSc level at department of Genetic Engineering and Biotechnology with distinction at SUST, Sylhet, Bangladesh. • Merit based University scholarship in 2011, 2012 based on undergraduate levels study performance.• Active member of Bangladesh Biotechnologist Association

• Active member of Bangladesh Biotechnologist Association• Member of Leibniz Institute, Germany Alumni Association.

1. Cancer Cell Biology2. Replication Error and Its Role on Human Aging Related Disorders3. Molecular Biology4. Enzyme Biotechnology

S M Rakib-Uz-Zaman

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Sabrina Shahrin Sharna appointed as a full time lecturer in Mathematics and Natural Sciences Department at BRAC University since January 2012. She has completed her graduation and masters from the Department of Geography and Environment, University of Dhaka, Bangladesh. In her MS level, she worked on land use land cover study using remote sensing data. Later on she engaged herself concentrated into study about Disaster. At present she is involved with research related to disaster of coastal region, their adaptation techniques and mitigation strategies for Bangladesh.

Level: 4, Room: 4G46Kha-224 Merul BaddaDhaka 1212. Bangladesh

Journals

Younus, M. A. F., & Sharna, S. S. (2014). Combination of community-based vulnerability and adaptation to storm surges in coastal regions of Bangladesh. Journal of Environmental Assessment Policy and Management, 16(4). https://doi.org/10.1142/S1464333214500367

Sharna, S. S., & Khan, A. U. (2010). Combination of community based vulnerability and adaptation to storm surges in coastal regions of Bangladesh. Dhaka University Journal of Earth and Environmental Sciences, 1(1).

Conferences

Younus, M., Sharna, S. S., & Rahman, T. (2014). Integrated assessment and decision-support tool for community-based vulnerability and adaptation to storm surges in four coastal areas in Bangladesh. In Australia New Zealand Society for Ecological Economics 2013 Conference. Australia.

Life Member – National Oceanographic and Maritime Institute (NOAMI)

Sabrina Shahrin Sharna

News & Events

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Brac University’s Office of Career Services and Alumni Relations (OCSAR) is a very significant department, vested with the duty to facilitate its students’ efficient and effective transition into their professional life which will ultimately lead to successful career for them.

It is an obvious fact that you will not remain students forever and will have to make the jump into the competitive world of work through an arduous and often grueling recruitment process; which can be made much more manageable, and even perhaps fun, if you use the support and resources that the OCSAR makes available to you.

The responsibilities to provide support and guidance lie at the core of the OCSAR, defining its function and criteria for success. The success of OCSAR is measured by the long-term professional accomplishments of you, its students, whom we are ultimately here to help.

Whether you need help writing a CV, or structuring a cover letter; require a better understanding of the ‘market’, 'corporate structures' or any other (possibly strange sounding) concepts; if your priority is winning an internship with a big firm, improving your interview technique, or gaining the confidence to convey your strengths intelligently to recruiters; even if you need a quick chat with one of our advisors to de-stress and make sense of the complex process of choosing a career, have no fear, we are here to help.

As a department responsible for helping you prepare for the recruitment process and realize your potential over the course of your career, our programs have been developed to offer solutions to the inevitable challenges you will face along the way.

Also, OCSAR organizes on campus recruitment sessions for graduating students alongside of internship placement to ensure the placement of our students in the job market.

After graduation, consider OCSAR as your home because you are proud alumni of Brac University. We are here to welcome you, address your issues & provide help, at the same time Brac University desires you to contribute to your very own institute with your valuable suggestions for academic or administrative process improvement, your engagement as resource person and your tangible or intangible contribution in various positive value addition programs of Brac University.

OCSAR is a platform where you prepare to end a journey as a student and begin a journey as proud alumni of Brac University.

We expect to work together to make Brac University a place where innovative, knowledgeable, academically sound, socially responsible leaders are groomed and help make the world a better place.

Our vision is to empower students and alumni to achieve success in the ‘world of work’ and ensure that they have the right competencies, skills set and overall approach for professional success. We facilitate opportunities to build lifelong relationships with and between our graduating students and alumni who are our most effective ambassadors.

Our mission is to provide an integrated range of high-quality career-focused services which meet the needs of our graduating students and prepares them for professional success. We aspire to create well rounded, socially aware, responsible human beings and conscious citizens.

• Professional Skills Development Program (PSDP)• Internship placement• Job placement• On-campus job/internship recruitment session• Seminar/webinar/workshop/networking session• Career fair• Career counseling• Help furnishing CV & cover letter

News Archive >>

Contact UsOffice of Career Services & Alumni Relations (OCSAR)Kha 224 Bir Uttam Rafiqul Islam AvenueMerul BaddaDhaka 1212. Bangladesh Tel: +88 09638464646Email: , , ,

, Website: ocsar.bracu.ac.bd

Contact Us

Office of Career Services & Alumni Relations (OCSAR)

Kha 224 Bir Uttam Rafiqul Islam AvenueMerul BaddaDhaka 1212. Bangladesh Tel: +88 09638464646

Email: , , , ,

Website: ocsar.bracu.ac.bd

News

BRACU Alumni-led Startup Shines in Shark Tank Bangladesh

BRAC University and United Group entertainment venture - Courtside Sign Memorandum of Understanding

BRAC University law alumna selected as UN Early Career Climate Fellow

Two-day career fair ends at BRAC University

The School of Architecture and Design is Brac University’s new venture.This follows from the tradition set by the Department of Architecture, which has established a name for itself in high quality design education.

The School will be comprehensive in its scope to include design disciplines that has relevance to the local context as well as international trends.

The school will initially offer certificate and degree programs in:

Following this more programs will be offered in:

New design disciplines are evolving all over the world and the School will gradually incorporate those which are relevant.

The School has started operation from Fall 2019 with a certificate course in Interior Design and will be launching more programs soon.

The School is being led by Prof. Fuad H Mallick as its Dean. Prof. Mallick is the founder of the Department of Architecture and the Post Graduate Programs in Disaster Management at Brac University, before going on to be the Pro Vice Chancellor.

For further information contact: +8801714129741soad[at]bracu.ac[dot]bd

News Archive >>

News

Workshop on Architecture Model Making Held at BRAC University

BRAC University architecture team recognized at international design competition

Students of The Department Architecture Excel at AIUB Architecture Design Charrette

Professor Fuad Hassan Mallick and Professor Zainab Faruqui Ali Speak at Bengal Institute’s Lecture Series on Exploring Muzharul Islam

Faculty and Research Staff

Fuad H. Mallick, PhD

At present, BracU has 4 schools and 3 institutes.

BracU also has seven departments offering different program disciplines.

Currently BracU has thirty two academic programs, of those, sixteen are undergraduate programs and the rest are postgraduate programs. BracU has a trimester system except for the Pharmacy Department, which runs on a semester system.

This Office of the Registrar is responsible for preparing academic scheduling, registration on courses, creation and maintenance of student academic records, academic transcripts, course information, student status, enrollment verification and graduation.

Brac University's Residential Campus offers a unique experience for university level students in Bangladesh.

The Registrar maintains the Academic Calendar of the University. So that students can find out when the semester begins and ends, when exams are scheduled etc.

You will find different centres and initiatives operated in Brac University.

Academic scheduling and registration unit is responsible for preparing university wide academic schedule both class and exams, and also manages academic calendar, registration process.

Useful Form:

Samanta Saha received her B.Sc. degree in Mechanical Engineering from Bangladesh University of Engineering and Technology, Dhaka, Bangladesh. Currently, she is pursuing her M.Sc. degree in Theoretical Physics from the University of Dhaka, Dhaka. Her research interests include Cosmology, General Relativity, Black hole, QCD, Computational fluid dynamics (CFD) and Heat Transfer, etc.

Level: Kha-224 Merul BaddaDhaka 1212. Bangladesh

M.Sc. - DU, running, M.Sc. in Theoretical Physics.B.Sc. - BUET, 2017, Bachelor's in Mechanical EngineeringH.S.C - Gaibandha Govt. College, Gaibandha, 2012S.S.C - Gaibandha Govt. Girls' High School, Gaibandha, 2010.

Lecturer, Dept. of MNSBrac University, Bangladesh.Duration: January 2020 - Present

Course Contractual Lecturer, Dept. of CSEBrac University, Bangladesh.Duration: September 2019 - December 2019.

[CSE 474 Lab] Simulation and Modeling.[CSE 330 Lab] Numerical Analysis.[PHY-111] Principles of Physics-111[PHY-112] Principles of Physics-112

“Numerical study of 2D axisymmetric wavy-walled turbulent pipe flow and heat transfer with Al2O3-water nanofluid”, Samanta Saha, Anwarul Islam Akash, Ahsan-Ul Hossain Abir, Sumon Saha, the 12th international conference on Mechanical Engineering (ICME 2017).

1. Cosmology,2. General Relativity,3. Black holes,4. QCD5. Computational fluid Dynamics (CFD) and Heat Transfer.

Samanta Saha

News & Events

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EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Fardousi Ara Begum completed her B.Sc (4 years Honours) as well as M. S (Thesis group) in Mathematics from the University of Dhaka in the years 2001 (held on 2003) and 2002 (held on 2005) respectively and secured first class in both. She completed her M. Phil in Mathematics from Bangladesh University of Engineering and Technology in 2016 and secured a CGPA of 3.69 out of 4.

FardousiAra Begum is now working as a senior lecturer in mathematics at the Department of Mathematics and Natural Sciences, BRAC University, Dhaka, Bangladesh. She joined BRAC University in 2006 as a lecturer.

Level: 4, Room: 4G44Kha-224 Merul BaddaDhaka 1212. Bangladesh

Journals

Ahmed, K. M., Ahmed, M., Khatun, G., Begum, F. A., Shahabuddiun, M., Das, T. K., & Saha, S. K. (2011). Study on cohomology of De Witt suppermanifolds. Jahangirnagar University Journal of Science, 34(2), 89–96.

• MAT101(Fundamentals of Mathematics)• MAT102(Introduction to Mathematics)• MAT103(Basic Concepts of Mathematics)• MAT104(Mathematics)•

MAT105(Calculus)• MAT110(Differential Calculus & Coordinate Geometry)• MAT120(Integral Calculus & Differential Equations)• MAT203(Matrices, Linear Algebra and Differential Equations)• MAT204(Complex Variables and Fourier Analysis)• MAT215(Complex Variables & Laplace Transformations)• MAT216(Linear Algebra & Fourier Analysis)• MAT211(Calculus II)• MAT221(Real Analysis I)• MAT 091: Basic Course in Mathematics(MBA)• MAT 121: Basic Algebra• MAT 222: Differential Equations I• MAT 314: Complex Analysis

1. Solving one-dimensional heat equation by control volume-method and then comparison with the exact solution, Room: UB1521, Department of Mathematics and Natural Sciences, BRAC University, Bangladesh, February 04, 2010.2. The study of the performance of a solar driven adsorption refrigeration chiller keeping chilled water outlet temperature fixed, 11th International Conference on Mechanical Engineering, Bangladesh, Dec 18-20, 2015.

Awarded a Scholarship for Studying M. S. in Mathematics at University of Dhaka, Dhaka, Bangladesh.

• Life member of Bangladesh Mathematical Society• Life member of NOAMI

Air cooling using solar thermal energy and waste heat energyNumerical solutionApplied Mathematics

Fardousi Ara Begum

News & Events

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BRAC University Architecture Faculty Achieves Top Honors in National Design

Competition

Workshop on Architecture Model Making Held at BRAC University

Sarah Umaymah Mahdiyah completed her school and college from Viqarunnisa Noon School & College. She finished both her MS and BSc. from the Department of Genetic Engineering and Biotechnology, Faculty of Biological Sciences, University of Dhaka in 2022 and 2020 respectively. Her strength lies in public speaking, creative problem solving and analytical thinking abilities. Her research interests involve a vast array of medical biotechnology topics such as Pharmacogenetics, Forensic Biotechnology, Oncology, Epigenetics and Public Health.

Level: 4Kha-224 Merul BaddaDhaka 1212. Bangladesh

Journals

Mahdiyah, S. U., Ahsan, T., Fatema, K., Shoily, S. S., & Sajib, A. A. (2022). Interactions of epitopic variants of epidermal growth factor receptor with therapeutic anti- EGFR antibodies. Dhaka University Journal of Biological Sciences, 30(3), 393–403. https://doi.org/10.3329/dujbs.v30i3.59032

• Lecturer, Biotechnology Program, Department of Mathematics and Natural Sciences, BRAC University• Vice President (Events) (2021-Present)Dhaka University Science Society (DUSS)• Quiz Creator (2020)For Robi 10 Minute School• Promotion Partner (2019)B2IN_Tutor in my hand

• Assistant Research & Training Secretary (2018-2019)Bangladesh Biology Olympiad

Medical Biotechnology (BTE310), Introduction to Biology (BIO101) and Biotech Lab 1 (BTE155)

Poster Presentation - National DNA Day Celebration-2016

• Honors Scholarship 2022 (General, University of Dhaka, Merit Position 112)Directorate of Secondary and Higher Secondary Education

• The Duke of Edinburgh’s International AwardBronze Standard• University

Pre-Graduation Scholarship (University of Dhaka)

• National Science and Technology Fellowship 2020-2021In the field of Biology and Medical Science

• Open Source Leader 2018 BadgeAwarded by Commonwealth 100

• FameLab National Finalist 2018 (April 2018)Created by CHELTENHAM Festivals

• Special Mention at PRE WHO SIMULATION WORKSHOP 102 (April 2018)Awarded by WHO Simulation Bangladesh

• Board Scholarship 2015 (General)Higher Secondary School Certificate (HSC) Examination, Dhaka Board

• Certificate of RecognitionFor commitment and participation in Active Citizens Youth Leadership Training

Network of Young Biotechnologists of Bangladesh (NYBB)

Pharmacogenetics, Forensic Biotechnology, Oncology, Epigenetics, Population Genetics, Epidemiology, Public Health and Biostatistics.

Sarah Umaymah Mahdiyah

News & Events

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Workshop on Architecture Model Making Held at BRAC University

The University provides Research Grant Proposal Application Form as a supportive environment in which researchers at every stage of their career can flourish. The University is dedicated to fostering research collaborations across the world with research institutions, research agencies, funding bodies, industrial and commercial partners, sponsors and benefactors.The research environment shall get a further boost when there will be close collaboration between the industry and academics. Already an MoU has been signed between the Computer Science and Engineering Department and Grameen Phone where such industry-academic collaboration will flourishResearch Grant Proposal Application Form

The University provides Research Grant Proposal Application Form as a supportive environment in which researchers at every stage of their career can flourish. The University is dedicated to fostering research collaborations across the world with research institutions, research agencies, funding bodies, industrial and commercial partners, sponsors and benefactors.

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Research Grant Proposal Application Form

Students will also write a multi-disciplinary dissertation, learning to draw across disciplines to focus on one specific problem, study the literature to develop an expertise in the area of inquiry, articulate arguments logically and coherently, and express themselves with clarity and eloquence. Each student will work closely with a faculty mentor who will guide them in their academic studies and professional development. Outstanding dissertations may take the form of a publication.

Thesis Template and Guidelines

Please see the link below for details:

http://library.bracu.ac.bd/thesis-help

Teaching

Each department, school and institute has its own academic community, dedicated to advancing knowledge in particular subject areas. They often work together on teaching joint or general education courses as well as interdisciplinary research projects. Basically the student is at the centre of the education provided by Brac University. Teachers are usually ready to provide support in terms of pedagogy and make themselves available to help the students in their creative thinking and to innovate.

By utilizing high quality, focused, and shorter than taught courses that many other universities teach, a BracU graduate course facilitates students' swift career progression. This is achieved in part through the close working relationships between supervisors and students utilizing teaching methodologies of small group teaching.

Teaching styles, formats and frequency of contact with faculty varies from course to course. All advanced courses require the student to take on a significant amount of case based study and this will normally be alongside a framework of seminars, workshops and lectures where students analyse and present to given problems and assignments.

Research

Academic staff and graduate students at Brac University are central to the University's research efforts, which span all areas of the globe and tackle issues of international significance. Research at the university has made an enormous impact on our fundamental understanding of the world and significant contribution to society’s needs, and our graduate students join our academic staff in tackling some of the major challenges facing the world today.

Students on research degrees work closely with an academic supervisor focusing on a specific research project to produce a thesis that represents a significant and substantial piece of work. Students are assessed on the basis of this thesis and a viva voce (an oral examination).

Year on year, BracU's total research expenditure is considerably higher than any other university in Bangladesh, hence a large number of research work in different disciplines are carried out.

The University provides a supportive environment in which researchers at every stage of their career can flourish. The University is dedicated to fostering research collaborations across the world with research institutions, research agencies, funding bodies, industrial and commercial partners, sponsors and benefactors.

The research environment shall get a further boost when there will be close collaboration between the industry and academics. Already an MoU has been signed between the Computer Science and Engineering Department and Grameen Phone where such industry-academic collaboration will flourish.

Both in the Undergraduate and Master program, the university promotes students to carry out research. In the field of research, Brac University promotes both innovation and entrepreneurship.

International Conference on Business and Management (ICBM 2017) Organized by BRAC Business School, BRAC University on the theme of 'Global Contemporary Practices in Business and Management' had two parallel sessions where the participants got the chance to present their papers before our distinguished chairs coming from the field of Academia. The day one's parallel session took place on September 21, 2017 at the Westin Hotel, Dhaka, which has been conducted in 3 different slots, where in each slot paper authors got the opportunity to concentrate on 5 different topics. On the first day, ICBM 2017 had presentation of seventy seven (77) Papers from Students, Academicians and Corporate Personnel's coming from both Public and Private sector of home and abroad.

First day's Parallel Session had 24 chairs among them a few names to be referred, including Prof. Ramayah Thurasamy (Universiti Sains Malaysia), Prof. Dr. Ali Ahsan (University of Dhaka), Prof. Prem Kumar (Vice-Chancellor, Malaysia University of Science and Technology), Mr. Chris Voegeli (General Manager, Hotel Radisson, Dhaka Bangladesh), Dr. Rabiul Bashar (BUP, Bangladesh), Prof. Mark GOH (Business School, National University of Singapore), Dr. Erne Suzila Kassim (UiTM, Malaysia), Prof. Dr. Abul Hasnat Md. Golam Azam (UIU, Bangladesh), Mr. Ismail Majumder (China), Commodore BN M Ziauddin Alamgir (Maritime University, Bangladesh), Mr. A. H. M Yeaseen Chowdhury (BUP, Bangladesh), Prof. Sharmistha Banerjee (University of Calcutta, India), Dr. Siti Noorsuriani Ma'on (UiTM, Malaysia), Prof. Rafiqul Islam (DIU, Bangaldesh), Dr. Pairach Piboonrungroj (Chiang Mai University, Thailand), Prof. M. A. Baten (SUST, Bangladesh), Dr. Mohammad Rafiqul Islam (BRAC University, Bangladesh), Dr. Saad Hasan (AIUB, Bangladesh), Prof. Dr. Abdul Hannan Chowdhury (V.C, Prime Asia University, Bangladesh), Assoc. Prof. Dr. Ireen Akhter (IBA, JU, Bangladesh), Mr. Ferdous Saleheen (LG Butterfly, Bangladesh).

List of topics on which papers were presented (along with its numbers):

The first day Parallel Session ended through the distribution of crest and certificates as a token of appreciation and encouragement among session chairs and paper presenters.

Abdul Quader is a lecturer in Mathematics at the Department of Mathematics and Natural Sciences at BRAC University. He has completed his B.Sc. (Honors) degree in Mathematics and M.Sc. (Thesis group) degree in Applied Mathematics from the Mathematics Discipline, Khulna University, Khulna, Bangladesh. His areas of interest at present are Applied Mathematics, Computational Fluid Dynamics and Numerical Analysis. He has publications in International journals and oral presentation in both International and national conferences.

Level: Kha-224 Merul BaddaDhaka 1212. Bangladesh

Journal:

Applied Mathematics, Computational Fluid Dynamics and Numerical Analysis

Abdul Quader

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

A seminar titled "Abed Chaudhury: A multi-decade journey in science covering Food, Climate and Genofax" was organized at BRAC University on 6 June 2024. The event was held as a part of a "Biotechnology Seminar Series" under the Biotechnology Program of the Department of Mathematics and Natural Sciences and in cooperation with the BRAC University Society for Biotechnology.

The guest speaker was Dr. Abed Chaudhury, a geneticist based both in Australia and Bangladesh who is currently working at Soil Company and whose expertise spans across crop breeding, molecular biology, human nutrition and biotechnology. He has previously taught and conducted research at the Massachusetts Institute of Technology, Commonwealth Scientific and Industrial Research Organisation, agrichemical company Syngenta and Vitagrain.

During the seminar, Dr. Abed Chaudhury spoke on his three primary projects, one of which was on ensuring food security through the development of a multi-harvest strain of rice named “Panchabrihi” which he has been cultivating at his birthplace in Kanihati of Sylhet.

This rice plant, with distinguishable roots, needs to be planted just once to provide harvests five times, which is in stark contrast with existing varieties which need to be planted for a harvest each time. Dr. Abed Chaudhury also gave a comparative analysis of the high yield of this variety with other hybrids. He also showed how he used drone technology to observe the growth of rice in his paddy fields.

He also said to have founded an organization to run a crop breeding operation in Bangladesh with innovations like rice varieties with an extended shelf life and which can help maintain low blood glucose.

Afterwards, he talked about his work on fertilization-independent seed development in Arabidopsis Thaliana, a small plant from the mustard family. His other research focuses on methods for the reduction of methane production in ruminants.

Faculty members and curious-minded students asked insightful questions about his works and relevant applications. Prior to the seminar, Dr. Abed Chaudhury conducted an interview on informing that he was looking for enthusiastic minds to engage in internships and research at Genofax, of which he was the co-founder and chief scientist. He informed that Genofax was a global biotechnology company that specializes in developing AI technologies to craft personalized medicine based on individual gut microbiome gene sequences.

This is in tune with the company’s vision to see a world where healthcare is tailor-made for every person with maximum precision to promote overall wellness. Several BRAC University students were able to crack the interview.

News Archive >>

Biotechnology seminar hosts Bangladeshi scientist who developed high-yielding “Panchabrihi” rice

Related News

Orientation held for Summer 2024 biotechnology students

Nanobiotechnology seminar organized by BUSB and Biotechnology Program

The biotechnology program and BUSB organized the inaugural Biotechnology Workshop Series on “Understanding and Designing Biosensors”

Orientation held for Spring 2024 biotechnology students

Brac University (BracU) established in 2001 is located in Dhaka Bangladesh. BracU follows a liberal arts approach to education which nurtures fresh ideas and gives new impetus to the field of tertiary education. It ensures a high quality of education and aims to meet the demands of contemporary times. Building on BRAC's experience of seeking solutions to challenges posed by extreme poverty, BracU hopes to instill in its students a commitment to working towards national development and progress. The medium of instruction and examination at Brac University is English. BracU is accredited by the University Grants Commission (UGC) and approved by the Ministry of Education, Government of Bangladesh.

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MISSION

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BracU Office of Communication (OoC) covers university’s academic or cultural activities, programs, seminars, initiatives etc. and circulates the news to respective print/visual media. Besides, OoC takes initiative to publish personal write-ups, academic articles or features of faculties, students, staffs and maintains individual Photo Archive and Press Release Archive. Externally, OoC coordinates between the university and government security authorities, institutions, bodies regarding foreign personnel’s visa, work permit and others. Additionally, OoC maintains relationship with the Honorable Chancellor’s Office, Prime Minister’s Office, Education Ministry, and University Grants Commission (UGC) relating to BRAC University affairs.

Download Resources:

BRACU Logo: PNG , AI

BRACU Anthem: MP3

Contact OoC for any kind of assistance:

Industry Talk Day 1

The first day of 1st International Conference on Business and Management, September 21, 2017 at the Westin Hotel, Dhaka, Bangladesh, Industry Talk session was chaired by Prof. Rahim B. Talukder, Advisor CED, BRAC University, Bangladesh and Prof. Dr. Anwar Hossain, Vice Chancellor, Northern University of Bangladesh. In the opening speech, Dr. Hossain stated that it is a big challenge for the universities to understand the industry expectation and prepare their students to hold the top positions in companies from the very first day of joining. Dr. Talukder added that expectations are changing with change in time and generation.

The first speaker of Industry Talk Session was Mr. A. E. Abdul Muhaimen, Managing Director, United Commercial Bank Ltd. Mr. Muhaimen introduced the ‘Finance Industry’ to his audience and talked about the Financial Institutions. In his speech he concentrated on the Economy of Bangladesh. To elaborate his idea he exemplified Bangladesh and said despite facing a lot of challenges such as natural disasters, economic downturn and political instability, it’s growing at an impressive rate of 6% which must be taken advantage of. Later he concluded his speech on an optimistic note saying that our growing population can be seen as our strength or weakness; how we will utilize that will solely depend on our viewpoint regarding that.

Afterward, Dr. Talukder invited Mr. Al-Amin, Director of Sales and Marketing, The Westin Dhaka and Ms. Farida Ansari, Owner’s Representative, Unique Hotel & Resorts Ltd. on behalf of Mr. Mohd. Noor Ali, Chairman, Unique Group of Companies. Both of them enlightened audience about the Hospitality Industry. Through their speech they tried to deliver the gist of how they started their journey in the Hospitality Industry, achieved success over the years and how they are planning to ensure its sustainability in the upcoming future, meeting customers’ expectations.

The third speaker was Mr. Syed Ferdous Raihan Kirmany, Executive Director, BBS Cables Ltd, a sister concern of Bangladesh Building Systems Ltd. (BBSL), a leading Pre-Engineered Steel Building Manufacturer in Bangladesh. Through his speech, he emphasized on the incorporation of advanced technology and innovation in their products and also explained why it is a bound necessity for them. Consequently, he illustrated the journey of BBS Cables which started with BBS fibers and later got diversified with wide ranges of products in their portfolio to compete with the industry demand. He also gave a brief idea about their product development process. Finally, he concluded his speech highlighting on the vigilance issue and said BBS Cables is really cautious of its product quality and hence the company is always striving for continuous development and innovation.

Tonima Fairooz Mouly commenced her role as a lecturer at BRAC University in September 2023. Prior to joining the university, she served as a research assistant at the Laboratory of Molecular Biology and Landscape Epidemiology at icddr,b.Tonima

Fairooz Mouly completed her school and college from Viqarunnisa Noon School & College. She completed her BSc. in Microbiology from Brac University. She achieved her MS Degree in Microbiology from Jagannath University. For her BSc. thesis, she focused on identifying potential point mutations in the gyrA gene of ciprofloxacin-resistant Pseudomonas aeruginosa isolates from patients with lung infections. Additionally, her Master's thesis centered on the molecular characterization of clinical Vibrio cholerae isolates, investigating amino acid substitutions in gyrA and parC genes associated with Ciprofloxacin resistance.Tonima Fairooz Mouly is particularly concerned about the global threat of antibiotic resistance and has chosen Antimicrobial Resistance (AMR) as her primary research area of interest.

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General Microbiology (MIC101), Microbial Chemistry (MIC201), Environmental Microbiology (MIC203), Food Microbiology (MIC302) and Laboratory Courses

• Antimicrobial Resistance (AMR) • Molecular Biology • Bioinformatics (gene sequencing, identifying mutations, WGS) • Clinical Microbiology • Microbial Epidemiology

Tonima Fairooz Mouly

News & Events

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EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Pharmacy is a multidisciplinary subject. The good practice of pharmacy today is one of the most crucial components in providing proper health services to people worldwide. It is a profession of honor and high esteem that demands huge responsibilities. Established in 2010, at the Department of Pharmacy, Brac University, we have designed an undergraduate programme to produce competent graduates, who will be capable of meeting the challenges of a rapidly evolving healthcare sector.

The Bachelor of Pharmacy (B. Pharm.) is divided into eight consecutive semesters approved by the University Grants Commission (UGC) and Pharmacy Council of Bangladesh (PCB). We have highly qualified faculty members, advanced research facilities and online library services containing large collection of books, articles and journals. Using innovative and interactive techniques, we make pharmacy education exciting for the students. The instructors undergo continuous training and assessment in order to uphold the quality of education. We are also proud of our sound disciplinary policies, which enable smooth operation of our department.

This program is designed to support those who wish to pursue a career in different areas of drug discovery, formulation development, drug manufacturing, pharmaceutical marketing, scientific research and higher education. Thus, our incessant efforts are directed towards creating highly qualified conscientious pharmacists who will actively contribute to all sectors of our society including in industrial, clinical, community, research or educational field.

Vision

To be established as a leading school for experiential and personalized learning, and be recognized for engagement and impact.

Mission

To deliver academic excellence that combines research with an impact on society.

Our Aims

Program Educational Objectives (PEOs)

The graduates of the program will be able to

PEO 1: Pursue careers as successful professionals in different fields of Pharmacy.

PEO 2: Pursue the aspiration, ethos, and competency required for lifelong learning contributing to the integrity of their métier.

PEO 3: Demonstrate leadership by accepting professional and societal responsibilities.

PEO 4: Adhere to the highest standards of ethical and professional behavior.

Program Outcomes (POs)

Upon successful completion of the B. Pharm. program, the students will be able to demonstrate the following outcomes:

POs

Description

PO1Knowledge

Demonstrate knowledge in fundamental concepts of pharmaceutical sciences

PO2Problem Solving

Solve multifaceted problems within the discipline of pharmacy

PO3Research

Exhibit general competencies, technical knowledge, and skills to conduct research

PO4Tools and Technology

Apply modern tools, resources, and appropriate techniques in relevant areas of pharmacy

PO5Ethics

Practice strong social, moral, and ethical values required to serve the pharmacy profession

PO6Communication

Effectively communicate with appropriate audiences in the field of pharmacy and pharmaceutical sciences

PO7Individual and Teamwork

Display interpersonal skills while working as an individual, a member, or a leader in diverse teams

PO8Lifelong Learning

Recognize the importance of continued lifelong learning to promote their growth as pharmacy professionals

Student Handbook 2019

Faculty and Research Staff

Eva Rahman Kabir, PhD

Hasina Yasmin, PhD

Mesbah Talukder, PhD

Sharmind Neelotpol, PhD

Raushanara Akter, PhD

Muhammad Asaduzzaman, PhD

Md. Jasim Uddin, PhD

Mohd. Raeed Jamiruddin, PhD

Md. Aminul Haque, PhD

Afrina Afrose, PhD

Farhana Alam Ripa, PhD

Md. Rabiul Islam, PhD

Sabrina Sharmin, PhD

Nishat Zareen Khair, PhD

Md. Samiul Alam Rajib

Mohammad Kawsar Sharif Siam

Md. Tanvir Kabir

Ashis Kumar Podder

Saif Shahriar Rahman

Namara Mariam Chowdhury

Tanisha Tabassum Sayka Khan

Easin Uddin Syed

Eshaba Karim

Faruque Azam

Tanisha Momtaz

Luluel Maknun Fariha

Nushrat Alam

Asef Raj

Mehedi Islam

Mushtahsin Ferdousi

Kazi Milenur Rahman Prattay

Pharmacy monthly newsletter

Pharma Highlights, July 2024

Pharma Highlights, June 2024

Pharma Highlights, May 2024

Pharma Highlights, April 2024

News and Events

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Dr. Sharmind Neelotpol of BRAC University Hosts ACU Gender Programme Symposium

School of Pharmacy Alumni Meet: First Ever Reunion at the New Campus

BRACU Research for Development Club Immerses in Pharmaceutical Innovation at Incepta’s Dhamrai Plant

Webinar on 'The Scope of Health Outcomes Research in the US'

Webinar on 'Developing Professional Competence: The Role of Adaptability and Integrity'

A Webinar on ‘Role of Clinical Pharmacists in COVID-19'

A Short Course on ‘Deciphering the Genetic Message: Transcriptome Analysis’

Course Requirements

The M.S. Biotech curriculum consists of core theory and lab courses and a wide variety of elective courses. In addition to these, a core course of 6 credits on research project (thesis) on an indigenous biotech-related problem is also included in the curriculum. The minimum credit requirement for the MS degree is 36 with a breakdown as given below:

Core courses: 15 CreditsElective courses: 15 CreditsResearch project (thesis): 6 Credits------------------------------------Total: 36 Credits

Each semester will concentrate on various fields of biotechnology so that at the completion of the MS degree the students will be acquainted with the science and techniques of biotechnology.

Core Courses

The students will be required to complete 15 credits distributed over 5 core courses. One of the core courses will be devoted to plant biotechnology keeping in mind that plant biotechnology has progressed a lot in our country. To teach students about commercialization of biotechnological research products and dangers of pollution, core courses like environment biotechnology and industrial biotechnology are designed. Apart from these, another core course, animal biotechnology will be offered to acquaint students with the progress in this field with special reference to human biotechnology. Research project preparation needs good training on literature review, experimental design and interpretation of results. Case studies will be included in this course.

Elective Courses

Students will be required to take 5 elective courses for a total of 15 credits.

List of Courses

Core Courses for the First Semester (9 credits):

BTC 501 Plant Biotechnology 3 Credits BTC 503 Animal Tissue Culture Techniques and Applications 3 Credits BTC 504 Fermentation and Industrial Biotechnology 3 Credits Total 9 Credits

Core and Elective Courses for the Second Semester (9 Credits):

Core Courses

BTC 505Environmental Biotechnology 3 Credits BTC 506 Research Project Preparation 3

Credits Subtotal 6 Credits

Elective Course (any one)

BTC 510 Fundamental and Applied Aspects of Plant Genetic Manipulation 3 Credits BTC 511 Commercial Production of Horticultural and Ornamental Plants 3 Credits BTC 512 Sex, Flowers and Biotechnology 3 Credits BTC 513 Gene Function and Its Regulation 3 Credits BTC 514 Gene Organization and Regulation 3 Credits BTC 515 Structural and Functional Genomic Studies 3 Credits Total 9 Credits

Elective Courses for the Third Semester (9 credits):

BTC 507 Biostatistics & Experimental Design (Theory and Lab) 3 Credits BTC 509 Genomics (Bioinformatics) 3 Credits BTC 517 Enzymology 3 Credits BTC 518 Recombinant DNA Technology 3 Credits BTC 519 Medical Biotechnology 3 Credits Total 9 Credits

Core and Elective Courses for the Fourth Semester (9 credits):

BTC 550 Core CourseResearch Project (Thesis)Elective Course (any one) 6 Credits BTC 502 Plant Biotechnology (Lab) 3 Credits BTC 508 Seminar 3 Credits BTC 516 Special Studies 3 Credits BTC 520 Cell Dynamics, Cell Cycle and Cell Death; Gene Mapping in Phages, and Bacteria 3 Credits BTC 521 Genetically Modified (GM) Crops, Biosafety and Intellectual Property Rights 3 Credits Subtotal 3 CreditsTotal 9 Credits

NOTE: The courses and curriculum are subject to changes to keep pace with changing requirements of local, regional and global educational and business environments.

Course Structure of Undergraduate Biotechnology Programme

A 4-year (12 semesters) biotechnology undergraduate programme has been designed including topics of recent trends in this new and emerging field with emphasis on applications of biotechnology in different areas of socio-economic activities. Once a student has completed this course he/she should be well equipped to fulfill the requirement for jobs in public sectors and private enterprises involved in different biotechnology efforts and to face the challenges of new developments in this field. The programme of study includes courses for improving communication skills, strengthening mathematical background and acquainting the student with socio-economic and historical background of Bangladesh. With this basic background the student should face no problem to find a suitable job in different universities, research and development organizations and private enterprises.The total credit requirements for the degree of Bachelor of Science in Biotechnology are 136. Out of these 21 credits is on general education, which is compulsory as per requirement of the BRAC University. The departmental core courses account for 76 credits including departmental core courses (63 credits) practicals (6 credits) , Biotechnology internship (3 credits) and thesis /project (4 credits). In addition a student has to undertake 12 credits on elective courses and 27 credits from outside the major specialization. For elective and outside major courses a student can choose from several courses outlined in the syllabus in consultation with the department. The student may also be required to take non-credit remedial course in English.

IV. Courses for Bachelor of Science in Biotechnology

a. General Education: (21credits)

1. BIO 101 Introduction to Biology2. CSE 101 Introduction to Computer Science3. DEV 101 Bangladesh Studies4. ENG 091 Foundation Course (non-credit)5. ENG 101 English Fundamentals6. ENG 102 Composition I7. HUM 103 Ethics and Culture8. MAT 101 Fundamentals of Mathematics

b. Departmental Core Courses: (63 credits)

1. BCH 101 Basic Biochemistry 2. BCH 102 Biophysical Chemistry3. BCH 201 Human Physiology4. BCH 202 Enzymes and Enzyme Kinetics 5. BTE 101 Introduction to Biotechnology and Genetic Engineering 6. BTE 102 Microbial World 7. BTE 103 Plants and People 8. BTE 201 Bioorganic Chemistry 9. BTE 202 Metabolism10. BTE 203 Introduction to Molecular Biology 11. BTE 204 Fundamentals of Genetic Engineering 12. BTE 302 Microbial Biotechnology 13. BTE 303 Basic Immunology14. BTE 304

Environmental Biotechnology 15. BTE 306 Industrial Biotechnology 16. BTE 307 Advanced Molecular Biology 17. BTE308 Plant Biotechnology and Genetic engineering 18. BTE 401 Bioinformatics 19. BTE 402 GMOs and Biosafety20. BTE403 DNA Fingerprinting and Molecular Diagnostics21. BTE404 Bioprocess Technology

c. Elective Courses (12 Credits)

1. BTE 309 Food Biotechnology 2. BTE 310 Medical Biotechnology3. BTE311 Aquaculture and Marine Biotechnology4. BTE 312 Computer Applications in Biotechnology5. BTE313 Biomass and Biofuels6. BTE 314 Livestock and Fishery Biotechnology7. BTE 315 Bioremediationn and Biodeterioration8. BTE 405 Advanced Immunology9. BTE 406 Genomics and Proteomics10. BTE 407 Waste Biotechnology11. BTE 408 Advanced Agricultural Biotechnology12. BTE 409 Advanced Bioinformatics13. BTE 410 Human Genome Project14. BTE 411 Virology and Oncology15. BTE 412 IPR and Biotechnology16. BTE 413 Bioethics17. BTE 414 Global Impact of Biotechnology 18. BTE 415 Entrepreneurship in Biotechnology19. BTE 416 Biochemical Engineering Principles20. BTE 417 Enzyme Biotechnology

d. Lab: (6 credits)BTE 105 Biotech Labs IBTE 205 Biotech Lab IIBTE 208 Biotech Lab IIIBTE 305 Biotech Lab IV

e. Biotechnology Internship ((3 Credits)

BTE 400 Industrial Attachments

f. Thesis/Project (4 Credits)

BTE 450 Biotech Project

g. Courses Outside Major Specialization (27 Credits)

1. ANT 101 Introduction to Anthropology2. ARC 292 Painting3. ARC 293 Music Appreciation4. BTE 316 Social Impact of Biotechnology5. BTE 317 Biostatistics 6. BUS 101 Introduction to Business7. CHE 101 Introduction to Chemistry8. CSE 110 Programming Language9. ENV 103 Elements of Environmental Sciences10. ECO 10 Principles of Economics11. HUM 101 World Civilizations and Culture12. HUM 102

Introduction to Philosophy13. HUM 111 History of Science14. MGT 211 Principles of Management15. PHY 101 Introduction to Physics16. POL 103 Introduction to Political Science17. PSY 101 Introduction to Psychology18. SOC 101 Introduction to Sociology 19. SOC 401 Gender and Development20. STA 201 Elements of Statistics and Probability

V. Semester wise Distribution of Courses for the Degree of Bachelor of Science in Biotechnology

1st Year

Semester Course No. Course Name CreditsFirst BI0 101 Introduction to Biology 3BCH 101 Basic Biochemistry 3BTE 101 Introduction to Biotechnology and Genetic Engineering 3DEV 101 Bangladesh Studies 3ENG 101 English Fundamentals 3BTE 105 Biotech Lab I 1.5Sub Total 16.5 Credits

Semester Course No. Course Name CreditsSecond CSE 101 Introduction to Computer Science 3ENG 102 Composition I 3MAT101 Fundamentals of Mathematics 3BTE 102 Microbial World 3BTE 103 Plants and People 3HUM 103 Ethics and Culture 3Sub Total 18 Credits

Total Credits in 1st Year = 16.5+18= 34.5

2nd Year

Semester Course No. Course Name CreditsThird BCH 102 Biophysical Chemistry 3BCH 201 Human Physiology 3BCH 202 Enzymes and Enzyme Kinetics 3Elective Course (c) 3Elective Course outside major(g) 3BTE 205 Biotech Lab1I 1.5Sub Total 16.5 Credits

Semester Course No. Course Name CreditsFourth BTE 201 Bioorganic Chemistry 3BTE 202 Metabolism 3BTE203 Introduction to Molecular Biology 3Elective Courses outside major (g) 6BTE 208 Biotech Lab III 1.5Sub Total 16.5 Credits

Total Credits in the 2nd year= 16.5+16.5=33

3rd Year

Semester Course No. Course Name CreditsFifth BTE 204 Fundamentals of Genetic

Engineering 3BTE 302 Microbial Biotechnology 3BTE 303 Basic Immunology 3BTE 304 Environmental Biotechnology 3Elective Courses outside major (g) 3BTE 305 Biotech Lab III 1.5Sub Total 16.5 Credits

Semester Course No. Course Name CreditsSixth BTE 306 Industrial Biotechnology 3BTE 307 Advanced Molecular Biology 3BTE 308 Plant Biotechnology and Genetic Engineering 3Elective Course (c) 3Elective Courses outside major(g) 6Sub Total 18 Credits

Total Credits in Third Year= 16.5+18=34.5

4th Year

Semester Course No. Course Name CreditsSeventh BTE 401 Bioinformatics 3BTE402 GMOs and Biosafety 3Elective Courses (c) 3Elective Courses (c) 6BTE 450 Biotech Project/Thesis 2Sub Total 17 Credits

Semester Course No. Course Name CreditsEighth BTE 403 DNA Fingerprinting and Molecular Diagnostics 3BTE 404 Bioprocess Technology 3Elective Course (c) 3BTE 400 Industrial Attachment 3Elective Course outside major(g) 3BTE 450 Biotech Project/Thesis 2Sub Total 17 Credits

Total Credits in 4th Year 17+17= 34

Total Credits in eight semesters/four years

1st Year= 34.52nd Year= 33.03rd year= 34.54th Year= 34.0

Total = 136

VI. Academic System & Evaluation Method

a. Academic Standards

In keeping with the mission and goals in mind, the Undergraduate programme in Biotechnology will strive to ensure high academic standards by implementing well-designed curricula, carefully selecting high quality students and faculty, utilizing modern and effective instructional methods and aides, and by continuously monitoring and rigorously evaluating all the pertinent activities and systems. A special feature of teaching will be the tutorial/ lab/ workshop sessions designed to assist students in learning application of the concepts and theories.

b. Courses and Credit Requirements

The Bachelor of Science programme in Biotechnology will follow the model of higher education consisting of semesters, courses, credit hours, continuous evaluation and letter grading as in all other courses of BRAC University. There are two regular semesters: Fall and Spring, each with a duration of 14 weeks and a Summer semester of 9 weeks. The undergraduate curriculum consists of general education courses, major courses, elective courses and non-major area courses.

Credit hours for a course are assigned on the basis of a 14 – week semester. One (1) credit hour means that the course meets for 60 minutes in a class each week; 3 credits mean that it will meet three times every week.

c. Examination, Evaluation and Grading

The grading process will undoubtedly be transparent. The performance of the students is evaluated throughout the semester through class tests, quizzes, assignments, and midterm exams. End of semester evaluation includes final examinations, term papers, project reports etc. Numerical scores earned by a student in tests, examinations, assignments etc. are cumulated and converted to letter grades.

d. Distribution of Marks & GPA Computation

The distribution of marks for the performance evaluation is as follows:

i. Theory Courses

Section Marks %1. Quizzes/ Class Tests/ Assignments/ Participation/ Attendance 302. Mid Term Examination 203. Final Examination (comprehensive), Projects 50Total Marks 100

ii. Lab Courses

Section Marks %1. Reports/ Class Tests/ Participation / Viva-Voce 302. Lab Experiments 303. Final Examination 40Total Marks 100

iii. Thesis / Project

Thesis / Project work will be spread over two semesters (7th & 8th). The mark distribution for the Thesis / Project will be as follows:

Section Marks%1. Viva-Voce (End of the 7th semester) 152. Thesis / Project Work 453. Defence (End of 8th semester) 40Total Marks 100

Class attendance is compulsory for every student. 5% of total marks in every course is allocated for attendance in classes including tutorials and labs. The basis for awarding marks for attendance is as follows:

Attendance Marks90% and above 585% to less than 90% 480% to less than 85% 375% to less than 80% 270% to less than 75% 1Less than 70% 0

If a student does not attend a minimum of 70% of the total classes including tutorials and labs, s/he will not be allowed to take the final exam.

Marks earned by the students in Class Tests, Quizzes, Assignments, Participation, Attendance, Midterm Exam , Final Exam, Projects, Term Papers etc are to be cumulated and the total is to be graded as per the scale given below:

Letter Grade GP 90 - 100 = A (4.0) Excellent85 - < 90 = A- (3.7) 80 - < 85 = B+ (3.3) 75 - < 80 = B (3.0) Good70 - < 75 = B- (2.7) 65 - < 70 = C+ (2.3) 60 - < 65 = C (2.0) Fair57 - < 60 = C- (1.7) 55 - < 57 = D+ (1.3) 52 - < 55 = D (1.0) Poor50 - < 52 = D- (0.7) <50 = F (0.0) Failure

P: PassI: IncompleteW: Withdrawal R: Retaken

GPA Computation

The Grade Point Average (GPA) is computed in the following manner:GPA = Sum of (Grade points x Credits)/ Sum of Credits Attempted.

Dr Nadia Sultana Deen is a biomedical scientist with a particular interest in microbiology and immunology. She has served in both teaching and research at different tertiary institutes in Bangladesh and Australia. Currently she is working as an associate professor in the microbiology program at Brac University. One of her academic goals is to prepare students to be globally competent. Dr Deen aspires to utilize and amplify her research knowledge to alleviate health-related issues in Bangladesh.

Level: 5, Room: 5B09Kha-224 Merul BaddaDhaka 1212. Bangladesh

Book Chapters

Deen, N. S., Lee, J. P., & Harris, J. (2020). Inducing and inhibiting autophagy to investigate its interactions with MIF. In J. Harris & E. F. Morand (Eds.), Macrophage Migration Inhibitory Factor: Methods and Protocols (pp. 147–158). Springer US. https://doi.org/10.1007/978-1-4939-9936-1_13

Abidi, J. H., Harris, J., & Deen, N. S. (2020). Co-immunoprecipitation of macrophage migration inhibitory factor. In J. Harris & E. F. Morand (Eds.), Macrophage Migration Inhibitory Factor: Methods and Protocols (pp. 115–122). Springer US. https://doi.org/10.1007/978-1-4939-9936-1_10

Flynn, J. K., Deen, N. S., & Harris, J. (2020). Flow cytometry phenotyping of bone marrow-derived macrophages from wild-type and Mif−/− Mice. In J. Harris & E. F. Morand (Eds.), Macrophage Migration Inhibitory Factor: Methods and Protocols (pp. 57–66). Springer US. https://doi.org/10.1007/978-1-4939-9936-1_6

Journals

Bujotzek, A., Tiefenthaler, G., Lariviere, L., D’Andrea, L., Marquez, E. A., Rudloff, I., Cho, S. X., Deen, N. S., Richter, W., Regenass-Lechner, F., Poehler, A., Whisstock, J. C., Sydow-Andersen, J., Reiser, X., Schuster, S., Neubauer, J., Hoepfl, S., Richter, K., Nold, M. F., … Ellisdon, A. M. (2022). Protein engineering of a stable and potent anti-inflammatory IL-37-Fc fusion with enhanced therapeutic potential. Cell Chemical Biology, 29(4), 586–596. https://doi.org/10.1016/j.chembiol.2021.10.004

Harris, J., VanPatten, S., Deen, N. S., Al-Abed, Y., & Morand, E. F. (2019). Rediscovering MIF: New tricks for an old cytokine. Trends in Immunology, 40(5), 447–462. https://doi.org/10.1016/j.it.2019.03.002

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Host-pathogen interaction, inflammation, clinical immunology, microbiota

Nadia Sultana Deen, PhD

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

1st International Conference on Business and Management (ICBM 2017), concluded on September 22, 2017, the two-day of the conference, with a call for becoming global citizens to give one's best in writing quality papers for publishing in peer reviewed journals so as to be of use to society as a whole.

Addressing a "Meet the Journal Editors" session at the conference organized by BRAC Business School (BBS) of BRAC University (BRACU) at The Westin Dhaka, BRACU Vice Chancellor Prof. Syed Saad Andaleeb, Ph.D. delineated on characteristics of quality papers and journals. On the journal of Bangladesh Studies, of which he is the editor, Prof. Andaleeb said it had a paper rejection rate of 80 per cent for plagiarism, insufficient definitions, improper rational flow, amateurship styles, inadequate designs, irrelevant topics, over-engineering and misaligned conclusions. Writers from this part of the world lack quality in writing and so should make use of close circles of "brutal reviews" and Turnitin software for better results, he said.

Moderating a corresponding question-answer session afterwards, BBS Assoc. Prof. Dr Md Mamun Habib, Editor-in-Chief of Scopus Indexed Journal, emphasized on having patience and not to be disheartened easily, as publishing a quality paper takes time. Therefore, young scholars must think peer-reviewed journal at first and then Scopus indexed journal as it would take about one year.

Prof. Dr Mark GOH of NUS Business School, National University of Singapore said with 1 billion PhDs rolling out every year, there was a “publish or perish” culture, generating tremendous pressure on getting papers published, which ultimately led to bad behavior.

On creating journals, Prof. Dr Pairach Piboonrungroj, Associate Dean, College of Maritime studies and Management, Chiang Mai University, Thailand urged on building brands, setting up special issues on specific topics, nurturing good personal relationships and knowing the audience for promotions.

Prof. Ramayah Thurasamy of USM, Malaysia and Editor-in-Chief of Journal of Applied Structural Equation Modeling, spoke on building tolerance on chances of papers being rejected and on staying alert of self- plagiarism and spam journal mails which hijack original logos.

Dr. Habib, session chair of “Meet-the-editors”, illustrated the basic format of the accepted journal papers and emphasized the contribution of the articles for the academicians, practitioners and finally the society. Ethical issues, including review process, rate of acceptance, similarity index of plagiarism software, namely Turnitin and iThenticate software, were discussed in this session. Finally, few queries were raised from the participants in a lively Q & A session.

Mehnaz Karim appointed as a full time lecturer of Mathematics in Mathematics and Natural Sciences Department at BRAC University since September 2011. She has completed her graduation from University of Toronto, Canada and post graduation from the RMIT University, Australia. During her post graduation, she developed her research in time series and demography and completed a study of demography based on classifying some of the Victorian local government areas of Australia for the comparison of their social economic states. Later on, after joining the BRAC University, she developed a couple of journals based on her thesis that developed from her research.

Level: 4, Room: 4G47Kha-224 Merul BaddaDhaka 1212. Bangladesh

Selected Publications

Journals

Karim, M. (2017). The birth registration rate in Bangladesh observed in rural and urban region. Advances in Social Sciences Research Journal, 4(18), 1–7. https://doi.org/10.14738/assrj.418.3476

Mosharraf, F. B., Anindita, M., Kalam, I. M. S., & Karim, M. (2016). Antibacterial resistance: A comparative study between clinical and environmental isolates. BRAC University Journal (Science and Engineering), XI (2), 71–84. Retrieved from http://hdl.handle.net/10361/8388

Karim, M. (2014). A study based on a survey of overall Australian labour force, occupation, industry of employment comparing to victorian (“Victoria” – Aprovincein Australia) suburbs. OmniScience: A Multi-Disciplinary Journal, 4(2), 18–25. Retrieved from http://sciencejournals.stmjournals.in/index.php/osmj/article/view/2254

MAT 091(Basic Course in Mathematics (Non-Credit)), MAT 092 (Remedial Course in Mathematics (Non-credit)), MAT 111 (Principles of Mathematics), MAT 121 (Basic Algebra), MAT 203 (Matrices, Linear Algebra & Differential Equations), MAT 211 (Calculus II), MAT 221 (Real Analysis I), MAT 312 (Numerical Analysis II), MAT 323 (Vector Mechanics).

Statistical data analysis, demography.

Mehnaz Karim

News & Events

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BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

With over 300 participants, including about 40 foreign delegates, and a total of 260 papers submissions, Brac Business School (BBS), Brac University successfully organised the 2nd International Conference on Business & Management (ICBM 2019) from 25 to 27 April, 2019 at The Westin, Dhaka, Bangladesh. The theme of the conference was “Industry Focused Global Research Trends in Business and Management”.

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After the huge success of the first edition of ICBM in 2017, this year's conference brought together academicians and renowned industrialists from all over the country and abroad under one roof. This conference was full of distinguished academicians, international dignitaries, business leaders, young scholars, and other participants. Foreign delegates from various continents of several countries, particularly Canada, Sweden, Singapore, Malaysia, Thailand, Taiwan, Indonesia, India, Nigeria, China, UK participated in the conference.

The three day event included the PhD Colloquium and Paper Producing Workshop on Day 1 followed by keynote presentations, academia-industry discussions, session with international journal editors, and paper presentations by the participants and invited guests on Day 2 and 3. The event concluded on the 3rd day with the recognition of the best papers followed by the closing ceremony. About 170 papers of 9 tracks in 24 parallel sessions were presented in 3 days conference. In addition, 3 keynote speeches, 8 invited talks, 9 industry speeches added values to the conference.

The PhD Colloquium and paper producing workshop demonstrated participants a detailed idea as to how they can start and continue their research, how and what to expect on their journey as a doctoral candidate, and how they can structure their academic papers to suit the needs of the modern day academia.

The keynote presentations, on the other hand, explored various topics in the fields of business, management, and information technology. Speakers from home and abroad shared their thoughts on different ideas and theories related to their respective field of expertise, mesmerising the audience throughout the sessions while keeping them informed about the latest trends in business and technology at the same time.

Industry talks and Academia-Industry discussion session reflect different flavor due to deliverables of top industry key persons. The conference also hosted a poster presentation session where the participants displayed their research work through their posters.

Apart from the participants and other invited speakers, the event was overseen by the Vice-Chancellor of Brac University, Prof. Dr Vincent Chang as the Honorary General Chair; Dean of Brac Business School (BBS), Prof. Dr Mohammad Mahboob Rahman as the General Chair; and Assoc. Prof. Dr Md. Mamun Habib from BBS as the Program Chair. The conference was also graced by the presence of Mr K M Khalid MP, State Minister, Ministry of Cultural Affairs, Government of the People's Republic of Bangladesh as the Chief Guest on the prize giving and closing ceremony.

Newsletter Archive >>

2nd International Conference on Business and Management (ICBM 2019)

Newsletter

Exams and Tests

The performance of the students is evaluated throughout the semester through class tests, quizzes, assignments, and midterm's exams. End of semester evaluation includes final exams, terms papers, project work etc. Numerical scores earned by a student in tests, exams, assignments etc. are cumulated and converted to letter grades. All Students are required to attend all tests, quizzes, assignments and exams.

Quizzes /Tests/Assignments

Quizzes /Tests/Assignments are scheduled in class hour or tutorial hour.

Mid Term & Final examsMidterm exam is held in the midterm exam week announced in the Semester Calendar. The duration of the midterm exam is between one to two hours.

Final Exam schedule is announced in the Semester Calendar. Final exams are comprehensive and the duration of the final exam is three hours for 3 credits course and two hours for a 2 credits course.

Students who attend final exams and are unable to complete them for any reason are assigned a failing grade (F). These students are not allowed to complete the exam at another time or appeal for a make-up exam. Students who do not withdraw or drop from courses and do not take their final exams are also assigned a failing grade for final exams.

Exam conflicts

An exam conflict occurs when a student has two exams or a class meeting and an exam scheduled for the same day and time. If the conflict involves two exams, the student will take both exams on the same day: one with the class and the second exam at a different time that day as scheduled by the instructor or department.

Make up Exams

All students are expected to take their exams on time and as scheduled.

Students who attend final exams and are unable to complete them for any reason are assigned a failing grade (F), for the course. These students are not allowed to appeal for a make up exams. Students who do not drop or withdraw from courses are also assigned a failing grade.

Make up tests, exams or quizzes may be organized for those students who could not appear in the tests/exam/quizzes for medical or other compelling/unavoidable reasons acceptable to the University. A student has to apply for makeup’s showing reasons with evidence for his failure to take the scheduled exams. Students seeking leave or makeup exams on medical grounds must contact the medical officer of BRAC University as soon as possible and obtain a medical certificate from him/her.

For make up of Quizzes, Class Tests, Assignments, Participation etc, s/he should apply to the course teacher and for makeup of Mid Term and Final Exams, s/he should apply to the chairperson of the parent department through the course teacher by filling the makeup exam form. The application form is available at the registrar’s office and on our website. Students must apply for a makeup exam within 10 days from the day of the exam.

The Registrar’s Office will announce and post the student list of those who are applying for makeup exam. Dates for makeup exams will be schedule by the respective departments and course teachers.

Students unable to appear in the final exams may sit for the final makeup exams on the dates decided by the department/school which be within 4 weeks of the following semester. Students failing to appear in the make up final exams shall be awarded an ‘F’ Grade and shall have to repeat the course. Students who will be permitted to appear in the makeup exams will be assigned an ‘I’ grade for that course and this grade will stay until the student appears in the makeup exams at the first available opportunity; if s/he fails to appear in the makeup exams the ‘I’ grade will automatically be converted to an ‘F’ grade.”

Directions for Makeup Exam:

• Collect the make up form from the registrar’s office• Verify your medical certificate/documents along with your health card from BRACU Medical Center (for medical ground)• Take the approval of the course teacher/s• Take the approval from the Dean/Chair of your respective Department• Pay the makeup exam fee and Collect a clearance from accounts office located on the 6th Floor of BU6 • Submit the form to the Registrar’s Office (located on the 4th Floor of BU6) with sufficient documents.

Exam Policies and Procedures

Re-checking of Examination Scripts and Re-grading

A proposal for a grade change may be initiated by the course teacher and must be approved by the Chair of Department and the Dean. A grade change may be initiated on the grounds of posting errors or errors in calculation, incomplete grades or other identified procedural shortfalls or errors. The grade change application will be submitted to the Examination Controller.

A student may apply to the Chair of Department or the Dean for the re-checking of an examination script. Such a request may be made on the grounds of evidence of procedural errors, arbitrariness or prejudice. The student must make any request within Seven (07) calendar days of the publication of the mark. The Chair of Department or the Dean may, after checking with the original grader, arrange for the evaluation of an examination script to be conducted by two members of the faculty not previously involved in the examination and marking process, supported by a member of the Examination Controller’s office. The rechecking process and submission of any revised grade to the Examinations Controller should be completed within 15 calendar days.

Transcript

Transcript policy would be effective to all students enrolled in Summer-2022 and onward as followed:

There was a proposal that the transcript should, in the interests of transparency, show the final grade achieved by the student and the number of attempts/repeats/retakes. It has been decided that full grading details of all retakes and repeats will be recorded in the transcripts, in order to encourage utmost transparency.

Exam offence

Policy on offences and related disciplinary actions

Brac University’s (BracU) Residential Semester (RS) at Savar campus is unique amongst higher education experiences in Bangladesh. The RS offers a holistic curriculum based on the principle of ‘experiential learning’, which cultivates a broad range of soft skills and qualities, to compliment the theoretical development that students undergo.The curriculum’s intensive programs (delivered mainly in English) help to develop rounded individuals, whose improved leadership skills, ability to independently face challenges, and proficiency in English will better prepare them to face the challenges of higher education (and eventually give them a competitive edge in the job market).The Residential Semester takes place at a specially designed campus in Savar, which provides a support system that aids students in becoming confident and self-reliant.

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Get prepared for Residential Semester Notice FAQsAcademic Courses RS Handbook

Notice FAQs

Academic Courses RS Handbook

Residential Semester emphasises four areas:Improving communications skills, especially in English.Creating a strong sense of social responsibility and an awareness of each other's rights and responsibilities as members of society and citizens of a nation.Developing firm principles that guide students in all of their actions, whether in the personal, social, or public spheres, and teaching them to respect one another.Providing a holistic education, through academic, extracurricular and experiential activities.

Away from home

Get prepared for Residential Semester

Residential Semester emphasises four areas:

News and Events

RS Observes National Martyrs Day and International Mother Language Day

Prize giving ceremony for RS Championship held at RS

BRAC University celebrates students’ artistic diversity through cultural program

Prize giving ceremony for RS Championship held at Residential Campus

Self Care through Yoga and Meditation Retreat

Residential Semester Day - Fall 2019

International Volunteers Day 2019

Residential Semester Day - Summer 2019

“Celebrating International Women’s Day 2023: Women in Science” organized jointly by the Bangladesh Society of Microbiologists (BSM) and Microbiology Program of BRAC University. We are honoured to have Professor Dr Malabika Sarker from BRAC University and Professor Dr Farhana Karabi Biswas from North South University (NSU) as the invited speakers in this event.

The event will take place on March 12, 2023 (Sunday) from 3 pm to 4:30 pm at the UB2 auditorium at BRAC University.

Celebrating International Women’s Day 2023: Women in Science

BRAC Institute of Governance and Development (BIGD) is a social science research and academic institute, grounded in developing country context and practices, with a mission to improve governance and development outcomes.

BIGD conducts rigorous, multi-method social science research, grounded in developing country operations, on a range of social and economic issues to promote practical, time-sensitive innovations for tackling governance and development challenges. Our research spans innovations taking place in government, non-profit, and private sectors for improving lives.

BIGD leads the research on the socioeconomic empowerment programs of BRAC—the global leader among NGOs focusing on poor and marginal people—which gives us a unique opportunity to find innovative, effective, and scalable approaches to complex development challenges through formative and large-scale field research on BRAC interventions in Bangladesh. Our affiliation with the Independent Evaluation and Research Cell (IERC) of BRAC International helps us reach countries in South Asia and Africa.

BIGD's priority areas include economic development and growth, focusing on poverty, skills and jobs, international migration, access to finance, and small and medium enterprises, with an aim to promote equitable, sustainable economic development. We also focus on themes of gender and social transformation, exploring discrimination and inequalities faced by women and other vulnerable groups and associated norms, values, and sociopolitical factors. Moreover, BIGD's research looks into issues of governance and politics, honing in on accountability, institutional reform, deliberative democracy, governance efficiency, and public service. Recently, BIGD has been expanding research on critical emerging questions—how to use digital technology for inclusive development and how to safeguard the lives and livelihoods of the population vulnerable to climatic shocks.

BIGD's academic programs are geared towards developing high-performing next-generation leadership in the public and development sectors. We are one of the leading academic institutions in Bangladesh in the areas of governance and development, and we are increasingly becoming international, both in terms of the quality of education and student profile. We offer three master's programs:

Master of Development Studies (MDS), a pioneering development studies program in Bangladesh, with a strong emphasis on research and practice for enhancing analytical and practical skills required to address ongoing and emerging development challenges, with alumni spread across South Asia and Africa

Master of Arts in Governance and Development (MAGD), one of its kind in Bangladesh, developing effective future leadership in the public sector

Masters in Procurement and Supply Management (MPSM), also one of its kind, building capacities to improve transparency and efficiency in public and, increasingly, development and private sector procurement

BIGD also offers specialized technical and managerial skills training for improving development-focused outcomes.

We also offer specialized technical and managerial skills training for improving development-focused outcomes. We are offering:

• Post Graduate Diploma in Knitwear Industry Management (PGD-KIM)• CIPS-UK Qualifications in Procurement and Supply• Certificate Course in Academy of Work

Realizing the need for research-based evidence and insights to support continuous innovation and improvement in contemporary public policy and management, BRAC University established the Centre for Governance Studies (CGS) in 2005.

CGS was upgraded to a full-fledged Institute of Governance Studies (IGS) in July 2007. It was a pioneering governance research institute in Bangladesh, undertaking research on governance issues of critical public interest alongside offering a post-graduate degree and professional training programs.

BRAC University also established the BRAC Development Institute (BDI) in 2008 for promoting research and developing knowledge on practical solutions to the issues of poverty, inequity, gender, and social justice.

But development is inextricably linked with governance. So, in 2013, the BRAC University Board of Trustees decided to merge these two institutes into what is known today as BRAC Institute of Governance and Development (BIGD) to transform the institute into a regional centre of excellence for governance and development research; to bring together academics and practitioners to raise critical questions on development; to provide lessons on good practices; and to advocate for pro-poor policies.

Dr Sultan Hafeez Rahman, a reputed economist and a former Director General (South Asia) of Asian Development Bank (ADB), took the helm as Executive Director (ED) of the newly-established institute in 2013.

He was succeeded by Dr Imran Matin, who has worked extensively at the intersection between international development and research in Bangladesh and globally. Dr Matin took charge of BIGD as the new ED in July 2018 and since then, has been guiding the BIGD family with a vision to transform the organization into a globally-recognized centre for rigorous research on policy and development interventions.

In 2018, BRAC transitioned its internal Research and Evaluation Division (RED) to three institutes of BRAC University—BRAC Institute of Educational Development (BRAC IED), BRAC Institute of Governance and Development (BIGD), BRAC James P Grant School of Public Health (JPGSPH)—to enrich research and learning on development interventions through close academic partnership and to provide a unique opportunity to the university to incorporate knowledge with learnings from development interventions in its research and knowledge base. BIGD has been working closely with BRAC from the beginning, but after this recent transition, BIGD now provides continuous research support to BRAC’s socioeconomic empowerment program.

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News

BIGD Alumna listed among BBC's 100 most powerful and inspirational women globally in 2023

BIGD, IPA sign collaboration deal for greater development research impact

BIGD organizes ‘Digitalization and New Frontiers of Service Delivery: Opportunities and Challenges’

Brac University student selected for Next Generation Leaders Program

Newsletter

a. Theoretical Courses: APE 101 Mechanics, Properties of Matter, Waves & Oscillations 3 credits Vectors & scalars, unit vectors, scalar and vector products, scalar and vector fields, gradient, divergence and curl, curvilinear co-ordinates, 1 D & 2 D motions, work and energy, conservation laws, conservative force, projectile motion, uniform circular motion, rotation of rigid bodies, angular momentum, gravitation, gravitational field, potential. Elasticity, Hooke's Law, adhesive and cohesive forces, molecular theory of surface tension, capillarity. Streamline & turbulent flow, Poiseulle's formula, streamline flow and turbulent flow, Reynold's Number, Equation of Continuity, Bernoulli's Theorem, Stokes' Law. Principle of superposition, interference of waves, phase velocity and group velocity, simple harmonic motion, combination of SHM, Lissajous figures, damped SHM, forced oscillations, resonance, power and intensity of wave motion, waves in elastic media, vibration of strings, beats, Doppler Effect, velocity of sound, ultrasonics, and their applications. APE 102 Thermal Physics, Radiation & Statistical Mechanics 3 credits Heat and temperature, thermal equilibrium, specific heat & calorimetry, Newton's Law of Cooling, Kinetic Theory of Gases, Boltzmann Distribution Law, Brownian motion, Law of equipartition of energy; Vander Waals' equation of state, heat transfer, conduction, convection and radiation, co-efficient of thermal conductivity and its measurement, First Law of thermodynamics, isothermal & adiabatic changes, reversible and irreversible processes, Carnot's cycle, Second Law of thermodynamics, entropy and disorder, absolute scale of temperature, Maxwell's relations, Clausius-Clapeyron Equation, Gibb's phase rule, Third Law of thermodynamics, Nernst heat theorem, radiation theory, black body radiation, Wien's Law, Stefan-Boltzman Law, Rayleigh Jeans Law, Planck's Law, variation of specific heat with temperature, Einstein's theory, Debye's theory, Joule-Thomson expansion, cryogenics, measurement of high temperature. Statistical Mechanics: Phase space, concept of state and ensemble, microcanonical, canonical and grand canonical ensembles, Boltzmann probability distribution, Maxwell velocity distribution, derivation of Bose-Einstein and Fermi-Dirac statistics, ideal Fermi gas, degenerate Fermi system, equation of state of ideal gases, ideal Bose gas. APE 103 Electrical Circuits I 3 credits Circuit variables and elements: Voltage, current, power, energy, independent and dependent sources, resistance. Basic laws: Ohm's law, Kirchhoff's current and voltage laws. Simple resistive circuits: Series and parallel circuits, voltage and current division, Wye-Delta transformation. Techniques of circuit analysis: Nodal and mesh analysis including supernode and super mesh. Network theorems: Source transformation, Thevenin's, Norton's and Superposition theorems with applications in circuits having independent and dependent sources, maximum power transfer condition and reciprocity theorem. Energy storage elements: Inductors and capacitors, series parallel combination of inductors and capacitors. Responses of RL and RC circuits: Natural and step responses. Magnetic quantities and variables: Flux, permeability and reluctance, magnetic field strength, magnetic potential, flux density, magnetization curve. Laws in magnetic circuits: Ohm's law and Ampere's circuital law. Magnetic circuits: series, parallel and series-parallel circuits. Prerequisite PHY 115 APE 201 Solid State Physics and Materials Science 3 Credits Crystalline state, Bravais lattices, crystal symmetry, point group & space group, unit cells, Miller indices, x-ray diffraction, Bragg's Law, reciprocal lattice, structure factor, interatomic force and classification of solids, ionic, covalent, molecular, hydrogen bonded crystals, lattice energy of ionic crystals, Madelung constant, lattice vibration, phonons, normal modes in monatomic and diatomic linear chains, theory of specific heat, Einstein and Debye models, thermal expansion, defects in crystals, dislocations, consequences of defects on mechanical properties, elastic properties. Amorphous, composite, fibrous materials, polymers, plastics, binding forces, thermal and electrical conductivity of metals, dielectric properties of solids , modes of dielectric polarisation, ferro electricity, piezo electricity, optical properties of solids ,classical and semi classical theory, free carrier effects, lattice absorption, electronic absorption, magnetic properties of solid, dia and paramagnatism, ferro & ferrimagnetism, antiferromagnetism, ferrites, magnetic resonance, superconductivity, liquid crystals.

APE 202 Electrodynamics & Electromagnetic Waves & Fields 3 Credits Solution of Laplace's equation and Poisson's equation and applications to electrostatic problems, dielectrics, electrostatic energy, Maxwell's equations, electromagnetic waves, propagation of electromagnetic waves in conducting and non-conducting media, reflection and refraction, polarization, dispersion, scattering, waves in the presence of metallic boundaries, Waves between parallel planes, attenuation, wave impedance, waves in coaxial lines & modes, waves in strip and micro-strip lines, waveguides and resonators, solution of the inhomogeneous wave equations, simple radiating system, antennas, accelerated charge, Cerenkov radiation, elements of plasma physics. Prerequisite PHY 115. APE 203 Electrical Circuits II 3 credits Sinusoidal functions: Instantaneous current, voltage, power, effective current and voltage, average power, phasors and complex quantities, impedance, real and reactive power, power factor. Analysis of single phase ac circuits: Series and parallel RL, RC and RLC circuits, nodal and mesh analysis, application of network theorems in ac circuits, circuits simultaneously excited by sinusoidal sources of several frequencies, transient response of RL and RC circuits with sinusoidal excitation. Resonance in ac circuits: Series and parallel resonance. Magnetically coupled circuits. Analysis of three phase circuits: Three phase supply, balanced and unbalanced circuits, power calculation. Prerequisite APE 103 APE 204 Digital Logic Design 3 credits An introduction to digital systems such as computer, communication and information systems. Topics covered include Boolean algebra, digital logic gates, combinational logic circuits, decoders, encoders, multiplexers. Asynchronous and synchronous counters. Registers, flip-flops, adders, Sequential circuit analysis and design. Simple computer architecture. APE 205 Electronic Devices and Circuits I 3 credits P-N junction as a circuit element: Intrinsic and extrinsic semiconductors, operational principle of p-n junction diode, contact potential, current-voltage characteristics of a diode, simplified dc and ac diode models, dynamic resistance and capacitance. Diode circuits: Half wave and full wave rectifiers, rectifiers with filter capacitor, characteristics of a zener diode, zener shunt regulator, clamping and clipping circuits. Bipolar junction transistor (BJT) as a circuit element: Basic structure. BJT characteristics and regions of operation, BJT as an amplifier, biasing the BJT for discrete circuits, small signal equivalent circuit models, BJT as a switch. Single stage BJT amplifier circuits and their configuarations: Voltage and current gain, input and output impedances. Metal-Oxide-Semiconductor Field-Effect-Transistor (MOSFET) as circuit element: structure and physical operation of MOSFETs, body effect, current-voltage characteristics of MOSFETs, biasing discrete and integrated MOS amplifier. Prerequisite APE 103 APE 302 Electronic Devices and Circuits II 3 credits Frequency response of amplifiers: Poles, zeros and Bode plots, amplifier transfer function, techniques of determining 3 dB frequencies of amplifier circuits, frequency response of single-stage and cascade amplifiers, frequency response of differential amplifiers. Operational amplifiers (Op-Amp): Properties of ideal Op-Amps, non-inverting and inverting amplifiers, inverting integrators, differentiator, weighted summer and other applications of Op-Amp circuits, effects of finite open loop gain and bandwidth on circuit performance, logic signal operation of Op-Amp, dc imperfections. General purpose Op-Amp: DC analysis, small-signal analysis of different stages, gain and frequency response of 741 Op-Amp. Negative feedback: properties, basic topologies, feedback amplifiers with different topologies, stability, frequency compensation. Active filters: Different types of filters and specifications, transfer functions, realization of first and second order low, high and bandpass filters using Op-Amps. Signal generators: Basic principle of sinusoidal oscillation, Op-Amp RC oscillators, LC and crystal oscillators. Power Amplifiers: Classification of output stages, class A, B and AB output stages. Prerequisite APE 205 APE 400 Thesis / Project 3 credits A student is required to carry out thesis/project work in the 7th and 8th semester in a chosen field. There will be a supervisor who will either be a BRAC University faculty or any other suitable expert from universities and R/D organizations of the country to guide the thesis/ project work .On completion of study and research s/he will have to submit the dissertation/report and face a viva board for the defence. APE 401 Measurements & Measuring Instruments 3 Credits Significance and methods of measurements, direct and indirect methods. Mechanical, electrical and electronic types of instruments, absolute and secondary instruments, analog and digital instruments, analog voltmeters and ammeters, AC transformer types, Flux gate magnetometer type. Accuracy and error of analog voltmeters and ammeters. Different types of Digital voltmeters, digital multimeters, Automation in multimeters. Oscilloscopes, signal generators. Transducers. Absorption and detection of radiation, Nucleonic instruments. Analytical & medical instruments, Industrial instruments, Instrument systems. Prerequisite APE 302 APE 402 Plasma Physics with Industrial Applications 3 credits General introduction to plasma physics, plasma as a fourth state of matter, definition, screening and Debye shielding, plasma frequency, ideal plasma, temperature and pressure of plasma, magnetic pressure and plasma drifts, plasma waves, Landau damping, collisions in plasmas, hydrodynamic description of plasma, one fluid model, two fluid model, Chew-Goldberg theory, low waves in maneto-hydrodynamics, description of plasma, dielectric tensor, longitudinal and transverse waves, plasma instabilities, transport in plasmas, plasma kinetic theory, Vlasov equation, linear waves, waves in magnetized plasma, electromagnetic waves, waves in hot plasmas, nonlinear waves, Landau damping, quasi linear theory, plasmas in fusion research, plasmas in industrial applications. APE 403 Control Engineering 3 credits Introduction to control systems, electric circuits and components, transfer function and block diagram, mechanical translation systems, analogous circuits, mechanical rotational systems, rotating power amplifiers, DC & AC servomotors. Inputs & responses, modeling of continuous systems; computer-aided solutions to systems problems; feedback control systems; stability, frequency response and transient response using root locus, frequency domain and state variable methods. Position control system, simulation diagrams, signal flow graphs, parallel state diagrams from transfer function. General frequency transfer function relationships, drawing the Bode plot, system type and gain as related to log magnitude curve, Nyquist's criterion and applications. Prerequisites ECE 220 and MAT 203 APE 404 Microprocessors and Assembly Language Programming 3 credits Introduction to different types of microprocessors. Microprocessor architecture, instruction set, interfacing/O operation, interrupt structure, DMA. Microprocessor interface ICs. RAM, ROM, PROM, EPROM and EPROM's. Advanced microprocessor concept of microprocessor based system design. Microcomputer systems, representation of numbers and characters, introduction to IBMPC assembly language. Logic, shift, multiplication & division instructions, arrays and addressing modes, string instructions, text display and keyboard programming, memory management. Prerequisite APE 204 APE 405 Computer Organization and Architecture 3 credits A systematic study of the various elements in computer design, including circuit design, storage mechanisms, addressing schemes, and various approaches to parallelism and distributed logic. Information representation and transfer; instruction and data access methods; CPU structure and functions processor and register organization, instruction cycles and pipe linings, the control unit; memory organisation. RISC and CISC machines. The course includes a compulsory 3 hour laboratory work each week. Prerequisite APE 204. APE 406 Radar Engineering 3 creditsThe course is oriented towards the understanding and design of radar systems. The contents will be radar principles & techniques, nature of radars, radar frequencies, radar target, radar equation, continuous & frequency modulated radars, detection & processing of radar signals, MTI radar, Pulse Doppler radar, tracking radar, radar indicators and displays, noise, ground & sea echoes & clutter, weather effect of radar, radar applications. Prerequisites: ECE 220 and MAT 203 APE 407 The Physics of Energy 3 credits Energy & development, energy consumption, world energy demand & future projection, energy units, earth's energy resource base, renewable and non-rewable sources of energy. Non-renewable energy sources-fossil fuels, coal, natural gas, petroleum, etc. and non-fossil fuels like uranium (fission energy). Renewable energy sources-solar energy, wind energy, tidal & wave & ocean energy, geothermal energy, biomass & hydropower, hydrogen energy. Advantages of renewable over conventional technologies, solar thermal conversion, radiation characteristics of materials, solar collectors, solar photovoltaic energy conversion, photovoltaic cells, design of PV systems, wind turbines, biopower, biofuels, integrated bioenergy systems, geothermal heat pumps, hydroelectricity & micro hydroelectric power, ocean thermal energy conversion. Energy, sustainability and environment, EIA. BI0 101 Introduction to Biology 3 credits An introduction to the cellular aspects of modern biology including the chemical basis of life, cell theory, energetics, genetics, development, physiology, behaviour, homeostasis and diversity, and evolution and ecology. This course will explain the development of cell structure and function as a consequence of evolutionary process, and stress the dynamic property of living systems. CHE 101 Introduction to Chemistry 3 credits The course is designed to give an understanding of basics in chemistry. Topics include nature of atoms and molecules; valence and periodic tables, chemical bonds, aliphatic and aromatic hydrocarbons, optical isomerism, chemical reactions. ENV 101 Introduction to Environmental Science 2 credits Fundamental concepts and scope of environmental science, Earth's atmosphere, hydrosphere, lithosphere and biosphere, men and nature, technology and population, ecological concepts and ecosystems, environmental quality and management, agriculture, water resources, fisheries, forestry and wildlife, energy and mineral energy sources; renewable and non renewable resources, environmental degradation; pollution and waste management, environmental impact analysis, remote sensing & environmental monitoring. ENV 103 Elements of Environmental Sciences 3 credits Fundamental concepts and scope of environmental science, Earth's atmosphere, hydrosphere, lithosphere and biosphere, men and nature, technology and population, ecological concepts and ecosystems, environmental quality and management, agriculture, water resources, fisheries, forestry and wildlife, energy and mineral energy sources; renewable and non renewable resources, environmental degradation; pollution and waste management, environmental impact analysis, remote sensing & environmental monitoring.

GEO 101 Introduction to Economic Geography 3 credits Introduction: The field and environment of economic Geography; Bases of Economic Geography: Relief, Climate, Vegetation, Soils and Population; Extractive resources and human-environment relations; Primary Activities: types and brief descriptions; Secondary Activities: types and factors of localization, Stages in growth; Tertiary Activities: Trade, Transportation, Utilities, Technical and Professional services; Regional Economy: classification, Growth and Development; Economic Geography of Bangladesh: A brief account. MAT 091 Basic Course in Mathematics No credit Topics including sets, relations and functions, real and complex numbers system, exponents and radicals, algebraic expressions; quadratic and cubic equations, systems of linear equations, matrices and determinants with simple applications; binomial theorem, sequences, summation of series (arithmetic and geometric), permutations and combinations, elementary trigonometry; trigonometric, exponential and logarithmic functions; co-ordinate geometry;

statics-composition and resolution of forces, equilibrium of concurrent forces; dynamics-speed and velocity, acceleration, equations of motion. No credit. MAT 101 Fundamentals of Mathematics 3 credits The real number system, exponents, polynomial, factoring, rational expression, radicals, complex number, linear equation, quadratic equation, variation, inequalities, coordinate system, functions, equations of line, equation of circle, exponential and logarithmic function, system of equations, system of inequalities properties of matrix, matrix solution of linear system, determinant, Cramer's rule, limit, rate of change, derivative. MAT 102 Introduction to Mathematics 3 credits Factorisation, Synthetic Division, Zeros (Roots) of Polynomials, Relation between Roots and Coefficients, Nature of Roots (Descarte's Rule of signs); Complex Number System, Graphical representation of Complex Numbers (Argand Diagram), Polar form of Complex Numbers; Conic Sections, Parabola, Circle, Ellipse, Hyperbola, Transformation of Coordinates and Applications; Exponential Growth & Decay. Applications; Mathematical Induction; Determinants, Fundamental Properties of Determinants, Minors and Cofactors, Application of Determinants to solve System of Linear Equations (Cramers, Rule); Introduction to Matrix Algebra, Matrix Multiplication, Augmented Matrix, Adjoint Matrix, Inverse Matrix, Application of Matrices-solution of System of Linear Equations (homogeneous & non-homogeneous), Consistency of System of Equations.

MAT 103 Basic Concepts in Mathematics 3 creditsThe real numbers, Absolute value of real numbers, Exponents, Polynomials, Basic operation and Factoring of polynomials, Rational expressions, Radicals. Linear Equations, Solution, graphs and applications. Variation, Linear inequalities. Exponential and Logarithmic Functions, Exponential growth and decay, Ratios, proportions, percent, application of simple and compound interest. Trigonometric Functions, The Sine and Cosine Functions, Cartesian coordinate systems, Graphing, Relations. Equations of a straight line its slope, Equation of a circle, Systems of Linear Equations, Matrix. Population, Sample, Variable, Raw data, Frequency distribution table, Graphical presentation, Measures of central tendency and measures of dispersion.   MAT 104 Mathematics 2 credits Calculus, definition of limit, continuity and differentiability, successive and partial differentiation, maxima and minima. Integration by parts, standard integrals, definite integrals. Solid geometry, system of coordinates. Distance between two points. Coordinate Transformation, Straight lines sphere and ellipsoid.   MAT 105 Calculus 3 credits Differential Calculus: Limits, continuity and differentiability, differentiation, Taylor's, Maclaurine's & Euler's theorems, indeterminate forms, tangent and normal, sub tangent and subnormal, maxima and minima, radius of curvature & their applications, introduction to calculus of function of several variables, Taylor's theorem, maxima and minima for function of several variables. Transformation of coordinates & rotation of axes, conic sections. Integral Calculus: Definition of integration, techniques of integration for definite & indefinite integrals, improper integrals, area, volume and surface integration, arc length and their applications, multiple integrals, Jacobian, line integrals, divergence theorem and Stokes' theorem, beta function and gamma function.   MAT 110 MATH I: Differential Calculus and Co-ordinate Geometry 3 credits Differential Calculus: Limits, Continuity and differentiability. Differentiation. Taylor's Maclaurine's & Euler's theorem. Indeterminate forms. Partial differentiation. Tangent and normal. Subtangent and subnormal. Maximum and minimum, radius of curvature & their applications. Co-ordinate Geometry: Transformation of coordinates & rotation of axis. Pair of straight lines. General equation of second degree. System of circles. Conics section. Tangent and normal, asymptotes & their applications.   MAT 111 Principles of Mathematics 3 credits Sets: Elementary idea of Set, Set notations, Set operations: union, intersection, complement, difference; Set operations and Venn diagrams. Set of Natural numbers, Integers, Rational numbers, Irrational numbers and Real numbers alongwith their geometrical representation, Idea of Open and Closed interval,. Idea of absolute value of real number, Variables and Constants, Product of two sets: Idea of product of sets, Product set of real numbers and their geometric representation, Axioms of real number system and their application in solving algebraic equations. Equation and Inequality, Laws of inequality, Solution of equations and inequalities. Variable and Functions: Variable of a set, Functions of single variable, Polynomial, Graph of Polynomial functions in single variable. Exponential, Logarithmic, Trigonometric functions and their graphs, Permutation and Combination. Binomial theorem. MAT 120 MATH II Integral Calculus and Differential Equations 3 credits Integral Calculus: Definitions of integration. Integration by the method of substitution. Integration by parts. Standard integrals. Integration by method of successive reduction. Definite integrals, its properties and use in summing series. Walli's formula. Improper integrals. Beta function and Gamma function. Area under a plane curve in Cartesian and polar coordinates. Area of the region enclosed by two curves in Cartesian and polar coordinates. Trapezoidal rule. Simpson's rule. Arc lengths of curves in Cartesian and polar coordinates, parametric and pedal equations. Intrinsic equations. Volumes of solids of revolution. Volume of hollow solids of revolutions by shell method. Area of surface of revolution. Ordinary Differential Equations: Degree of order of ordinary differential equations. Formation of differential equations. Solution of first order differential equations by various methods. Solutions of general linear equations of second and higher order with constant coefficients. Solution of homogeneous linear equations. Applications. Solution of differential equations of the higher order when the dependent and independent variables are absent. Solution of differential equations by the method based on the factorisation of the operators. [Students will be expected to attend a 3 hour tutorial class, once each week and submit tutorial worksheets.] Prerequisite: MAT 110 MAT 121 Basic Algebra 3 credits Elements of logic: Mathematical statements, Logical connectives, Conditional and biconditional statements, Truth tables and tautologies, Quantifications, Logical implication and equivalence, Deductive reasoning. The concept of sets: Sets and subsets, Set operations, Family of Sets. Relations and functions:

Cartesian product of two sets, Relations, Order relation, Equivalence relations, Functions, Images and inverse images of sets, Injective, surjective, and bijective functions, Inverse functions. Real number system: Field and order properties, Natural numbers, integers and rational numbers. Absolute value, Basic inequalities. (Including inequalities of means, powers, Weierstrass, Cauchy) Complex number system: Field of complex numbers, De Moivre's theorem and its applications. Elementary number theory: Divisibility, Fundamental theorem of arithmetic, Congruence. Summation of finite series: Arithmetic-geometric series, Method of difference, Successive differences, Use of mathematical induction. Theory of equations: Synthetic division, Number of roots of polynomial equations, Relations between roots and coefficients, Multiplicity of roots, Symmetric functions of roots, Transformation of equations. MAT 122 Analytic and Vector Geometry 3 credits Two dimensional geometry Coordinates: Cartesian and polar Coordinates, Transformations of coordinates and its applications. Reduction of second degree equations to standard forms, Pairs of straight lines, Circles, Identification of conics, Equations of conics in polar Coordinates. Three dimensional geometry Coordinates in three dimensions, Direction cosines, and Direction ratios. Planes, straight lines, shortest distance, sphere, orthogonal projection and conicoids. Vector geometry Vectors in plane and space, Algebra of vectors, scalar and vector products, Triple scalar products, its applications to Geometry. MAT 123 Calculus I 3 credits Differential Calculus: Real number system and its geometrical representation, real variable, function of single (real) variable, parametric equations, limit, continuity and differentiability, derivatives of different types of functions, geometrical significance of derivative, Rolle's Theorem, Mean Value Theorem, Taylor's Theorem; maxima, minima, point of inflexion, concavity and convexity, sketching of curves using concepts of calculus; Indeterminate Form, L'Hospital Rule, Successive Differentiation, Leibnitz's Theorem, tangent, normal and related formulas, curvature. Integral Calculus: Indefinite integrals of different types of functions, various methods of integrations, definite integrals, Fundamental Theorems of Definite Integrals, properties of definite integrals, Reduction formulas, Arc Length, Area under the curves, Surface area and Volume of a 3-D objects. Improper Integrals and applications. Prerequisite: MAT122 MAT 124 Scientific Programming Problem-solving techniques using computers: Flowcharts, Algorithms, Pseudocode. Programming in FORTRAN: Syntax and semantics, data types and structures, input/output, loops, decision statements, arrays, user-defined functions, subroutines and recursion. Computing using FORTRAN: Construction and implementation of FORTRAN programs for solving problems in mathematics and sciences. MAT 203 Matrices, Linear Algebra & Differential Equations 3 credits Matrices: Types of matrices, algebraic operation on matrices, determinants, adjoint & inverse matrix, orthogonality & diagonalization of matrix. Linear Algebra: System of linear equations, vector space; 2D- space, 3D- space, Euclidean nD- space, sub space, linear dependence, basis and dimension, row space, column space, rank and nullity, linear transformation, eigen value and eigen vector, matrix diagonalization and similarity, application of linear algebra. Ordinary Differential Equations: Introduction to differential equations, first-order differential equations and applications, higher order differential equations and applications, series solutions of linear equations, systems of linear first-order differential equations. Prerequisite MAT 105 MAT 204 Complex Variables & Fourier Analysis 3 credits Complex Variables: Complex number systems, general functions of a complex variable, limits and continuity of a function of complex variables and related theorems, complex differentiation and Cauchy-Riemann equations, mapping by elementary functions, line integral of a complex function. Cauchy's integral theorem, Cauchy's integral formula, Liouville's theorem, Taylor's and Laurent's theorem, singular points, residue, Cauchy's residue theorem, evaluation of residues, contour integration and conformal mapping. Fourier analysis: Real and complex form, finite Fourier transform, Fourier integrals, Fourier transforms and their use in solving boundary value problems. Prerequisite MAT 105 MAT 205 Introduction to Numerical Methods 3 credits Computer arithmetic: floating point representation of numbers, arithmetic operations with normalized floating point numbers; iterative methods, different iterative methods for finding the roots of an equation f (x) = 0 and their computer implementation; solution of simultaneous algebraic equations by various methods, solution of tri-diagonal system of equations, interpolation for equispaced and non-equispaced nodes, least square approximation of functions, curve fitting, Taylor series representation, Chebyshev series, numerical differentiation and integration and numerical solution of ordinary differential equations & partial differential equations. Prerequisite MAT 203 MAT 211 Calculus II 3 credits Functions of several variables, concept of surface, sketching of , contour sketch for surface, limit and continuity, partial derivative and its geometrical significance, chain rule of partial differentiation, concept of gradient, divergence and curl, directional derivative and tangent plane, concept of differential and perfect differential, linear approximation and increment estimation, maxima, minima and saddle point, Lagrange multiplier, higher order derivatives, Taylor's theorem of function of several variables. Multiple integrals: Double integrals, Double integrals in Polar coordinates, Triple integrals, Triple integrals in Cylindrical and Spherical coordinates, Change of variables in Multiple integrals, Jacobian, Line integrals, Green's theorem, Surface integrals, Applications of Surface integrals, Divergence theorem, Stoke's theorem. Prerequisite: MAT123. MAT 212 Linear Algebra 3 credits Introduction to matrix, different types of matrices, equivalent matrices, determinants, properties of determinants, minors, cofactors, evaluation of determinants, adjoint matrix, inverse matrix, method for finding inverse matrix, elementary row operations and echelon form of matrix, system of linear equations (homogeneous and non-homogeneous equations) and their solutions; Vector, vector spaces and subspaces, linear independence and dependence, basis and dimension, change of bases, rank and nullity, linear transformation, kernel and images of a linear transformation and their properties, eigenvalues and eigenvectors, diagonalization, Cayley Hamilton theorem, MAT 215 MATH III Complex Variables and Laplace Transformations 3 credits Complex Variables: Complex number systems. General functions of a complex variable. Limits and continuity of a function of complex variables and related theorems. Complex differentiation and Couchy-Riemann equations. Mapping by elementary functions. Line integral of a complex function. Cauchy's integral theorem. Cauchy's integral formula. Liouville's theorem. Taylor's and Laurent's theorem. Singular points. Residue. Cauchy's residue theorem. Evaluation of residues. Contour integration. And conformal mapping Laplace Transforms: Definition. Laplace transforms of some elementary functions. Sufficient conditions for existence of Laplace transforms. Inverse Laplace transforms. Laplace transforms of derivatives. The unit step function. Periodic function. Some special theorems on Laplace transforms. Solutions of differential equations by Laplace transformations. Evaluation of improper integrals. Prerequisite: MAT120 MAT 216 MATH IV Linear Algebra and Fourier Analysis 3 Credits Linear Algebra: Basic subject on matrix theory and linear algebra, emphasizing topics useful in other discipline, including systems of equations, vector spaces, determinants, Eigenvalues, similarity, and positive definite matrices, Applications to Gauss elimination with pivoting. Fourier Analysis: Real and complex form. Finite transform. Fourier integral. Fourier transforms and their uses in solving boundary value problems. Multiple integrals; surface and volume integrals, divergence and Stoke's theorem. Prerequisite: MAT 120 MAT 221 Real Analysis I 3 credits Real number system: Completeness of real numbers, supremum and infimum principles and their consequences, Dedekind's theorems, Bolzano-Weierstrass theorem. Sequences of Real Numbers: Infinite sequence, Convergent sequences, Monotone sequences, subsequences, Cauchy sequence, Cauchy criteria for convergence of sequences. Infinite Series: Concept of sum and convergence, series of positive terms, alternating series, absolute and conditional convergence, test for convergence, Convergence of sequences and series of functions. Continuity and Limits: Properties of continuous functions, Extreme Value Theorem and Intermediate Value Theorem, Uniform continuity concepts, Limits, Heine-Borel theorem. Integration: Necessary and sufficient conditions for integrability, Darboux Sums and Riemann Sums, Improper integral and their tests for convergence. MAT 222 Differential Equations I 3 credits Ordinary differential equations and their solutions: Classification of differential equations, Solutions, Implicit solutions, Singular solutions, Initial Value Problems, Boundary Value Problems, Basic existence and uniqueness theorems (statement and illustration only), Solution of first order equations: Separable equations, Linear equations, Exact equations, Special integrating factors, Substitutions and transformations, Modeling with first order differential equations, Model solutions and interpretation of results, Orthogonal and oblique trajectories, Solution of higher order linear equations: Linear differential operators, Basic theory of linear differential equations, Solution space of homogeneous linear equations, Fundamental solutions of homogeneous solutions, Reduction of orders, Homogeneous linear equations with constant coefficients, Non homogeneous equation, Method of undetermined coefficients, Variation of parameters, Euler Cauchy differential equation, Modeling with second order equations, Initial Value Problems and Boundary Value problems, reduction of order, Euler equation, generating functions, eigenvalue problems. Inhomogeneous linear difference equations (variation of parameters, reduction of order, Series solutions of second order linear equations: Taylor series solutions about an ordinary point. Frobenious series solutions about regular singular points. Series solutions of Legendre, Bessel, Laguerre and Hermite equations. Systems of linear first order differential equations: Elimination method. Matrix method for homogeneous linear systems with constant coefficients. Variation of parameters. Matrix exponential. Prerequisite: MAT211

MAT 223 Numerical Analysis I 3 credits Solution of equation of single variable: Fixed point iteration, Bisection algorithm, Method of False position, Newton-Raphson's method, Error Analysis for Iterative methods, Accelerating limit of convergence. Interpolation: Interpolating polynomials for equispaced and nonequispaced nodes, Lagrange's polynomial, Newton-Gregory's Interpolating polynomials, curve fitting with Least Square method, Iterated interpolation, Extrapolation. Differentiation and Integration: Numerical differentiation, Single point and (n+1)-point formulae of differentiation, Richardson's extrapolation, Numerical Integration, Gaussian quadrature formula, Trapezoidal, Simpson's, Weddle's Rules. Prerequisite: MAT124

MAT 301 Group Theory 3 Credits Definition and various examples of groups, subgroups, cosets, normal subgroups, quotient (factor) groups, permutation groups, cyclic groups, generator of a cyclic group, centre of a group, Abelian group, normalizer and centralizer of an element/ subset of a group and its application to physics, group homomorphism, isomorphism and automorphism and related theorems, symmetry groups, SU (3), SU (6), application of group theory in solid state physics & elementary particles. MAT 303 Tensor Analysis 3 Credits Definition of tensor, tensor density, affine tensor and geometrical object, properties of tensor symmetry, criteria of tensor properties, metric tensor, Kronecker symbol and LeviCivita's symbol, determinant of metric tensor, connection between metric tensor and Dirac's matrices in the Sommerfeld representation, evolution of square root from four dimensional interval in matrix sense, transformation properties of vector partial derivatives by coordinates, connection coefficients and covariant derivatives, Christoffel's symbols, geodetic lines (geodetics) as a generalization of notion of straight line, variation principle for geodetics, parallel transport, connection between geodetics and covariant differentiation, transport along closed line curvature tensor of the 4th rank, curvature tensor of the 2 D rank, scalar curvature, equations of geodetic deviation, curvature expression in terms of Dirac's matrices, Bianchi's identity, Einstein's conservative tensor, integral operations and corresponding theorems. MAT 311 Abstract Algebra 3 credits Equivalence relation and residue classes modulo n. Groups and subgroups, Cyclic groups, Symmetric groups. Cosets and Lagrange's Theorem: Normal subgroups, Quotient groups, Permutation groups, Homomorphism, Isomorphism and Automorphism of groups with related theorems and problems, Cayley's Theorem, centralizer and normalizer of an element/ subset in a group. Rings, Ideals, and Quotient Rings, Prime and Maximal Ideals. Integral Domain, Field of fractions. Principal Ideal Domain, Euclidian Domain, Unique Factorization Domain. Polynomial Rings, Primitive polynomials, Gauss Theorem, Eisenstein's criterion for irreducibility. Prime Fields, characteristic of Fields. Prerequisite: MAT121 MAT 312 Numerical Analysis II 3 credits Part A: Theory Solutions of linear system of equations: Gaussian Elimination method with pivoting, Matrix inversion, Direct factorization of matrices, Iterative Techniques for solving linear system of equations: Jacobi's and Gauss-Seidel Method. Solution of tridiagonal system, Eigen values and Eigen vectors (Power Method). Numerical solution of Nonlinear system: Fixed point for functions of several variables, Newton's method, Quasi-Newton's method. Initial Value Problem for ODE: Euler's method, Higher order Taylor's method, Runge-Kutta methods, Multistep methods, Variable Stepsize Multistep methods. Boundary Value Problem: Linear Shooting method, Shooting method for non linear BVP. Boundary Value problem involving elliptic, parabolic and hyperbolic equations, explicit and implicit Finite Difference method. Part B: Numerical Analysis Lab Construction and implementation of FORTRAN / C, C++ programs of techniques in Numerical Analysis. There will be at least 15 lab assignments. Prerequisite: MAT 223

MAT 313 Differential Geometry 3 credits Curves in space: Vector functions of one variable, space curves, unit tangent to a space curve, equation of a tangent line to a curve, Osculating plane (or plane of curvature), vector function of two variables, tangent and normal plane for the surface , Principal normal, binormal and fundamental planes, curvature and torsion, Serret Frenet's formulae, theorems on curvature and torsion, Helices and its properties, Circular helix. Spherical indicatrik, Curvature and torsion. Curvature and torsion for spherical indicatrices. Involute and Evolute of a given curve, Bertrand curves. Surface: Curvilinear coordinates, parametric curves, Metric (first fundamental form), geometrical interpretation of metric, relation between coefficients

E, F, G. properties of metric, angle between parametric curves, elements of area, second fundamental form. Derivatives of surface normal M (Weingarten equations), Third fundamental form, Principal sections, Principal sections, direction and curvature, first curvature, mean curvature, Gaussian curvature, normal curvature, lines of curvature, centre of curvature, Rodrigues's formula, condition for parametric curves to be line of curvature, Euler's Theorem, Elliptic, hyperbolic and parabolic points, Dupin Indicatrix. MAT 314 Complex Analysis 3 credits Introduction to complex numbers and their properties, complex functions, limits and continuity of complex functions, Analytic functions, Cauchy Riemann equations, harmonic functions, Rational functions, Exponential functions, Trigonometric functions, Logarithmic functions, Hyperbolic functions. Contour integration: Cauchy's Theorem, Simply and Multiply connected domain, Cauchy integral formula, Morera's theorem, Liouville's theorem. Convergent series of analytic functions, Laurent and Taylor series, Zeroes , Singularities and Poles, residues, Cauchy's Residue theorem and its applications, Conformal Mapping. Prerequisite: MAT123 MAT 315 History of Mathematics 3 credits A Survey of the development of mathematics beginning with the history of numeration and continuing through the development of the calculus. The study of selected topics from each field is extended to the 20th century. Biographical and historical aspects will be reinforced with studies of procedures and techniques of earlier mathematical cultures. MAT 316 Operations Research I 3 credits Convex sets and related theorems, Introduction to linear programming, Formulation of linear programming problems, Graphical solutions, Simplex method, Duality of linear programming and related theorems, Sensitivity. Unconstrained optimization: Newton's method, Trust region algorithms, Least Squares and zero finding. Constrained optimization: linear/nonlinear equality/inequality constraints, Duality, working set methods. Linear programming: Simplex method, primal dual interior point methods. MAT 321 Real Analysis II 3 credits Metric spaces: definition and some examples, open sets, closed sets, Convergence, Completeness, Baire's theorem. Connected set: Compact sets, locally compact sets and related theorems, connected sets, locally connected sets, continuity and compactness. Sequence in metric space: Convergent and Cauchy sequence, Completeness, Banach Fixed Point theorem with applications, sequence and series of functions, pointwise and uniform convergence, differentiation and integration of series. Continuous function on metric space: Boundedness, Intermediate Value Theorem, uniform continuity. Differentiation in Ñ : Jacobian , implicit and inverse function theorems. Integration in Ñ : contents and integrals, Fubini's theorem, change of variables. Prerequisite: MAT221

MAT 322 Differential Equations II 3 credits ODE: Existence and uniqueness theory: Fundamental existence and uniqueness theorem. Dependence of solutions on initial conditions and equation parameters. Existence and uniqueness theorems for systems of equations and higher order equations. Eigen value problems and Strum-Liouville boundary value problems: Regular Strum-Liouville boundary value problems. Solution by eigenfunction expansion. Green's functions. Singular Strum-Liouville boundary value problems. Oscillation and comparison theory. Nonlinear differential equations: Phase plane, paths and critical points. Critical points and paths of linear systems. PDE: First order equations: complete integral, General solution. Cauchy problems. Method of characteristics for linear and quasilinear equations. Charpit's method for finding complete integrals. Methods for finding general solutions. Second order equations: Classifications, Reduction to canonical forms. Boundary value problems related to linear equations. Applications of Fourier methods (Coordinates systems and separability. Homogeneous equations.) Boundary value problems involving special functions. Transformation methods for boundary value problems, Applications of the Laplace transform. Application of Fourier sine and cosine transforms. Prerequisite: MAT222.

MAT 323 Dynamical Systems 3 credits Statics: Fundamental concept and principle of Mechanics. Statics of Particles: Review of vectors, vector addition of forces, resultant of several concurrent forces, resolution of forces into components, equilibrium of particles in a plane and in space. Rigid bodies: momentum of a force and a couple, Varignon's theorem, equivalent system of forces and vectors, reduction of system of forces. Equilibrium of rigid bodies: reactions at supports and connections of rigid bodies in two dimensions. Centroid and center of gravity: CG of two and three dimensional bodies, centroids of areas, lines and volumes. Moment of inertia, moments and products of inertia, radius of gyration, parallel axis theorem, principal axis and principal moments of inertia. Dynamics: Kinematics of particles: rectilinear and curvilinear motion of particles. Kinematics of particles: Newton's second law of motion, linear and angular momentum of a particle, conservation of energy and momentum, principle of work and energy and their applications, motion under a central force and conservative central force, impulsive motion. System of particles: Newton's law, effective forces, linear and angular momentum, conservation of momentum and energy, work energy principle. Kinematics of rigid bodies: translation, rotation, and plane motion relative to rotating frame, Coriolis force. Plane motion of a rigid body: motion in two dimensions, Euler's equation of motion about a fixed point. Prerequisite: MAT122. MAT 324 Graph Theory 3 credits Number System: Numbers with different bases, their conversion and arithmetic operations, normalized scientific notations. Logic: Introduction to logic, logical operations, application of logic to sets. Mathematical Reasoning: Methods of proof, Mathematical induction, recurrence relations, generating functions. Boolean Algebra: Ordered sets, lattices, Boolean algebra and operations, Boolean expressions, logic gates, minimization of Boolean expressions, Karnaugh maps, Karnaugh map algorithm. Graphs: introduction and definitions, representing graphs, graph isomorphism, connected graph, planar graph, path and circuit, shortest path algorithm, Eulerian path, Euler's theorem, Seven Bridges of Problem, graph coloring. Application of graph: tree, tree reversal, trees and sorting, spanning trees, minimum spanning trees: related algorithms. Search trees: binary search tree, leaves on a rooted tree, spanning trees and the MST problems, network and flows, the max flow and the min-cut theorem.

Binary trees. MAT 325 Mathematical Methods 3 credits Series solution: singularity of a rational function, series solution of linear differential equations at nonsingular and regular singular points. Fourier series: Introduction to orthogonal functions and Integral transform, Fourier integral, Fourier transform and their applications. Laplace transformation method: Definition, existence and properties of Laplace transform, Inverse Laplace transform, Transforms of discontinuous and periodic functions. Convolution. Impulses and Dirac delta function. Solving initial value problems. Solving linear systems, Harmonic functions: Laplace equation in different coordinates and its applications. Special functions: Legendre functions of first and second kinds, Hermite polynomials, generating function, Hypergeometric functions, Laguerre functions, Bessel function and their properties. Prerequisite: MAT 322 MAT 326 Hydrodynamics 3 credits Preliminaries: Concept of viscosity; Inviscid fluid; stream line, path line and streak lines; steady and unsteady motion. Equation of motion: Equation of continuity; Euler's equation of motion, conservative forces, Bernoulli's equation; circulation and Kelvin's circulation theorem; vorticity, irrotational and rotational motion, velocity potential; energy equation, Kelvin's minimum energy theorem. Two dimensional motion: vorticity, stream function and velocity potential function, streaming motion, complex potential and complex velocity, stagnation points, motion past a circular cylinder, circle theorem, motion past a cylinder, Joukowaski transformation, Blasius theorem; two dimensional source, sink and doublets, source and sink in a stream, the method of image. Vortex motion: vortex line, tube and filament, rectilinear and circular vortices, kinetic energy of system of vortices, vortex sheet, Karman's vortex street.

MAT 400 Project /Thesis 3 Credits A student is required to carry out project / thesis work in the last two semesters in her/his chosen field. There will be a supervisor who will either be a BRAC University faculty or any other suitable expert from universities and R/D organizations of the country to guide the project / thesis work .On completion of study and research s/he will have to submit the dissertation report and to face a viva board for the defence of the dissertation. MAT 411 Topology 3 credits Metric Spaces: Definition and some examples. Open sets. Closed sets. Convergence. Completeness. Baire's theorem. Continuous mappings. Spaces of continuous functions. Euclidean and unitary spaces. Topological Spaces: Definition and some examples. Elementary concepts. Open bases and open subbases. Weak topologies. Function algebras. Compactness: Compact spaces. Product spaces. Tychonoff's Theorem. Locally compact spaces. Compactness for metric spaces. Separation: T1-spaces and Hausdorff spaces. Completely regular spaces and normal spaces. Urysohn's lemma. Connectedness: Connected spaces. Locally connected spaces. Pathwise connectedness. Banach Spaces: Definition and some examples. Continuous linear transformations. Hahn-Banach theorem. Natural embedding. Open mapping theorem. Conjugate of an operator. Hilbert Spaces: Definition and some simple properties. Orthogonal complements. Orthogonal sets. Conjugate spaces. Adjoint and self-adjoint operators. Fixed point theory: Banach contraction principle. Schauder Principle. Applications. Prerequisite: MAT 221 MAT 415 Finite Element Methods 3 credits Basic concept of finite element method, approximate solution of BVP, direct approach to Finite Element Methods. Galerkin's weighted residual method for one-dimensional BVP, the modified Galerkin's technique. Shape functions for one-dimensional elements. Division of a region into elements, linear and quadratic elements, numerical integration over elements. Finite element solution of one dimensional BVPs. Finite Element approximations of line and double integrals: line integral using quadratic elements, double integrals using triangular and quadrilateral elements, double integrals using curved elements. Finite Element solution of two-dimensional BVP: Galerkin formulation, matrix formulation for 2–D finite elements. Three-nodded triangular elements. Variational formulation of BVP: construction of variational functions, the Ritz method and finite elements, matrix formulation of the Ritz procedure, solution of two-dimensional problems. Pre-processing and solution assembly: mesh generation in one and two dimensions, techniques of assembly and solutions. Prerequisite: MAT312. MAT 416 Tensor Calculus 3 credits Tensor: Coordinates, Vectors and tensors: Curvilinear coordinates, Kronecker delta, summation convention, space of N dimensions, Euclidean and Riemannian space, coordinate transformation, Contravariant and covariant vectors, the tensor concept, symmetric and skewsymmetric tensor. Riemannian metric and metric tensors: Basis and reciprocal basis vectors, Euclidean metric in three dimensions, reciprocal or conjugate tensors, Conjugate metric tensor, associated vectors and tensor's length and angle between two vector's, The Christoffel symbols. Covariant Differentiation of Tensors and applications: Covariant derivatives and its higher rank tensor and covariant curvature tensor. Prerequisite: MAT313 MAT 421 Fluid Mechanics 3 credits Preliminaries: Real and ideal fluids, Viscosity, Reynolds number, laminar and turbulent flows, boundary layers. Stress and rate of strain, General stress state of deformable bodies, General state of deformation of flowing fluid, Relation between stress and rate of deformation in general orthogonal coordinates. Equations of motion: Thermodynamic equation of state, Equation of continuity, Navier-Stokes equations, Energy equation, Equations of motion in different coordinate system. Exact solution of Navier-Stokes equations, Steady plane flow, Couette-Poiseuille flow, Plane stagnation-point flow, flow past parabolic body and circular cylinder, Steady axisymmetric flow, Circular pipe flow (Hagen-Poiseuille flow), Flow between two concentric rotating cylinders, Flow at a rotating disc, Unsteady plane flow, First Stokes problem, Second Stokes problem, Startup of Couette flow, Unsteady plane stagnation-point flow. Similarity analysis: Reynolds law of similarity, Dimensional analysis and theorem, Important non-dimensional quantities. Very slow motion: Equations of slow motion, Motion of a sphere in a viscous fluid, Theory of lubrication. Laminar boundary layer: Introduction to boundary layer, boundary layer equations in two dimensions, Dimensional representation of boundary layer equations, Displacement thickness, Friction drag, Flat plate boundary layer, Momentum thickness, Energy thickness, Similar solutions of boundary layer equations: Derivation of ODE, Wedge flow, Flow in a convergent channel, Integral relations of the boundary layer: Momentum-Integral equation, Energy-Integral equation. Prerequisite: MAT 326 MAT 422 Theory of Numbers 3 credits Arithmetic in Ù. Euclidean algorithm. Continued fractions. The ring Ù and its group of units. Chinese Remainder Theorem. Linear Diophantine equations. Arithmetical functions. Dirichlet convolution. Multiplicative function. Representation by sum of two and four squares. Arithmetic of quadratic fields. Euclidean quadratic Fields. MAT 423 Mathematical Modelling 3 credits Modelling in Biology Continuous population models for single species: Continuous growth models, Delay models, Periodic fluctuations, Harvesting models. Discrete population models for single species: Simple models, Cobwebbing, Discrete logistic models, Stability, Periodic fluctuations and Bifurcations, Discrete Delay models, Continuous models for interacting populations: Predator-prey models, Lotka-Volterra systems, Complexity and stability, Periodic behaviour, Competition Models, Mutualism. Discrete growth models for interacting populations: Predator-prey models. Epidemic models and dynamics of infectious diseases: Simple epidemic models and practical applications. Modelling in Economics Theory of the household: Preference and indifference relations, Utility function, Order conditions of optimization, Stutsky equation, Demand functions, Revealed Preference hypothesis, Von Neumann-Morgenstern utility. Theory of the firm: Production function, Laws of production and scale, Optimizing behaviour, Cost curves and cost functions, Constrained output maximization. Theory of factor demand: Optimal input mix, Factor demand and supply curves, Elasticity of derived demand. Market structures and equilibrium: Market Economy and equilibrium, Stability of equilibrium, Dynamic stability. Prerequisite: MAT 322. MAT 424 Operations Research II 3 credits Transportation and Assignment Problem: Introduction and formulation; relationship with linear programming. Network models: shortest route problems, minimal spanning, maximal-flow problem. Sequencing problem: Minimax-maximin strategies, mixed strategies, expected pay-off, solution of and games, games by linear programming and

Brown's algorithm. Dynamic programming: Investment problem, Production Scheduling problem, Stagecoach problem, Equipment replacement problem. Non-linear programming: Introduction, unconstrained problem, Lagrange method for equality constraint problem, Kuhn-Tucker method for inequality constraint problem. Quadratic programming. Prerequisite: MAT 316   MAT 425 Advanced Numerical Methods 3 credits Non-iterative (Newton's, steepest descent) methods for solution of a system of equation(linear and non-linear). Approximation theory: discrete least square applications, Chebyshev polynomial applications, rational function approximation, trigonometric polynomial approximation, Fast Fourier Transform. Approximating Eigenvalues: Honseholder's method, QR algorithm. BVP involving ODE: shooting method for linear and nonlinear problems, finite difference method for linear and nonlinear problems. PDE: Finite difference methods for elliptic, parabolic and hyperbolic problems. Prerequisite: MAT 312

MAT490: Special Topics in Mathematics

This course contains an introduction to a selection of mathematical topics that are not covered in traditional undergraduate mathematics curricula but are suitable for coverage at this level, such as Advanced Analysis, Advanced Linear Algebra, Algebraic Topology, Category Theory, Calculus of Variations, Differential Geometry, Functional Analysis, Graph Theory, Introduction to Riemannian Geometry, Lie Algebras, Lie Groups, Measure Theory, Optimization Techniques, Quantum Computation, Symplectic Geometry, Mathematical Foundation of Quantum Mechanics, etc. The topics covered in any particular semester depend on the interest of the students and instructor. Emphasis is on basic ideas and on applications in physics, economics, engineering. Notes attached to the course in a particular semester will describe the actual contents.

PHY 101 Introduction to Physics 3 Credits Vectors and scalars, Newton's Laws of motion, inertia, force, momentum, conservation of linear momentum, work, energy, conservation of energy, power, gravitation, escape velocity, projectile motion, simple harmonic motion, uniform circular motion. Structural properties of matter, elasticity, Hooke's Law, viscosity, surface tension. Heat and temperature, different scales of temperature, thermal expansion, specific heat, gas laws, heat transfer. Waves and oscillations, longitudinal and transverse waves, sound waves, velocity of sound, ultrasonic waves & their applications. Reflection and refraction of light, mirrors and lenses, total internal reflection, interference, diffraction. Coulomb's Law, ohm's law; resistance, potential difference, capacitance. Magnetic force on a moving charge, electromagnetic spectrum, velocity of light. Atoms and nuclei, mass number and atomic number, isotopes, isobars & isotones, atomic theory, Planck's Law, Photo-electric effect, wave-particle duality, special theory of relativity, radioactive decay, nuclear fission & nuclear fusion, nuclear energy, fossil fuels & other sources of energy. Structure & vastness of the universe, big bang theory, light year, solar system, Kepler's Laws of planetary motion, cosmological principle, Hubble's Law, red shift, stellar energy, neutron stars, quasars, supernovae, pulsars, black holes. PHY 102 Fundamentals of Physics 2 Credits Vectors and scalars, Newton's Laws of motion, principles of conservation of linear momentum and energy, gravitation, projectile motion, simple harmonic motion, rotation of rigid bodies. Elasticity, Hooke's Law, viscosity, Stokes' Law , surface tension. Heat & temperature, specific heat, gas laws, Newton's Law of cooling, First and Second Laws of thermodynamics, kinetic theory of gases, heat transfer. Wave motion, stationary waves, sound waves, Doppler Effect, beats, acoustics, ultrasonic & applications. Huygens' principle, electromagnetic waves, reflection, refraction, interference, diffraction. PHY 110 Mechanics and Properties of Matter 3 credits Mechanics: Vectors & scalars, vector addition and subtraction, unit vectors, scalar and vector products, scalar & triple vector product, scalar and vector fields, gradient, divergence and curl, curvilinear co‑ordinates, motion in one dimension, motion in a plane, work and energy, conservation laws, conservative force, projectile motion, uniform circular motion, simple harmonic motion, rotational motion, moment of inertia, radius of gyration, angular momentum, Kater's pendulum, Newton's Law of gravitation, gravitational field, potential, escape velocity. Properties of Matter: Hooke's Law, elastic modulii, adhesive and cohesive forces, molecular theory of surface tension, capillarity, variation of surface tension with temperature. Streamline flow, Poiseulle's formula, streamline flow and turbulent flow, Reynold's Number, Equation of Continuity, Bernoulli's Theorem, Stokes' Law. PHY 111 Principles of Physics I 3 Credits Vectors and scalars, unit vector, scalar and vector products, static equilibrium, Newton's Laws of motion, principles of conservation of linear momentum and energy, friction, elastic and inelastic collisions, projectile motion, uniform circular motion, centripetal force, simple harmonic motion, rotation of rigid bodies, angular momentum, torque, moment of inertia and examples, Newton's Law of gravitation, gravitational field, potential and potential energy. Structure of matter, stresses and strains, Modulii of elasticity Poisson's ratio, relations between elastic constants, work done in deforming a body, bending of beams, fluid motion and viscosity, Bernoulli's Theorem, Stokes' Law, surface tension and surface energy, pressure across a liquid surface, capillarity. Temperature and Zeroth Law of thermodynamics, temperature scales, isotherms, heat capacity and specific heat, Newton's Law of cooling, thermal expansion, First Law of thermodynamics, change of state, Second Law of thermodynamics, Carnot cycle, efficiency, kinetic theory of gases, heat transfer. Waves & their propagation, differential equation of wave motion, stationary waves, vibration in strings & columns, sound wave & its velocity, Doppler effect, beats, intensity & loudness, ultrasonics and its practical applications. Huygens' principle, electromagnetic waves, velocity of light, reflection, refraction, lenses, interference, diffraction, polarization. PHY 112 Principles of Physics II 3 Credits Electric charge, Coulomb's Law, electric field & flux density, Gauss's Law, electric potential, capacitors, steady current, Ohm's law, Kirchhoff's Laws. Magnetic field, Biot-Savart Law, Ampere's Law, electromagnetic induction, Faraday's Law, Lenz's Law, self inductance and mutual inductance, alternating current, magnetic properties of matter, diamagnetism, paramagnetism and ferromagnetism. Maxwell's equations of electromagnetic waves, transmission along wave- guides. Special theory of relativity, length contraction and time dilation, mass-energy relation. Quantum theory, Photoelectric effect, x-rays, Compton effect, dual nature of matter and radiation, Heisenberg uncertainty principle. Atomic model, Bohr's postulates, electron orbits and electron energy, Rutherford nuclear model, isotopes, isobars and isotones, radioactive decay, half-life, alpha, beta and gamma rays, nuclear binding energy, fission and fusion.Fundamentals of solid state physics, lasers, holography. PHY 113 Waves, Oscillation &Acoustics 3 Credits Principle of superposition, interference of waves, phase velocity and group velocity, simple harmonic motion, combination of SHM, Lissajous figures, damped SHM, forced oscillation, resonance, power and intensity of wave motion, waves in elastic media, vibration of strings, beats, Doppler Effect, acoustics, stroboscopy, velocity of sound, ultrasonics, and their applications. PHY 114 Thermal Physics & Radiation 3 Credits Heat and temperature, thermal equilibrium, Zeroth Law of thermodynamics, specific heat & calorimetry, Newton's Law of cooling, Kinetic Theory of Gases, idea of pressure due to collisions of molecules, mean free path, Boltzmann Distribution Law, Brownian motion, Law of equipartition of energy; Vander Waals' equation of state, heat transfer, conduction, convection and radiation, conduction of heat in solids, co􀀀efficient of thermal conductivity and its measurement, First Law of thermodynamics, isothermal & adiabatic changes, reversible and irreversible processes, Carnot's cycle, efficiency of heat engines, Second Law of thermodynamics, entropy and disorder, absolute scale of temperature, thermodynamic functions, Maxwell's relations, Clausius-Clapeyron Equation, Gibb's phase rule, Third Law of thermodynamics, Nernst heat theorem, radiation theory, black body radiation, Wien's Law, Stefan-Boltzman Law, Rayleigh Jeans Law, Planck's Law, variation of specific heat with temperature, Einstein's theory, Debye's theory, conduction of heat in solids, measurement of conductivity, Joule􀀀Thomson expansion, refrigeration, heat engines, Rankine cycles, cryogenics, measurement of high temperature. PHY 115 Electricity and Magnetism 3 Credits Charge, quantization of charge, Coulomb's Law, electric field and potential. Gauss's Law, electric dipole, dielectrics, capacitance, energy of charged systems, electrical images, magnetic dipole, energy in a magnetic field. Direct current and electromotive force, Ohm's Law, Kirchhoff's Laws, Wheatstone Bridge, Lorentz force, magnetic field of a current and Ampere's Law, Biot-Savart Law, electromagnetic induction, Faraday's Law, self induction, mutual induction, alternating current, RMS value, power factor, CR, LR and LCR circuits, resonance. PHY 201 Solid State Physics 3 Credits Crystalline state, Bravais lattices, crystal symmetry, point group & space group, unit cells, Miller indices, x ray diffraction, Bragg's Law, reciprocal lattice, structure factor, interatomic force and classification of solids, ionic, covalent, molecular, hydrogen bonded crystals, lattice energy of ionic crystals, Madelung constant, lattice vibration, phonons, normal modes in monatomic and diatomic linear chains, theory of specific heat, Einstein and Debye models, thermal expansion, defects in crystals, dislocations, consequences of defects on mechanical properties. PHY 202 Optics 3 Credits Laws of reflection and refraction, total internal reflection, Huygens' Principle, velocity of light, Young's experiment, Fresnel's bi prism, Newton's rings, Michelson's interferometer, multiple reflections, Fabry Perot interferometer, diffraction of light, Fresnel and Fraunhoffer diffraction, single, double and multiple slit diffraction, diffraction grating, spectrometer, resolving power of a grating, polarization of light, production of polarized light, plane, circular and elliptically polarized light, optical activity, double refraction, optic axis, half wave and quarter wave plate, nicol prism, dispersion of light, scattering of light, Thomson scattering. PHY 204 Classical Mechanics and Special Theory of Relativity 3 Credits Classical Mechanics: Newtonian equations of motion, conservation laws of a system of particles, variable mass, generalized co ordinates, generalized force, D' Alembert's Principle, variational method, Euler Lagrange equations of motion, Hamilton's principles, two body central force problem, elliptic orbit, scattering in a central field, Rutherford formula, kinematics of rigid body motion, Euler angles, rotating co-ordinates, Coriolis force, wind motion, principal axis transformation, top motion, principle of least action, Hamiltonian equations of motion, small oscillations, normal co-ordinates, normal modes. Special Theory of Relativity: Galilean relativity, Michelson-Morley experiment, postulates of special theory of relativity, Lorentz transformation, length contraction, time dilation, twin paradox, variation of mass, relativistic kinematics, mass energy relation. PHY 205 Statistical Mechanics 3 Credits Statistical Mechanics: Phase space, concept of state and ensemble, microcanonical, canonical and grand canonical ensembles, Boltzmann probability distribution, Maxwell velocity distribution, derivation of Bose-Einstein and Fermi-Dirac statistics, ideal Fermi gas, degenerate Fermi system, equation of state of ideal gases, ideal Bose gas, application of Statistical mechanics in various fields in physical, biological, social sciences, economics, finance and in engineering & ICT. PHY 210 Quantum Physics of Atoms, Solids and Nuclei 3 Credits Special Theory of Relativity: Michelson-Morley Experiment, Special Theory of Relativity, Lorentz Transformations, Time Dilation, Length Contraction, Mass-Energy Relation. Quantum Phenomena: Blackbody Radiation, Planck's Law, Photoelectric Effect, Bohr Atomic Model, Energy Levels & Atomic Spectra, Correspondence Principle, Dual Nature of Matter & Waves. Introductory Quantum Mechanics: Wave Function, Operators, Expectation Values, Schrödinger's Wave Equation, Particle in Box, Schrödinger Equation for Hydrogen Atom, Energy Levels, Magnetic & Orbital Angular Momentum, Concept of Quantum Numbers. Solid State Physics: Crystal Structure, Crystal Diffraction, Bragg Law, Lattice Vibrations & Phonons, Free Electron Model, Energy Levels & Density of States, Fermi-Dirac distribution function, Free Electron gas in Three dimension, Electrical conductivity & Thermal Conductivity, Hall Effect, Band Theory of Solids, Band Diagrams of Insulator, Semiconductor & Metals, Superconductivity, Lasers & Holography. Nuclear Physics: Rutherford Nuclear Model, Radioactivity, Half life & Mean life, Nuclear Binding Energy, Fission & Fusion, Particle Accelerator, Elementary Particles & Nuclear Interactions, Quarks, Lepton & Hadrons, Big Bang & Origin of the Universe.

PHY 301 Classical Electrodynamics 3 Credits Solution of Laplace's equation and Poisson's equation and applications to electrostatic problems, dielectrics, electrostatic energy, Maxwell's equations, electromagnetic waves, propagation of electromagnetic waves in conducting and non-conducting media, reflection and refraction, polarization, dispersion, scattering, waves in the presence of metallic boundaries, waveguides and resonators, solution of the inhomogeneous wave equations, simple radiating system, antennas, accelerated charge, Cerenkov radiation, elements of plasma physics. Prerequisite PHY 115. PHY 302 Fluid Mechanics 3 Credits Fluid properties, fluid statics, manometry, force on submerged planes and curved surface, buoyancy and floatation, one dimensional flow of fluid, equation of continuity, Euler's equation, flow of fluid in pipes, Bernoulli's equation, flow through orifice, mouthpiece, venturimeter, fundamental relations of compressible flow, frictional losses in pipes and fittings, types of fluid machinery, impulse and reaction turbines, centrifugal and axial flow pumps, deep well turbine pumps, specific speed, unit power, unit speed, unit discharge, performance and characteristics of turbines and pumps, design of pumps, reciprocating pumps. Prerequisite PHY 110 PHY 303 Quantum Mechanics I 3 Credits Breakdown of classical physics, quantum nature of radiation, Planck's Law, photoelectric effect, Einstein's photon concept and explanation of photoelectric effect, de Broglie wave, wave particle duality, electron diffraction, Davisson-Germer experiment, emergence of quantum mechanics, Schrodinger equation, basic postulates of quantum mechanics, physical interpretation of wave function, wave packets, Heisenberg's uncertainty principle, linear operators, Hermitian operators, eigenvalue equation, one-dimensional potential problem, harmonic oscillator, orbital angular momentum, rotation operator, spherical harmonics, spin angular momentum, addition of angular momenta, solution of the Schrodinger equation for hydrogen atom, matrix formulation of quantum mechanics. PHY 304 Quantum Mechanics II 3 Credits Rutherford scattering experiment, Discovery of the nucleus, Bohr quantization rules, hydrogen atom spectra, Franck-Hertz experiment, Sommerfeld-Wilson quantization rules, electron spin, Stern – Gerlach experiment, Pauli exclusion principle, electronic configuration of atoms, vector atom model, coupling schemes, Hund's rule, multiplet structure, fine structure in hydrogen spectral lines, Zeeman effect, Paschen-Beck effect, production of X-rays, measurement of X-ray wavelength, X-ray scattering, Compton Effect, Mosely's Law, molecular spectra, rotational and vibrational levels, Raman Effect and its applications, lasers. PHY 305 Quantum Mechanics III 3 Credits Basic properties of nuclei, constituents of nuclei, nuclear mass, charge, size and density, nuclear force, spin, angular momentum, electric and magnetic moments, binding energy, separation energy, semi-empirical mass formula, radioactive decay law, transformation laws of successive changes, measurement of decay constant, artificial radioactivity, radioisotopes, theory of alpha decay, gamma radiation, energy measurement, pair spectrometer, classical treatment of gamma emission, internal conversion, Mossbauer Effect, beta decay, energy measurement, conservation of energy and momentum in beta decay, neutrino hypothesis, orbital electron capture, positron emission, interaction of radiation in matter, ionisation, multiple scattering, range determination, bremsstrahlung, pair production, annihilation. Discovery of neutrons, production and properties of neutrons, nuclear reactions, elastic and inelastic scattering, Q-value of a reaction and its measurements, nuclear cross-section, compound nucleus theory, direct reaction and kinematics. Prerequisite PHY 304 PHY 306 Basic Electronics 3 Credits Network theorems, filters, transmission line, basic semiconductor concepts, energy bands, electrons and holes, semiconductor diode, rectification, regulators, Zener diode, diode circuits, unijunction transistor, FET and its characteristics, transistor amplifier, FET amplifier; amplifier circuits, voltage amplifiers, RC coupled amplifiers and tuned amplifiers, frequency response, bandwidth, power amplifier, push-pull amplifier, feedback and amplifier stability, operational amplifier and its characteristics, oscillators, modulation and demodulation, digital electronics, digital logic, logic gates, Boolean algebra, logic circuits, information registers, flip￾flop circuit. PHY 308 Methods of Experimental Physics & Instrumentation 3 Credits Optical and spectroscopic instruments, defects of images and their remedies, optical blooming, phase contrast and polarizing microscope, spectrophotometers, optical transmittance, reflectance and absorption, application of interferometry, production and measurement of high and ultrahigh vacuum. Rotary pump, diffusion pump, ion pump and turbo pump, pirani, penning and ionisation gauges, measurement of current and voltages, potentiometer, VTVM, oscilloscope, D.C. amplifier, lock-in amplifier, frequency meter and counter, four point probe, flux meter and Hall probed transducers , piezoelectric, thermistor, photo-transducers, voltage regulator, SCR type temperature controllers. Prerequisites PHY 202 and PHY 306 PHY 309 Introduction to Materials Science 3 Credits Crystalline solids, amorphous, composite, fibrous materials, polymers, plastics, binding forces, elastic properties, dislocations, defects etc, specific heat, thermal expansion, thermal conductivity and electrical conductivity of metals, dielectric properties of solids , modes of dielectric polarisation, ferro electricity, piezo electricity, optical properties of solids ,classical and semi classical theory, free carrier effects, lattice absorption, electronic absorption, magnetic properties of solid, atomic magnetic moments, dia and paramagnatism, ferro & ferrimagnetism, antiferromagnetism, ferrites, magnetic resonance, superconductivity, type-1, type –2 superconductors, liquid crystals. Prerequisite PHY 201 PHY 310 Advanced Solid State Physics 3 Credits Free electron theory, transport properties, Sommerfeld theory, Hall Effect, box quantization, density of states, Fermi surface, Fermi energy, electrical conductivity, Wiedmann Franz law, band theory of solids, electron in a periodic potential, Schrödinger equation, Bloch function, LCAO and OPW methods, dielectric properties of insulators, Clausius-Mosotti relations, dielectric loss, relaxation time, polarization mechanism, direct & indirect band gap semiconductors, extrinsic semiconductors, charge carrier concentration, recombination process of p-n junction, superconductivity, Meissner Effect, London equation, BCS theory, introduction to high temperature superconductivity, magnetic materials, quantum theory of diamagnetism and paramagentism, theory of ferromagnetic, ferrimagnetic and anti‐ferromagnetic orders, magnetic resonance. Prerequisite PHY 201 PHY 311 X-Rays 3 Credits Continuous and Characteristic X-rays, Bremsstrahlung, Properties of X-rays, X-ray technique, Weissenberg and precession methods, identification of crystal structure from powder photograph and diffraction traces, Laue photograph for single crystal, geometrical and physical factors affecting X-ray intensities, analysis of amorphous solids and fibre textured crystal. Prerequisite PHY 201 PHY 312 Nuclear Physics II 3 Credits Determination of nuclear size by scattering methods and electromagnetic methods, mirror nuclei, electron scattering, nuclear shapes, electric and magnetic multiple moments, isotopic spin formalism, two-nucleon problem, nuclear forces, exchange force, meson theory of nuclear forces. Shell model, refinement of extreme single particle model, collective model, nuclear reactions, compound nucleus model, concept of optical potential, energy averaged cross section and the optical model at low energies, phenomenological optical model, direct reactions, parity violation in beta decay, nuclear fission and nuclear reactor, nuclear fusion, nuclear liquid drop model & shell model, magic numbers (qualitative) accelerators, Van de Graaff generator, linear accelerator, cyclotron, synchrotron, detection of charged particles, photons and neutrons, nuclear pulse counting systems, elementary particles. Prerequisite PHY 305 PHY 313 Physics for Development 3 Credits Twenty first century development issues, physics and break through technologies, ICT, fibre optics, quantum information theory, physics in genetics engineering and molecular biology, physics and health issues, bio and medical physics, materials science and physics, high temperature superconducting materials, space physics, microgravity experiments, econo-physics, physics principles applied in sociology. PHY 400 Thesis / Project 4.5 Credits A student is required to carry out thesis/project work in the 7th and 8th semester in a chosen field. There will be a supervisor who will either be a BRAC University faculty or any other suitable expert from universities and R/D organizations of the country to guide the thesis/ project work .On completion of study and research s/he will have to submit the dissertation paper and to face a viva board for the defence. PHY 401 Particle and Reactor Physics 3 Credits Interactions of neutrons with matter, cross – sections for neutron reactions, thermal neutron cross-sections, nuclear fission, energy release in fission, neutron multiplication, nuclear chain reaction, steady state reactor theory, criticality condition, homogeneous and heterogeneous reactor systems, neutron moderation, neutron diffusion, control of nuclear reactions, coolant, types of nuclear reactors: power reactor, research reactor, fast reactor, breeder reactor, reactor shielding, waste disposal. Prerequisite PHY 305 PHY 402 Atmospheric Physics 3 Credits Structure of the atmosphere, elementary ideas about the sun and the laws of radiation, definitions and units of solar radiation, depletion of solar radiation in the atmosphere, terrestrial radiation, radiation transfer, heat balance in the atmosphere, heat budget, vertical temperature profile, radiation charts and their uses, composition of the atmosphere, mean molecular weight, humidity, mixing ratio, density and saturation vapour pressure. Fundamental equations of atmospheric motion, approximations of the equation, circulation and vorticity and their equations. Introduction to atmospheric thermodynamics. PHY 403 Cosmology & Astrophysics 3 Credits General introduction to plasma physics, plasma as a fourth state of matter, definition, screening, and Debye shielding, plasma frequency, ideal plasma, temperature and pressure of plasma, magnetic pressure and plasma drifts, plasma waves, Landau damping, collisions in plasmas, hydrodynamic description of plasma, one fluid model, two fluid model,Chew-Goldberg theory, low waves in magneto-hydrodynamics, description of plasma, dielectric tensor, longitudinal and transverse waves, plasma instabilities, transport in plasmas, plasma kinetic theory, Vlasov equation, linear waves, waves in magnetized plasma, electromagnetic waves, waves in hot plasmas, nonlinear waves, Landau damping, quasi linear theory, plasmas in fusion research, astrophysical plasmas. Introduction to astrophysics, formation of stars and galaxies, evolution of stars, the notion of cosmology, Cosmological Principle, various cosmological models of the universe, expansion of universe, Hubble's Law, problem of singularity in time, solutions of Friedmann, de Sitter and others, density of matter in the universe, cosmological term, self screening effect for matter. Prerequisites PHY 301 and PHY 304 PHY 404 Electronic Devices and Circuits 3 Credits Modelling and application of Semiconductor devices and integrated circuits, advanced transistor amplifier analysis, including feedback effects. Design for power amplifiers, operational amplifiers (OPAMP), analog filters, oscillators, A/D and D/A converters and power converters. Introduction to transistor level design of CMOS digital circuits. Prerequisite PHY 306 PHY 405 Mathematical Physics 3 Credits Series solution of 2nd order ordinary differential equations about ordinary and singular points, orthogonal set functions, Sturm-Liouville boundary value problem (SLP), eigen values and eigen functions of different SLP, series of orthogonal set of function. Laplace transforms: definition, Laplace transformations of some elementary functions, inverse Laplace transformations, Laplace transformations of derivatives, Dirac delta function, some special theorem on Laplace transformations, solution of differential equations by Laplace transformations, evaluation of improper integrals; finite Fourier series, Fourier transforms, Fourier integrals, Fourier transform and application to solution BVP, beta and gamma functions, Legendre functions, Bessel functions, solution of boundary value problem by method of separation of variables, solution PDE of mathematical physics: Helmholtz equation, wave equation: vibrating string, vibrating membrane, diffusion equation, Laplace equation, Hermite polynomials, Laguerre polynomials, hyper-geometric functions. Prerequisite MAT 203 PHY 406 Medical Physics & Instrumentation 3 Credits Ultrasound imaging, A-scan, B-scan, M-scan, clinical applications, rectilinear scanner, gamma camera, CAT scanner, MRI, clinical applications, audiology, hearing aids, vascular measurements, blood pressure, blood flow, blood velocity, cardiac measurements; ECG, ECG planes, elementary ideas on heart disorders, defibrillators, pacemakers, neuromuscular measurements; EEG, EMG, stimulation of neural tissue, nerve conduction measurements, bio-electric amplifiers, patient safety, radiopharmaceuticals, radiotherapy, radiation protection, radiation dosimetry. PHY 407 Mathematical Modelling in Physics 3 Credits Basic concept of mathematical modelling, formulation and solution, overview of computational methods of classical and quantum physics, numerical procedure for special functions, Random numbers generator, Brownian motion simulation, linear system of equations, sparse linear system, eigen value problems, BVP involving ODE, Sturm-Liouville problems, BVP involving PDE: elliptic, parabolic and hyperbolic problems using finite difference and other methods, Monte Carlo integration and simulation, mathematical modelling of problems of physics using above techniques. Prerequisite MAT 205 PHY 408 Advanced Quantum Mechanics 3 Credits Heisenberg and Dirac or interaction pictures, time-independent perturbation theory, degenerate perturbation theory, variation method, hydrogen atom and helium atom, WB approximation method, Sommerfeld-Wilson quantisation condition, time-dependent perturbation theory, Fermi's golden rule, applications, identical particles, parity, Pauli principle, applications, non-relativistic scattering theory, partial wave expansion, optical theorem, S matrix, solution of the wave equation by the method of Green's function, Lippmann Schwinger equation, Neumann series, Born approximation, applications, Klein-Gordon and Dirac equations, existence of electron spin, magnetic moment, plane wave solution of the Dirac equation, hole theory; prediction of the positron. Prerequisite PHY 303 PHY 409 Physics of Radiology 3 Credits The production and properties of X-rays, diagnostic and therapeutic X-ray tubes, X-ray circuit with rectification, electron interaction, characteristic radiation, bremsstrahlung, angular distribution of X-rays, quality of

X-rays, beam restricting devices, the grid, radiographic film, radiographic quality, factors affecting the image, image modification, image intensification, contrast media, modulation transfer function, exposure in diagnostic radiology, fluoroscopy, computed tomography, ultrasound, magnetic resonance imaging (MRI). PHY 410 Laser Physics 3 Credits Spontaneous and stimulated emission, absorption, pumping schemes, characteristic properties of laser beam, laser speckle, grain size calculation for free space propagation, semi classical treatment of absorption and stimulated emission, spontaneous emission, results of QED treatment, electric dipole, allowed and forbidden transitions, Einstein's A and B coefficient, radiation trapping, superfluorescence, superradiane and amplified spontaneous emission, nonradiative decay, homogeneous and inhomogeneous broadening, linewidth calculations for naturally, collisionally and Doppler broadened line, two level and four, level saturation, saturation of absorption & inhomogeneously broadened line, passive optical resonators, continuous wave and transient laser behaviour, laser beam transformation, types of lasers, their construction and use, applications of lasers, optical communications, laser in fusion research, holography. Prerequisite PHY 304 PHY 411 Geophysics 3 Credits Solar system, the planets, meteorites, cosmic ray exposures of meteorites, Poynting-Robertson effect, compositions of the terrestrial planet, pre-radioactivity age problem, radioactive elements and the principle of radiometric dating, growth of constituents and of atmospheric argon, age of the earth and of meteorites, dating the nuclear synthesis, figure of the earth, precession of the equinoxes, the Chandler- wobble, tidal friction and the history of the earth moon system, fluctuation in rotation and excitation of the wobble, seismology of the earth, elastic wave and seismic rays, travel time and velocity depth curves for body waves, shockwave, internal pressure of earth core, internal density and composition, free oscillation, earthquake prediction problem, terrestrial magnetism, earth magnetic field, geophysical prospecting; seismic, gravitational, magnetic, electrical and nuclear methods. PHY 412 Dynamical & Tropical Meteorology

3 Credits Geophysical fluid dynamics, Navier-Stokes' equation, rotating and stratified flow, scale analysis, hydrostatic approximation, Coriolis force, geopotential etc., gradient and thermal wind, vorticity and circulation theorems, Proudmen-Taylor theorem, atmospheric wave, atmospheric turbulence, barotroic and baroclinic instabilities, numerical weather forecasting, quasi-geotropic approximation, barotropic vorticity equation, primitive equation, multilayered models, tropical cyclones, norwesters and tornadoes, the monsoons, dynamical climatology, physics of upper atmosphere: geomagnetism, neutral atmosphere, ionosphere and magnetosphere. Prerequisite PHY 402 PHY 413 General Theory of Relativity 3 Credits Gravitation, Lagrangian Einstein equations, approximation of weak field and Hilbert's auxiliary conditions, comparison of corresponding relations with those of Newton's theory of gravitation, source of gravitation field, Schwarzschild's solution in isotropic and other coordinate systems, analogy between gravitation and electromagnetism, motion of test mass and geodetic lines, motion in Schwarszchild's field, equations of motion in general relativistic mechanics as a consequence of Einstein's equation of gravitational field, gravitational waves in weak field approximation, problem of energy transfer, exact wave solutions in the case of gravitational field, waves of matrices or wave of curvature, locally plane gravitational waves, Weber's and Braginski's experiments, prospects of future gravitational experiments. PHY 414 Field Theory 3 Credits Equation of motion, quantization, conservation laws, construction of Hilbert space, Lagrangian, equation of motion, quantization of neutral and charged Klein-Gordon fields, Dirac equation, spinors, quantization of Dirac field, Maxwell fields, Gupta-Bleuler formalism, theory of gauge fields, invariant functions propagators for Klein-Gordon field, Dirac fields and electromagnetic fields, symmetries of interactions, interaction picture; U and S matrices, Feynman diagrams, Wick's theorem, Feynman rules, lowest order, amplitude and cross section for Compton scattering, GSW model of electroweak interactions, elements of QCD, path integral in field theory, introduction to string

theory. PHY 415 Neutron Scattering 3 Credits Neutron sources, continuous and pulsed sources, monochromatization, collimation and moderation of neutrons, neutron detectors, scattering of neutrons and its advantages, elastic scattering of neutrons, magnetic scattering and determination of magnetic structure, inelastic scattering, thermal vibration of crystal lattices, lattice dynamics and phonons, neutron polarization, polarized neutron applications, scattering by liquids and molecules, Van Hove correlation formalism, some experimental results of scattering by liquids and molecules, small angle neutron scattering and its applications in the study of biological molecules and defects, experimental techniques of scattering measurements, time-of-flight method, crystal diffraction techniques, neutron diffractometer and triple-axis spectrometer, constant Q-method. Prerequisite PHY 305 PHY 416 Radiation Biophysics 3 Credits Nucleus, ionizing radiations, radiation doses, interaction of radiation with matter, cell structure, radiation effects on independent cell systems, oxygen effect, hyperthermia, LET and RBE, lethal, potentially lethal and sub-lethal radiation damage, dose-rate effect, acute effects of radiation, somatic effects, late effects, non-specific life shortening and carcinogenesis, genetic changes, nominal standard dose (NSD), time dose fractionation (TDF), Standquist curve. Prerequisite PHY 305 STA 101 Introduction to Statistics 3 credits Introduction to statistics, graphical displays, frequency distribution, mean, median, mode and other measures of central tendency, standard deviation and other measures of dispersion, measure of skewness and Whisker-Box plot, correlation and regression analysis, elementary probability theory, conditional and joint probability, sample survey, simple random sampling, stratified random sampling, systematic sampling, index number, time series and quality control. STA 201 Elements of Statistics and Probability 3 credits Frequency distribution, mean, median, mode and other measures of central tendency, standard deviation and other measures of dispersion, measure of skewness and Whisker-Box plot, correlation and regression analysis, elementary probability theory, conditional and joint probability, Bayes' theorem, discrete probability distributions, binomial, hypergeometric, Poisson, geometric and negative binomial distributions, continuous probability distributions, normal and exponential distributions, sampling distributions for relevant statistics (Normal, t, chi-square, and F distributions), central limit theorem, confidence intervals and hypothesis testing for parameter (mean and proportion). STA 301 Modern Probability Theory & Stochastic Processes 3 Credits Stochastic Processes: Definition of different types of stochastic processes, recurrent events, renewal equation, delayed recurrent events, number of occurrence of a recurrent event. Markov Chain: Transition matrix, higher transition probabilities, classification of sets and chains, ergodic properties. Finite Markov Chain: General theory of random walk with reflecting barriers, transient states, absorption probabilities, application of recurrence time, gambler's ruin problem. Homogeneous Markov Processes: Poisson process, simple birth process, simple death process, simple birth death process, general birth process, effect of immigration, nonhomogeneous birth death process, Queuing theory. Modern Probability Theory: Probability of a set function, Borel field and extension of probability measure, probability measure notion of random variables, probability space, distribution functions, expectation and moments. Prerequisite: STA201 b. Practical Courses: APE 104 APE Lab I 1.5 Credits List of Experiments: EXP 1: Determination of the Modulus of Rigidity of a Wire by the Method of Oscillations EXP 2: Determination of Surface Tension of Mercury and the Angle of Contact by Quincke's Method EXP 3: Determination of the Specific Heat of a Liquid by the Method of Cooling EXP 4: Determination of the Thermal Conductivity of a Bad Conductor by Lee's Method EXP 5: Determination of the Specific Resistance of a Wire using a Meter Bridge EXP 6: Determination of the High Resistance of a Suspended Coil Galvanometer by the Method of Deflection EXP 7: Determination of the Temperature Co-efficient of Resistance of the Material of a Wire EXP 8: Determination of the Line Frequency by Lissajous Figure using an Oscilloscope and a Function Generator and Verification of the Calibration of Time/Div

Knob at a Particular Position for Different Frequencies EXP 9: Charging and Discharging of Capacitors and Study of Their Various Characteristics. EXP 10: Verification of Thevenin's and Norton's Theorem. EXP 11: Verification of Maximum Power Transfer Theorem. EXP 12: Verification of Current Division Rule (CDR), KVL and KCL EXP 13: Conversion of Galvanometer into Voltmeter. EXP 14: Conversion of Galvanometer into Ohmmeter. EXP 15: Determination of the e/m of Electron Using Helmholtz Coil. EXP 16: Determination of the Threshold Frequency for Photoelectric Effect of a Photo-Cathode and the Value of Planck's Constant by Using a Photoelectric Cell. APE 206 APE Lab II 1.5 Credits List of Experiments: EXP 1: Determination of the Refractive Index of the Material of a Prism by using a Spectrometer. EXP 2: Determination of the Radius of Curvature of a Lens by Newton's Rings Method EXP 3: Determination of the Wavelengths of Various Spectral Lines by Spectrometer by using Plane Diffraction Grating EXP 4: Study of the Frequency Responses of Series and Parallel LRC Series Circuit and the Variation of Q-factor with Resistance. EXP 5: Study of the Variation of Electrical Conductivity of a Semiconductor and Determine of its Energy Gap. EXP 6: Study of the Characteristics of a PN Junction and Zener Diode. EXP 7: Study of the Characteristics of a NPN Bipolar Junction Transistor (BJT) in Common Base configuration. EXP 8: Study of the Characteristics of Junction Field Effect Transistor (JFET) in Common source configuration. EXP 9: Design and construction of a 4-diode Full Wave Rectifier power supply and study the effect of Shunt Capacitor filter. EXP 10: Implementation of AND, OR, NOT logic gates. EXP 11: Design of S-R flip-flop EXP 12: To design Code converters (Decimal-to-BCD, BCD-to-Decimal) EXP 13: To design Ripple, Ring and Decade Counters using JK-FFs. EXP 14: To study the characteristics of IC MUX, to realization of combinational circuits and generation of complex wavesforms. EXP 15: Use of IC 74138 decoder as DEMUX, realization of 1-to-16 line DEMUX using 74138. APE 301 APE Lab III 1.5 Credits List of Experiments: EXP 1: Study of the characteristics of a Uni-junction Transistor. EXP 2: Study of the characteristics of a Silicon Controlled

Rectifier Transistor. EXP 3: To draw and study the I-V characteristics of a solar cell. EXP 4: Design and construction of a BJT CE single-stage amplifier using potential divider biasing. EXP 5: Study the frequency response characteristics of a two stage RC coupled BJT amplifier. EXP 6: Study of the characteristics of 741 Operational Amplifier. EXP 7: Design of Inverting and Non-inverting amplifiers. EXP 8: Design and construction of active low pass and high pass filters using Op-Amps. EXP 9: Design and construction of active Butterworth Band pass filter. EXP 10: Design and construction of a Summing Amplifier using 741 Op-Amp. EXP 11: Study of the percentage distortion and power output of a complimentary symmetry push-pull power amplifier. EXP 12: Design and Construction of a Colpitts Oscillator. EXP 13: Design and Construction of Astable and Monostable Multividrators using BJTs EXP 14: Expt with minority carrier. EXP 15: Design and Construction of a Crystal Oscillator.

APE 303 APE Lab IV 1.5 Credits List of Experiments: EXP 1: Design and Construction of an Amplitude modulator and a demodulator. EXP 2: Design and Construction of a Frequency modulator and a demodulator. EXP 3: Design and Construction of a Phase-Shift-Keying and its detection. EXP 4: Design and Construction of a Pulse Amplitude modulation and its detection. EXP 5: Design and Construction of a Pulse Width modulation and its detection. EXP 6: Design and Construction of a Pulse Code modulation and its detection. EXP 7: To study Time Division Multiplex System EXP 8: Expt. with DSP (using DSP trainer) EXP 9: To measure microwave standing wave ratio. EXP 10: To measure microwave Frequency and Wavelength. EXP 11: Expt. with microwave antenna. EXP 12: Expt. with PLL. EXP 13: Expt. with Fiber-Optic Communication. EXP 14: To Study the TV Composite Video signal. EXP 15: Design, construction and testing of an Astable, Monostable and Voltage Controlled Oscillator using 555 Timer. EXP 16: Expt. with microprocessor (8086). MAT 250 Mathematics Lab I 2 Credits Introduction to the computer algebra package MATHEMATICA/Matlab. Evaluation and graphical representation of function. Solution of linear and nonlinear equations by using False- Position, Bisection, Newton Raphson methods. Solution of system of linear equations by using Gaussian Elimination method. Interpolation and extrapolation. Numerical differentiations. Curve fitting. Trapezoidal and Simpson's rules for numerical integration. Problem solving in concurrent courses (e.g. Calculus, Linear Algebra and Geometry), using FORTRAN, MATHEMATICA and Matlab. Lab Assignments: There shall be at least 15 lab assignments MAT 350 Mathematics Lab II 2 Credits Solution of initial value or boundary value problems for Ordinary differential equations. Solution of initial value or boundary value problems for partial differential equations. Problem solving in concurrent courses (e.g; Calculus, Advanced linear algebra problems, Differential Equations and Numerical Analysis, Complex Analysis, Linear Programming, Numerical Analysis and Applied Mathematics) using FORTRAN and MATHEMATICA/Matlab. Lab Assignments: There shall be at least 15 lab assignments. PHY 116 Physics Lab I 1.5 Credits List of Experiments: EXP 1: Determination of the Young's Modulus of a Short Wire by Searle's Dynamic Method EXP 2: Determination of the Modulus of Rigidity of a Wire by the Method of Oscillations EXP 3: Determination of g by means of a Compound Pendulum EXP 4: Determination of the Moment of Inertia of a Flywheel about its Axis of Rotation EXP 5: Determination of the Spring Constant and Effective Mass of a given Spiral Spring EXP 6: Determination of Surface Tension of Water by Capillary Tube Method EXP 7: Determination of Surface Tension of Mercury and the Angle of Contact by Quincke's Method EXP 8: Determination of the Viscosity of Glycerine by Applying Stokes' Law. EXP 9: Determination of the Specific Heat of a Liquid by the Method of Mixture EXP 10: Determination of the Specific Heat of a Liquid by the Method of Cooling EXP 11: Determination of the Thermal Conductivity of a Bad Conductor by Lee's Method EXP 12: Determination of the Pressure Co-efficient of a Gas at Constant Volume by Constant Volume Air Thermometer EXP 13: Determination of the Stefan's Constant EXP 14: Study of Variation of the Frequency of a Tuning Fork with the Length of a Sonometer (n-l curve) under given Tension and Hence to Determine the

Unknown Frequency EXP 15: Determination of the Frequency of a Tuning Fork by Melde's Experiment EXP 16: Determination of Velocity of Sound by Kundt's Tube. PHY 203 Physics Lab II 1.5 Credits List of Experiments: EXP 1: Determination of the Focal Length and Hence the Power of a Convex Lens by Displacement Method with the Help of an Optical Bench EXP 2: Determination of the Refractive Index of a Liquid by Plane Mirror and Pin Method using a Convex Lens EXP 3: Determination of the Radius of Curvature of a Lens by Newton's Rings Method EXP 4: Determination of the Refractive Index of the Material of a Prism by using a Spectrometer EXP 5: Determination of the Wavelengths of Various Spectral Lines by Spectrometer by using Plane Diffraction Grating EXP 6: Determination of the Value of an Unknown Resistance and Verification of the Laws of Series and Parallel Resistances by Means of a Post Office Box EXP 7: Determination of the Internal Resistance of a Cell by a Potentiometer EXP 8: Determination of the Specific Resistance of a Wire using a Meter Bridge EXP 9: Determination of the Resistance of a Galvanometer by the Half-Deflection Method EXP 10: Determination of the High Resistance of a Suspended Coil Galvanometer by the Method of Deflection EXP 11: Comparison of the EMF of Two Cells with a Potentiometer EXP 12: Determination of the Resistance per Unit Length of a Meter Bridge EXP 13: Determination of the Temperature Co-efficient of Resistance of the Material of a Wire EXP 14: Determination of the Value of J by Electrical Method EXP 15: Determination of the Line Frequency by Lissajous Figure using an Oscilloscope and a Function Generator and Verification of the Calibration of Time/Div Knob at a Particular Position for Different Frequencies EXP 16: Determination of the Self-Inductance of a Coil by Anderson's Method. EXP 17: Charging and Discharging of Capacitors and Study of Their Various Characteristics. PHY 307 Physics Lab III 1.5 Credits List of Experiments: EXP 1: Determination of the Excitation and Ionization Potentials (of mercury) by Frank-Hertz Experiment. EXP 2: Determination of the e/m of Electron Using Helmholtz Coil. EXP 3: Determination of the Threshold Frequency for Photoelectric Effect of a Photo-Cathode and the Value of Planck's Constant by Using a Photoelectric Cell. EXP 4: Determination of the Plateau of a Geiger-Muller Counter and Hence to Find its Operating voltage. EXP 5: Study of the Variation of Electrical Conductivity of a Semiconductor and Determine of its Energy Gap. EXP 6: Study of the Characteristics of a PN Junction and Zener Diode. EXP 7: Study of the Characteristics of PNP and NPN Transistors. EXP 8: Study of the Frequency Response Characteristics of an RC Low pass, RC High pass, a Band pass and a Parallel T Filter. EXP 9: Study of the Frequency Response in LRC Series Circuit and the Variation of Q-factor with Resistance. EXP 10: Determination of the Frequency Response in LRC Parallel Circuit and Determination of Q-factor. EXP 11: Study of Variation of Reactance due to L and C with Frequency. EXP 12: Designing and Construction a Summing Amplifier Using 741 Operation Amplifier (OPAMP). EXP 13: Construction of Full Wave Bridge Rectifier Using Semiconducting Diodes and Study of the Effect of Filters. EXP 14: Determination of Transistor Characteristics in Common Emitter Configuration and Determination of Hybrid Parameter. EXP 15: Determination of the Coefficient of Mutual Inductance Between Two Coils and Hence to Show its Variation with the Separation Between the Coils. EXP 16: Determination of the Absorption Coefficients of Different Materials for the Radiation Emitted by a Radioactive Source by Using a Geiger-Mueller Counter.

The Board of Trustees is the highest policy making body of BracU. It is responsible for ensuring that the best educational and administrative standards are set and maintained at BracU.

The Syndicate is the highest executive body that exercises administrative and supervisory control over the academic, administrative and management activities of the University.

The Academic Council recommends the educational policies of the University and also determines the curricula and courses that can help achieve high educational standards.

Board of Trustees

Syndicate

Academic Council

Counseling and Wellness Centre

A place for healing and self-empowerment

Every individual has the potential to rejoice in the beauty of life and can overcome challenges. University life's challenges can often be difficult in several ways, for example, facing problems with new environments, confusion in deciding on a major or career path, struggling in relationships, or making friends. In such circumstances, we may feel anxious, depressed, confused, or overwhelmed. Talking with a professional counselor/psychotherapist can help to deal with such difficulties, and explore potentialities to reach the peak of self-empowerment. Counselors maintain confidentiality and assist the client with the help of different therapeutic tools and techniques.

The Counseling and Wellness Centre of Brac University provides professional psychotherapy services for free to all the stakeholders of the university alongside workshops, seminars, and other programs.

You have every right to take care of yourself and we can provide you the support you need for quality self-care.

The fourth and final keynote presentation for ICBM 2019 took place after the lunch on the final day (April 27, 2019) of the conference. This presentation was managed by the session chairs Prof. Dr. Mohammad Tamim, Pro-Vice Chancellor, Brac University and Prof. Dr. Mirza Azizul Islam from Brac Business School (BBS), Brac University.

The speakers for this session were Mr. Anis A. Khan, Managing Director & CEO, Mutual Trust Bank Limited, Prof. Dr. Akbar Ali Khan, BBS, Mr. Sanjeev Gathani, CEO, Better Business Governance (BBG), Singapore and Mr. Ishtiaq Ahmed, CEO, Kazi Farms Group.

Prof. Khan started off the session with his presentation on technology and its impacts on the recent developments in business. He stated that technology is changing the world and no one can deny it. Dr. Khan then went on to talk about how technology has gone on to revolutionise businesses over the years and how it will continue to do so for decades to come.

The floor was then given to Mr. Anis, who gave a brief talk on achieving the Sustainable Development Goals (SDGs) through entrepreneurship, commerce and investment. He was also full of praise for BRAC and its continuous contribution in bringing development in different sectors of society including healthcare financing and health kiosks. Mr Anis is confident that by building stronger partnerships, and renewing focus on technology, innovation and challenges faced by the urban community, we can provide our citizens with the tools needed to achieve financial inclusion in Bangladesh.

Mr. Sanjeev Gathani talked about cyber security, the contemporary challenges faced due to cyber-attacks and their possible remedies. Addressing the audience, Mr. Gathani pointed out that our habit of casually using free Wi-Fi without considering the potential security threats is extremely dangerous to our personal information. “Money is not important today, it is information that is important”, said Mr Gathani. “Protection of information is very important for universities and the corporate sector. Technology knows more about you than you know about technology itself”. He also advised us that we should be proactive when it comes to dealing with such threats and not be reactive.

Mr. Ishtiaq Ahmed focused his lecture on bridging strategy between the academia and industry. He emphasised on the importance of solving localised case studies that portray the business and industrial scenario of Bangladesh. Students cannot know what is going on in this country if they only work on foreign case studies that are worked out in universities like Harvard. Mr. Ishtiaq also pointed out how majority of our faculties lack hands-on industry experience in Bangladesh and that a paradigm shift is needed that will lead the transformation of bridging the gap between academia and industry.

The session was then brought to an end with the distribution of appreciation certificates among the speakers and session chairs.

When do students go to Savar for the Residential Semester?Students are eligible to attend the RS in their third semester, after completing their first and second semester at Mohakhali to a satisfactory standard. This requires students to have - attained a minimum CGPA of 1.5 and - Passed the ENG091 course.

Is the Residential Semester compulsory? Yes, completing the RS is compulsory for all BRACU students, as it is a requirement for every degree. Students who fail to attend this program will not be able to graduate from BRACU.

Where is the Savar Campus located? Away from urban congestion, BRACU's Savar Campus is located in the rural village of Khagan on the Ashulia-Birulia road in Savar. It is about an hour's drive from Dhaka. Students must use BRACU's transport on the first day to join Savar Campus.

How many academic courses must students take during the Residential Semester? Compulsory courses include Bangla, English, Bangladesh Studies, and Ethics & Culture. In addition, there are compulsory study tours and field visits to rural areas that give hands-on understanding of rural life, the development process and the rural culture of Bangladesh. Students need to buy course materials from Ayesha Abed Library at Savar Campus (course materials will be available days before the course starts).

What percentage of attendance is needed in order to pass?A student will need to attend 90 percent of classes in order to be allowed to take the final examinations.

What are the Experiential Learning Activities (ELA) and their purposes? The RS curriculum gives Experiential Learning Activities that develop students' social skills and academic coursework equal importance.

These activities include on and off-campus initiatives such as social learning labs, study tours, institution and organization visits, co-curricular activities and village immersion programs. Participation in all co-curricular activities is mandatory for all students.

Why are activities outside of the campus important?   Activities outside of the campus are important for students, to help them acquire firsthand knowledge of their subject matter, outside the classroom e.g. students will take a day trip to a neighbouring village.

How is participation in activities rewarded?Students who perform exceptionally are awarded with the Vice Chancellor’s certificates. Certificates and prizes are also awarded to students who complete a Residential Semester activity e.g. dance, or yoga. Students who miss two or more activity classes will not receive the certificate.

What are the dormitory opening and closing times?The dormitory gate will open at 6:30am in the morning and close at 10:00pm sharp. Students are responsible for reaching their respective dormitory in time.

How frequently can students leave Savar Campus to visit home?The RS semester is designed to be entirely residential, so students are discouraged from leaving campus. Students will be allowed a short leave in emergencies, and possibly in exceptional circumstances, provided they submit a leave application that must be approved by the concerned authority. Any overstay of leave without a valid reason will be considered a breach of discipline and the offending student will not be granted further leave for the rest of the semester.

When and how frequently can students receive outside guests? Only nominated visitors, preferably parents, are allowed to visit the students on Parents’ Day: May 11 & 12, 2018, Friday and Saturdays from 3:00pm-5:00 pm. They may also visit the campus from 3.30pm-5.00pm on the following Fridays thereafter: June 01, July 06 and July 27, 2018

What personal items should students bring with them? Students should bring formal and casual wear, some sportswear, two sets of medium or small locks (for their room door), five passport-sized photos and a photocopy of their health card. They should also bring personal toiletries, stationary for class, and any required medicines. Two or three seated rooms with beds, bedcovers, pillows, pillow covers and mosquito nets are provided. Students are discouraged from bringing expensive cell phones, cameras or ornaments.

Strictly prohibited items include desktop PCs, sound systems, electric heaters, irons, playing cards and illegal drugs. Management cannot be held responsible for the loss of personal belongings or valuable items. Students are requested to bring light luggage so that it can be carried to the dorms easily. Students are also requested to submit the payment/deposit slip and grade sheet to their respective dorm tutors during dorm registration.

Student conduct Students should also keep from plagiarism, eve teasing, bullying, sexual harassment, unfair means in the examination, misbehaving with the teachers & staffs, destroying the assets of the campus, any kind of theft/stealing, any kind of fighting, any kind of political activism, violation of rules, using a fake visitor's profile, socially unaccepted relationships with other students, entrance in the opposite gender's dorms, leaving the campus without permission, racism, use of slang, gambling, viewing adult movies, smoking in the non-smoking area, writing on walls or bringing any kind of arms onto the campus. Strict action will be taken on any student transgressing these rules.

Strictly ProhibitedBringing, taking or sharing of drugs on Savar Campus is strictly prohibited. Any student caught doing any of these things will face BRACU's Disciplinary Committee, and may be expelled from the university. Suspected drug abusers may also be asked at any time to take a urine/blood test, and, if tested positive, will face the same rigorous disciplinary procedure. Only prescription drugs recommended by a qualified physician are exempt.

What is the protocol for students facing problems with fellow students or the dorm?Students should report any problems to the house tutors, faculty members or counsellors. If the matter remains unresolved, the problem can be reported to the Assistant Campus Superintendent (ACS) and, in extreme cases, to the Campus Superintendent (CS).

What facilities are available on Savar Campus?The academic facilities on the Residential Campus include well-equipped classrooms, spacious conference rooms, a rich library with reading areas, and modern computer labs with internet connection. There are comfortable dormitories for students and faculty members, four large dining halls, an auditorium for presentations and meetings, and grounds for outdoor games. Savar Campus also offers medical services, psychosocial support, and sports facilities.

Medical servicesThe campus has a resident doctor and staff nurse. The university has also made institutional arrangements with Savar’s Enam Medical College and Hospital for medical emergencies. However, students have to bear their own medical costs. Parents are encouraged to share any concerns for their child’s healthcare, which is not included on their health cards, such as drug use. In addition, students are required to mention any proposed doctor’s appointments, along with the doctor’s name (where possible), in the health card, otherwise s/he will not be permitted any leave during the semester in this regard.

Psychosocial supports Working with a counselor may help students to see their problems with greater clarity, acquire insights into its origins, and think about how to move forward constructively. Counseling can also help students in making decisions that are right for them. Therefore to help students in this regard Savar Campus has one male and one female full-time psychosocial counsellor to support students undergoing any kind of personal or social difficulties.

Sports facilities Savar campus has the necessary infrastructure for students who wish to play football, cricket, table tennis, carom, handball, volleyball, or badminton. However, students will have to bring their own sports equipment, such as balls, bats, racquets etc.

Mathematics is one of the oldest intellectual pursuits of mankind. In modern society there are few areas of knowledge and practice in which mathematics does not have a role to play. Mathematics, as a subject of study possesses a great diversity and offers a wide range of opportunities. MNS Department started BS in Mathematics program (127 credits) in Spring, 2012. Mathematics is also offered as a minor for students doing major in physics, CS/CSE, ECE, EEE, ESS or any other relevant discipline. Such major-minor combination will stand the students in good stead in the job market. It may be mentioned that undergraduate course in mathematics is only offered in a very few of the private universities in Bangladesh.

News Archive >>

News

Biotechnology seminar hosts Bangladeshi scientist who developed high-yielding "Panchabrihi" rice

Orientation held for Summer 2024 biotechnology students

Nanobiotechnology seminar organized by BUSB and Biotechnology Program

The biotechnology program and BUSB organized the inaugural Biotechnology Workshop Series on "Understanding and Designing Biosensors"

On the basis of the admission test results, a student will have to take one or more of the following courses:

a. ENG 101 (Credit course)b. ENG 102 (Credit course)c. ENG 091 (Non credit course)d. MAT 091 (Non credit course)e. MAT 092 (Non credit course)

The exact number of course(s) a student is required to take will be shown against each name of the student in the test result.

Initial admission fee includes a fee one course. Any additional course up to a maximum of four (4) in total, will incur a fee of Tk. 22,500 (all program except LAW) for each subsequent course.

In case of LAW Tk. 24,000 will be incurred for each subsequent course.

If a freshman is prescribed two (02) non-credit courses, the third course for him/her is optional.

*Fees can be changed as per the decision of the university

Other Fees

For Foreign StudentsAccount Name: Brac University (Collection A/C)Account Number: 1501200132106002Bank Name: BRAC Bank Ltd.Bank Address: 1, Gulshan Avenue, Gulshan, Dhaka-1212Swift: BRAKBDDH

For Local StudentsBRAC Bank Ltd. (Any Branch Except Agent Banking)Dhaka Bank Ltd. (Any Branch)IFIC Bank Ltd. (Mohakhali Branch)ONE Bank Ltd. (Any Branch)Southeast Bank Ltd. (Any Branch)Prime Bank Limited (Any Branch)

*May increase with a notice before a semester

Fees for New Postgraduate Students effective from Fall 2023*

Fees for Postgraduate Students (all students enrolled till Summer 2023)*

Mode of Payment:

Fees for Foreign Students*

Fees for Visiting Student (Undergraduate)*

Fees for Visiting Student (Postgraduate)*

BRAC University offers its own affordable, safe and comfortable transport service (buses with air conditioning) for students on nine routes across Dhaka city.

Service Availability: From Sunday to Thursday (excluding weekends, government holidays and semester breaks)

The following schedule is applicable from 28 May 2024.

Route-01: Abdullahpur to BRAC University

Stoppages

2nd Trip (Pickup time)

Abdullahpur (Polwel Carnation Shopping Centre)

6:40 AM

8:50 AM

House Building (Labaid Diagnostic)

6:43 AM

8:53 AM

Azampur (in front of Uttara East Police Station)

6:46 AM

8:56 AM

Jasimuddin (Popular Diagnostic Centre)

6:49 AM

8:59 AM

Airport (in front of police box)

6:51 AM

9:02 AM

Kawla (near foot over-bridge)

6:52 AM

9:04 AM

Khilkhet (first foot over-bridge coming from Uttara)

6:56 AM

9:07 AM

Biswa Road (near foot over-bridge)

6:59 AM

9:09 AM

Shewrah (near foot over-bridge)

7:00 AM

9:10 AM

BRAC University (arrival time)

7:30 AM

10:40 AM

Route-02: Mirpur-A to BRAC University

Stoppages

1st Trip (Pickup time)

2nd Trip (Pickup time)

Mirpur Bangla College

6:30 AM

8:40 AM

Sony Cinema Hall (near KFC)

6:34 AM

8:45 AM

Mirpur College Gate (Mirpur 2)

6:36 AM

8:47 AM

Swimming Federation Gate (Mirpur stadium)

6:38 AM

8:49 AM

Popular Diagnostic Centre (Mirpur 6)

6:41 AM

8:53 AM

Pallabi Post Office (near Purobi Cinema Hall)

6:44 AM

8:57 AM

Mirpur Ceramics (near CNG filling station)

6:46 AM

9:00 AM

Mirpur DOHS Gate (near Trust bus stop)

6:50 AM

9:05 AM

Pallabi Thana (Shagufta mour)

6:53 AM

9:07 AM

Kalshi Bus Stand

6:55 AM

9:09 AM

ECB Chattar

6:58 AM

9:12 AM

BRAC University (arrival time)

7:30 AM

10:30 AM

Route-02: Mirpur-B to BRAC University

Stoppages

1st Trip (Pickup time)

2nd Trip (Pickup time)

Mirpur 14 Bus Stand (opposite to Marks Medical College Hospital)

6:45 AM

9:00 AM

In front of NAM Garden Officers Quarter, Mirpur 13 (near BRTA 2nd Gate)

6:48 AM

9:03 AM

In front of Adarsha High School (Mirpur 10)

6:51 AM

9:06 AM

Al Helal Specialized Hospital (Mirpur 10)

6:55 AM

9:10 AM

Kazipara (near Bank Asia and foot over-bridge)

6:58 AM

9:13 AM

Shewrapara (near Jamuna Plaza and metro rail station)

7:01 AM

9:16 AM

Taltola Bus Stand (opposite to Glory School & College)

7:04 AM

9:19 AM

Shadhinata Tower (next to Bir Srestha Shaheed Jahangir Gate)

7:11 AM

9:24 AM

BRAC University (arrival time)

7:30 AM

10:30 AM

Route-03: Jigatola to BRAC University

Stoppages

1st Trip (Pickup time)

2nd Trip (Pickup time)

Jigatola Bus Stand (in front of Japan-Bangladesh Friendship Hospital)

6:35 AM

9:00 AM

Shankar Bus Stand

6:40 AM

9:05 AM

Mohammadpur Bus Stand (in front of Allah Karim Mosque)

6:45 AM

9:10 AM

West end of Manik Mia Avenue (opposite to Aarong)

6:50 AM

9:18 AM

BRAC University (arrival time)

7:30 AM

10:30 AM

Route-04: Azimpur to BRAC University

Stoppages

1st Trip (Pickup time)

2nd Trip (Pickup time)

Azimpur Etimkhana Mour

6:15 AM

8:30 AM

Azimpur Chowrasta

6:18 AM

8:35 AM

New Market (gate no 2, Janata Bank)

6:21 AM

8:40 AM

Happy Arcade Shopping Mall (Beside City College)

6:24 AM

8:45 AM

Dhanmondi Road No 7 (ARA Center Shopping Mall)

6:27 AM

8:50 AM

Kalabagan Krira Chokro (near foot over-bridge)

6:30 AM

8:55 AM

Sobhanbag (near foot over-bridge)

6:33 AM

9:00 AM

West End of Manik Mia Avenue (opposite to Aarong)

6:36 AM

9:05 AM

Khejur Bagan Mour (Manik Mia Avenue)

6:38 AM

9:08 AM

BRAC University (arrival time)

7:30 AM

10:45 AM

Route-05: Jatrabari to BRAC University

Stoppages

1st Trip (Pickup time)

2nd Trip (Pickup time)

Doyaganj Mour (near mandir)

6:30 AM

8:40 AM

Ittefaq Mour

6:40 AM

8:50 AM

Mugda Foot Over-bridge

6:50 AM

9:05 AM

Boudhha Mandir

6:51 AM

9:07 AM

Bashabo Bus Stand

6:52 AM

9:08 AM

Khidmah Hospital

6:55 AM

9:15 AM

Malibagh Rail Gate

6:57 AM

9:18 AM

BRAC University (arrival time)

7:30 AM

10:30 AM

Route-06: Baldha Garden to BRAC University

Stoppages

1st Trip (Pickup time)

2nd Trip (Pickup time)

Baldha Garden

6:40 AM

8:45 AM

Motijheel (foot over-bridge)

6:42 AM

8:50 AM

Arambagh (near Janata Bank)

6:43 AM

8:52 AM

Fakirapool Mour (Apex showroom)

6:45 AM

8:56 AM

Purana Paltan (Hatil showroom)

6:46 AM

8:58 AM

Kakrail Mour (near Eastern Bank)

6:47 AM

9:02 AM

Shantinagar Mour (One Bank)

6:48 AM

9:04 AM

Malibagh Mour (City Bank)

6:49 AM

9:05 AM

Mouchak Mour (near IFIC Bank)

6:52 AM

9:07 AM

Moghbazar T&T Office

6:53 AM

9:08 AM

Moghbazar Mour (NCC Bank)

6:56 AM

9:13 AM

BRAC University (arrival time)

7:40 AM

10:30 AM

Route-07: Mohammadpur to BRAC University

Stoppages

1st Trip (Pickup time)

2nd Trip (Pickup time)

Shia Masjid (near Top Ten shop)

6:30 AM

8:50 AM

Opposite to Suchona Community Center

6:32 AM

8:52 AM

Shampa Market

6:34 AM

8:55 AM

Shyamoli (Bata Mour)

6:37 AM

8:58 AM

Opposite to Shyamoli Cinema Hall

6:39 AM

9:01 AM

Agargaon Radio Office

6:42 AM

9:03 AM

Opposite to Agargaon flower shops

6:44 AM

9:05 AM

BRAC University (arrival time)

7:35 AM

10:45 AM

Route-08: Narayanganj to BRAC University

Stoppages

Pickup time

Narayanganj Central Shaheed Minar (Chashara)

6:30 AM

Himaloy Chinese & Community Centre (opposite to Narayanganj Zila Parishad)

6:34 AM

Shibu Market (in front of Dutch Bangla Bank)

6:37 AM

Jalkuri Bus Stand, Fatullah (in front of First Security Islami Bank)

6:40 AM

Bhuighar (near Darussunnah Islamia Kamil Madrasah)

6:43 AM

Signboard (near Islamia Eye Hospital)

6:48 AM

Shamsul Hoque School & College (under foot over-bridge)

6:51 AM

Sanarpar Bus Stand, Siddhirganj

6:54 AM

Madaninagar Fazil Madrasah, Siddhirganj (near foot over-bridge)

6:57 AM

Khankaye Jame Masjid, Narayanganj (near foot over-bridge)

7:02 AM

Opposite to Rani Mahal Cinema Hall, Demra

7:04 AM

IFIC Bank Ltd (opposite to Sarulia Bazar)

7:05 AM

In front of Demra Ideal College

7:11 AM

BRAC University (arrival time)

7:35 AM

Route-09: Bashundhara Residential Area to BRAC University

Stoppages

1st Trip (Pickup time)

2nd Trip (Pickup time)

Ebenezer International School, Bashundhara R/A

7:00 AM

9:02 AM

Block-F, Road No 08, Bashundhara R/A (opposite to IFIC Bank)

7:05 AM

9:25 AM

In front of Aga Khan Academy, Bashundhara R/A

7:10 AM

9:30 AM

Kuril Intersection (in front of Sanji CNG Cylinder Retest Center)

7:15 AM

9:35 AM

Jamuna Future Park (under foot over-bridge)

7:20 AM

9:40 AM

Vatara Police Station

7:25 AM

9:50 AM

BRAC University (arrival time)

7:40 AM

10:35 AM

1st Outbound Timing (drop off) from Basement 01:

BRAC University to Abdullahpur: Route No-01

2:05 PM

BRAC University to Mirpur-A: Route No-02

2:10 PM

BRAC University to Mirpur-B: Route No-02

2:05 PM

BRAC University to Jigatola: Route No-03

2:05 PM

BRAC University to Azimpur: Route No-04

2:10 PM

BRAC University to Jatrabari: Route No-05

2:05 PM

BRAC University to Baldha Garden: Route No-06

2:10 PM

BRAC University to Mohammadpur: Route No-07

2:10 PM

BRAC University to Bashundhara: Route No-09

2:05 PM

2nd Outbound Timing (drop off) from Basement 01:

BRAC University to Abdullahpur: Route No-01

5:10 PM

BRAC University to Mirpur-A: Route No-02

5:15 PM

BRAC University to Mirpur-B: Route No-02

5:10 PM

BRAC University to Jigatola: Route No-03

5:10 PM

BRAC University to Azimpur: Route No-04

2:10 PM

BRAC University to Jatrabari: Route No-05

5:15 PM

BRAC University to Baldha Garden: Route No-06

5:15 PM

BRAC University to Mohammadpur: Route No-07

5:10 PM

BRAC University to Narayanganj: Route No-08

5:15 PM

BRAC University to Bashundhara: Route No-09

5:10 PM

Fare (except for Narayanganj and Bashundhara routes): BDT 100 (Taka One Hundred Only) for each passenger for each trip. For round trip (coming and going), BDT 200 (Taka Two Hundred Only).

For each trip on the Narayanganj route, the fare for each passenger is BDT 150 (Taka One Hundred and Fifty Only). For a round trip (coming and going), it is BDT 300 (Taka Three Hundred Only).

For each trip on the Bashundhara route, the fare for each passenger is BDT 50 (Taka Fifty Only). For round trip (coming and going), it is BDT 100 (Taka One Hundred Only).

Fare needs to be paid in cash to service attendants inside the buses. bKash accepted on Route-02: Mirpur-A to BRAC University

General Instructions

For any official query, please contact, Md Lutfor Rahman: 01711-859787 and Mijan Hosen: 01827-407161

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* Subject to appearance of the moon and Subject to decision by the Government regarding the official holidays.

BTC 501 Plant Biotechnology 3 Credits

The plant biotechnology course covers principles and different aspects of plant biotechnology.

Topics include:

- Plant cell cultures; growing tissue-, axillary bud, root- and meristem cultures- Protoplast culture, somatic hybridisation and importance, cybridization- Application of mass propagation (micropropagation) for virus-free vegetatively propagated crops such as potatoes, ornamentals, forest trees and medicinal plants- Plant growth regulators: auxin, cytokinin, gibberellin, zeatin, 2iP and their role and putative mode of action- Regeneration pathways: organogenesis vs somatic embryogensis; concepts and applications- Plant transformation- application of the technique in transferring useful genes such as genes for disease-, insect resistance, those that add nutritional values to the crops of interest across wide genetic barriers- Production of genetically modified (transgenic) plants: indirect and direct methods, selectable markers, transient and stable expression, merits and demerits of the respective methods- A critical assessment of GM crops containing genes for herbicide-, virus-, bacterial-, fungal-, nematode- and insect pests resistance. evaluation of GM crops for their adoption in developing countries- Germplasm preservation-ex situ and in situ preservation; gene bank- Plant secondary metabolites of medicinal importance- Commercial exploitation of plant tissue culture and possibilities in Bangladesh

Recommended Books:

1. An Introduction to Plant Tissue Culture, M. K. Razdan, 19982. Plant Tissue Culture: Theory and Practice, S. S. Bhojwani and M. K. Razdan, 19963. Biochemistry & Molecular

Biology of Plants, Bob B. Buchanan, 20004. Biotechnology: Theory and Techniques: Plant Biotechnology, Animal Cell Culture, Immunobiotechnology (Vol I), Jack G Chirikjian (Ed.), 19955. Plant Biotechnology: Perspectives and Prospects, P. C. Trivedi, 20066. Plant Tissue Culture & Biotechnology, P. C. Trivedi, 20077. Principles and Practices in Plant Science, P. D. Walton, 1988

BTC 502 Plant Biotechnology (Lab) 3 Credits

The Plant Biotechnology (Lab) course will cover the following experiments:- Setting up tissue culture experiments for callusing - Differentiation as well as different aspects of molecular biology beginning from DNA and RNA isolation, running them in gel for their characterization based on their kb length- Use of restriction enzymes for DNA and RNA fragmentation at predetermined sites- Ligation of different pieces of DNA in a suitable plasmid vector such as pBluescripts- Plant cell cultures; media: sterilization techniques- Initiation of primary cultures; morphogenesis and phytohormones- DNA extraction from E. coli plasmids, total RNA isolation from model plants- Construction of cDNA library and isolating cDNA clones; minipreparation of plasmid DNA- Computer analysis of DNA and protein sequence- Plant genomic DNA isolation- Restriction digestion of DNA and Southern transfer; RNA gel electrophoresis and Northern transfer; non-radioactive hybridisation of Southern and Northern Blots.- Agrobacterium-mediated gene transfer (vector construction, co-cultivation); RAPD and microsatellite analyses for confirmation of hybridity/ DNA fingerprinting, biolistics, analyses of transgenic plants (PCR and RT-PCR)- Southern analyses, chromosome preparations and physiological analyses of transgenic plants

Recommended Books:

1. Plant Tissue Culture: Theory and Practice, Bhojwani, S. S. & Razdan, M. K., 19962. Biotechnology: Theory and Techniques: Plant Biotechnology, Animal Cell Culture, Immunobiotechnology (Vol I), Jack G Chirikjian (Ed.), 19953. Plant Tissue Culture & Biotechnology, Trivedi, P. C. (Ed.), 20074. Plant Tissue Culture: Application and

Limitations, Bhowjwani, S.S. 19905. Plant Cell Culture: A Practical Approach, Dixon, 1994

BTC 503 Animal Tissue Culture Techniques and Applications 3 Credits

The course has been designed to impart basic knowledge in animal biotechnology so that students may take up a topic on animal biotechnology leading to cloning of useful genes from important animals such as cattle in Bangladesh.

Topics include: - Definition, principle and significance of animal tissue culture, basic differences between plant and animal cell cultures- Mycoplasm and viral contaminants of animal cell culture; maintenance of sterility and use of antibiotics- Various systems of tissue culture: their distinguishing features, advantages and limitations- Culture medium: logic of formulation (natural media, synthetic media and sera) - Methodology- i) primary culture: behavior of cells, properties, utility ii) explant culture iii) suspension culture- Nutrient media: obligatory and optional constituents- Incubation systems: static agitated culture systems- Hormone signaling and mechanisms of signal transduction, fertilization, early embryogenesis- Applications of gamete and embryo manipulations for biomedical purposes, tissue-specific gene expression and tumorigenesis- Development and preparation of vaccines against infecting organisms- Artificial animal breeding; in vitro fertilization - Cloning: techniques and technologies; Dolly, the cloned sheep - Mutant cell lines: significance in biomedical research, identification and isolation of mutants- Application of genetic manipulation, medicinally important compounds; screening of cell lines for novel variations: disease resistance, stress tolerance- Gene transfer to animal cell, viral vector (development and use), gene therapy; transgenic animal- Cytotoxicity and diagnostic tests; development and preparation of vaccines against infecting organisms

Recommended Books:1. Biotechnology: Theory and Techniques: Plant Biotechnology, Animal Cell Culture, Immunobiotechnology (Vol I), Jack G Chirikjian (Ed.), 19952. Animal Biotechnology, M. M.Ranga, 2004 3. Molecular Biotechnology, B. R. Glick and J. J.

Pasternak, 20034. Molecular Biology, Primrose, 20035. Animal Cell Biotechnology, R.E. Spier and J.B. Griffiths, 1988

BTC 504 Fermentation and Industrial Biotechnology 3 Credits

The course outlines the process of fermentation and industrial biotechnology from organism and environmental points of view.

Topics include:- Industrially important microorganisms-bacteria, fungi, actinomycetes and yeasts - Major classes of microbial products and processes- Growth of microorganisms in batch, semibatch and continuous culture- Microbial growth kinetics for primary and secondary metabolites in batch and continuous culture- Industrial applications of continuous culture-potentials and limitations- Upstream and downstream processing of microbial fermentations- Twin faceted core of industrial biotechnology-genetic manipulation including application of genetic engineering techniques for microbial strain improvement- Optimization of media and fermentation conditions for maximizing productivity- Scale up of fermentation processes; criteria for scale up; physical, chemical and sterilization factors- Bioreactor design-batch, semi-batch and continuous bioreactor operations- Aeration and agitation; sterilization: batch and continuous- Process monitoring- off line and on-line sensors; and process control- Unit operations in product recovery and purification- Principles and methods of cell immobilization and their industrial applications- Biotechnological production of representative metabolites: organic acid, amino acids, alcohol, industrial enzymes, antibiotics, recombinant proteins; biopharmaceuticals and vaccines- Anaerobic fermentations- industrial production of alcohol, acetone, butanol etc.- Scope of industrial biotechnology in Bangladesh

Recommended Books:1. Modern Industrial Microbiology and Biotechnology, Nduka Okafor, 20072. Principles of Fermentation Technology, Peter F. Stanbury, Stephen J. Hall, and Allan Whitaker, 19993. Prescott & Dunn's Industrial Microbiology, 4th Edition, G. Reed, (Ed.), 1999

4. Microbial Biotechnology-Fundamentals of Applied Microbiology, Alexander N. Glazer, Hiroshi Nikaido, 20075. Fermentation Microbiology and Biotechnology, 2nd Edition, E.M.T. El-Mansi, C.F.A. Bryce, A.L. Demain and A.R. Allman, (Eds), 20076. Microbial Biotechnology Principles and Applications, 2nd Edition, Lee Yuan Kun (Ed.), 2008

BTC 505 Environmental Biotechnology 3 Credits

This course details the dangers of pollution; how the environment is rapidly being polluted due to setting up of industries all over the world and in Bangladesh as well; how the effluents from factories along the shores of rivers and water bodies are killing the aquatic life in and around the cities and possible remedial measures.

Topics include: - Microbes and metabolism; genetic blueprint for metabolic capability; metabolic pathways of particular relevance to environmental biotechnology- Fundamentals of biological interventions: extremophiles with diverse degradative capabilities; - Pollution and pollution control biotechnology: types of pollution and pollution control strategies and available technologies- Contaminated land and in situ and ex situ techniques for remediation- Aerobes and effluents-sewage treatment tank; activated sludge systems; oxidation ditch; membrane bioreactors- Phytotechnology- metal phytoremediation, rhizofiltration; organic phytoremediation; algal treatment systems- Biotechnology of wastes- composition of wastes; biological waste treatment; composting and application in waste treatment; anaerobic digestion; biogas and other technologies- Bioremediation; suitability and factors affecting bioremediation; essential features of biological treatment systems- Environmentally transmitted pathogens; risk assessment; microorganisms and metal pollutants- Biosensors- Concept of viable but non-culturable cells (VBNC); present status of VBNC; molecular genetic methods for detection and identification of VBNC; implication and significance of VBNC in environment and health- Use of commercial blends of micro-organisms and enzymes in wastewater treatment; immobilized cells in the waste treatment; potential application of recombinant DNA technology in waste treatment- Xenobiotic degrading bacteria and their catabolic genes in bioremediation: in situ analysis of microbial community and activity in bioremediation, DNA- and RNA-based methods; genetic finger printing techniques; recent powerful sensitive techniques for detection of specific compounds- Integrated approaches in environmental biotechnology

Recommended Books:1. Environmental Biotechnology: Theory and Applications, GM Evans and J.C Furlog, 20032. Environmental Biotechnology: Principles and Applications, B.E Rittman and P.L. McCarty, 2001 3. Environmental Biotechnology, Alan Scragg, 20024. An Introduction to Environmental Biotechnology, Milton Wainwright, 20045. Environmental Biotechnology, Hemant Rawat, 20086. Environmental Biotechnology: Concepts and Applications, Hans-Joachim Jördening, Josef Winter (Ed.), 2005

BTC 506 Research Project Preparation 3 Credits

This course gives research orientation to students so that they can carry out research projects with efficiency and confidence.

Topics include:- Project design, planning, implementation and evaluation: writing project proposals; essential components of a project- overall development objectives, specific objectives, output, verifiable indicators, means of verification, assumptions; logical framework matrix; project monitoring and evaluation methods- Research project planning: literature review through search engines, such as Google, HINRARI, Altavista, PubMed, and other online literature survey to dig out information relevant to a research project - Design of methodology and experiments supported by knowledge in biostatistics, record observations, interpret results- Writing a scientific paper following the format of a particular journal including bibliography- Use of search engines to be up to date in the line of research - Selection of a suitable research topic, steps to be followed-idea and concept, objective analysis, theme development and chronological design of different plans of activities- Selection of a suitable guide for research; communication skill for interacting with the guide at different stages of work- Writing a thesis- different components to be included; relevance of objectives to literature review, methodology, overview of study design and results, comparative analysis with the existing information in the selected project; analysis of uniqueness of findings; discussion and interpretation- Writing a research paper for journal publication, structure of a scientific paper- the IMRAD format- selection of suitable title, key words, abstract , introduction materials and methods, arrangements of tables and figures in a self-explanatory way, results and discussion and references all based on thematic approach- Convenient order of reading or writing a paper; modern methods of communication- electronic submission, e- journals etc.- Presentation of paper- oral presentation, poster presentation; logical organization of powerpoint presentation-structuring the presentation largely as a story

Recommended Books:

1. Thesis and Dissertation Accomplished: A Comprehensive CD-Rom, Carter W. Tada F. 2006, http://www.tadafinallyfinished.com/2. Writing and Publishing Your Thesis, Dissertation, and Research: A Guide for Students in the Helping Professions, Heppner, P. P., & Heppner, M. J., 20043. The Elements of Style (4th ed.), William Strunk, Jr., E. B. White, 20004. . Dissertations and Theses from Start to Finish: Cone, J. D., Foster, S. L., 2006

BTC 507 Biostatistics and Experimental Design (Theory and Lab) 3 Credits

This course has been tailored to the need of molecular biologists who are often confronted with the problem of making a valid conclusion for want of properly organized experimental layout.

Topics include: - Definition and scope of biostatistics, measures of central value; mean, median, mode, measures of dispersion, range, quartile deviation, mean deviation, variance, standard deviation, standard error, coefficient of variation- Sampling distribution, confidence limit; correlation and regression: calculation of correlation coefficient and test for its significance, regression coefficient, regression line, multiple regressions- Concept of probability, probability rules, conditional probability and independence- Probability distributions: binomial, Poisson and normal distributions and their applications; hypothesis testing, null hypothesis; level of significance; comparison of two means, t-test, t-tests for small and large samples, paired t-test, chi-squared test, goodness of fit test, test of independence, contingency tables- Analysis of variance; one-way and two-way classifications, comparison of three or more samples, F-test; concepts of experimental design, principles of experimental design, completely randomized design (CRD), randomized bock design (RBD), latin square design, factorial experiments, split-plot design- Multiple comparisons, least significant difference test (LSD test), Duncun's multiple range test

Recommended Books:1. The Design of Experiments, R. A. Fisher, 19752. Statistical Table for Biology, Agriculture and Medical Research, R. A. Fisher and T. Yates, 1965

BTC 508 Seminar 3 Credits

The course aims at development of presentation skill of the students about topics chosen from literatures and their own findings

Topics include: - Presentation of the contents of a complex article in a short, concise and comprehensive form- General structure of a seminar talk: title, list of contents, conceptual background as to why the topic is interesting, hypothesis or research objectives, presentation of materials and methods to depict only what is needed to understand the results (short), presentation of results preferably with clearly understood more figures and less tables (little long), discussion to explain if the objectives of the study have been achieved, what are the alternative explanations or whether are more experiments are needed (little long), summary containing the most interesting part to attract the audience- Seminar talk on the selected published paper keeping in mind the following pertinent points:problem investigated; research question answered; primary motivation for investigating the problem; background information (context); relevance to understanding the problem; key points of the paper; method(s) or experimental design used; results reported in the paper; unexpected results, if any and if so whether such results are explained sufficiently- The paper’s conclusions, implications of the investigation, how does the investigation influence future work?- Weaknesses if any of the paper, personal evaluation - Points to consider about the audience: why the audience should be interested in this particular topic; what they already know about it; what terms or definitions are new to them; which key points or concepts will be the most interesting and which ones will be difficult for them to grasp

Recommended Books:

1. Survival Skills for Scientists, Federico Rosei, T. W. Johnston, 20062. Speaking Your Mind: Oral Presentation and Seminar Skills

, Rebecca Stott, Tory Young, Cordelia Bryan, 2001 3. The Craft of Scientific Presentations: Critical Steps to Succeed, Michael, 2003

BTC 509 Genomics (Bioinformatics) 3 Credits

The course outlines a number of software tools used for characterizing an unknown gene or a part of it on the basis of stored data of DNA base sequences of a similar gene. Topics include: - Web based methods in molecular genetics, computer aided analysis of genetic sequences- Genome analysis, identification and characterization of important functional genes with the help of NCBI data base and suitable software- Modern applications of genetic mapping and importance of genome synteny between species revealing their relationship on phylogenic trees- Current research in molecular genetics and genome analysis, with particular emphasis on modern applications of genetic mapping and the importance of genomic synteny between species- Gene tagging, plant transposons, gene banks and genome databases- Gene cloning based on genome maps, sequencing programs and protein sequence motifs- Principles of algorithms and software for sequence alignment, similarity search of biological databases and DNA sequence analysis- Motif discovery, estimation of molecular phylogenetic trees-

Structural prediction and functional inference; predicting the structure and evolution of macromolecules

Recommended Books:

1. From Molecular to Modular Cell Biology, L H. Hartwell, J. J. Hopfield, S. Leibler and A. W. Murray, Nature, 402: 47-52, 1999 2. Genomics, Gene expression and DNA Arrays, Lockhart, D.J. and Winzeler, E.A. Nature 405: 827-836, 20003. Automated DNA Sequencing and Analysis, M.D. Adams, C. Fields, J. C. Venter (Eds.), 19944. Bioinformatics: a Practical Guide to the Analysis of Genes and Proteins, A. Baxevanis and F. B. F. Ouellette (Eds.), 19985. Algorithms on Strings, Trees, and Sequences: Computer Science and Computational Biology, D. Gusfield and A. M. Lesk (Ed.), 19976. The Internet and the New Biology: Tools for Genomic and Molecular Research, L. F. Peruski Jr., and A. Harwood Peruski, 19977. Molecular Bioinformatics: Algorithms and Applications, S. Schulze-Kremer, 19958. Introduction to Computational Molecular Biology, J. Setubal and J. Meidanis, 19969. Molecular Databases for Protein Sequences and Structure Studies: an Introduction, J. Sillince and M. Sillince, 199210. Biocomputing, Informatics and Genome Projects, D.W. Smith (Ed.), 1994

BTC 510 Fundamental and Applied Aspects of Plant Genetic Manipulation 3 Credits

The course aims at giving biotech students an in-depth knowledge in fundamental and applied aspects of plant genetic manipulation.

Topics include: - Introduction to plant genetic manipulation- Plant tissue culture, transgenic plant production (factors and elements)- GM technologies and GMOs: recent development and future trends- Agrobacterium-mediated transformation- Chloroplast transformation method- Biotic stress resistance transformation- resistant molecules- Fungus resistant transgenic plant production- Abiotic stress resistant transgenic plant production- Innovative techniques for genetic manipulation of plants against a background of a continuing need for plant improvement in agriculture, horticulture and forestry- Cell fusion technology for novel somatic hybrid production- Development of plant transformation systems comprising Agrobacterium-mediated gene delivery, direct DNA uptake and biolistics; vector design; molecular methods in crop improvement alongside the value of gene mapping and genetic fingerprinting for germplasm evaluation- Biopharming- Biofuel

Recommended Books

1. Plants, Genes, and Crop Biotechnology, J. Chrispeels, Maarten and David E. Sadava, 20032. Plant Biotechnology: The Genetic Manipulation of Plants, Slater Adrian, 20063. Genes IX, B. Lewin, 20084. Transgenic Plants (Vol. 1 and 2), Kung and R. Wu, 19935. Plant Biotechnology in Agriculture, K. Lindsey and M.G.K. Jones, 1990

BTC 511 Commercial Production of Horticultural and Ornamental Plants 3 Credits

This course has been designed to train students to work in biotech farms dedicated to commercial production of horticultural, ornamental, timber and medicinal plants through tissue culture techniques.

Topics include:- Cost saving devices in tissue culture: used glass jars from hotels, locally made unscrewed plastic caps to replace cotton plugs, preparation of distilled, deionized water, making of various kinds of culture media under sterile conditions- Adoption of extra precautionary measures to eliminate bacterial and fungal contamination- Different techniques used for preparing explants for callusing and differentiation, procedure for hardening plantlets after they are taken out of culture bottles- Maintenance of plants inside the greenhouse in accordance with their requirements for light, temperature and moisture- Lessons on construction of greenhouses creating microclimate for the growth of different types of horticultural plants, installation of misting and ventilation system including the cost involved- Packing of tissue culture derived material for marketing, without damage during the transport of the material- Status of commercial production of horticultural and ornamental plants in Bangladesh and other neighbouring countries- Potentials of tissue culture technology

Recommended Books:

1. Commercial Production of Horticultural Crops, Kunal Mitra, 2008 2. Scientific Horticulture (Vol. 10), S.P. Singh (Ed.), 20073. Advances in Arid Horticulture (Vol. I) Present Status, P.L. Saroj, B.B. Vashishtha, D.G. Dhandar, 20044. Advances in Horticulture Science Research, R.C. Upadhyaya, 20085. Advances in Ornamental Horticulture (6 Vols), Supriya Kumar Bhattacharjee, 20066. Biotechnology of Horticultural Crops (3 vols), V.A. Parthasarathy (Ed.), T.K. Bose, P.C. Deka, P. Das, 20017. Clonal Tissue Culture of Important Fruit Crops, Atul Kumar, Vandana A. Kumar, 19988. Floriculture : From Greenhouse Production to Floral Design, Ronald J. Biondo, Dianne A. Noland, 20069. Ornamental Crop Production Technology, Supriya Kumar Bhattacharjee (Ed.), 200610. Ornamental Plant Propagation in the Tropics, Carmine Damiano, Giovanni Ferraiolo, W.O. Baudoin, 2005

BTC 512 Sex, Flowers and Biotechnology 3 Credits

The course will outline floral development, senescence and incompatibility status.

Topics include: - Introduction to plant alternation of generation- Reproduction and senescence- Haploid production through tissue culture- In vitro pollination and fertilization- Embryo culture under in vitro condition- Flower colour modification through genetic manipulation- Terminator seed technology- RNAi and PTGS- Gene silencing for the crop improvement- Hybrid seed technology- Methods and achievements in the genetic engineering of crops by modifying floral development- Reproduction in higher plants and the implications for horticulture and crop production- Genetic control of floral development; applied aspects of flowering and reproduction- Molecular basis of self incompatibility, floral senescence, seed storage proteins and the physiology- Biochemistry and molecular biology of fruit ripening

Recommended Books:1. The Plant: the Flower, Better Farming Series 3, Food and Agriculture Organization of the United Nations-Rome, 19972. Plant Propagation: Principles and Practices, Hudson T. Hartmann, Dale E Kester, Fred T Davies, Robert L Geneve, 20023. A Text book of Biotechnology, R.C.Dubey, 1995 4. Biotechnology:

Theory and Techniques: Plant Biotechnology, Animal Cell Culture Immunobiotechnology (Volume I), Jack G Chirikjian (Ed.), 19955. Plant Tissue Culture & Biotechnology, P. C.Trivedi, (Ed.), 20076. Gene Transfer to Plants, Portykns, 1995

BTC 513 Gene Function and Its Regulation 3 Credits

This course will give the students an in depth knowledge about how a gene controls different steps in a biochemical pathway leading to the formation of an end product.

Topics include:

- Molecular structure of gene and chromosomes, mobile DNA- Chromosomal DNA: functional rearrangements and organelle DNAs- Regulation of gene expression: prokaryotic system- Prokaryotic gene expression regulation-II and eukaryotic gene expression regulation- Transcription control at termination and RNA processing step-Gene expression fidelity and cancer- Environmental effect and gene expression-Mutation and repair of DNA, DNA damage and repair regulations- Modern concept of homologous recombination- Operon model; signal transduction, cyclic nucleotides and hormones in gene regulation- Genetic recombination in vivo, homologous recombination by hybrid DNA formation- Site-specific recombination- Transposons and non-homologous recombination, retro-transposons- Mutation: site-directed mutagenesis and 'protein engineering'- Mutations in human genetic diseases and clinical medicine-DNA amplification in vitro: Polymerase chain reaction (PCR), applications of PCR in research, clinical medicine and forensic science- DNA diagnostics: ribotyping; pulsed field gel electrophoresis- DNA fingerprinting ; non-radioactive DNA probe technology-Molecular biology in animal biotechnology: tissue-specific gene expression, gene transfer in animal cells: viral vectors, embryonic stem cells, gene knock-out organisms

Recommended Books

1. Molecular Biology of the Cell, Bruce Alberts, Alexander Johnson, Julian Lewis, Martin Raff, Keith Roberts, Peter Walter, 5th Edition, 2007 2. Principles of Genetics, D. P. Snustad, M. J. Simmons and J. B. Jenkins, 19973. Genes IX, B. Lewin, 20084. Molecular

Cell Biology, J. Darn ell, H. Lodesh and D. Baltimore, 1986 5. Molecular Biotechnology. B. R. Glick and J. J. Pasternak, 20036. DNA Cloning 1 and 2, D. M. Glover, and B. D. Hames, 19957. Molecular Cell Biology (5th Edition). Lodish, Berk, Matsudaira, Kaiser, Krieger, Scott, Zipersky and Darnell, 2003

BTC 514 Gene Organization and Regulation 3 Credits

The course will outline the recent developments in gene regulations, protein folding etc. Topics include:- Structures of genes and chromosomes in relation to regulation of gene expression- Regulation by transcription factors and enhancers/repressors- Co-transcription regulation and the effects of chromatin structure- Details of mRNA processing including the splicesomes, auto-catalysis, polyA addition, differential splicing and RNA editing- Adenosine deaminases acting on RNAs (ADARS)- Transcriptional RNA silencing: small RNAs and insights into a new level of gene regulation- Non-coding RNAs, processes affected by non-coding RNAs- Post-transcriptional RNA silencing (PTGS) - components of PTGS e.g. dicer, RISC (RNA induced silencing complex), RdRP (RNA dependent RNA polymerase)- Molecular steps in RNA silencing, RNA silencing as a tool for knocking out genes- RNA viruses and RNA silencing- Use of various expression systems for the production of recombinant proteins including strategies for protein isolation and refolding including the use of molecular chaperons- Epigenetic control of gene regulation- environmental interplay with genetic elements - Web-based methods in molecular genetics, computer-aided analysis of genetic sequences

Recommended Books:

1. An Introduction to Genetic Analysis. W. H. Freeman, 7th Edition2. Gene Regulation and Metabolism: Post-Genomic Computational Approaches-Computational Molecular Biology, Ralf Hofestädt (Ed.), 20043. Anatomy of Gene Regulation, Panagiotis A. Tsonis, 20034. Gene Regulation. David Lachman, 20055. Epigenetics, C. David Allis, Thomas Jenuwein, Danny Reinberg, Marie-Laure Caparros, 20076. Epigenetic Mechanisms of Gene Regulation, Vincenzo E.A. Russo (Ed.), 1996

BTC 515 Structural and Functional Genomics Studies 3 Credits

The course is intended to give a clear concept about genome, its location and functions of the major genes in normal congenital disease-conditions in living systems.

Topics include:- Structural genomics and functional genomics- Genome elements: genes - their sites, location and function; mitochondrial genome- Genome mapping: genetic mapping; physical mapping - low resolution and high resolution- Structural genomics: molecular modeling: method for unraveling protein structure and function- Human genome: the physical structure, variations, evolution, genetic disorder- Human genome project: goal and achievements so far- Genome sequencing, ESTs, SNPs, microarrays- Plant genome: general features, chloroplast genome- Comparative use of RAPD, RFLP, AFLP, ISSR and micro-satellite markers in plant gene mapping and selection- Model plant genome: Arabidopsis thaliana and Oryza sativa- Identification and characterization of important plant genes: salt-, insect, disease, submergence resistance genes - Identification and characterization of genes controlling (a) flowering, (b) vernalization (c) photoperiod (d) circadian clock- Use of different web sites in the identification of important genes and their functions- DNA finger printing and its utilization as molecule markers in selection of important agronomic characters

Recommended Books:

1. From Structural Genomics to Functional genomics Methods and Applications, Ioannis Ragoussis, to be published 2010 2. Sequence - Evolution - FunctionComputational Approaches in Comparative Genomics, Koonin, Eugene V., Galperin, Michael Y. 20024. Genomes. Sussman, Hillary (Ed.), Smit, Maria (Ed.), 2006 5. Introduction to Genomics, A. Lesk, 2004 6. The Implicit Genome Caporale, Lynn (Ed.), 2006 ,7. Quantitative Genetics, Genomics, and Plant Breeding. Manjit S. Kang, 20028. The Dictionary of Gene Technology : Genomics, Transcriptomics, Proteomics, Gunter Kahl, Volume 3rd, 2007

BTC 516 Special Study 3 Credits

The course has been designed to acquaint the students with a number of important websites containing valuable information about genomics and proteomics of important plants and organisms.

Websites include:

- Hinari: http://www.who.int/hinari/en/ - The National Center for Biotechnology Information (NCBI) runs the most useful website for bioinformatics- Bioinformatics Web, offering links and information on bioinformatics and computational biology- Bio-computing.org, covers recent literature, tutorials, a bioinformatics lab registry, links, bioinformatics database, jobs, and news - updated daily - BioMedNet, biomedical and bioinformatics resources - CAMDA: Critical Assessment of Techniques for Microarray Data Analysis, includes data and papers from CAMDA conferences - Ensembl, up-to-date sequence data and the best possible automatic annotation for many genomes- Ergito, high-quality books and information on molecular biology and related topics, some sections free- GeneCards, a database of human genes, their functions, and related diseases- Genome KnowledgeBase, a collection of mini-reviews in the field of basic molecular biology- icademic.org, Bioinformatics, Genetics, and other Life Sciences Information Sources. - ISCB, International Society for Computational Biology - Lipi's Bioinformatics World, tutorials, tools, databases, and more - PubMed (Free MEDLINE Search) - UniGene from NCBI- UCSC Genome Bioinformatics, contains the reference sequence for the human genome and the working drafts for the mouse and rat genomes- Y. F. Leung genomics and bioinformatics site

The students will study these websites critically and thoroughly. Familiarity with these websites will enable the students to use similar sites to retrieve relevant information in their respective area of research. Students choosing this course will study all the sections of selected websites and prepare their summaries bringing out salient points of these sections. During the examination, students will be allowed to use these websites to find answers to the questions such as location of a gene in a particular chromosome, its length, sequence and promoter and other particulars.

BTC 517 Enzymology 3 Credits

The study of enzymatic properties will enable a biotechnologist to plan his experiments on biodegradation and bioremediation to be applicable on bioconversion; softening of basal jute stem cuttings or softening of hides of sacrificed animals, bioremediation etc.

Topics include:- Three-dimensioal structure of enzyme, active site, cofactors, activators, prosthetic groups, coenzymes, enzyme-substrate complex, energy of activation- Factors affecting rate of enzyme reaction, regulations of enzyme reaction- Basic aspects of chemical kinetics, molecular interpretation of rate constants- Activation free energies, enthalpies and entropies, kinetics of enzyme-catalyzed reaction; significance of Km and Vm values - Allosteric sites, homotropic effects, cooperativity, heterotropic effects, allosteric effect- Enzyme inhibition- kinetics of competitive non-competitive and uncompetitive inhibition; partially competitive inhibitors- Enzyme immobilization: different methods of immobilization, advantages and disadvantages of immobilization; applications of immobilized enzymes in industry- Enzyme technology in industries: biological detergent, baby food, brewery industry, baking industry, fruit juice, dairy industry, starch industry, rubber industry, paper industry, photographic industry; applications of enzymes in bio-conversion and biotransformation.- Enzymes as biosensors, enzyme technology in biodegradation of industrial toxic pollution: role of lignocellulosic enzymes in removing industrial toxic pollution- Purification and characterization of an enzyme: (1) gel-filtration - determination of molecular weight (size exclusion chromatography) (2) affinity chromatography (ion-exchange chromatography). (3) gel electrophoresis. - Assaying different enzymes: laccase, cellulase, pectinase, xylanase, a-amylase, test for presence of enzymes in different plant materials, applications of enzymes in industries (visit to different industries to observe the applications), applications of enzymes in biodegradation

Recommended Books

1. Enzymes: Biochemistry, Biotechnology, Clinical Chemistry, Trevor Palmer and L.

Philip, 20072. Handbook of Enzyme Biotechnology, A. Wiseman, 19953. Microbial Enzymes and Biotransformations (Methods in Biotechnology), Jose Luis Barredo, 20054. Industrial Enzymes: Structure, Function and Applications, Julio Polaina and Andrew P. MacCabe, 20075. Immobilization of Enzymes and Cells (Methods in Biotechnology), Gordon F. Bickerstaff, 19976. Biotechnology Processing Steps for Enzyme Manufacturing, R. C. Tripathi, 20077. Microbial Enzymes and Biotechnology, W.M. Fogarty and C.T. Kelly, 19908. Applications of Enzyme Biotechnology, Jeffery W. and Thomas O. Baldwin, 20049. Biocatalysts and Enzyme Technology, Klaus Buchholz, Volker Kasche and Uwe Theo Bornscheuer, 200510. White Biotechnology, R. Ulbeer, D. Sell and P. Baiakeds, 2007

BTC 518 Recombinant DNA Technology 3 Credits

This course outlines the basis of modern techniques in the applications of recombinant DNA technology, its recent progress and application in plant and animal development.

Topics include:- Introduction to recombinant DNA technology, enzymes used in recombinant DNA technology - Cloning vectors, gene cloning and transformation techniques- Analysis of recombinant DNA techniques, gene targeting and site specific recombination- Genetic transformation of prokaryotes, manipulation of gene expression of prokaryotes- Principles, techniques and applications of recombinant DNA technology relevant to medical research, and the investigation and therapy of infectious and inherited diseases- Site-directed mutagenesis, protein engineering; DNA sequencing- Production of protein from cloned genes: production of recombinant protein in E. coli- Production of recombinant protein by eukaryotic cells (yeast), mammalian cell expression vectors- Special vectors for expression of foreign genes in E. coli, using animal cells for recombinant protein production; recombinant proteins from plants- Molecular enzymology and protein engineering: the alteration of a protein structure by site directed mutagenesis of the DNA coding for that protein- Molecular basis of binding specificity, catalysis, subunit interactions etc., examined by physico-chemical methods on proteins and enzymes mutated at key amino acid residues- Construction of genomic library and cDNA library, isolation of gene

Recommended Books

1. Recombinant DNA: Genes and Genomes - a Short Course, James D. Watson, Richard M. Myers, Jan Witkowski, Amy A. Caudy, 2006 2. Recombinant DNA, Watson, 1992. An Introduction to Genetic Engineering, Desmond S. T. Nicholl, 20083. Principles of Genetics, D. P. Snustad, M. J. Simmons and J. B. Jenkins, 19974. Genes IX, B. Lewin, 2008 5. Molecular Biotechnology, B. R. Glick and J. J. Pasternak, 20036. DNA Cloning 1 and 2, D. M. Glover and B. D. Hames, 19957. Molecular Biology, Primrose, 2003

BTC 519 Medical Biotechnology 3 Credits

This course will focus on the recent biotechnological and molecular development in disease diagnosis and treatment emphasizing on the immunological aspects of human.

Topics include:- Laboratory aspects of research in biotechnology related to health- Important areas of biotechnology including biochemistry, pathology, microbiology, molecular biology, immunology and immunobiology- Aspects of specimen handling, storage and biosafety issues- Research in medical biotechnology with special reference to developing countries- Routine diagnostics, modern diagnostic tools- Immunological aspects of host defense- innate immunity: both humoral and cellular- Immunological aspects of host defense- adaptive immunity: both humoural and cellular- Molecular methods for diagnosis of infectious agents common in developing countries- Immunological methods for diagnosis of infectious agents common in developing countries- High throughput techniques for studies of genetics of host and pathogens in infectious diseases- Genetic disorders- inborn error of metabolism, metabolic disorders- Gene therapy- ethical, legal and social implications- Vaccine development and production

Recommended Books:

1. Medical Biotechnology, S.N. Jogdand, 20002. Medical Biotechnology, Judit Pongracz

and Mary Keen, 20093. Medical Biotechnology: Achievements, Prospects and Perceptions, Albert Sasson, 20064. Biotechnology and Biopharmaceuticals: Transforming Proteins and Genes into Drugs,

Rodney J. Y. Ho, Milo Gibaldi, 2003

BTC 520 Cell Dynamics, Cell Cycle and Cell Death; Gene Mapping in Phages, and Bacteria 3 Credits

The course is designed to impart in-depth knowledge about the cell-cell interaction and cell communication and factors that contribute to the death of cells known as apoptosis, physiological changes that occur in a transformed cell.

Topics include:- Role of chromatin in gene expression and DNA damage detection at molecular level- Cell dynamics, cytoskeleton and cell surface, extra-cellular matrix, cell-cell interaction and cell matrix interaction- Cell differentiation- Hormones and growth factors, apoptosis- Cell cycle regulation and cancer: the Biology of oncology; role of p53, p21 and other oncogenes and suppressors; treatment approaches for cancer - Transformed cell, gene mapping in phages, bacteria: conditional lethal and suppressor mutations- Control of gene expression in bacteria, genetics of biosynthetic pathways- Transposons in prokaryotes and eukaryotes, detection of DNA damage at molecular level

Recommended Books:

1. Molecular Biology of the Cell, Bruce Alberts, Alexander Johnson, Julian Lewis, Martin Raff, Keith Roberts, Peter Walter, 5th Edition, 2007 2. Molecular Cell Biology. Harvey Lodish, Paul Matsudaira, Arnold Berk, Hidde Ploegh, 20042. Animal Cell Biotechnology: Methods and Protocols, R. Portner, (Ed.), 2nd Edition, Humana press, 20073. Animal Cell Culture and Technology, 2nd Edition. M. Butler, BIOS Scientific, 20045. Basic Cell Culture: A Practical Approach, Davis, (Ed.), 2nd edition, 20026. Cell Growth,

Differentiation and Senescence: A Practical Approach. G.P. Studzinski, (Ed.), 19997. Cell-Cell Interactions: A Practical Approach, 2nd edition, T. Fleming, 20028. Culture of Animal Cells: A Manual of Basic Techniques, 5th Edition, R. Ian Freshney, 20059. Dynamics of Cell Division, S. Endow and D. Glover, (Ed.), 199810. Molecular Cell Biology for the 21st Century, 5th Edition, 200811. The Biological Basis of Cancer, Robert G. McKinnell, Ralph E. Parchment, Alan O. Perantoni, G. Barry Pierce, Ivan Damjanov, 2nd Edition, 200612. Recent Advances in Clinical Oncology, Frank Branicki, Abdu Adem (Ed.), Saad Ghazal-Aswad (Ed.), Fawaz Chikh Torab (Ed.), 2008

BTC 521: Genetically Modified (GM) Crops, Biosafety and Intellectual Property Rights 3 credits

The course gives emphasis on global differences in acceptance or lack of it of GM food crops in the backdrop of national culture and history, economic conditions, and government initiatives or responses related to the issue.

Topics include:- Risk perception related to the precautionary approach, benefits of GMOs, public acceptance, case study: the Monarch Butterfly, StarLinkTM corn - Introduction to biosafety concept and its risk, transgenesis in plants- Risk for animal or human health - toxicity and food quality/safety, allergies, pathogen drug resistance (antibiotic resistance)- Risk for agriculture - weeds or superweeds, alteration of nutritional value (attractiveness of the organism to the pests), reduction of cultivars (increase of susceptibility) and loss of biodiversity- Risk for the environment - persistence of gene or transgene or transgene products, resistance/tolerance of target organism or susceptibility of non-target organisms, increased use of chemicals in agriculture, unpredictable gene expression or transgene instability- General concerns - loss of familiarity, higher cost of agriculture, field trials not planned for risk assessment, ethical issues (labelling)- Critical assessment of biosafety rules operating in developing countries and the importance of its strict enforcement to protect the population of the third world countries from harmful effects of indiscriminate introduction of GM products-

Global status of acceptance or lack of it regarding GM food crops - Critical evaluation of "Golden" rice and its Bangladesh version BRRI Dhan-29 evolved at IRRI by Bangladeshi scientists- Case studies to examine in-depth the interplay of these factors particularly in the context of the developing world- Convention on biological diversity (CBD), Cartagena Protocol- Biosafety regulations to protect nature, growers and consumers interest and national interest, biosafety regulations in Europe, Canada, EU and different countries in the Asia and the Pacific region- National biosafety guidelines of Bangladesh, biosafety framework, biosafety clearing house- Public perception and biotechnology- Intellectual property rights-categories of IP, patents, international legal framework, WTO- TRIPS-the significance of horizontal provisions in the TRIPS agreement- Existing TRIPS flexibilities and exceptions: checks and balances- Indigenous community knowledge and TRIPS- Importance of IPR for protecting biodiversity and biotech products

Recommended Books1. Genetically Modified Organisms and Biosafety: A Background Paper for Decision-Makers and Others to Assist in Consideration of GMO Issues, Tomme R. Young, IUCN Policy and Global Change Group, 20042. Modern Food Biotechnology, Human Health & Development: An Evidence Based Study, Food Safety Department, World Health Organization, 20053. Genetic Engineering in Agriculture and the Environment: Assessing Risk and Benefits, Maurizio G. Paoletti and David Pimentel, http:\\www.ag.auburn.edu/biotech/genetic.html4. Molecular Biotechnology, Principles and Applications of Recombinant DNA, B. R. Glick and Jack J Peterson, 2003 5. Plants, Genes and Crop Biotechnology, Chrispeels, J. Maarten and E. David Sadava, 20036. National Biosafety Framework, GoB, 20087. Focus on Intellectual Property Rights, US Department of State, Bureau of International Information Programs, http/usinfo.state.gov/8. Intellectual Property Rights: An Introduction for Scientists and Technologists, M.B.E Faez, 2005

BTC 550 Research Project (Thesis) 6 credits

All students will be required to do a research project of 6 credits during one full semester. The research project can be carried out either at BRACU or at any other university or research institutes under the joint supervision of a BRACU biotech faculty and a recognized professor/biotechnologist of the concerned institution. A student will have the option of choosing her/his potential guide and the problem she/he will undertake in consultation with the Coordinator, Biotechnology Programme of BRACU and the final decision about her/his choice of guide(s) and the topic will rest on the Chairperson of the Department.

Tasmin Kamal Tulka is a full time lecturer in Mathematics and Natural Sciences Department at BRAC University since January 2019. She has completed her graduation from University of Dhaka, Bangladesh and post graduation from University of New South Wales (UNSW), Australia. Her research interests include renewable energy, power, wireless communication etc.

Level: 5, Room: 5B16Kha-224 Merul BaddaDhaka 1212. Bangladesh

PHY 111: Principles of Physics IPHY 306: Basic ElectronicsCSE 251: Electronic Devices and CircuitsPHY 307: Physics Lab III

Renewable energy, Power, Wireless communication etc.

Tasmin Kamal Tulka

News & Events

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The Syndicate is the highest executive body that exercises administrative and supervisory control over the academic, administrative and management activities of the University.

Members

Professor Mohammad Mahboob Rahman, PhDTreasurerBrac University

Professor Ahmed Mushtaque Raza Chowdhury, PhDFormer Vice ChairpersonBRAC

Professor Mustafizur Rahman, PhD.Distinguished FellowCentre for Policy Dialogue (CPD) and Former ProfessorUniversity of Dhaka

Professor Abul BarkatProfessor and Former Chairman, Department of Economics; &Professor and Founding Chairman, Department of Japanese Studies,University of Dhaka

Ms. Khaleda AkhterAdditional Secretary (University)Secondary and Higher Education DivisionMinistry of Education

Professor Arshad M. Chowdhury, PhDProfessor and Dean, BSRM School of EngineeringBRAC University

Professor Firdous Azim, PhDChairpersonDepartment of English and Humanities Brac University

Dr. Samia HuqProfessor (ESS) and DeanGeneral EducationBrac University

Member Secretary

Dr David DowlandRegistrar & Controller of Examinations (acting)Brac University

Innovation | Collaboration | Social impact BracU Mobile" is the official application and one-touch experience. "BracU Mobile" application is part of a broader initiative to improve the mobile experience of students, faculty, staff and visitors who interact with BRAC University community. We will continue to develop, expand and improve the mobile application to access BRACU facilities, resources and information. We always welcome your ideas and feedback. Please email your questions, suggestions and ideas to wmt[at]bracu.ac[dot]bd

"BracU Mobile" provides instant access to information on University resources from a web-enabled Android device. Currently the app is available for Android operated devices only. Tap into the campus wherever you go with the official BRACU Mobile app.

Get it now for your Android OS:

Download the APK file

Key Features of version 2.0 includes:

- New User-Interface with some new functionalities and better User-Experience as well as some bug fixing
- Splash Screen: Implementation of Splash Screen
- Students Login: Profile, Class/Exam Schedules, Notifications, Notice, Advised Courses, Result
- Faculty Login: Profile, View Student’s Short Profile, Send Notifications to Students
- Announcements: View latest University announcements
- Academic Calendar: View academic calendar
- News Updates: Read the latest news of Brac University
- Events: Browse a wide variety of events from BracU Events Calendar
- Admissions: Direct link for applying into the University
- Library: Search library collections and resources in the University Libraries
- Location map: Locate BRAC University and see your relative location and explore BracU campus by zooming
- Social Media: Live Facebook feeds, Follow BRACU on - Facebook, Twitter, YouTube,

Instagram, Flickr and LinkedIn

- Contacts: All important contacts in one place

- Important links: List of important links and website

- LMS: Learning Management System

- DSPACE: Browse tons of papers in DSpace

- Video: University video stream

- Photos: University photo stream

- Cafeteria: Services of University Cafeteria

- BRACU Anthem: Listen to BRACU anthem

- QR Reader: Included QR code (Quick Response Code) reader will convert QR code to human readable form

- Push Notification: Get live notifications instantly and get connected with what is happening around BRACU

- Feedback: Mail us your feedback and suggestions on the BracU Mobile app

- About: Basic information about the application and Brac University

BracU offers excellent facilities to International students for studying in Bangladesh in various programs at the undergraduate level. Visit the link for detailed admissions programs: https://www.bracu.ac.bd/academics/programs/undergraduate

Application deadlines:

BracU offers three semesters in each academic year - Spring, Summer and Fall. International students can apply any time during the year. There is no application deadline for International Applicant.

Application for admission

A member of BracU International Student Recruitment Team will assist you to complete your admission application. We aim to make your admissions process as simple and easy as possible.

Minimum qualifications for applying

Students have to submit all necessary academic documents to the International Office or may email scanned copies of these to info.international-office[at]bracu.ac[dot]bd. Students will be informed about their eligibility for being admitted once these documents have been scrutinized and evaluated.

Documents Required

The academic and other related documents required for admission to BracU are as follows:

Academic Qualifications for Undergraduate Admission

Minimum GPA of 3.0 in O-Levels in five subjects and A-Levels in two subjects according to the Brac University scale (A=5, B=4, C=3 & D=2). Subjects with E grade will not be considered.

Candidates who have completed higher secondary education (12 years of schooling) under a system different from 0/A Level or equivalent will be considered.

Candidates with a break of study of not more than three years may apply for admission.

Candidates must have a minimum aggregate GPA of 8.00 out of 10 (on a scale of 10), with no grade lower than 3.5 Secondary/Equivalent & Higher Secondary/Equivalent. In addition, they must achieve a minimum GPA of 4.00 in Physics, Chemistry and Biology in both examinations. They also need a minimum GPA of 3.5 in Mathematics (on a scale of 5 in Higher Secondary or equivalent). For “O” levels & "A" level candidates, they must have a B grade in Physics, Chemistry and Biology, and minimum C grade in Mathematics.

Candidates having no Mathematics or failed in the Higher Secondary/ “A” level or recognized equivalents may be admitted, but they need to take an extra 3 (three) credits course on Mathematics relevant to B. Pharm. curriculum.

Candidate must pass Higher Secondary/ "A" level or recognized equivalent examination in the current year or one previous year. For example, the 2024 January session candidate must pass their Higher Secondary/ "A" level or recognized equivalent examination in 2022 or 2023.

Foreign students must have passed 12 education years and got the same grades in recognized equivalent examinations from foreign recognized institutions and must need to get an equivalence certificate from the Pharmacy Council before admission.

After completing the B. Pharm foreign students will not get professional registration from Bangladesh Pharmacy Council.

Please visit the link for details: https://www.bracu.ac.bd/admissions/undergraduate

Admission Test:

All eligible candidates will have to sit for a written admission test and an interview. Those who qualify in the written test will be called for interview for final selection.

Written Admission test for the international students will be taken according to current practice, which is administered by Brac International Offices. Where there is no Brac office, partnership with local academic institutions will be made for administering the admission test.

Applicants of EMBA are not required to give written test but must pass a viva for admission.

Interview

All qualified applicants of written admission test will have to give their viva on line on a specified date and time.

Admission test exemption

Applicants will get written admission test exemption if they have a minimum SAT score of 1550 (Old)/SAT score 1050 (New) for ANT, ENG, LLB, BBA, ECO, BIO, MIC, CS, CSE, ECE, EEE, MAT, APE, PHY and TOEFL paper based score of 550/ TOEFL IBT score of 79 for ANT, ENG, LLB or IELTS score of 6.5 for ANT, ENG, LLB. They must provide proper evidence to get written test exemption through email minimum 15 days before the admission test.

Provisional admission

Some students wishing to join the University may not have the required standard of proficiency in English language. If students fail to attain the minimum standard of English proficiency required by the University, they may be asked to attend remedial English courses before admission. At the end of this course they will have to take an English proficiency test and if successful will be admitted to BracU.

Credit Transfer

Credit transfer from an educational institution with a system similar to Brac University may be considered after admission. Such candidates will have to apply with the required documents and are subject to the credit transfer rules of BRAC University.

The total credits transferred by a student from other universities should not exceed 50 (or 65 credits for students of Architecture Department). The student must meet the residency requirement of at least two years at Brac University.

Thank you for showing interest in Brac University (BracU). Please complete the contact details box.

Inquiry

Sukriti Kundu is a lecturer in Mathematics at the Department of Mathematics and Natural Sciences (MNS) at BRAC University. She has completed both her Bachelor’s and Master’s degrees in Mathematics from the Department of Mathematics, Jahangirnagar University, Bangladesh. Before joining BRAC University she was lecturer in the Department of Mathematics, Fareast International University. She has interest on Applied Mathematics, Theory of Numbers and Differential equations.

Level: 4Kha-224 Merul BaddaDhaka 1212. Bangladesh

• MAT101• MAT110

Presentation:Sukriti Kundu, Laek Sazzad Andallah, Presentation on “Estimation of Air Pollution by Numerical Simulation of Advection Diffusion Equation” at 18th International Mathematics Conference, Independent University of Bangladesh, 20-22th

March,2014.

- BADHON, JU Branch.
- Interest on Applied Mathematics, Differential Equation and Theory of Numbers.

Sukriti Kundu

News & Events

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Workshop on Architecture Model Making Held at BRAC University

Ahmed Rakin Kamal completed his schooling from Chittagong Cantonment Public College. He completed his Bachelor's degree (BSc in Physics) from BRAC University. His undergraduate thesis work was with Professor Mahbub Majumdar of BRAC University and it addressed the famous Blackhole Information paradox as proposed by Stephen Hawking. Subsequently he completed his masters in Mathematical Physics from Université de Bourgogne-Franche Comté. His master thesis was on the recent developments in 2 dimensional quantum gravity models and it was supervised by Professor Taro Kimura of Institute of Mathematics at Burgundy. After completing his masters, Ahmed worked as a research intern at the Institute of Mathematics at Burgundy. During his internship, he worked in 2 dimensional quantum gravity, Matrix Models and Higher Spin theory.

Level:Kha-224 Merul BaddaDhaka 1212. Bangladesh

BSc. in Physics(2017-2019)MSc. in Mathematical Physics (2020-2021)

● Vice-Chancellor's fellow, BRAC University, Dhaka, Bangladesh ● Research Intern, Institute of Mathematics at Burgundy (IMB), Dijon, France (2021)● Lecturer, BRAC University, Dhaka, Bangladesh (October 2021-Present)

Calculus 1, Calculus 2, Random Matrix Theory, Differential Equations, Fundamentals of Physics.

● Talk on "Black hole Information Paradox and Correlations between Hawking Radiation" at SPARQ workshops.● Talk on "AdS Black holes" at SPARQ workshops. ● Gave two talks on Blackholes at Bangladesh Astronomy Research Collaboration (BARC).

Quantum Gravity, String Theory, Random Matrix Theory, Holographic Duality, Higher Spin theory.

Ahmed Rakin Kamal

News & Events

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Brac University’s General Education program provides a common education foundation to all students before they take courses specific to their major and minor fields of study. The program promotes the core aims of the liberal arts curricula: to develop well-rounded individuals who have a broad range of knowledge on a variety of subjects and a mastery of skills that can be used across disciplines. Students graduate from the program with the ability to think critically, learn creatively and communicate skilfully.

Students take a variety of courses in the fields of humanities, social, natural and formal sciences; and interdisciplinary studies. A unique part of the GenEd program is experiential learning. As part of their course work, students spend time in rural and urban communities to learn first-hand from the people themselves, their history, the challenges they face and the resilience they demonstrate in combating social, economic and natural adversity. Participation in workshops, conferences and seminars is also part of the experiential learning experience.

GenEd courses expose students to new ideas and engage them to think about a variety of critical modern-day issues, ranging from governance structures and failures, to the environment and businesses. The program provides multiple approaches to analytical and quantitative reasoning, with the idea that students will explore and express complex ideas in both academic and non-academic settings.

Through the General Education Program, students develop skills to gather, organize, refine and critically evaluate information and ideas. They learn how to identify and think critically about socially relevant problems and provide creative and sustainable solutions to these problems. Students are able to reason and solve problems from a wide array of contexts and everyday life situations; understand and create logical arguments supported by quantitative evidence; and clearly communicate those arguments in a variety of formats as appropriate. Students are able to articulate their value systems, understand the ethical implications of their actions based on those values, and develop skills consistent with having a positive impact on individuals, groups, or communities by understanding the foundation of empathy and ethical values. This will allow them to identify areas of difficulty in responding to situations demanding ethical inquiry and understand and evaluate the causes of societal problems and potential solutions.

A great emphasis is placed on the importance of diversity. Students develop an understanding of how social, cultural, linguistic, artistic, religious, philosophical, and historical contexts have shaped society, and the thoughts and actions of people worldwide. They also analytically compare the influences of community, institutions, and other constructions such as class, gender, and race on the ways of thinking, believing, and acting in cultural and historical settings other than one's own. Students also learn to promote an inclusive culture that accepts and appreciates human diversity in race, religion, gender, ethnicity, economic and class backgrounds and sexual preferences. Hence they graduate with a strong moral compass and ethical outlook.

Dear Brac U students and Family, Welcome to what will undoubtedly be amongst the best years of your life. Every student's career at Brac U will begin with the General Education track, equipping him/her with the requisite skills to learn, understand, analyze and innovate. These skills will not only train you in university level learning and prepare you for the degrees you will pursue, they will also help you think as critical and conscientious students. Our wish at Brac U is that you learn in great depth and breadth, with curiosity and compassion, and carry that forward into your productive lives as citizens that are with the times globally, yet rooted to the needs of our communities and our country at large.

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Dean's Message

Core Team

Samia Huq, PhD

Syed M Hashemi, PhD

Riaz Partha Khan, PhD

Fahmida Rahman

This office process students’ results, transcript requests for current students and alumni of all Departments, Institutes and Schools across the university, and produce certification letters for enrollment, registration, graduation, and academic standing. We can certify upon request any information that is based on and can be supported by your academic record. We work with students, parents, alumni, faculty, and staff to offer assistance in all areas of the academic experience.

Useful Forms

The contents of the undergraduate courses offered by the MNS Department for its BS programme in Physics and also for other departments of BRACU are given in the following:

BI0 101: Introduction to Biology

An introduction to the cellular aspects of modern biology including the chemical basis of life, cell theory, energetics, genetics, development, physiology, behaviour, homeostasis and diversity, and evolution and ecology. This course will explain the development of cell structure and function as a consequence of evolutionary process, and stress the dynamic property of living systems.3 credits CHE 101: Introduction to Chemistry

The course is designed to give an understanding of basics in chemistry. Topics include nature of atoms and molecules; valence and periodic tables, chemical bonds, aliphatic and aromatic hydrocarbons, optical isomerism, chemical reactions.3 credits ENV 101: Introduction to Environmental Sciences

Fundamental concepts and scope of environmental science, Earth's atmosphere, hydrosphere, lithosphere and biosphere, men and nature, technology and population, ecological concepts and ecosystems, environmental quality and management, agriculture, water resources, fisheries, forestry and wildlife, energy and mineral energy sources; renewable and non renewable resources, environmental degradation; pollution and waste management, environmental impact analysis, remote sensing & environmental monitoring. 3 credits GEO 101: Introduction to Economic Geography

Introduction: The field and environment of economic Geography; Bases of Economic Geography: Relief, Climate, Vegetation, Soils and Population; Extractive resources and human-environment relations; Primary Activities: types and brief descriptions; Secondary Activities: types and factors of localization, Stages in growth; Tertiary Activities: Trade, Transportation, Utilities, Technical and Professional services; Regional Economy: classification, Growth and Development; Economic Geography of Bangladesh: A brief account.3 credits MAT 091: Basic Course in Mathematics

Topics including sets, relations and functions, real and complex numbers system, exponents and radicals, algebraic expressions; quadratic and cubic equations, systems of linear equations, matrices and determinants with simple applications; binomial theorem, sequences, summation of series (arithmetic and geometric), permutations and combinations, elementary trigonometry; trigonometric, exponential and logarithmic functions; co-ordinate geometry; statics-composition and resolution of forces, equilibrium of concurrent forces; dynamics-speed and velocity, acceleration, equations of motion. No credit. MAT 101: Fundamentals of Mathematics

The real number system, exponents, polynomial, factoring, rational expression, radicals, complex number, linear equation, quadratic equation, variation, inequalities, coordinate system, functions, equations of line, equation of circle, exponential and logarithmic function, system of equations, system of inequalities, properties of matrix, matrix solution of linear system, determinant, Cramer's rule, limit, rate of change, derivative. 3 credits MAT 102: Introduction to Mathematics

Factorisation, Synthetic Division, Zeros (Roots) of Polynomials, Relation between Roots and Coefficients, Nature of Roots (Descarte's Rule of signs); Complex Number System, Graphical representation of Complex Numbers (Argand Diagram), Polar form of Complex Numbers; Conic Sections, Parabola, Circle, Ellipse, Hyperbola, Transformation of Coordinates and Applications; Exponential Growth & Decay. Applications; Mathematical Induction; Determinants, Fundamental Properties of Determinants, Minors and Cofactors, Application of Determinants to solve System of Linear Equations (Cramers, Rule); Introduction to Matrix Algebra, Matrix Multiplication, Augmented Matrix, Adjoint Matrix, Inverse Matrix, Application of Matrices-solution of System of Linear Equations (homogeneous & non-homogeneous), Consistency of System of Equations.3 credits MAT 103: Basic Concepts in Mathematics

The real numbers, Absolute value of real numbers, Exponents, Polynomials, Basic operation and Factoring of polynomials, Rational expressions, Radicals. Linear Equations, Solution, graphs and applications. Variation, Linear inequalities. Exponential and Logarithmic Functions, Exponential growth and decay, Ratios, proportions, percent, application of simple and compound interest. Trigonometric Functions, The Sine and Cosine Functions, Cartesian coordinate systems, Graphing, Relations. Equations of a straight line its slope, Equation of a circle, Systems of Linear Equations, Matrix. Population, Sample, Variable, Raw data, Frequency distribution table, Graphical presentation, Measures of central tendency and measures of dispersion. 3 credits MAT 104: Mathematics

Calculus, definition of limit, continuity and differentiability, successive and partial differentiation, maxima and minima. Integration by parts, standard integrals, definite integrals. Solid geometry, system of coordinates. Distance between two points. Coordinate Transformation, Straight lines sphere and ellipsoid. 2 credits MAT 105: Calculus

Differential Calculus: Limits, continuity and differentiability, differentiation, Taylor's, Maclaurine's & Euler's theorems, indeterminate forms, tangent and normal, sub tangent and subnormal, maxima and minima, radius of curvature & their applications, introduction to calculus of function of several variables, Taylor's theorem, maxima and minima for function of several variables. Transformation of coordinates & rotation of axes, conic sections. Integral Calculus: Definition of integration, techniques of integration for definite & indefinite integrals, improper integrals, area, volume and surface integration, arc length and their applications, multiple integrals, Jacobian, line integrals, divergence theorem and Stokes' theorem, beta function and gamma function.3 credits MAT 110: MATH I Differential Calculus and Co-ordinate Geometry

Differential Calculus: Limits, Continuity and differentiability. Differentiation. Taylor's Maclaurine's & Euler's theorem. Indeterminate forms. Partial differentiation. Tangent and normal. Subtangent and subnormal. Maximum and minimum, radius of curvature & their applications. Co-ordinate Geometry: Transformation of coordinates & rotation of axis. Pair of straight lines. General equation of second degree. System of circles. Conics section. Tangent and normal, asymptotes & their applications.3 credits MAT 120: MATH II Integral Calculus and Differential Equations

Integral Calculus: Definitions of integration. Integration by the method of substitution. Integration by parts. Standard integrals. Integration by method of successive reduction. Definite integrals, its properties and use in summing series. Walli's formula. Improper integrals. Beta function and Gamma function. Area under a plane curve in Cartesian and polar coordinates. Area of the region enclosed by two curves in Cartesian and polar coordinates. Trapezoidal rule. Simpson's rule. Arc lengths of curves in Cartesian and polar coordinates, parametric and pedal equations. Intrinsic equations. Volumes of solids of revolution. Volume of hollow solids of revolutions by shell method. Area of surface of revolution. Ordinary Differential Equations: Degree of order of ordinary differential equations. Formation of differential equations. Solution of first order differential equations by various methods. Solutions of general linear equations of second and higher order with constant coefficients. Solution of homogeneous linear equations. Applications. Solution of differential equations of the higher order when the dependent and independent variables are absent. Solution of differential equations by the method based on the factorisation of the operators. [Students will be expected to attend a 3 hour tutorial class, once each week and submit tutorial worksheets.]3 credits Prerequisite: MAT 110 MAT 203: Matrices, Linear Algebra and Differential Equations 3 credits

Matrices: Types of matrices, algebraic operation on matrices, determinants, adjoint & inverse matrix, orthogonality & diagonalization of matrix. Linear Algebra: System of linear equations, vector space; 2D-space, 3D-space, Euclidean nD-space, sub space, linear dependence, basis and dimension, row space, column space, rank and nullity, linear transformation, eigen value and eigen vector, matrix diagonalization and similarity, application of linear algebra. Ordinary Differential Equations: Introduction to differential equations, first-order differential equations and applications, higher order differential equations and applications, series solutions of linear equations, systems of linear first-order differential equations. Prerequisite MAT 105 MAT 204: Complex Variables and Fourier Analysis 3 credits

Complex Variables: Complex number systems, general functions of a complex variable, limits and continuity of a function of complex variables and related theorems, complex differentiation and Cauchy-Riemann equations, mapping by elementary functions, line integral of a complex function. Cauchy's integral theorem, Cauchy's integral formula, Liouville's theorem, Taylor's and Laurent's theorem, singular points, residue, Cauchy's residue theorem, evaluation of residues, contour integration and conformal mapping.

Fourier analysis: Real and complex form, finite Fourier transform, Fourier integrals, Fourier transforms and their use in solving boundary value problems. Prerequisite MAT 105 MAT 205: Introduction to Numerical Methods

Computer arithmetic: floating point representation of numbers, arithmetic operations with normalized floating point numbers; iterative methods, different iterative methods for finding the roots of an equation f (x) = 0 and their computer implementation; solution of simultaneous algebraic equations by various methods, solution of tri-diagonal system of equations, interpolation for equispaced and non-equispaced nodes, least square approximation of functions, curve fitting, Taylor series representation, Chebyshev series, numerical differentiation and integration and numerical solution of ordinary differential equations & partial differential equations.3 credits Prerequisite MAT 203 MAT 215: MATH III Complex Variables and Laplace Transformations

Complex Variables: Complex number systems. General functions of a complex variable. Limits and continuity of a function of complex variables and related theorems. Complex differentiation and Couchy-Riemann equations. Mapping by elementary functions. Line integral of a complex function. Cauchy's integral theorem. Cauchy's integral formula. Liouville's theorem. Taylor's and Laurent's theorem. Singular points. Residue. Cauchy's residue theorem. Evaluation of residues. Contour integration. And conformal mapping Laplace Transforms: Definition. Laplace transforms of some elementary functions. Sufficient conditions for existence of Laplace transforms. Inverse Laplace transforms. Laplace transforms of derivatives. The unit step function. Periodic function. Some special theorems on Laplace transforms. Solutions of differential equations by Laplace transformations. Evaluation of improper integrals.3 credits Prerequisite: MAT120 MAT 216: MATH IV Linear Algebra and Fourier Analysis

Linear Algebra: Basic subject on matrix theory and linear algebra, emphasizing topics useful in other discipline, including systems of equations, vector spaces, determinants, Eigenvalues, similarity, and positive definite matrices, Applications to Gauss elimination with pivoting. Fourier Analysis: Real and complex form. Finite transform. Fourier integral. Fourier transforms and their uses in solving boundary value problems. Multiple integrals; surface and volume integrals, divergence and Stoke's theorem.3 Credits Prerequisite: MAT 120 MAT 301: Group Theory

Definition and various examples of groups, subgroups, cosets, normal subgroups, quotient (factor) groups, permutation groups, cyclic groups, generator of a cyclic group, centre of a group, Abelian group, normalizer and centralizer of an element/subset of a group and its application to physics, group homomorphism, isomorphism and automorphism and related theorems, symmetry groups, SU (3), SU (6), application of group theory in solid state physics & elementary particles.3 Credits MAT 303: Tensor Analysis

Definition of tensor, tensor density, affine tensor and geometrical object, properties of tensor symmetry, criteria of tensor properties, metric tensor, Kronecker symbol and Levi-Civita's symbol, determinant of metric tensor, connection between metric tensor and Dirac's matrices in the Sommerfeld representation, evolution of square root from four dimensional interval in matrix sense, transformation properties of vector partial derivatives by coordinates, connection coefficients and covariant derivatives, Christoffel's symbols, geodetic lines (geodetics) as a generalization of notion of straight line, variation principle for geodetics, parallel transport, connection between geodetics and covariant differentiation, transport along closed line curvature tensor of the 4th rank, curvature tensor of the 2 D rank, scalar curvature, equations of geodetic deviation, curvature expression in terms of Dirac's matrices, Bianchi's identity, Einstein's conservative tensor, integral operations and corresponding theorems.3 Credits PHY 101: Introduction to Physics 3 Credits

Vectors and scalars, Newton's Laws of motion, inertia, force, momentum, conservation of linear momentum, work, energy, conservation of energy, power, gravitation, escape velocity, projectile motion, simple harmonic motion, uniform circular motion. Structural properties of matter, elasticity, Hooke's Law, viscosity, surface tension. Heat and temperature, different scales of temperature, thermal expansion, specific heat, gas laws, heat transfer. Waves and oscillations, longitudinal and transverse waves, sound waves, velocity of sound, ultrasonic waves & their applications. Reflection and refraction of light, mirrors and lenses, total internal reflection, interference, diffraction.

Coulomb's Law, ohm's law; resistance, potential difference, capacitance. Magnetic force on a moving charge, electromagnetic spectrum, velocity of light. Atoms and nuclei, mass number and atomic number, isotopes, isobars & isotones, atomic theory, Planck's Law, Photo-electric effect, wave-particle duality, special theory of relativity, radioactive decay, nuclear fission & nuclear fusion, nuclear energy, fossil fuels & other sources of energy. Structure & vastness of the universe, big bang theory, light year, solar system, Kepler's Laws of planetary motion, cosmological principle, Hubble's Law, red shift, stellar energy, neutron stars, quasars, supernovae, pulsars, black holes. PHY 102: Fundamentals of Physics

Vectors and scalars, Newton's Laws of motion, principles of conservation of linear momentum and energy, gravitation, projectile motion, simple harmonic motion, rotation of rigid bodies. Elasticity, Hooke's Law, viscosity, Stokes' Law, surface tension. Heat & temperature, specific heat, gas laws, Newton's Law of cooling, First and Second Laws of thermodynamics, kinetic theory of gases, heat transfer. Wave motion, stationary waves, sound waves, Doppler Effect, beats, acoustics, ultrasonic & applications. Huygens' principle, electromagnetic waves, reflection, refraction, interference, diffraction.2 Credits PHY 110: Mechanics and Properties of Matter 3 credits

Mechanics: Vectors & scalars, vector addition and subtraction, unit vectors, scalar and vector products, scalar & triple vector product, scalar and vector fields, gradient, divergence and curl, curvilinear co-ordinates, motion in one dimension, motion in a plane, work and energy, conservation laws, conservative force, projectile motion, uniform circular motion, simple harmonic motion, rotational motion, moment of inertia, radius of gyration, angular momentum, Kater's pendulum, Newton's Law of gravitation, gravitational field, potential, escape velocity. Properties of Matter: Hooke's Law, elastic modulii, adhesive and cohesive forces, molecular theory of surface tension, capillarity, variation of surface tension with temperature. Streamline flow, Poiseulle's formula, streamline flow and turbulent flow, Reynold's Number, Equation of Continuity, Bernoulli's Theorem, Stokes' Law. PHY 111: Principles of Physics I

Vectors and scalars, unit vector, scalar and vector products, static equilibrium, Newton's Laws of motion, principles of conservation of linear momentum and energy, friction, elastic and inelastic collisions, projectile motion, uniform circular motion, centripetal force, simple harmonic motion, rotation of rigid bodies, angular momentum, torque, moment of inertia and examples, Newton's Law of gravitation, gravitational field, potential and potential energy. Structure of matter, stresses and strains, Modulii of elasticity Poisson's ratio, relations between elastic constants, work done in deforming a body, bending of beams, fluid motion and viscosity, Bernoulli's Theorem, Stokes' Law, surface tension and surface energy, pressure across a liquid surface, capillarity. Temperature and Zeroth Law of thermodynamics, temperature scales, isotherms, heat capacity and specific heat, Newton's Law of cooling, thermal expansion, First Law of thermodynamics, change of state, Second Law of thermodynamics, Carnot cycle, efficiency, kinetic theory of gases, heat transfer. Waves & their propagation, differential equation of wave motion, stationary waves, vibration in strings & columns, sound wave & its velocity, Doppler effect, beats, intensity & loudness, ultrasonics and its practical applications. Huygens' principle, electromagnetic waves, velocity of light, reflection, refraction, lenses, interference, diffraction, polarization.3 Credits PHY 112: Principles of Physics II

Electric charge, Coulomb's Law, electric field & flux density, Gauss's Law, electric potential, capacitors, steady current, Ohm's law, Kirchhoff's Laws. Magnetic field, Biot-Savart Law, Ampere's Law, electromagnetic induction, Faraday's Law, Lenz's Law, self inductance and mutual inductance, alternating current, magnetic properties of matter, diamagnetism, paramagnetism and ferromagnetism. Maxwell's equations of electromagnetic waves, transmission along wave-guides. Special theory of relativity, length contraction and time dilation, mass-energy relation. Quantum theory, Photoelectric effect, x-rays, Compton effect, dual nature of matter and radiation, Heisenberg uncertainty principle. Atomic model, Bohr's postulates, electron orbits and electron energy, Rutherford nuclear model, isotopes, isobars and isotones, radioactive decay, half-life, alpha, beta and gamma rays, nuclear binding energy, fission and fusion.Fundamentals of solid state physics, lasers, holography.3 Credits PHY 113: Waves, Oscillation and Acoustics

Principle of superposition, interference of waves, phase velocity and group velocity, simple harmonic motion, combination of SHM, Lissajous figures, damped SHM, forced oscillation, resonance, power and intensity of wave motion, waves in elastic media, vibration of strings, beats, Doppler Effect, acoustics, stroboscopy, velocity of sound, ultrasonics, and their applications.3 Credits PHY 114: Thermal Physics and Radiation

Heat and temperature, thermal equilibrium, Zeroth Law of thermodynamics, specific heat & calorimetry, Newton's Law of cooling, Kinetic Theory of Gases, idea of pressure due to collisions of molecules, mean free path, Boltzmann Distribution Law, Brownian motion, Law of equipartition of energy; Vander Waals' equation of state, heat transfer, conduction, convection and radiation, conduction of heat in solids, co-efficient of thermal conductivity and its measurement, First Law of thermodynamics, isothermal & adiabatic changes, reversible and irreversible processes, Carnot's cycle, efficiency of heat engines, Second Law of thermodynamics, entropy and disorder, absolute scale of temperature, thermodynamic functions, Maxwell's relations, Clausius-Clapeyron Equation, Gibb's phase rule, Third Law of thermodynamics, Nernst heat theorem, radiation theory, black body radiation, Wien's Law, Stefan-Boltzman Law, Rayleigh Jeans

Law, Planck's Law, variation of specific heat with temperature, Einstein's theory, Debye's theory, conduction of heat in solids, measurement of conductivity, Joule-Thomson expansion, refrigeration, heat engines, Rankine cycles, cryogenics, measurement of high temperature. 3 Credits PHY 115: Electricity and Magnetism

Charge, quantization of charge, Coulomb's Law, electric field and potential. Gauss's Law, electric dipole, dielectrics, capacitance, energy of charged systems, electrical images, magnetic dipole, energy in a magnetic field. Direct current and electromotive force, Ohm's Law, Kirchhoff's Laws, Wheatstone Bridge, Lorentz force, magnetic field of a current and Ampere's Law, Biot-Savart Law, electromagnetic induction, Faraday's Law, self-induction, mutual induction, alternating current, RMS value, power factor, CR, LR and LCR circuits, resonance.3 Credits PHY 201: Solid State Physics

Crystalline state, Bravais lattices, crystal symmetry, point group & space group, unit cells, Miller indices, x-ray diffraction, Bragg's Law, reciprocal lattice, structure factor, interatomic force and classification of solids, ionic, covalent, molecular, hydrogen bonded crystals, lattice energy of ionic crystals, Madelung constant, lattice vibration, phonons, normal modes in monatomic and diatomic linear chains, theory of specific heat, Einstein and Debye models, thermal expansion, defects in crystals, dislocations, consequences of defects on mechanical properties.3 Credits PHY 202: Optics

Laws of reflection and refraction, total internal reflection, Huygens' Principle, velocity of light, Young's experiment, Fresnel's bi-prism, Newton's rings, Michelson's interferometer, multiple reflections, Fabry-Perot interferometer, diffraction of light, Fresnel and Fraunhoffer diffraction, single, double and multiple-slit diffraction, diffraction grating, spectrometer, resolving power of a grating, polarization of light, production of polarized light, plane, circular and elliptically polarized light, optical activity, double refraction, optic axis, half-wave and quarter-wave plate, nicol prism, dispersion of light, scattering of light, Thomson scattering.3 Credits PHY 204: Classical Mechanics and Special Theory of Relativity

Classical Mechanics: Newtonian equations of motion, conservation laws of a system of particles, variable mass, generalized co-ordinates, generalized force, D' Alembert's Principle, variational method, Euler-Lagrange equations of motion, Hamilton's principles, two-body central force problem, elliptic orbit, scattering in a central field, Rutherford formula, kinematics of rigid body motion, Euler angles, rotating co-ordinates, Coriolis force, wind motion, principal axis transformation, top motion, principle of least action, Hamiltonian equations of motion, small oscillations, normal co-ordinates, normal modes.Special Theory of Relativity: Galilean relativity, Michelson-Morley experiment, postulates of special theory of relativity, Lorentz transformation, length contraction, time dilation, twin paradox, variation of mass, relativistic kinematics, mass energy relation.3 Credits PHY 205: Statistical Mechanics

Statistical Mechanics: Phase space, concept of state and ensemble, microcanonical, canonical and grand canonical ensembles, Boltzmann probability distribution, Maxwell velocity distribution, derivation of Bose-Einstein and Fermi-Dirac statistics, ideal Fermi gas, degenerate Fermi system, equation of state of ideal gases, ideal Bose gas, application of Statistical mechanics in various fields in physical, biological, social sciences, economics, finance and in engineering & ICT.3 Credits PHY 210: Quantum Physics of Atoms, Solids and Nuclei

Special Theory of Relativity: Michelson-Morley Experiment, Special Theory of Relativity, Lorentz Transformations, Time Dilation, Length Contraction, Mass-Energy Relation. Quantum Phenomena: Blackbody Radiation, Planck's Law, Photoelectric Effect, Bohr Atomic Model, Energy Levels & Atomic Spectra, Correspondence Principle, Dual Nature of Matter & Waves. Introductory Quantum Mechanics: Wave Function, Operators, Expectation Values, Schrodinger's Wave Equation, Particle in Box, Schrodinger Equation for Hydrogen Atom, Energy Levels, Magnetic & Orbital Angular Momentum, Concept of Quantum Numbers. Solid State Physics: Crystal Structure, Crystal Diffraction, Bragg Law, Lattice Vibrations & Phonons, Free Electron Model, Energy Levels & Density of

States, Fermi-Dirac distribution function, Free Electron gas in Three dimension, Electrical conductivity & Thermal Conductivity, Hall Effect, Band Theory of Solids, Band Diagrams of Insulator, Semiconductor & Metals, Superconductivity, Lasers & Holography. Nuclear Physics: Rutherford Nuclear Model, Radioactivity, Half life & Mean life, Nuclear Binding Energy, Fission & Fusion, Particle Accelerator, Elementary Particles & Nuclear Interactions, Quarks, Lepton & Hadrons, Big Bang & Origin of the Universe.3 Credits PHY 301: Classical Electrodynamics

Solution of Laplace's equation and Poisson's equation and applications to electrostatic problems, dielectrics, electrostatic energy, Maxwell's equations, electromagnetic waves, propagation of electromagnetic waves in conducting and non-conducting media, reflection and refraction, polarization, dispersion, scattering, waves in the presence of metallic boundaries, waveguides and resonators, solution of the inhomogeneous wave equations, simple radiating system, antennas, accelerated charge, Cerenkov radiation, elements of plasma physics.3 Credits Prerequisite PHY 115. PHY 302: Fluid Mechanics 3 Credits

Fluid properties, fluid statics, manometry, force on submerged planes and curved surface, buoyancy and floatation, one dimensional flow of fluid, equation of continuity, Euler's equation, flow of fluid in pipes, Bernoulli's equation, flow through orifice, mouthpiece, venturimeter, fundamental relations of compressible flow, frictional losses in pipes and fittings, types of fluid machinery, impulse and reaction turbines, centrifugal and axial flow pumps, deep well turbine pumps, specific speed, unit power, unit speed, unit discharge, performance and characteristics of turbines and pumps, design of pumps, reciprocating pumps.3 Credits Prerequisite PHY 110 PHY 303: Quantum Mechanics I

Breakdown of classical physics, quantum nature of radiation, Planck's Law, photoelectric effect, Einstein's photon concept and explanation of photoelectric effect, de Broglie wave, wave particle duality, electron diffraction, Davisson-Germer experiment, emergence of quantum mechanics, Schrodinger equation, basic postulates of quantum mechanics, physical interpretation of wave function, wave packets, Heisenberg's uncertainty principle, linear operators, Hermitian operators, eigenvalue equation, one-dimensional potential problem, harmonic oscillator, orbital angular momentum, rotation operator, spherical harmonics, spin angular momentum, addition of angular momenta, solution of the Schrodinger equation for hydrogen atom, matrix formulation of quantum mechanics.3 Credits PHY 304: Quantum Mechanics II

Rutherford scattering experiment, Discovery of the nucleus, Bohr quantization rules, hydrogen atom spectra, Franck-Hertz experiment, Sommerfeld-Wilson quantization rules, electron spin, Stern-Gerlach experiment, Pauli exclusion principle, electronic configuration of atoms, vector atom model, coupling schemes, Hund's rule, multiplet structure, fine structure in hydrogen spectral lines, Zeeman effect, Paschen-Beck effect, production of X-rays, measurement of X-ray wavelength, X-ray scattering, Compton Effect, Mosely's Law, molecular spectra, rotational and vibrational levels, Raman Effect and its applications, lasers.3 Credits PHY 305: Quantum Mechanics III

Basic properties of nuclei, constituents of nuclei, nuclear mass, charge, size and density, nuclear force, spin, angular momentum, electric and magnetic moments, binding energy, separation energy, semi-empirical mass formula, radioactive decay law, transformation laws of successive changes, measurement of decay constant, artificial radioactivity, radioisotopes, theory of alpha decay, gamma radiation, energy measurement, pair spectrometer, classical treatment of gamma emission, internal conversion, Mossbauer Effect, beta decay, energy measurement, conservation of energy and momentum in beta decay, neutrino hypothesis, orbital electron capture, positron emission, interaction of radiation in matter, ionisation, multiple scattering, range determination, bremsstrahlung, pair production, annihilation. Discovery of neutrons, production and properties of neutrons, nuclear reactions, elastic and inelastic scattering, Q-value of a reaction and its measurements, nuclear cross-section, compound nucleus theory, direct reaction and kinematics.3 CreditsPrerequisite PHY 304

PHY 306: Basic Electronics

Network theorems, filters, transmission line, basic semiconductor concepts, energy bands, electrons and holes, semiconductor diode, rectification, regulators, Zener diode, diode circuits, unijunction transistor, FET and its characteristics, transistor amplifier, FET amplifier; amplifier circuits, voltage amplifiers, RC coupled amplifiers and tuned amplifiers, frequency response, bandwidth, power amplifier, push-pull amplifier, feedback and amplifier stability, operational amplifier and its characteristics, oscillators, modulation and demodulation, digital electronics, digital logic, logic gates, Boolean algebra, logic circuits, information registers, flip-flop circuit.3 Credits PHY 308: Methods of Experimental Physics and Instrumentation

Optical and spectroscopic instruments, defects of images and their remedies, optical blooming, phase contrast and polarizing microscope, spectrophotometers, optical transmittance, reflectance and absorption, application of interferometry, production and measurement of high and ultrahigh vacuum. Rotary pump, diffusion pump, ion pump and turbo pump, pirani, penning and ionisation gauges, measurement of current and voltages, potentiometer, VTVM, oscilloscope, D.C. amplifier, lock-in amplifier, frequency meter and counter, four point probe, flux meter and Hall probed transducers, piezoelectric, thermistor, photo-transducers, voltage regulator, SCR type temperature controllers. 3 CreditsPrerequisites PHY 202 and PHY 306 PHY 309: Introduction to Materials Science

Crystalline solids, amorphous, composite, fibrous materials, polymers, plastics, binding forces, elastic properties, dislocations, defects etc, specific heat, thermal expansion, thermal conductivity and electrical conductivity of metals, dielectric properties of solids, modes of dielectric polarisation, ferro electricity, piezo electricity, optical properties of solids ,classical and semi classical theory, free carrier effects, lattice absorption, electronic absorption, magnetic properties of solid, atomic magnetic moments, dia and paramagnatism, ferro & ferrimagnetism, antiferromagnetism, ferrites, magnetic resonance, superconductivity, type-1, type-2 superconductors, liquid crystals.3 CreditsPrerequisite PHY 201 PHY 310: Advanced Solid State Physics

Free electron theory, transport properties, Sommerfeld theory, Hall Effect, box quantization, density of states, Fermi surface, Fermi energy, electrical conductivity, WiedmannFranz law, band theory of solids, electron in a periodic potential, Schrodinger equation, Bloch function, LCAO and OPW methods, dielectric properties of insulators, Clausius-Mosotti relations, dielectric loss, relaxation time, polarization mechanism, direct & indirect band gap semiconductors, extrinsic semiconductors, charge carrier concentration, recombination process of p-n junction, superconductivity, Meissner Effect, London equation, BCS theory, introduction to high temperature superconductivity, magnetic materials, quantum theory of diamagnetism and paramagentism, theory of ferromagnetic, ferrimagnetic and anti-ferromagnetic orders, magnetic resonance.3 Credits Prerequisite PHY 201 PHY 311: X-Rays

Continuous and Characteristic X-rays, Bremsstrahlung, Properties of X-rays, X-ray technique, Weissenberg and precession methods, identification of crystal structure from powder photograph and diffraction traces, Laue photograph for single crystal, geometrical and physical factors affecting X-ray intensities, analysis of amorphous solids and fibre textured crystal.3 Credits Prerequisite PHY 201 PHY 312: Nuclear Physics II

Determination of nuclear size by scattering methods and electromagnetic methods, mirror nuclei, electron scattering, nuclear shapes, electric and magnetic multiple moments, isotopic spin formalism, two-nucleon problem, nuclear forces, exchange force, meson theory of nuclear forces. Shell model, refinement of extreme single particle model, collective model, nuclear reactions, compound nucleus model, concept of optical potential, energy averaged cross section and the optical model at low energies, phenomenological optical model, direct reactions, parity violation in beta decay, nuclear fission and nuclear reactor, nuclear fusion, nuclear liquid drop model & shell model, magic numbers (qualitative) accelerators, Van de Graaff generator, linear accelerator, cyclotron, synchrotron, detection of charged particles, photons and neutrons, nuclear pulse counting systems, elementary particles.3 Credits Prerequisite PHY 305 PHY 313: Physics for Development

Twenty first century development issues, physics and break through technologies, ICT, fibre optics, quantum information theory, physics in genetics engineering and molecular biology, physics and health issues, bio and medical physics, materials science and physics, high temperature superconducting materials, space physics, microgravity experiments, econo-physics, physics principles applied in sociology.3 Credits PHY 400: Thesis/Project

A student is required to carry out thesis/project work in her/his last two semesters in a chosen field. There will be a supervisor who will either be a BRAC University faculty or any other suitable expert from universities and R/D organizations of the country to guide the thesis/project work. On completion of study and research s/he will have to submit the dissertation paper and to face a viva board for the defence.4.5 Credits PHY 401: Particle and Reactor Physics

Interactions of neutrons with matter, cross-sections for neutron reactions, thermal neutron cross-sections, nuclear fission, energy release in fission, neutron multiplication, nuclear chain reaction, steady state reactor theory, criticality condition, homogeneous and heterogeneous reactor systems, neutron moderation, neutron diffusion, control of nuclear reactions, coolant, types of nuclear reactors: power reactor, research reactor, fast reactor, breeder reactor, reactor shielding, waste disposal. 3 CreditsPrerequisite PHY 305 PHY 402: Atmospheric Physics

Structure of the atmosphere, elementary ideas about the sun and the laws of radiation, definitions and units of solar radiation, depletion of solar radiation in the atmosphere, terrestrial radiation, radiation transfer, heat balance in the atmosphere, heat budget, vertical temperature profile, radiation charts and their uses, composition of the atmosphere, mean molecular weight, humidity, mixing ratio, density and saturation vapour pressure. Fundamental equations of atmospheric motion, approximations of the equation, circulation and vorticity and their equations. Introduction to atmospheric thermodynamics.3 Credits PHY 403: Cosmology & Astrophysics 3 Credits

General introduction to plasma physics, plasma as a fourth state of matter, definition, screening, and Debye shielding, plasma frequency, ideal plasma, temperature and pressure of plasma, magnetic pressure and plasma drifts, plasma waves, Landau damping, collisions in plasmas, hydrodynamic description of plasma, one fluid model, two fluid model,Chew-Goldberg theory, low waves in magneto-hydrodynamics, description of plasma, dielectric tensor, longitudinal and transverse waves, plasma instabilities, transport in plasmas, plasma kinetic theory, Vlasov equation, linear waves, waves in magnetized plasma, electromagnetic waves, waves in hot plasmas, nonlinear waves, Landau damping, quasi linear theory, plasmas in fusion research, astrophysical plasmas. Introduction to astrophysics, formation of stars and galaxies, evolution of stars, the notion of cosmology, Cosmological Principle, various cosmological models of the universe, expansion of universe, Hubble's Law, problem of singularity in time, solutions of Friedmann, de Sitter and others, density of matter in the universe, cosmological term, self screening effect for matter.3 Credits Prerequisites PHY 301 and PHY 304. PHY 404: Electronic Devices and Circuits

Modelling and application of Semiconductor devices and integrated circuits, advanced transistor amplifier analysis, including feedback effects. Design for power amplifiers, operational amplifiers (OPAMP), analog filters, oscillators, A/D and D/A converters and power converters. Introduction to transistor level design of CMOS digital circuits.3 Credits Prerequisite PHY 306 PHY 405: Mathematical Physics

Series solution of 2nd order ordinary differential equations about ordinary and singular points, orthogonal set functions, Sturm-Liouville boundary value problem (SLP), eigen values and eigen functions of different SLP, series of orthogonal set of function. Laplace transforms: definition, Laplace transformations of some elementary functions, inverse Laplace transformations, Laplace transformations of derivatives, Dirac delta function, some special theorem on Laplace transformations, solution of differential equations by Laplace transformations, evaluation of improper integrals; finite Fourier series, Fourier transforms, Fourier integrals, Fourier transform and application to solution BVP, beta and gamma functions, Legendre functions, Bessel functions, solution of boundary value problem by method of separation of variables, solution PDE of mathematical physics: Helmholtz equation, wave equation: vibrating string, vibrating membrane, diffusion equation, Laplace equation, Hermite polynomials, Laguerre polynomials, hyper-geometric functions.3 CreditsPrerequisite MAT 203 PHY 406: Medical Physics and Instrumentation

Ultrasound imaging, A-scan, B-scan, M-scan, clinical applications, rectilinear scanner, gamma camera, CAT scanner, MRI, clinical applications, audiology, hearing aids, vascular measurements, blood pressure, blood flow, blood velocity, cardiac measurements; ECG, ECG planes, elementary ideas on heart disorders, defibrillators, pacemakers, neuromuscular measurements; EEG, EMG, stimulation of neural tissue, nerve conduction measurements, bio-electric amplifiers, patient safety, radiopharmaceuticals, radiotherapy, radiation protection, radiation dosimetry.3 Credits

PHY 407: Mathematical Modelling in Physics

Basic concept of mathematical modelling, formulation and solution, overview of computational methods of classical and quantum physics, numerical procedure for special functions, Random numbers generator, Brownian motion simulation, linear system of equations, sparse linear system, eigen value problems, BVP involving ODE, Sturm-Liouville problems, BVP involving PDE: elliptic, parabolic and hyperbolic problems using finite difference and other methods, Monte Carlo integration and simulation, mathematical modelling of problems of physics using above techniques.3 Credits

Prerequisite MAT 205 PHY 408: Advanced Quantum Mechanics

Heisenberg and Dirac or interaction pictures, time-independent perturbation theory, degenerate perturbation theory, variation method, hydrogen atom and helium atom, WB approximation method, Sommerfeld-Wilson quantisation condition, time-dependent perturbation theory, Fermi's golden rule, applications, identical particles, parity, Pauli principle, applications, non-relativistic scattering theory, partial wave expansion, optical theorem, Smatrix, solution of the wave equation by the method of Green's function, LippmannSchwinger equation, Neumann series, Born approximation, applications, Klein-Gordon and Dirac equations, existence of electron spin, magnetic moment, plane wave solution of the Dirac equation, hole theory; prediction of the positron.3 CreditsPrerequisite PHY 303 PHY 409: Physics of Radiology

The production and properties of X-rays, diagnostic and therapeutic X-ray tubes, X-ray circuit with rectification, electron interaction, characteristic radiation, bremsstrahlung, angular distribution of X-rays, quality of X-rays, beam restricting devices, the grid, radiographic film, radiographic quality, factors affecting the image, image modification, image intensification, contrast media, modulation transfer function, exposure in diagnostic radiology, fluoroscopy, computed tomography, ultrasound, magnetic resonance imaging (MRI).3 Credits PHY 410: Laser Physics

Spontaneous and stimulated emission, absorption, pumping schemes, characteristic properties of laser beam, laser speckle, grain size calculation for free-space propagation, semi classical treatment of absorption and stimulated emission, spontaneous emission, results of QED treatment, electric dipole, allowed and forbidden transitions, Einstein's A and B coefficient, radiation trapping, superfluorescence, superradiane and amplified spontaneous emission, nonradiative decay, homogeneous and inhomogeneous broadening, linewidth calculations for naturally, collisionally and Doppler broadened line, two level and four, level saturation, saturation of absorption & inhomogeneously broadened line, passive optical resonators, continuous wave and transient laser behaviour, laser beam transformation, types of lasers, their construction and use, applications of lasers, optical communications, laser in fusion research, holography.3 CreditsPrerequisite PHY 304 PHY 411: Geophysics

Solar system, the planets, meteorites, cosmic ray exposures of meteorites, Poynting-Robertson effect, compositions of the terrestrial planet, pre-radioactivity age problem, radioactive elements and the principle of radiometric dating, growth of constituents and of atmospheric argon, age of the earth and of meteorites, dating the nuclear synthesis, figure of the earth, precession of the equinoxes, the Chandler-wobble, tidal friction and the history of the earth moon system, fluctuation in rotation and excitation of the wobble, seismology of the earth, elastic wave and seismic rays, travel time and velocity depth curves for body waves, shockwave, internal pressure of earth core, internal density and composition, free oscillation, earthquake prediction problem, terrestrial magnetism, earth magnetic field, geophysical prospecting; seismic, gravitational, magnetic, electrical and nuclear methods.3 Credits

PHY 412: Dynamical and Tropical Meteorology

Geophysical fluid dynamics, Navier-Stokes' equation, rotating and stratified flow, scale analysis, hydrostatic approximation, Coriolis force, geopotential etc., gradient and thermal wind, vorticity and circulation theorems, Proudmen-Taylor theorem, atmospheric wave, atmospheric turbulence, barotroic and baroclinic instabilities, numerical weather forecasting, quasi-geotropic approximation, barotropic vorticity equation, primitive equation, multilayered models, tropical cyclones, norwesters and tornadoes, the monsoons, dynamical climatology, physics of upper atmosphere: geomagnetism, neutral atmosphere, ionosphere and magnetosphere.3 CreditsPrerequisite PHY 402 PHY 413: General Theory of Relativity

Gravitation, Lagrangian Einstein equations, approximation of weak field and Hilbert's auxiliary conditions, comparison of corresponding relations with those of Newton's theory of gravitation, source of gravitation field, Schwarzschild's solution in isotropic and other coordinate systems, analogy between gravitation and electromagnetism, motion of test mass and geodetic lines, motion in Schwarszchild's field, equations of motion in general relativistic mechanics as a consequence of Einstein's equation of gravitational field, gravitational waves in weak field approximation, problem of energy transfer, exact wave solutions in the case of gravitational field, waves of matrices or wave of curvature, locally plane gravitational waves, Weber's and Braginski's experiments, prospects of future gravitational experiments.3 Credits PHY 414: Field Theory

Equation of motion, quantization, conservation laws, construction of Hilbert space, Lagrangian, equation of motion, quantization of neutral and charged Klein-Gordon fields, Dirac equation, spinors, quantization of Dirac field, Maxwell fields, Gupta-Bleuler formalism, theory of gauge fields, invariant functions propagators for Klein-Gordon field, Dirac fields and electromagnetic fields, symmetries of interactions, interaction picture; U and S matrices, Feynman diagrams, Wick's theorem, Feynman rules, lowest order, amplitude and cross section for Compton scattering, GSW model of electroweak interactions, elements of QCD, path integral in field theory, introduction to string theory.3 Credits PHY 415: Neutron Scattering

Neutron sources, continuous and pulsed sources, monochromatization, collimation and moderation of neutrons, neutron detectors, scattering of neutrons and its advantages, elastic scattering of neutrons, magnetic scattering and determination of magnetic structure, inelastic scattering, thermal vibration of crystal lattices, lattice dynamics and phonons, neutron polarization, polarized neutron applications, scattering by liquids and molecules, Van Hove correlation formalism, some experimental results of scattering by liquids and molecules, small angle neutron scattering and its applications in the study of biological molecules and defects, experimental techniques of scattering measurements, time-of-flight method, crystal diffraction techniques, neutron diffractometer and triple-axis spectrometer, constant Q-method.3 CreditsPrerequisite:

PHY 305 PHY 416: Radiation Biophysics

Nucleus, ionizing radiations, radiation doses, interaction of radiation with matter, cell structure, radiation effects on independent cell systems, oxygen effect, hyperthermia, LET and RBE, lethal, potentially lethal and sub-lethal radiation damage, dose-rate effect, acute effects of radiation, somatic effects, late effects, non-specific life shortening and carcinogenesis, genetic changes, nominal standard dose (NSD), time dose fractionation (TDF), Standquist curve.3 Credits Prerequisite: PHY 305 STA 101: Introduction To Statistics

Frequency distribution. Mean, median, mode and other measures of central tendency. Standard deviation and other measures of dispersion. Moments, Skews and kurtosis. Elementary probability theory and discontinuous probability distribution, e.g. binomial, Poisson and negative binomial. Continuous probability distribution, e.g. normal and exponential. Characteristics of distributions. Hypothesis testing and regression analysis. Basic concepts and applications of probability theory and statistics. Chi-square test.3 credits STA 201: Elements of Statistics and Probability

Frequency distribution, mean, median, mode and other measures of central tendency, standard deviation and other measures of dispersion, moments, skewness and kurtosis, elementary probability theory and discontinuous probability distribution, binomial, Poisson and negative binomial distribution, continuous probability distributions, normal and exponential, characteristics of distributions, hypothesis testing and regression analysis, basic concepts and applications of probability theory and statistics, chi-squared test.3 credits LAB COURSES PHY 116: Physics Lab I 1.5 Credits List of Experiments:EXP 1: Determination of the Young's Modulus of a Short Wire by Searle's Dynamic MethodEXP 2: Determination of the Modulus of Rigidity of a Wire by the Method of OscillationsEXP 3: Determination of g by means of a Compound PendulumEXP 4: Determination of the Moment of Inertia of a Flywheel about its Axis of RotationEXP 5: Determination of the Spring Constant and Effective Mass of a given

Spiral Spring EXP 6: Determination of Surface Tension of Water by Capillary Tube MethodEXP 7: Determination of Surface Tension of Mercury and the Angle of Contact by Quincke's MethodEXP 8: Determination of the Viscosity of Glycerine by Applying Stokes' Law.EXP 9: Determination of the Specific Heat of a Liquid by the Method of Mixture EXP 10: Determination of the Specific Heat of a Liquid by the Method of CoolingEXP 11: Determination of the Thermal Conductivity of a Bad Conductor by Lee's MethodEXP 12: Determination of the Pressure Co-efficient of a Gas at Constant Volume by Constant Volume Air ThermometerEXP 13: Determination of the Stefan's ConstantEXP 14: Study of Variation of the Frequency of a Tuning Fork with the Length of a Sonometer (n-l curve) under given Tension and Hence to Determine the Unknown Frequency EXP 15: Determination of the Frequency of a Tuning Fork by Melde's ExperimentEXP 16: Determination of Velocity of Sound by Kundt's Tube. PHY 203: Physics Lab II 1.5 Credits List of Experiments:EXP 1: Determination of the Focal Length and Hence the Power of a Convex Lens by Displacement Method with the Help of an Optical BenchEXP 2: Determination of the Refractive Index of a Liquid by Plane Mirror and Pin Method using a Convex LensEXP 3: Determination of the Radius of Curvature of a Lens by Newton's Rings MethodEXP 4: Determination of the Refractive Index of the Material of a Prism by using a SpectrometerEXP 5: Determination of the Wavelengths of Various Spectral Lines by Spectrometer by using Plane Diffraction GratingEXP 6: Determination of the Value of an Unknown Resistance and Verification of the Laws of Series and Parallel Resistances by Means of a Post Office BoxEXP 7: Determination of the Internal Resistance of a Cell by a PotentiometerEXP 8: Determination of the Specific Resistance of a Wire using a Meter BridgeEXP 9: Determination of the Resistance of a Galvanometer by the Half-Deflection MethodEXP 10: Determination of the High Resistance of a Suspended Coil Galvanometer by the Method of Deflection EXP 11: Comparison of the EMF of Two Cells with a PotentiometerEXP 12: Determination of the Resistance per Unit Length of a Meter BridgeEXP 13: Determination of the Temperature Co-efficient of Resistance of the

Material of a WireEXP 14: Determination of the Value of J by Electrical MethodEXP 15: Determination of the Line Frequency by Lissajous Figure using an Oscilloscope and a Function Generator and Verification of the Calibration of Time/Div Knob at a Particular Position for Different FrequenciesEXP 16: Determination of the Self-Inductance of a Coil by Anderson's Method.EXP 17: Charging and Discharging of Capacitors and Study of Their Various Characteristics.

PHY 307: Physics Lab III 1.5 Credits List of Experiments:EXP 1: Determination of the Excitation and Ionization Potentials (of mercury) by Frank-Hertz Experiment.EXP 2: Determination of the e/m of Electron Using Helmholtz Coil.EXP 3: Determination of the Threshold Frequency for Photoelectric Effect of a Photo-Cathode and the Value of Planck's Constant by Using a Photoelectric Cell.EXP 4: Determination of the Plateau of a Geiger-Muller Counter and Hence to Find its Operating voltage.EXP 5: Study of the Variation of Electrical Conductivity of a Semiconductor and Determine of its Energy Gap.EXP 6: Study of the Characteristics of a PN Junction and Zener Diode.EXP 7: Study of the Characteristics of PNP and NPN Transistors.EXP 8: Study of the Frequency Response Characteristics of an RC Low pass, RC High pass, a Band pass and a Parallel T Filter.EXP 9: Study of the Frequency Response in LRC Series Circuit and the Variation of Q-factor with Resistance.EXP 10: Determination of the Frequency Response in LRC Parallel Circuit and Determination of Q-factor.EXP 11: Study of Variation of Reactance due to L and C with Frequency.EXP 12: Designing and Construction a Summing Amplifier Using 741 Operation Amplifier (OPAMP). EXP 13: Construction of Full Wave Bridge Rectifier Using Semiconducting Diodes and Study of the Effect of Filters.EXP 14: Determination of Transistor Characteristics in Common Emitter Configuration and Determination of Hybrid Parameter.EXP 15: Determination of the Coefficient of Mutual Inductance Between Two Coils and Hence to Show its Variation with the Separation Between the Coils. EXP 16: Determination of the Absorption Coefficients of Different Materials for the Radiation Emitted by a Radioactive Source by Using a Geiger-Mueller

Counter.

The Biotechnology Program under the Department of Mathematics and Natural Sciences and the BRAC University Society for Biotechnology (BUSB) successfully arranged the first session of the informative workshop series on Thursday, 29th February 2024. This interactive session shed light on a revolutionary area of Biotechnology named “Biosensors”. Dr. Munima Haque, Program Director and Associate Professor in the Biotechnology program, inaugurated this event with her insightful words. Dr. Munima introduced the speaker Subarno Hossain Turja. Subarno is a lecturer under the Biotechnology Program at BRAC University who completed his BSc. in Biomedical Engineering from Bucknell University, Pennsylvania, USA and MSc. also, in Biomedical Engineering from Drexel University, Pennsylvania, USA. Subarno's single-minded devotion to the biomedical field is a valuable addition to the biotechnology program as well as BRAC University.

During the session, Subarno enlightened the audience with fascinating contexts of Biosensors and associated applications. Biosensors are considered the key to the future of biotechnological advancement. The research area related to Biosensors assembles intellectuals of different academic backgrounds. BUSB arranged this session for the curious minds of all departments. Apart from the biotechnology students, CSE, and EEE major students also joined this session and quenched their thirst for knowledge on Biosensors.

The workshop provided a comprehensive understanding of biosensors, highlighting their multifaceted applications. Highlighted examples include glucose biosensors for diabetes management, environmental biosensors for pollutant detection, and DNA biosensors for genetic analysis. The workshop also shed light on the engineering principles related to biosensors, which emphasize safety, functionality, design, innovation, and sustainability. Participants explored the critical components of biosensors, primarily transducers and electronic circuits, and gained insights into their functionality in detecting changes. The interactive session not only promoted active participation but also allowed students to apply their newly acquired knowledge creatively.

The workshop on biosensors proved to be an enlightening experience, equipping students with both theoretical knowledge and practical skills, laying a foundation for their future endeavors in biotechnology.

News Archive >>

The biotechnology program and BUSB organized the inaugural Biotechnology Workshop Series on "Understanding and Designing Biosensors"

Related News

Biotechnology seminar hosts Bangladeshi scientist who developed high-yielding "Panchabrihi" rice

Orientation held for Summer 2024 biotechnology students

Nanobiotechnology seminar organized by BUSB and Biotechnology Program

Orientation held for Spring 2024 biotechnology students

MNS department of Brac University are going to organize an International virtual Webinar to Celebrate International Women's Day on 24 March, 2021 (07:00 pm – 8:30 pm) bd Time.

Honorable Lady Syeda Sarwat Abed will deliver her speech as a Chief Guest and International and National Scientists will deliver speeches to motivate and inspire our students and will share important information about grants, scholarship, etc.

Programme(PDF)

A Virtual Webinar Celebrating International Women's Day

a. Theoretical Courses

ANT 101 Introduction to Anthropology 3 credits

Humans in nature, human evolution, history of culture, rise of early civilizations in the old and new world, organizations of pre-industrial society environment, resources and their distribution; gender, kinship and descent, religion, economics, politics, survival of indigenous groups, forms of culture and society among contemporary peoples, Comparative study of traditional and changing Third World societies, impact of modern world on traditional societies, power and social order; custom and law, conflict and change, Cultural and ethnic diversity.

Recommended Books:1. Cultural Anthropology, a Global Perspective: R Scupin2. Anthropology: The Exploration of Human Diversity: Conrad P Kottak 3. Anthropology: C Ember and M Ember4. Cultural Anthropology: W A Haviland5. Anthropology - Social and Cultural: Kedar Nath, Ram Nath 6. An Introduction to Anthropology: Victor Barnouw.

APE 101 Mechanics, Properties of Matter, Waves & Oscillations 3 credits

Vectors & scalars, unit vectors, scalar and vector products, scalar and vector fields, gradient, divergence and curl, curvilinear co-ordinates, 1 D & 2 D motions, work and energy, conservation laws, conservative force, projectile motion, uniform circular motion, rotation of rigid bodies, angular momentum, gravitation, gravitational field, potential. Elasticity, Hooke's Law, adhesive and cohesive forces, molecular theory of surface tension, capillarity. Streamline & turbulent flow, Poiseulle's formula, streamline flow and turbulent flow, Reynold's Number, Equation of Continuity, Bernoulli's Theorem, Stokes' Law.Principle of superposition, interference of waves, phase velocity and group velocity, simple harmonic motion, combination of SHM, Lissajous figures, damped SHM, forced oscillations, resonance, power and intensity of wave motion, waves in elastic media, vibration of strings, beats, Doppler Effect, velocity of sound, ultrasonics, and their applications.

Recommended Books:1. Vector Analysis: M. Spiegel, Schaum's Outline Series2. Fundamentals of Physics: D. Halliday, R. Resnick & J. Walker3. Gases, Liquids and Solids: D. Tabor4. The Mechanical Properties of Matter: M.T. Sprackling5. The General Properties of Matter: F.H. Newman and V.H.L. Searle6. Elements of Properties of Matter: D.S. Mathur7. University Physics: Francis W. Sears, Mark W. Zemansky & Hugh D. Young8. Fundamentals of Vibrations and Waves: S.P. Puri9. Vibrations and Waves: A.P. French10. The Physics of Vibration & Waves: H.J. Pain11. Vibrations & Waves: I.G. Main

APE 102 Thermal Physics, Radiation & Statistical Mechanics 3 Credits

Heat and temperature, thermal equilibrium, specific heat & calorimetry, Newton's Law of Cooling, Kinetic Theory of Gases, Boltzmann Distribution Law, Brownian motion, Law of equipartition of energy; Vander Waals' equation of state, heat transfer, conduction, convection and radiation, co-efficient of thermal conductivity and its measurement, First Law of thermodynamics, isothermal & adiabatic changes, reversible and irreversible processes, Carnot's cycle, Second Law of thermodynamics, entropy and disorder, absolute scale of temperature, Maxwell's relations, Clausius-Clapeyron Equation, Gibb's phase rule, Third Law of thermodynamics, Nernst heat theorem, radiation theory, black body radiation, Wien's Law, Stefan-Boltzman Law, Rayleigh Jeans Law, Planck's Law, variation of specific heat with temperature, Einstein's theory, Debye's theory, Joule-Thomson expansion, cryogenics, measurement of high temperature.Statistical Mechanics: Phase space, concept of state and ensemble, microcanonical, canonical and grand canonical ensembles, Boltzmann probability distribution, Maxwell velocity distribution, derivation of Bose-Einstein and Fermi-Dirac statistics, ideal Fermi gas, degenerate Fermi system, equation of state of ideal gases, ideal Bose gas.

Recommended Books:1. Fundamentals of Physics: D. Halliday, R. Resnick & J. Walker2. Heat and Thermodynamics: M.W.Zemansky and R.H. Dittman3. An Introduction to Thermodynamics: F.W. Sears4. Fundamentals of Statistical and Thermal Physics: F. Reif5. Thermal Physics: C. Kittel and H. Kroener6. A Treatise on Heat: M.N. Saha and B.N. Srivastava7. Statistical Mechanics: Kerson Huang8. Statistical Physics: F. Mandl9. Elementary Statistical Physics: C. Kittel

APE 103 Electrical Circuits I 3 credits

Circuit variables and elements: Voltage, current, power, energy, independent and dependent sources, resistance. Basic laws: Ohm's law, Kirchhoff's current and voltage laws. Simple resistive circuits: Series and parallel circuits, voltage and current division, Wye-Delta transformation. Techniques of circuit analysis: Nodal and mesh analysis including supernode and super mesh. Network theorems: Source transformation, Thevenin's, Norton's and Superposition theorems with applications in circuits having independent and dependent sources, maximum power transfer condition and reciprocity theorem. Energy storage elements: Inductors and capacitors, series parallel combination of inductors and capacitors. Responses of RL and RC circuits: Natural and step responses. Magnetic quantities and variables: Flux, permeability and reluctance, magnetic field strength, magnetic potential, flux density, magnetization curve. Laws in magnetic circuits: Ohm's law and Ampere's circuital law. Magnetic circuits: series, parallel and series-parallel circuits. Prerequisite PHY 115

Recommended Books:

1. Engineering Circuit Analysis: W. H. Hayt, J. Kemmerly and S. M. Durbin 2. Electric Circuits: J. W. Nilsson and S. Riedel 3. Basic Engineering Circuit Analysis: J. D. Irwin 4. Introduction to Electric Circuits: R. C. Dorf and J. A. Svoboda5. Electric Circuit Analysis: D. E. Johnson, J. R. Johnson, J. L. Hilburn and P. D. Scott6. The Analysis and Design of

Linear Circuits: R. E. Thomas and A. J. Rosa

APE 201 Solid State Physics and Materials Science 3 Credits

Crystalline state, Bravais lattices, crystal symmetry, point group & space group, unit cells, Miller indices, x-ray diffraction, Bragg's Law, reciprocal lattice, structure factor, interatomic force and classification of solids, ionic, covalent, molecular, hydrogen bonded crystals, lattice energy of ionic crystals, Madelung constant, lattice vibration, phonons, normal modes in monatomic and diatomic linear chains, theory of specific heat, Einstein and Debye models, thermal expansion, defects in crystals, dislocations, consequences of defects on mechanical properties, elastic properties. Amorphous, composite, fibrous materials, polymers, plastics, binding forces, thermal and electrical conductivity of metals, dielectric properties of solids , modes of dielectric polarisation, ferro electricity, piezo electricity, optical properties of solids ,classical and semi classical theory, free carrier effects, lattice absorption, electronic absorption, magnetic properties of solid, dia and paramagnatism, ferro & ferrimagnetism, antiferromagnetism, ferrites, magnetic resonance, superconductivity, liquid crystals.

Recommended Books:1. Introduction to Sold State Physics: Charles Kittel2. Introduction to Solid State Physics: A. J. Dekker3. Solid State Physics: M.A. Omar.4. Solid State Physics by B.S. Saxena, R. C. Gupta & P. N. Saxena5. Materials: Principles & Practice: C. Newey & G. Weaver6. Properties of Materials: Mary Anne White7. Mechanical Metallurgy: George Dieter8. Fundamentals of Materials Science and Engineering: William D. Callister, Jr., William D. Callister9. The Coming of Materials Science: Robert W. Cahn10. Introduction to Materials Science for Engineers: James F. Shackleford11. Engineering Materials (Vol 1 & 2): M.F. Ashby & David R.H. Jones

APE 202 Electrodynamics & Electromagnetic Waves & Fields 3 Credits

Solution of Laplace's equation and Poisson's equation and applications to electrostatic problems, dielectrics, electrostatic energy, Maxwell's equations, electromagnetic waves, propagation of electromagnetic waves in conducting and non-conducting media, reflection and refraction, polarization, dispersion, scattering, waves in the presence of metallic boundaries, Waves between parallel planes, attenuation, wave impedance, waves in coaxial lines & modes, waves in strip and micro-strip lines, waveguides and resonators, solution of the inhomogeneous wave equations, simple radiating system, antennas, accelerated charge, Cerenkov radiation, elements of plasma physics. Prerequisite PHY 115.

Recommended Books:1. Foundations of Electromagnetic Theory: John Reitz, F.J. Milford and R. W. Christy2. Electromagnetic Fields and Waves: Paul Lorrain and Dale Corson3. Classical Electrodynamics: J.D. Jackson4. Introduction to Electrodynamics: D.J. Griffiths

APE 203 Electrical Circuits II 3 credits

Sinusoidal functions: Instantaneous current, voltage, power, effective current and voltage, average power, phasors and complex quantities, impedance, real and reactive power, power factor. Analysis of single phase ac circuits: Series and parallel RL, RC and RLC circuits, nodal and mesh analysis, application of network theorems in ac circuits, circuits simultaneously excited by sinusoidal sources of several frequencies, transient response of RL and RC circuits with sinusoidal excitation. Resonance in ac circuits: Series and parallel resonance. Magnetically coupled circuits. Analysis of three phase circuits: Three phase supply, balanced and unbalanced circuits, power calculation. Prerequisite APE 103

Recommended Books:1. Engineering Circuit Analysis: W. H. Hayt, J. Kemmerly and S. M. Durbin 2. Electric Circuits: J. W. Nilsson and S. Riedel 3. Basic Engineering Circuit Analysis: J. D. Irwin 4. Introduction to Electric Circuits: R. C. Dorf and J. A. Svoboda5. Electric Circuit Analysis: D. E. Johnson, J. R. Johnson, J. L. Hilburn and P. D. Scott6. The Analysis and Design of Linear Circuits: R. E. Thomas and A. J. Rosa

APE 204 Digital Logic Design 3 credits

An introduction to digital systems such as computer, communication and information systems. Topics covered include Boolean algebra, digital logic gates, combinational logic circuits, decoders, encoders, multiplexers. Asynchronous and synchronous counters. Registers, flip-flops, adders, Sequential circuit analysis and design. Simple computer architecture.

Recommended Books:1. Digital Systems: Principles and Application: R. J. Tocci, N. S. Widmer and G. L. Moss

APE 205 Electronic Devices and Circuits I 3 credits

P-N junction as a circuit element: Intrinsic and extrinsic semiconductors, operational principle of p-n junction diode, contact potential, current-voltage characteristics of a diode, simplified dc and ac diode models, dynamic resistance and capacitance. Diode circuits: Half wave and full wave rectifiers, rectifiers with filter capacitor, characteristics of a zener diode, zener shunt regulator, clamping and clipping circuits. Bipolar junction transistor (BJT) as a circuit element: Basic structure. BJT characteristics and regions of operation, BJT as an amplifier, biasing the BJT for discrete circuits, small signal equivalent circuit models, BJT as a switch. Single stage BJT amplifier circuits and their configuarations: Voltage and current gain, input and output impedances. Metal-Oxide-Semiconductor Field-Effect-Transistor (MOSFET) as circuit element: structure and physical operation of MOSFETs, body effect, current- voltage characteristics of MOSFETs, biasing discrete and integrated MOS amplifier. Prerequisite APE 103

Recommended Books:1. Microelectronic Circuits: A. S. Sedra and K. C. Smith 2. Engineering Circuit Analysis: W. H. Hayt, J. Kemmerly and S. M. Durbin J.

APE 302 Electronic Devices and Circuits II 3 credits

Frequency response of amplifiers: Poles, zeros and Bode plots, amplifier transfer function, techniques of determining 3 dB frequencies of amplifier circuits, frequency response of single-stage and cascade amplifiers, frequency response of differential amplifiers. Operational amplifiers (Op-Amp): Properties of ideal Op-Amps, non-inverting and inverting amplifiers, inverting integrators, differentiator, weighted summer and other applications of Op-Amp circuits, effects of finite open loop gain and bandwidth on circuit performance, logic signal operation of Op-Amp, dc imperfections. General purpose Op-Amp: DC analysis, small-signal analysis of different stages, gain and frequency response of 741 Op-Amp. Negative feedback: properties, basic topologies, feedback amplifiers with different topologies, stability, frequency compensation. Active filters: Different types of filters and specifications, transfer functions, realization of first and second order low, high and bandpass filters using Op-Amps. Signal generators: Basic principle of sinusoidal oscillation, Op-Amp RC oscillators, LC and crystal oscillators. Power Amplifiers: Classification of output stages, class A, B and AB output stages. Prerequisite APE 205

Recommended Books:1. Microelectronic Circuits: A. S. Sedra and K. C. Smith 2. Engineering Circuit Analysis: W. H. Hayt, J. Kemmerly and S. M. Durbin J.3. Digital Computer Electronics: A. P. Malvino and J. A. Brown

APE 401 Measurements & Measuring Instruments 3 Credits

Significance and methods of measurements, direct and indirect methods. Mechanical, electrical and electronic types of instruments, absolute and secondary instruments, analog and digital instruments, analog voltmeters and ammeters, AC transformer types, Flux gate magnetometer type. Accuracy and error of analog voltmeters and ammeters. Different types of Digital voltmeters, digital multimeters, Automation in multimeters. Oscilloscopes, signal generators. Transducers. Absorption and detection of radiation, Nucleonic instruments. Analytical & medical instruments, Industrial instruments, Instrument systems. Prerequisite APE 302

Recommended Books:1. Instrumentation, Measurements and Feedback: B.E Jone 2. Electronic Instrumentation and Measurement Techniques: W.d. Copper3. Instrument Technology: Jones E.B. Butterworths4. Understanding Electrography: E.G Zailis and M.H Conover 5. Industrial Instrumentation Fundamentals: A.E. Fribance

APE 402 Plasma Physics with Industrial Applications 3 credits

General introduction to plasma physics, plasma as a fourth state of matter, definition, screening and Debye shielding, plasma frequency, ideal plasma, temperature and pressure of plasma, magnetic pressure and plasma drifts, plasma waves, Landau damping, collisions in plasmas, hydrodynamic description of plasma, one fluid model, two fluid model, Chew-Goldberg theory, low waves in maneto-hydrodynamics, description of plasma, dielectric tensor, longitudinal and transverse waves, plasma instabilities, transport in plasmas, plasma kinetic theory, Vlasov equation, linear waves, waves in magnetized plasma, electromagnetic waves, waves in hot plasmas, nonlinear waves, Landau damping, quasi linear theory, plasmas in fusion research, plasmas in industrial applications.

Recommended Books: 1. Principles of Plasma Physics: Nicholas A. Krall and Alvin W. Trivelpiece2. Industrial Applications of Plasma Physics: Francis Chen

APE 403 Control Engineering 3 credits

Introduction to control systems, electric circuits and components, transfer function and block diagram, mechanical translation systems, analogous circuits, mechanical rotational systems, rotating power amplifiers, DC & AC servomotors. Inputs & responses, modeling of continuous systems; computer-aided solutions to systems problems; feedback control systems; stability, frequency response and transient response using root locus, frequency domain and state variable methods. Position control system, simulation diagrams, signal flow graphs, parallel state diagrams from transfer function. General frequency transfer function relationships, drawing the Bode plot, system type and gain as related to log magnitude curve, Nyquist's criterion and applications. Prerequisites ECE 220 and MAT 203

Recommended Books: 1. Linear Control System Analysis and Design (Fourth edition): John J. D. Azzo and Constantine H. Houpis2. Modern Control Engineering, 4th ed., Prentice Hall, 2001: K. Ogata3. Feedback Control of Dynamic Systems, 4th ed., Addison-Wesley, 2002: G. E. Franklin, J. D. Powell, and A. Emami-Naeni

APE 404 Microprocessors and Assembly Language Programming 3 credits

Introduction to different types of microprocessors. Microprocessor architecture, instruction set, interfacing/O operation, interrupt structure, DMA. Microprocessor interface ICs. RAM, ROM, PROM, EPROM and EPROM's. Advanced microprocessor concept of microprocessor based system design. Microcomputer systems, representation of numbers and characters, introduction to IBMPC assembly language. Logic, shift, multiplication & division instructions, arrays and addressing modes, string instructions, text display and keyboard programming, memory management. Prerequisite APE 204

Recommended Books:1. Microcomputer Systems: The 8086/8088 Family Architecture, Programming Design, 2nd ed., Prentice-Hall, 1986: Y. Liu and G. A. Gibson 2. Microprocessors: Theory and Applications: Intel and Motorola, Revised ed., Prentice Hall, 1992: M. Rafiquzzaman 3. Microprocessors and Interfacing: Programming and Hardware 2nd ed., Gloence McGraw Hill, 1991: Douglas V. Hall4. Assembly Language Programming: Maruti

APE 405 Computer Organization and Architecture 3 credits

A systematic study of the various elements in computer design, including circuit design, storage mechanisms, addressing schemes, and various approaches to parallelism and distributed logic. Information representation and transfer; instruction and data access methods; CPU structure and functions processor and register organization, instruction cycles and pipe linings, the control unit; memory organisation. RISC and CISC machines. The course includes a compulsory 3 hour laboratory work each week. Prerequisite APE 204.

Recommended Books: 1. Computer Organization and Design: The Hardware/Software Interface, 3rd ed., Morgan Kauffmann, 2004: D. A. Patterson, J. L. Hennessy, P. J. Ashenden J. R. Larus and D. J. Sorin2. Computer Architecture and Organization, 3rd ed., McGraw Hill, 1997: J. P. Hayes3. Computer Organization and Architecture, 6th ed., Prentice Hall, 2002: W. Stallings4. Computer Organization & Programming, McGraw Hill: C.W. Gear5. Computer Organization, McGraw Hill: V. Hamacher, Z. Vranesic and S. Zaky6. Introduction toComputer Organization: I. Tomok & Pitman

APE 406 Radar Engineering 3 credits

The course is oriented towards the understanding and design of radar systems. The contents will be radar principles & techniques, nature of radars, radar frequencies, radar target, radar equation, continuous & frequency modulated radars, detection & processing of radar signals, MTI radar, Pulse Doppler radar, tracking radar, radar indicators and displays, noise, ground & sea echoes & clutter, weather effect of radar, radar applications. Prerequisites: ECE 220 and MAT 203

Recommended Books:1. Detection of Signals in Noise 2nd edition, Academic Press 1995: R. N. McDonough & A.D.Whalen2. Radar Principles for the Non-specialist: John C, T.P. Hannen3. Radar Signal Processing, McGraw Hill: Marks Richards4. Introduction to Radar Systems: Merril I. Skolnik5. Radar Signals: Introduction to Theory of Application: C.E. Cook and Marvin Bernfeld

APE 407 The Physics of Energy 3 credits

Energy & development, energy consumption, world energy demand & future projection, energy units, earth's energy resource base, renewable and non-rewable sources of energy.Non-renewable energy sources-fossil fuels, coal, natural gas, petroleum, etc. and non-fossil fuels like uranium (fission energy). Renewable energy sources-solar energy, wind energy, tidal & wave & ocean energy, geothermal energy, biomass & hydropower, hydrogen energy. Advantages of renewable over conventional technologies, solar thermal conversion, radiation characteristics of materials, solar collectors, solar photovoltaic energy conversion, photovoltaic cells, design of PV systems, wind turbines, biopower, biofuels, integrated bioenergy systems, geothermal heat pumps, hydroelectricity & micro hydroelectric power, ocean thermal energy conversion. Energy, sustainability and environment, EIA.

Recommended Books:1. Review of Renewable Energy Resources ed. M. S. Soda, S.S. Mathur and M.A. S. Malik2. Solar Enegry & Thermal Processes: Duffie3. Applied Solar Energy: An Introduction: A. B. Mienel4. Energy Resources of Bangladesh: Badrul Imam5. Renewable Energy: Sources of Fuels & Electricity: ed. T.B. Johansson;R.H. Williams, L.Burnhem, A.K.N.Reddy, H. Kelly6. Prospects for Sustainable Energy: A Critical Assessment: Edward S.Cossdy

ARC 292 Painting 2 credits

Painting as a form of artistic and architectural expression. Introduction to various media in painting. Still life sketches and painting. Study of forms in painting. Landscapes and cityscapes. Colour pencils, crayons, pastels and watercolour. Mixed media. Computers in painting.

ARC 293 Music Appreciation 2 credits

Musical form. Ingredients of music: sound and time. Indian and Western music: melody and harmony. Foundations of sub-continental music: raga system. Presentation of vocal and instrumental music. Modern Bengali music and works of major composers and demonstrations. Western classical music and works of major composers. Music and its rhythm, composition etc.

BI0 101 Introduction to Biology 3 credits

An introduction to the cellular aspects of modern biology including the chemical basis of life, cell theory, energetics, genetics, development, physiology, behaviour, homeostasis and diversity, and evolution and ecology. This course will explain the development of cell structure and function as a consequence of evolutionary process, and stress the dynamic property of living systems.

Recommended Books:1. Biology: P.H. Raven and G.B. Johnson2. Biological Science: G. W. Stout and D. J. Taylor3. Advanced Biology: J. Simpkins and J. J. Williams4. Biology: A Fundamental Approach: M. B. Roberts

CHE 101 Introduction to Chemistry 3 credits

The course is designed to give an understanding of basics in chemistry. Topics include nature of atoms and molecules; valence and periodic tables, chemical bonds, aliphatic and aromatic hydrocarbons, optical isomerism, chemical reactions.

Recommended Books:1. Introduction to Modern Inorganic Chemistry: S. Z. Haider2. Physical Chemistry: Haque & Nawab 3. Organic Chemistry: R. T. Morrison & R. N. Boyd4. General Chemistry: Raymond Chang

CSE 101 Introduction to Computer Science 3 credits

Introduction to the use of computer hardware and software as tools for solving problems. Automated input devices and output methods (including pre-printed stationery and turnaround documents) as part of the solution. Using personal computers as effective problem solving tools for the present and the future. Theory behind solving problems using common application software including word processing, spreadsheets, database management, and electronic communications. Problem solving using the Internet and the World Wide Web. Programming principles and use of macros to support the understanding of application software. The course includes a compulsory 3 hour laboratory work each week.

Recommended Books:1. Computer Science – A modern introduction: Goldschlager and Lister2. Fundamentals of Computers: V Rajaraman 3. Work Out Computer Studies GCSE (Macmillan Work Out S.): Graham Taylor

CSE 110 Programming Language I 3 credits

An introduction to the foundations of computation and purpose of mechanised computation, techniques of problem analysis and the development of algorithms and programs, principles of structured programming and corresponding algorithm design, Topics will include data structures, abstraction, recursion, iteration as well as the design and analysis of basic algorithms, (language C is primarily used), introduction to digital computers and programming algorithms and flow chart construction, information representation in digital computers, writing, debugging and running programs (including file handling) on various digital computers using C. The course includes a compulsory 3 hour laboratory work each week. Prerequisite CSE 101

Recommended Books:1. Working with C: Y Kanetkar2. Schaums Outline of Theory and Problems of Programming With C: Byron S. Gottfried

CSE 111 Programming Language II 3 credits

Introduction to data structures. Formal specification of syntax. Elements of language theory: mathematical preliminaries. Formal languages. Structured programming concepts. Survey of features of existing high level languages. Appropriate application using C++. The course includes a compulsory 3 hour laboratory work each week. Prerequisite CSE 110

Recommended Books:1. C++ How to Program: H. M. Deitel and P. J. Deitel, 2. The C++ Programming Language: B. Stroustrup, 3. Thinking in C++, Volume 1: Introduction to Standard C++: B. Eckel

CSE 350 Digital Electronics and Pulse Techniques 3 Credits

Diode logic gates, transistor switches, transistor gates, MOS gates, Logic families: TTL, ECL, IIL and CMOS logic with operation details. Propagation delay, product and noise immunity. Open collector and High impedance gates. Electronic circuits for flip flops, counters and register, memory systems. PLA's (A/D, D/A converters with applications, S/H circuits) LED, LCD and optically coupled oscillators. Non-linear applications of OPAMPs. Analog switches. Linear wave shaping: diode wave shaping techniques, clipping and clamping circuits, comparator circuits, switching circuits. Pulse transformers, pulse transmission. Pulse generation: monostable, bistable and stable multivibrations, Timing circuits. Simple voltage sweeps, linear circuit sweeps. Schmitrigger, blocking oscillators and time base circuit. Prerequisite APE 204

Recommended Books:1. Digital Computer Electronics: A. P. Malvino and J. A. Brown R. J.

CSE 421 Computer Networks 3 credits

An introduction to the basics of transport connections and sessions. The protocol hierarchy, design issues in transport and session layer protocol, end to end protocols, message handling protocols, terminal and file transfer protocols, Internet TCP/IP protocols. End to end data networks, congestion control networks, wireless networks, mobile computing, high speed networks. Concurrent programming, data link layer, framing and error control, media access control. Models of distributed computation, management and resource control of networks and distributed operating systems, distributed file systems, caching scheduling, process migration. Fault tolerance, network security and privacy, algorithm for deadlock detection. Synchronization and concurrency control in distributed systems. The course includes a compulsory 3 hour laboratory work alternate week.

Recommended Books:1. Computer Networks: Protocols Standard and Interfaces, 5th ed., Prentice Hall, 1987: U. D. Black2. Computer Network (3rd edition): Andrew S. Tanenbaum

CSE 428 Image Processing 3 credits

Digital image fundamentals, perception, representation; image transforms; Fast Fourier Transform (FFT), Discrete Cosine Transform (DCT), Karhumen and Loeve Transform (KLT), Wavelet transform and sub-band decomposition; image enhancement and restoration techniques, image compression techniques, image compression standards: JPEG, MPEG, H.261, and H.263. Prerequisite MAT 204

Recommended Books:1. The Image Processing Handbook: J. C. Russ2. Handbook of Image and Video Processing: A. Bovik3. Practical Algorithms for Image Analysis: Descriptions, Examples, and Code: M. Seul, L. O'Gorman, M. J Sammon

DEV 101 Bangladesh Studies 3 credits

Socio-economic profile of Bangladesh, agriculture, industry, service sector, demographic patterns, social aid and physical infrastructures. Social stratification and power, power structures, government and NGO activities in socio-economic development, national issues and policies and changing society of Bangladesh.

Recommended Books:1. Bangladesh: National Cultures and Heritage: An Introductory Reader: A.F. Salahuddin Ahmed & Bazlul Mobin Chowdhury2. The History of Bengal (Vol.1 &Vol.2) : R.C. Majumdar3. Banglapedia, 2003: Asiatic Society of Bangladesh4. Bangladesh Arthaniti: Khan, Md. Shamsul Kabir5. Bangladesh on the Threshold of the Twenty-First Century, Asiatic Society of Bangladesh, 2002: A.M Chowdhury and Fakrul Alam 6. Poverty Reduction & Strategy: What, Why & for Whom in Asit Biswas et.al Contemporary Issues in Development : M.M Akash 7. Bangladesh 2020: A long-run perspectives study: The World Bank

ECE 220 Signals and Systems 3 credits

Classification of signals and systems: signals - classification, basic operation on signals, elementary signals, representation of signals using impulse function; systems – classification. Properties of Linear Time Invariant (LTI) systems: linearity, causality, time invariance, memory, stability, invertibility. Time domain analysis of LTI systems: Differential equations - system representation, order of the system, solution techniques, zero state and zero input response, system properties; impulse response - convolution integral, determination of system properties; state variable - basic concept, state equation and time domain solution. Frequency domain analysis of LTI systems: Fourier series- properties, harmonic representation, system response, frequency response of LTI systems; Fourier transformation- properties, system transfer function, system response and distortion-less systems. Applications of time and frequency domain analyses: solution of analog electrical and mechanical systems, amplitude modulation and demodulation, time-division and frequency-division multiplexing. Laplace transformation: properties, inverse transform, solution of system equations, system transfer function, system stability and frequency response and application. Pre requisites APE 103 and MAT 204.

Recommended Books:1. Signals and Systems: A.V. Oppenheim and A. S. Willsky 2. Linear Systems and Signals: B. P. Lathi

ECE 230 Semiconductor Devices and Materials 3 credits

Semiconductor fundamentals, crystal structure, Fermi level, energy-band diagram, intrinsic and extrinsic semiconductor, carrier concentration, scattering and drift of electrons and holes, drift current, diffusion mechanism, Hall effect, generation, recombination and injection of carriers, transient response, basic governing equations in semiconductor, physical description of p-n junction, deletion approximation, biasing, transition capacitance, varactor diodes, junction breakdown, space charge effect and diffusion approximation, current–voltage characteristics and temperature dependence, tunnelling current, optical absorption in a semiconductor, photovoltaic effect, semiconductor lasers. Pre requisites APE 201 and APE 205.

Recommended Books:1. Solid State Electronic Devices: B. G. Streetman.

ECE 310 Introduction to Communication Engineering 3 Credits

Basic introduction to Fourier analysis and its application to communication systems, Overview of current communication systems (cellular, radio, and TV broadcasting, satellites, Internet), Fourier series and Fourier transforms, filtering and signal distortion, time domain and frequency domain analysis, analog modulation (AM and FM), digital modulation, noise in communication systems. Overview of current systems: the public-switched telephone network, radio and TV broadcasting, cellular and cordless telephones, satellite communications and paging. Prerequisites APE 202, APE 203 and APE 205.

Recommended Books:1. Communication Systems: Simon Haykin2. Electronic Communication Systems: G Kennedy 3. Principles of Communication Systems: Taub and Schilling4. Communication Systems: A.Bruce Carlson 5. Electronic Communication: D Roody and J Coolen

ECE 320 Microwave Engineering 3 credits

Advanced analysis of wave guides, stripline, and microstrip; microwave circuit and device theory including ferrites, junctions and resonators; high frequency generation and amplification, microwave systems. Basic antenna concepts, Radiation Patterns, Beam solid angle, radiation intensity, directivity, effective aperture, antenna field zones, Polarization, impedance, cross field, Poynting vector. Antenna and transmission lines, Radiation from a dipole antenna, antenna temperature. Pre requisite ECE 310.

Recommended Books:1. Microwave Engineering: D. M. Pozar 2. Fields and Waves in Communication Electronics: S. Ramo, J. R. Whinnery and T. V. Duzer 3. Microwave Engineering: A. Das and S. K. Das 4. Antennas: J. D. Krauss and R. J. Marhefka 5. Antenna Theory: Analysis and Design Technology: C. A. Balanis6. Antennas and Radio Propagation: P. E. Collins7. Electromagnetic Waves and Radiating Systems: E. C. Jordan and K. G. Balmain

ECE 328 Digital Signal Processing 3 credits

Introduction to Digital Signal Processing : Discrete-time signals and systems, analog to digital conversion, aliasing, impulse response, difference equation, correlation and convolution, transient and steady state response. Discrete transformations: discrete-time Fourier series (DTFS), discrete-time Fourier transform (DTFT), discrete Fourier transform (DFT) and their properties, fast Fourier transform (FFT). Z transformation - properties, transfer function, and inverse Z transform. Application of Digital Signal Processing. Digital Filters: FIR filters - linear phase filters, filter specifications, designing FIR filter using window, optimal and frequency sampling methods; IIR filters – specifications, designing IIR filters using impulse invariant, bi-linear Z transformation, least-square methods and finite precision effects. MATLAB application to DSP. Prerequisite APE 203

Recommended Books:1. Digital Signal Processing: Principles, Algorithms and Applications: G. Proakis and D. Manolakis 2. Digital Signal Processing: R. A. Roberts and C. T. Mullis 3. Understanding Digital Signal Processing: R. G. Lyons

ECE 330 Telecommunication Switching Systems 3 credits

Evolution of telecommunication switching and circuits: Evolution of Public Switched

Telecommunication Networks Strowger exchange, Crossbar exchange, Stored programme exchange. Digital exchange – Basic Telecommunication equipment – Telephone handset, Hybrid circuit, Echo suppressors and cancellors, PCM coders, Modems and Relays. Electronic switching: Circuit Switching, Message switching, Centralized stored programme switching, Time switching, Spare switching, Combination switching – Digital switching system hardware configuration, Switching system software, Organization, Switching system call processing software, Hardware software integration. Telecommunication signaling and traffic: Channel associated signaling, Common channel signaling, SS7 signaling protocol, SS7 protocol architecture, Concept of Telecommunication traffic, Grade of service, Modeling switching systems, Blocking models and Delay systems. Integrated digital networks: Subscriber loop characteristics, Local access wire line and wire less PCM / TDM carrier standards transmission line codes, Digital multiplexing techniques, Synchronous, Asynchronous, Plesiocronous multiplexing techniques, SONET / SDH, Integrated Digital Network (IDN) environment – Principles of Integrated Services Digital Network (ISDN) – Cellular Mobile Communication Principles. Prerequisite ECE 310.

Recommended Books:1. Digital Switching Systems: Syed R. Ali 2. Telecommunication Switching System and Networks: Viswanathan 3. Introduction to Data Communication and Networking: B. Forouzan 4. Integrated Digital Networks: L.S. Lawton

ECE 340 Optoelectronic Devices 3 credits

Elements of Light and Solid State Physics: Wave nature of light, Polarization, Interference, Diffraction, Light Source, review of Quantum Mechanical concept, Review of Solid State Physics, Review of Semiconductor Physics, Semiconductor Junction Device, Review. Display Devices and Lasers: Introduction, Photo Luminescence, Cathode Luminescence, Electro Luminescence, Injection Luminescence, LED, Plasma Displaycs, Liquid Crystal Displays, Numeric Display, Laser Emission, Absorption, Radiation, Population Inversion, Optical feedback, Threshold condition, Laser Modes, Classes of Lasers, Mode Locking, Laser applications. Optical detection devices: Photo detector, Thermal detector, Photon Devices, Photo Conductors, Photo diodes, Detector Performance. Optoelectronic modulator and switching devices: Introduction, Analog and Digital Modulation, Electro-optic modulators, Magneto Optic Devices, Optical, Switching and Logic Devices. Optoelectronic integrated circuits: Introduction, hybrid and Monolithic Integration, Applications of Opto Electronics Integrated Circuits, Integrated transmitters and Receivers, Guided wave devices. Pre requisite ECE 230.

Recommended Books:1. Optoelectronics and Photonics Principles and Practices: Kasap 2. Optoelectronics – An Introduction: J.Wilson and J.Haukes 3. Semiconductor Optoelectronic Devices: P. Bhattacharya 4. Optoelectronics – An Introduction to materials and Devices: J. Singh

ECE 424 Power Electronics 3 credits

Power semiconductor devices: Power transistors, Fast recovery diodes, Thyristors, Power TRIAC, MOSFET, IGBT, GTO, UJT and DIAC-characteristics, rating, Protection circuits, Driver Circuits. Power supplies: Single Phase and Three Phase Controlled rectifiers, Design of Trigger circuits, Switching mode regulators – Boost, Buck, Buck-Boost and Cuk regulators, AC voltage regulator. Inverters: Voltage and current source inverters, Resonant, Series inverter, PWM inverter. Choppers: Type A, B, C and D choppers, Pulse width modulation - Gating requirements. Motor control: DC motor drives, Induction and Synchronous motor drives, Stepper motor control, Switched reluctance and brushless motor drives. Prerequisite APE 302

Recommended Books:1. Power Electronics: Circuits, Devices and Application: Muhamed H. Roshid 2. Power Electronics: M. D.Singh, K. B. Khanchandani 3. Power Electronics: Converters, Applications and Design: N. Mohan, T. M.Undeland, W. P.Robbins4. Modern Power Electronics: B. K.Bose5. Power Electronics: Sen

ECE 430 Satellite Communications 3 credits

Orbital parameters: Orbital parameters, Orbital perturbations, Geo stationary orbits.

Low Earth and medium Earth orbits. Frequency selection, Frequency co-ordination and regulatory services, Sun transit outages, Limits of visibility, Attitude and Orientation control, Spin stabilization techniques, Gimbal platform. Link calculations: Space craft configuration, Payload and supporting subsystems, Satellite up link-down link, Link power budget, C/No, G/T, Noise temperature, System noise, Propagation factors, Rain and Ice effects, Polarization calculations. Access techniques: Modulation and Multiplexing: Voice, Data, Video, Analog and Digital transmission systems, Multiple access techniques: FDMA, TDMA, T1- T2 carrier systems, SPADE, SS-TDMA, CDMA, Assignment Methods, Spread spectrum communication, Compression techniques. Earth station parameters: Earth station location, Propagation effects of ground, High power transmitters. Receivers: Low noise front-end amplifiers, MIC devices, Antennas: Reflector antennas, Cassegranian feeds, Measurements on G/T and Eb/No. Satellite applications, Mobile Satellite services. Prerequisite ECE 310.

Recommended Books:1. Satellite Communication Systems Engineering: W. L. Pritchard, G. H. Suyderhood, R. A. Nelson 2. The Satellite Communication Applications Hand Book: B. R. Elbert 3. Satellite Communications: D. Roddy 4. Digital Satellite Communication: T. T. Ha 5. Digital Communication Satellite / Earth Station Engineering: K. Feher

ECO 103 Principles of Economics 3 credits

A study of the fundamentals of micro and macroeconomics, nature and method of economics, individual markets, demand and supply, elasticity of demand and supply. Production and cost, market structures with special focus on perfect competition and monopoly, economic efficiency and market failure, determination of national income. The aggregate supply model, unemployment, inflation, unemployment-inflation trade-off, government budget and fiscal policy, money creation and monetary policy, business cycles, economic growth, theory of comparative advantage, free trade versus protection, balance of payments and exchange rate policies.

Recommended Books:1. Economics: John Solman 2. Principles of Macroeconomics:

Robert H Frank3. Modern Economic Theory: K.K. Dewett4. International Economics: Appleyard & Field

ENG 091 Foundation Course (non-credit)

The English Foundation Course is designed to enable students to develop their competence in reading, writing, speaking, listening and grammar for academic purposes. The students will be encouraged to acquire skills and strategies for using language appropriately and effectively in various situations. The approach at all times will be communicative and interactive involving individual, pair and group work.

Recommended Books:1. J. C. Richards, J. Hull, and S. Proctor, "New Interchange: Student 's Book 3-A", Cambridge University Press, 20022. J. Nadel, B. Johnson, and P. Langan, "Vocabulary Basics", Townsend Press, 1998.3. A. Hogue, "First steps in Academic Writing", Longman, 19964. K. Blanchard, C. Root, "Get Ready to Write", Longman, 1998.

ENG 101 English Fundamentals 3 credits

Developing basic writing skills: mechanics, spelling, syntax, usage, grammar review, sentence and essay writing.

Recommended Books:1. Fundamentals of English: Jack C. Richards

ENG 102 Composition I 3 credits

The main focus of this course is writing. The course attempts to enhance students' writing abilities through diverse writing skills and techniques. Students will be introduced to aspects of expository writing: personalized/ subjective and analytical/persuasive. In the first category, students will write essays expressing their subjective viewpoints. In the second category students will analyze issues objectively, sticking firmly to factual details. This course seeks also to develop students' analytical abilities so that they are able to produce works that are critical and thought provoking.

Recommended Books:1. Composition I: The Pearl; John Steinbeck

ENV 101 Introduction to Environmental Science 3 credits

Fundamental concepts and scope of environmental science, Earth's atmosphere, hydrosphere, lithosphere and biosphere, men and nature, technology and population, ecological concepts and ecosystems, environmental quality and management, agriculture, water resources, fisheries, forestry and wildlife, energy and mineral energy sources; renewable and non renewable resources, environmental degradation; pollution and waste management, environmental impact analysis, remote sensing & environmental monitoring.

Recommended Books:1. Something New Under the Sun: An Environmental History of the Twentieth-Century World: J. R. McNeil & Paul Kennedy 2. Principles of Ecology: R Brewer3. Fundamentals of Ecology: E. P. Odum

HUM 101 World Civilization and Culture 3 credits

A brief view of the major civilizations and cultural aspects in different continents covering ancient, medieval and modern civilizations. Topics include renaissance, reformation, and the beginning of the modern world, scientific revolution, industrial revolution, the age of democratic revolutions, nineteenth century Europe, Asia-Pacific Region, Africa, World Wars, South Asia: colonization, decolonization and after; contemporary world: Cold War and after.

Recommended Books:1. World Civilization: Bums & others2. Civilization: T Walter Walbank and others3. A History of World Civilization: J. E. Swain4. Western Civilization: : Robert E. Lerner & Standish Meachem

HUM 102 Introduction to Philosophy 3 credits

Philosophy: Concept of philosophy; science and philosophy; religion, literature and philosophy; sources of knowledge: empiricism, rationalism and criticism; concepts of value, ethics and sources of ethical standards.

Recommended Books:1. A Modern Introduction to Philosophy: P Edwards and A. Pap 2. Philosophy: R. J. Hirst 3. Introduction to Modern Philosophy: C.E.M. Joad 4. An Introduction to Philosophical Analysis: J. Hospers5. An Outline of Philosophy: A. Matin6.

Introduction of Philosophy: T. W. Patrick 7. Living Issues in Philosophy : H.H. Titus

HUM 103 Ethics and Culture 3 credits

This course introduces the students to principles and concepts of ethics and their application to our personal life. It establishes a basic understanding of social responsibility, relationship with social and cultural aspects, and eventually requires each student to develop a framework for making ethical decision in his work. Students learn a systematic approach to moral reasoning. It focuses on problems associated with moral conflicts, justice, the relationship between rightness and goodness, objective vs. subjective, moral judgment, moral truth and relativism. It also examines personal ethical perspectives as well as social cultural norms and values in relation to their use in our society. Topics include: truth telling and fairness, objectivity vs. subjectivity, privacy, confidentiality, bias, economic pressures and social responsibility, controversial and morally offensive content, exploitation, manipulation, special considerations (i.e. juveniles, courts) and professional and ethical work issues and decisions. On conclusion of the course, the students will be able to identify and discuss professional and ethical concerns, use moral reasoning skills to examine, analyze and resolve ethical dilemmas and distinguish differences and similarities among legal, ethical and moral perspectives.

Recommended Books:1. Ethics, Culture and Psychiatry - International Perspectives: Ahmed Okasha, Julio Arboleda2. Ethics and HRD: A New Approach to Leading Responsible Organizations: Tim Hatcher3. The Ethical Challenge: How to Lead with Unyielding Integrity: Noel M. Tichy and Andrew R. 4. Dignity of Difference: How to Avoid the Clash of Civilizations: Jonathan Sacks 5. Culture and Ethics: Michel Labour, Charles Juwah, Nancy White and Sarah Tolley6. Culture Matters: How Values Shape Human Progress: Samuel P. Huntington

HUM 111 History of Science 3 credits

This course will present a general overview of the development of scientific knowledge from ancient to modern times. It will examine how our modern scientific worldview developed over the ages in the fields of astronomy, physics, biology, chemistry, medicine, geology and other science disciplines. Focus will be on significant discoveries, the major scientists responsible for these revolutions, and the interrelation between science and society over the centuries. The course will contain the following:Science & philosophy, development of science in the ancient times, Greek & Egyptian science, science in the Orient, medieval science, science in the Islamic world, Western renaissance & industrialization, evolutionary theory, science in the modern ages. Science & religion, nature of scientific truth, validation of scientific theories.

Recommended Books:1. Reader's Guide to the History of Science: A. Hessenbruch2. Scientific Laws, Principles, and Theories: a Reference Guide: Robert E Krebs3. The History of Science: an Annotated Bibliography: G Miller4. A Guide to the History of Science: a First Guide for the Study of the History of Science, with Introductory Essays on Science and Tradition: G. Sarton5. Knowledge & the World: Challenges Beyond the Science Wars: M. Cavrier, J. Roggenhofer, G. Kuppers & P. Blanclard 6. The Forgotten Revolution: How Science was Born in 300 BC and Why It Had to be Reborn: Lucio Russo7. Hitler's Scientists: Science, War and the Devil's Fact: J. Cornwell

MAT 102 Introduction to Mathematics 3 credits

Factorisation, Synthetic Division, Zeros (Roots) of Polynomials, Relation between Roots and Coefficients, Nature of Roots (Descarte's Rule of signs); Complex Number System, Graphical representation of Complex Numbers (Argand Diagram), Polar form of Complex Numbers; Conic Sections, Parabola, Circle, Ellipse, Hyperbola, Transformation of Coordinates and Applications; Exponential Growth & Decay. Applications; Mathematical Induction; Determinants, Fundamental Properties of Determinants, Minors and Cofactors, Application of Determinants to solve System of Linear Equations (Cramers, Rule); Introduction to Matrix Algebra, Matrix Multiplication, Augmented Matrix, Adjoint Matrix, Inverse Matrix, Application of Matrices-solution of System of

Linear Equations (homogeneous & non-homogeneous), Consistency of System of Equations.

Recommended Books:

1. Algebra and Trigonometry: Lial and C.D. Miller2. College Algebra: Michael Sullivan

MAT 105 Calculus 3 credits

Differential Calculus: Limits, continuity and differentiability, differentiation, Taylor's, Maclaurine's & Euler's theorems, indeterminate forms, tangent and normal, sub tangent and subnormal, maxima and minima, radius of curvature & their applications, introduction to calculus of function of several variables, Taylor's theorem, maxima and minima for function of several variables. Transformation of coordinates & rotation of axes, conic sections. Integral Calculus: Definition of integration, techniques of integration for definite & indefinite integrals, improper integrals, area, volume and surface integration, arc length and their applications, multiple integrals, Jacobian, line integrals, divergence theorem and Stokes' theorem, beta function and gamma function.

Recommended Books:1. Calculus: Howard Anton2. Calculus with analytic Geometry: E. W. Swokowski

MAT 203 Matrices, Linear Algebra & Differential Equations 3 credits

Matrices: Types of matrices, algebraic operation on matrices, determinants, adjoint & inverse matrix, orthogonality & diagonalization of matrix. Linear Algebra: System of linear equations, vector space; 2D- space, 3D- space, Euclidean nD- space, sub space, linear dependence, basis and dimension, row space, column space, rank and nullity, linear transformation, eigen value and eigen vector, matrix diagonalization and similarity, application of linear algebra. Ordinary Differential Equations: Introduction to differential equations, first-order differential equations and applications, higher order differential equations and applications, series solutions of linear equations, systems of linear first-order differential equations. Prerequisite MAT 105

Recommended Books:1. Elementary Linear Algebra: Howard Anton2. Introductory

Linear Algebra with Application: Bernard Kolman3. First Course in Linear Algebra: P.B. Bhattacharya & S.K. Jain4. A First Course in Differential Equations: D.G. Zill 5. Introduction to Differential Equations: L. Ross

MAT 204 Complex Variables & Fourier Analysis 3 credits

Complex Variables: Complex number systems, general functions of a complex variable, limits and continuity of a function of complex variables and related theorems, complex differentiation and Cauchy-Riemann equations, mapping by elementary functions, line integral of a complex function. Cauchy's integral theorem, Cauchy's integral formula, Liouville's theorem, Taylor's and Laurent's theorem, singular points, residue, Cauchy's residue theorem, evaluation of residues, contour integration and conformal mapping. Fourier analysis: Real and complex form, finite Fourier transform, Fourier integrals, Fourier transforms and their use in solving boundary value problems. Prerequisite MAT 105

Recommended Books:1. Complex Variable and Applications: James W Brown and Ruel V Churchill2. Complex Variables: M R Speigel

MAT 205 Introduction to Numerical Methods 3 credits

Computer arithmetic: floating point representation of numbers, arithmetic operations with normalized floating point numbers; iterative methods, different iterative methods for finding the roots of an equation f (x) = 0 and their computer implementation; solution of simultaneous algebraic equations by various methods, solution of tri-diagonal system of equations, interpolation for equispaced and non-equispaced nodes, least square approximation of functions, curve fitting, Taylor series representation, Chebyshev series, numerical differentiation and integration and numerical solution of ordinary differential equations & partial differential equations. Prerequisite MAT 203

Recommended Books:1. Introductory Methods of Numerical Analysis: S.S. Sastry2. Numerical Analysis: R.L. Burden and J.D. Faires3. Numerical Recipes: W.H. Press, S.A.

Teukolsky, W.T. Vetterling & B.P. Flannery

MGT 211 Principles of Management 3 credits

Meaning and importance of management, evolution of management thoughts; managerial decision making; Environmental impact, corporate social responsibility, planning, setting objectives, implementing plans, organizing; organization design, managing change, directing, motivation, leadership, managing work groups, controlling: principles, process and problems and managers in changing environment.

Recommended Books:1. Management: Stephen P. Robbins and Mary Coulter2. Management: James A. F. Stoner, Edward R. Freeman & Daniel R. Gilbert

PHY 115 Electricity and Magnetism 3 Credits

Charge, quantization of charge, Coulomb's Law, electric field and potential. Gauss's Law, electric dipole, dielectrics, capacitance, energy of charged systems, electrical images, magnetic dipole, energy in a magnetic field. Direct current and electromotive force, Ohm's Law, Kirchhoff's Laws, Wheatstone Bridge, Lorentz force, magnetic field of a current and Ampere's Law, Biot-Savart Law, electromagnetic induction, Faraday's Law, self-induction, mutual induction, alternating current, RMS value, power factor, CR, LR and LCR circuits, resonance.

Recommended Books:1. Foundations of Electromagnetic Theory: John R. Reitz, F. J. Milford andR. W. Christy2. Physics: R. Resnick and D. Halliday3. Electromagnetic Fields and Waves: Paul Lorrain and Dale Corson4. Electricity and Magnetism with Electronics: K.K. Tewari5. Fundamentals of Electricity & Magnetism: A. Kip6. University Physics: Francis W. Sears, Mark W. Zemansky, Hugh D. Young

PHY 202 Optics 3 Credits

Laws of reflection and refraction, total internal reflection, Huygens' Principle, velocity of light, Young's experiment, Fresnel's bi-prism, Newton's rings, Michelson's interferometer, multiple reflections, Fabry-Perot interferometer, diffraction of light, Fresnel and Fraunhoffer diffraction, single, double and multiple-slit diffraction,

diffraction grating, spectrometer, resolving power of a grating, polarization of light, production of polarized light, plane, circular and elliptically polarized light, optical activity, double refraction, optic axis, half-wave and quarter-wave plate, nicol prism, dispersion of light, scattering of light, Thomson scattering.

Recommended Books:1. Vibrations and Waves: A. P. French2. Fundamentals of Optics: Francis A. Jenkins and Harvey E. White3. Optics: Eugene Hecht and Alfred Zajac4. Fundamentals of Physics: D. Halliday, R. Resnick & J. Walker5. University Physics: Francis W. Sears, Mark W. Zemansky, Hugh D. Young

PHY 204 Classical Mechanics and Special Theory of Relativity 3 Credits

Classical Mechanics: Newtonian equations of motion, conservation laws of a system of particles, variable mass, generalized co‐ordinates, generalized force, D' Alembert's Principle, variational method, Euler-Lagrange equations of motion, Hamilton's principles, two‐body central force problem, elliptic orbit, scattering in a central field, Rutherford formula, kinematics of rigid body motion, Euler angles, rotating co-ordinates, Coriolis force, wind motion, principal axis transformation, top motion, principle of least action, Hamiltonian equations of motion, small oscillations, normal co‐ordinates, normal modes.Special Theory of Relativity: Galilean relativity, Michelson-Morley experiment, postulates of special theory of relativity, Lorentz transformation, length contraction, time dilation, twin paradox, variation of mass, relativistic kinematics, mass energy relation.

Recommended Books:1. Classical Mechanics: H. Goldstein2. Special Relativity: A.P. French3. Perspectives of Modern Physics: A. Beiser4. Special Relativity from Einstein to Strings: Patricia Schwarz5. An Introduction to Special & General Relativity: Hans Stephain

PHY 303 Quantum Mechanics I 3 Credits

Breakdown of classical physics, quantum nature of radiation, Planck's Law, photoelectric effect, Einstein's photon concept and explanation of photoelectric effect, de Broglie wave, wave particle duality, electron diffraction, Davisson-Germer experiment, emergence of quantum mechanics, Schrodinger equation, basic postulates of quantum mechanics, physical interpretation of wave function, wave packets, Heisenberg's uncertainty principle, linear operators, Hermitian operators, eigenvalue equation, one-dimensional potential problem, harmonic oscillator, orbital angular momentum, rotation operator, spherical harmonics, spin angular momentum, addition of angular momenta, solution of the Schrodinger equation for hydrogen atom, matrix formulation of quantum mechanics.

Recommended Books:1. Quantum Mechanics: John L. Powell and B. Craseman2. Quantum Mechanics: L.I. Schiff3. Quantum Mechanics: E. Merzbacher4. Quantum Mechanics: A.M. Harun ar Rashid5. Introduction to Quantum Mechanics: P.T. Matthews6. Modern Quantum Mechanics: J.J. Sakurai7. Quantum Mechanics: D.R. Bes

PHY 304 Quantum Mechanics II 3 Credits

Rutherford scattering experiment, Discovery of the nucleus, Bohr quantization rules, hydrogen atom spectra, Franck-Hertz experiment, Sommerfeld-Wilson quantization rules, electron spin, Stern – Gerlach experiment, Pauli exclusion principle, electronic configuration of atoms, vector atom model, coupling schemes, Hund's rule, multiplet structure, fine structure in hydrogen spectral lines, Zeeman effect, Paschen-Beck effect, production of X-rays, measurement of X-ray wavelength, X-ray scattering, Compton Effect, Mosely's Law, molecular spectra, rotational and vibrational levels, Raman Effect and its applications, lasers.

Recommended Books:1. Perspectives of Modern Physics: A. Beiser2. Concepts of Modern Physics: A. Beiser3. Atomic Physics: Rajam4. Atomic & Molecular Spectroscopy: S. Svanberg5. Essentials of Modern Physics: Virgilio Acosta, Clyde L. Cowan and B. J. Graham6. Modern Physics : Kenneth S. Krane7. Modern Physics by Robert L. Sproull & W. Andrew Phillips

PHY 305 Quantum Mechanics III 3 Credits

Basic properties of nuclei, constituents of nuclei, nuclear mass, charge, size and density, nuclear force, spin, angular momentum, electric and magnetic moments, binding energy, separation energy, semi-empirical mass formula, radioactive decay law, transformation laws of successive changes, measurement of decay constant, artificial radioactivity, radioisotopes, theory of alpha decay, gamma radiation, energy measurement, pair spectrometer, classical treatment of gamma emission, internal conversion, Mossbauer Effect, beta decay, energy measurement, conservation of energy and momentum in beta decay, neutrino hypothesis, orbital electron capture, positron emission, interaction of radiation in matter, ionisation, multiple scattering, range determination, bremsstrahlung, pair production, annihilation. Discovery of neutrons, production and properties of neutrons, nuclear reactions, elastic and inelastic scattering, Q-value of a reaction and its measurements, nuclear cross-section, compound nucleus theory, direct reaction and kinematics. Prerequisite PHY 304

Recommended Books:1. Introduction to Nuclear Physics: H.A. Enge2. Concepts of Nuclear Physics: B.L. Cohen3. Elements of Nuclear Physics: W.E. Meyerhof4. Nuclear Physics: W.E. Burcham5. Nuclear Physics: Irving Kaplan6. Fundamentals of Nuclear Physics: N.A. Gelly7. Introductory Nuclear Physics: Kenneth S. Krane

PHY 310 Advanced Solid State Physics 3 Credits

Free electron theory, transport properties, Sommerfeld theory, Hall Effect, box quantization, density of states, Fermi surface, Fermi energy, electrical conductivity, Wiedmann Franz law, band theory of solids, electron in a periodic potential, Schrödinger equation, Bloch function, LCAO and OPW methods, dielectric properties of insulators, Clausius-Mosotti relations, dielectric loss, relaxation time, polarization mechanism, direct & indirect band gap semiconductors, extrinsic semiconductors, charge carrier concentration, recombination process of p-n junction, superconductivity, Meissner Effect, London equation, BCS theory, introduction to high temperature superconductivity, magnetic materials, quantum theory of diamagnetism and paramagentism, theory of ferromagnetic, ferrimagnetic and anti￾ferromagnetic orders, magnetic resonance. Prerequisite APE 201.

Recommended Books:1. Introduction to Sold State Physics: Charles Kittel2. Fundamentals of Statistical and Thermal Physics: F. Reif3. Introduction to Solid State Physics: A. J. Dekker4. Solid State Physics: M.A. Omar5. Solid State Physics: N.Y. Ashcroft and N..D. Mermin

PHY 311 X-Rays 3 Credits

Continuous and Characteristic X-rays, Bremsstrahlung, Properties of X￾rays, X￾ray technique, Weissenberg and precession methods, identification of crystal structure from powder photograph and diffraction traces, Laue photograph for single crystal, geometrical and physical factors affecting X￾ray intensities, analysis of amorphous solids and fibre textured crystal. Prerequisite APE 201

Recommended Books:1. Elements of X-ray Diffraction: B.D. Cullity2. X-ray Structure Determination: G.H. Stout and L.H. Jensen3. Crystal, X-ray and Proteins: D. Sherwood4. Elements of X-ray Crystallography: E.V. Azroff

PHY 313 Physics for Development 3 Credits

Twenty first century development issues, physics and break through technologies, ICT, fibre optics, quantum information theory, physics in genetics engineering and molecular biology, physics and health issues, bio and medical physics, materials science and physics, high temperature superconducting materials, space physics, microgravity experiments, econo-physics, physics principles applied in sociology.

Recommended Books:1. Oxford Companion to the History of Modern Science: J.L. Heilbron.2. The Evolution of Technology (Cambridge Studies in the History of Science): George Basalla & Owen Hannaway 3. Technology in World Civilization: A Thousand-Year History: Arnold Pacey 4. Zoological Physics: B.K. Ahlborn5. New Directions in Statistical Physics: Econophysics, Bioformatics & Pattern Reorganization: L.T. Willie6. Knowledge & the World: Challenges Beyond the Science Wars: M. Cavrier, J. Roggenhofer, G. Kuppers

& P. Blanchard

PHY 406 Medical Physics & Instrumentation 3 Credits

Ultrasound imaging, A scan, B scan, M scan, clinical applications, rectilinear scanner, gamma camera, CAT scanner, MRI, clinical applications, audiology, hearing aids, vascular measurements, blood pressure, blood flow, blood velocity, cardiac measurements; ECG, ECG planes, elementary ideas on heart disorders, defibrillators, pacemakers, neuromuscular measurements; EEG, EMG, stimulation of neural tissue, nerve conduction measurements, bio electric amplifiers, patient safety, radiopharmaceuticals, radiotherapy, radiation protection, radiation dosimetry.

Recommended Books:

1. Medical Physics and Physiological Measurements: B.H. Brown and R.H. Smallwood2. Medical Physics: J.R. Cameron and J.G. Skofronick3. Introduction to Health Physics: H. Cember

PHY 409 Physics of Radiology 3 Credits

The production and properties of X-rays, diagnostic and therapeutic X-ray tubes, X-ray circuit with rectification, electron interaction, characteristic radiation, bremsstrahlung, angular distribution of X-rays, quality of X-rays, beam restricting devices, the grid, radiographic film, radiographic quality, factors affecting the image, image modification, image intensification, contrast media, modulation transfer function, exposure in diagnostic radiology, fluoroscopy, computed tomography, ultrasound, magnetic resonance imaging (MRI).

Recommended Books:

1. Introduction to Health Physics: H. Cember2. Radiobiology for the Radiologist: E.J. Hall3. The Physics of Radiology: Harold Elford Johns

PHY 410 Laser Physics 3 Credits

Spontaneous and stimulated emission, absorption, pumping schemes, characteristic properties of laser beam, laser speckle, grain size calculation for free space propagation, semi classical treatment of absorption and stimulated emission, spontaneous emission, results of QED treatment, electric dipole, allowed and forbidden transitions, Einstein's A and B coefficient, radiation trapping, superfluorescence, superradiane and amplified spontaneous emission, nonradiative decay, homogeneous and inhomogeneous broadening, linewidth calculations for naturally, collisionally and Doppler broadened line, two level and four, level saturation, saturation of absorption & inhomogeneously broadened line, passive optical resonators, continuous wave and transient laser behaviour, laser beam transformation, types of lasers, their construction and use, applications of lasers, optical communications, laser in fusion research, holography. Prerequisite PHY 304

Recommended Books:1. Fundamentals of Quantum Electronics: R.H. Pantell and H.E. Putoff2. Principles of Laser: Orazio Svelto3. Laser Electronics: J.T. Verdyen4. Laser Fundamentals: W Silfvast

POL 103 Introduction to Political Science 3 Credits

A study of political systems and process with special reference to Bangladesh. Topics include nature and origin of state, sovereignty of state, forms of political units, liberty, law, process of politics, political structure, political ideas- democracy, socialism, nationalism, peoples' behaviour in politics. Political system, process and problems of Bangladesh.

Recommended Books:1. A History of Political Thought: Plato to Marx: Subrata Mukherjee & Sushila Ramaswamy 2. An Introduction to Political Science: Rand Dyck3. A Social Political History of Bengal and the Birth of Bangladesh: Kamruddin Ahmed4. Radical Politics and the Emergence of Bangladesh: Talukder Manizzaman 5. India and Pakistan: A Political Analysis: Hugh Tinker6. Politics and Policy Making in Developing Countries: Perspective on the New Political Economy: Gerald M. Meier7. Political Culture, Political Parties and Democratic Transition in Bangladesh: Shamsul I. Khan, S. Aminul Islam & Imdadul Haque 8. Involvement in Bangladesh's Struggle for Freedom: T.

Hossain9. History of Bangladesh 1704-1971: Political History: Sirajul Islam10. Conflict and Compromise: An Introduction to Political Science: H.R. Winter

POL 245 Women, Power & Politics 3 Credits

A critical examination of the impact of gender on forms and distributions of power and politics, with primary reference to the experience of Women in South Asia. Three major concerns will be addressed. First, what do we: mean by "sex", "gender". "women", "power" and "politics"? Second, how do issues of class and race/ethnicity inform our understanding of women and politics? Third, what is the relationship between women and the state? How can women organise collectively to challenge state policies, how does the state respond to organised women? Prerequisite POL 103

Recommended Books:1. The Elusive Agenda: Mainstreaming Women in Development, UPL: Raunaq Jahan.2. Persistent Inequalities: Women and World Development, 1990: Frene Tinker3. Women Leaders in Development Organizations and Institutions: Sayeda Rawsan Kadir.4. Population Policy and Women Rights: Transforming Reproductive Choice: Prayer, Ruth Dixon-Mueller.5. Women in Politics, 1994, Dhaka Women for Women: Najma Chowdhury, Hamida A Begum, Mahmuda Islam and Nazmunnessa Mahtab.

PSY 101 Introduction to Psychology 3 Credits

The objective of this course is to provide knowledge about the basic concepts and principles of psychology pertaining to real-life problems. The course will familiarize students with the fundamental process that occur within organism-biological basis of behaviour, perception, motivation, emotion, learning, memory and forgetting and also to the social perspective-social perception and social forces that act upon the individual.

Recommended Books:1. Introduction to Psychology: C.T. Morgan 2. Introduction to Psychology: R.F. Crider 3. Understanding Psychology: Robert Feldman

SOC 101 Introduction to Sociology 3 credits

Perspectives on society, culture, and social interaction, Topics include community, class, ethnicity, family, sex roles, and deviance. Social problems and sociological problems. Problems, theories, and the nature of sociological explanation. Explanation, evidence and objectivity. Sociology as a comparative study of social action and social systems. Some models of sociological thinking as applied to the study of the following: aspects of social ranking; forms of interpersonal and personal relationships; the changing nature of the relationship between economy and society; the sociology of development; the origins and spread of capitalism and socialism; ideology and belief systems; religion and society; rationality and non-rationality; conformity and deviance.

Recommended Books:

1. Sociology: Anthony Giddens 2. Sociology: Richard T. Schaefer 3. Sociology: Rao and C.N. Shankar 4. Sociology: Neil J. Smelser

SOC 401 Gender and Development 3 credits

Position & role of women in society, contemporary issues, analysis of various aspects of gender relations, gender discrimination, societal attitude, different forms of feminism, women in higher education, employment of women & discrimination, workplace harassment, contribution of women in development: world picture I position in Bangladesh. Prerequisite SOC 101

Recommended Books:1. Women and Social Security: Progress Towards Equality of Treatment, 1990 Geneva International Labor Office: Anne- Marie Brocas, Anne-Marie Cailloux and Virgine Oget.2. Impact of Women in Development Projects on Women Status and Fertility in Bangladesh, 1993, Dhaka, Development Researchers and Associates: M. Kabir, Rokeya Khatun, Ishrat Ahmed.3. Integration of Women in Development: Why, When and How: Ester Boseup, Cristine Liljencrantz.4. Women in the Third World: Gender Issue in Rural and Urban areas: Hants5. Women, Man and Society: The Sociology of Gender: Allyn and Bacon, Claire M Renzetti, Daniel J Curran.6. Race, Ethnicity, Gender and Class: the Sociology of Group Conflict and Change, London:

Joseph F Healey

STA 201 Elements of Statistics and Probability 3 credits

Frequency distribution, mean, median, mode and other measures of central tendency, standard deviation and other measures of dispersion, moments, skewness and kurtosis, elementary probability theory and discontinuous probability distribution, binomial, Poisson and negative binomial distribution, continuous probability distributions, normal and exponential, characteristics of distributions, hypothesis testing and regression analysis, basic concepts and applications of probability theory and statistics, chi-squared test.

Recommended Books:

1. Probability and Random Processes: G.R. Grimmett and D.R. Stirzaker2. Elementary Probability Theory with Stochastic Processes: K.L. Chung

b. Practical Courses:

APE 104 APE Lab I 1.5 Credits

List of Experiments:

EXP 1: Determination of the Modulus of Rigidity of a Wire by the Method of OscillationsEXP 2: Determination of Surface Tension of Mercury and the Angle of Contact by Quincke's MethodEXP 3: Determination of the Specific Heat of a Liquid by the Method of CoolingEXP 4: Determination of the Thermal Conductivity of a Bad Conductor by Lee's MethodEXP 5: Determination of the Specific Resistance of a Wire using a Meter BridgeEXP 6: Determination of the High Resistance of a Suspended Coil Galvanometer by the Method of DeflectionEXP 7: Determination of the Temperature Co-efficient of Resistance of the Material of a WireEXP 8: Determination of the Line Frequency by Lissajous Figure using an Oscilloscope and a Function Generator and Verification of the Calibration of Time/Div Knob at a Particular Position for Different FrequenciesEXP 9: Charging and Discharging of Capacitors and Study of Their Various Characteristics.EXP 10: Verification of Thevenin's and Norton's Theorem.EXP 11:

Verification of Maximum Power Transfer Theorem.EXP 12: Verification of Current Division Rule (CDR), KVL and KCLEXP 13: Conversion of Galvanometer into Voltmeter.EXP 14: Conversion of Galvanometer into Ohmmeter.EXP 15: Determination of the e/m of Electron Using Helmholtz Coil.EXP 16: Determination of the Threshold Frequency for Photoelectric Effect of a Photo-Cathode and the Value of Planck's Constant by Using a Photoelectric Cell.

APE 206 APE Lab II 1.5 Credits

List of Experiments:

EXP 1: Determination of the Refractive Index of the Material of a Prism by using a Spectrometer. EXP 2: Determination of the Radius of Curvature of a Lens by Newton's Rings MethodEXP 3: Determination of the Wavelengths of Various Spectral Lines by Spectrometer by using Plane Diffraction GratingEXP 4: Study of the Frequency Responses of Series and Parallel LRC Series Circuit and the Variation of Q-factor with Resistance.EXP 5: Study of the Variation of Electrical Conductivity of a Semiconductor and Determine of its Energy Gap.EXP 6: Study of the Characteristics of a PN Junction and Zener Diode.EXP 7: Study of the Characteristics of a NPN Bipolar Junction Transistor (BJT) in Common Base configuration.EXP 8: Study of the Characteristics of Junction Field Effect Transistor (JFET) in Common source configuration.EXP 9: Design and construction of a 4-diode Full Wave Rectifier power supply and study the effect of Shunt Capacitor filter.EXP 10: Implementation of AND, OR, NOT logic gates. EXP 11: Design of S-R flip-flopEXP 12: To design Code converters (Decimal-to-BCD, BCD-to-Decimal)EXP 13: To design Ripple, Ring and Decade Counters using JK-FFs. EXP 14: To study the characteristics of IC MUX, to realization of combinational circuits and generation of complex wavesforms.EXP 15: Use of IC 74138 decoder as DEMUX, realization of 1-to-16 line DEMUX using 74138.

APE 301 APE Lab III 1.5 Credits

List of Experiments:

EXP 1: Study of the characteristics of a Uni-junction Transistor.EXP 2: Study of the characteristics of a Silicon Controlled Rectifier Transistor.EXP 3: To draw and study the I-V characteristics of a solar cell.EXP 4: Design and construction of a BJT CE single-stage amplifier using potential divider biasing.EXP 5: Study the frequency response characteristics of a two stage RC coupled BJT amplifier.EXP 6: Study of the characteristics of 741 Operational Amplifier.EXP 7: Design of Inverting and Non-inverting amplifiers.EXP 8: Design and construction of active low pass and high pass filters using Op-Amps.EXP 9: Design and construction of active Butterworth Band pass filter.EXP 10: Design and construction of a Summing Amplifier using 741 Op-Amp.EXP 11: Study of the percentage distortion and power output of a complimentary symmetry push-pull power amplifier.EXP 12: Design and Construction of a Colpitts Oscillator.EXP 13: Design and Construction of Astable and Monostable Multividrators using BJTsEXP 14: Expt with minority carrier.EXP 15: Design and Construction of a Crystal Oscillator.

APE 303 APE Lab IV 1.5 Credits

List of Experiments:

EXP 1: Design and Construction of an Amplitude modulator and a demodulator.EXP 2: Design and Construction of a Frequency modulator and a demodulator.EXP 3: Design and Construction of a Phase-Shift-Keying and its detection.EXP 4: Design and Construction of a Pulse Amplitude modulation and its detection.EXP 5: Design and Construction of a Pulse Width modulation and its detection.EXP 6: Design and Construction of a Pulse Code modulation and its detection.EXP 7: To study Time Division Multiplex SystemEXP 8: Expt. with DSP (using DSP trainer)EXP 9: To measure microwave standing wave ratio.EXP 10: To measure microwave Frequency and Wavelength.EXP 11: Expt. with microwave antenna.EXP 12: Expt. with PLL.EXP 13: Expt. with Fiber-Optic Communication.EXP 14: To Study the TV Composite Video signal.EXP 15: Design, construction and testing of an Astable, Monostable and Voltage Controlled

Oscillator using 555 Timer.EXP 16: Expt. with microprocessor (8086).

Recommended Books:

1. Practical Physics: Giasuddin Ahmed and M. Sahabuddin2. Advanced Practical Physics for Students: B.L. Worsnop and H.T. Flint

c. Dissertation/Report

APE 400 Thesis / Project 3 credits

A student is required to carry out thesis/project work in the 7th and 8th semester in a chosen field. There will be a supervisor who will either be a BRAC University faculty or any other suitable expert from universities and R/D organizations of the country to guide the thesis/ project work .On completion of study and research s/he will have to submit the dissertation/report and face a viva board for the defence.

d. Internship 1 credit

The internship aims at providing on the job exposure to the students and an opportunity for translating theoretical concepts to real life situation. Students will be placed at R/D institutions, electronics & IT firms etc for their internship programme.

IX. Facilities & Logistics

BRAC University has a Mathematics & Natural Science Department and Physics is an integral part of that. At present an under-graduate programme for the degree of B. Sc. in Physics is offered and different relevant physics courses are also being offered to undergraduate students of Computer Science and Engineering, Computer Science, Electronics & Communication Engineering, Architecture, Business Administration, Economics, Law & English departments.

BRAC University has the competent teaching staff, classrooms, seminar rooms, conference rooms & well-equipped laboratories. To run the proposed undergraduate applied physics & electronics programme more faculties will be recruited. Experimental facilities in the labs will also be expanded.

A unique aspect of ICBM-2019 was the interactive “Meet the Editors” session organized on the last day of the conference, 27th of April at The Westin, Dhaka. This session consisted of a highly distinguished panel of speakers consisting of renowned and experienced editors who have published multiple papers each while also being in charge of international academic publications. The speakers for this session were Prof. Dr. Hui-Ming Wee, Chung Yuan Christian University, Taiwan and Prof. Dr. Tzong-Ru Lee, National Chung Hsing University, Taiwan with Brac University's Assoc. Prof. Dr. Md. Mamun Habib, the Program Chair of ICBM-2019, as the session chair. This session provided the audience with the chance to find out how to submit papers for conferences and journals successfully by knowing first hand from the editors what they look for in a paper. This session progressed in a lively question-answer manner with an audience member posing a question which was then discussed by the panel.

Prof. Dr. Tzong Ru Lee advised the scholars to focus on the contribution of their research and how their ideas can solve a particular problem. If the problem pertains to a particular industry then that should be properly highlighted in the paper. When asked about which publications have higher quality, Prof. Dr. Lee mentioned that those with peer review processes and index citations are highly respected by scholars. Speaking on the topic of choosing which publication to submit a paper in, he advised scholars to research the Editorial Board of that publication to see how diversified in terms of race, nationality the editors are, as greater diversity leads to fairer evaluation. Talking about local cases, Prof. Lee mentioned that BRAC being a successful social enterprise would make a good topic.

Prof. Dr. Hui Ming Wee suggested the researchers to give an attractive title to the paper and clearly communicate the message of the research through the abstract. Prof. Dr. Wee encouraged new scholars to send their papers to multiple conferences in order to gain experience and suggested them to apply to those with no participation fee in the beginning years of their research so as to make participating more economical. He also mentioned that submitting to high index value publications is extra beneficial since even if the paper does not get accepted, the researcher gets a very valuable feedback and this review can be used to improve the research and the approach applied by the researcher. In case of Asian scholars, Prof. Dr. Wee said that an expert in English should be consulted to proof read the document since most international publications are written in English. When asked about providing references for a research, Prof. Wee said a good practice is to provide reference to the works of the Editor(s) in the Editorial Board. He also spoke about the scientific data presented I the paper saying that the result must be genuine and the methodology clearly indicated.

Dr. Md. Mamun Habib, an Editor-in-chief of UK based Scopus indexed journal, emphasized the importance of mentioning the novelties and contribution of the paper. He also suggested that research paper should focus on contemporary issues and that the timing of any research can be the key to its acceptance in society. The session chair further advised the participants that significant modification to conference papers, at least 50% is necessary in case the researchers want to submit the same paper to a journal.

All three editors spoke of how business focused cases can help industries and would be a good trend to pursue in coming years.

Tabassum Rahman Sunfi completed her school and college from Viqarunnisa Noon School and College. She finished her BS and MS from University of Dhaka in 2017 and 2018 respectively.

Level:Kha-224 Merul BaddaDhaka 1212. Bangladesh

BS and MS in Biochemistry and Molecular Biology -Undergraduate completed in 2017 and Graduated in 2018

Genetic engineering and Biotechnology, Microbial world, Basic Biochemistry,Industrial Biotechnology, Food Biotechnology

• Motiur Rahman Memorial Scholarship for academic excellence in Bachelor study at the Department of Biochemistry and Molecular Biology, University Of Dhaka(2018)•

Bangladesh Government Scholarship for outstanding result in Higher Secondary School Certificate Exam (2011)• Bangladesh Government Scholarship for outstanding result in Secondary School Certificate Exam (2009)

Research Assistant, IRCMS Program, Hematology and Leukemiagenesis Laboratory, Kumamto University, Japan (January 2019)

Tabassum Rahman Sunfi

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Registration is a continuous process at BRAC University, beginning with the student's first day to register and continuing until the first day of classes for the semester. Once the semester begins, the process of changing the student's course schedule is referred to as dropping/adding courses. A student has several opportunities within a semester to drop a course, but different consequences apply at each stage. Before dropping a course, students should consult with their academic advisors. Dropping courses not only affects a student's academic progress, but also may have consequences for financial aid. There is no refund of tuition for individual courses dropped after the last day of the change of program period.

Ceasing/stopping to attend classes or simply notifying the instructor does not constitute dropping a course or a semester. If a student wants to drop a course (or courses), s/he must formally/ officially drop the course(s) from the semester other wise it will appear as an “F” grade on the transcript and affect the CGPA. If a student can not continue a semester for any reason must drop the semester.

Dropping a course during the change of academic calendar

A student may drop a course with or without refund within the calendar period of add/ drop of the semester. Courses dropped within this period do not appear on a student’s transcript and does not affect CGPA.

Dropping with W grade

After the late drop deadline, students may drop a course. Courses dropped after the late drop deadline but prior to the final drop deadline will be recorded on the transcript with the notation “W” (withdrew). Students are charged full tuition for individual courses from which they selectively withdraw. “W” grade appear on a student’s transcript but does not affect CGPA.

Dropping a course after the deadline

Student applies to drop a course(s) after the final drop deadline may be granted with the permission of the department only with documentation of extenuating circumstances (such as a serious illness or a crisis beyond the student's control or even for a wrong advising). Some courses can also be dropped with the permission of head the department. BRACU has specific policy regarding courses dropping or replacing in this regard:

Students dropping a course after the deadline will have to fill up a Course Drop Form with sufficient documents. Course dropped in such situation will not affect the CGPA and will not appear on transcript. Students have to follow the following directions for course drop:

Leave of Absence:

A student may be granted leave of absence or semester drop for a period of up to three consecutive semesters or one academic year, subject to the student meeting academic requirements. This arrangement does not apply to students who received academic probation or who have been expelled from the university on disciplinary grounds or excluded on academic grounds. The decision to grant leave of absence will rest with the Dean or Head of Department in consultation with the Registrar. A student granted leave of absence must register in the following semester immediately after the expiry of the leave period.

Students have to apply through the Semester Drop Form with sufficient documents for semester drop. Dropping a whole semester does not affect the CGPA and will not appear on the transcripts. Students have to follow the following directions for semester drop:

A current Lecturer of Biotechnology at MNS, Abu Nasim Haider, has been employed in Industry and Academia. As a result, he has substantial experience in the wet lab and biopharmaceutical industry affairs.

He completed his BS and MS from the Department of Genetic Engineering and Biotechnology, University of Dhaka. His MS thesis focused on developing cheap, non-invasive diagnostic technologies for cancer by utilizing cfDNA. Ever passionate about science, he has been an active member of the science communication circle of Bangladesh for half a decade.

He started his professional career at Healthcare Pharmaceuticals Ltd. Bangladesh. In the Biotech R&D Division, his role was to develop the manufacturing and testing of new and existing biotech drugs as part of the team. His skill set includes but is not limited to - sterile manufacturing, mammalian cell culture, cell-based assays, and molecular assays.

He finds the role of a faculty very rewarding as he is very passionate about learning and helping others learn. Cancer biology, molecular diagnostics, and pharmaceutical affairs are some of his interests.

Level:Kha-224 Merul BaddaDhaka 1212. Bangladesh

• Master of Science (MS) in Genetic Engineering and Biotechnology, University of Dhaka (2019)• Bachelor of Science (BS) in Genetic Engineering and Biotechnology, University of Dhaka (2018)

• Lecturer, Department of Mathematics and Natural Sciences, Brac University (October 2021 - Present)• Officer, Biotech R&D, Healthcare Pharmaceuticals Ltd. (September 2020 -October 2021)

• BTE404: Bioprocess Technology• BTE406: Genomics and Proteomics• BTE155: Biotech Lab I• BTE255: Biotech Lab II

• National Science and Technology (NST) Fellowship for MS research, Session 2019-20; Ministry of Science and Technology, Government of the People's Republic of Bangladesh• Dean's Award for Academic Excellence in BS Examination 2018, Faculty of Biological Sciences, University of Dhaka• Govt. Honours Scholarship for BS Examination 2018, Government of the People's Republic of Bangladesh

Federation of Asian Biotech Associations – Bangladesh Chapter

Molecular Diagnostics, Cancer Biology, Computational Biology

Abu Nasim Haider

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Akash Ahmed is a lecturer in Microbiology Program at the Department of Mathematics and Natural Sciences, Brac University, Dhaka. He has completed his Bachelor’s in Microbiology from the Department of Mathematics and Natural Sciences, Brac University, Dhaka and Master’s in Microbiology from the Department of Microbiology, Jagannath University Dhaka. He did his B.S. thesis on antibiotic resistant bacteria: A prospective study on the efficiency of ciprofloxacin in combination with chloramphenicol and probiotics against multiple antibiotic resistant bacteria in Microbiology laboratory in the department of Mathematics and Natural Science, Brac University. During his master degree he continued working on antibiotic resistant bacteria and his research title was “Isolation of qnrB and gyrA gene from levofloxacin resistant Pseudomonas aeruginosa and identification of potential mutation in gyrA gene.” Currently, he is focusing on the effect of biocides and pesticides on bacterial metabolism.

Level: 5, Room: 5B18Kha-224 Merul BaddaDhaka 1212. Bangladesh

Akash Ahmed

News & Events

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EEE student selected for Huawei program in China

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Nanoscale science is a fascinating world for the upcoming advanced technology-driven century. Keeping that in mind, BRAC University Society for Biotechnology (BUSB) along with the Biotechnology program, BRAC University had arranged the third episode of the Biotechnology Seminar Series on “Nanobiotechnology” last 28th March 2024. The purpose of this seminar was to open the magic door of nanobiotechnology to the participants. This seminar mainly focusing on basic aspects of Nanobiotechnology was attended by MNS Chairperson Professor Dr. Yusuf Haider, faculties and BRACU students from different majors such as Biotechnology, CSE, Microbiology, and Pharmacy.

The speaker Dr. Munima Haque was introduced by Assistant Professor Dr. Jebunnessa Chowdhury of the Biotechnology program. Dr. Munima Haque, Associate Professor of Biotechnology Program, conducted an extremely engaging session on Nanobiotechnology as her research areas involved Nanobiotechnology, Radiation Biotechnology, Environmental Biotechnology, Medical Physics, and Biomedical Data Science.

Dr. Munima completed B.Sc. (honors) and M.Sc. in Physics with minors in Mathematics and Statistics from Rajshahi University, Bangladesh. Then she did another M.S. in Nuclear Radiation Physics from Southern Illinois University (SIU), Carbondale, IL, USA. She obtained her Ph. D. in Agricultural and Biological Engineering from the University of Illinois at Urbana-Champaign (UIUC), Urbana, IL, USA with a thesis titled “A colloidal science approach to characterize the nanoscale aggregation behavior of corn protein zein”. Her postdoctoral research was in the Micro and Nanotechnology Laboratory, University of Illinois at Urbana Champaign, IL, USA in Biomedical Engineering on Targeted Drug delivery using hybrid nanocarriers for breast and prostate cancer detection and treatment in addition to research on gene therapy.

The seminar discueesd that the fusion of nanotechnology and biotechnological advancement unveiled current possibilities of learning to participants. Nanobiotechnology focuses on studying and applying biological and biochemical processes found in nature to create innovative devices such as biosensors for disease detection. Targeted drug delivery is such an approach to create medication for a patient in a way that enhances the concentration of the medication in specific parts of the body compared to others. Moreover, Nanopharmacology promises targeted treatments with minimized side effects by combining nanotechnology with pharmacology and enhancing drug delivery with precision and efficacy.

The discussion flowed from medical applications of nanoparticles to the textile industry, food preservation, the agricultural sector, nanobioelectronics, and many more. The interactive format of the seminar encouraged participants to perform critical analysis and integrate nanomaterials with biological systems.

The technical session enlightened students about the potential of nanobiotechnology and the importance of achieving skills and insights for future biotechnological endeavors.

News Archive >>

Nanobiotechnology seminar organized by BUSB and Biotechnology Program

Related News

Biotechnology seminar hosts Bangladeshi scientist who developed high-yielding “Panchabrihi” rice

Orientation held for Summer 2024 biotechnology students

The biotechnology program and BUSB organized the inaugural Biotechnology Workshop Series on “Understanding and Designing Biosensors”

Orientation held for Spring 2024 biotechnology students

Mr. M. H. Murad received his B. Sc. in Mathematics and MS in Mathematics (Applied) from the University of Chittagong, Bangladesh in 2006 and 2007, respectively. He did his Master’s thesis on some exact static spherically symmetric perfect fluid solutions of Einstein’s gravitational field equations. To his credit, he has published several research articles in internationally prestigious and recognized peer reviewed journals like Astrophysics and Space Science, International Journal of Theoretical Physics, European Physical Journal C, Annals of Physics, etc. His current research focuses on the anisotropic stellar models of electrically neutral/charged compact stars. Before joining Brac University, in Summer 2014 as a Lecturer, he had worked at Daffodil International University, Dhaka, Bangladesh for over two years as a Lecturer, and nearly one year as a Senior Lecturer of Mathematics.

Additional Profile Links

ResearchGate

https://www.researchgate.net/profile/Mohammad_Murad

Google Scholar

https://scholar.google.com/citations?hl=en&user=MD9wTdIAAAAJ&view_op=list_works&sortby=pubdate

Journals

Fatema, S., Murad, M. H., & Singh, K. N. (2019). New exact anisotropic static spherically symmetric stellar models satisfying the Eiesland condition. Annals of Physics, 402, 1–17. https://doi.org/10.1016/j.aop.2019.01.007

Murad, M. H. (2018). Some families of relativistic anisotropic compact stellar models embedded in pseudo-Euclidean space E5 : an algorithm. European Physical Journal C, 78(4). https://doi.org/10.1140/epjc/s10052-018-5712-5

Singh, K. N., Murad, M. H., & Pant, N. (2017). A 4D spacetime embedded in a 5D pseudo-Euclidean space describing interior of compact stars. European Physical Journal A, 53(2). https://doi.org/10.1140/epja/i2017-12210-1

Bhar, P., & Murad, M. H. (2016). Relativistic compact anisotropic charged stellar models with Chaplygin equation of state. Astrophysics and Space Science, 361(10). https://doi.org/10.1007/s10509-016-2923-9

Bhar, P., Murad, M. H., & Pant, N. (2015). Relativistic anisotropic stellar models with Tolman VII spacetime. Astrophysics and Space Science, 359(1). https://doi.org/10.1007/s10509-015-2462-9

Murad, M. H., & Fatema, S. (2015). Anisotropic charged stellar models in Generalized Tolman IV spacetime. European Physical Journal Plus, 130(1). https://doi.org/10.1140/epjp/i2015-15003-y

Murad, M. H., & Fatema, S. (2015). Some new Wyman–Leibovitz–Adler type static relativistic charged anisotropic fluid spheres compatible to self-bound stellar modeling. European Physical Journal C, 75(11), 1–21. https://doi.org/10.1140/epjc/s10052-015-3737-6

Pant, N., Pradhan, N., & Murad, M. H. (2015). A class of relativistic anisotropic charged stellar models in isotropic coordinates. Astrophysics and Space Science, 355(1), 137–145. https://doi.org/10.1007/s10509-014-2156-8

Murad, M. H. (2015). Some analytical models of anisotropic strange stars. Astrophysics and Space Science, 361(1), 20. https://doi.org/10.1007/s10509-015-2582-2

Pant, N., Pradhan, N., & Murad, M. H. (2014). A class of super dense stars models using charged analogues of Hajj-boutros type relativistic fluid solutions. International Journal of Theoretical Physics, 53(11), 3958–3969. https://doi.org/10.1007/s10773-014-2147-0

Murad, M. H., & Pant, N. (2014). A class of exact isotropic solutions of Einstein's equations and relativistic stellar models in general relativity. Astrophysics and Space

Science, 350(1), 349–359. https://doi.org/10.1007/s10509-013-1713-x

Murad, M. H., & Fatema, S. (2014). Some static relativistic compact charged fluid spheres in general relativity. Astrophysics and Space Science, 350(1), 293–305. https://doi.org/10.1007/s10509-013-1722-9

Mahbubur Rahman, A. H. M., & Murad, M. H. (2014). Some electrically charged relativistic stellar models in general relativity. Astrophysics and Space Science, 351(1), 255–265. https://doi.org/10.1007/s10509-014-1823-0

Pant, N., Pradhan, N., & Murad, M. H. (2014). A family of exact solutions of Einstein-Maxwell field equations in isotropic coordinates: an application to optimization of quark star mass. Astrophysics and Space Science, 352(1), 135–141. https://doi.org/10.1007/s10509-014-1904-0

Murad, M. H., & Fatema, S. (2013). A family of well behaved charge analogues of Durgapal’s perfect fluid exact solution in general relativity II. Astrophysics and Space Science, 343(2), 587–597. https://doi.org/10.1007/s10509-012-1277-1

Murad, M. H. (2013). A new well behaved class of charge analogue of Adler’s relativistic exact solution. Astrophysics and Space Science, 343(1), 187–194. https://doi.org/10.1007/s10509-012-1258-4

Murad, M. H., & Fatema, S. (2013). A family of well behaved charge analogues of Durgapal’s perfect fluid exact solution in general relativity. Astrophysics and Space Science, 343(2), 587–597. https://doi.org/10.1007/s10509-012-1277-1

Fatema, S., & Murad, M. H. (2013). An Exact Family of Einstein–Maxwell Wyman–Adler Solution in General Relativity. International Journal of Theoretical Physics, 52(7), 2508–2529. https://doi.org/10.1007/s10773-013-1538-y

Murad, M. H., & Fatema, S. (2013). Some exact relativistic models of electrically charged self-bound stars. International Journal of Theoretical Physics, 52(12), 4342–4359. https://doi.org/10.1007/s10773-013-1752-7

• January 2020–presentUndergraduate Mathematics Program Coordinator Department of Mathematics and Natural Sciences (MNS) Brac UniversityDhaka, Bangladesh

• July 2019–presentAssistant ProfessorDepartment of Mathematics and Natural Sciences (MNS)Brac UniversityDhaka, Bangladesh

• July 2016–June 2019Senior LecturerDepartment of Mathematics and Natural Sciences (MNS)Brac UniversityDhaka, Bangladesh

• May 2014–June 2016LecturerDepartment of Mathematics and Natural Sciences (MNS) Brac UniversityDhaka, Bangladesh

• August 2013–May 2014Senior LecturerDepartment of Natural SciencesDaffodil International UniversityDhaka, Bangladesh

• May 2011–August 2013LecturerDepartment of Natural SciencesDaffodil International UniversityDhaka, Bangladesh

• MAT092 Remedial Course in Mathematics• MAT101 Fundamentals of Mathematics• MAT103 Basic Concepts in Mathematics• MAT104 Mathematics• MAT105 Calculus• MAT110 MATH I: Differential Calculus & Coordinate Geometry• MAT120 MATH II: Integral Calculus & Differential Equations• MAT203 Matrices, Linear Algebra & Differential Equations• MAT204 Complex Variables and Fourier Analysis• MAT211 Calculus II• MAT215 MATH III: Complex Variables & Laplace Transformations• MAT216 MATH IV: Linear Algebra & Fourier Analysis• MAT221 Real Analysis I• MAT301 Group Theory• MAT303 Tensor Analysis• MAT311 Abstract Algebra• MAT321 Real Analysis II• MAT322 Differential Equations II• MAT325 Mathematical Methods

• Analysis • Differential Equations• Linear Algebra• Mathematical Physics• General Theory of Relativity• Cosmology• Exact solutions of Einstein’s gravitational field equations• Relativistic compact stellar structure.

Mohammad Hassan Murad

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Ms. Moumita Datta Gupta joined as a Lecturer of Statistics at the Department of Mathematics and Natural Sciences of Brac University in the Spring Semester, 2020. She has completed his B.S. (Hons) and M.S. degrees from the Department of Statistics of the University of Dhaka. Prior to joining Brac University she was a statistical officer in icddr,b. Before joining icddr,b she worked as an adjunct Lecturer of Statistics at the Department of Mathematics and Natural Sciences of Brac University in Summer semester, 2019. Her research interests are in the areas of Biostatistics, Public Health, Epidemiology, Econometrics and Financial Statistics.

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• M.S. (Master of Science) in Statistics, Biostatistics and Informatics from the Department of Statistics, University of Dhaka – 2014.• B.S. (Bachelor of Science) in Statistics, Biostatistics and Informatics from the Department of Statistics, University of Dhaka – 2013.

• Statistical Officer, Maternal and Child Health Division, ICDDR’B, Dhaka. June, 2019 to January, 2020.• Adjunct Faculty (Statistics) Mathematical and Natural Science Department, Brac University (Summer Semester- 2019).

• Introduction to Statistics• Elements of Statistics & Probability

• Gupta MD, Khanam M. The Temporal Causal Relationship among Energy Consumption, Real GDP and Industry Value Added of Bangladesh. Second International

Conference on Theory and Applications of Statistics, 2015.• Speaker on several round table discussions jointly organized by UNFPA & Daily Star on HIV & Adolescent Reproductive Health.

• Life Member, Alumni Association (DUSSDA), Department of Statistics, Biostatistics and Informatics, University of Dhaka.• Life Member, Viqarunnisa Noon Science Club.• Life Member, UNFPA Youth Forum for RH.

• Biostatistics, • Epidemiology • Public Health • Econometrics and • Financial Statistics

Moumita Datta Gupta

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Nourin Ferdausi completed her school and college from Bangladesh International School, Dammam, Saudi Arabia. She finished her BS and MS in Biochemistry and Molecular Biology from University of Dhaka in 2018 and 2019 respectively. Her MS thesis was based on the evaluation of different enzymatic and non-enzymatic biomarkers of oxidative stress in patients with Acute Coronary Syndrome. She was also involved in a research project on computer-aided drug discovery.

Level:Kha-224 Merul BaddaDhaka 1212. Bangladesh

• Master of Science in Biochemistry and Molecular Biology, University of Dhaka, Dhaka-1000, Bangladesh -Graduated in 2019• Bachelor of Science in Biochemistry and Molecular Biology, University of Dhaka, Dhaka-1000, Bangladesh- Graduated in 2018

Lecturer (Part-time) -January 2020 – September 2021Department of Mathematics and Natural Sciences (Biotechnology Program)BRAC University, Bangladesh

Bioorganic Chemistry, Industrial Biotechnology, Introduction to Biology, Microbial Biotechnology, Biotech Lab I, Biotech Lab II, Biotech Lab III

• Oral Presentationo Ferdausi, N., Anik, M., Binti, N. and Islam, L. (2019) Assessment of Enzymatic and Non-enzymatic Biomarkers of Oxidative Stress in Patients with Acute Coronary Syndrome. BSBMB Conference, University of Chittagong, Chittagong.

• Poster Presentationso Binti, N., Anik, M., Ferdausi, N. and Islam, L. (2019) Evaluation of Oxidative Status and Protein Damage in patients with Acute Coronary Syndrome. BSBMB Conference 2019, University of Chittagong, Chittagongo “CRISPR-Cas system” (2017) at Biochemistry Olympiad, University of Dhaka.o “Nanobots- An Inspirational Future in Medicine” (2016) at Biochemistry Olympiad, University of Dhaka.

Global Outreach Member of the American Society for Microbiology

Cardiovascular Diseases, Biomarkers of Oxidative Stress and Inflammation, Bioinformatics, Computer-Aided Drug Discovery.

Nourin Ferdausi

News & Events

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Competition

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After being born in Tangail, Shamira Tabrejee completed her SSC and HSC from Viqarunnisa Noon School and College and later pursued BS degree from University of Dhaka. Her research interest lies in the field of transcriptional regulation of genes where she wants to combine the wet lab research with computational biology.

Level: Kha-224 Merul BaddaDhaka 1212. Bangladesh

• MS in Genetics, genomics and bioinformatics from The State University of New York at Buffalo, USA (2018-2020).• MS from Genetic Engineering and Biotechnology, University of Dhaka, Bangladesh (2012-2013).• BS (4 years Honours) from Genetic Engineering and Biotechnology, University of Dhaka, Bangladesh (2009-2012).

Journals

Tabrejee, S., & Hossain, M. M. (2019). In silico approach to design a B-cell epitope based vaccine target against yellow fever virus. Bangladesh Journal of Microbiology, 35(1). https://doi.org/10.3329/bjm.v35i1.39801

Tabrejee, S., Chowdhury, M. M. K., Bari, L., & Yeasmin, S. (2018). Isolation and characterization of enterococci from yoghurts of Bangladesh. Modern Care Journal, 15(3). https://doi.org/10.5812/modernc.74460

Rahman, T., Heickal, H., Tabrejee, S., Chowdhury, M. M., Sarwar, S., & Shoyaib, M. (2016). SeqDev: An algorithm for constructing genetic elements using comparative assembly. Plant Tissue Culture and Biotechnology, 26(1), 105–121. https://doi.org/10.3329/ptcb.v26i1.29772

Conferences

Easmin, R., Tabrejee, S., Rowshon, L., Gias, A. U., Chowdhury, M. M. K., & Shah, K. (2014). In silico analysis of salt responsive survival associated motifs in rice promoter. In 8th International conference on software, knowledge, information management and applications (SKIMA). Dhaka, Bangladesh.

Katha, T. R., Tabrejee, S., Chowdhury, T., Chowdhury, M. M. K., & Shoyaib, M. (2014). BLASTAssemb: An approach to construct genetic element using BLAST. In 8th International conference on software, knowledge, information management and applications (SKIMA). Dhaka, Bangladesh.

Worked as a member of the Internal Quality Assurance Cell (IQAC), Brac University, Dhaka, Bangladesh

Genomics and Proteomics, Advanced Molecular Biology, Basic Biochemistry, Metabolism, Microbial World, Fundamentals of Genetic Engineering, Industrial Biotechnology, Microbial Biotechnology, Bioprocess Technology, Biotechnology LabI and Lab II.

• J. William Foreign Fulbright Scholarship, U.S Department of State's Bureau Educational and Cultural affairs. 2018-2019.• National Science and Technology Fellowship (NST), Ministry of Science, Government of Bangladesh, 2014.• Scholarships for brilliant academic results in B.S. and M.S by University of Dhaka.

• Dean's Award, Faculty of Biological Sciences, University of Dhaka, 2014.

• Sumitomo Corporation Scholarship for brilliant academic results, 2010-2013.

• Dhaka Board Talent pool Scholarship for brilliant result in both HSC and SSC examinations.

Genomics, gene regulation, transcriptional regulation, DNA-protein interactions, bioinformatics

Shamira Tabrejee

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9:00 AM to 9:00 PM

Department

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Office of the Controller of Examinations

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Medical Centre

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Student Information Centre

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Student Life

studentlife[at]bracu.ac[dot]bd

Office of Career Services and Alumni Relations (OCSAR)

csoadmin[at]bracu.ac[dot]bd

Office of Academic Advising (OAA)

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Counseling and Wellness Centre

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International and Scholarship Office

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Ayesha Abed Library

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Institutional Quality Assurance Cell (IQAC)

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Office of Communication

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Operations

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Accounts and Finance

queries-accounts[at]bracu.ac[dot]bd

Human Resources Department

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IT Department

support[at]bracu.ac[dot]bd

General Information

info[at]bracu.ac[dot]bd

BRAC University Campus

Residential Campus

K.M. Mazharul Alam, a newly appointed lecturer in Microbiology, completed both his school and college from Pabna Cadet College. He accomplished his graduation and post-graduation from Department of Microbiology, University of Dhaka. Before joining BRAC University, he was associated with a number of research organizations. He was a laboratory analyst in Center for Advanced Research in Sciences, Dhaka University and also a research assistant in Genomics and Bioinformatics Laboratory, Department of Microbiology, University of Dhaka. Recently, for Covid-19 Emergency Response Pandemic Preparedness, Mr. Mazharul Alam has been recommended as a lab consultant officer (Microbiologist) by Directorate General of Health Services (DGHS). Besides academic achievements, he cultivates a cultural personality in him and also is engaged to various social activities. He was an award winning musician at college and university level, won multiple prizes for extra curricular activities (Public Speaking, English Debate an and member of football, cricket and runners up basketball team and also a member of Dhaka University Microbiology Blood Donation Club too. With all his efforts and excellence, Mr. Mazhar is expected to bring dynamicity in this university through his work and activities.

Level: 5Kha-224 Merul BaddaDhaka 1212. Bangladesh

Advanced Molecular Biology (MIC 310)Analytical Microbiology (MIC 402)Microbial Lab I (MIC 155)Microbial Lab III (MIC 355)

Oral presentation at 34th BSM Conference (2020) on “Isolation and molecular characterization of virulent Aeromonas hydrophila (vAh) from Ompok pabda (Hamilton, 1822) fishes infected with Motile Aeromonas Septicemia (MAS)” organized by Bangladesh Society of Microbiologists.Oral presentation at 35th BSM Conference (2021) on “Development of VP1-based sero-diagnostic kit for post-vaccination immuno-surveillance of Foot-and-Mouth Disease Virus (FMDV)” organized by Bangladesh Society of Microbiologists

Oral presentation at International Colloquium on Authentic Scientific Publications (2022) on “Heterologous Expression of VP1 Capsid Protein of Foot and Mouth Disease Virus (FMDV) in Escherichia coli and its Evaluation as a Capture Antigen for Serotype-dependent Detection of Anti-FMDV Antibodies” organized by National Young Academy of Bangladesh.

i. Talent pool and General Scholarship from Directorate of Secondary and Higher Secondary Education in 2013 and 2015ii. Best Oral Presenter at 34th BSM Conferenceiii. Best Oral Presenter at 35th BSM Conferenceiv. Best Oral Presenter at International Colloquium on Authentic Scientific Publications (2022)v. National Science and Technology (NST) Fellowship,2021-22.vi. Scholarship from University of Dhaka for academic excellence in Bachelor of Sciencevii. Best All-rounder cadet, Pabna Cadet College (2015)

Dhaka University Microbiology Blood Donation Club

Mr. Mazharul has performed his thesis on developing a Novel diagnostic kit for three circulating serotypes of Foot and Mouth Disease Virus (FMDV) by recombinant DNA technology. He has also been engaged to evolutionary analysis of emerging lineages and sub-lineages of FMDV and reported emergence of a novel sub-lineage of serotype O of the virus (Manuscript submitted). He is also interested in Bacterial pathogenesis and done his research on identifying hypervirulent strain of Aeromonas hydrophila (vAh) from Bangladesh and placed it in the global phylogeny. He has profound interest in Virology, Immunology, Bacterial Pathogenesis, Viral Evolution and Bioinformatics.

K.M. Mazharul Alam

News & Events

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Afia Maheda Prema completed her BSc. in Biochemistry and Molecular Biology from Shahjalal University of Science and Technology, Sylhet, Bangladesh. After that in January 2021, she got admission into East Tennessee State University, Tennessee, USA in the Biological Sciences MSc. program with a prestigious scholarship(full-funded) for a Graduate Teaching Assistantship. At ETSU, she worked as a graduate teaching assistant for two and a half years. Moreover, she achieved level 1(High Contact) in the US University-based oral proficiency test, so she was given full responsibility for grading, lab instructions, and classroom lectures. She taught undergraduate biology students on various topics such as neuro-biochemistry, organismal biology, protein chemistry, bioinformatics, microscopy, etc.

She also worked as a Graduate Research Assistant in the Biological Sciences Department. Her whole thesis was conducted under the Centre of Excellence for Inflammation, Infectious Disease and Immunity, Quillen College of Medicine, ETSU. During her thesis work and graduate program studies, she gathered expertise in cloning and recombinant plasmid construction, mutagenic primer designing using OLIGO software, both atypical and conventional PCR-Site-Directed mutagenesis with the unique mutant generation, CHROMAS, DNA isolation, XL1B, DH5α, BL21 competent cell transformation, and cell culture, agarose DNA gel electrophoresis, SDS-PAGE, BCA assay, Ni-NTA column chromatography mediated protein isolation and dialysis, surface plasmon resonance assay and Kinetic characterization of membrane protein binding interactions using ForteBio Octet BLI system & SPR, Western blotting, ELISA, protein quantification, flow cytometry and cell adhesion assay, 3D Protein structure homology modeling, DNA quantification by real-time PCR, biomolecule separation by HPLC, pathogenicity island analysis with Artemis, SPSS and statistical data analysis.

Afia´s master´s thesis focused on Mapping the binding site within human Integrin aDb2 for carboxyethyl (CEP)-modified proteins which aimed to identify the specific amino acids of integrin aDb2 that participate in binding to 2-(ω-carboxyethyl) pyrrole or CEP, a ligand at inflammatory sites. Investigation of this interaction is a major step toward treatment and antibody development with the potential to prevent human chronic inflammation-mediated diseases. Her thesis work is a combination of molecular biology, immunology, microbiology, protein biochemistry, bioinformatics, and biostatistics.

During her graduate studies, she conducted a semester-long project under the Department of English, at ETSU where she supervised students of diverse nationalities about the history, culture, and traditions of Bangladesh. Respective students completed the project successfully under Afia´s supervision.

In her undergrad, she was awarded for placing third position in the National Biochemistry Olympiad (Bio Park Poster category) arranged by Dhaka University for two consecutive years where she presented her innovative ideas and thoughts on genetic engineering with the aim of chronic disease prevention.

Currently, she is working on nanotechnology topics, however, she has an immense interest in working on human chronic diseases as well.

Level: 4Kha-224 Merul BaddaDhaka 1212. Bangladesh

Human Chronic Diseases, Cancer Biology, Nanobiotechnology, Immunology.

Afia Maheda Prema

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Available Program For Admission

Centre for Inclusive Architecture and Urbanism (Ci+AU) at Brac University is multidisciplinary research institute and offers research-based consultancy on urban issues – the exploration of conditions, processes, and policies that affect the livability, resilience, and sustainability of the built environment. The Centre's mission is to create a dynamic culture of architecture and urbanism that is inclusive, empathetic, environmentally conscious, and humanistic. The centre is headed by Prof. Adnan Zillur Morshed, PhD, Executive Director.

By integrating rigorous research with applied knowledge, Ci+AU seeks to engender a system of bottom-up thinking about how to build inclusive environments and livable cities. The Centre's collaborative and action-based research activities contribute to the development of new ideas and offer innovative design solutions and policy guidance for both the public and private sectors. It serves policy- and decision-makers, professional agencies and organizations, development agencies, stakeholders in urban development, and academia, with a view to creating a platform for critical thinking around community-driven architecture and urbanism in the Global South and beyond. Founded by Professor Adnan Zillur Morshed (Brac University, Dhaka, and Catholic University of America, Washington, DC) in 2017, the Centre's inaugural advisory board included late Sir Fazle Hasan Abed KCMG (Founder of BRAC), Professor Mark Jarzombek (MIT), Professor Haider A. Khan (University of Denver), Professor Howard Davis (University of Oregon), Professor Rahul Mehrotra (Harvard University), and Professor Syed Saad Andaleeb (Penn State University). A new advisory board will soon be announced.

The Centre is a multi-tiered collaborative for the study of architecture and urbanism by bringing together a multi-disciplinary group of educators, researchers, policymakers, urban administrators, students, and practitioners. It will:

Ongoing consultancy work includes the design of BRAC regional offices across Bangladesh; interior design for the Brac University Residential Campus at Savar, and School for Rohingya children in Cox's Bazar. Recently completed works include: Participation in the "Historic Preservation and Landscape Design of the Old Dhaka Central Jail and Redevelopment of Its Surrounding Area," National Design Competition, May – Oct. 2017 (received honorable mention) and Master plan and pavilion design for the exhibition of Aarong's 40th Anniversary. Current research work includes a study of the effect of personal automobiles on the traffic conditions of Dhaka city and an urban/environmental assessment of plot-based housing in Dhaka.

News Archive >>

News

Ci+AU holds exhibition titled 'Architecture as Freedom' in Washington DC

Research on socio-economic impact of the residual spaces under the flyover(s) and elevated expressways of Dhaka

Ci+AU Team

Professor Adnan Zillur Morshed, PhD

Md. Faysal Kabir

S.M Shafaiet Mahmud

Syed Tawsif Munawar

Nuzhat Shama

Carlos A. Reimers. PhD

Paco Mejias Villatoro

Junjie Xi

Sumaiya Ara Simi

Tahseen Reza Anika

The graduation programme for Applied Physics & Electronics will contain the courses as appended below indicating course numbers and divisions. However, the courses listed below may change as the need be in future.

a. General Education: (21 credits)

1. APE 101 Mechanics, Properties of Matter, Waves & Oscillations 2. CSE 101 Introduction to Computer Science 3. DEV 101 Bangladesh Studies4. ENG 091 Foundation Course (non-credit)5. ENG 101 English Fundamentals6. ENG 102 Composition I7. HUM 103 Ethics and Culture8. MAT 102 Introduction to Mathematics

b. Departmental Core Courses: (57 credits)

1. APE 102 Thermal Physics, Radiation & Statistical Mechanics 2. APE 103 Electrical Circuits I 3. APE 201 Solid State Physics & Materials Science 4. APE 202 Electrodynamics & Electromagnetic Waves & Fields 5. APE 203 Electrical Circuits II 6. APE 204 Digital Logic Design7. APE 205 Electronic Devices and Circuits I 8. APE 302 Electronic Devices and Circuits II 9. APE 401 Measurements & Measuring Instruments 10. ECE 220 Signals and Systems11. ECE 230 Semiconductor Devices and Materials12. ECE 310 Introduction to Communication Engineering13. ECE 320 Microwave Engineering14. MAT 105 Calculus 15. MAT 203 Matrices, Linear Algebra & Differential Equations 16. MAT 204 Complex Variables & Fourier Analysis 17. PHY 115 Electricity and Magnetism 18. PHY 202 Optics 19. STA 201 Elements of Statistics and Probability

c. Elective Courses: (15 credits)

1. APE 402 Plasma Physics with Industrial Applications2. APE 403 Control Engineering 3. APE 404 Microprocessors and Assembly Language Programming4. APE 405 Computer Organization and Architecture 5. APE 406 Radar Engineering6. APE 407 The Physics of Energy7. CSE 350 Digital Electronics and Pulse Techniques8. CSE 421 Computer Networks9. CSE 428 Image Processing10. ECE 328 Digital Signal Processing11. ECE 330

Telecommunication Switching Systems12. ECE 340 Optoelectronic Devices13. ECE 424 Power Electronics14. ECE 430 Satellite Communications 15. MAT 205 Introduction to Numerical Methods 16. PHY 204 Classical Mechanics and Special Theory of Relativity 17. PHY 303 Quantum Mechanics I18. PHY 304 Quantum Mechanics II 19. PHY 305 Quantum Mechanics III 20. PHY 310 Advanced Solid State Physics21. PHY 311 X-Rays22. PHY 406 Medical Physics & Instrumentation23. PHY 409 Physics of Radiology 24. PHY 410 Laser Physics

d. Lab: (6 credits)

1. APE 104 APE Lab I2. APE 206 APE Lab II3. APE 301 APE Lab III4. APE 303 APE Lab IV

e. Thesis/Project and Internship: (4 credits)

1. APE 400 Thesis/Project: 3 credits2. Internship: 1 credit

f. Courses Outside Major Specialization: (27 credits)*

1. ANT 101 Introduction to Anthropology2. ARC 292 Painting**3. ARC 293 Music Appreciation**4. BI0 101 Introduction to Biology5. CHE 101 Introduction to Chemistry6. CSE 110 Programming Language I7. CSE 111 Programming Language II8. ECO 103 Principles of Economics9. ENV 101 Introduction to Environmental Science10. HUM 101 World Civilization and Culture 11. HUM 102 Introduction to Philosophy12. HUM 111 History of Science13. MGT 211 Principles of Management14. PHY 313 Physics for Development 15. POL 103 Introduction to Political Science16. POL 245 Women, Power & Politics17. PSY 101 Introduction to Psychology18. SOC 101 Introduction to Sociology19. SOC 401 Gender and Development

* Other courses may also be taken from the approved courses of BU with the permission of the student’s advisor and the Chair of the relevant Department.** 2 Credit Courses

The Institutional Quality Assurance Cell (IQAC) of BracU was established in July 2015 in order to ensure quality education at the tertiary level. IQAC was established as a project for a period of 3 years with financial assistance from HEQEP, UGC and the World Bank.

IQAC will become a permanent organ of the University after three years and its operations will be sustained and maintained under the revenue budget of the University.

To promote a quality enhancement culture within the University by ensuring that the academic entity adopts and implements Quality Assurance (QA) standards and benchmarked quality criteria.

The IQAC was set up following an administrative order of the University authority. It is organized in a manner which befits the size, existing structure and capacity of the University as delineated in the Act. The role of IQAC is to:

Thirteen (13) programs of Brac University were selected by Quality Assurance Unit (QAU) of UGC for Self-Assessment, financed by World Bank through HEQEP. The selected programs are as follows:

All the thirteen programs have completed their Self-Assessment (SA) exercises following SA manual prescribed by QAU of UGC.

The main exercises in a nutshell were:

(i) A team building workshop to create awareness on SA and Quality Assurance (QA) in higher education among the members of the program offering entity.

(ii) Preparation of Self-Assessment Report (SAR) mainly from the survey results of five (5) stakeholders. The stakeholders are: (a) Existing students (b) Academic staff (c) Non-academic staff (d) Alumni and (f) Employers.Each program organized at least one workshop to share survey results with the stakeholders and considering the feedback from the stakeholders finalized the SAR.

(iii) The program SAR's were validated and reviewed by External Peer Reviewer Teams (EPRT). Each EPRT consist of 3 members, one was an international QA expert, one local QA, and one local subject expert. Each team stayed at BRACU physically for 3-days and submitted exit report on the basis of their findings on the 3rd day evening. The honorable teams submitted their final External Peer Reviewer Report (EPRR) in stipulated time.

(iv) Benchmarking mainly on SAR and EPRR, each program prepared a draft Post Self-Assessment Improvement Plan (PSAIP). The program organized a workshop and shared the draft PSAIP with the stakeholders and finalized the PSAIP.

The Self-Assessment Committee (SAC) of each program with the help of respective faculty members/stakeholders completed the rigorous tasks of SA activities. The IQAC of BracU helped them to complete the SA activities.

Latest News and Events

Outcome-based education a rearranging of norms: speaker tells workshop

Two BracU staffs participated in a day-long workshop titled QS Masterclass on University Rankings & Ratings at BUET

A Two-Day Workshop on OBE Organized by ENH and IQAC

External peer review team of IQAC visited BRACU as a part of evaluation of self-assessment report of CSE

The Proctors’ Office is an impartial and discrete institution within the University. The Proctor and its administration are trusted to ensure the safeguarding of the University. In summary, the Proctor’s office oversees the following responsibilities;● Ensure all members of the University comply with existing University policies, statutes, and code of conduct through governance and administration,● Investigate violation and breach of University policies and code of conduct both academic and non-academic,● Ensure administration of the disciplinary management process,● Address all grievances and complaints,● Oversee disciplinary issues with regards to examinations and academic activities,● Serve on a number of University bodies and committees,● Oversee registration of student bodies and clubs. The Proctor has general responsibility for students and staff safety and may work with the law enforcement authorities and other partners to ensure that our students and employees are safe and feel safe.

We ensure all disciplinary actions are dealt with relevant and genuine evidence. Our area of duty includes both Academic and Non-academic matters. Upon obtaining, valid complaints will be processed through our designated departments for further investigation. We make sure every case is processed with integrity and in alignment with the University policies and guidelines.

Academic Conduct – Assures that each and every student is indemnified in an area free of academic dishonesty and plagiarism. Protect students from any sort of academic discrimination targeted towards them.Student Complaints (Academic) – To give appropriate advice and support to the students and faculties wishing to make complaints about cheating, plagiarism, and misdemeanor.

Student & Staff Conduct - Ensures that all students and staff can go about their activities and lives within the University environment without being subjected to inappropriate behavior by other students and members.Student & Staff Complaints (Non-Academic) – Safeguards the students and staff regarding bullying, sexual harassment, ragging, threat, bribing, and serious criminal offenses.

For any queries: proctor[at]bracu.ac[dot]bd

For Complaints: please use the complaint form only for lodging complaints.

Complaints Form

Staff

Rubana Ahmed, PhD

Farhan Haq

Md. Khalid Amirul Islam

Sabrina Jahan

Mehjabin Hossain

Farzana Faiza

Professor A F M Yusuf Haider was a selection grade professor of physics of the University of Dhaka (retired, in June 2016). His latest field of research was Laser-induced Breakdown Spectroscopy (LIBS), Laser Raman Spectroscopy (LRS) and nano-particle fabrication by Laser ablation technique. Professor Haider had been a teacher and a researcher (joined as lecturer in 1972) in the physics department of the University of Dhaka for 43.5 years. Professor Haider was the Vice-Chancellor for two months and Pro Vice-Chancellor for six and a half years, University of Dhaka. Professor Haider was the director of semiconductor Technology Research Centre for more than seven years. Professor Haider was also the Chairman of the physics department for three years, the Dean of the faculty of science and Provost of Surja Sen Hall (student dormitory, for more than six years).

Level: 5, Room: 5B15Kha-224 Merul BaddaDhaka 1212. Bangladesh

Book Chapters

Haider, A. F. M. Y. (2013). Gold nanoparticles: Fabrication by laser ablation technique and their optical and morphological studies. In A. Jarnagin & L. Halshauser (Eds.), Gold nanoparticles: Synthesis, optical properties and applications for cancer treatment (pp. 39–76). New York, USA: Nova Science Pub Inc. Retrieved from http://www.novapublishers.org/catalog/product_info.php?products_id=32220

Haider, A. F. M. Y., & Talukder, A. I. (2013). Fabrication of silver nanoparticles by laser ablation technique: Their characterization and uses. In I. Armentano & J. M. Kenny (Eds.), Silver nanoparticles: Synthesis, uses and health concerns (pp. 35–58). New York, USA: Nova Science Pub Inc. Retrieved from https://novapublishers.com/shop/silver-nanoparticles-synthesis-uses-and-...

Abedin, K. M., Rahman, S. M. M., & Haider, A. F. M. Y. (2012). Generating near-field fresnel diffraction patterns by iterative fresnel integrals method: Acomputer simulation approach. In Computer Simulations: Technology, Industrial Applications and Effects on Learning (pp. 55–97). New York, USA: Nova Science Publishers, Inc.

Haider, A. F. M. Y., & Abedin, K. M. (2011). Laser-induced breakdown spectroscopy and its applications in mineral and pollution analyses. In W. T. Arkin (Ed.), Advances in laser & optics research (Vol. 9). New York: Nova Science Publishers, Inc.

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Haider, A. F. M. Y. (1992). Intensifier and cathode-ray tube technologies. In M. A. Karim (Ed.), Electro-optical displays (pp. 1–18). New York, USA: Marcell Dekker Inc.

Journals

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Ali, M. M., Hadi, M. A., Rahman, M. L., Haque, F. H., Haider, A. F. M. Y., & Aftabuzzaman, M. (2020). DFT investigations into the physical properties of a MAB phase Cr4AlB4. Journal of Alloys and Compounds, 821, 153547.

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Wahadoszamen, M., Rahaman, A., Hoque, N. M. R., I Talukder, A., Abedin, K. M., & Haider, A. F. M. Y. (2015). Laser raman spectroscopy with different excitation sources and extension to surface enhanced raman spectroscopy. Journal of Spectroscopy, 2015, 895317. https://doi.org/10.1155/2015/895317

Haider, A. F. M. Y., Hedayet Ullah, M., Khan, Z. H., Kabir, F., & Abedin, K. M. (2014). Detection of trace amount of arsenic in groundwater by laser-induced breakdown spectroscopy and adsorption. Optics & Laser Technology, 56, 299–303. https://doi.org/10.1016/j.optlastec.2013.09.002

Haider, A. F. M. Y., Ira, M. K., Khan, Z. H., & Abedin, K. M. (2014). Radiative lifetime measurement of excited neutral nitrogen atoms by time resolved laser-induced breakdown spectroscopy. Journal of Analytical Atomic Spectrometry, 29(8), 1385–1392. https://doi.org/10.1039/C3JA50199J

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Abedin, K. M., Haider, A. F. M. Y., Rony, M. A., & Khan, Z. H. (2011). Identification of multiple rare earths and associated elements in raw monazite sands by laser-induced breakdown spectroscopy. Optics and Laser Technology, 43(1), 45–49. https://doi.org/10.1016/j.optlastec.2010.05.003

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Farhad, S. F. U., Abedin, K. M., Islam, M. R., Talukder, A. I., & Haider, A. F. M. Y. (2009). Determination of ratio of unsaturated to total fatty acids in edible oils by laser raman spectroscopy. Journal of Applied Sciences, 9(8), 1538–1543. https://doi.org/10.3923/jas.2009.1538.1543

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Haider, A. F. M. Y. (1998). A dual-frequency dye laser oscillator-amplifier system. Ind. Journal of Physics, 62B, 552–555.

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Das, K. K., Arif, M., Halder, A. F. M. Y., & Alam, M. S. (1994). Interferometric refractive index profile measurement of an optical fiber. Microwave and Optical Technology Letters, 7(18), 821–823. https://doi.org/10.1002/mop.4650071802

Pini, R., Salimbeni, R., Vannini, M., Haider, A. F. M. Y., & Lin, C. (1986). High conversion efficiency ultraviolet fiber Raman oscillator--amplifier system. Appl. Opt., 25(7), 1048–1050. https://doi.org/10.1364/AO.25.001048

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Ahmed, M. F., Abedin, K. M., Sadat, M. E., Talukder, A. I., Wahadoszamen, M., & Haider, A. (1970). Elemental profiling of surface water around Dhaka city by laser induced breakdown spectroscopy. Journal of Bangladesh Academy of Sciences, 33(2), 209–218. https://doi.org/10.3329/jbas.v33i2.4104

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Conferences

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Haider, A. F. M. Y. (2010, March 27-29). Temperature measurement of laser-induced plasma as a function of input laser pulse power at two different wavelengths. In International Conference on Recent Advances in Physics, RAP 2010. Dhaka, Bangladesh: University of Dhaka.

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Magazine / Newspaper Articles

Haider, A. F. M. Y. (2007, June). Podarthabiggane 2008 shaler nobel purushker bijoy: Bisyabromanda pratishammo vongger rohossho uddghaton. Dhaka Bisshobiddalaya Patrika, 88, 129–140.

Teaching:

Experience of Academic Administration:

a.Director, Semiconductor Technology Research Centre, University of Dhaka (7 Years: 1994-2001).

b. Provost, Surja Sen Hall (student dormitory), University of Dhaka, From Jan 1996 to Feb 2002 (Six years).

c. Dean, Faculty of Science, University of Dhaka, 2002.

d .Pro-Vice Chancellor , University of Dhaka, from 25 July 2002 to 23 January 2009 (Six and a half years).

e.Vice Chancellor (Acting), University of Dhaka, From 1 Aug 2002 to 23 Sept. 2002.

f. Treasurer (additional responsibility), University of Dhaka, from 15 Aug 2003 to 08 Sept. 2003.

g. Chairman of Physics, University of Dhaka, June 30, 2010 to June 29, 2013 (three years).

h. Coordinator, Nonlinear optics and laser spectroscopy, Centre for Advanced Research in Sciences, University of Dhaka, Jan. 2009 to Jan 2014.

i. Currently the chairperson of the Mathematics and Natural Sciences (MNS) department of BRAC University.

Academic / Research Infrastructure Development as Pro vice- chancellor of DU

a)Introduced Broadband Internet Network in D.U.b)Introduced Digital Library / Online Access to Journals for Dhaka University (DU).c)Optical Fibre Backbone and Digital PABX of the University.d) Established the centre for advanced Research in Sciences (CARS).

Taught in the Physics Department, University of Dhaka (DU), for more than 43 years (1st Dec 1972 to 30 th June 2016).

Current courses at MNS BRACU:Electricity ,magnetism, special theory of relativity, heat , radiation and thermodynamics, waves and oscillations and acoustics.Atomic and molecular physics.Courses taught:Courses taken at undergraduate level: Mechanics; Electricity & Magnetism; Thermodynamics; Waves & Optics; Electrical Circuits; Electronics; Atomic & Molecular Physics; Modern Physics; Quantum Mechanics; Solid State Physics; Nuclear Physics; Methods of Experimental Physics; Introductory Course on Optical Fibre; Lasers and Photonics.

Mechanics, Thermodynamics, Waves & Optics at Daffodil International University, DIU, as Adjunct professor.

Courses taken at postgraduate level: Laser Physics; Solid State Physics.

Courses taught at A level (year 11 and 12): Mechanics, electricity and magnetism, waves, optics, nuclear physics and modern physics. (1984-2002).

FELLOW: Bangladesh Physical Society (BPS)

Honours and Achievements:

Celebrated Personality of the World: Enlisted in the WHOs WHO in the world since 1999.

I was invited to submit nominations for the award of the Nobel Prize in Physics for the years 1994, 2013 and also for 2016 by the NOBEL COMMITTEE FOR PHYSICS, Royal Swedish Academy of Sciences.

Two of my published papers (please see my list of publications,) were amongst the Top 25 Hottest Articles as the most read Paper for a number of quarters.

Razzak-Shamsun research award for the year 2009.

Deans award for writing a book chapter in 2011.

(i) Optics and Laser Technology, Elsevier (U.K.)

(ii) IEEE/OSA Journal of Light wave Technology. (U.S.A.)

(iii) Optics and Lasers in Engineering, Elsevier (U.K.)

(iv) Analytical Methods, RSC Publishing (U.K.)

(v) Journal of Optical Society of America (JOSA), (U.S.A.)

(vi) IEEE Journal of Quantum Electronics (U.S.A.)

(vii) Energy & fuels, American Chemical Society (ACS) (U.S.A.)

(viii) Spectroscopy Letters, Taylor & Francis Publications (U.K., U.S.A.)

(ix) Analytical Methods, RSC Publications (U.K.)

Development of Research Laboratories and Research Experience:

To farther the research capability of the University of Dhaka, a modern research laboratory had been developed (2005) as the Centre for Advanced Research in Sciences, (Centre of Excellence) in the University of Dhaka, with financial support of the Government of the Peoples' Republic of Bangladesh and Japan Debit Cancellation Fund (JDCF) with my initiative and direct supervision when I was the Pro-Vice Chancellor of the university of Dhaka .Established the Modern Optics and Laser laboratory and introduced the courses on laser physics (in 1984) at undergraduate and post graduate levels in the physics department of the University of Dhaka.Research activities:The latest research interest:

Major areas of research involvement in the past:

Nuclear Physics: Inelastic scattering of neutrons by 165 Ho and 89Y nuclei by using the Dhaka 3 MeV Van de Graaff Accelerator, 512 multi-channel analyser and NaI (Tl) scintillation detector.

Solid-state spectroscopy (both theory and experiment): Electron paramagnetic resonance (EPR) spectroscopy,

optical (UV, ViS & IR) and Mossbauer spectroscopy of 3dn ions doped in SrO and MgTiO3 crystals and theoretical and experimental study of Off-Centre behaviour of transition metal ions in SrO. Energy levels of some transition metal ions in different charge states in different crystalline environments were also calculated theoretically.

Ionic thermo-current measurements: Mn Doped NaCl crystals.

Laser Physics: (a) Coherent light generation by lasing: design, fabrication and operation of a moderate power N2 laser,

(b)design, fabrication and operation of a dual frequency dye laser oscillator-amplifier system using a XeCl Excimer laser.

Study of Non-linear Optics: (a) Stimulated Raman Scattering (SRS), Four Photon Mixing (FPM), Stimulated Four Photon Mixing (SFPM) in optical fibres using Excimer lasers and Optical Multi-channel Analyser (OMA), (b) SRS in pressurized H2 gas, (c) Theoretical and Experimental Study of Laser Raman Fibre Oscillator-Amplifier system using Excimer lasers.

Study on Holography: Recording and reconstruction of Holograms (3-D images) using HeNe laser and Diode Lasers.

Study on Optical signal processing: Image processing and character identification using Fourier transform techniques and spatial filters.

Fibre Optics: Optical fibre characterization, refractive index profile measurement by interferometric method and calculation of optical fibre bandwidth and study of elementary optical communication.

Soliton propagation in optical fibres (Theoretical).

Study on Electronic Speckle Pattern Interferometry (ESPI): Measurement of In-Plane motions and micro Rotations and micro deformation of surfaces.

Research Supervision:

Research work of a number of M.S., M.Phil. and Ph.D.(one, thesis not yet submitted) students in different areas of

Physics such as:

Experimental solid state Physics: Transport properties.

Design and operation of high power pulsed nitrogen laser.

Coherent optical signal processing by spatial filtering.

Optical fibre characterization.

Soliton propagation in optical fibres and different dispersive media.

ESPI and its applications.

LIBS and its applications.

Laser Raman Spectroscopy (LRS) and its applications.

Fabrication of metal nanoparticles by Laser ablation technique and their characterization.

a) Played important part in introducing the health insurance for teachers and officers as the president of the Dhaka University Teachers' Association (DUTA) and as Pro-Vice Chancellor.

b) Active patronization of debating societies of the University of Dhaka as the chief advisor of DUDS (Dhaka University Debating Society) during the tenure of my office as the Pro-Vice Chancellor.

A F M Yusuf Haider, PhD

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

BRAC University students Nafira Nayeem Ahmad, Umme Kulsum Ananna, and Mushfiqur Rahman presented their innovative civic engagement projects at the 2024 Get Engaged: Student Action and Youth Leadership conference, held in Berlin from June 24 to July 1, 2024. This conference emphasizes the development of community leadership skills, and these students effectively represented their initiatives: Nafira Nayeem Ahmad with "Amplitude," Umme Kulsum Ananna with "Project Pother School," and Mushfiqur Rahman with "Eco Flow Revive.

They were accompanied by their Civic Engagement Coordinator, Ms. Fahmida Rahman, who is a Senior Lecturer and Program Coordinator at the School of General Education.

This year, Ms. Dina Hosssain, also a Senior Lecturer at the School of General Education, along with Partho Protim Chowdhury, a student from BRAC University, participated as part of the OSUN Film Core team for capturing footage and producing films featuring civic engagement student leaders at the OSUN Get Engaged conference in Berlin in June 2024.

The conference highlighted the project "Global Educator Fellowship Program (GEFP)," in which BRACU's Nafira Nayeem Ahmad won the Ideas Incubator round, receiving the highest number of votes for its impactful concept.

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School of General Education hosts “BRAC University and OSUN Film Festival 2024”

BRACU student receives "UNHCR Best Film" award

Dr. Samia Huq, receives invitation to join the Scientific Advisory Board (SAB) of the Paris Institute for Advanced Study

Ms. Fahmida Rahman Appointed Liberal Arts and Sciences Advisor in LAS Collaborative

Ms. Fahmida Rahman, Senior Lecturer and Program Coordinator from the School of General Education has been offered to be a Liberal Arts and Sciences Advisor as a part of the LAS Collaborative. LAS Collab is a partnership between seven higher educational institutions (Al Quds Bard College, American University of Afghanistan, American University of Central Asia, Ashesi University, Bard College, BRAC University, and Parami University).

As a part of her advisory role, she will serve as a guiding resource, offering her expertise to those keen on enhancing their LAS education delivery through curriculum, and/or pedagogy. Anyone interested in consulting, please make an appointment from here.

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Dr. Samia Huq, receives invitation to join the Scientific Advisory Board (SAB) of the Paris Institute for Advanced Study

Dr. Samia Huq, Dean and Professor of the School of General Education, received an invitation to join the Scientific Advisory Board (SAB) of the Paris Institute for Advanced Study. The SAB members guide the Director on the Institute's scientific policies and the selection of Fellows. They are elected by the Governing Board for a three-year term.

The Paris Institute for Advanced Study is an independent research organization that collaborates with 13 renowned universities and research institutions in the Paris region, including CNRS, ENS, EHESS, and Sorbonne Université.

The Institute offers a supportive environment for top international researchers, fostering collaboration and innovative research projects within the local research ecosystem. This ecosystem represents the largest community of scientists in Europe. The Institute prioritizes projects that tackle groundbreaking and interdisciplinary research that addresses significant scientific and societal questions.

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ENH lecturer attains Fulbright teaching assistantship

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BRAC University's Centre for Entrepreneurship Development Wins Global Impact Award at ICSB World Congress

Law Alumni Association of BRAC University Hosts Meeting at Supreme Court

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BRACU Alumni-led Startup Shines in Shark Tank Bangladesh

BRAC University and United Group entertainment venture - Courtside Sign Memorandum of Understanding

Awareness sessions on “Campus Safeguarding” at BRAC University’s Residential Campus conducted by the Office of the Proctor

BRAC Business School Hosts Closing Ceremony for ‘Uddami Ami’ Women Entrepreneurship Training Program

Pages

Dhaka, July 8, 2024 – BRAC University and Courtside have signed a Memorandum of Understanding (MoU) today at the United Group Head Office in Dhaka. This partnership aims to provide exclusive benefits and opportunities to the university's stakeholders.

The MoU establishes a framework for collaboration, offering BRAC University’s faculty, employees, and students special discounts and promotional packages on various recreational activities at Courtside’s facilities. Additionally, Courtside will offer engagement opportunities such as internships, career sessions, and campus ambassador programs for BRACU students.

Dr. David Dowland, Registrar of BRAC University, Tahsina Rahman Joint, Director, Student Life; Prantar Das Anoy, Assistant Manager, OCSAR and Anisur Rahman, General Manager, United Group, Md. Arif Hossain, Assistant General Manager & Wahid Sorwar, Assistant Manager signed the MoU on behalf of their respective organizations. Dr. Dowland expressed his enthusiasm, stating, "This collaboration will not only provide recreational benefits to our community but also open doors for our students to gain valuable industry experience."

The MoU is effective from July 8, 2024, to July 7, 2025, with the possibility of renewal based on mutual agreement. Both organizations look forward to a fruitful partnership that enhances the overall experience for BRAC University’s stakeholders while promoting Courtside’s services within the academic community.

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BRAC University law alumna selected as UN Early Career Climate Fellow

Two-day career fair ends at BRAC University

OCSAR organized On-campus Interview for Therap (BD) Ltd.

A team of the Department of Architecture at BRAC University achieved an “Honorable Mention” at an international design competition recently.

The team is comprised of Golam Ahammad Sunny, Teaching Assistant and an alumni, Monishita Mumtahina, an alumna, and Hasibul Islam Jahin and Ayman Tahmid, who are both in their fourth year.

361BIT, an education and community organization of the architecture, engineering and construction industry, organized the competition, titled “Presence”, on imagining a space to reflect and celebrate the journey of Indian architecture. Teams from 45 universities of over 17 countries took part in the contest.

In a joint message, the team said, “This accomplishment proves our dedication to innovative problem-solving and addressing significant cultural and social issues. We extend our gratitude to all team members for their diligent work, creativity, and perseverance throughout the competition.”

“Looking ahead, we remain committed to driving positive change and making meaningful contributions to communities worldwide,” they added.

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BRAC University architecture team recognized at international design competition

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Faculty of the Department of Architecture attended the ARCC Conference in Aarhus

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These actions contradict the positive goals of recent student movements aimed at addressing inequality and promoting a fair and just society. As a civilised society, we must come together to oppose and prevent these destructive behaviours proactively.

We urge everyone to stay vigilant and work together to protect and support each other. By uniting against these terrible acts, we can ensure that the hard-earned freedom leads to a peaceful, safe, and respectful environment for all.

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BRAC University Hosts Orientation for Summer 2024 Semester

On June 27, 2024, BRAC University Proctor's Office, in collaboration with the Savar Residential Campus authority, organized and conducted three sessions on "Campus Safeguarding" to raise awareness among students and staff, with over 1400 students, faculty, and staff members in attendance. The Office of the Proctor has been implementing robust and comprehensive safeguarding policies and processes as part of its broader Campus Safeguarding Awareness Campaign to ensure a safe environment for all members of the BRAC University community.

The awareness sessions were conducted by Advocate Farhan Haq, Deputy Proctor and Safeguarding Lead, BRAC University, and the team at the Office of the Proctor including Junior Proctors and Officers- Md. Khalid Amirul Islam, Advocate Sabrina Jahan, Mehjabin Hossain, Farzana Faiza and University Proctor Dr. Rubana Ahmed in collaboration with

Mir Sadik Faisal, Campus Superintendent, Savar Residential Campus along with other staff members of BRAC University. The Experiential Learning Coordinator of Savar Residential Campus, BRAC University, Ms. Mitali moderated the sessions and ensured necessary support during the sessions.

Throughout the sessions, the organizers attempted to strengthen commitments to raising awareness and promoting a safe and inclusive environment for all members of the BRAC University community. The sessions focused on introducing the concept of safeguarding and campus safeguarding, as well as discussing various academic and non-academic infractions, their consequences, the responsible use of social media, and information regarding the channels for raising any concern. The discussions covered BRAC University's disciplinary policies, procedures, and support protocols, emphasizing that personal and institutional safeguarding is a shared responsibility.

First of its kind in Bangladesh, BRAC University’s Office of the Proctor has developed and is currently implementing an internal "Campus Safeguarding Framework'' with comprehensive safeguarding protocols in place. A dedicated team, comprising of Junior Proctors and officers, is actively working to enforce policies and procedures in collaboration with and support from different departments, schools, and university operations team members, to create an improved learning environment for all members of BRACU community. The Goal is to foster a safe, inclusive, and supportive atmosphere for both students and staff. The primary objective of the Proctor's Office is to design and implement efficient campus security processes and protocols, allowing students and staff to voice their concerns and obtain the necessary resources to ensure their safety.

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Faculty member of Department of Architecture at BRAC University has garnered significant accolades in the prestigious National Architectural Design Competition organized by the Institute of Architects Bangladesh (IAB) in collaboration with Northern Electricity Supply Company (NESCO). Mahia Mustary Nushin, a lecturer and her team secured the first prize, while Tanveer Ahamed Bin Ali Naser, also a lecturer in the Department of Architecture and his team awarded an honorable mention for their outstanding work.

From the left Tanveer Ahamed Bin Ali Naser, Mahia Mustary Nushin

The competition, aimed at designing the Main Control Center building for NESCO in Rajshahi, saw participation from prominent architects and architectural firms across Bangladesh. The innovative proposals demonstrated the high level of creativity and expertise present in the country's architectural community.

On May 16, 2024, a total of five teams were recognized for their outstanding design innovations at an announcement event. The Honorable State Minister for the Ministry of Power, Energy & Mineral Resources, Mr. Nasrul Hamid, graced the occasion as the chief guest, highlighting the importance of architectural excellence in national development.

Mahia Mustary, along with her team proposed a Green building for the electricity supply company NESCO in a campus concept where all the existing and new buildings will create an urban space for the people of Rajshahi. Other team members of this project are Mohammad Naimul Ahsan Khan, Farzana Rahman, Aqib Mohammad Nibir, Mahjabin Noor, Md Rafiul Islam, Shohanur Rahman and others.

Proposal from Mahia Mustary and her team

Tanveer Ahamed and his team proposed a Sustainable Urban Icon which would not be a replica of a generic style of modernism but rather a regional representation of local practice, climate, history, culture & identity. At the core of the Harmonic Square's design philosophy lies the elegant simplicity of the square form. Other team members of this project are Joy Dutta, Tansen Alam, Rafia Rukhsat, Shampad Khalifa, Sharmin Jahan, Rubayet Ridoy and Riju Sarker.

Proposal from Tanveer Ahamed and his team

This recognition underscores the exceptional talent and dedication of BRAC University's faculty and their commitment to advancing architectural design in Bangladesh. The Department of Architecture, BRAC University, congratulated the winners on this commendable achievement.

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BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

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Dr. Sharmind Neelotpol of BRAC University Hosts ACU Gender Programme Symposium

Pharmacy alumnus gets fund support for PhD at The University of Texas at Arlington

School of Pharmacy Alumni Meet: First Ever Reunion at the New Campus

BRACU Research for Development Club Immerses in Pharmaceutical Innovation at Incepta’s Dhamrai Plant

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BRAC University Hosts Orientation for Summer 2024 Semester

BRAC University's admission tests, both undergraduate and postgraduate, scheduled to be held on 26 July and 2 August respectively have been postponed.

New test dates and application deadlines will be announced later.

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Excellence at Akij Food and Beverages Ltd

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BRAC University architecture students have excelled at the “AIUB Architecture Design Charrette” securing both the second and third prizes.

The team of Julkarnain, Nafisa Anzum, and Farhan Shahriar won the second prize, while Abdullah Al Mamun Pranto, Biproy Kanti Das, and Rubaiya Ruhi Saba secured the third prize. Their innovative designs and creative solutions impressed the judges, highlighting the high standards of BRAC University’s architecture program.

Congratulations to the winning teams for their outstanding achievements!

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Workshop on Architecture Model Making Held at BRAC University

BRAC University architecture team recognized at international design competition

Professor Fuad Hassan Mallick and Professor Zainab Faruqui Ali Speak at Bengal Institute’s Lecture Series on Exploring Muzharul Islam

Faculty of the Department of Architecture attended the ARCC Conference in Aarhus

Department of Architecture of BRAC University hosted a hands-on workshop on Architecture Model Making for the summer 2024 students of Department of Architecture. The workshop held at the university’s Exhibition Space, was part of the Department of Architecture's ongoing effort to enhance skills among its students and foster a collaborative learning environment.

The daylong workshop was led by Tanveer Ahamed Bin Ali Naser, Lecturer in the Department of Architecture. Fardin Rahman, lecturer, Md. Rakibul Hasan, Teaching assistant and Mumtahinah Momo, an alumna of the department, also assisted in the workshop.

The team provided participants with valuable insights into the intricacies of architectural model making, from conceptualization to execution. Participants were introduced to various techniques and materials used in model making, Hands-on sessions allowed students to apply techniques in real-time.

Prof. Zainab Faruqui the Head of the Department of Architecture, highlighted the importance of such workshops in the academic curriculum said ‘‘This is an integral part of architectural education aiding students to excel in model making and it needs to be continued’’

The discussion helped peer-learning. Students expressed their enthusiasm and appreciation for the workshop. In total 73 students from different design studios participated on the workshop.

Model Exhibition

The event concluded with an exhibition of the models made by the participants, showcasing their skills they had acquired during the workshop. Department of Architecture plans to continue offering similar workshops in the future, aiming to provide its students with a well-rounded education.

Workshop Poster

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The Bengal Institute continues its enlightening lecture series, "Exploring Muzharul Islam," with a lecture by Professor Fuad Hassan Mallick and Professor Zainab Faruqui Ali titled "Sun, Wind, Rain: The Architecture of Muzharul Islam." The lecture delved into the environmental and climatic considerations in the works of the legendary architect Muzharul Islam, the pioneer of modern architecture of the country.

Fuad Hassan Mallick, Dean of the School of Architecture and Design at BRAC University, and Zainab Faruqui Ali, Chair of the Department of Architecture at BRAC University, bring a wealth of experience to the discussion. Both speakers have over 30 years of academic and professional expertise, having extensively published on the environmental performance of buildings, low-income settlements, and the impacts of climate change.

Their collaborative work the book titled "Muzharul Islam Architect," provides a comprehensive look at Islam's architectural philosophy, which harmoniously integrates environmental elements. The lecture highlighted how Muzharul Islam's designs ingeniously utilize natural resources like sunlight, wind, and rain to create sustainable and aesthetically pleasing buildings.

Fuad Hassan Mallick emphasized Islam's foresight in addressing climate responsiveness long before it became a global architectural priority with tropics. Zainab Faruqui Ali discussed specific examples of Islam's works, comparing them to the approaches of renowned architects like Louis Kahn and Le Corbusier in tropical climates.

The lecture, part of the "Exploring Muzharul Islam" series continues to offer valuable insights into the architectural genius of Muzharul Islam, fostering a deeper appreciation of his contributions to sustainable and environmentally conscious design. They showed with technical drawings including climatic data, solar angle calculations, etc. how the climatic modifying elements of Islam’s designs addressed the problems of the specific climate.

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Dubotech Digital Limited, a startup led by BRAC University alumni, achieved significant success by participating in "Shark Tank Bangladesh," the local edition of the prestigious business reality show. They secured an investment of BDT 5 million in their pre-seed round, highlighting their innovation and potential. The investment was in exchange for 10% equity and was provided by Sami Ahmed, Managing Director and CEO of Startup Bangladesh Limited, and Ahmed R Ali, founder and chairman of TISTA Science and Technology Corporation. This milestone sets a strong foundation for Dubotech's future growth, and they are working to raise additional funding while collaborating with prominent industry names.

Five years ago, the team began developing underwater robots called "BRACU DUBURI." They initiated research and development to create an autonomous underwater robot capable of solving underwater problems wirelessly using artificial intelligence. The team has won several awards both nationally and internationally.

Dubotech is now a deep-tech startup, serving the most technologies that they have for solving local problems. The company has built its own engineering team, showcasing engineering excellence by creating world-class robots in Bangladesh. It has received consistent support from BRAC University. For the research and development of BRACU DUBURI, BRAC University provided funding and support under the supervision of Dr. Md. Khalilur Rahman, a professor at the university.

BRAC University's Centre for Entrepreneurship Development (CED) supported Dubotech from its inception. Through the university's Business Incubation Center (BIC), established and managed by CED, Dubotech received comprehensive support, including strategic guidance, business incubation, mentorship, networking opportunities, co-working space, acceleration, and utilities

“Innovative startups like Dubotech will undoubtedly inspire our students and alumni to launch even more groundbreaking ventures in the future." as mentioned by CED’s Joint Director Afshana Choudhury.

Dubotech has a dream that one day people in Bangladesh will experience the beauty of the Bay of Bengal with their service, "Dubo Ride." It's a virtual way to explore underwater beauty and experience things like diving underwater with VR technology.

Dubotech CEO Nayem Hossain Saikat said, "We are discovering underwater possibilities to ensure Bangladesh gets the highest benefits from our technology, and we are also capable of serving the global market as our research is already recognized around the world. We are thankful to BRAC University, the ICT division, CED, and government organizations who are supporting us to make the best products from Bangladesh.".

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BRAC University Hosts First Coding Competition for Non-CS Students

BUCC holds webinar on “Pathway to Research and Industry in the USA”

BRAC University law alumna selected as UN Early Career Climate Fellow

Department of Architecture of BRAC University hosted a hands-on workshop on Architecture Model Making for the summer 2024 students of Department of Architecture. The workshop held at the university’s Exhibition Space, was part of the Department of Architecture's ongoing effort to enhance skills among its students and foster a collaborative learning environment.

The daylong workshop was led by Tanveer Ahamed Bin Ali Naser, Lecturer in the Department of Architecture. Fardin Rahman, lecturer, Md. Rakibul Hasan, Teaching assistant and Mumtahinah Momo, an alumna of the department, also assisted in the workshop.

The team provided participants with valuable insights into the intricacies of architectural model making, from conceptualization to execution. Participants were introduced to various techniques and materials used in model making, Hands-on sessions allowed students to apply techniques in real-time.

Prof. Zainab Faruqui the Head of the Department of Architecture, highlighted the importance of such workshops in the academic curriculum said ‘‘This is an integral part of architectural education aiding students to excel in model making and it needs to be continued’’

The discussion helped peer-learning. Students expressed their enthusiasm and appreciation for the workshop. In total 73 students from different design studios participated on the workshop.

Model Exhibition

The event concluded with an exhibition of the models made by the participants, showcasing their skills they had acquired during the workshop. Department of Architecture plans to continue offering similar workshops in the future, aiming to provide its students with a well-rounded education.

Workshop Poster

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On Thursday, 4 July 2024, the Executive Committee (2024-2025) of the Law Alumni Association of BRAC University (LAABRACU) convened a meeting at the Supreme Court cafeteria. This event facilitated introductions between the new committee members and the president and general secretary of the Supreme Court Bar Association.

The meeting discussed plans for upcoming events, including the 20-year celebration of the School of Law, Alumni Day, and other activities. Furthermore, it was discussed that the Alumni Relations office of the BRAC University would assist in promoting the association across various sectors. In line with this, a preliminary decision was made to hold the next General Body Meeting (GBM) at the BRAC University campus.

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School Of Law Organizes Workshop On ‘Moot Court Competition and Its Fundamentals’

Uddami Ami, BRAC Business School’s flagship women entrepreneurship training program concluded Cohort 6 on July 6, 2024 at BRAC University New Campus, Merul Badda. The Closing Ceremony celebrated completion of the training by 20 vibrant and diverse women entrepreneurs across varying sectors of Bangladesh. Uddami Ami has successfully trained nearly 120 women entrepreneurs in Bangladesh since its inception in September 2021.

The event was graced by Mohammad Ashiqur Rahman, Director, SME & Special Program Department, Bangladesh Bank; Md. Rezaul Karim, Company Secretary, HATIL; Dr. David Dowland, Registrar, BRAC University; and Mohammad Mujibul Haque, PhD, Professor and Associate Dean, Acting Dean, BRAC Business School, BRAC University.

Bangladesh Bank Director, Mohammad Ashiqur Rahman, congratulated the participants on their courage to pursue entrepreneurship. Mr. Rahman reminded the participants of the government support available to women entrepreneurs through 35,000 crore taka allocation by Bangladesh Bank. He encouraged the participants to expand their business models and products to avail these funds.

Md. Rezaul Karim, Company Secretary, HATIL was delighted to host the women entrepreneurs at HATIL Factory for the Industry Visit session of Uddami Ami. Mr. Rezaul earnestly requested BRAC Business School and BRAC University to undertake developing local business cases so that students and entrepreneurs can learn the local context of dynamics of business in Bangladesh.

Dr. David Dowland continued the praise on the courageous entrepreneurs and thanked BRAC Business School, BRAC Bank PLC, Bangladesh Bank and Hatil Furnitures on their collaborative effort to make the training program a success.

Mohammad Mujibul Haque, PhD, Professor and Associate Dean (Acting Dean), BRAC Business School, BRAC University emphasized that BRAC Business School is actively engaged in upskilling entrepreneurs and professionals in Bangladesh. This includes developing local business cases through the 1st International Case Conference on Business and Management (ICCBM 2023), and conducting research and training programs that offer collaborative opportunities for all stakeholders at the Uddami Ami Closing Ceremony.

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BRAC University students get Inspiring Women Award

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BRAC University is proud to announce that its Centre for Entrepreneurship Development (CED) has been honored with the Entrepreneurship Center Global Impact Award at the 2024 International Council for Small Business (ICSB) World Congress held in Germany.

The ICSB, founded in 1955, is the oldest and largest nonprofit organization dedicated to advancing small businesses worldwide, with representation in over 85 countries. Each June, ICSB's World Congress brings together global educators, researchers, policymakers, and practitioners to share knowledge through various programs and certifications. Designed in collaboration with the Association to Advance Collegiate Schools of Business (AACSB), the ICSB Entrepreneurship Center Global Impact Award recipients must demonstrate their merit with respect to the following criteria: Innovativeness regarding programs and promotion, Impact on target population, Impact on society and Replicability across industries and borders.

CED at BRAC University, since its inception in April 2011, has been unwavering in its mission to nurture Bangladeshi entrepreneurs by equipping them with essential entrepreneurial skills and enabling them to strategically develop and expand their businesses. CED was established under the visionary leadership of Sir Fazle Hasan Abed and has since emerged as a beacon of sustainable innovation, guiding entrepreneurs from ideation to tangible impact.

With a proven track record, the CED has empowered over 2,000 aspiring entrepreneurs, nurtured more than 30 innovative ventures, and provided critical incubation support to over 15 startups. These accomplishments underscore CED’s pivotal role as a catalyst for change in Bangladesh’s entrepreneurial ecosystem.

Professor Syed Mahfuzul Aziz, Pro Vice-Chancellor and Acting Vice-Chancellor of BRAC University stated, "This prestigious award highlights our CED team's dedication. Their commitment to support innovation and entrepreneurship has transformed lives and contributed to boosting Bangladesh's economy. We are proud of this achievement and will continue to support our entrepreneurs in building sustainable and impactful businesses."

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Faculty member of Department of Architecture at BRAC University has garnered significant accolades in the prestigious National Architectural Design Competition organized by the Institute of Architects Bangladesh (IAB) in collaboration with Northern Electricity Supply Company (NESCO). Mahia Mustary Nushin, a lecturer and her team secured the first prize, while Tanveer Ahamed Bin Ali Naser, also a lecturer in the Department of Architecture and his team awarded an honorable mention for their outstanding work.

From the left Tanveer Ahamed Bin Ali Naser, Mahia Mustary Nushin

The competition, aimed at designing the Main Control Center building for NESCO in Rajshahi, saw participation from prominent architects and architectural firms across Bangladesh. The innovative proposals demonstrated the high level of creativity and expertise present in the country's architectural community.

On May 16, 2024, a total of five teams were recognized for their outstanding design innovations at an announcement event. The Honorable State Minister for the Ministry of Power, Energy & Mineral Resources, Mr. Nasrul Hamid, graced the occasion as the chief guest, highlighting the importance of architectural excellence in national development.

Mahia Mustary, along with her team proposed a Green building for the electricity supply company NESCO in a campus concept where all the existing and new buildings will create an urban space for the people of Rajshahi. Other team members of this project are Mohammad Naimul Ahsan Khan, Farzana Rahman, Aqib Mohammad Nibir, Mahjabin Noor, Md Rafiul Islam, Shohanur Rahman and others.

Proposal from Mahia Mustary and her team

Tanveer Ahamed and his team proposed a Sustainable Urban Icon which would not be a replica of a generic style of modernism but rather a regional representation of local practice, climate, history, culture & identity. At the core of the Harmonic Square's design philosophy lies the elegant simplicity of the square form. Other team members of this project are Joy Dutta, Tansen Alam, Rafia Rukhsat, Shampad Khalifa, Sharmin Jahan, Rubayet Ridoy and Riju Sarker.

Proposal from Tanveer Ahamed and his team

This recognition underscores the exceptional talent and dedication of BRAC University's faculty and their commitment to advancing architectural design in Bangladesh. The Department of Architecture, BRAC University, congratulated the winners on this commendable achievement.

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BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

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Mohammad Fasiul Abedin Khan, a student in his 8th semester at the Department of Electrical and Electronic Engineering at BRAC University, has recently been selected, alongside nine others, to take part in a Huawei Seeds for the Future 2024 program in China this September.

Launched in 2008, Seeds for the Future is Huawei's flagship CSR program which selects young talents from around the world to participate in short-term trainings, global competitions and ongoing alumni activities centered around digital technology.

The overall aim is to provide them with an important platform to understand industry innovation, experience cross-cultural exchanges, and cultivate an entrepreneurial spirit.

The Bangladesh rounds were launched in March this year. It was designed to help students learn about the latest trends in digitalisation and explore how digital technologies can address common social issues.

Following interviews and CV evaluations, 50 university students from Bangladesh were chosen to participate in bootcamps and training sessions. They were divided into six teams to generate ideas on using technology to make the world a better place.

Khan’s team secured second place with an idea on using machine learning, which is a subset of artificial intelligence, to solve problems faced by fish farmers.

Afterwards, 24 students moved to the next round, when they were divided into three teams to again generate ideas. Khan’s team took the top spot with an idea on using machine learning and the internet of things to monitor water quality and conduct filtering and data analysis.

Finally, the 10 members of Khan’s team, named Lal Sobuj, were selected to move on to the regional round in China.

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EEE student selected for Huawei program in China

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EEE students win Sheikh Jamal Innovation Grant 2024

BRAC University student becomes “Ethics Champion”

EEE lecturer honoured with the prestigious 2023 IAF Emerging Space Leader Award

Samirah Tabassum, a lecturer of the Department of English and Humanities at BRAC University, has been chosen to participate in a Fulbright Foreign Language Teaching

Assistant (FLTA) Program sponsored by the United States Department of State Bureau of Educational and Cultural Affairs.

The program is designed to develop Americans' knowledge of foreign cultures and languages by supporting teaching assistantships in more than 35 languages at hundreds of U.S. institutions of higher education.

The program offers educators from over 55 countries the opportunity to develop their professional skills and gain first-hand knowledge of the U.S., its culture and its people.

Tabassum will start teaching at the Department of Asian Studies of Cornell University from Fall 2024 under a Fulbright FLTA grant for one academic year (approximately 9-10 months).

She will also be representing Bangladesh as a cultural ambassador in the United States.

Tabassum is the only person from Bangladesh chosen to participate in the program for the 2024-2025 academic year.

Her accomplishment continues the legacy of impactful Fulbright alumni from Bangladesh, including Nobel laureate Dr. Muhammad Yunus.

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BRAC University architecture students have excelled at the “AIUB Architecture Design Charrette” securing both the second and third prizes.

The team of Julkarnain, Nafisa Anzum, and Farhan Shahriar won the second prize, while Abdullah Al Mamun Pranto, Biproy Kanti Das, and Rubaiya Ruhi Saba secured the third prize. Their innovative designs and creative solutions impressed the judges, highlighting the high standards of BRAC University’s architecture program.

Congratulations to the winning teams for their outstanding achievements!

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BRAC University students Nafira Nayeem Ahmad, Umme Kulsum Ananna, and Mushfiqur Rahman presented their innovative civic engagement projects at the 2024 Get Engaged: Student Action and Youth Leadership conference, held in Berlin from June 24 to July 1, 2024. This conference emphasizes the development of community leadership skills, and these students effectively represented their initiatives: Nafira Nayeem Ahmad with "Amplitude," Umme Kulsum Ananna with "Project Pother School," and Mushfiqur Rahman with "Eco Flow Revive.

They were accompanied by their Civic Engagement Coordinator, Ms. Fahmida Rahman, who is a Senior Lecturer and Program Coordinator at the School of General Education.

This year, Ms. Dina Hosssain, also a Senior Lecturer at the School of General Education, along with Partho Protim Chowdhury, a student from BRAC University, participated as part of the OSUN Film Core team for capturing footage and producing films featuring civic engagement student leaders at the OSUN Get Engaged conference in Berlin in June 2024.

The conference highlighted the project "Global Educator Fellowship Program (GEFP)," in which BRACU's Nafira Nayeem Ahmad won the Ideas Incubator round, receiving the highest number of votes for its impactful concept.

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BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

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School of General Education hosts “BRAC University and OSUN Film Festival 2024”

BRACU student receives "UNHCR Best Film" award

Dr. Samia Huq, receives invitation to join the Scientific Advisory Board (SAB) of the Paris Institute for Advanced Study

Ms. Fahmida Rahman Appointed Liberal Arts and Sciences Advisor in LAS Collaborative

The Bengal Institute continues its enlightening lecture series, "Exploring Muzharul Islam," with a lecture by Professor Fuad Hassan Mallick and Professor Zainab Faruqui Ali titled "Sun, Wind, Rain: The Architecture of Muzharul Islam." The lecture delved into the environmental and climatic considerations in the works of the legendary architect Muzharul Islam, the pioneer of modern architecture of the country.

Fuad Hassan Mallick, Dean of the School of Architecture and Design at BRAC University, and Zainab Faruqui Ali, Chair of the Department of Architecture at BRAC University, bring a wealth of experience to the discussion. Both speakers have over 30 years of academic and professional expertise, having extensively published on the environmental performance of buildings, low-income settlements, and the impacts of climate change.

Their collaborative work the book titled "Muzharul Islam Architect," provides a comprehensive look at Islam's architectural philosophy, which harmoniously integrates environmental elements. The lecture highlighted how Muzharul Islam's designs ingeniously utilize natural resources like sunlight, wind, and rain to create sustainable and aesthetically pleasing buildings.

Fuad Hassan Mallick emphasized Islam's foresight in addressing climate responsiveness long before it became a global architectural priority with tropics. Zainab Faruqui Ali discussed specific examples of Islam's works, comparing them to the approaches of renowned architects like Louis Kahn and Le Corbusier in tropical climates.

The lecture, part of the "Exploring Muzharul Islam" series continues to offer valuable insights into the architectural genius of Muzharul Islam, fostering a deeper appreciation of his contributions to sustainable and environmentally conscious design. They showed with technical drawings including climatic data, solar angle calculations, etc. how the climatic modifying elements of Islam’s designs addressed the problems of the specific climate.

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A team of the Department of Architecture at BRAC University achieved an “Honorable Mention” at an international design competition recently.

The team is comprised of Golam Ahammad Sunny, Teaching Assistant and an alumni, Monishita Mumtahina, an alumna, and Hasibul Islam Jahin and Ayman Tahmid, who are both in their fourth year.

361BIT, an education and community organization of the architecture, engineering and construction industry, organized the competition, titled “Presence”, on imagining a space to reflect and celebrate the journey of Indian architecture. Teams from 45 universities of over 17 countries took part in the contest.

In a joint message, the team said, “This accomplishment proves our dedication to innovative problem-solving and addressing significant cultural and social issues. We extend our gratitude to all team members for their diligent work, creativity, and perseverance throughout the competition.”

“Looking ahead, we remain committed to driving positive change and making meaningful contributions to communities worldwide,” they added.

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Dubotech Digital Limited, a startup led by BRAC University alumni, achieved significant success by participating in "Shark Tank Bangladesh," the local edition of the prestigious business reality show. They secured an investment of BDT 5 million in their pre-seed round, highlighting their innovation and potential. The investment was in exchange for 10% equity and was provided by Sami Ahmed, Managing Director and CEO of Startup Bangladesh Limited, and Ahmed R Ali, founder and chairman of TISTA Science and Technology Corporation. This milestone sets a strong foundation for Dubotech's future growth, and they are working to raise additional funding while collaborating with prominent industry names.

Five years ago, the team began developing underwater robots called "BRACU DUBURI." They initiated research and development to create an autonomous underwater robot capable of solving underwater problems wirelessly using artificial intelligence. The team has won several awards both nationally and internationally.

Dubotech is now a deep-tech startup, serving the most technologies that they have for solving local problems. The company has built its own engineering team, showcasing engineering excellence by creating world-class robots in Bangladesh. It has received consistent support from BRAC University. For the research and development of BRACU DUBURI, BRAC University provided funding and support under the supervision of Dr. Md. Khalilur Rahman, a professor at the university.

BRAC University's Centre for Entrepreneurship Development (CED) supported Dubotech from its inception. Through the university's Business Incubation Center (BIC), established and managed by CED, Dubotech received comprehensive support, including strategic guidance, business incubation, mentorship, networking opportunities, co-working space, acceleration, and utilities

“Innovative startups like Dubotech will undoubtedly inspire our students and alumni to launch even more groundbreaking ventures in the future." as mentioned by CED's Joint Director Afshana Choudhury.

Dubotech has a dream that one day people in Bangladesh will experience the beauty of the Bay of Bengal with their service, "Dubo Ride." It's a virtual way to explore underwater beauty and experience things like diving underwater with VR technology.

Dubotech CEO Nayem Hossain Saikat said, "We are discovering underwater possibilities to ensure Bangladesh gets the highest benefits from our technology, and we are also capable of serving the global market as our research is already recognized around the world. We are thankful to BRAC University, the ICT division, CED, and government organizations who are supporting us to make the best products from Bangladesh.".

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BRACU Alumni-led Startup Shines in Shark Tank Bangladesh

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BRAC University and United Group entertainment venture - Courtside Sign Memorandum of Understanding

BRAC University Hosts First Coding Competition for Non-CS Students

BUCC holds webinar on “Pathway to Research and Industry in the USA”

BRAC University law alumna selected as UN Early Career Climate Fellow

BRAC University's admission tests, both undergraduate and postgraduate, scheduled to be held on 26 July and 2 August respectively have been postponed.

New test dates and application deadlines will be announced later.

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Announcement regarding admission test

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Urgent Action Needed to Protect Our Community

BRAC University conducts Fire Drill at its campus

BRAC University Research for Development Club Explores Marketing and Manufacturing Excellence at Akij Food and Beverages Ltd

BRAC University Hosts Orientation for Summer 2024 Semester

The School of General Education (GenEd) of BRAC University organized a “BRAC University and OSUN Film Festival 2024” on the Merul Badda campus recently showcasing documentaries created by students.

The documentaries were produced as a part of a “Documentary Film: Theory and Practice" course, highlighting the students’ creativity, commitment to civic engagement and the power of storytelling in addressing social issues.

The course was taught by award-winning filmmaker and GenEd Senior Lecturer Dina Hossain and Assistant Professor Dr. Ratan Kumar Roy.

It is also run as a “Visual Storytelling for Civic Engagement” course under Open Society University Network (OSUN), a platform for universities, institutes and think tanks worldwide to collaborate on teaching, research and other forms of exchanges.

The festival has also been held at several OSUN-member universities, including BARD College, Al-Quds University, American University of Central Asia, and European Humanities University, to promote cross-cultural understanding.

The course imparts practical lessons on making documentaries, including through the use of smartphones and in groups, focusing on ethics and truthfulness and tackling important social issues.

The documentaries are intended to educate and motivate people towards bringing about a positive change in society.

The festival was inaugurated by Khairul Basher, Director of Communications. Notable guests included Dr. Dave Dowland, Registrar, Samia Huq, GenEd Dean, Ariful Islam, Chief Financial Officer, and Montaha Chowdhury, an alumna of bioinformatics from the Asian University for Women.

Of the documentaries screened at the festival, three had earlier won OSUN awards.

Of them, “Fading Tides, Rising Hope - The Fight to Save Dhaka’s Water Bodies” by MD Tahjid Ahsan, Nazmul Hasan, Fatin Nur Ishraq and Omar Ibtesam Nibir won an OSUN Civic Engagement Award.

“Road Safety Movement 2018: Are We Safe Yet?” by Prodipta Hasin, Anindita Hossain Rhine, Imranul Ahsan Moonim and Md Imtiaz Sarker Shihab won an OSUN Best Script Award.

“Food Waste in Bangladesh” by Sakib Sadi, Saklain Nizam Thakur, Muhtasim Billah Nahin and Tanzila Parveen received an OSUN Environmental Stewardship Award.

The other documentaries included "I am NOT Alone: Women and STEM in Bangladesh", "Day Care Centers: Transforming the Lives of Working Women in Bangladesh", "Beyond Boundaries - Shaping Tomorrow's Bangladesh", and "The Journey of Hope and Recognition - Geneva Camp Mohammadpur".

Moreover, two computer science and engineering students, Nowrin Islam Rio and Tanvir Islam Sayem, were presented awards for outstanding “research” and “effort” respectively.

During a subsequent Q&A session, students discussed challenges in filmmaking such as negotiating politically sensitive issues, finding archival footage and convincing people to appear on camera.

News Archive >>

School of General Education hosts “BRAC University and OSUN Film Festival 2024”

Related News

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

BRACU student receives "UNHCR Best Film" award

Dr. Samia Huq, receives invitation to join the Scientific Advisory Board (SAB) of the Paris Institute for Advanced Study

Ms. Fahmida Rahman Appointed Liberal Arts and Sciences Advisor in LAS Collaborative

Nowrin Islam Rio, a student at BRAC University, recently achieved the "UNHCR Best Film" award at the 10th Dhaka International Mobile Film Festival (DIMFF). She successfully completed the CST 304 course on Documentary Filmmaking: Theory and Practice, offered by the School of General Education. Her acclaimed film, 'The Unbreakable Wall,' competed in the 'Forced to Flee' category. The prestigious award was presented during the closing ceremony of DIMFF 2024, which took place on March 3 at the Star Cineplex in Bashundhara City Shopping Complex

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Since its inception in 2001, Brac University has become one of the most reputed educational institutions in Bangladesh. We have focused on generating new knowledge and promoting critical thinking amongst our students, graduating more than 7,000 young men and women during this time. At present, the total number of students is around 8,000.

Our goal at Brac University is not only to provide the highest quality teaching in the region, but also to inculcate the values essential for tomorrow's leaders. Through a liberal education, we endeavour to produce graduates with the attributes needed to achieve personal success and make a valuable contribution to society. Our talented and dedicated faculty play a critical role in both these processes.

I am pleased to report that the construction of the new University campus, comprising a 14-storey building on 5 acres of land, is well under way. We hope to relocate by mid-2020

Sir Fazle Hasan Abed, KCMGChairperson Emeritus, Board of TrusteesBrac University

This circular is applicable only for the students of the School of Pharmacy, BRAC University.

BRAC University understands the challenges that some students may face in managing their finances for the Summer 2024 semester due to the unforeseen situation in our country . In an effort to support our students during these times, we are pleased to extend the deadline to apply for installment payment for the School of Pharmacy, BRAC University. This initiative aims to provide convenient alternatives for those experiencing financial difficulties, ensuring that everyone has the opportunity to pursue their education without unnecessary stress.

In addition, BRAC Bank and ONE Bank have agreed to provide student loan facilities for BRAC University students. Students who want to avail student loan, can contact BRAC Bank or ONE Bank directly.

This circular for installment payment options is only applicable for the students of the School of Pharmacy, BRAC University, Summer 2024 semester.

Those who have already applied for Installment Payment for Summer 2024 semester are requested not to apply again.

Payment in Installments:

To apply for Payment in Installments, students need to complete the Application Form (link given below)

Application Form:

https://forms.gle/CaBnB9npZPgzZ8LJA

Submit the filled-up form with valid supporting documents (Pls. see in the form) - On or before 29 August 2024 (Thursday) by 11:59 pm (for both new and continuing students).

Students applying for installment payments will be required to pay the tuition fees of the Summer 2024 semester in three equal installments as per the following deadlines:

*Students need to submit their application forms with valid supporting documents. Any application without valid supporting documents will be rejected.

Results of Summer 2024 semester will be withheld until all dues of that semester are cleared and students need to clear the installment payments in time to avoid late fees.

Students will not be able to complete pre-registration or registration for Spring 2025 semester if the full payments of the Summer 2024 semester, and previous semester/s are not completed.

For any queries about the Installment Plan, please contact: finaid[at]bracu.ac[dot]bd. Any queries or requests sent to any individual(s) will not be entertained.

FAQ:

Please refer to the FAQ shared in the link below:

FAQ_PHR_Summer'24

If you need any further clarification which is not clearly articulated in this FAQ, please contact Financial Aid, BRAC University: finaid[at]bracu.ac[dot]bd

Thanking You,Financial Aid Team,BRAC University.

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Extension of the Deadline to Apply for Installment Payment Options, Summer 2024 Semester, School of Pharmacy, BRAC University Policy & Procedure

Extension of the Deadline to Make Payment-Summer 2024 Semester Policy & Procedure

Student Inquiry Email Address

Circular to Apply for Installment Payment Options, Summer 2024 Semester, Department of Pharmacy, BRAC University, Policy & Procedure

Extension of Deadline to Apply for Installment Payment Options, Summer 2024 Semester till 08 June 2024 (Saturday) Policy & Procedure

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sic[at]bracu.ac[dot]bd

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This circular is applicable only for the students of BRAC University except the Department of Pharmacy, and School of Law, BRAC University.

BRAC University understands the challenges that some students may face in managing their finances for the Summer 2024 semester due to the unforeseen situation in our country . In an effort to support our students during these times, we are pleased to extend the deadline to make payments of installments. In addition, for those who have not applied for any installment payments, the deadline to make payment for the Summer 2024 Semester has also been extended. This initiative aims to provide convenient alternatives for those experiencing financial difficulties, ensuring that everyone has the opportunity to pursue their education without unnecessary stress.

Students who got approval for installment payments are requested to pay the tuition fees of the Summer 2024 semester in three equal installments as per the following deadlines:

For the students, who did not apply for installment payments, the last date of payment for Summer 2024 Semester (except the Department of Pharmacy, and School of Law, BRAC University) is 29 August 2024 (Thursday). Their late fine will remain the same from 15 July 2024 to 29 August 2024. The late fine will be applied as usual after 29 August 2024.

For any payment related queries, you can communicate: queries-accounts[at]bracu.ac[dot]bd For any queries about Installment Plan, please contact: finaid[at]bracu.ac[dot]bd. Any queries or requests sent to any individual(s) will not be entertained.

We appreciate your patience and understanding of our process. Please continue to stay safe and healthy.

Regards, Financial Aid Team BRAC University

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Extension of the Deadline to Make Payment-Summer 2024 Semester Policy & Procedure

Student Inquiry Email Address

Circular to Apply for Installment Payment Options, Summer 2024 Semester, Department of Pharmacy, BRAC University, Policy & Procedure

Extension of Deadline to Apply for Installment Payment Options, Summer 2024 Semester till 08 June 2024 (Saturday) Policy & Procedure

Orientation-Summer 2024

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Orientation-Summer 2024

Extension of the Deadline to Make Payment-Summer 2024 Semester Policy & Procedure

Student Inquiry Email Address

Circular to Apply for Installment Payment Options, Summer 2024 Semester, Department of Pharmacy, BRAC University, Policy & Procedure

Extension of Deadline to Apply for Installment Payment Options, Summer 2024 Semester till 08 June 2024 (Saturday) Policy & Procedure

This circular is applicable only for all the students of BRAC University except the Department of Pharmacy and the School of Law.

BRAC University understands that some students are facing challenges in managing their finances for the Summer 2024 semester. To support the students facing financial distress, we are pleased to extend installment payment options for the Summer 2024 semester.

Those who have already applied for Summer 2024 are requested not to apply again.

In addition, BRAC Bank and ONE Bank have agreed to provide student loan facilities for BRAC University students. Students who want to avail student loan, can contact BRAC Bank or ONE Bank directly.

This circular is only applicable for all the students of BRAC University except the Department of Pharmacy and the School of Law. For the Department of Pharmacy, BRAC University, a separate circular for installment plans will be announced later.

To apply for Payment in Installments, students need to complete the Application Form (link given below)

https://forms.gle/ZyoCTChFTY9yoDa19

Submit the filled-up form with valid supporting documents (Pls. see in the form) - On or before 08 June 2024 (Sunday) by 11:59 pm (for both new and continuing students).

Students applying for installment payments will be required to pay the tuition fees of the Summer 2024 semester in three equal installments as per the following deadlines:

*Students need to submit their application forms with valid supporting documents. Any application without valid supporting documents will be rejected.

Results of Summer 2024 semester will be withheld until all dues of that semester are cleared and students need to clear the installment payments in time to avoid late fees.

Students will not be able to complete pre-registration or registration for Fall 2024 semester if the full payments of the Summer 2024 semester, and previous semester/s are not completed.

For any queries about the Installment Plan, please contact: finaid[at]bracu.ac[dot]bd . Any queries or requests sent to any individual(s) will not be entertained.

Please refer to the FAQ shared in the link below:

https://docs.google.com/document/d/1t2cjsNnd5-05xJOQiwbFWRRpqIY6m2PB94c4YwJ_F-s/edit

If you need any further clarification which is not clearly articulated in this FAQ, please contact Financial Aid, BRAC University: finaid[at]bracu.ac[dot]bd

Thanking You,Financial Aid Team,BRAC University.

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Extension of Deadline to Apply for Installment Payment Options, Summer 2024 Semester till 08 June 2024 (Saturday) Policy & Procedure

Extension of the Deadline to Make Payment-Summer 2024 Semester Policy & Procedure

Student Inquiry Email Address

Circular to Apply for Installment Payment Options, Summer 2024 Semester, Department of Pharmacy, BRAC University, Policy & Procedure

Orientation-Summer 2024

As physical student service is not available for the time being, an email address has been created so that students/ guardians can reach us for any inquiry.

sic[at]bracu.ac[dot]bd

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Student Inquiry Email Address

Extension of the Deadline to Make Payment-Summer 2024 Semester Policy & Procedure

Circular to Apply for Installment Payment Options, Summer 2024 Semester, Department of Pharmacy, BRAC University, Policy & Procedure

Extension of Deadline to Apply for Installment Payment Options, Summer 2024 Semester till 08 June 2024 (Saturday) Policy & Procedure

Orientation-Summer 2024

This circular is applicable only for the students of the Department of Pharmacy, BRAC

University.

BRAC University understands the challenges that some students may face in managing their finances for the Summer 2024 semester. In an effort to support our students during these times, we are pleased to offer installment payment options. This initiative aims to provide convenient alternatives for those experiencing financial difficulties, ensuring that everyone has the opportunity to pursue their education without unnecessary stress.

In addition, BRAC Bank and ONE Bank have agreed to provide student loan facilities for BRAC University students. Students who want to avail student loan, can contact BRAC Bank or ONE Bank directly.

This circular for installment payment options is only applicable for the students of the Department of Pharmacy, BRAC University, Summer 2024 semester.

Payment in Installments:

To apply for Payment in Installments, students need to complete the Application Form (link given below)

Application Form:

https://forms.gle/3iNXzDJYKDNuKzN66

Submit the filled-up form with valid supporting documents (Pls. see in the form) - On or before 29 July 2024 (Monday) by 11:59 pm (for both new and continuing students).

Students applying for installment payments will be required to pay the tuition fees of the Summer 2024 semester in three equal installments as per the following deadlines:

*Students need to submit their application forms with valid supporting documents. Any application without valid supporting documents will be rejected.

Results of Summer 2024 semester will be withheld until all dues of that semester are cleared and students need to clear the installment payments in time to avoid late fees. Students will not be able to complete pre-registration or registration for Spring 2025 semester if the full payments of the Summer 2024 semester, and previous semester/s are not completed.

For any queries about the Installment Plan, please contact: finaid[at]bracu.ac[dot]bd . Any queries or requests sent to any individual(s) will not be entertained.

FAQ:

Please refer to the FAQ shared in the link below:

FAQ_PHR_Summer'24

If you need any further clarification which is not clearly articulated in this FAQ, please contact Financial Aid, BRAC University: finaid[at]bracu.ac[dot]bd

Thanking You,Financial Aid Team,BRAC University.

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Circular to Apply for Installment Payment Options, Summer 2024 Semester, Department of Pharmacy, BRAC University, Policy & Procedure

Extension of the Deadline to Make Payment-Summer 2024 Semester Policy & Procedure

Student Inquiry Email Address

Extension of Deadline to Apply for Installment Payment Options, Summer 2024 Semester till 08 June 2024 (Saturday) Policy & Procedure

Orientation-Summer 2024

BRAC University's admission tests, both undergraduate and postgraduate, scheduled to be held on 26 July and 2 August respectively have been postponed.

New test dates and application deadlines will be announced later.

News Archive >>

Announcement regarding admission test

Extension of the Deadline to Make Payment-Summer 2024 Semester Policy & Procedure

Student Inquiry Email Address

Circular to Apply for Installment Payment Options, Summer 2024 Semester, Department of Pharmacy, BRAC University, Policy & Procedure

Extension of Deadline to Apply for Installment Payment Options, Summer 2024 Semester till 08 June 2024 (Saturday) Policy & Procedure

By donating to BracU Scholarship/Financial Aid program, you are empowering young leaders who will contribute to the communities in a way that is responsive to society's needs.

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"It was not possible for my parents to fund the cost of food and my accommodation. Moreover, I had a responsibility to finance my family partly as well. The reality would have been different without the scholarship as I would have had to get a full-time employment for my family."

- Mohsin applied and received the BRAC Scholarship in 2003.

“Without the scholarship, I never would have been able to attend a school like Brac University. Since I enrolled, I have maintained a perfect GPA. I'm involved in a lot of extracurricular activities, but my passion is biotechnology. I'm currently studying bacteria in the lab. It's not just about myself It's for the community, for the world, for my family. I believe if I can work hard enough, and if I'm supported enough, I can make a difference."

"I extend my sincere and heartfelt gratitude for the scholarship awarded to me. Engineering has been my dream from the age of eight up to date. Having narrowly missed the government scholarship [in Uganda], I had no more hopes left for higher education. Thanks to God who restored my hopes by bringing the Brac University opportunity to me."

PoliciesScholarship/Financial Aid will be awarded to all undergraduate and graduate students who meet the eligibility criteria*. All applications for scholarship/tuition waiver must be submitted within the deadline announced by the university.Only one of the waivers will be awarded to a particular student.All categories of tuition fee waiver will depend on the availability of funds.Students applying for renewal of their tuition fee waiver in any category will have to take minimum 12 (18 for pharmacy) credit course load.*The financial aid policy is subject to change as guided by exigencies.

Scholarship/Financial Aid will be awarded to all undergraduate and graduate students who meet the eligibility criteria*.

To keep pace with BRAC's focus on adolescents and youth as a special group of the country, Brac University (BracU) offers quality tertiary education, as well as financial support to meritorious students who come from disadvantaged communities. Brac University promotes the values of lifelong learning as part of its social responsibility initiative for the development of the country. Student recruitment, retention and graduation are key components of BracU's strategic plan. Hence, scholarship and tuition waivers are offered to attract students with high potential, to remove financial barriers and provide access to qualified students.

Each semester, Brac University is proud to honor academically talented and exceptionally skilled students with a variety of scholarships and awards. The university annually awards more than 100 million takas as scholarships to both undergraduate and postgraduate students.

Eligibility Criteria

The process is designed to attract a high quality student body, add diversity to the student population, remove financial barriers for qualified but financially disadvantaged students and assist in meeting institutional and regulatory compliance requirements. For any queries or for sending Scholarship application please email to: scholarship[at]bracu.ac[dot]bd

Read eligibility in detail (Undergraduate) Read eligibility in detail (Postgraduate) Merit Based Scholarship Policy (Updated)

Application form for Local students (Undergraduate) Application form for Local students (Postgraduate)

Application form for Foreign students

20-40% Tuition fees waiver for postgraduate programsView More

View More

The impact and opportunity of the Brac Scholarship program during the last fourteen years has been substantial. Financial support under the program significantly improved students' chances of graduating from Brac University. Students also reflect the collective belief that higher education will lead to a stronger and more vibrant community and ultimately greater achievement in the job world. Year 2013 2014 2015 201620172018 Number of Awardees 570 597 609 690846864

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Policies

20-40% Tuition fees waiver for postgraduate programs

News and Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

5th Intra University Moot Court Competition- 2024

Exploring New Contextualism 2.0: An Approach to Architecture and Urban Design

LSDB’s Regional Education Award 2023 (LREA-2023)

Certificate Course on ‘Intellectual Property for Innovation, Creativity and Branding’

Center for Learning Intellectual Property (CLIP) in collaboration with the School of Law, BRAC University is going to organize a certificate course titled ‘Intellectual Property for Innovation, Creativity and Branding’. The course will be held on August 4 and 5 (Friday and Saturday) 2023. The two-day long course is divided into five sessions and will focus primarily on Intellectual Property in the context of Bangladesh. The course is offered to students of undergraduate and graduate levels from diverse backgrounds including law, business, economics, pharmacy, engineering, technology, fashion design, etc. Young professionals from the said industries are also the target group of this course. A group of young and dynamic facilitators, including academics, attorneys, activists, and government IP officials will conduct the sessions.

The course will be held on the premise of BRAC University. Participants are expected to complete their registration on or before July 31, 2023, through the submission of the Google form. The registration will be completed only upon the payment of the full course fee. The course fee will cover all course materials, notepads, stationary etc. The Course fee will also cover lunch, snacks and water for two days.

After the completion of the course, participants will be entitled to a certificate of participation issued jointly by Center for Learning Intellectual Property (CLIP) and the School of Law, BRAC University. To achieve the certificate participants must attend all five sessions.

The course outline, detailed schedule and necessary reading materials will be sent to each participant through email provided in the Google form after the completion of the registration.

Sessions:

1. Introduction to Intellectual PropertyMahua Zahur, Director CLIP; Legal Consultant IP Chronicles
2. Role of IP in Branding: A Myth or Reality?MD. Belal Hossen, Assistant Director (Trademarks); Additional Charge in WTO & International Affairs Wing, Department of Patent, Design & Trademarks, Ministry of Industries, Bangladesh
3. Protecting Creativity in Digital Age: From Concepts to Practical ApplicationsDr. Muhammed Atiqur Rahman, Senior Lecturer and Academic Coordinator, School of Law, BRAC University

4. Role of Patents for Pharmaceutical InnovationsMohammad Ataul Karim, DPhil in IP Law Researcher, University of Oxford; Founder and Executive Director, CLIP

5. The Enforcement Regime of Intellectual Property Rights in BangladeshNuran Choudhury, Legal Advisor, US Embassy, Dhaka; Advocate, Supreme Court of Bangladesh

Date: 4-5 August (Friday and Saturday), 2023

Venue: GDLN Center, Level 18, Building 2, BRAC University.

Course Fee: Tk. 4000/- for Students   Tk. 4500/- for Professionals

Deadline for Registration: July 31, 2023Link for registration: https://forms.gle/W4jPRoa8VPfXig8W9

Payment mode: bkash to 01817716933 or directly to Department Coordination Officer of School of Law, BRAC University

Contact: e-mail to clipbangladesh23[at]email[dot]com Call to 01817716933

Certificate Course on ‘Intellectual Property for Innovation, Creativity and Branding’

Services

Helpline

Email

Contact Hours

Student Information CentreRegistryExam ControllerAdmission

+88 09638464646 (IVR press 2)

sic[at]bracu.ac[dot]bd

9:00 AM to 9:00 PM

University Medical Centre

+8801322821534

medicalcenter[at]bracu.ac[dot]bd

9:00 AM to 9:00 PM

Office of the Proctor

+8801313049111 +8801313049105 +8801729071209

proctor[at]bracu.ac[dot]bd

9:00 AM to 9:00 PM

Counseling and Wellness Centre

+8801322917314 +8801322917315

counseling[at]bracu.ac[dot]bd

9:00 AM to 9:00 PM

Department	Email
Office of the Registrar	registrar[at]bracu.ac[dot]bd
Office of the Proctor	proctor[at]bracu.ac[dot]bd
Office of the Controller of Examinations	academic.records[at]bracu.ac[dot]bd
Medical Centre	doctor[at]bracu.ac[dot]bd
Student Information Centre	sic[at]bracu.ac[dot]bd
Student Life	studentlife[at]bracu.ac[dot]bd
Office of Career Services and Alumni Relations (OCSAR)	csoadmin[at]bracu.ac[dot]bd

Office of Academic Advising (OAA)

bracu-oaa[at]bracu.ac[dot]bd

Counseling and Wellness Centre

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International and Scholarship Office

scholarship[at]bracu.ac[dot]bd

Ayesha Abed Library

librarian[at]bracu.ac[dot]bd

Institutional Quality Assurance Cell (IQAC)

iqac[at]bracu.ac[dot]bd

Office of Communication

communications[at]bracu.ac[dot]bd

Operations

operations[at]bracu.ac[dot]bd

Accounts and Finance

queries-accounts[at]bracu.ac[dot]bd

Human Resources Department

hrd[at]bracu.ac[dot]bd

IT Department

support[at]bracu.ac[dot]bd

General Information

info[at]bracu.ac[dot]bd

BRAC University Campus

Residential Campus

Join us at the Department of Architecture, Brac University, for an engaging workshop titled "Exploring New Contextualism 2.0: An Approach to Architecture and Urban

Design." This workshop promises to delve into the philosophy of New Contextualism and its significance in the fields of architecture and urban design. Discover a fresh framework for addressing contemporary built environment challenges and opportunities.

Event Details:

Date: February 17, 2024

Time: 9:00 AM - 5:00 PM

Reporting Time: 8:30 AM

Location: Block C, Floor 5, Department of Architecture, KHA-228, 1 Bir Uttam Rafiqul Islam Ave, Dhaka 1212

Workshop Highlights: Delve into hands-on exercises and discussions covering:

Facilitators:

In addition to our esteemed facilitators, we will be joined by ten accomplished alumni from the Department of Architecture, BRAC University, who will serve as mentors during the hands-on exercises.

Please note that seats are limited while the lectures are open to all. Kindly register now to secure your spot and indicate your interest in attending the lectures. We look forward to your participation in this enriching event.

Here is the registration link

https://docs.google.com/forms/d/e/1FAIpQLSf92NVgB-nu6LDsnXU-4FTCPBadKNATlMJdkUa1NHGAjHT4tA/viewform

Exploring New Contextualism 2.0: An Approach to Architecture and Urban Design

BRAC University Law Society (BULS) in collaboration with the School of Law, BRAC University, (upon hosting a successful Workshop for its students titled Moot Court and its Fundamentals on March 27, 2024) is preparing to arrange the “5th Intra-University Moot Court Competition – 2024” scheduled to be held on 25 - 27 April 2024 (Thursday - Saturday) to encourage and motivate the eager students of the School of Law to engage in mooting. The primary objective of this competition is to help interested students become competent Mooters and to contribute their successes to Moot Court contests. The competition is offered to all School of Law, BRAC University students. This competition is anticipated to cover topics such as Memorial drafting, Oral submission, and Courtroom Etiquette. Mooting is an element of a law student's education, and the objective of Mooting is to prepare students before they become professionals and begin practising law in Courtrooms.

We are honoured to have Mr Justice Zafar Ahmed, Honorable Judge of the High Court of Division, Supreme Court of Bangladesh, and Mr Justice A. K. M. Rabiul Hassan, Honorable Judge of the High Court of Division, Supreme Court of Bangladesh are invited as Judges in the Final Round of the Competition.

The Final Round of the Competition is scheduled to be held on Saturday, 27 April 2024 from 2:00 pm to 6:30 pm at the BRAC University, School of Law -Moot Court, 09 D – 19C and Prize Giving Ceremony is scheduled to be held at the BRAC University, Lecture Theater, 09C – 16T.

This event will be open for all, we would be grateful if you could grace us with your presence. We hope that your presence will motivate and inspire the participants and law learners.

Best regards,School of LawBRAC University& BRAC University Law Society

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Certificate Course on ‘Intellectual Property for Innovation, Creativity and Branding’

Please be informed that the London School of DIGITAL BUSINESS, United Kingdom is inviting you to highly anticipated LSDB Regional Education Awards (LREA), where they will be celebrating and honoring the unwavering dedication and remarkable leadership of Professors, Principals and Directors like yourselves.

Below is the detailed participation criteria and process:

Registration Link- https://www.lsdb.co.uk/events/registrations

Registration Deadline: 20th Oct, 2023, Friday

Note- There are no registration charges for the nomination and for the event.

Platform: Zoom Video Conference (Virtual Ceremony)

This event is a testament to the pivotal role you play in shaping the future of education and your unwavering commitment to the success of your students and staff. We believe it's essential to honor your contributions and showcase exemplary leadership.

We look forward to your presence and participation in this prestigious event. If you have any inquiries or require further information, please do not hesitate to reach out at growth[at]lsdb.co[dot]uk or WhatsApp at +44 7721 067721.

LSDB’s Regional Education Award 2023 (LREA-2023)

Services

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sic[at]bracu.ac[dot]bd

9:00 AM to 9:00 PM

University Medical Centre

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Human Resources Department

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IT Department

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General Information

info[at]bracu.ac[dot]bd

BRAC University Campus

Residential Campus

New Extended Submission Date: On request from several national and international prospective participants, the submission date has been extended till 15th July 2024

There is a recent shift towards critiquing existing preoccupations with logo-centrism and anthropocentrism in language studies, advocating for a post-humanist approach. This approach considers multisensory, multimodal resources, and ecological systems in meaning-making processes. With the transformation in the ontology and epistemology of language, language education has also undergone a notable metamorphosis in recent years.In light of this phenomenon, in the international conference entitled Language Metamorphosis: Implications for Language Education in Decolonial Contexts (LMLEDC), BRAC Institute of Languages (BIL), BRAC University will highlight the transformation in language education that encourages the development of students and teachers' critical language awareness, use of mother tongues, multi-model semiotic resources, collaborative learning and diverse sociocultural pedagogical approaches. BIL invites papers, panels, and posters based on interdisciplinary studies on languages and their roles in society with reference to sociology, history, political science, anthropology, education, culture and media studies, critical geography, linguistics, and literature. BIL expects that LMLEDC will create an opportunity for scholars, researchers, language practitioners and post-graduate students to problematise linguicism and linguistic imperialism and generate ideas for ‘decoloniality in/as praxis’ in language and language-education policies and practices.

Professor Suresh Canagarajah, Evan Pugh University, Professor of Applied Linguistics, English, and Asian Studies, Pennsylvania State University, USA

Professor Stephen May, School of Māori and Indigenous Education, Faculty of Education and Social Work, University of Auckland, New Zealand

Professor Sender Dovchin, Senior Principal Research Fellow, Curtin University, Australia

Professor Robert Phillipson, PhD, Professor Emeritus, Copenhagen Business School, Denmark

Professor Alastair Pennycook, PhD, Professor Emeritus, University of Technology Sydney, Australia

Professor Sirpa Leppänen, PhD, Professor Emeritus, Department of Language and Communication Studies, University of Jyväskylä, Finland

● Contact information - the first and last name, designation, name of affiliated institution, name of the country of affiliation, phone number, and email address● For paper and poster presentation and colloquium - Full title

● For paper and poster presentation: Abstract 250-word

● For the colloquium, convenors are requested to separately (1) collect 4 abstracts, (2) find a discussant, (3) write an abstract for the colloquium, and (4) submit the full document with 4 abstracts.

● To submit, Click Here and fill out the form.

● Please be aware that the information you enter when completing the form will be used for the Official Letter of Acceptance, Letter of Invitation, and certificate and in the conference brochure. Therefore, please ensure the correct spelling of names, affiliated institutions, and other relevant information.

● Choose one of the presentation formats and submit your abstract (250 words), and a short bio (150 words).

● Your abstract will undergo peer review and the results will be returned to you generally within four weeks.

● If your abstract is accepted, you will be invited to register for the conference.

● Upon payment of the registration fee, you will receive a confirmation email containing your official receipt.

Obaid Hamid, PhD Associate Professor, School of Education The University of Queensland

Obaid Hamid is an Associate Professor & Integrity Officer, School of Education, at the University of Queensland. He received his education in English literature, applied linguistics, and TESOL (Teaching English to Speakers of Other Languages). He considers TESOL his home. Read More

Manjet Kaur, PhD Associate Professor &; Deputy Dean School of Languages, Literacies, and Translation Universiti Sains Malaysia, Malaysia

Assoc. Prof. Dr. Manjet Kaur Mehar Singh completed her B.A. (Education) Hons., (1995), Master in Education (TESOL) (2000), and Ph.D specialising in Academic Literacies and International Students (2013) at the University of Science Malaysia. She started her academic career as a language instructor at the Center of Languages and Translation, USM in 1996. Read More

Pramod K. Sah, PhD Assistant Professor, Department of English Language Education The Education University of Hong Kong, Hong Kong

Dr. Pramod K. Sah is an Assistant Professor of English Language Education at The Education University of Hong Kong. He also holds an Honorary Norham Fellow position at the Department of Education, University of Oxford. Read More

Anik Nandi, PhD Director, Centre for Languages and Multicultural Communication (CLMC) Woxsen University, India

Dr. Anik Nandi earned his PhD in 2017 as a prestigious James Watt Fellow from the Heriot-Watt University, Edinburgh, United Kingdom. He is the Director of the Centre for Languages and Multicultural Communication and Joseph Lo Bianco Chair for Language Policy Studies at Woxsen University, Telangana, India. His current postdoctoral research project investigates the intergenerational transmission of Basque (in Spain) and Santali (in India). Read More

Beth Trudell English Language Specialist US State Department, USA

Beth Trudell was an English Language Fellow at the BRAC Institute of Languages (BIL) from 2006 to 2008. As a Fellow, she did teacher training and curriculum design. She led the team of teachers who designed the Institute's Pre-University Program, and she co-authored an article on the program. Read More

Professor Muhammad Kamarul Kabilan, PhD The School of Educational Studies Universiti Sains Malaysia

Professor Dr. Muhammad Kamarul Kabilan is a Professor at the School of Educational Studies, University Sains Malaysia, Penang. Read More

Professor Asifa Sultana, PhD Department of English and Humanities BRAC University, Bangladesh

Dr. Asifa Sultana joined ENH as a faculty member in 2008. She completed her PhD in the language development of Bangladeshi children from the University of Canterbury, New Zealand in 2015. Read More

Farzana Yesmen Chowdhury, PhD Director &; Associate Professor, Institute of Modern Languages University of Chittagong, Bangladesh

Dr Farzana Yesmen Chowdhury is a social science researcher with a keen interest in all aspects of sociolinguistics and cultural studies. Read More

Rafi Saleh, PhD Associate Professor, Department of English and Humanities University of Liberal Arts Bangladesh, Bangladesh

Dr Abu Saleh Mohammad Rafi is a critical applied linguist with a proven track record of publishing in top-tier journals and editing special issues. He has also been invited to contribute to collections edited by leading international academics, delivered keynote addresses in multiple countries, and has teaching experience in universities in both the global south and the global north. Read More

Professor Firdous Azim, PhD Chairperson, Department of English and Humanities BRAC University, Bangladesh

Firdous Azim is a Professor of English and is chair of the Department of English and Humanities at BRAC University. She is also a member of Naripokkho, a woman's activist group in Bangladesh. She obtained her Ph.D. from the University of Sussex, UK in 1989 and was awarded a fellowship by the University as one of its fifty most distinguished alumni and academics in 2012. Read More

Mashrur Shahid Hossain Professor, Department of English Jahangirnagar University, Bangladesh

Mashrur Shahid Hossain is a Professor of English, Jahangirnagar University, Dhaka, Bangladesh. Read More

Professor Sayeedur Rahman, PhD Director, Institute of Modern Languages University of Dhaka, Bangladesh &; President TESOL Society of Bangladesh

Sayeedur Rahman is a Professor and former Chairman of the Department of English Language at the Institute of Modern Languages (IML), University of Dhaka. He received PhD in ELT from Jawaharlal Nehru University, India, as an Indian Council for Cultural Relations (ICCR) Scholar. Read More

Professor Mahmud Hasan Khan, PhD Department of English and Modern Languages Independent University, Bangladesh

Mahmud Hasan Khan is a Professor, teaches English in the Department of English and Humanities, University of Liberal Arts Bangladesh. He has two PhDs from the International Islamic University Malaysia and Macquarie University, Sydney, Australia. Read More

Professor Md Zulfeqar Haider, PhD Senior Specialist, National Curriculum and Textbook

Board (NCTB), Bangladesh

Md. Zulfeqar Haider is a Professor and Senior Specialist (Curriculum), National Curriculum and Textbook Board (NCTB), Dhaka. He received his PhD in Teaching English as a Second Language (TESL) from Monash University, Australia, where he did his MEd (TESOL) too. Read More

Sayeedur Rahman is a Professor and former Chairman of the Department of English Language at the Institute of Modern Languages (IML), University of Dhaka. He received PhD in ELT from Jawaharlal Nehru University, India, as an Indian Council for Cultural Relations (ICCR) Scholar. He has extensively worked as an ESL/EFL teacher and researcher for the last 20 years and has published widely in the areas of SLA and Sociolinguistics. He has worked as a consultant and led several ELT projects with Save the Children, the Open University-UK, UNICEF, Dhaka Ahsania Mission (DAM), British Council and English in Action (EIA) project funded by the UK Government and Mott MacDonald. He was awarded the Researcher Links fellowships from the British Council, UK in 2014 as a Visiting Fellow with the OU, UK. He is also a certified Master Trainer by the University Grants Commission (UGC), Bangladesh, and the British Council Bangladesh. His research interests include sociolinguistics, EFL syllabus design and materials development, socio-psychological study of EFL, individual differences in language learning, and EAP/ESP.

Institute of Modern Languages University of Dhaka, Bangladesh&;PresidentTESOL Society of Bangladesh

Professor Sayeedur Rahman, PhD

News & Events

BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Dr. Pramod K. Sah is an Assistant Professor of English Language Education at The Education University of Hong Kong. He also holds an Honorary Norham Fellow position at the Department of Education, University of Oxford. Before joining EduHK, he worked as a Postdoctoral Associate at the University of Calgary's Werklund School of Education. He serves as an Editor-in-Chief of the Journal of Education, Language, and Ideology. His primary area of scholarly focus revolves around investigating how colonial and liberal ideologies in language policy and practices create socioemotional and educational disparities among diverse students. The central goal of his research is to conceptualize anti-oppressive and asset-based alternatives for pedagogies (e.g., linguistically and culturally sustaining pedagogies and critical translanguaging), research, and policies.

The Education University of Hong Kong, Hong Kong

Pramod K. Sah, PhD

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Assoc. Prof. Dr. Manjet Kaur Mehar Singh completed her B.A. (Education) Hons., (1995), Master in Education (TESOL) (2000), and Ph.D specialising in Academic Literacies and International Students (2013) at the University of Science Malaysia. She started her academic career as a language instructor at the Center of Languages and Translation, USM in 1996. As a scholar, Dr.Manjet Kaur’s field of research interests is wide-ranging, which include language, literacy, multiculturalism and research in higher education. She is very active in doing research and publishing and is also a member of the editorial board of journals at national and international level.

Other than teaching at the undergraduate level, she is also currently supervising Master and PhD students. She has also widely presented at international conferences worldwide. For her contribution to the university, Dr. Manjet Kaur was awarded excellent service award by USM in 2008. As for her contribution to the state of Penang, Dr. Manjet Kaur was awarded the “P.J.K.” award by the Governor of Penang in 2015.

School of Languages, Literacies, and Translation Universiti Sains Malaysia, Malaysia

Manjet Kaur, PhD

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Mashrur Shahid Hossain is a Professor of English, Jahangirnagar University, Dhaka, Bangladesh. He teaches, researches, publishes, and presents papers on a host of areas, ranging from Comparative Literature and Postcolonialism/Decoloniality to Critical Pedagogy, Cultural Studies, and Queer Studies. He is presently working on two book-length projects. The first one examines 'extreme' theatre and music to explore the utopian potential of media, and the second one forms a framework for teaching 'English' literature in Bangladesh.

Department of English

Jahangirnagar University, Bangladesh

Mashrur Shahid Hossain

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Obaid Hamid is an Associate Professor & Integrity Officer, School of Education, at the

University of Queensland. He received his education in English literature, applied linguistics, and TESOL (Teaching English to Speakers of Other Languages). He considers TESOL his home. His research has focused on the policy and practice of TESOL education in Asia. He is particularly interested in the role of English in individual mobility and social development. This interest has led him to examine English as a medium of instruction (EMI) policy and practice and its academic and social outcomes. He uses qualitative, quantitative, and textual data. His work is underpinned by critical perspectives, his multidisciplinary backgrounds, and his life experiences as a confused transnational.

School of Education, The University of Queensland

Obaid Hamid, PhD

News & Events

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Dr. Anik Nandi earned his PhD in 2017 as a prestigious James Watt Fellow from the Heriot-Watt University, Edinburgh, United Kingdom. He is the Director of the Centre for Languages and Multicultural Communication and Joseph Lo Bianco Chair for Language Policy Studies at Woxsen University, Telangana, India. His current postdoctoral research project investigates the intergenerational transmission of Basque (in Spain) and Santali (in India). In addition, Dr. Anik Nandi is also a Senior Research Associate (Hon.) at the Sociolinguistic Seminar of the Royal Galician Language Academy (Spain), a Visiting Researcher at the Leiden Centre for Linguistics (The Netherlands), and a Collaborator at the research centre attached to the UNESCO Chair on World Language Heritage in the University of the Basque Country (Spain). Dr. Nandi specialises in language policy and literacy practices in multilingual settings and his research papers appeared in many prestigious international journals linked to Cultural Studies and Applied Linguistics. Since 2021, he has worked as the Book Reviews Editor of the Sociolinguistic Studies journal (ISSN: 1750-8649; Scopus and Q2), Equinox Publishing, United Kingdom.

Woxsen University, India

Anik Nandi, PhD

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Beth Trudell was an English Language Fellow at the BRAC Institute of Languages (BIL) from 2006 to 2008. As a Fellow, she did teacher training and curriculum design. She led the team of teachers who designed the Institute’s Pre-University Program, and she co-authored an article on the program. She created BIL’s undergraduate academic writing program and the curriculum for the first advanced academic writing courses.

Beth was part of the team that designed BIL’s MA TESOL graduate program.

Beth was a Senior English Language Fellow in Pakistan and Afghanistan, where she was part of the team that designed the country’s first MEd for TESOL. She was an English Language Specialist for the American English Radio Project Bangladesh, and the project team presented the project at the TESOL International Convention. Also, she was a Specialist in Tanzania, China, Thailand, and Russia.

She has an MA in English for TESOL from California State University. During her graduate work, she specialized in academic writing, and she was a Graduate Writing Associate for Writing Across the Curriculum. She taught Writing for Proficiency classes at the university.

US State Department, USA

Beth Trudell

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In recent research studies, the preoccupation with logo-centricism in the post-structuralist approach to language has been empirically challenged. The post-humanist approach critiques the overt and covert fetishisation of language and suggests space, multisensory and multi-modal resources, spatial arrangements that include both human participants and nonhuman entities, such as objects, bodies, actions, activities, technologies, and larger societal and ecological systems indispensable for exploring meaning-making processes.

With the change in the ontology and epistemology of language, education has been expected to undergo a notable metamorphosis in recent years too. A significant change in educational philosophy encourages the use of the mother tongue and other linguistic and semiotic resources available in the linguistic ecology and the collaboration of students with teachers and community members for language learning and teaching. Language education, thus, challenges linguicism and linguistic imperialism that promote specific official or national languages, while marginalising minority languages and disempowering communities based on their languages, religion, nationality, ethnicities, races, or sexual orientation. The metamorphosis of language education is also tied up with the advancements in technology that encourage the use of the Internet, gadgets, devices and various pieces of software for language teaching and education research.

Considering the pervasiveness of logo-centrism and anthropocentrism in language education, BRAC Institute of Languages, BRAC University invites papers to configure and contest the hegemonic definition and status of languages in various domains of life and their interrelationships with language education. The international conference entitled Language Metamorphosis: Implications for Language Education in Decolonial Contexts (LMLEDC), scheduled for 28-30th November 2024, will hence offer a platform where a new understanding of languages and innovative pedagogies will be explored. Bringing together language practitioners and professionals from diverse backgrounds, the conference will create opportunities to learn about innovative practices, techniques, methods, and activities for language teaching and learning.

In addition, the international LMLEDC will bring scholars, researchers, language practitioners and post-graduate students together to enhance their networking and professional development and foster a sense of community among themselves. Further, this platform will encourage language practitioners to present their diverse socio-cultural pedagogical approaches and methods of utilising digital literacy for intercultural communication in language education. The conference will also be interdisciplinary and showcase papers, panels, and posters on languages and their roles in society with reference to sociology, history, political science, anthropology, education, culture and media studies, critical geography, and linguistics and literature. We expect scholars, researchers and professionals to advance and interchange ideas about critical language awareness for social justice, equality and decoloniality in/as praxis.

The possible topics include, but are not limited to, the following:

• Ontology and epistemology of language• Epistemology and ontology in language teaching and learning • Linguistic imperialism• Social justice, decoloniality, and Southern epistemologies • Language as a local practice • Trans-movement and language education • Translingual accentism, racism• Language and sense of (dis)placement – migrants, refugees, ethnic minorities• Language in the workplace, court, hospital, media• Medium of instruction• Innovation in language education• Language Acquisition and Learning (L2, L3, L4, Lx)• CLIL• EAP/ESP/ EGP • Language policy and planning• Language and language education for Sustainable Development • Linguistic landscape and multimodality• Language, language teaching, and technology • AI, digital spaces, language education• Sociolinguistics (e.g., Language & socioeconomic class, gender, style, varieties, identity etc.)• Comparative linguistics• Language and media• Language and mobility• Language and translation • Literature and New Englishes• Postcolonial writing in English

The language of the conference is English.

Paper presentations (General and Post-graduate sessions):

Paper presentations will be allocated 20 minutes plus 5 minutes for questions.

Posters:

Poster presentations will be allocated their time slot during the conference.

Colloquiums:

Convenors are requested to separately (1) collect 4 abstracts, (2) find a discussant, (3) write an abstract for the colloquium, and (4) submit the full document with 4 abstracts.

Colloquiums will be allocated 50 minutes, plus 10 minutes for Q & A at the end of the colloquiums.

Abstracts for general papers, posters and colloquiums of max. 250 words in MS Word including references should be uploaded by 15th July, 2024.

Abstract Submission:

To submit, Click Here and fill out the form.

Abstracts for general papers, posters and colloquiums must be of max. 250 words in MS Word (excluding references).

New Extended Submission Date: On request from several national and international prospective participants, the submission date has been extended till 15th July 2024.

Email:

conference.bil[at]bracu.ac[dot]bd

The outcome of the abstract submission notification: 16th August 2024

Early bird registration: 20th August 2024 and end on 14th September 2024

Regular registration will start on 15th September 2024 and will be closed on 15th October 2024

Registration fees:

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BRAC University Campus

Residential Campus

Professor Suresh Canagarajah Evan Pugh University Professor of Applied Linguistics, English, and Asian Studies, Pennsylvania State University, USA

Professor Suresh Canagarajah, a Sri Lankan Tamil scholar in the fields of sociolinguistics, literacy, and English language teaching, is currently the Evan Pugh University Professor of Applied Linguistics, English, and Asian Studies at Pennsylvania State University. He is best known for introducing orientations to language and education from traditions and practices in the Global South to diversify dominant norms and policies in higher education and academia. Read More

Professor Stephen MayProfessor, Te Puna Wānanga, University of Auckland, New Zealand

Professor Stephen May is a Professor in Te Puna Wānanga (School of Māori and Indigenous Education) in the Faculty of Education and Social Work and the Director of the cross-faculty and interdisciplinary Master of Regional Development program at the University of Auckland. He is also an Honorary Professor at the University of Tartu, Estonia, and at Taiyuan University of Technology, China. Read More

Professor Sender Dovchin Senior Principal Research Fellow, Curtin University, Australia

Professor Sender Dovchin is a Senior Principal Research Fellow and an Australian Research Council Fellow. Her Australian Research Council DECRA (Discovery Early Career Researcher Award) project focuses on empowering vulnerable youth in Australia by combatting linguistic racism. Read More

Professor Robert Phillipson, PhD Professor Emeritus, Copenhagen Business School, Denmark

Robert Phillipson studied at Cambridge and Leeds Universities, UK, and has a doctorate from the University of Amsterdam. He worked for the British Council in Algeria, Yugoslavia, and London, before emigrating to Denmark in 1973. He is an emeritus Professor at Copenhagen Business School. Read More

Professor Alastair Pennycook, PhD Professor Emeritus, University of Technology Sydney, Australia

Alastair Pennycook is an Emeritus Professor of Language, Society, and Education at the University of Technology Sydney. He won the prestigious BAAL Book Prize for his books Posthumanist Applied Linguistics, Language and Mobility: Unexpected Places, and

Global Englishes and Transcultural Flows. Read More

Professor Sirpa Leppänen Professor Emeritus, Department of Language and Communication Studies University of Jyväskylä, Finland

Sirpa Leppänen is Professor Emerita of English at the University of Jyväskylä, Finland. She has directed several extensive research projects focusing on multilingualism in Finnish society, with a particular focus on English and Finnish. Read More

Alastair Pennycook is an Emeritus Professor of Language, Society, and Education at the University of Technology Sydney. He won the prestigious BAAL Book Prize for his books Posthumanist Applied Linguistics, Language and Mobility: Unexpected Places, and Global Englishes and Transcultural Flows. His most recent book is Innovations and Challenges in Applied Linguistics from the Global South (Routledge).

University of Technology Sydney, Australia

Professor Alastair Pennycook, PhD

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Robert Phillipson studied at Cambridge and Leeds Universities, UK, and has a doctorate from the University of Amsterdam. He worked for the British Council in Algeria, Yugoslavia, and London, before emigrating to Denmark in 1973. He is an emeritus Professor at Copenhagen Business School. His books have been published in 14 countries, among them: Learner language and language learning (with Claus Færch, and Kirsten Haastrup, Gyldendal & Multilingual Matters), Linguistic imperialism (Oxford UP), English-only Europe? Challenging language policy (Routledge), Linguistic imperialism continued (Orient Blackswan and Routledge), Why English? Confronting the Hydra (edited with Pauline Bunce, Vaughan Rapatahana, and Ruanni Tupas, Multilingual Matters), and Handbook of Linguistic Human Rights (edited with Tove Skutnabb-Kangas, Wiley Blackwell). He has lectured in over 40 countries, and given postgraduate courses in language policy and language rights in Denmark, Hyderabad/India, and Shanghai. He was awarded the UNESCO Linguapax prize in 2010, and the TESOL International President’s Award in 2024.

Copenhagen Business School, Denmark

Professor Robert Phillipson, PhD

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Professor Sender Dovchin is a Senior Principal Research Fellow and an Australian Research Council Fellow. Her Australian Research Council DECRA (Discovery Early Career Researcher Award) project focuses on empowering vulnerable youth in Australia by combatting linguistic racism. The project investigates how culturally and linguistically diverse young Australians experience discrimination in their daily lives because of how they speak. Previously, she was an Associate Professor at the Centre for Language Research, The University of Aizu, Japan, and was awarded Young Scientist (Kakenhi) by the Japan Society for the Promotion of Science.

Professor Dovchin is also an Editor-in-Chief of the Australian Review of Applied Linguistics. She was identified as the “Top Researcher in the field of Language & Linguistics” under The Humanities, Arts & Literature of Australia's 2021 Research Magazine and Top 250 Researchers in Australia in 2021. Her research pragmatically contributes to the second language education of the young generation living in the peripheries, providing a pedagogical view to accommodate the multiple co-existences of linguistic diversity in a globalised world. She has authored articles in top-tier international peer-reviewed journals, such as Applied Linguistics, Journal of Sociolinguistics, System, TESOL Quarterly, International Journal of Multilingualism, World Englishes, Asian Englishes, English Today, International Journal Bilingualism and Bilingual Education, International Journal of Multilingual Research, Journal of Multicultural Discourses, International Journal Bilingualism, Ethnicities, Multilingual, Linguistics, and Education, among others. She has authored six books with international publishers such as Routledge, Springer, Palgrave Macmillan, and Multilingual Matters. Professor Dovchin has had notable research funding success, including four external international and national research grants as the lead and co-investigator.

Curtin University, Australia

Professor Sender Dovchin

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Professor Stephen May is a Professor in Te Puna Wānanga (School of Māori and Indigenous Education) in the Faculty of Education and Social Work and the Director of the cross-faculty and interdisciplinary Master of Regional Development program at the University of Auckland. He is also an Honorary Professor at the University of Tartu, Estonia, and Taiyuan University of Technology, China.

Professor Stephen May is an international authority on language rights, language policy, language revitalisation, language education (especially, Indigenous, bilingual, and multilingual education), critical multiculturalism, and the "multilingual turn". To date, he has published 27 books and over 100 articles and chapters in these areas and since 2022 has been named in the top two percent of scientists globally in the field of language education in Stanford University's citation index. He is the editor-in-chief of the 10-volume Encyclopedia of Language and Education (3rd ed. Springer, 2017) and a Founding Editor of the international, interdisciplinary journal, Ethnicities.

In 2023, Professor May received the Royal Society of New Zealand, Te Apārangi, Mason

Durie Medal awarded to New Zealand's pre-eminent social scientist, and in 2018, he was awarded the McKenzie Award by the New Zealand Association of Educational Research (NZARE), the preeminent award for educational research in New Zealand. In 2015 he became an American Educational Research Association (AERA) fellow and was elected a Te Apārangi/Royal Society of New Zealand Fellow (FRSNZ) in 2016.

In 2008, Professor May was a Fulbright Senior Scholar, based at Arizona State University and City University New York, and in 2016 he was a Visiting Scholar at the University of the Basque Country, San Sebastian. From 2005 to 2015, he was an Associate Editor of Language Policy. He began his professional career as a secondary teacher of English and ESL in New Zealand and has subsequently taught in universities in New Zealand, Britain, USA, and Canada. The latter include the Sociology Department of the University of Bristol, UK, where he remains affiliated with the Centre for the Study of Ethnicity and Citizenship, and the University of Waikato, where he was a Foundation Professor of Language and Literacy Education.

University of Auckland, New Zealand

Professor Stephen May

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Professor Suresh Canagarajah, a Sri Lankan Tamil scholar in the fields of sociolinguistics, literacy, and English language teaching, is currently the Evan Pugh University Professor of Applied Linguistics, English, and Asian Studies at Pennsylvania State University. He is best known for introducing orientations to language and education from traditions and practices in the Global South to diversify dominant norms and policies in higher education and academia. He has played a leading role in empirically studying, theorizing, and defining the notion of translingual practice, which introduces a way of looking at communication as exceeding bounded languages and involving negotiation of diverse semiotic repertoires, including words, multimodal resources, objects, and artifacts, and material structures. He treats this ecological, ethical, and inclusive orientation to speaking and writing as part of his South Asian heritage and ancient practices in the Global South, which were later suppressed by European colonization.

Pennsylvania State University, USA

Professor Suresh Canagarajah

News & Events

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Sirpa Leppänen is Professor Emerita of English at the University of Jyväskylä, Finland.

She has directed several extensive research projects focusing on multilingualism in Finnish society, with a particular focus on English and Finnish. As part of the Center of Excellence in Research Unit for Variation, Contacts, and Change, her research group conducted a nation-wide survey on the roles and meanings of, and attitudes to English in Finnish society, as well as investigated the uses and functions of English in various institutional, everyday and mediatized settings. Her publications include 8 books or edited collections and over 80 research articles. Her most recent work has focused on multilingualism and multimodality as sociolinguistic resources for identification, gender, belonging, and counterspeech in social media.

University of Jyväskylä, Finland

Professor Sirpa Leppänen, PhD

News & Events

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The Westin Dhaka

View Map

https://www.marriott.com/en-us/hotels/dacwi-the-westin-dhaka/overview/

Renaissance Dhaka Gulshan Hotel

View Map

https://www.marriott.com/en-us/hotels/dacbr-renaissance-dhaka-gulshan-hotel/overview/

Holiday Inn Dhaka City Centre

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https://www.ihg.com/holidayinn/hotels/us/en/dhaka/dachi/hoteldetail

Inter Continental Dhaka

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Crowne Plaza Dhaka Gulshan

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Radisson Blu Water Garden Hotel

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https://www.radissonhotels.com/en-us/hotels/radisson-blu-dhaka

Lakeshore Hotels Banani

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https://lakeshorebanani.com.bd/

Amari Dhaka

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https://www.amari.com/dhaka

Hotel Sarina Dhaka

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https://www.sarinahotel.com/

Shopping Malls

Aarong

View Map

https://aarong.com/

Jamuna Future Park

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https://jamunafuturepark.com/

Bashundhara City Shopping Mall

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https://www.bashundhara-city.com/about-bashundhara-city

Unimart

View Map

Unimart Gulshan 2

Unimart Gulshan 1

https://www.united.com.bd/our-concerns/unimart-ltd/

Police Plaza Concord Shopping Plaza

View Map

Places to Visit

Ahsan Manzil Museum

Ahsan Manzil Museum

https://www.ahsanmanzil.org.bd/site/

Lalbagh Fort

Lalbagh Fort

https://beautifulbangladesh.gov.bd/district-destination/dhaka/heritage/4

Bangladesh National Zoo

Bangladesh National Zoo

https://bnzoo.org/

Bangladesh National Parliament House

Bangladesh National Parliament House

https://www.parliament.gov.bd/Architect-and-Parliament

Botanical Garden

Botanical Garden

Museum of Independence

Museum of Independence

Shahid Minar

Central Shaheed Minar

https://beautifulbangladesh.gov.bd/loc/dhaka/92

Bangabandhu Military Museum

Bangabandhu Military Museum

https://bangabandhumilitarymuseum.com/

Professor Dr. Muhammad Kamarul Kabilan is a Professor at the School of Educational Studies, University Sains Malaysia, Penang. His research interests include ICT and English language education and, professional development and critical practices of teachers. He has published widely in his area of research in reputable journals such as TESOL Quarterly, British Journal of Educational Technology, Computer and Education, the Internet and Higher Education, and Professional Development in Education.

The School of Educational StudiesUniversiti Sains Malaysia

Professor Muhammad Kamarul Kabilan, PhD

News & Events

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Dr Farzana Yesmen Chowdhury is a social science researcher with a keen interest in all aspects of sociolinguistics and cultural studies. She has also worked as an educationalist in the fields of TESOL Studies and Applied Linguistics since 2005. She obtained her PhD and Master of Applied Linguistics in TESOL Studies from the University of Queensland, Australia. She has a strong preference for research of an ethnographic and narrative nature.

Director &; Associate Professor, Institute of Modern Languages

University of Chittagong, Bangladesh

Farzana Yesmen Chowdhury, PhD

News & Events

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Dr Abu Saleh Mohammad Rafi is a critical applied linguist with a proven track record of publishing in top-tier journals and editing special issues. He has also been invited to contribute to collections edited by leading international academics, delivered keynote addresses in multiple countries, and has teaching experience in universities in both the global south and the global north. Before joining ULAB, Dr Rafi conducted doctoral research on translanguaging pedagogies in the context of Bangladesh higher education at James Cook University, Australia. In Bangladesh, he taught at East West University, North South University, Daffodil International University, and Jahangirnagar University.

Dr Rafi authored articles in top international journals including the International Journal of Multilingualism; Language, Identity & Education; Teaching in Higher Education; Classroom Discourse, and Critical Inquiry in Language Studies. Furthermore, he published in the Bilingual-Multilingual Education Interest Section (B-MEIS) of the

TESOL International Association and contributed invited chapters to several edited volumes published/forthcoming from Springer Nature, Routledge, De Gruyter, and Multilingual Matters. To date, he has delivered 11 invited talks at universities in Australia, Bangladesh, Pakistan, India, and the Philippines. Currently, Dr Rafi is co-editing special issues for three top journals and a handbook on professional learning with leading international academics, as well as writing an invited critical review for an Oxford University Press journal.

Associate Professor, Department of English and Humanities

University of Liberal Arts Bangladesh, Bangladesh

Rafi Saleh, PhD

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Workshop on Architecture Model Making Held at BRAC University

Firdous Azim is a Professor of English and is chair of the Department of English and Humanities at BRAC University. She is also a member of Naripokkho, a woman’s activist group in Bangladesh. She obtained her Ph.D. from the University of Sussex, UK in 1989 and was awarded a fellowship by the University as one of its fifty most distinguished alumni and academics in 2012.

Firdous Azim has published widely both in the field of post-colonialism and literature, as well as on feminist issues. Her books include The Colonial Rise of the Novel (Routledge, 1993) and Infinite Variety: Women in Society and Literature (University Press Limited, Dhaka, 1996). She is a contributing editor for Feminist Review, for which she has edited a special issue entitled South Asian Feminisms: Negotiating New Terrains (March 2009). She has edited a book entitled Complex Terrains: Islam, Culture, and Women in Asia (Routledge, March 2013) and co-edited several books including Infinite Variety: Women in Society and Literature (UPL, 1994) and Other Englishes: Essays on Commonwealth Writing (UPL, 1991).

She is a member of the editorial board of the Journal of Inter-Asia Cultural Studies as well as a member of the Board of the Inter-Asia Cultural Studies Society.

Her current work is based on research and writing about the cultural history of women in Bangladesh.

Department of English and Humanities

BRAC University, Bangladesh

Professor Firdous Azim, PhD

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BRAC University Students Showcase Civic Engagement Projects at the 2024 Get Engaged: Student Action and Youth Leadership conference

EEE student selected for Huawei program in China

BRAC University Architecture Faculty Achieves Top Honors in National Design Competition

Workshop on Architecture Model Making Held at BRAC University

Md. Zulfeqar Haider is a Professor and Senior Specialist (Curriculum), National

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National Curriculum and Textbook Board (NCTB), Bangladesh

Professor Md Zulfeqar Haider, PhD

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Dr. Asifa Sultana joined ENH as a faculty member in 2008. She completed her PhD in the language development of Bangladeshi children from the University of Canterbury, New Zealand in 2015. She is currently involved in a range of research projects on child language development that connects Bangla child language data to the vast crosslinguistic body of work conducted in this area. She teaches courses in the Applied

Linguistics and ELT stream and supervises theses of both undergraduate- and graduate-level students at the department.

Department of English and HumanitiesBRAC University, Bangladesh

Professor Asifa Sultana, PhD

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Mahmud Hasan Khan is a Professor, teaches English in the Department of English and Humanities, University of Liberal Arts Bangladesh. He has two PhDs from the International Islamic University Malaysia and Macquarie University, Sydney, Australia. He examines the debates around the medium of instruction policy and politics/formation of identity from a discourse analytical perspective. He has taught and published scholarly articles and book chapters in his areas of interest. His recent publications have appeared in journals like Discourse: Studies in the Cultural Politics of Education, Asia Pacific Journal of Education, Higher Education, and Multilingua: Journal of Cross-Cultural Studies.

Department of English and Modern Languages

Independent University, Bangladesh

Professor Mahmud Hasan Khan, PhD

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Available Program For Admission

BRAC University has been ranked No. 1 nationally alongside three other universities and 801–1000 globally in the Times Higher Education (THE) World University Rankings 2024.This achievement is particularly significant because BRAC University has participated in THE World University Rankings for the very first time this year. The Times Higher Education World University Rankings rely on various calibrated performance indicators, evaluating institutions across five vital domains: teaching quality, research environment, research excellence, industry engagement, and international outlook. THE ranking assessed 1,904 universities across 108 countries and regions in its 2024 edition.“BRAC University's achievement in overall ranking is a testament to the quality of our teaching, research and impact. Our global ranking in Research Quality is 533 and our Citation Impact score stands at an impressive 99.8, reflecting the significant impact of our research” said Professor Syed Mahfuzul Aziz, Pro Vice-Chancellor and Acting Vice-Chancellor, BRAC University.He also mentioned that BRAC University remains committed to enhancing its position as an impact making institution of higher learning. “We shall continue to take steps to ensure we build on our success through targeted strategies to enhance our teaching, research, industry engagement and internationalization. You will be pleased to know that we are working towards this goal in collaboration with renowned institutions around the world”, he added.Ranking

Link- https://www.timeshighereducation.com/world-university-rankings/2024/world-ranking#!/length/25/locations/BGD/sort_by/rank/sort_order/asc/cols/stats

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BRAC University ranks country’s no. 1 University in Times Higher Education (THE) World University Rankings 2024

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BRAC University conducts Fire Drill at its campus

BRAC University Research for Development Club Explores Marketing and Manufacturing Excellence at Akij Food and Beverages Ltd

BRAC University (BRACU), in collaboration with the Bangladesh Fire Service and Civil Defence, successfully conducted a comprehensive fire drill at its campus recently. The exercise aimed to test and enhance the university's emergency preparedness and response capabilities.

The drill commenced at 11:00 AM with the activation of a siren, followed by the simulation of smoke emanating from top floors. Within minutes, firefighters from Baridhara Fire Station arrived on the scene, demonstrating their rapid response proficiency. They executed a series of operations, including the evacuation and treatment of simulated casualties, who were then transported by ambulance.

Subsequently, a ladder truck was deployed to douse an imaginary flame, showcasing the effective use of firefighting equipment.

Personnel from BRACU's Operations department and Medical Centre actively participated in the drill, performing their designated roles under the supervision of firefighters. This collaborative effort ensured that all procedures were meticulously followed, underscoring the importance of preparedness in such critical situations.

The fire drill provided a valuable training opportunity, reinforcing the necessity of coordinated efforts during emergencies. BRAC University is committed to conducting such drills regularly to ensure the safety and well-being of its students, faculty, and staff, while fostering a proactive approach to disaster management.

Mohammed Sajedul Karim, Director of Operations at BRAC University, expressed his satisfaction with the drill’s execution, stating, "Regular drills are essential in ensuring that everyone knows how to respond in an emergency, and today's exercise has demonstrated our preparedness and coordination."

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BRAC University Hosts Orientation for Summer 2024 Semester

BRAC University warmly welcomed its new students for the Summer 2024 semester with an engaging orientation program held today, Monday, 27 May 2024, at the Multipurpose Hall of its Merul Badda climate-friendly inner-city campus.

The event was graced by the presence of Professor Syed Mahfuzul Aziz, Pro Vice-Chancellor of BRAC University, who inspired the new students with his address. He emphasized that learning is a lifelong journey essential for adapting to change. He encouraged students to focus not only on their studies but also on developing life skills, critical thinking, and maintaining a strong moral character. Professor Aziz highlighted the importance of being constructive, grateful for the support they receive, and eager to share knowledge, particularly to uplift those who are disadvantaged.

Rizwan Bin Farouq, Honorary Secretary to the Board of Trustees and Chairman of the Executive Committee of The Duke of Edinburgh’s Award Foundation Bangladesh, was the guest of honor. He shared insights about the award program, explaining how the six-month initiative builds confidence, promotes physical activity, facilitates skill acquisition, and encourages community service.

The event also featured speeches from Dr. Dave Dowland, Registrar; Professor Arshad Mahmud Chowdhury, Dean of the BSRM School of Engineering; and Dr. Rubana Ahmed, Proctor. The program was moderated by Tahsina Rahman, Joint Director of Student Life.

In addition to the inspiring speeches, BRAC University presented special recognition awards to several students for their outstanding accomplishments, highlighting the university's commitment to celebrating excellence and achievement.

The orientation marked the beginning of an exciting new chapter for the Summer 2024 students, setting a positive tone for their academic journey at BRAC University.

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On May 25, 2024, the BRAC University Research for Development Club embarked on an educational visit to the Akij Food and Beverages Ltd (AFBL) plant in Dhamrai. The visit provided club members an opportunity to delve into the intricacies of modern marketing and manufacturing practices, showcasing Bangladesh’s commitment to producing top-quality snacks and beverages.

Led by Club Advisor Dr. Tarnima Warda Andalib and Club President Abtahi Noor, the group included members Saif Safayat, Marjana Maria, Abrar Jahin, Faria Alam, Nafisha Tanjim, Fahim Muntasir, Ishmam Jahan Oyshi, Mubassir Rahman, Mashrafi Azad, Mubashira Mahboob, Ragib Shahriyar, and Tasnuva Salvin Anushka.

The visit to the AFBL plant revealed a range of state-of-the-art machinery, smart and hygienic manufacturing processes, and complex systems. These elements underscore the company's dedication to delivering high-quality products to consumers. The club members gained insights into the processes, workflows, and strategic decisions that have propelled AFBL to the forefront of the beverage industry, both domestically and internationally, with exports reaching 47 countries worldwide.

This excursion highlighted the role of innovative manufacturing and strategic marketing in establishing and maintaining a leading position in the competitive global market.

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BRAC University Hosts Orientation for Summer 2024 Semester

BRACU Duburi, an autonomous underwater vehicle developed by BRAC University students, has been adjudged runner-up of Robosub 2023, one of the world’s most prestigious robotics competitions. The team also received Ingenuity Special Award in this competition.

Hosted by Robonation and co-hosted by Naval Information Warfare Centre (NIWC) Pacific of the USA in conjunction with the Office of Naval Research at Transdec, California, USA, RoboSub is an international competition that invites participants to tackle simplified versions of challenges facing the underwater maritime industry. This year 34 teams from globally reputed universities and education institutions took part in the competition in California, US from 31 July to 6 August. The first place was secured by the National University of Singapore while University of Alberta got the third place.

Pioneering underwater robotics research in Bangladesh, BRACU Duburi is capable of autonomous underwater navigation and machine learning powered data collection. By analyzing the collected data, it can provide sustainable solutions for water pollution reduction, search-and-rescue missions and autonomous monitoring and research in areas where manned missions are quite impossible. Built keeping the Bangladeshi scenario in mind, its low light camera can operate in opaque water and low light situations while its 8 state-of-the-art thrusters enables it to traverse turbulent waters.

Dr. Md. Khalilur Rahman, Professor, Computer Science and Engineering Department of BRAC University was the Advisor and Sayantan Roy Arko, Research Assistant and Md Nayem Hossain Saikat, Research Assistant were Co-advisors from BRAC University. The BRACU Duburi team consists of around 50 student members, with A. T. M. Masum Billah Mishu as team lead and Md Mahfujul Haque as vice-team lead. Subteam leads are S.M Minoor Karim, Nazmul Haque Omi, Shaownak Md Ibne Shahriar Talukder, Partho Protim Sarker Joy, Samia Abdullah, Umama Tasnuva Aziz. Including contributing members Mahdi Uddin Ahmed, Soumik Hasan Shranto, Md. Musfique Billah and Md Shazzad Hossain Chowdhury.

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BRACU Duburi emerges runner-up of Robosub 2023

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BRACU Alumni-led Startup Shines in Shark Tank Bangladesh

BRAC University Hosts First Coding Competition for Non-CS Students

The Computer Science and Engineering (CSE) Department at BRAC University held its first-ever coding competition for students from non-computer science backgrounds. The event concluded with an awards ceremony where the top three teams were recognized with prizes and certificates.

With 327 students enrolled in the course "Introduction to Computer Science (CSE101/CSE103)," the competition saw enthusiastic participation, showcasing impressive coding skills. The event was successfully organized with the assistance of CSE101/CSE103 faculty members. During the ceremony, CSE Chairperson Dr. Sadia Hamid Kazi distributed the certificates to the winners. The winning groups were:

1st Place:

Rafayet IslamAbdullah AnsariAdiya Ashraf

2nd Place:

Miskatul Jahan OisheZayed HassanSuraiya Hossain

3rd Place:

Sajia AfsaraAnwesha Dutta

The event was graced by the presence of Dr. Ashraful Alam, Dr. Aniqua Nusrat Zereen, the course coordinator, and other esteemed faculty members. This initiative not only highlighted the diverse talent within the university but also encouraged interdisciplinary learning and collaboration.

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BRAC University Hosts First Coding Competition for Non-CS Students

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BUCC holds webinar on “Pathway to Research and Industry in the USA”

CSE Faculty wins Best Presentation Award at ICHMI 2024

BRAC University signs MoU with BRAC

BRAC and BRAC University signed a Memorandum of Understanding (MoU) on May 9th, 2024, to strengthen the skills training offered by BRAC Institute of Skills Development

(BRAC ISD) . This collaboration aims to revamp BRAC ISD's curriculum, incorporating digital trends, industry needs, and diverse learning styles. The MoU signifies a commitment to quality training by evaluating current trainers as well as curriculum. Joint efforts will focus on developing self-paced training modules, expanding the content repository, and attracting a wider range of youth, particularly for technical and digital courses.

The MoU also includes the way for collaborative research and international opportunities leveraging BRAC's extensive network. This strategic partnership signifies BRAC's dedication to equipping Bangladesh's youth with the skills needed to thrive in the digital age. Signatories to the MoU included Tasmiah Tabassum Rahman, Associate Director of BRAC Skills Development Programme, and Sadia Hamid Kazi, Chairperson and Associate Professor of BRAC University.

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Dr. Aniqua Nusrat Zereen, Assistant Professor, Department of Computer Science and Engineering, BRAC University, has attained the best presentation award in the International Conference on Human–Machine Interaction (ICHMI 2024) held in Xi'an, China. The paper titled "A Two-Stage Framework for Dynamic Two-Hand On-Screen Keyboard-Mouse Interaction", co-authored with Sadman Sakib Alif and Nasim Anzum Promise.

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BUCC holds webinar on “Pathway to Research and Industry in the USA”

BRAC University signs MoU with BRAC

BRAC University Computer Club (BUCC) of BRAC University organized a webinar titled “Pathway to Research and Industry in the USA” on 4 June 2024 to share insights on undertaking PhD research and securing careers in the US tech industry.

The guest speaker of the event was Dr. Swapnil Sayan Saha, an algorithm development engineer at multinational technology company STMicroelectronics. Istiak Zaman Shuvo, Senior Executive, Human Resource Department, BUCC, moderated the webinar.

Dr. Swapnil started off by highlighting the necessity of determining a purpose when undertaking research for a PhD. He firmly believed that identifying a problem and the extent to which it affects people can lead to better research outcomes. He discussed the stages of problem formulation, purpose of a solution and solution improvement cycle which are crucial aspects when it comes to humanizing the problem.

Afterwards, Dr. Swapnil shared the benefits of having work experience along with relevant learning for fruitful research. He strongly suggested focusing more on reading, especially conference papers, journals and newspapers, saying it helped attain a better understanding of problems and generates ideas for data collection. Later, he shared ways to undertake comparative analysis.

On preparing for careers in the US tech industry, Dr. Swapnil said it mainly involves three stages -- preparation, application and interviews. He recommended focusing on creating a good portfolio and LinkedIn profile and learning about the fields involving the jobs targeted to prepare well for the interviews. Moreover, he suggested applying for jobs at multiple places as it increases the chance of a callback and adding keywords to make the application appear better.

Regarding interviews, Dr. Swapnil said there could be multiple stages like research interviews and problem solving with multiple solutions. It is very important to communicate well with the interviewer by relating to problems solved in the past and presenting oneself in a way that leaves a good impression and increases one’s chance to get to the next stage, he said.

In response to queries, Dr. Swapnil shared different methods of tackling the application process for PHD programs. Overall, he highlighted the importance of engaging in extracurricular activities alongside academic endeavors, working smarter and not harder and keeping in touch with peers and professors even if in a different field of work.

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Extension of the Deadline to Make Payment-Summer 2024 Semester Policy & Procedure

Student Inquiry Email Address

Circular to Apply for Installment Payment Options, Summer 2024 Semester, Department of Pharmacy, BRAC University, Policy & Procedure

Extension of Deadline to Apply for Installment Payment Options, Summer 2024 Semester till 08 June 2024 (Saturday) Policy & Procedure

BRAC University unveiled its new campus through a “Meet and Greet with Media” event under the theme “□□□□□□□ □□□□□□□ □□□□□□□ □□□□□□□□”. The event took place on Thursday, 29 Fe green campus located in Merul Badda which features environmental sustainability. The occasion highlighted BRAC University's commitment to academic excellence, cultural enrichment, and environmental conservation. Various aspects of BRAC University's dedication to environmental sustainability were showcased during the event. The university also shared its aspiration to lead in providing a comprehensive educational experience by engaging in environmentally friendly initiatives, showcasing local performances, and fostering discussions on pertinent issues.

In this “Meet and Greet with Media” event, Tamara Hasan Abed, Chairperson, Board of Trustees, BRAC University, along with other members of the Board of Trustees, Professor Syed Mahfuzul Aziz, Acting Vice-Chancellor, deans of schools, department chairpersons, and journalists from various media outlets were present.

Covering 7 acres of land in Merul Badda, this expansive BRAC University campus is the first environmentally friendly and sustainable advanced city campus in Bangladesh. Inspired by the ecosystem of the Sundarbans, the campus has been constructed to offer a blend of nature and architecture. The design, construction, and environmental aspects of this building involve experts from various countries, including Singapore, China, and Germany, contributing to its various unique features.

The building utilizes modern technologies such as cross-ventilation and hybrid thermal management systems, allowing light and air to enter from all sides. The use of Aerodynamic Fins ensures the optimal flow of air within the building. The greenery, resembling a green carpet on the building's facade, provides oxygen, and the hybrid cooling system brings fresh air into the classrooms. As a result, even if students sit for long hours in class, fatigue will be minimized. These features contribute to reduced dependency on air conditioning, leading to the building's 40% energy self-sufficiency.

Numerous rain chains have been installed to allow rainwater to reach trees and vegetation within the building and use the excess to replenish water bodies on the ground floor. The building also houses an advanced sewage treatment plant for waste management.

On the roof, there are solar panels generating 1.5 megawatts of electricity, providing 25% of the energy required for the building. The majority of the space in this modern and aesthetic building has been kept open for the students. In these open spaces, they can engage in mutual interactions and participate in various extracurricular activities. The campus is designed with universal accessibility so that students with special needs can easily access the campus and move around all areas comfortably.

Tamara Hasan Abed, Chairperson, Board of Trustees, BRAC University, shared the inspiring story behind the construction of the university's new eco-friendly campus and the educational philosophy of Sir Fazle Hasan Abed. She said, "Sir Fazle's vision was to establish a university where learning and pedagogy would flourish while coexisting with nature. The new campus is designed to combine nature and modern architecture to inspire future generations to think differently about life. While urbanization is an inevitable consequence of development, we must discover innovative solutions to safeguard nature. This new campus will serve as a benchmark for students and professionals in urban planning and environmental conservation."

BRAC University's new campus is envisioned to serve as a laboratory for knowledge and science, as expressed by Professor Syed Mahfuzul Aziz, Acting Vice-Chancellor, BRAC University. He states, "Essentially, we want to establish this new campus of BRAC University as a hub for knowledge and science. Our aspiration is to make this university a flagship institution in Bangladesh through ethical research and the provision of high-quality education. We are working towards that goal."

At the conclusion of the event, the closing remarks were delivered by Salahdin Imam, a member of BRAC University's Board of Trustees and the Chair of the Campus Development Committee. The event was hosted by Khairul Basher, the Director of the Office of Communications, BRAC University.

Following the program, invited journalists were given a tour of BRAC University's new campus by Ariful Islam, Chief Financial Officer of BRAC University, Sajedul Karim Chowdhury, Director of Operations of BRAC University, and Shafiqul Islam, Architect of the BRAC University campus project.

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